Real Analysis

— *EYNTKA* Series —

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The goal of this document is to encapsulate all the notes and intuition I will gain taking MAT457. The preliminary chapter will be a quick reminder on why we care to generization our notion of integrability.

Assumptions in reading this text

1. basic definition and notions in topology are assumed, covering at least:
   (a) open/closed sets, closure and interior, Hausdorff space, limit definition of closure (given the space if Hausdroff, or at least T_1), continuity
   (b) Compactness, connectedness, Heine-Borel Bolazno-Weiestass in $\mathbb{R}^n$, Path-connectedness
   (c) subspace topology, product topology, Quotient topology
   (d) Metric spaces, many equivalent metric on product spaces,
2. A good understanding of Riemann integration
3. A basic understanding of point-wise convergence and uniform convergence. For point-wise convergence, we'll use $f_n \to f$ notation. For uniform convergence, we'll use $f_n \rightrightarrows f$.
4. Defining and understanding basic properties of sup, inf, lim sup, and lim inf. In particular, remember that if S is a set and $M = \sup S$, for all $\epsilon > 0$, there exists an $s \in S$ such that

   $$M < s + \epsilon$$

   And similarly for the infimum, with $s < m + \epsilon$.

Since we are doing analysis, the axiom of choice is assumed. There are some fascinating results we get by assuming the axiom of choice which we will point out.

What is Analysis?

Now that I've taken the course and have done some reserach in PDE's I feel like I've gained a perspective on how to think about analysis. Fundamentally, Analysis is about *estimates* and *approximation*, especially on functions; a lot of the fundamental results in anlysis is centered around these two concepts. Since we are working over $\mathbb{R}$, we have the privelge of having an *ordering*, allowing us make precise the notion of a number being bigger than another (something we can't do over $\mathbb{C}$ or many other fields). The following bullet points elaborates on this:

1. Many times, we will want to give a numerical value to a function or a space of functions. To accomplish this, we will often define a functional (or subfounctional) a space of functions. For example, the integral $\int : C^\infty(\mathbb{R}^n) \to \mathbb{R}_{>0}$ takes in functions and assigns them numerical value giving us some idea of how fast the function grows and the total "energy", "work", or "total value" the function has. Notie that the function $\int$ as we've defiend it above is not well-defined, there are function whose integral is infinite. Conversely, we are missing many integrable function (recall if a function is merely almost everywehre continuous, it is integrable). We will refine the domain in this book.

   The integral is not the only functional. We will soon find there are in fact many functions, letting us control many properties of a function such as:

(a) the total boundedness of the function (i.e. that it nowhere goes to infinity)

(b) the "total work" of the function (i.e. a finite integral)

(c) the speed decay of a function (i.e. that it goes to zero fast enough)

(d) the "frequency" of a function (i.e. that the function doesn't start oscillating too rapidly)

(e) the regularity of a function (i.e. that the derivative or derivatives dont' explode too fast or oscilate to fast)

2. interpolation: (I'm thinking of things like $L^p + L^q$ or $L^p \cap L^q$ and how useful this is and how it shows up)

3. "perfect approximatinos" via sequences: At some point, we all must have wondered how in the world can we find the area of a curved shape(?!). We found out when learning through Riemann integrals that the key idea is that we can approximate it with rectangles, and then make the approximation "perfect". This idea of being able to find some simpler objects and use them to approximate more complicated ones is at the heart of analysis. For example, those rectangles we used in Riemann integration can be thought of as characteristic functions. Thus, something is Riemann integrable if the "upper" and "lower" characteristic functions, when taking the limit of the partitions, are equal.

   More generally, we will often be working over many different types of functions (most of this book will in fact be about how to classify different function space). One key part of working with function spaces will often be to find a subset of those functions that we can use to approximate any other function: such a set is called a *dense set*. Dense sets are sort of like a basis in vector-spaces: just like any vector in a k-vector space is a k-linear combination of basis elements, elements in most function space will be the result of a *sequence of functions* where each function in the sequence comes from the dense set.

   Two major examples of this is in the theory of integration and Fourier analysis. In integration, we will approximate any integrable function via *characteristic functions*, and in the theory of Fourier transforms, we will approximate integrable functions via its *fourier series*, where each function in the fourier series will be part of a special dense set we will define later.

4. Estimates: having given numerical values to function and having interpreted them through interpolation or a good sequence, we can now often solve problems by bounding the "badness" of the thing we are measuring. A super-simple example is that if f is bounded, then there exists a constant C such that
$$|f(x)| \leq C$$
   for all x in the codomain. Naturally, as we do more analysis, the estimates will be more sophisticated, but the idea remains the same: we will simplify our problems by bounding what we is bad and trying to "eliminate" it. One of the ways we will do this is by figuring out how common operations (such as $\frac{d}{dx}$, $\int$, $+, \cdot$, and so forth) which usually "complicate" our problem can be eliminated with some appropriate estiamtes. For example, let's say f is a bounded function, and we want to work with:
$$\left| \int_D f \right|$$
   then this can be a hard value to find. Howver, we can eliminte the complexity by doing:
$$\left| \int_D f \right| \leq \int_D |f| \leq \int_D C$$

which naturally now is much easier to work with. Of course, we might have over-simplified our problem with this estimate, which is why we will have a lot of different estimates playing different roles depending on our situation.

In terms of the theory of differentiation, real differentiable functions are very different from complex differentaible function. Real differentiable function are a strictly local property, while being complex-differentiable (holomorphic) entails many global properties (namely, they are *harmonic*, and give *closed forms*). Real smooth functions also out-number holomorphic functions (leading to their prominence in Distribution Theory), again due to the harmonic properties of holomorhpic functions being so restrictive (all holomorphic functions are locally a power series with coefficients given by contour integrals of any size, showing how much "global" information is captured in local representatives). This also means that holomorphic functions have some nicer properties (for example, they are all analytic functions), and lead to important results in PDE's with wave equations and algebraic geometry, and they can also be *factored* into an infinite polynomial up to an entire holomorphic function[1] (an impossible thoerem for real smooth functions due to thier much greater flexibility). But these reaons also show why complex analysis is a separate field of study, moving more towards sheaf theory and analytic number theory rather than PDE's and harmonic analysis[2].

[1]see Weiestrass Factorization Theorem

[2]Though there will be some interesting motivation for harmonic analysis coming form elementary complex analysis

Preliminary Integration

In pre-university education, it is taught how to find the area of some simple shapes like rectangles, triangles, cubes, spheres, and so on. Later on, you learn how many of these shapes can be captured as the area under a particular graph (the graph usually being continuous). This means that the notion of area can be represented by functions (in particular the graph of a functions), and so it becomes a meaningful question to ask about how to find the area under the graph.

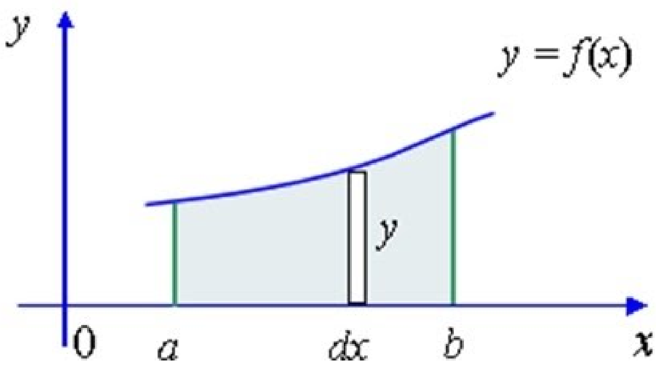


Figure 1: integrating under a graph

Let's say that f is the 1 (or more) dimensional continuous function over $\mathbb{R}$ (or $\mathbb{R}^n$), and we are trying to find the area under the graph. The first attempt at this might be the following (called Darboux sums): Given a partition the domain $P = \{x_1, x_2, ..., x_n\}$, we can make many rectangles which are all easy to measure. Then, as the partition gets finner (size$(P) \to 0$, $|P| \to \infty$), we will get a better an better approximation of the area.

If we take this approach, we will introduce some level of choice: there are many ways in which this partition can be formed, as well as a choice of the "height" of the rectangle. For example, you can choose the lowest or highest point:

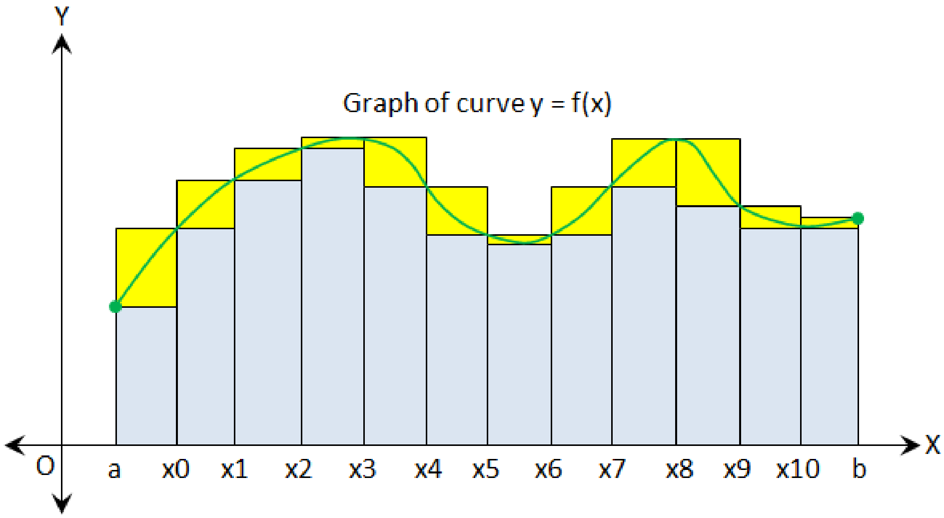


Figure 2: lower and upper sums

You can also choose any points in between, though what "in between" means starts to get hard to define in higher dimensional spaces where different orders can be put onto the "surfaces" (or hyper-surfaces) of the top of the rectangles (i.e. hyper-rectangles). However, maximal and minimal point is always well defined on these surfaces (or better, inf and sup always being well defined on these bounded surfaces), and even better, choosing the maximal point for each rectangle height upper-bounds all possible other choices of Darboux sums, and the minimal point for each rectangle's height lower-bounds all possible choices of Darboux sums. To be more precise, we'll replace the maximum and minimum condition with sup and inf, in which case the resulting sum is called the *Riemann sum*.

Intuitively, the upper and lower Riemann sums must match up and exist as size$(P) \to 0$ and $|P| \to \infty$. It would be strange if they don't match up, and so we can (as a start) define a function that is not "too weird" as one where the upper and lower Darboux sums match as size$(P) \to 0$, that is

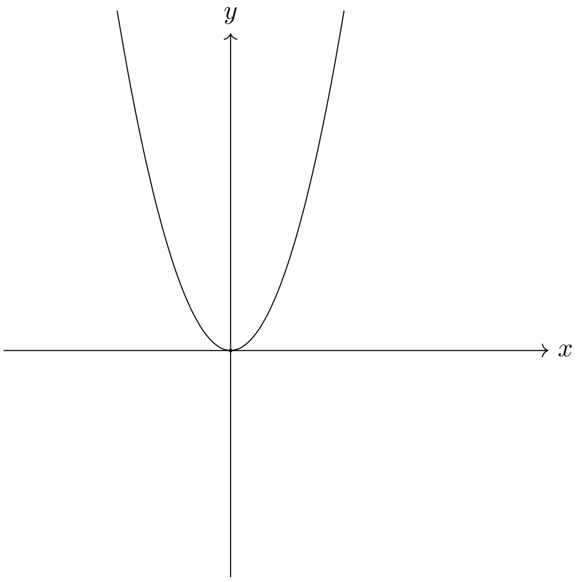
$$\lim_{n\to\infty} \sum_{i=1}^{n-1} \left(\inf_{[x_i, x_i+1]} f(x) \right) \cdot (x_{i+1} - x_i) = \lim_{n\to\infty} \sum_{i=1}^{n-1} \left(\sup_{[x_i, x_i+1]} f(x) \right) \cdot (x_{i+1} - x_i)$$

If this is the case, f is said to be *Riemann integrable*. Fortunately, it turns out that if f is Riemann integrable on one partition, it is Riemann integrable on all partitions, and so we don't need to "remember" our initial choice of partition when we say f is Riemann integrable (we don't say "f is Riemann integrable on partition p, while f is not Riemann integrable on partition q").

All continuous functions over a bounded domains will be integrable (so trigonometric functions, polynomials, log, $\sqrt{}$ over the appropriate domain, and so forth). Since any geometric shape learnt in pre-university education can be bounded by a continuous function, we have just generalized the idea of measuring those areas[3]. In fact, being continuous *almost* encapsulates all possible functions. To understand the generalization, consider

$$f_{\text{no }1} : [-1,1] \to \mathbb{R}, \qquad f_{\text{no }1}(x) \begin{cases} x^3 & x \neq 1 \\ 0 & x = 1 \end{cases}$$

Notice that there is a gap in the line



however, this gap has "zero measure", we would think that the result should be the same. And in fact it is! The function $f_{\text{no }1}$ is also Riemann integrable (as can be checked) and will have the same value as x^3 on $[-1,1]$. In general, if finitely (or even countably) many points are ommited, then f will still have the same measure. Since $f_{\text{no }1}$ is no longer continuous, but it as just off by some finite number of points, we say that f is continuous *almost everywhere* [4]. It should be reviewed that f is Riemann integrable if and only if f is continuous almost everywhere.

Stepping aside from measuring area under a graph for a moment, we turn to the study of another very common notion in analysis: limits. Limits are of fundamental importance in Analysis. They are the tool used to construct or define many different types of analytical objects (ex. $\mathbb{R}$, continuity, differentiability, Fourier transforms, sequences and series). More generally, the concept (and formalisation) of a limit is how mathematicians get to study infinites which are "well-defined". In my mind, limits are one of the distinguishing feature between algebra and analysis: many algebraic properties "break down" or are different when we try to bring in more than finitely many objects (ex. products of infinite groups cannot be direct products in **Grp** since $1+1+\cdots$ is not defined (hence they are direct sums), the dimension of $\mathbb{Q}[x]$ and $\mathbb{Q}[[x]]$ as $\mathbb{Q}$-algebras differ, etc.). A good definition of limit is important to be able to define objects "at infinity" (or similarly, at the "infinitely small"), since if it is not well defined, then we can make the "result" non-unique.

Here is a silly example of that: what is $\infty - \infty$? Maybe we can say that the answer is 0? or ∞? If we are not careful with our definition, it can actually be any number. First, "notice" that

[3]though we won't go into the details of finding those areas

[4]Later, almost every continuous will mean continuous up to a zero measure set once we properly define a measure

$1 + 1/2 + 1/3 + \cdots \text{“} ='' \infty$, or more precisely, the series $S_n = \sum 1/n$ is unbounded from above. Since these two are “equal”, we can treat them as the same. Then assuming the usual algebraic properties of distributivity work over countably many components, we hvae

$$(1 + 1/2 + 1/3 + \cdots) - (1 + 1/2 + 1/3 + \cdots) = (1 + 1/2 + 1/3 + \cdots) - 1 - 1/2 - 1/3 - \cdots$$

Now, pick any $x \in \mathbb{R}$. Then this sequence can be “rearranged” to converge to that number. Start by picking 1. If $x > 1$, pick $1/2$, if it's less pick -1. Keep doing this, and this will evidentially converge to x! Thus $\infty - \infty$ can be any real number we want!

This is why the idea of a limit is important: If the limit exists, we mean that there is a unique value for which the limit will converge to. This is why limits are used to define objects with some “countably infinite” properties [5].

One important consequence of limits being unique is when we use limits to construct new functions from a limit of functions! As review, you should check the construction of a *continuous but nowhere differentiable function*. Another example is with Fourier series and Fourier transformations which are used to encode and decode signals. Another example the definition of compact exhaustion for imperfect integrals. There are many more yet to come as well!

Now, returning to integrals for a moment, we can ask ourselves if integrals commute with limits, that is

$$\lim_{n\to\infty} \int f_n = \int \lim_{n\to\infty} f_n$$

where the limit here is *pointwise limit*. Note that uniform limits (if it exist) will sort of commute:

$$\text{unif}\lim_{n\to\infty} \int f_n = \int \lim_{n\to\infty} f_n$$

However, point-wise limit doesn't generally do

Example 0.0.1: Limit Doesn't Commute

Take $\chi_{\mathbb{Q}} : [0,1] \to [0,1]$ to be the indicator function on $\mathbb{Q}$, that is

$$\chi_{\mathbb{Q}}(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [0,1] \\ 0 & \text{otherwise} \end{cases}$$

Notice that $\chi_{\mathbb{Q}}$ is nowhere continuous, and hence is not Riemann integrable. We can see this more clearly by noticing that the upper Riemann sum is 1 and lower Riemann sum is 0. However

$$g_n(x) = \begin{cases} 1 & \text{if } \frac{p}{q},\ p, q \in \mathbb{N}, p \leq n \\ 0 & \text{otherwise} \end{cases}$$

is integrable, and $g_n \to \chi_{\mathbb{Q}}$, that is $\lim_{n\to\infty} g_n = \chi_{\mathbb{Q}}$. Notice that each g_n are integrable, but $\chi_{\mathbb{Q}}$ is not, that is

$$\lim_{n\to\infty} \int g_n \neq \int \lim_{n\to\infty} g_n$$

Since the right hand side is not defined.

[5] Object with some “uncountably infinite” or larger properties use a generalisation of a limit called an *ultra-filter* which we don't need to study in this course

Thus, our notion of integrability does not work will with limits! Thus, since integrals gives us useful ways of analyzing our functions, and many functions are constructed as the limit of functions, we are going to spend lots of time upgrading our notion of integrability to make it commute (under appropriate circumstances) with limits.To upgrade the notion of integration, there must be some requirements we must keep in mind while finding a suitable replacement. In particular

1. The area under the graph should be the same as the Riemann Integral
2. Geometric intuition's should still be consistent, in particular
   (a) if $f \leq g$ then $\int f \leq \int g$
   (b) $\int af + bg = a \int f + b \int g$
3. A set A can be thought to be "integrable" if we define

$$\chi_A(x) \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

   and $\int \chi_A$ is defined.

This last criterion is actually very interesting as it allows us to generalize the question from finding the area's under graphs (or measuring the area under the graph) to figuring out how to measure sets in general! In fact, starting by defining how to measure sets and then defining how to integrate function turns out to be a fruitful order in which we can learn about this generalisation of integration. The reason this order is as actually an insight by Lebesgue: defining the properties of something being "measurable" will allow us to re-define the integral with more rigour. Even better, many different spaces can have different notions of a "measure" (for example, finite spaces where we measure points, or probability spaces where sets represent the probability of an event, or physics where we might want to measure the average density of a set), which the integral is the wrong concept (it's no longer strictly about "area"), but follows the same geometric intuitions as these other spaces (as we will show). Therefore, we will start by defining the spaces over which we will want to define a way to measure sets.

Part I

Measure Theory

(I feel like I should split the book in two, with all the measure theory/integration build up, followed by the more function-analysis stuff afterwards)

1

Measures

As a start, we might want to define a reasonable definition of what it means to measure any subset of $\mathbb{R}^n$. Such a function would be of the form $\mu : P(\mathbb{R}^n) \to [0, \infty]$. If μ were to exist, we would want our common geometric intuitions to hold, that is:

1. If $E_1, E_2, ..., E_n, ...$ is a countable collection of pair-wise disjoint subsets, then

$$\mu\left(\bigcup_i E_i\right) = \sum_i \mu(E_i)$$

2. If E and F are sets such that there is a function that only translates, rotates, and reflects E to get to F (i.e. a rigid motion), then $\mu(E) = \mu(F)$

3. If C is a unit cube in $\mathbb{R}^n$, then $\mu(C) = 1$

However, these three axioms are inconsistent with one another! We have actually shown this by showing if μ is defined on $P(\mathbb{R}^n)$ (even when $n = 1$), then we can construct *Vitali sets* that will have to be both 0 and ∞ in measure – a contradiction.

Theorem 1.0.1: Vitali Sets Are Unmeasruable

Vitali sets are un-measurable

***Proof*:**
Take $[0, 1]$. Partition this set as follows:

$$x \sim y \iff x - y \in \mathbb{Q}$$

In a sense, this is the set of cosets of $\mathbb{R}/\mathbb{Q}$ restricted to $[0, 1]$; we will rely on this intuition to label the equivalence classes. Since $\mathbb{R}/\mathbb{Q}$ is the set of irrational numbers modulo rational numbers, let N_q be an equivalence class of contains x that is equivalent to the irrational number q. Notice that there are uncountably many such equivalence relations.

Now, pick one $p \in N_q$ for each equivalence class, and let N be the collection of all these p's, so N contains one point from each equivalence class (notice that the axiom of choice must be used to create a non-empty N). We'll show that N is unmeasurable.

First, for each rational number $r \in \mathbb{Q} \cap [0,1)$, define:

$$N + r = \{x + r \mid x \in N \cap [0, 1-r)\} \cup \{x + r \mid x \in N \cap (1-r, 1)\}$$

Then clearly, $N + r \subseteq [0,1)$ for each $r \in \mathbb{Q} \cap [0,1)$, and furthermore, for each $x \in [0,1)$ there exists an r such that $x \in N + r$. To show this, let's say $y \in N$ is the element that was picked from the equivalence class of of $[x]$. Then by our choice of y, $x \in N + r$ since $r = x - y$ if $x \geq y$, or $r = x - y + 1$ if $x < y$. Next, notice that the $N + r$ also forms a partition: if $(N+r) \cap (N+s) \neq \emptyset$ then $x - r$(or $x - r + 1$) would equal $x - s$ (or $x - s + 1$), but they belond to distinct equivalence classes, and so $(N + r) \cap (N + s) = \emptyset$.

We have no reached the point where we can declare our contradiction. If μ satisfies the three wanted properties, by (1) and (2):

$$\mu(N) = \mu(N \cap [0, 1-r)) + \mu(N \cap (1-r, 1)) = \mu(N + r)$$

for any $r \in \mathbb{Q} \cap [0,1)$. Since $\mathbb{Q} \cap [0,1)$ is countable and $[0,1)$ is the disjoint union of the $N + r$'s:

$$1 = \mu([0,1)) = \sum_{r \in \mathbb{Q} \cap [0,1)} \mu(N + r)$$

where $1 = \mu([0,1))$ comes from (3). Now, we have two possibilities:

1. Either $\mu(N) = 0$, in which case each $\mu(N + r) = 0$ but then $0 = 1$ – a contradiction
2. Either $\mu(N) \neq 0$ (say k, so $\mu(N + r) = k$, but then $0 = \infty$ – a contradiction

In other words, no mater what $\mu(N)$ is, we will run into a contradiction!

As you can verify, this construction required that we take advantage of the countability condition (the 1st property). You might ask if weakening it to only finitely many will fix the issue, but sort of crazily it does not! In 1924, Banach and Tarski proved the following paradox now known as the *Banach Tarski Paradox*:

> Let U and V be open subsets of $\mathbb{R}^n$, $n \geq 3$. Then there exists a $k \in \mathbb{N}$ and subsets $E_1, E_2, ..., E_k$, $F_1, F_2, ..., F_k$ such that
>
> 1. all E_j's are disjoint and their union is U
> 2. all F_j's are disjoint and their union is V
> 3. There exists a rigid motion from E_j to F_j for $j = 1, ..., k$

This is kind of insane: this means that we can take a pea-size ball, and make into the size of the earth! This means that the notion of "geometry" is not actually a consistent concept in general set theory!

Fortunately for us, The sets E_j and F_j are very strange sets; most sets that are in fact measurable. To be more precise, it is consistent to say that the axiom of choice fails and that all sets are Lebesgue measurable (which will be the measure that will replace the Riemann integrability). For the purposes of this course, that means that any set we construct without the axiom of choice (and just doing normal ZF without any sneaky logical cheating) will be measurable. For more details, see

https://math.stackexchange.com/questions/1847052/most-functions-are-measurable

1.1 Sigma-Algebras and Properties

Since the axiom of choice is the source of the problem of unmeasurability in the previous problems, we start by defining a space on which we avoid unmeasurable sets, namely, we want to limit ourselves to *countable* choices. The definition should be pretty general, as it should accommodate measures in more general spaces. It should also be "arithmetically expandable", as in we should be able to measure smaller parts of the space (perhaps in more convenient orientations) and add them all up. Thus, we define an *algebra*:

Definition 1.1.1: Algebra and σ-Algebra

Let X be an arbitrary set. Then let $\mathcal{A} \subseteq P(X)$ be a non-empty collection of subsets such that

1. for any finite set of sets $A_1, A_2, ..., A_n \in \mathcal{A}$, $\bigcup_i A_i \in \mathcal{A}$
2. For any $A \in \mathcal{A}$, $A^c \in \mathcal{A}$.

If it is closed under countable union, we call it a σ-algebra. We will usually denote σ-algebra on X as $\mathcal{M}(X)$.

The σ-algebra will be our natural space in which we will show in the next section we can define a "measure function" without the previous contradiction.

I like to think of the finite closure as a more general way of saying that a space is "closed under a function", the function here being $\cup$. In this way, it's sort of the most general algebraic structure. Adding the fact that it's closed under countable union brings us further way from the realm of algebra, and hence it's a σ-algebra. The term sigma is very common to mean countable, for example, a space σ-compactness if it is the countable union of compact subspaces.

Definition 1.1.2: Measurable Space

Let X be a set and $\mathcal{M} \subseteq P(X)$ be a σ-algebra. Then $(X, \mathcal{M})$ is called a *measurable space* and a set in $\mathcal{M}$ is called a *measurable set*

Proposition 1.1.3: Properties of σ-algebra

Some immediate properties of a σ-algebra:

1. It is closed under countable intersection, since $\bigcap_i X_i = \left(\bigcup_i X\right)^c$, and by difference, since $X \setminus Y = X \cap Y^c$.
2. $\emptyset$ and $X \in \mathcal{A}$, since for any $E \in \mathcal{A}$, $E \cap E^c = \emptyset \in \mathcal{A}$ and $E \cup E^c = X \in \mathcal{A}$
3. Any countable collection of subsets $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ can be turned into a countable disjoint union, where
$$F_k = E_k \setminus \left(\bigcup_{i=1}^{k-1} E_i\right) = E_k \cap \left(\bigcup_{i=1}^{k-1} E_i\right)^c$$
and by what we've just shown, every $F_k \in \mathcal{A}$, and by construction $\cup_i E_i = \cup_i F_i$. We will use this technique often very soon (note that not all sets are nonempty, that proof requires more set-theory manipulation).
4. Given any family of σ-algebras on X, their intersection is also a σ-algebra. Because of this, given $Y \subseteq X$, there is always a unique smallest σ-algebra $\mathcal{M}(Y)$ that contains Y (there is at least always one, $P(Y)$, and so this concept is well-defined). $\mathcal{M}(Y)$ is called the σ-algebra generated by Y.

A lemma we'll point out right now to simplify future proofs is the following:

Lemma 1.1.4

If $Y \subseteq \mathcal{M}(X)$, then $\mathcal{M}(Y) \subseteq \mathcal{M}(X)$

***Proof*:**
Since $\mathcal{M}(X)$ is a σ-algebra that contains Y, then it will contain all countable unions and compliment of Y, and hence it will contain $\mathcal{M}(Y)$ (i.e. proposition 1.1.3[4]).

Example 1.1.5: σ-Algebras

1. If X is any set, then $\{\emptyset, X\}$ and $P(X)$ are trivially σ-algebras. Once we define measurable function, these will be the initial and final object in measurable spaces.
2. If X is any set, then $P(X)$ is also a σ-algebra. Usually, we will use proposition 1.1.3[5] and other tools to construct a smaller σ-algebra to not work with $P(X)$ since it's too big for most cases.
3. If X is an uncountable set, define
$$\mathcal{A} = \{E \subseteq X \mid E \text{ is countable or } E^c \text{ is countable}\}$$
this set is called the σ-algebra of countable and co-countable sets. It is useful when we want to make sure that the countable union of sets is still countable (or it's compliment is) [a]

[a] Sneakily, we use the axiom of choice for countably many choices to prove this is true

Another common σ-algebra generated on a set X is related to a topology on X, that is, if $\mathcal{T}(X)$ is a topology, of X, we will define the following σ-algebra:

Definition 1.1.6: Borel σ-Algebra

Let X be a set and $\mathcal{T}(X)$ be a topology on X. Define the *Borel σ-algebra* to be

$$\mathcal{B}_X := \mathcal{M}(\{U \subseteq X \mid U \text{ is an open set contained in } X\}$$

The elements of $\mathcal{B}_X$ are called *Borel sets*

The set $\mathcal{B}_X$ will include the open sets, closed sets, countable union and intersection of closed sets, and the countable union of countable intersections of open/closed sets, et cetera.

Since the countable intersection of open sets and countable union of closed sets are not necessarily open or closed sets respectively[1], but are members in $\mathcal{B}_X$, we will give them names: A countable intersection of open sets is called a G_δ set (G for *Gebeit* in German), and a countable union of closed sets is called a F_σ set (F for *fermé* in french). The σ is for *Summe* (sum in German) since it's a union and the δ is for *Durchschnitt* (intersection in German) since it's an intersection. We can continue this pattern and define $G_{\delta\sigma}$ to be the countable union of countable intersection of open sets, $F_{\sigma\delta}$ to be the countable intersection of countable union, and so and so on and so forth for any length of interchanging σ's and δ's. In general, this "chain" of unions and intersections does not need to terminate, and a $F_{\sigma\delta}$ (as a simple example for a chain of length 2, $\mathbb{R} \setminus \mathbb{Q}$ is a $F_{\sigma\delta}$ set, but not a F_σ set; use Baire's Categories theorem). There will be cases in which it will terminate (in fact, the majority of measure's we'll work with will have this terminate after at just F_σ or G_δ "up to a zero set", see theorem 1.4.5, and example ref:HERE for a measure that does not terminate)

One of the most common Borel sets we will be working with is $\mathcal{B}_\mathbb{R}$ with the usual euclidean topology on $\mathbb{R}$. It is useful to have an of different generating sets for $\mathcal{B}_\mathbb{R}$:

Proposition 1.1.7: Generators For $\mathcal{B}_X$

Let $\mathbb{R}$ be the real numbers and $\mathcal{B}_\mathbb{R}$ be the Borel σ-aglebra on $\mathbb{R}$ given the standard euclidean topology $\mathcal{T}(\mathbb{R})$. Then the following generate $\mathcal{B}_\mathbb{R}$:

1. The set of open intervals
2. the set of closed intervals
3. the set of half-open intervals
4. the set of open rays
5. the set of closed rays

***Proof*:**
These proofs are quite easy to prove, and should be done as an exercise. A fact that you might need to recall is that every open set in $\mathbb{R}$ can be written as a countable union of open intervals (hint: $\mathbb{R}$ has a countable basis).

[1] it is possible that the countable intersection/union of open/closed sets are open/closed, and so the word "necessarily" is required

Products and σ-Algebras

Recall that the product between two sets is defined as $A \times B := \{(a, b) \mid a \in A, b \in B\}$, or more generally $\{X_\alpha\}_{\alpha\in A}$ for a collection of non-empty sets X_α[2], with $X = \prod_{\alpha\in A} X_\alpha$,. As we know from Category Theory, the product comes equipped with set functions π_α that satisfy the universal property of products (or more generally, this is a limit over the discrete category). Though we have yet to give a category where σ-algebras are the objects (we need to define what the morphisms are[3]), we can already construct our set $\prod_\alpha X_\alpha$ that will be the product of the X_α in the category of measurable spaces.

Definition 1.1.8: Product σ-Algebra

Let $\{X_\alpha\}_{\alpha\in A}$ be a collection of non-empty sets and $X = \prod_{\alpha\in A} X_\alpha$. Let

$$\bigotimes_{\alpha\in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in M_\alpha,\ \alpha \in A\})$$

to be the *product σ-Algebra* on $\{M_\alpha\}_{\alpha\in A}$, i.e., it is the σ-algebra generated by all of the pre-images of the sets in M_α.

It is not difficult to check that the product σ-algebra is indeed a σ-algebra. If $|A|$ is countable we usually write $\bigotimes_{i=1}^\infty M_i$, and if $|A|$ is finite, we can write $\bigotimes_{i=1}^n M_i$ or $M_1 \otimes M_2 \otimes \cdots \otimes M_n$. Notice that since we allow for countable unions, it is not so easy to represent $\bigotimes_{\alpha\in A} M_\alpha$ as the product of the elements of M_α, (just like for Groups in the category of **Grp**), and hence why we do not use the $\prod$ or $\bigoplus$ notation. On the other hand, the fact that only a countable union of elements is permitted will still limit the product in the number of non-trivial components, hence justifying the $\bigotimes$ notation instead of $\prod$. In fact, it is the same as $\bigoplus$, but instead of "all but finitely many", we have "all but countably many":

Proposition 1.1.9: Countable Product σ-algebra

Let $M_n = \mathcal{M}(\mathcal{E}_n)$ be a σ-algebra for $n \in \mathbb{N}$ where $\mathcal{E}_n \subseteq P(X_n)$. If $X_n \in \mathcal{E}_n$, then

$$\bigotimes_{n=1}^\infty M_n = \mathcal{M}\left(\left\{\prod_{i=1}^n E_i \,\middle|\, E_i \in M_i\right\}\right)$$

***Proof*:**
The $\supseteq$ direction is easy (Take $\bigcap_{i=1}^n \pi_i(E_i)$ for appropriate values). For $\subseteq$, Since $X_i \in M_i$, we have that every element is inside the right hand side. Make sure you see why $X_n \in \mathcal{E}_n$ makes a difference. By lemma 1.1.4, the result follows.

We can further simplify our representation of $\bigotimes_{\alpha\in S} M_\alpha$ given a generating set for each M_α (such a set always exists, namely M_α itself is always a generating set)

[2] If they were empty, then the product would be empty
[3] this is done in chapter 2

Proposition 1.1.10: Generating set for Product σ-Algebra

Let $\{M_\alpha\}_{\alpha\in A}$ be a non-empty collection of σ-algebras, and let $\mathcal{E}_\alpha \subseteq P(X_\alpha)$ be a generating set for M_α. If $\mathcal{F} = \cup_{\alpha\in A}\mathcal{E}_\alpha$, then:

$$\bigotimes_{\alpha\in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha,\ \alpha \in A\}) = \mathcal{M}(\mathcal{F})$$

Furthermore, if $|A|$ is countable and $X_\alpha \in \mathcal{E}_\alpha$ for each $\alpha \in A$, then

$$\bigotimes_{\alpha\in A} M_\alpha = \mathcal{M}\left(\left\{\prod_{\alpha\in A} E_\alpha \,\middle|\, E_\alpha \in \mathcal{E}_\alpha,\ \alpha \in A\right\}\right)$$

***Proof*:**
Dealing with the first case, the $\supseteq$ is clear since $E_\alpha \in \mathcal{E}_\alpha$ implies $E_\alpha \in M_\alpha$, so $E_\alpha \in \bigotimes_{\alpha\in A} M_\alpha$. For the $\subseteq$ direction, If $E_\alpha \notin \mathcal{E}_\alpha$, then since $\mathcal{M}(\mathcal{E}_\alpha) = M_\alpha$, some countable unions and intersections of elements of $\mathcal{E}_\alpha$ equal E_α, proving the $\subseteq$ direction.

The countable case follows proposition 1.1.9

Borel Spaces and Products

A topological space of much study in Analysis is metric spaces. As a reminder, if $\{X_\alpha\}_{i=1}^n$ is a collection of metric spaces, then $X = \prod_{i=1}^n X_i$ has the product metric which is the generalization of the euclidean metric. As was seen in a topology class, the following metrics on a product space all define the same product topology (for notational simplicity, let's work with two metric spaces X and Y and consider $X \times Y$):

1. $d((x_1, x_2), (y_1, y_2)) = \max\{d_X(x_1, y_1), d_Y(x_2, y_2)\}$

2. $d((x_1, x_2), (y_1, y_2)) = \sqrt{d_X(x_1, y_1)^2 + d_Y(x_2, y_2)^2}$

3. $d((x_1, x_2), (y_1, y_2)) = d_X(x_1, y_1) + d_Y(x_2, y_2)$

We can ask about establishing Borel sets $\mathcal{B}_X$ on $X = \prod_{i=1}^n X_i$ with X having the product measure. It is tempting to say that this naturally splits into $\bigotimes \mathcal{B}_{X_i}$, however, this is not always the case! This is mainly due to the fact that not all topological spaces have a countable basis, and since we're allowing countable intersection of open sets, this will actually create more sets than the if we take the product of the Borel spaces:

Proposition 1.1.11: Borel sets on product Metric Space

Let $X_1, ..., X_n$ be metric spaces, and $X = \prod_{i=1}^{n} X_i$ be the product metric space. Then

$$\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$$

If the spaces X_i are separable (i.e. has a countable dense subset, and since it's a metric space it means there is a countable basis), then

$$\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} = \mathcal{B}_X$$

***Proof*:**
The proof of the $\subseteq$ direction rather simple. By proposition 1.1.10, $\bigotimes_{i=1}^{n} \mathcal{B}_{X_i}$ is generated by $\mathcal{E} = \left\{\pi_i^{-1}(U_j) \;\middle|\; U_j \text{ is open in } X_j,\ 1 \leq j \leq n\right\}$. Since the sets of $\mathcal{E}$ are open in X, by lemma 1.1.4, $\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$.
For the converse, since each X_j is separable, each have a countable dense subset. Since X_j is a metric space, we can form a countable basis with open balls with rational radii around each points in the countable dense subsets.

As a consequence, $\bigotimes_{i=1}^{n} \mathcal{B}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R}^n}$, since $\mathbb{R}$ has a countable dense subset.

Elementary Sets

This is a technical result needed for later (maybe introduce it as exercises?)

Definition 1.1.12: Elementary Family

Let X be a set and $\mathcal{E} \subseteq P(X)$. Then $\mathcal{E}$ is called an *elementary family* if

1. $\emptyset \in \mathcal{E}$
2. If $E, F \in \mathcal{E}$ then $E \cap F \in \mathcal{E}$
3. If $E \in \mathcal{E}$, then E^c is a finite disjoint union of elements of $\mathcal{E}$

Proposition 1.1.13: Elementary Families and Algebra

Let $\mathcal{E}$ be an elementary family. Then the collection of finite disjoint unions of element of $\mathcal{E}$ forms an algebra

***Proof*:**
(TBD) Let $U, V \in \mathcal{E}$, and consider $V^c = \coprod_{j=1}^{J} F_j$ (for disjoint $F_j \in \mathcal{E}$). Then

$$U \setminus V = U \cap V^c = \coprod_{j=1}^{J} (U \cap F_j)$$

notice that each $U \cap F_j \in \mathcal{E}$. So

$$U \cup V = (U \setminus V) \cup V = \coprod_{j=1}^{J} (U \cap F_j) \coprod V$$

where $\coprod_{j=1}^{J} (U \cap F_j) \in A$
Continuing inductively, can check Folland for proof, or just do it yourself.

(also, remember that separable means countable dense subset) Once the course starts

1. There is an interseting example ith logical statements!!

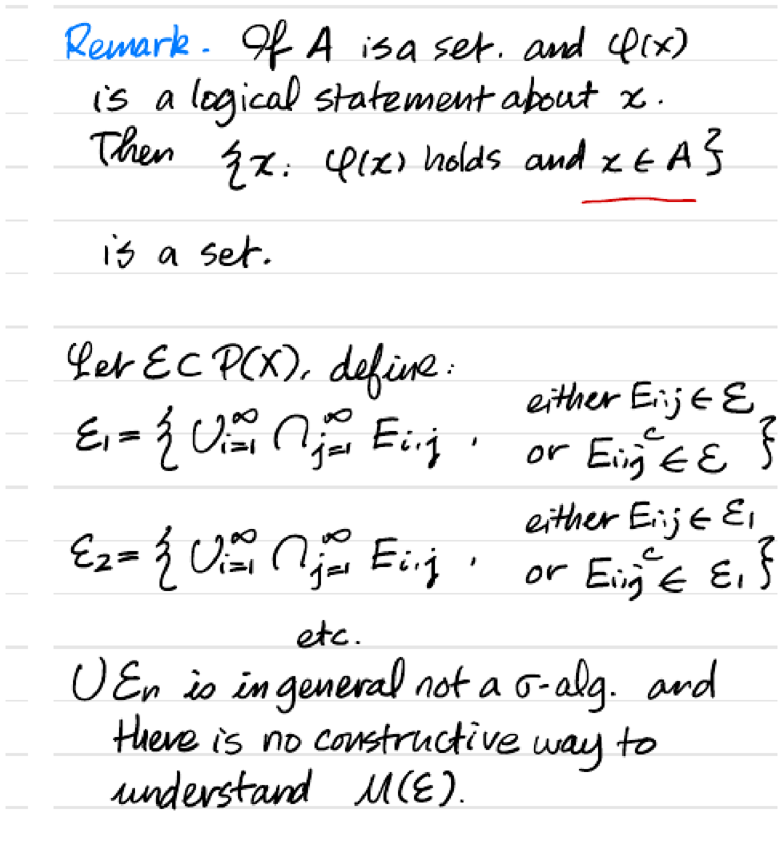


Figure 1.1: fascinating!

1.2 Measure

We can finally define the function which will give us an idea of the "size" of sets. The key idea behind the idea of notion is that the "bigger" a set is, the "larger" the the area. In other words, if μ is a measure on X, then (abusing notation a bit):

$$\mu(A \subseteq B) \;\Leftrightarrow\; \mu(A) \leq \mu(B) \tag{1.1}$$

This also means that we want the codomain of a measure to be *totally ordered*, and since we want limits to work, it also have to be *complete*. Furthermore, if we have a countable number of disjoint areas, it should possible to measure them all separately and or all together at once and get the same measure (a sort of "invariance of space"). In fact, if we just assume $\mu(\emptyset) = 0$ (i.e. "nothing" should have no area of measure), then using this "invariance of space" notion, we can deduce equation (1.1), as we'll show in the next proposition, and hence we would want the codomain of a measure to be $[0, \infty]$. We thus define a measure on these intuitions:

Definition 1.2.1: Measure

Let X be a set along with a σ-algebra $\mathcal{M}$. A function $\mu : \mathcal{M} \to [0, \infty]$ is called a *measure* on $\mathcal{M}$ (or $(X, \mathcal{M})$ or even X if the measure is clear from context) if

1. $\mu(\emptyset) = 0$
2. **(Countably Additivity)** If $\{E_i\}_{i=1}^{\infty}$ is a sequence of disjoint sets, then $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$

Given that $(X, \mathcal{M})$ is a measurable space, along with a μ we can say the space has a measure:

Definition 1.2.2: Measure Space

Let $(X, \mathcal{M})$ be a measurable space. Then if μ is a measure on X, then $(X, \mathcal{M}, \mu)$ is a *measure space*

If the 2nd condition of a measure was limited to a finite sequence (or more generally, all but countably many of the sets in the sequence or nonempty), then we would say that it respects finite additivity. If μ respects finite but not countable additivity, μ is called a *finite additive measure*.

Most measure's that we will work with have some finiteness condition associated with it:

Definition 1.2.3: Finiteness Condition on μ

Let $(X, \mathcal{M}, \mu)$ be a measure space.

1. If $m(X) < \infty$, then μ is called a *finite measure*
2. if there exists sets $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ such that $\bigcup_i^{\infty} E_i = X$ and $\mu(E_i) < \infty$ for all E_i, then μ is called a *σ-finite measure*. More generally, if $E = \bigcup_{i=1}^{\infty} E_i$ for some sequence of E_i, then E is called σ-finite on μ (some say that E is said to be σ-finite on μ)
3. if for for all $E \in \mathcal{M}$ such that $\mu(E) = \infty$, there exists a $D \subseteq E$, $D \in \mathcal{M}$ such that $\mu(D) < \infty$, then μ is called a *semi-finite measure*

If $\mu(X) < \infty$, then this implies for all $S \in \mathcal{M}$, $\mu(S) < \infty$ (as can quickly be checked). Naturally, all finite measure's are σ-finite, and all σ-finite measure are semi-finite, but the converses are not true (as we'll explore in the following examples). Note too that most measure's we will be working over will be σ-finite (think of $\mathbb{R}$ with the intuitive notion of length of an interval).

Example 1.2.4: Measures

1. Let X be nonempty, $\mathcal{M} = P(X)$, and $f : X \to [0, \infty]$. Then we can define

$$\mu_f(E) = \sum_{e \in E} f(e)$$

where the sum can be infinite (the domain of f is the positive part of the extended real numbers). Notice that f is semifinite if and only if $f(x) < \infty$ for all $x \in X$, and μ is σ-finite if and only if μ is semi-finite and $\{x \mid f(x) > 0\}$ if countable.

If $f(x) = 1$ for all $x \in X$, then μ_f is called the *counting measure*. On the other hand, if for some $x_0 \in X$, $f(x_0) = 1$ and $f(x) = 0$ if $x \neq x_0$, then μ_f is called a *point mass* or *Dirac measure* at x_0

2. Let X be an uncountable set an $\mathcal{M}$ be the σ-algebra of countable or co-countable sets (recall example 1.1.5). Define

$$\mu(E) = \begin{cases} 0 & E \text{ is countable} \\ 1 & E \text{ is co-countable} \end{cases}$$

Then μ is a measure!

3. An example of an finite additive measure that is not a measure is the following: let X be some infinite set and $\mathcal{M} = P(X)$. Define μ to be

$$\mu(E) = \begin{cases} 0 & E \text{ is finite} \\ \infty & E \text{ is infinite} \end{cases}$$

Then μ is an finite additive measure, but not a measure

Proposition 1.2.5: Properties of Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then

1. **(Monotonicity)** Let $E, F \in \mathcal{M}$ and $E \subseteq F$. Then $\mu(E) \leq \mu(F)$
2. **(Subadditivity)** if $\{E_i\}_{i=1}^\infty \subseteq \mathcal{M}$, then $\mu\left(\cup_i \mu(E_i)\right) \leq \sum_i \mu(E_i)$
3. **(Continuity from Below)** if $\{E_i\}_{i=1}^\infty \subseteq \mathcal{M}$, and $E_1 \subseteq E_2 \subseteq \cdots$, then

$$\mu\left(\cup_i E_i\right) = \lim_{i \to \infty} \mu(E_i)$$

4. **Continuity from Above** if $\{E_i\}_{i=1}^\infty \subseteq \mathcal{M}$ and $E_1 \supseteq E_2 \supseteq$ *and* $\mu(E_1) < \infty$. Then

$$\mu\left(\cap_i E_i\right) = \lim_{i \to \infty} \mu(E_i)$$

***Proof*:**
here. Do them as exercises! I also like this solution by Hytham Farah for continuity from above:

$$
\begin{aligned}
\mu\Big(\bigcup_{n\geq 1} F_n\Big) &= \mu\Big(\bigcup_{n\geq 1} E_n\Big) \\
&= \mu\Big(\big(\bigcap_{n\geq 1} E_n^c\big)^c\Big) \\
&= \mu(X) - \mu\Big(\bigcap_{n\geq 1} E_n^c\Big) \\
&= \mu(X) - \lim_{n\to\infty} \mu(E_n^c) \\
&= \mu(X) - \lim_{n\to\infty} \big(\mu(X) - \mu(E_n)\big) \\
&= \lim_{n\to\infty} \mu(E_n) \\
&= \sum_{i=1}^{\infty} F_n.
\end{aligned}
$$

Notice that for (d), we can instead limit it to $\mu(E_i) < \infty$ for some i, and the proof remains the same. We still require that some i exists such that $\mu(E_i) < \infty$, for it's possible that all $\mu(E_i)$ is infinite, but $\mu(\cap_i E_i) < 0$. For example, take the measurable space $(\mathbb{N}, P(\mathbb{N}))$ with the counting measure. Then the sets

$$E_i = \{n \in \mathbb{N} \mid n \geq i\}$$

then every E_i is infinite but their intersection is empty and thus has measure 0 in the counting measure.

zero measure sets

Frequently, there are sets in our measure that have a "trivial size" in the current measure. This is akin to when we integrate, the border has no effect on the value of the integral. Such sets are said to be zero sets or null set:

Definition 1.2.6: Null Set

Let $(X, \mathcal{M}, \mu)$ be a measure space and $E \in \mathcal{M}$. Then if $\mu(E) = 0$, E is said to be a *null set* or *zero set*.

By countably subadditivity, the countable union of null sets is a null set, hence null sets are "invariant" under countable union. Due to this, we can work with measurable sets in $\mathcal{M}$ *up to null sets*. We shall often state in proofs that a statement is true *almost everywhere* (often abbreviated to a.e.) if it is true on the sets except for the null sets.

If we take $E \in \mathcal{M}$ such that $\mu(E) = 0$, then by monotonicity, if $F \subseteq E$ the $\mu(F) = 0$, *provided that* $F \in \mathcal{M}$. This is not always the case:

Example 1.2.7: Not Complete Space

Partition $[0, 1]$ into 10 (really any number) of subintervals, $\left[\frac{i}{n}, \frac{i+1}{n}\right]$ for $0 \leq i \leq 9$, and let each of their measures be 0. Take $\mathcal{M}$ to be the σ-algebra defined on this set. Then is clearly an uncountable number of sets that do not have a measure defined on them that are subset of zero-measure sets.

However, adding all of these F's that are subsets of null sets does not break our measure. A measure who's domain includes all subsets of null sets is a *complete measure* (i.e. the σ-algebra is closed under subsets of null sets). Proving that this is still a measure is useful to eliminate future "trivial obstructions" to our proofs:

Theorem 1.2.8: Completing a Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} \mid \mu(N) = 0\}$ and

$$\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subseteq N \text{ for some } N \in \mathcal{N}\}$$

Then $\overline{\mathcal{M}}$ is also a σ-algebra, and there is a *unique* extension $\overline{\mu}$ of μ that is a complete measure on $\overline{\mathcal{M}}$ that restricts to the measure of μ on $\mathcal{M}$

***Proof*:**
First of all, since $\mathcal{M}$ and $\mathcal{N}$ are both closed under countable union, so $\overline{\mathcal{M}}$ is closed under countable union. To show it's closed under compliments, consider some $E \cup F \in \overline{\mathcal{M}}$ ($E \in \mathcal{M}$, $F \subseteq N \in \mathcal{N}$). Without loss of generality, assume $E \cap N = \emptyset$ (if not, take $F \setminus E$ and $N \setminus E$ and continue the proof, then this will have he same value as our original $E \cap N$). Then

$$E \cup F = (E \cup N) \cap (N^c \cup F)$$

So we get

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F)$$

Since $E \cup N \in \mathcal{M}$, $(E \cup N)^c \in \mathcal{M}$. Since $N \setminus F \in \mathcal{N}$, Since it's closed under union, $(E \cup F)^c \in \overline{\mathcal{M}}$. Thus $\overline{\mathcal{M}}$ is a σ-algebra.

Since $\overline{\mathcal{M}}$ is a σ-algebra, we can try to define a measure, and indeed

$$\overline{\mu}(E \cup F) = \mu(E)$$

is a well-defined measure. If

$$E_1 \cup F_1 = E_2 \cup F_2$$

Then $E_1 \subseteq E_2 \cup F_2$, and so by monotonicity:

$$\mu(E_1) \leq \mu(E_2 \cup F_2) \leq (E_2 \cup N_2) = \mu(E_2) + \mu(N_2) = \mu(E_2)$$

Similarly, $\mu(E_1) \geq \mu(E_2)$. Thus $\overline{\mu}(E_1 \cup F_1) = \overline{\mu}(E_2 \cup F_2)$, and so $\overline{\mu}$ is well-defined. Since $\overline{\mu}$ has all subsets of zero-sets in its domain, it is a complete measure.

Furthermore, if $\overline{\nu}$ was another complete measure on $\mathcal{M}$, then it is easy to see that $\overline{\nu} = \overline{\mu}$, (since if it were different, then some subset of a zero-set must have non-zero measure, a contradiction to monotonicity) showing that $\overline{\mu}$ is unique.

Definition 1.2.9: Completion of Measure

Let $(X, \mathcal{M}, \mu)$ be a measurable space. Then $\overline{\mathcal{M}}$ and $\overline{\mu}$ is called the *completion of* $\mathcal{M}$ *with respect to* μ and the *completion of* μ respectively

1.3 Outer Measure

We now move on to constructing a particular type of measure which is draws inspiration from how we measured area's when we calculated Riemann Integrals. In the abstract, when we tried defining a Jordan measure (or content[4]), we approximated the inner area of a shape E in $\mathbb{R}^n$ by the limit of sums of rectangles contained in E, and the outer-area of E in $\mathbb{R}^n$ by the limit of the sums of rectangles containing E. The two numbers are called the inner and outer area of E. If the two matched, we would call the result the area of E. This intuition can be generalized to the countable case, except we can replace rectangles with some other "basic sets" that we might want that form an algebra. Furthermore, the intuition of having the "inner area" and "outer-area' is actually a bit restricting to work with in the general measure case, and so that condition will be replaced with a different measureability condition we will soon introduce.

For the sake of generality, we will start by definition a more general notion of the measure called the outer-measure, and then build-up on how form the outer-measure, we can get a measure that satisfies this generalization of Riemann integration:

Definition 1.3.1: Outer Measure

A function $\mu^* : P(X) \to [0, \infty]$ is called an *outer measure* if

1. $\mu^*(0) = \emptyset$
2. if $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$
3. $\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (even if the sets are disjoint!)[a]

[a] We will return to see when equality holds in definition 1.3.3

Note that we have not yet proven that an outer-measure is a measure; in particular notice that the domain is the powerset. What we will do is define a generalization of the Riemann integral that will be our "base-line" for the outer-measure:

[4] since "Jordan measures" only allow finite additivity, some don't like to use the word measure and opt for the word content. In this book, we would say *finite additive measure* as has already been stated earlier

Proposition 1.3.2: Outer Measure Construction

Let $\mathcal{E} \subseteq P(X)$ where $\emptyset, X \in \mathcal{E}$, and let $\rho : \mathcal{E} \to [0,1]$ where $\rho(\emptyset) = 0$. For any $A \subseteq X$, define

$$\mu^*(A) = \inf \left\{ \sum_1^\infty \rho(E_j) \;\middle|\; E_j \in \mathcal{E} \text{ and } A \subseteq \bigcup_1^\infty E_j \right\}$$

Then μ^* is an outer measure

***Proof*:**
First, this measure is well-defined since for any $A \subseteq X$, If we take $E_i = X$, then $A \subseteq \cup_i E_i$. Next, we verify the 3 properties

1. Clearly, $\mu^*(\emptyset) = 0$ by taking $E_i = \emptyset$ for all i
2. If $A \subseteq B$, then let $\mathcal{A}$ and $\mathcal{B}$ sets that correspond to $\mu^*(A)$ and $\mu^*(B)$. Then since all of the elements of $\mathcal{A} \supseteq \mathcal{B}$, by the property of the infimum $\inf \mathcal{A} \leq \inf \mathcal{B}$, which transferring notations gives us $\mu^*(A) \leq \mu^*(B)$
3. Let $\{E_i\}_{i=1}^\infty \subseteq P(X)$ be a collection of (not necessarily disjoint) subsets. Let $\epsilon > 0$. Consider $\mu^*(E_k)$ for some $E_k \in \{E_i\}_{i=1}^\infty$. By the property of the infimum, there must exist some $\{F_j^k\}_{j=1}^\infty$ such that $E_i \subseteq \cup_j F_j^k$ where $\sum_{j=1}^\infty \rho(F_j^k) \leq \mu^*(E_k) + \epsilon 2^{-k}$.

   Now, let $E = \cup_i E_i$. By construction, $E \subseteq \cup_{i,k=1}^\infty F_j^k$, and

   $$\sum_{j,k} \rho(F_j^k) \leq \mu^*(A) + \epsilon$$

   Since ϵ can be arbitrarily small, this completes the proof.

As mentioned before, an outer-measure is not necessarily a measure. What we will do to allow this to be a measure is single out sets that have the following property

Definition 1.3.3: μ^*-measureable

Let $E \subseteq X$ be a set and μ^* be an outer measure on X. Then E is called μ^**-measurable* if

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

In other word, if E can naturally "split" a set S into a part that's "inside" E or "outside" E. The set $S \subseteq X$ is called the *test set*.

By definition of the outer measure, since $S = (S \cap E) \cup (S \cap E^c)$:

$$\mu^*(S) \leq \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

Therefore, what is more important is to check the $\geq$ direction. Furthermore, if $\mu^*(A) = \infty$. Then $\geq$ holds trivially, so to show a set is μ^* measurable, it is equivalent to show that for all finite measure sets A, $\mu^*(S) \geq \mu(S \cap E) + \mu(S \cap E^c)$.

As mentioned in the definition, this gives us a sense of being able to measure the "inside" and "outside" of a set. If $E \subseteq A$, that is the μ^*-measurable set belong in a better set, then $\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \setminus E)$ which can be thought of as saying that we are measuring S from the "inside" of E and the "outisde" of E, and getting the same result. In fact, if $\mu^*(X) < \infty$, then we can define $\mu_*(E) = \mu^*(X) - \mu(E^c)$, and have that E is μ^*-measurable if and only if $\mu_*(E) = \mu^*(E)$ aligning with our intuition for inside equalling outside (see exercise ref:HERE). We do not do this method since it requires some finiteness condition on μ. We can conversely define μ_* in terms of supremums and need that E contains the collection of $\cup_i A_i$, and that the elements of $\{A_i\}_{i=1}^{\infty}$ must be disjoint (see exercise ref:HERE).

The big thing that is in the way of saying that an outer measure is a measure is that its domain is $P(X)$, which can cause problems when X is uncountable. However, all sets that satisfy μ^*-measureability turn out form a σ-algebra and make μ^* a complete measure!

Theorem 1.3.4: Carathéodory's Theorem

Let μ^* be an outer measure on X, and let $\mathcal{M}$ be the collection of μ^*-measurable sets. Then $\mathcal{M}$ forms a σ-algebra, and the restriction of μ^* to $\mathcal{M}$ is a complete measure

Notice that in the process of defining this measure, we started with the powerset, and then simply choose all sets that restrict (which we will show forms a σ-algebra). In this way proving a function is an outer-measure is a little less finicky since we don't need to be so careful in defining over which σ-algebra we want to define it over: the σ-algebra comes with μ^*

***Proof*:**
Let $\mathcal{M}$ be the collection of μ^*-measurable sets. Clearly, $\emptyset \in \mathcal{M}$ since for all $S \subseteq X$:

$$\mu^*(S \cap \emptyset) + \mu^*(S \cap X) = 0 + \mu^*(S)$$

implying μ^*-measureability. Next, $\mathcal{M}$ is closed under compliments, since if $E \in \mathcal{M}$, then

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \mu^*(S \cap E^c) + \mu^*(S \cap E)$$

which also shows that $X \in \mathcal{M}$. Finally, we need to show that $\mathcal{M}$ is closed under countable unions. We'll start by showing finite unions, which then means it suffices to show the union of two sets is closed (due to induction). Let $A, B \in \mathcal{M}$. As we remarked, it suffices to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. Since A and B are μ^* measurable, we have

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) && E \text{ is the test set} \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) \\ &\quad + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) && E \cap A \text{ and } E \cap A^c \text{ are the test sets} \end{aligned}$$

Then notice we can write

$$A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$$

Thus, using subadditivity, we can re-write 3 of the terms in the previous expression as:

$$\mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) \geq \mu^*(E \cap A \cup B)$$

Since $E \cap A^c \cap B^c = E \cap (A \cup B)^c$, we get:

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

showing that $\mathcal{M}$ is closed under finite union. Since $\mathcal{M}$ is closed under compliment and contains $\emptyset$, $\mathcal{M}$ is an algebra. Even better, If we let A and B be disjoint, and let $A \cup B$ be the test set, then since A is μ^* measurable:

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B)$$

showing that $\mu^*|_{\mathcal{M}}$ is a finitely additive measure.

Next, we must show that $\mathcal{M}$ is a σ-algebra, which as a consequence will show that $\mu^*|_{\mathcal{M}}$ is a measure. Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$. To simplify our task, re-construct $\{E_i\}_{i=1}^{\infty}$ to be the set of disjoint sets $\{D_i\}_{i=1}^{\infty}$. Let $B_n = \bigcup_{i=1}^{n} D_i$ and $B = \bigcup_{i=1}^{\infty} D_i$. Our goal is to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c)$. We will start by considering $\mu^*(E \cap B_n)$:

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap B_n \cap D_n) + \mu^*(E \cap B_n \cap D_n^c) && E \cap B_n \text{ is the test set} \\ &= \mu^*(E \cap D_n) + \mu^*(E \cap B_{n-1}) \end{aligned}$$

Notice that on the right hand side we have diminished to the case of $E \cap B_{n-1}$ from $E \cap B_n$. Using induction, we can see that the expression simplifies to

$$\mu^*(E \cap B_n) = \sum_{i=1}^{n} \mu^*(E \cap D_i)$$

Continuing, since B_n is measurable

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \overset{!}{\geq} \mu^*(E \cap B_n) + \mu^*(E \cap B^c) = \sum_{i=1}^{n} \mu^*(E \cap D_i) + \mu^*(E \cap B_n^c)$$

where $\overset{!}{\geq}$ comes from monotonicity after switching B_n^c to B^c. Since this equality holds true for all n, we get:

$$\begin{aligned} \mu^*(E) &\geq \sum_{i=1}^{\infty} \mu^*(E \cap D_i) + \mu^*(E \cap B^c) \\ &\geq \mu^* \left(\bigcup_i E \cap D_i \right) + \mu^*(E \cap B^c) && \text{subadditivity} \\ &= \mu^*(E \cap B) + \mu^*(E \cap B^c) && \text{by construction of } E \cap B \\ &\geq \mu^*(E) && \text{subadditivity} \end{aligned}$$

Since we got $\mu^*(E) \geq \mu^*(E)$, we see that in fact all inequalities are equalities, getting us that

$$\mu^*(E) = \mu^*(E \cap B) + \mu^*(E \cap B^c)$$

showing us that $B \in \mathcal{M}$. Furthermore, replacing $E = B$ in the previous proof shows us that

$$\mu^*(B) = \sum_{i=1}^{\infty} \mu^*(D_i)$$

giving us that μ^* is a measure on $\mathcal{M}$.

Finally, we must show that $\mathcal{M}$ is complete, that is, if $E \in \mathcal{M}$ such that $\mu^*(E) = 0$, then all subsets of E are in $\mathcal{M}$. Notice that since μ^* is defiend on all of $P(X)$, it is equivalent to show that if $\mu^*(A) = 0$, then $A \in \mathcal{M}$. Then for any test set E, by monotonicity:

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) = 0 + \mu^*(E \cap A^c) \leq \mu^*(E)$$

meaning equality holds, showing that $A \in \mathcal{M}$. Thus, $\mu^*|_{\mathcal{M}}$ is a complete measure, as we sought to show.

Carathéodory's Theorem is quite useful being able to state the existence of a measure given an outer-measure, but it's not good to construct a measure (for example, can you tell me a non-trivial element in $\mathcal{M}$ without directly appealing to the definition?). The following concept is used to more systematically construct the set $\mathcal{M}$ and the measure μ induced by μ^*:

Definition 1.3.5: Premeasure

Let $(X, \mathcal{A})$ be an algebra. Then μ_0 is called a *premeasure* if

1. $\mu_0(\emptyset) = 0$
2. **"Respects" countable union**: If $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ is a collection of disjoint such that $\cup_i A_i \in \mathcal{A}$, then $\mu_0(\cup_i A_i) = \sum_i \mu_0(A_i)$

Note that the premeasure is automatically finitely additive since we can let all but finitely many of the terms in (2) be $\emptyset$ (and hence are a stronger condition than finitely additive measure). However, it is not yet a measure since it's domain is not necessarily closed under countable union. However, if it is closed for countable union for a particular collection, that collection must satisfy the "countable" condition. Interestingly, by the definition of an algebra, this implies that the countable union can be represented as a finite union and intersection of elements from $\mathcal{A}$.

As before, we can define semifinite and σ-finite in a similar way. Using μ_0, we can define the following outer-measure:

$$\mu^* : P(X) \to [0, \infty] \quad \mu^*(E) = \inf \left\{ \sum_{i}^{\infty} \mu_0(A_i) \;\middle|\; A_i \in \mathcal{A},\ E \subseteq \cup_i A_i \right\}$$

Proposition 1.3.6: Premeasure To Measure

Let μ_0 be a premeasure on $\mathcal{A}$, and let μ^* be defined as above. Then

1. $\mu^*|_{\mathcal{A}} = \mu_0$
2. Every set in $\mathcal{A}$ is μ^*-measurable

Essentially, μ^* is an extension of μ_0 and the associated restriction of μ^* to a measure will contain $\mathcal{A}$

***Proof*:**

1. Let's show $\leq$ and $\geq$ for any $E \in \mathcal{A}$. The $\leq$ direction is obvious since $\mu^*(E) \leq \mu_0(E)$ since $E \subseteq \bigcup_{i=1}^{\infty} A_i$ where $A_1 = E$ and $A_i = \emptyset$ for $i > 1$, and so it is part of the set over which

we take the infimum, and so the result *has* to be smaller.

For the $\geq$ direction, let's say $E \subseteq \bigcup_{i=1}^{\infty} A_i$ for any $\{A_i\}_{i=1}^{\infty} \in \mathcal{A}$. Let $B_n = E \cap \left(A_n \setminus \cup_1^{n-1} A_i\right)$. Then the collection $\{B_i\}_{i=1}^{\infty}$ is a disjoint collection of members of $\mathcal{A}$ whose union is E. Thus, by the property of premeasure

$$\mu_0(E) = \sum_1^{\infty} \mu_0(B_i) \overset{!}{\leq} \sum_1^{\infty} \mu_0(A_i)$$

where the $\overset{!}{\leq}$ inequality comes from the fact that $\mu_0(B_i) \leq \mu_0(A_i)$. Since this is true for any arbitrary collection $\{A_i\}_{i=1}^{\infty}$, it follows that

$$\mu_0(E) \leq \mu^*(E)$$

Thus establishing equality

2. We need to show that for any $E \subseteq X$ and $A \in \mathcal{A}$, $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$. As mentioned earlier, it suffice to prove $\geq$ where $\mu^*(E) < \infty$.

   Let $\epsilon > 0$. By the property of the infimum, there exists a subset $\{B_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \cup_i B_i$ such that $\sum_i^{\infty} \mu_0(B_i) \leq \mu^*(E) + \epsilon$. By part (1), μ^* is [countably] additive on $\mathcal{A}$ (since it equals the pre-measure)

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{i=1}^{\infty} \mu_0(B_i \cap A) + \sum_{i=1}^{\infty} (B_i \cap A^c) && \text{additivity} \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) && \text{monotinicity} \end{aligned}$$

   and since ϵ was arbitrary, A must be μ^*-measurable, completing the proof

We can in fact be more precise with the measures that extend a premeasure:

Theorem 1.3.7: Premeasure Extensions

Let $\mathcal{A} \subseteq P(X)$ be an algebra, μ_0 a premeasure on $\mathcal{A}$, and $\mathcal{M} = \mathcal{M}(\mathcal{A})$ (i.e. $\mathcal{M}$ is the σ-algebra generated by $\mathcal{A}$). Let μ be the extension of μ_0 from proposition 1.3.6. Then if ν is another measure on $\mathcal{M}$ that extends μ_0, then

$$\nu(E) \leq \mu(E) \qquad \forall E \in \mathcal{M}$$

with equality when $\mu(E) < \infty$ (i.e., it possible that μ is infinite while ν is finite).
If μ_0 is σ-finite, then μ is the unique extension of μ_0 to a measure on $\mathcal{M}$

In other words, we can extend a pre-measure to a measure by knowing what it does on the algebra (which in most cases will be a collection of simpler sets, like intervals or rectangles), and then what happens on the rest of the measurable sets we use the infimum definition. If we define another extension of μ_0, then that measure will be *finer* than the μ given by μ^*, i.e, μ is the *coarsest* extension. If μ_0 is σ-finite, then this will in fact be the only extension of μ_0

***Proof*:**
Let's say ν is an extension of μ_0 so that $(X, \mathcal{M}(\mathcal{A}), \nu)$ is a measure space. To show that $\nu(E) \leq \mu(E)$, let's say $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \cup_i A_i$. Then since ν is a measure, it is sub-additive on non-disjoint sets, and so

$$\nu(E) \leq \sum_i \nu(A_i)$$

Since ν extends μ_0, by proposition 1.3.6 $\nu|_{\mathcal{A}} = \mu_0$, so

$$\nu(E) \leq \sum_i \nu(A_i) = \sum_i \mu_0(A_i)$$

Since $\{A_i\}_{i=1}^{\infty}$ was an arbitrary choice, this works for any collection, and so

$$\nu(E) \leq \mu(E) \quad [= \mu^*(E)]$$

We show that $\nu(E) = \mu(E)$ if $\mu(E) < \infty$. Let $A = \cup_i A_i$ for some collection $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$. Then

$$\nu(A) = \lim_{n\to\infty} \nu\left(\bigcup_{i=1}^{n}(A_i)\right) = \lim_{n\to\infty} \mu\left(\bigcup_{i=1}^{n}(A_i)\right) = \mu(A)$$

If μ is not σ-finite, then there can be more than one extension:

Example 1.3.8: 2 extensions of pre-measure

1. Let $\mathcal{A}$ be the collection of finite unions of the sets of the form $(a, b] \cap \mathbb{Q}$ with $-\infty \leq a \leq b \leq \infty$. Then $\mathcal{A}$ is an algebra on $\mathbb{Q}$

***Proof*:**
Recall proposition 1.1.13 where we can construct an algebra using an elementary family. Thus, we will show that the collection $\mathcal{A}$ satisfies the 3 properties of an elementary family:

(a) To get the empty-set in $\mathcal{A}$, we will understand the notion of "finite union" to also include "no union". Otherwise, since $\mathbb{Q}$ is dense in $\mathbb{R}$, all intervals $(a, b]$ intersected with $\mathbb{Q}$ is nonempty (by MAT157 knowledge). Since $a < b$, the singleton interval is not admissible, and so we will have to assume this "no union" property

(b) Let $E, F \in \mathcal{A}$. We want to show that $E \cap F \in \mathcal{A}$. By definition, $E = (a, b] \cap \mathbb{Q}$ and $F = (c, d] \cap \mathbb{Q}$. Then by some basic set-theory properties, we know that the intersection of two left-open/right-closed intervals is again a left-open/right-closed interval or $\emptyset$. If their intersection is $\emptyset$, then we're done. If not, let $(e, f] = (a, b] \cap (c, d]$. Then $E \cap F = (e, f] \cap \mathbb{Q}$. Thus, $E \cap F \in \mathcal{A}$

(c) Let $E \in \mathcal{A}$, so that $E = (a, b] \cap \mathbb{Q}$. Let $i \in \mathbb{N}$ $(0 \notin \mathbb{N})$ and:

$$E_i = (a - i, a - (i + 1)] \cap \mathbb{Q} \quad F_i = (b + (i - 1), b + i] \cap \mathbb{Q}$$

then each $E_i, F_i \in \mathcal{A}$, $E_i \cap E_j = F_i \cap F_j = \emptyset$ if $i \neq j$ and $E_i \cap F_j = \emptyset$ for all $i, j \in \mathbb{N}$. Let $E = \{E_i\}_{i=1}^{\infty}$ and $F = \{F_i\}_{i=1}^{\infty}$ and let $G = E \cup F$. Then

$$\bigcup_{g \in G} G = E^c$$

since G is a countable union of disjoint elements of $\mathcal{A}$, this completes the proof.

Thus, by proposition 1.7, $\mathcal{A}$ is an algebra.

2. The σ-algebra generated by $\mathcal{A}$ is $P(\mathbb{Q})$

***Proof*:**
Let $x \in \mathbb{Q}$ Notice that

$$\{x\} = \bigcap_{k=1}^{\infty} \left(x - \frac{1}{k}, x\right] \cap \mathbb{Q}$$

hence, every singleton is inside the σ-algebra of $\mathbb{Q}$. Since every subset of $\mathbb{Q}$ is countable, every subset is the countable union of singletons. Since every singleton is inside the σ-algebra, we have that the σ-algebra is $P(\mathbb{Q})$.

3. Define μ_0 on $\mathcal{A}$ by $\mu_0(\emptyset) = 0$ and $\mu_0(A) = \infty$ for $A \neq \emptyset$. Then μ_0 is a premeasure on $\mathcal{A}$, and there is more than one measure on $P(\mathbb{Q})$ whose restriction to $\mathcal{A}$ is μ_0

***Proof*:**
μ_0 is trivially a pre-measure. By definition, $\mu_0(\emptyset) = 0$. For any nonempty $A \in \mathcal{A}$, we have $\mu_0(A) = \infty$. When checking countable additivity, unless all the sets are empty, we get that both sides are infinite.

Thus, we can define μ to be the μ^* induced by μ_0 restricted to all measurable sets. By construction, all sets are measurable. If a sets is non-empty, then it will always be covered by a nonempty set, which has measure ∞. Thus, If $E \in P(X)$ for any nonempty test set $S \subseteq \mathbb{Q}$

$$\infty = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \infty$$

and if $S = \emptyset$

$$0 = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = 0$$

Thus, $\mathcal{M}_\mu = P(\mathbb{Q})$.

We can define another extension of μ_0 using the counting measure, in particular, let μ_C be the counting measure on $P(\mathbb{Q})$. Then since every $A \in \mathcal{A}$ has countable cardinality

$$\mu_C(A) = \infty$$

Thus, $\mu_C|_{\mathcal{A}} = \mu_0$. However

$$\mu_C(\{1\}) = 1 \neq \infty = \mu(\{1\})$$

Thus, we have defined two separate extensions of μ_0.

1.4 Borel Measure on the Real Line

The full title of this section should really be "Borel Measures on the Real line defined through Distributions", but that title seems to cumbersome. Nevertheless, this is exactly what we will be doing in this section. We will extensively concentrating on these types of measures, since they are, in a sense, the focus of "real analysis". From a high-level perspective, the fact that we are studying "real analysis" justifies why we will take so much time understanding this measure. More concretely, these measure are pervasive in analysis in general: they are central to defining integration, to the study of Fourier series, to studying L^p spaces, to linking differentiation to integration, to generalizing the notation of integration of groups (in particular, it will make sense to integrate locally compact topological groups), and so much more. Another way of looking at it is that the euclidean topology is central to the study of many fields of mathematics and physics, and is the basis for our notion of manifolds (which are locally euclidean). Therefore, getting a sense of how to define notions of "size" in $\mathbb{R}$ (and then generalize to $\mathbb{R}^n$, $\mathbb{R}^\infty$, or manifolds) gives us a solid first step in gaining an intuition on measures.

One way to define a measure in $\mathbb{R}$ is by using our intuition of the fact that the length of (a, b) is $b - a$. This is a good intuition and will be the basis to the generalization of this idea we'll do in order to be able to define a larger class of measures based on this idea which will also have more uses outside of $\mathbb{R}$. Here is how we will motivate the construction of these measures: let μ be a finite measure on Borel sets (a measure define on Borel sets is called the *Borel Measure*). Define:

$$F(x) = \mu((\infty, x])$$

F is sometimes called the *distribution function* of μ. Intuitively, it shows the "accumulated size" of the space. Clearly, F is an increasing function, and is right-continuous since

$$(\infty, x] = \bigcap_k (\infty, x + 1/k]$$

or any decreasing function, $x_n \searrow x$ (i.e. limits are preserved from the right). Furthermore, if we take $a < b$, then $(-\infty, b] = (-\infty, a] \cup (a, b]$, thus

$$\mu((a, b]) = F(b) - F(a)$$

Notice how similar this is to the usual result from Riemann integration! In the following, instead of defining F in terms of μ, we will define μ from an increasing right-continuous function F by showing how F define's a pre-measure and using Carathéodory. The special case of $F(x) = x$ will give us our usual notion of length in $\mathbb{R}$.

Throughout this section, we will use sets of the form $(a, b]$ where $a < b$, along with the edge cases of (a, ∞) and $\emptyset$. Let all intervals of this type be called h-intervals (h for half open). It is clear that the union or compliment of two h-intervals are h-intervals (or two disjoint h-intervals) and that h-intervals are closed under finite operations, and so h-intervals from a algebra $\mathcal{A}$. By proposition 1.1.7, the σ-algebra by the set $\mathcal{A}$ is $\mathcal{B}_{\mathbb{R}}$. Using this, we'll first show that we can define a premeasure μ based off of an F:

Proposition 1.4.1: Distribution and Premeasure

Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right continuous. If $(a_i, b_i]$ is a set of disjoint h-intervals. If we define μ_0 to be:

$$\mu_0 \left(\bigcup_i^{\infty} (a_i, b_i] \right) = \sum_{i=1}^{\infty} F(b_i) - F(a_i)$$

and $\mu_0(\emptyset) = 0$, then μ_0 is a premeasure on the algebra $\mathcal{A}$

***Proof*:**
We will first show that μ_0 is well-defined, then we simply must show that $\mu\left(\bigcup_i (a_i, b_i]\right) = \sum_i \mu_0((a_i, b_i])$.

To show it's well-defined, we must show that for any finite representation of $(a, b]$, $\mu_0((a, b]) = F(b) - F(a)$. To that end, let $\{(a_i, b_i]\}_{i=1}^n$ be disjoint where $(a, b] = \coprod i = 1^n [a_i, b_i)$. Since the co-domain is commutative, we can re-arrange the order that we are taking the union (i.e., re-index the set) so that

$$a < a_1 < b_1 = a_2 < b_2 = a_3 < \cdots < b_n = b$$

Then the resulting μ_0 is simply an alternating series:

$$\sum_{j=1}^{n} F(b_j) - F(a_j) = F(b) - F(a)$$

showing that it is independent of representation of the intervals. Thus, if we have any disjoint collection $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^n$ such that $\cup_i I_i = \cup_j J_j$, since we can always form alternative series:

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j)$$

and so μ_0 is well-defined. Since it is well-defined, then we also get that μ_0 is finitely additive (essentially by definition), and since $\mu_0(\emptyset) = 0$, it is at least a finite measure.

Next, we need to show that it is a premeasure. Since $\mu_0(\emptyset) = 0$, it remains to check that if $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $\cup_i^{\infty} A_i \in \mathcal{A}$, then

$$\mu_0\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu_0(A_i)$$

Since $\cup_i^{\infty} I_i = I \in \mathcal{A}$, there exists (at least one) subsets $\{A_i\}_{i=1}^{n} \subseteq \mathcal{A}$ such that $\cup_i^{\infty} I_i = \cup_i^{n} A_i$. Therefore the elements from I can be partitioned so that the union of each partition represents a single h-interval. By finite additivity of μ_0, it suffices to consider what happens at one of these partitions. So, let J be the union of one of the families so that $(a,b]J = \cup_{j=1}^{\infty} I_{i_j}$. We will show that $\mu_0(\cup_j^{\infty} J_j) = \sum_j^{\infty} \mu_0(J_j)$ by showing $\leq$ and $\geq$. For the $\geq$ direction, notice that

$$\mu_0(J) = \mu_0\left(\bigcup_j^n J\right) + \mu_0\left(J \setminus \bigcup_j^n J\right) \geq \mu_0\left(\bigcup_j^n J\right) = \sum_j^n \mu_0\left(J\right)$$

Thus, as $n \to \infty$ we have $\mu_0(J) \geq \sum_j^{\infty} \mu_0(J_j)$.

For the $\leq$ direction, we will take advantage of the right-continuity of F.

(Might not be the case TBD p.34 Folland) First, if either b is infinite, the result holds automatically. If a is infinite, then b would have to also be infinite, and so we have the zero-intervals. So, let a and b be finite.

Let $\epsilon > 0$. Then since F is right-continuous, there exists a δ such that $F(a+\delta) - F(a) < \epsilon$ or similarly $-F(a) < -F(a+\delta) + \epsilon$. Similarly, for the same epsilon, there exists a δ_j such that $F(b_j + \delta_j) - F(b_j) < \frac{\epsilon}{2^n}$. Notice that $(a_j, b_j] \subseteq (a_j, b_j + \delta_j)$, so the open intervals cover $[a,b]$, and since $[a,b]$ is compact, there exists a finite sub-cover! If $\{(a_j, b_j + \delta_j)\}_{i=1}^{N}$ is the finite subcover of $[a,b]$, if we discard all sets that are contained in a bigger set in this family, then we get that

$$b_j + \delta_j \in (a_{j+1}, \beta_{j+1} + \delta_{j+1}) \quad \text{for } j = 1, \ldots N-1$$

With this, we get the following sequence of inequalities:

$$\begin{aligned}
\mu_0(J) &= F(b) - F(a) \\
&\leq F(b) - F(a+\delta) + \epsilon \\
&\leq F(b_N + \delta_N) - F(a_1) + \epsilon && \text{extrema of cover} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(a_{j+1}) - F(a_j))\epsilon && \text{fancy 0} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(b_{j+1} + \delta_j) - F(a_j))\epsilon && \text{expanding} \\
&< \sum_{j=1}^{N} [f(b_j - f(a_j) + \epsilon 2^{-n}] + \epsilon \\
&< \sum_{j=1}^{\infty} \mu_0(J_j) + 2\epsilon
\end{aligned}$$

and since ϵ was arbitrary, the result follows.

This result let's us prove that *any* increasing right-continuous function defines a measure!

Theorem 1.4.2: Distributions define Measure

If $F : \mathbb{R} \to \mathbb{R}$ is an increasing right-continuous function, there is a unique Borel measure μ_F on $\mathbb{R}$ such that $\mu_F((a,b]) = F(b) - F(a)$ for all $a, b \in \mathbb{R}$. This function is unique up to the following condition: if G is another such function, then $F - G$ is constant.
Conversely, if μ is a Borel measure that is bounded on all finite Borel sets, we can define

$$F(x) = \begin{cases} \mu((0,x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((-x,0]) & \text{if } x < 0 \end{cases}$$

Then F is an increasing right-continuous function

***Proof*:**
First, by the previous proposition, F induces a premeasure on $\mathcal{A}$. If $F - G$ is constant, then the sums will cancel out in $F(b) - F(a)$, and so $\mu_F = \mu_G$. Similarly, if the two are equal, F and G can only differ up to a constant and that these measures are σ-finite (recall that $\mathbb{R} = \bigcup_{-\infty}^{\infty} (-i, i+1])$.

For the converse, note that if μ is monotonic, then so is F. Next, above/bellow continuity of μ will imply right-continuity of F for $x \geq 0$ and $x < 0$. Therefore, by construction, $\mu = \mu_F$ on $\mathcal{A}$, and so by proposition 1.4.1, $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$ by uniqueness in theorem 1.3.7.

Note that F need not be left-continuous. Something with the point-density measure that proves the counter-case.

We have been using intervals of the form $(a, b]$ and right-continuous function. The same theory can be developed with $[a, b)$ intervals and left-continuous functions. If we restrict μ to being a finite Borel measure, then we get $\mu = \mu_F$ where $F(x) = \mu((-\infty, x])$ (i.e. it differs up to a constant of the μ_F when μ is a Borel measure). Next, by the construction of the measure from the pre-measure, the resulting measure is in fact a complete measure. Even better, taking completion of μ_F and then applying Carathéodory is the same as just taking completion of μ after applying Carathéodory to μ_F, and so taking the completion on μ_0 or μ_F does not make a difference! The complete measure is also usually denoted by μ_F and called the *Lebesgue-Stieljes measure* of F.

As a consequence of the Lebesgue-Stieljes measure μ being complete, if we have a measure on a Borel space (which we usually call a Borel measure), then then then the set of measurable sets $\mathcal{M}^*$ might be larger than the number of Borel sets (in particular in that it may contain some zero sets that are not Borel sets).

The Lebesgue-Stieljes measure has a couple of nice properties worth exploring. Let μ be a complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to F, and let $\mathcal{M}_\mu$ be the associated μ^*-measurable sets. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\left\{\sum_1^\infty F(b_i) - F(a_i) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i]\right\} = \inf\left\{\sum_1^\infty \mu((a_i, b_i]) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i]\right\}$$

Notice that the contribution of the right end points of every h intervals is clearly negligable and so we can replace it with open intervals

Lemma 1.4.3: h-Intervals to Open Intervals

Let μ be the complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to F. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\left\{\sum_1^\infty \mu((a_i, b_i)) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i)\right\}$$

***Proof*:**
This is the same epsilon trick you have seen before, right it down here as an exercise later

In fact, we can write down $\mu(E)$ in even simple terms

Theorem 1.4.4: Lebesgue-Stieljes in Terms of Open and Compact sets

Let μ be the complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to $\mathbb{F}$. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\{\mu(U) \mid U \supseteq E \text{ and } U \text{ is open}\}$$

$$\mu(E) = \sup\{\mu(K) \mid K \subseteq E \text{ and } U \text{ is compact}\}$$

***Proof*:**
By lemma 1.4.3, for all $\epsilon > 0$, there exists open intervals (a_i, b_i) such that $E \subseteq \cup_i^\infty (a_i, b_i)$ and $\mu(E) \leq \sum_i^\infty (a_i, b_i) + \epsilon$. If we let $U = \cup_i (a_i, b_i)$, we have that $\mu(U) \leq \mu(E) + \epsilon$. Conversely, since $E \subseteq U$, $\mu(U) \geq \mu(E)$, and so the result for the first statements follows.

For the 2nd equivalence, let's first suppose E is bounded. If E is closed, then it is compact and so E is contained in the set that we are taking the sup over, and so equality is immediate. If E was open, then we can choose an $\epsilon > 0$ such that $\overline{E} \setminus E \subseteq U$ and $\mu(U) \leq \mu(\overline{E} \setminus E) + \epsilon$. Set $K = \overline{E} \setminus U$. Then K is compact, and so

$$\begin{aligned} \mu(K) &= \mu(E) - \mu(E \cap U) \\ &= \mu(E) - [\mu(U) - \mu(U \setminus E)] \\ &\geq \mu(E) - [\mu(U) - \mu(\overline{E} \setminus E)] \\ &= \mu(E) - \mu(U) + \mu(\overline{E} \setminus E) \\ &\geq \mu(E) - \epsilon \end{aligned}$$

showing that any compact set arbitrarily approximates K.

If E is unbounded, let $E_i = E \cap (i, i+1]$ so that $E = \cup_i E_i >$ Then by what we've just shown, for all $\epsilon > 0$, there exists a $K_j \subseteq E_j$ such that $\mu(K_j) \leq \mu(E_j) - \frac{\epsilon}{2^n}$. Take $H_n = \cup_{-n}^{n} K_i$. Then H_n is compact, $H_n \subseteq E$, and

$$\mu(H_n) \geq \mu(\cup_{-n}^{n} E_i) - \epsilon$$

Since this equality holds when $n \to \infty$, the result follows

This gives us a strikingly easy representation of representing every measurable subset of $\mathbb{R}$!

Theorem 1.4.5: $E \subseteq \mathbb{R}$ Representation

Let $E \subseteq \mathbb{R}$. Then the following are equivalent

1. $E \in \mathcal{M}_\mu$
2. $E = V \setminus N_1$ where V is a G_δ set and $\mu(N_1) = 0$
3. $E = W \cup N_2$ where W is a H_σ set and $\mu(N_2) = 0$

***Proof*:**
Clearly, (2) and (3) imply (1). For (1) implies (2) or (3), I would highly recommend you try and solve this in the $\mu(E) < \infty$ case to jog your memory.

For the $\mu(E) = \infty$, TBD (exercise 25 p.37 Folland)

This means that all borel sets, or more generally sets in $\mathcal{M}_\mu$ (also contain all unions of borel sets with measure zero sets). Note that this theorem and proof would be much harrier if we did not assume the measure was complete. ‘

I will leave this section with another way of interpreting finite measurable sets that Folland left as an exercise (we'll use the notation of $A \triangle B = (A \setminus B) \cup (B \setminus A)$)

Proposition 1.4.6: Diminishing Difference

Let $E \in \mathcal{M}_\mu$ and $\mu(E) < \infty$. Then for every $\epsilon > 0$, there exists a set A which is the finite union of open intervals and

$$\mu(E \triangle A) < \epsilon$$

1.4.1 Lebesgue Measure

We will study the case where the measure μ is defined through the distribution $F(x) = x$, i.e., when F represents our usual intuition of length! This is arguably the most important measure on $\mathbb{R}$, and most certainly the most used. For that reason, it has some special symbology: we will denote the measure by m instead of μ and the set of measurable sets by $\mathcal{L}$. If we restrict m to $\mathcal{B}_{\mathbb{R}}$, we will also write it as m.

The first properties worth establishing is to see that measurable sets in $\mathbb{R}$ inherit some of our intuitions how a "shape" should act under translation and dilation, that it's invariant under translation (just like we assumed for the construction of Vitali sets!) and it scales linearly with dilation:

Proposition 1.4.7: Scaling and Dilation

Let $(X, \mathcal{L}, m)$ be a Lebesgue measure space. Define

$$E + s := \{e + s \mid e \in E\} \qquad rE := \{re \mid e \in E\}$$

Then $E + s, rE \in \mathcal{L}$ for all $r, s \in \mathbb{R}$, and:

$$m(E + s) = m(E) \qquad m(rE) = |r|m(E)$$

***Proof*:**
We essentially notice that open intervals are invariant under translation and operate in the form described in the proposition, then apply theorem 1.3.7 (which is a really cool application of this theorem!!)

For any $E \in \mathcal{B}_{\mathbb{R}}$ define $m_s(E) = m(E+s)$ and $m^r(E) = M(rE)$. Then m_s and m^r clearly agree with m and $|r|m$ on finite unions of intervals, and so by theorem 1.3.7 they agree on all of $\mathcal{B}_{\mathbb{R}}$. As a consequence, if $m(E) = 0$, then $m(E + s) = m(rE) = 0$. Thus, for all sets in $\mathcal{L}$ (i.e. all unions of borel sets and Lebesgue zero-sets) must be preserved and linearly scaled (respectively) under translation and dilation:

$$m(E + s) = m(E) \qquad m(rE) = |r|m(E)$$

as we sought to show

Since we are working over $\mathcal{M}(\mathcal{B}_{\mathbb{R}})$, the next interesting questions we may ask about m is how it interacts with the topological sets on $\mathbb{R}$. For starters, it is clear (or should be immediately proven) that any singleton has measure 0. Therefore, by countable additivity, every countable set has measure zero; in particular $m(\mathbb{Q}) = 0$. Notice that the rational are dense which is a "topologically large" concept, however it is measure-theoretically small. In fact, we can even have a collection of *open dense* sets, and still be measure-theoretically small. Take some enumeration $\{r_i\}_{i=1}^{\infty}$ of the rational numbers in $[0, 1]$ and let I be the indexing set for this enumeration $i \in I$. Let I_n be the interval of size $\frac{\epsilon}{2^n}$ which is centered at r_n. Then set $U = \bigcup_n I_n$. By construction, U must be dense and open in $(0, 1)$, however

$$\mu(U) \leq \sum_{i=1}^{\infty} \mu(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon$$

and hence can be as measure-theoretically small as we want. From this we can also find the opposite, where we have something topologically small but measure-theoretically big, by taking $K = [0,1] - U$. By monotonicity

$$\mu(K) \geq 1 - \epsilon$$

and yet K is nowhere dense.

There is at least one intuition that can remain: a nonempty *open* set will have non-zero measure. For more on how open/closed sets interact with the measure, see the exercises.

Note that the converse that a non-zero measure set will have an open intervals, is not actually true! For that, we can explore the concept of *cantor sets*

Cantor Set

(was gonna look through: (commented so that this complies

Definition 1.4.8: Classical Cantor Set

Let $I = [0,1]$. From this, we'll construct the cantor set iteratively

1. Start by removing the middle third $I_1 = I - \left(\frac{1}{3}, \frac{2}{3}\right)$
2. Given I_i, remove the open middle third of each connected compoennt

Let $C = \cap_{n=0}^{\infty} I_n$ be the Cantor set. In particular, it is the *standard middle thirds Cantor set.* We'll usually refer to this set as "the" Cantor set.

Notice that C is measurable by continuity from above since $C_1 \subseteq C_2 \subseteq \cdots \subseteq C_n \subseteq$.

Proposition 1.4.9: Properties Of Cantor Sets

Give C the standard subspace metric of $[0,1]$. Then C is:

1. nonempty and uncountable
2. compact
3. nowhere dense ($\text{int}(\overline{A}) = \emptyset$ for all $A \subseteq C$
4. $m(C) = 0$
5. totally disconnected

***Proof*:**
Using some topological properties, since C is the intersection of compact spaces, it is compact.

Next, the cantor set is nonempty since $0 \in C_k$ for all k, and so $0 \in C$. To show C is uncountable, Notice that we can represent every element of the uncountable set $[0,1]$ in it's base 3 expansion, namely for appropriate $a_k \in \{0,1,2\}$:

$$x = \sum_{k=1}^{\infty} a_k 3^{-k}$$

Then take:

$$f : [0,1] \to C \qquad x = \sum_{k=1}^{\infty} a_k 3^{-k} \mapsto$$

To show total disconnectedness, let's take the same I as before. I is closed in $\mathbb{R}$, and thus in C^n. Note that I^c is the union of finite closed sets, and thus is also closed. Thus, I is clopen in C^n. Thus, $C \cap I$ is also a clopen neigborhood of x in C. Since x was arbitrary, C is totally disconnected.

Finally uncountablenss come from C being compact and complete, which makes C uncountable (exercise).

This construction does not rely on the size of the middle-third; in fact, we can make it converge to any value we want. To see this, first notice that To do this, we first prove a lemma:

Lemma 1.4.10

Suppose $\{\alpha_i\}_{i=1}^{\infty} \subseteq (0,1)$.

1. $\prod_1^{\infty}(1-\alpha_i) > 0$ if and only if $\sum_1^{\infty} \alpha_i < \infty$ (Compare $\sum_1^{\infty} \log(1-\alpha_i)$ to $\sum_1^{\infty} \alpha_i$)
2. Given $\beta \in (0,1)$, there always exists a sequence $\{\alpha_i\}_{i=1}^{\infty}$ such that $\prod_1^{\infty}(1-\alpha_i) = \beta$

***Proof*:**
First, we see that we can convert $\prod_i^{\infty}(1-\alpha_i)$ to $\sum_i^{\infty} \log(1-\alpha_j)$ by considering

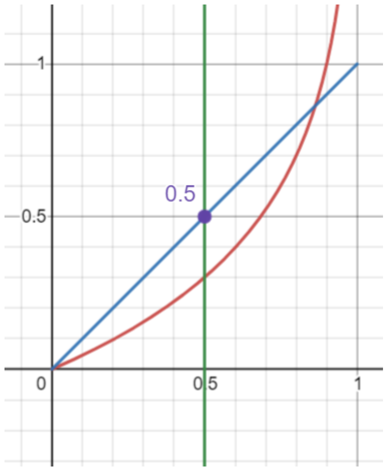
$$\begin{aligned}
\log\left(\prod_i^{\infty}(1-\alpha_i)\right) &= \log\left(\lim_{n\to\infty}\prod_i^{n}(1-\alpha_i)\right) \\
&\overset{!}{=} \lim_{n\to\infty} \log(\prod_i^{n}(1-\alpha_i)) \\
&= \lim_{n\to\infty} \sum_i^{n} \log(1-\alpha_i) \\
&= \sum_{i=1}^{\infty} \log(1-\alpha_i)
\end{aligned}$$

where $\overset{!}{=}$ comes from continuity of log for $1-\alpha_i > 0$. Notice that it's impossible that $1-\alpha_i \leq 0$ form some α_i terms since $\alpha_i \in (0,1)$, and so this proof works. Thus, we get the equivalent condition of

$$\sum_i^{\infty} \log(1-\alpha_i) > -\infty$$

With this setup, we proceed with the proof

($\Leftarrow$) Let $\sum_i^{\infty} \alpha_i < \infty$. Then we know that $\alpha_i \to 0$. Thus, pick an N such that for all $i > N$, $\alpha_i < 0.5$. Then notice that $\log(1-\alpha_i) < \alpha_i$. This is easier to see with a visual aid:



In particular, for all $0 < x < 1/2$, we have $|\log(1-x)| - x > 0$, showing that x is greater than $\log(1-x)$ for all $x \in (0, 1/2)$.

Then from here, this is a simple application of the Weiestrass M-test. Let $K = \sum_i^N \log(1-\alpha_i)$ to eliminate the irregular part. Then since $|\log(1-\alpha_i)| < \alpha_i$, for all $i > N$ and $\sum_i^\infty \alpha_i$ converges, by the Weiestrass M-test the $\sum_{i=N+1}^\infty \log(1-\alpha_i)$ converges. Let's say it converges to L. Then

$$\sum_{i=1}^\infty \log(1-\alpha_i) = K + L > -\infty$$

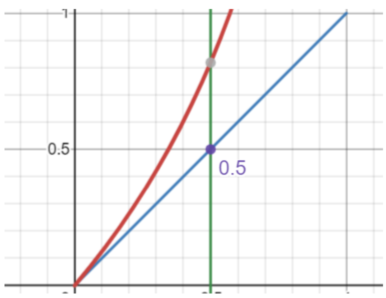
as we sought to show.

($\Rightarrow$) let $\prod(1-\alpha_i) > 0$ so equivalently $\sum_i^\infty \log(1-\alpha_i) > -\infty$ so $\sum_i^\infty \log(1-\alpha_i) = K$ for appropriate $K \in \mathbb{R}$. Then since the series converges, we have that

$$eK = e\sum_i^\infty \log(1-\alpha_i) = \sum_i^\infty e\log(1-\alpha_i)$$

i.e., we can scale the terms by e. We will show why this scaling is important in a moment; before we do we must setup a little more.

Since the series converges, it must be that $e\log(1-\alpha_i) \to 0$. So pick N such that for all $i > N$, $e\log(1-\alpha_i) < 1/2$. Then notice that $\alpha_i < e\log(1-\alpha_i)$. This is easier to see with a visual aid:



Or, to carry out some of the computations, if you raise $e\log(1-\alpha_i) = \log(1-\alpha_i)e$ to the power of e, you get

$$(e^{\log(1-\alpha_i)})^e = (1-\alpha_i)^e \geq \alpha_i^e \quad \forall \alpha_i \in (0, 1/2]$$

and so the inequality holds. Then the proof continues as for the $\Leftarrow$ direction by setting up an appropriate constant term to bound the irregularities, then using the Weiestrass M test bound every α_i by $\log(1-\alpha_i)$, and since $\sum_i^\infty \log(1-\alpha_i)$ converges, we get that $\sum_{i=N}^\infty \alpha_i$ also converge, and since it will be the sum of a constant plus the convergent value, $\sum_{i=1}^\infty \alpha_i$ converges, as we sought to show.

Now, let β be given. Let $x_i = \prod_{i=1}^n (1-\alpha_i)$. We'll construct a sequence where

$$x_1 = \beta + \frac{1-\beta}{2} \qquad x_2 = \beta + \frac{1-\beta}{4} \quad \cdots \quad x_n = \beta + \frac{1-\beta}{2^n} \cdots$$

to accomplish this, let

$$\alpha_i = 1 - \frac{\beta + \frac{1-\beta}{2^i}}{\beta + \frac{1-\beta}{2^{i-1}}} \tag{1.2}$$

This construction comes from solving the following equation:

$$\left(\beta + \frac{1-\beta}{2}\right) y_2 = \beta + \frac{1-\beta}{4} \ \Rightarrow\ y_2 = \frac{\beta + \frac{1-\beta}{4}}{\beta + \frac{1-\beta}{2}}$$

so $\alpha_2 = 1 - y_2$, and clearly $\alpha_2 \in (0,1)$ since the denominator is bigger than the numerator in the above calculations. Generalizing this pattern, we get the that α_i would be of the from presented in equation (1.2) and $\alpha_i \in (0,1)$. Thus, we get that

$$\lim_{n\to\infty} \prod_i^n (1-\alpha_i) = \lim_{n\to\infty} \beta + \frac{1-\beta}{2^n} = \beta$$

as we sought to show

Now, let's say we are starting with $[0,1]$ but on each iteration eliminates intervals that sum to α_i. Let K_i be the ith step in this itteration. Notice that:

$$m(K_{i+1}) = (1-\alpha_i)m(K_i)$$

Then by what we've just shown, we can make $K = \cap_i^\infty K_i$ be any value up to and including 0 and 1.

Proposition 1.4.11: All Cantor Set Homeomorphism

Let C_n be a cantor set where $m(C_n) = n$. Then $C_n \cong C_0$

***Proof*:**
section 9 of Pugh – will get back to it

You can also prove uncountability directly by making a string represent which side of the split interval you go to.

Next, you might've seen many uncountable sets which are dense in $\mathbb{R}$. Here we have an absolutely facsinating result: C is *nowhere dense in* $[0, 1]$.

Definition 1.4.12: Dense, Somewhere Dense, Nowhere Dense

If $S \subseteq M$ and $\overline{S} = M$, then S is dense in M. The set S is somewhere dense if there exists an open nonempty set $U \subseteq M$ such that $\overline{S \cap U} \supset U$. If S is not somewhere dense, then it is *nowhere dense*

Proposition 1.4.13: C Nowhere Dense

The Cantor set contains no interval and is nowhere dense in $\mathbb{R}$

***Proof*:**
To prove it doesn't contain an interval, assume it does, then chose an n large enough, that is $(\frac{1}{3})^n < b - a$, so that it will not be in the interval.

For nowhere dense, assume ti's somewhere then, then $\overline{C \cap U} \supset U \supset (a, b)$ – a contradiction.

We know present how Cantor sets are almost initial objects in compact metric space

Theorem 1.4.14: Cantor Surjection Theorem

Given a compact nonempty metric space M. there is a continuous surjection of C onto M

***Proof*:**
exercise

Finally, we present how all cantor sets are homeomorphic.

Theorem 1.4.15: Moore-Kline Theorem

We say M is a Cantor space if it is compact, nonempty, perfect, and totally disconnected (like the cantor set). Every cantor space M is homeomorphic to the standard middle-thirds Cantor set C.
In other words, those properties *define* Cantor sets.

***Proof*:**
Pugh. p.112

Corollary 1.4.16: Cantor And Fat Cantor Set

The fact Cantor set is homeomorphic to the standard Cantor set

***Proof*:**
Using 1.4.15, the result is immediate.

This is intersting, since it shows that meausre is *not* a topological poperty! This should makes sense with $(0,1)$ and $\mathbb{R}$ being homeomorphic with the same measures, but different, but this is an even more extreme example.

Corollary 1.4.17: Product Of Cantor Sets

A cantor set is homeomorhic to its own cartesian product; $C \equiv C \times C$.

***Proof*:**

A fact that Folland added that I thought was cool is that since every subset of the cantor set is Lebesgue Measurable (since the cantor set has measure zero and the Lebesgue measure is complete), we have that

$$|\mathcal{L}| = |P(\mathbb{R})| > \aleph_1 \quad \text{but} \quad |\mathcal{B}_{\mathbb{R}}| = \aleph_1$$

so there are a *tone* of zero-measure sets, however, we can essentially ignore them because they are zero measure sets!!

For those who are Categorically inclined, Cantor sets are almost like the initial objects of the category of compact metric spaces. They are not initial because there does not exist a unique uniformally continuous function from the cantor set to any compact metric space, but there always will exist at least 1 such function.

Exercise 1.4.1

1. If $E \in \mathcal{L}$ and $m(E) > 0$, for any $\alpha < 1$, there is an open interval I such that $m(E \cap I) > \alpha m(I)$

2. If $E \in \mathcal{L}$, and $m(E) > 0$, the set $E - E = \{x - y \mid x, y \in E\}$ contains an interval centered at 0 (If I is as in the previous exercise with $\alpha > \frac{3}{4}$, then $E - E$ contains $(-\frac{1}{2}m(I), \frac{1}{2}m(I))$).

3. Let E be a Lebesgue measurable set (so $E \in \mathcal{L} = \overline{\mathcal{M}}(\mathcal{B}_{\mathbb{R}})$)

   1. If $E \subseteq N$ where N is a non-measurable set described in section 1.1, then $m(E) = 0$
   2. if $m(E) > 0$, then E contains a non-measurable set (it suffices to assume $E \subseteq [0,1]$. In notation of section 1.1 $E = \cup_{r \in R} E \cap N_r$

2

Integration

In this chapter, we now work with function's between measure spaces that preserve measure structure! Measurable functions feel like the types of functions we "usually" work with, the same way measurable sets are the sets we "usually" work with. Being continuous can be a bit strong (ex. being a monotone function doesn't imply continuous), but measurability rectifies that by in a sense expanding what are the possible "pre-images" we allow to take (ex. Monotone functions *are* always measurable, as we will demonstrate).

After defining measurable functions, we will focus on real and complex functions (i.e. functions with codomain $\mathbb{R}$ or $\mathbb{C}$). In the real case, I like to think of real functions as given every point in the domain X a value or a weight, in contrast to a measure which is a real function which simply assigns a constant value at each point (this will become more clear as we introduce the definitions). For the complex case, each point is given a weight and a *phase* or *state*[1]. In terms of computation, we simply split the problem into the real and imaginary component. Then, we will define how to "measure" a real (resp. complex) measurable function. We then focus on measurable functions which have "finite measure", and will call them *(Lebesgue) integrable functions*; integrable functions will allow us to have some regularity in how we can combine measurable functions without running into some problems with infinities (as we'll soon show). One result of this is that we can think of the integral as being a "measure" for a function, and since integrable function will have "finite measure", it will make it more meaningful to see which functions are "close" (in the appropriate sense of the word close) to a certain function.

A word must be said about Lebesgue integrability vs. Riemann integrability. The key difference is the allowed measurable sets. For Riemann integrability, all our measurable sets are *rectangles*: every partition of the area of integration can be seen as a sum of indicator functions over pair-wise disjoint

[1]you can think of every input having a natural magnitude that periodically shifts. The phase $e^{i\theta}$ tracks this. So if $f(a) = 10$ and $f(b) = 10e^{-i\pi} = -10$, then a is at the beginning of its phase with magnitude 10, while b is half way through its phase which shifted it's magnitude to -10. Addition adds the interference (ex. $(2+i) + (-2-2i) = -i$) while

rectangles. On the other hand, Lebesgue integration is over *measurable* sets. The strategy shall shift so that instead of partitioning the domain based off of its geometry, we shall take partition it using $f^{-1}([-k,k])$ (or really any measurable covering of $\mathbb{R}$, and appropriately modified for $\mathbb{C}$) where we shall show the pre-image of a measurable set is measurable if f is measurable. This shall solve many headache the Riemann integral imposes (ex. any rectangle of $\chi_{\mathbb{Q}}$ on $[0,1]$ will have have rational and irrational points, making it unfeasible to measure).

After we have defined integrable functions, we will have to take a moment to compare different types of convergences, since at that stage we will have 4 different ways functions can converge. We will then move on to trying to bring the theory of the product of σ-algebras to measure and integrable functions more generally, allowing us to construct larger measures and integrable functions from smaller components. Importantly, we will show that we can find the a measure $\mu \times \nu$ by separating any problem to only measuring "slices" of the set with respect to μ, then "slices" of the set with respect to ν (this is Fubini's Theorem).

2.1 Measurable Functions

Recall that if $f : X \to Y$ is a function, then $f^{-1} : P(Y) \to P(X)$ is a function where $f^{-1}(E) = \{x \in X \mid f(x) \in E\}$ which preserves unions, intersections, and compliments. Thus, if $\mathcal{N}$ is a σ-algebra on Y, $f^{-1}(\mathcal{N})$ is a σ-algebra on X. If X and Y are measure space, if $f^{-1}(\mathcal{N})$ maps into $\mathcal{M}$, we will call f *measurable*:

Definition 2.1.1: Measurable Function

Let $f : X \to Y$ be a function and $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. Then f is called a $(\mathcal{M}, \mathcal{N})$-*measurable function* (or just measurable) if for all $E \in \mathcal{N}$

$$f^{-1}(E) \in \mathcal{M}$$

Just like in topology, this could be thought of as a surjection of the measurable space on X onto the measurable space on Y. It should be clear that the composition of two measurable functions is measurable (in particular, the composition of an $(\mathcal{M}, \mathcal{N})$-measurable function with $(\mathcal{N}, \mathcal{O})$-measurable function is $(\mathcal{M}, \mathcal{O})$-measurable), and that the identity function $\mathrm{id} : X \to X$ is clearly a measurable function, and so the collection of measurable functions and measurable spaces from a category, usually denoted **Meas**.

Just like lemma 1.1.4 was useful for simplifying proofs, we proof the following:

Lemma 2.1.2: Measurable Function and Generating Sets

Let $\mathcal{N}$ be generated by $\mathcal{E}$. Then $f : X \to Y$ is a $(\mathcal{M}, \mathcal{N})$-measurable function if and only if for all $E \in \mathcal{E}$, $f^{-1}(E) \in \mathcal{M}$

***Proof*:**
The $\Rightarrow$ direction is *a-fortiori* true by definition. The $\Leftarrow$ direction comes from f^{-1} preserving unions and compliments, so $\left\{E \subseteq Y \;\middle|\; f^{-1}(E) \in \mathcal{M}\right\}$ is a σ-algebra that contains $\mathcal{E}$, and so contains $\mathcal{N}$.

From this, there is a very nice corrollary when $\mathcal{M}$ is a collection of Borel sets:

Corollary 2.1.3: Continuous Function ⇒ [Borel] Measurable Function

Let $f : X \to Y$ be a continuous function and Y have the borel σ-algebar. Then f is a measurable function.

***Proof*:**
Since f is continuous, the pre-image of an open set (in particular, open intervals) is open, and hence by proposition 1.1.7 and lemma 2.1.2 it is measurable

Naturally, in analysis, the most important measurable functions will be of the form

$$f : \mathbb{R} \to \mathbb{R} \qquad f : \mathbb{R} \to \mathbb{C}$$

This also means that the domain can either have the $\mathcal{B}_{\mathbb{R}}$ σ-algebra or $\mathcal{L}$ σ-algebra.

Example 2.1.4: Measurable Functions
Most functions you have worked with will in fact be measurable

1. As already shown, all continuous functions are measurable
2. All monotone functions are measurable: the pre-image of closed rays will either be open or closed rays, and so by proposition 1.1.7, they are measurable.
3. All indicator functions over measurable sets are measurable (hence, bump functions are essentially measurable)
4. All Riemann integrable functions are measurable (we will show this this in section 2.3.1)
5. We will soon show that the point-wise limit of measurable functions is also measurable, hence almost all examples of functions that shall be studied are measurable.

It is in practice rare to work with non-measurable functions unless a contrived example is used, like a measurable function purposefully built to map to non-measurable sets (like in the case of the composition of two Lebesgue measurable functions)

2.1.5 Convention e will *always* assume the codomain will have $\mathcal{B}_{\mathbb{R}}$ (resp. $\mathcal{B}_{\mathbb{C}}$) σ-algebara. In fact, if we have a function $f : X \to \mathbb{R}$ (resp. $f : X \to \mathbb{C}$), then we will sometimes say f is $\mathcal{M}$-measurable instead of $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable (resp. $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable). If the domain is also Borel, we will say that f is *Borel measurable* (and ommit the borel measure if it's $\mathbb{R}$ or $\mathbb{C}$). Similarly, if the domain is $\mathcal{L}$, we will call f Lebesgue measruable.

Becasue of the convention of the co-domain being Borel, it is possible that the composition of two Lebesgue measurable functions is *not* Lebesgue measurable: if $f : \mathbb{R} \to \mathbb{R}$, $g : \mathbb{R} \to \mathbb{R}$ are two Lebesgue measruable functions where $E \in \mathcal{B}_{\mathbb{R}}$, $g^{-1}(E) \in \mathcal{L}$ but $g^{-1}(E) \notin \mathcal{B}_{\mathbb{R}}$, it is possible that $f^{-1}(g^{-1}(E)) \notin \mathcal{L}$

Example 2.1.6: Composition Not Lebesgue Measurable
Let $f : [0, 1] \to [0, 1]$ be the cantor function form section 1.5, and let $g(x) = f(x) + x$.

1. g is a bijection from $[0, 1]$ to $[0, 2]$ and $h = g^{-1}$ is continuous from $[0, 2]$ to $[0, 1]$

   ***Proof*:**
   First, since f is increasing (though not injective) and x is strictly increasing, then $f + x$ is injective. Since f and x are both continuous, so is $g = f + x$. Hence the pre-image of a closed set must be closed. If we consider $g^{-1}([0, 2]) = D$, then using the fact that g is a monotonic functions, since $g(0) = 0$ and $g(1) = 2$, it must be that $D = [0, 1]$, showing that g is indeed surjective, and hence bijective.

   To show that $g^{-1} : [0, 2] \to [0, 1]$, is continuous, bijective, and the codomain is Hausdorff, then g is homeomorphic (recall that these imply that g is an open map: if $U \subseteq [0, 1]$ then $S = [0, 1] - U$ is closed, and since g is continuous $g([0, 1] - U) = [0, 2] - g(U)$ is closed, so $g(U)^c$ is closed, so $g(U)$ is open), and hence g^{-1} is continuous.

2. If C is the cantor set, then $m(g(C)) = 1$

   ***Proof*:**
   We will take advantage of translation invariance of the Lebesgue measure. Let C be the cantor set on $[0, 1]$. Then C can be thought of as the interval $[0, 1]$ minus all of the open sets that are eliminated during construction:

   $$C = [0, 1] - I_1 - I_{21} - I_{22} - \cdots - I_{ik} - \cdots$$

   Let $I = \coprod_{i,k} I_{ik}$ collection of these sets so that $C = [0, 1] - I$. Since $m(C) = 0$, $m(I) = 1$. Consider now $g(I)$. Since g is injective and I is the disjoint union of sets, then there exists sets $\{J_j\}_{j=1}^{\infty}$ such that $J_j = g(I_{ik})$. We now examine the measure of the sets J_j.

   By definition, I_{ik} does not contain any points of the cantor set, and so for appropriate $c_{ik} \in C$,

   $$g_{I_{ik}}(x) = f(c_{ik}) + x$$

   that is, g on the interval I_{ik} is of the form $x + f(c_{ik})$ for constant $f(c_{ik})$! Since the Lebesgue measure is translation invariant, we get:

   $$m(g(I_{ik})) = m(g(I_{ik}) - f(c))$$

   and so, since the I_{ik} intervals are all disjoint and g is injective:

   $$g(\coprod I_{ik}) = \coprod g(I_{ik})$$

   and so the measure of $g(I)$ is in fact exactly the measure of I, that is, $m(g(I)) = m(J) = 1$. Thus:

   $$2 = g(m([0, 2])) = m(g(C) \coprod J) = m(C) + m(J) = m(g(C)) + 1$$

   and so $m(g(C)) = 1$, as we sought to show.

3. by exercise 29, $g(C)$ contains a Lebesgue non-measurable set A. Let $B = g^{-1}(A)$. Then B is Lebesgue measurable, but not Borel

> ***Proof*:**
> Since $A \subseteq g(C)$, $B \subseteq g^{-1}(g(C)) = C$. Since $m(C)$ is zero, by completeness, $m(A) = 0$, and so A is Lebesgue measurable.
>
> Next, to show that B is not Borel-measurable, For the sake of contradiction let's assume it was. Then since g^{-1} is continuous, it is Borel-measurable, meaning the pre-image (with respect to g^{-1}) of a measurable set is measurable. However, $g(B) = A$, which is not measurable, and hence B is not Borel measurable.

4. There exists a Lebesgue measurable function F and a continuous function G on $\mathbb{R}$ such that $F \circ G$ is not Lebesgue measurable

> ***Proof*:**
> Let F be the indicator function on B, I_B, and $G = g^{-1}$. Then F and G are clearly Lebesgue measurable, However, G^{-1} does not map all Lebesgue measurable sets to Lebesgue measurable sets (in particular, B maps to a Borel set), and so
>
> $$(F \circ G)^{-1}(B) = (G^{-1} \circ F^{-1})(B) = (g(I_B^{-1}(B)) = g(B) = A$$
>
> However, A is not Lebesgue measurable set, and so $F \circ G$ is not Lebesgue measurable.

Just like for Borel sets earlier, we have the following equivalence for measurable functions with codomain $\mathbb{R}$:

Proposition 2.1.7: Borel sets and Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $f : X \to \mathbb{R}$. Then the following are equivalent:

1. f is $\mathcal{M}$-measurable
2. $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
3. $f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
4. $f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
5. $f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$

> ***Proof*:**
> This follows from proposition 1.1.7 and lemma 2.1.2.

Similarly, we will soon be encountering co-domains which are product spaces (ex. $\mathbb{R}^n$ or $\mathbb{C}^n$). In particular, let's say we had some set X, and some family of measurable spaces $\{(Y_\alpha, \mathcal{M}_\alpha)\}_{\alpha \in A}$, and $f : X \to Y_\alpha$ is a map for each $\alpha \in A$. Then we can impose a unique smallest σ-algebra on X that makes each f_α a measurable function, in particular, $\mathcal{M}_X = \left\{ f_\alpha^{-1}(E_\alpha) \;\middle|\; E_\alpha \in \mathcal{M}_\alpha,\ \alpha \in A \right\}$ [2]. This σ-algebra on X is called the *σ-algebra generated by* $\{f_\alpha\}_{\alpha \in A}$.

[2]A similar strategy is used in the constrution of *weak* topologies. If you're more categorical, this constructions makes X satisfy the universal property of products in the category of **Meas**

Proposition 2.1.8: Product of Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $(Y_\alpha, \mathcal{N}_\alpha)$ be a collection of measurable spaces, so $Y = \prod_{\alpha \in A} Y_\alpha$, $\mathcal{N} = \bigotimes_{\alpha \in A} \mathcal{N}_\alpha$, and $\pi_\alpha : Y \to Y_\alpha$ is the coordinate map. Then $f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable if and only if $f_\alpha = \pi_\alpha \circ f$ is $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable.

***Proof*:**
Let's say f is $(\mathcal{M}, \mathcal{N})$-measurable. Then since the composition of two measurable functions is measurable, so are each f_α is measurable (since π_α is measurable by definition of $\bigotimes$).

Conversely, if every f_α is $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable, then for each $E_\alpha \in \mathcal{N}_\alpha$, $f^{-1}(\pi_\alpha^{-1}(E_\alpha)) = f^{-1}(E_\alpha) \in \mathcal{M}$, and since these sets generate $\mathcal{M}$, we have that f is $(\mathcal{M}, \mathcal{N})$-measurable by proposition 2.1.7

This proposition also let's us define what it means to take the "product" of two functions. As a consequence, we can give an easy criterion for when complex functions are measurable:

Corollary 2.1.9: Complex Function Measureability

Let $f : X \to \mathbb{C}$ be a complex function. Then f is $\mathcal{M}$-measurable if and only if $\mathrm{Re} f$ and $\mathrm{Im} f$ are both measurable.

***Proof*:**
This is simply taking advantage of the topology of $\mathbb{C}$, namely

$$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} \overset{!}{=} \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$$

where the $\overset{!}{=}$ equality comes from the fact that $\mathbb{R}$ is separable (see proposition 1.1.11). And so proposition 2.1.8 with the fact that Re and Im can be considered projections completes the proof.

Sometimes, we want to work with the extended real line (or in general some compactification of a topological space). If $\overline{\mathbb{R}} = [-\infty, \infty]$, Then let

$$\mathcal{B}_{\overline{\mathbb{R}}} = \left\{ E \subseteq \overline{\mathbb{R}} \;\middle|\; E \cap \mathbb{R} \in \mathcal{B}_{\mathbb{R}} \right\}$$

Notice that is we put the metric ρ on $\overline{\mathbb{R}}$ to be $\rho(x, y) = |\tan^{-1}(x) - \tan^{-1}(y)|$, then $\mathcal{B}_{\overline{\mathbb{R}}}$ is indeed the usual definition of a Borel σ-algebra (exercise). It should be verified that $\mathcal{B}_{\overline{\mathbb{R}}}$ can be generated by open or closed rays, and so we will say $f : X \to \overline{\mathbb{R}}$ is $\mathcal{M}$-measurable if it is $(\mathcal{M}, \mathcal{B}_{\overline{\mathbb{R}}})$-measurable.

Next, if the codomain has some notion of $+$ and $\cdot$, we establish that the basic "$\mathbb{C}$-algebra operations" work on measurable functions:

Proposition 2.1.10: Arithmetics on Measurable Functions

Let $f : X \to \mathbb{C}$ and $g : X \to \mathbb{C}$ be $\mathcal{M}$-measurable functions. Then $f + g$, fg, and zf for $z \in \mathbb{C}$ are $\mathcal{M}$-measurable functions

Note that we have limited ourselves to $\mathbb{C}$. This still works with $\overline{\mathbb{R}}$ with the appropriate care of the $\infty - \infty$ and $0 \cdot \infty$ case (exercise 2 in book)

***Proof*:**
The proof for zf being measurable is similar to that of $+$ and $\cdot$, so is left as an exercise.

Since f and g, are measurable, then their "cartesian product" is measurable by proposition 2.1.8

$$F : X \to \mathbb{C} \times \mathbb{C} \qquad F(x) = (f(x), g(x))$$

In particular, F is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathbb{C}})$-measurable. Next, consider $p : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, $p(x, y) = x + y$ and $t : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, $t(x, y) = xy$ (p for plus, t for times). Then since p and t are continuous, by corollary 2.1.3 p and t are $(\mathcal{B}_{\mathbb{C}\times\mathbb{C}}, \mathcal{B}_{\mathbb{C}})$-measurable. Thus, $p \circ F$ and $t \circ F$ is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable, and by definition

$$f + g = p \circ F \qquad fg = t \circ F$$

completing the proof

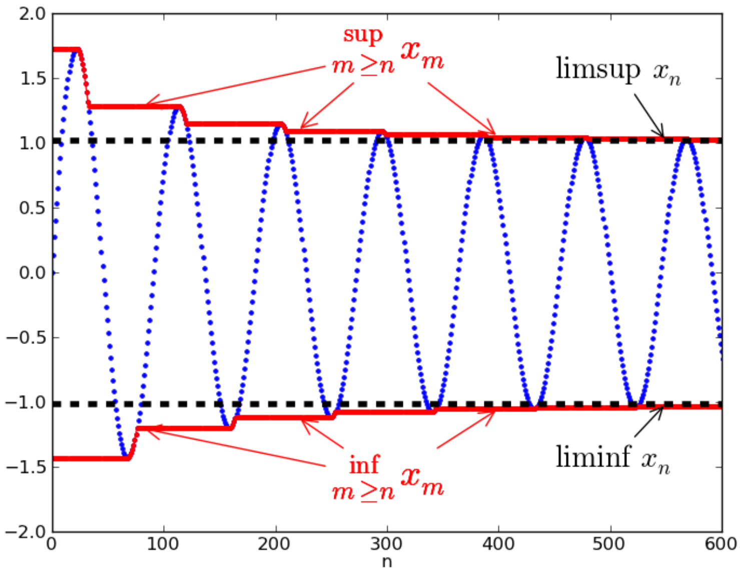
The next operations we'll show are measurable are sup, inf, lim sup and lim inf. This is perhaps one of the most important functions to be measurable since they are at the heart of many constructions in analysis. Recall that

$$\sup_i f_i(x) = \sup \{f_i(x) \mid i \in \mathbb{N}\}$$

and so this expression means we're taking the set $f_i(x)$ for each i and taking the sup of that set. In other words, for every point x we're taking the supremum of the set $\{f_1(x), f_2(x), ...\}$. Symmetrically the same for inf. For lim sup, recall that:

$$\limsup_i f_i(x) = \lim_{i\to\infty} \sup \{f_k(x) \mid k \geq i\}$$

that is, we are constantly shrinking the set over which we are doing the supremum, which in a sense means we're trying to "hug" the value from above as close as possible in the limit. Symmetrically the same for lim inf. If you're visual, I like to keep this image in mind: let's say the blue line is the value of $f_i(x_0)$ for a specified x_0. Then:



Comment here for future reference:

Proposition 2.1.11: Measurable functions and Limit Bounding

Let $f_i : X \to \overline{\mathbb{R}}$ be $\mathcal{M}$-measurable functions for each i and consider $\{f_i\}_{i=1}^{\infty}$. Then

$$g_1(x) = \sup_i f_i(x) \qquad g_2(x) = \inf_i f_i(x)$$

$$g_3(x) = \limsup_{i\to\infty} f_i(x) \qquad g_4(x) = \liminf_{i\to\infty} f_i(x)$$

The codomain $\overline{\mathbb{R}}$ was used so that the sup and inf is still considered defined if sup $= \pm\infty$ or inf $= \pm\infty$.

***Proof*:**
To show $\sup_i f_i$ and $\inf_i f_i$ are measurable, since the codomain domain is $\overline{\mathbb{R}}$, which is equivalent measure-theoretically to $\mathbb{R}$, we will show the pre-image of open rays is the union of measurable sets:

$$\begin{aligned} x \in g_1^{-1}((a,\infty]) &\Leftrightarrow g_1(x) > a \\ &\Leftrightarrow f_i(x) > a && \text{for some } i \\ &\Leftrightarrow x \in f_i^{-1}((a,\infty]) && \text{for some } i \\ &\Leftrightarrow x \in \bigcup_{i=1}^{n} f_i^{-1}((a,\infty]) \end{aligned}$$

and similarly for g_2, except considering $(-\infty, a]$, therefore:

$$g_1^{-1}((a,\infty]) = \bigcup_i^{\infty} f_i^{-1}((a,\infty]) \qquad g_2^{-1}(-\infty,a])) = \cup_i^{\infty} f_i^{-1}([-\infty,a))$$

since each f_i is measurable, we have a union of measurable sets, so g_1 and g_2 are measurable by proposition 1.1.11. Next, let

$$h_k(x) = \sup_{i>k} f_i(x) = \sup\{f_k(x) \mid k > i\}$$

then by what we've just established, h_k is measurable for each k. Then by of $\limsup$:

$$g_3 = \inf_k h_k$$

and so g_3 is measurable since we just established the inf of measurable functions is measurable. Similarly for g_4.

as an immediate corollary:

Corollary 2.1.12: Measurability of Max and Min

Let f, g be measurable functions. Then so is $\max\{f, g\}$ and $\min\{f, g\}$

Corollary 2.1.13: Measurable Preserved under Pointwise limit

Let $\{f_i\}$ be a sequence of measurable function. If $\lim_{i\to\infty} f_i(x)$ exists for every x, and we define $f : X \to \overline{\mathbb{R}}$ to be

$$f(x) = \lim_{i\to\infty} f_i(x)$$

then f is measurable (i.e. point-wise convergence preserves measureability)

Recall that point-wise convergence *does not preserve continuity* (the steeple function's are the usual example). However, measureability is sufficiently general to be preserved under this weaker form of convergence! Naturally, if the sequence of functions $\{f_i\}$ are all $\mathcal{M}$-measurable and uniformally convergent, they are measurable

***Proof*:**
If f exists (i.e., every $f_n(x)$ converges pointwise to $f(x)$), then by proposition 2.1.11, $f = g_3 = g_4$, completing the proof.

2.1.14 Remark Note how important it is that *all* points converge! It is possible that a single point doesn't converge (ex. $\lim f_i(0)$ doesn't converge), and so the limit is *not* a measurable function, in fact it doesn't even need to be a function. Take for example the collection of functions $f_k : \mathbb{R} \to \mathbb{R}$ with the properties:

1. f_k is continuous (it can even be smooth)
2. $\int_{\mathbb{R}} f(x)dx = 1$ (In the Riemann sense), or really any constant $c \in \mathbb{R}_{>0}$
3. $\text{supp} f \subseteq \left[\frac{-1}{k}, \frac{1}{k}\right]$

In other words, the collection of f_k are all smooth bump functions with volume 1, but the area in which it has support is shrinking, and so the "bump" of each f_k must be getting higher and higher (to define such functions, consider starting with $f_1 = e^{-\frac{1}{1-x^2}}$ where $x \in (-1, 1)$ and 0 otherwise). Then $f_k \to 0$ at all but 1 point, namely $f_k(0)$ does *not* converge. As a consequecne, this limit is *not* a function!
This example shows the important of somehow "bounding" your area; we will get back to this observation in section 2.2. Some might think that this example is a bit cheeky, it still feels like the function is measurable if we just circumnavigate this problem by making the domain of the function $\mathbb{R} \cup \{\infty\}$, there is the trouble of defining a measure on the 1-point compactification of $\mathbb{R}$ while extending the measure of $\mathbb{R}$[a]. If the reader wonders if there is a way of defining some generlization of a function that will accept the fact that the f_k converge to something that still somehow preserves the fact that $\int f = 1$, then we will show in in chapter 11 that this converges to something called a *distribution*[b]

[a] In particular, $\mu(\{\infty\}) = \infty$, as can be shown using continuity form above, and so somehow we have "gained" an infinite amount of area. On the other-hand, if we just ignore this and consider that ∞ is just some point like any other point, then we have "lost" area

[b] The example named here is a distribution called the *delta distribution*, or sometimes *delta function* for reasons that is covered in chapter 11

Now that we proved that these operations preserving measurability, we will introduce some new notation to decompose real and complex functions which will soon become useful when defining integrable functions.

Definition 2.1.15: Real and Complex Decomposition

Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{C}$ be real and complex measurable functions respectively. Then

$$f^+ = \max\{f, 0\} \qquad f^- = \max\{-f, 0\}$$

and

$$\text{sgn}(g(x)) = \begin{cases} \frac{g(x)}{|g(x)|} & g(x) \neq 0 \\ 0 & g(x) = 0 \end{cases} \qquad |g(x)| = |z| = |x + iy| = \sqrt{x^2 + y^2}$$

which gives the decomposition of:

$$f = f^+ - f^-, \ g = (\text{sgn} g)|g|$$

The functions f^+ and f^- are clearly well-defined and measurable. For the complex case, $|\cdot|$ is continuous everywhere and $z \mapsto \text{sgn} z$ is continuous everywhere except 0. It is measurable since if $U \subseteq \mathbb{C}$ is open, then $\text{sgn}^{-1}(U)$ is either V for some open set V, or $V \cup \{0\}$, meaning sgn is Borel measurable. In some textbooks, arg is used instead of sgn.

2.1.1 Standard Representation

Recall that every measurable function is the point-wise limit of a bunch of easy to work with function, in particular, linear combination of characteristic functions! As a reminder, if $\chi_E : X \to \mathbb{R}$ (or $\chi_E : X \to \{0, 1\}$) where $E \subseteq X$, then:

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

Then χ_E is called the characteristic (or indicator) function. It is sometimes also denoted as 1_E or I_E.

Definition 2.1.16: Simple function

Let $\{E_i\} \subseteq P(\mathbb{C})$ be some finite collection of sets. Then the finite linear combination of characteristic functions on E_i multiplied by a complex constant:

$$f = \sum_{i=1}^{n} z_i \chi_{E_i}$$

is called a *simple function*

Example 2.1.17: Simple Functions

1. Show that $|f(X)| < \infty$ then f is in fact a simple function (hint: let $E_i = z_i$ for each $z_i \in f(X)$). In fact, each coefficient can also be distinct (let it be the respective z_i from the range)
2. Show that if f and g are simple functions, so are $f + g$ and fg

Using these, we can now show that any measurable function is simply the limit of simple functions!

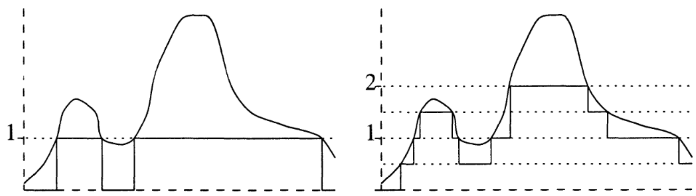
Theorem 2.1.18: Simple Function Approximation

Let $(X, \mathcal{M})$ be a measurable space. Then

1. If $f : X \to [0, \infty]$ is measurable, there exists a sequence of functions $\{\varphi_i\}_{i=1}^{\infty}$ such that
$$0 \le \varphi_1 \le \varphi_2 \le \cdots \le f$$
where $\lim_{n\to\infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \to f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformally converges)

2. If $f : X \to \mathbb{C}$ is measurable, there exists a sequence of functions $\{\varphi_i\}_{i=1}^{\infty}$ such that
$$0 \le |\varphi_1| \le |\varphi_2| \le \cdots \le |f|$$
where $\lim_{n\to\infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \to f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformally converges)

***Proof*:**
Essentially, you will get characteristic functions to converge to the value like so:



In particular, for $n \in \mathbb{N}_{\ge 0}$ and $0 \le k \le 2^{2n} - 1$, let
$$E_n^k = f^{-1}\left(\left(\frac{k}{2^n}, \frac{k+1}{2^n}\right)\right) \qquad F_n = f^{-1}((2^n, \infty])$$
and
$$\varphi_n = \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n}\chi_{E_n^k} 2^n \chi_{F_n}$$
then we get that $\varphi_n \le \varphi_{n+1}$ for all n and $0 \le f - \varphi_n \le \frac{1}{2^n}$ where $f \le 2^n$. The rest is left to check.

Generalizing to $\mathbb{C}$, decomposing to $f = g + ih = (g^+ + g^-) = i(h^+ + h^-)$, where the $+$ and $-$ functions are the positive and negative parts of f and g, then we can repeat the process for each of these and get $\psi_n^+, \psi_n^-, \zeta_n^+, \zeta_n^-$ and let $\varphi_n = (\psi_n^+ - \psi_n^-) + i(\zeta_n^+ - \zeta_n^-)$.

This shows that measurable functions are a very natural type of function to work with in analysis: they are all a pointwise-limit (and on a bounded domain a uniform limit) of simple functions. The particular sequence of simple functions will be called the *standard representation of f*. Using this result, we can define a new definition of equality up to zero-measure sets:

Proposition 2.1.19: Measurability up to Zero Set

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be two measure with ν being a complete measure, and let $f : X \to Y$ be $(\mathcal{M}, \mathcal{N})$-measurable. Then:

1. If $f = g$ up to a zero measure set with respect to μ (shorthanded to μ-a.e.), then g is measurable
2. If f_n is measurable for each $n \in \mathbb{N}$ and $f_n \to f$ μ-a.e, then f is measurable

***Proof*:**

1. Let $N = \{x \in X \mid f(x) \neq g(x)\}$. Let $B \in \mathcal{N}$ and consider $g^{-1}(B)$. Then by definition:

$$\begin{aligned} g^{-1}(B) &= \{x \in M \mid g(x) \in B\} \\ &= \{x \in \mathcal{M} \mid g(x) \in B,\ g(x) = f(x)\} \cup \{x \in \mathcal{M} \mid g \in B,\ g(x) \neq f(x)\} \end{aligned}$$

the latter set is a subset of N, and so is measurable by the completeness of f. The former set is of the form:

$$\begin{aligned} \{x \in \mathcal{M} \mid g(x) \in B,\ g(x) = f(x)\} &= \{x \in \mathcal{M} \mid f(x) \in B,\ g(x) = f(x)\} \\ &= f^{-1}(B) \cap \{x \mid f(x) = g(x)\} \\ &= f^{-1}(B) \cap (X - \{x \mid f(x) \neq g(x)\}) \\ &= f^{-1}(B) \cap (X - N) \end{aligned}$$

and since f measurable the pre-image of a measurable set is too, and since N is measuable, $X - N$ is too, so we have that $g^{-1}(B)$ is measurable.

2. Let's say N is the set on which $f_n \not\to f$. Define

$$g_n = f_n \chi_{X-N} + f\chi_N$$

Then notice that $g_n = f_n$ μ-a.e., and so by part 1 are measurable, and by construction $g_n \to f$ everywehre, and so f is measurable!

However, if μ is not complete, it turns out to make little difference since we can complete the measure, produce the result, and restrict back to the original measure without changing the result:

Proposition 2.1.20: Extending To Complete Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space and $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. if f is a $\overline{\mathcal{M}}$-measurable function on X, there is a $\mathcal{M}$-measurable function g such that $f = g$ $\overline{\mu}$-a.e.

***Proof*:**
If f is a simple function, then we see that the completion will at most remove finitely many points when we restrict every χ_E ($E \in \overline{\mathcal{M}}$) to g, and so $f = g$ $\overline{\mu}$-a.e.

Now let f be a $\overline{\mathcal{M}}$-measurable function, and by theorem 2.1.18 choose a sequence of $\overline{\mathcal{M}}$-measurable functions $\{\varphi_i\}_{i=1}^{\infty}$ that converge pointwise to f. The idea is that each $\varphi_i = \psi_i$ $\overline{\mu}$-a.e., that is, there exists a E_i where $\varphi_i(x) \neq \psi_i(x)$ for all $x \in E_i$ but $\overline{\mu}(E_i) = 0$, and the countable union of these will still be measure 0.

In particular Let $E = \bigcup_i^{\infty} E_i$. By definition of completion, there must exist some N such that $\mu(N) = 0$ and $E \subseteq N$. Let $g = \lim_{\chi_{X-N}} \psi_n$. Then g is measurable by corollary 2.1.13, and clearly $f = g$ on N^c, completing the proof.

2.2 Integration of non-negative Function

We are now in a position to define how to integrate functions (integrable functions coming soon)! Throughout this section, let $(X, \mathcal{M}, \mu)$ be a measure space

Definition 2.2.1: L^+

Let
$$L^+ = \{f : X \to [0, \infty] \mid f \text{ is a } \mathcal{M}\text{-measurable function}\}$$

Sometimes, L^+ is written as $L^+(X)$, $L^+(\mu)$, or $L^+(\mu, X)$ to emphasize which space of functions we are dealing with. If it is unambiguous, we use L^+.

Definition 2.2.2: Integral of Simple Function

Let $f = \sum_1^n a_i \chi_{E_i}$ be a simple function. Then define the *integral of f with respect to μ* to be
$$\int f d\mu = \sum_i^n a_i \mu(E_i)$$

In other words, we simply took the measure of the domain of χ_{E_i} for each i. As usual, we take the convention that $0 \cdot \infty = 0$. We will also allow $\int f = \infty$ to be a valid result. If there is no confusion, we will write $\int f$ instead of $\int f d\mu$. It should also be emphasized that it is the measure of the *domain* and the coefficient from the *codomain* that we are taking.

If we want to integrate over just some $A \subseteq X$ where $A \in \mathcal{M}$, then $\varphi|_A$ is also a simple function (simply take $\chi_{E \cap A}$ for each characteristic function). We usually denote this by
$$\int_A f d\mu = \int f \chi_A \, d\mu$$

Because of this notation, we can write $\int f d\mu$ as
$$\int f d\mu = \int_X f d\mu$$

Proposition 2.2.3: properties of Integrals of Simple Functions

Let φ and ψ be simple functions in L^+. Then

1. if $c \geq 0$, $\int c\varphi d\mu = c\int \varphi d\mu$
2. $\int(\varphi+\psi)d\mu = \int \varphi d\mu + \int \psi d\mu$
3. if $\varphi \leq \psi$ then $\int \varphi \leq \int \psi$
4. The map $A \mapsto \int_A 1 d\mu$ is a measure on $\mathcal{M}$ (more generally, $A \mapsto \int_A \varphi d\mu$ is a meaure on $\mathcal{M}$ for a simple function φ)

***Proof*:**
Do these as midterm exercises

With this definition, we can define integrable function more generally:

Definition 2.2.4: Integral of Measurable Function

Let $f \in L^+$. Then define

$$\int f d\mu := \sup\left\{\int \varphi d\mu \;\middle|\; \varphi \text{ is a simple function, } 0 \leq \varphi \leq f\right\}$$

It's important to see why the integral of a simple function will give the same result as the "simple integral" (i.e. integral over simple functions from earlier), in particular since the set over which we are containing the supremum will contain the simple function in question, and so the maximum will be achieved by itself (and proposition2.2.3?). Furthermore, results 1 and 3 from proposition 2.2.3 holds for the definition of integration just given. The other results, in particular linearity, will be established soon. We can also see how the definition of the integral solidifies the idea that the integral is a sort of "weighted measure": the integral of a simple function is just a sum of measure of sets weighted by some constant, and we take the supremum of all such sets. We will in fact take a closer look at defining measure in terms of integration of functions in chapter 3.

Given the supremum definition of the integral, it could be rather hard to find the integral of a function in L^+. We essentially never check this, and treat the following theorem as the de-facto way of finding the integral:

Theorem 2.2.5: Monotone Convergence Theorem (MCT)

If $\{f_n\}_{i=1}^{\infty} \subseteq L^+$ is a sequence such that $f_i \leq f_{i+1}$ for all $i \in \mathbb{N}$ and $f = \lim_{n\to\infty} f_n$ $(= \sup_n f_n)$, then

$$\int f = \lim_{n\to\infty} \int f_n$$

or, since $f = \lim_{n\to\infty} f_n$:

$$\int \lim_{n\to\infty} f = \lim_{n\to\infty} \int f_n$$

i.e. the limit commutes

It might be tempting to then say that $\int$ is "continuous" in the appropriate domain since it commutes with limits. However, the f_i must be *increasging* for the monotone convergence theorem to apply, and hence it is better to say that $\int$ is continuous from bellow. We will later show that the limit need not always commute (and hence it will be innapropriate to say that the integral is continuous on the space of measurable functions).

2.2.6 Remark any times in Analysis, when the limit can commute with something (ex. integral, another limit, functions), then the proof that it commutes is usually given a name (ex. sequential continuity, MCT, DCT, Fatou lemma, Moore-Osgood theorem, etc.) see:

wikipedia: Interchanging Limits

***Proof*:**
We'll show $\leq$ and $\geq$. For $\leq$, notice that $\{\int f_n\}$ is an increasing sequence, and so converges (possibly to ∞). Furthermore, $f_n \leq f$ for all n, and so by proposition 2.2.3 updated for general integral functions, we have:

$$\int f_n \leq \int f$$

and so

$$\lim_{n\to\infty} \int f_n \leq f$$

Conversely, we'll show that if we "shrink" the left hand side by a factor of $\alpha \in (0,1)$ so that we get $\geq$, then since this will be true for all α, we will get the result we want.

Fix some $\alpha \in (0,1)$, and choose a simple function where $0 \leq \varphi \leq f$ (such a function exists by theorem 2.1.18). Let

$$E_n = \{x \in \overline{\mathbb{R}} \mid f_n(x) \geq \alpha\varphi(x)\}$$

Then $\{E_n\}_{n=1}^{\infty}$ is an increasing sequence (via inclusion) of measurable sets who's union is X (since $\lim_{n\to\infty} f_n = f$ and the f_n's are increasing). Then

$$\int f_n \geq \int_{E_n} f_n \geq \alpha \int_{E_n} \varphi$$

Since the map $E_n \mapsto \int_{E_n} \varphi$ is measurable by proposition 2.2.3 and the E_n's are continuous from above, we have

$$\lim_{n\to\infty}\int_{E_n}\varphi=\int\varphi$$

and so

$$\lim_{n\to\infty}\int f_n\geq\alpha\int\varphi$$

Since this is true for all $\alpha<1$, it is true for $\alpha=1$, and so $\int f_n\geq\int\varphi$. Taking the supremum over all φ, we get that

$$\lim_{n\to\infty}\int f_n\geq\int f$$

And since we showed the $\leq$ direction, we in fact have:

$$\lim_{n\to\infty}\int f_n=\int f$$

as we sought to show

Thus, to find the value of $\int f$, it suffices to find some sequence of simple functions $\{\varphi_n\}$ that converges to f, which always exists since every measurable function has a standard representation (theorem 2.1.18).

An immediate consequence of the Monotone Convergence Theorem is that linearity also applies to integrals:

Proposition 2.2.7: Linearity of Integrals

Let $\{f_i\}_{i=1}^N\subseteq L^+$ where $N\in\mathbb{N}\cup\{\infty\}$ and let $f=\sum_i^N f_i$. Then

$$\int f=\sum_i^N\int f_n$$

***Proof*:**
Let's first consider the case where $n=2$ so that we have f_1 and f_2. Let $\{\varphi_i\}$ and $\{\psi_i\}$ be a sequence for the standard representation of f_1 and f_2. Then clearly $\{\varphi_i+\psi_i\}$ is a sequence for a standard representation of f_1+f_2. Then by the Monotone Convergence Theorem and linearity

of integrals with simple functions:

$$\begin{aligned}
\int f_1 + f_2 &= \lim_{i\to\infty} \int \varphi_i + \psi_i \\
&\overset{\text{MCT}}{=} \int \lim_{i\to\infty} \varphi_i + \psi_i \\
&= \int \lim_{i\to\infty} \varphi_i + \lim_{i\to\infty} \psi_i \\
&= \int \lim_{i\to\infty} \varphi_i + \int \lim_{i\to\infty} \psi_i \\
&\overset{\text{MCT}}{=} \lim_{i\to\infty} \int \varphi_i + \int \lim_{i\to\infty} \psi_i \\
&= \int f_1 + \int f_2
\end{aligned}$$

By induction, it holds that

$$\int \sum_i^n f_i = \sum_i^n \int f_i$$

Letting $n \to \infty$, we can once again apply the Monotone Convergence Theorem and get

$$\begin{aligned}
\int \sum_i^\infty f_i &= \int \lim_{n\to\infty} \sum_i^n f_i \\
&\overset{\text{MCT}}{=} \lim_{n\to\infty} \int \sum_i^n f_i \\
&= \lim_{n\to\infty} \sum_i^n \int f_i \\
&= \sum_i^\infty \int f_n
\end{aligned}$$

completing the proof

The next result is a generalization from the same result with Riemann integrals, except we replace the condition of "finitely many points" with "null-set amount of points". It follows the trend of integration "forgetting" up to a zero-measure amount of points:

Proposition 2.2.8: Zero Integral

Let $f \in L^+$. Then $\int f = 0$ if and only if $f = 0$ a.e.

***Proof*:**
Starting with simple functions, this is evident in both directions. If φ was our simple function and $\int \varphi = 0$, then $\sum_i^n a_i \mu(E_i) = 0$, so either each $a_i = 0$ or each $\mu(E_i) = 0$ which immediately implies φ is zero a.e. (not necessarily everywehre since it's possible that $E_i \neq \emptyset$ but $\mu(E_i) = 0$).

Conversely, if each $\mu(E_i) = 0$ a.e., then clearly $\int \varphi = 0$.

Moving onto to a measurable function f, the $\Leftarrow$ comes almost immediately. If $f = 0$ a.e., then for every simple function $\varphi \leq f$, then $\varphi = 0$ a.e., which we have established means $\int \varphi = 0$. Then by definition of $\int f$:

$$\int f = \sup_{\varphi \leq f} \int \varphi = 0$$

For the $\Rightarrow$ direction, assume that $\int f = 0$, and for the sake of contradiction let's say $f \neq 0$ a.e. The idea is that we can then construct a simple function φ where $\varphi \leq f$ whose integral is greater than zero, but $\int f = \sup_{\varphi \leq f} \int \varphi$. To that end take:

$$\bigcup_{i}^{\infty} E_n = \bigcup_{i}^{\infty} \left\{ x \;\middle|\; f(x) > \frac{1}{n} \right\} = \{x \mid f(x) > 0\} = E$$

Since $f \neq 0$ a.e., it must be that $\mu(E) > 0$, and so by construction for some n, $\mu(E_n) > 0$. But then

$$f \geq n^{-1}\chi_{E_n} \qquad \Leftrightarrow \qquad \int f \geq n^{-1}\mu(E_n) > 0$$

so the supremum of values in f contain simple functions of nonzero measure, showing that $\int f > 0$, a contradiction to our original assumption.

This theorem also tells us that if $f = g$ a.e., then $\int f = \int g$, since $f - g = 0$ a.e., so $\int f - g = 0$, so $\int f = \int g$. Using this, we can upgrade the Monotone Conversely Theorem by needing the convergence to happen up to a measure zero set:

Corollary 2.2.9: MCT a.e.

Let $\{f_n\}_{i=1}^{\infty} \subseteq L^+$ and $f \in L^+$. Then if $f_1 \leq f_2 \leq \cdots \leq f$ and $f_n \to f$ a.e., then

$$\int f = \lim_{n\to\infty} \int f_n$$

***Proof*:**
Let $f_n(x) \uparrow f(x)$ for all $x \in E$ such that $\mu(E^c) = 0$. Then $f - f\chi_E = 0$ a.e. which implies $f_n - f_n\chi_E = 0$ a.e., so by the Monotone Conversely Theorem

$$\int f = \int f\chi_E \overset{\text{MCT}}{=} \lim_{n\to\infty} \int f_n\chi_E = \lim_{n\to\infty} \int f_n$$

as we sought to show.

We next address the fact that the sequence $\{f_i\}$ must be increasing (at least a.e.) for the MCT to work (recall we relied on the sequence being increasing in the construction of one of the direcitons of the inequality). If it were not, then it need not be the case that the limit passes through the integral!

Example 2.2.10: not increasing $\not\Rightarrow$ Limit commutes with $\int$

Let $X = \mathbb{R}$ and μ be the Lebesgue measure. Consider the two following examples:

$$\{n\chi_{(0,1/n)}\}_{n=1}^{\infty}$$

then it is clear that $n\chi_{(0,1/n)} \to 0$. However, notice that $\int n\chi_{(0,1/n)} = 1$ for each element in the sequence. Thus:

$$0 = \int \lim_{n\to\infty} n\chi_{(0,1/n)} \neq \lim_{n\to\infty} \int n\chi_{(0,1/n)} = 1$$

Even if the function is bounded, the problem remains: take $\chi_{(n,n+1)}$ which also converges pointwise to 0 but the sequence of the integrals of the functions is a sequence of constants 1 and so converges to 1 which is clearly not 0.

This is interesting if we interpret it from a sort of "physics" perspective: somehow, we have "lost our mass" in the process of taking our limit. There is in fact essentially 3 ways we can "lose mass', the first two begin the counter-examples above, the last one being a function that oscillates faster and faster (this last one requires a function that is also defined in the negative, and so we will get back to this)

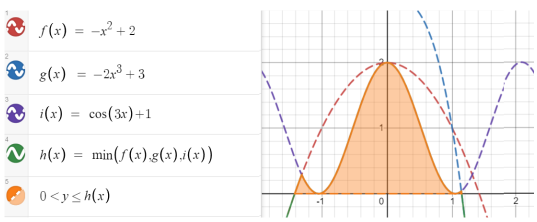
Notice that in both examples, there is some "escaping to infinity" going on. Perhaps if we have some overarching "bound" to the sequence, then we can pass the limit through, and that will indeed be the case. We address this when talking about the Dominated Convergence Theorem in the next section. However, not all hope is lost for the general case; there is weakening to only the lim inf and the equality to $\leq$ which works in the general case:

Theorem 2.2.11: Fatou's Lemma

If $\{f_n\} \subseteq L^+$ is *any* sequence, then

$$\int \liminf_n f_n \leq \liminf_n \int f_n$$

To visualize this, consider:



The integral of the orange region weill be less than or equal to the lim inf of those functions over that region. In this case, the integral would be equal, but in the case where the mass somehow "escapes", we will have an inequality.

***Proof*:**
This comes directly from properties of the lim inf. Let $k \geq 1$. Then

$$\inf_{i \geq k} f_i \leq f_j \quad \forall\, j \geq k$$

Then by theorem 2.2.3

$$\int \inf_{i \geq k} f_i \leq \int f_j \quad \forall j \geq k$$

and since this holds for all $j \geq k$, we get:

$$\int \inf_{i \geq k} f_i \leq \inf_{j \geq k} \int f_j$$

Since the lim inf defines an increasing sequence of function, by letting $k \to \infty$ and applying the Monotone Conversely Theorem:

$$\int \liminf f_n = \lim_{k \to \infty} \int \inf_{n \geq k} f_n \leq \liminf \int f_n$$

as we sought to show

Like before, the result holds if we loosen the condition so that $f_n \to f$ a.e.

Corollary 2.2.12: Fatou's Lemma a.e.

let $\{f_n\} \subseteq L^+$ be any sequence and $f \in L^+$. If $f_n \to f$ a.e. Then

$$\int f \leq \liminf_n \int f_n$$

***Proof*:**
Using proposition 2.2.9, make $f_n \to f$ everywhere arguing that it is equivalent to do so, and then the rest is Fatou's Lemma.

Finally, we show that if $\int f < \infty$, we can guarantee that f cannot "explode" too much, and that even better, the set over which the value of f is non-zero is σ-finite, i.e., it is not "pathological"

Proposition 2.2.13

Let $f \in L^+$. If $\int f < \infty$, then $\{x \mid f(x) = \infty\}$ is a null set and $\{x \mid f(x) > 0\}$ is σ-finite

***Proof*:**
was left as an exercise to the reader in the book

2.3 Integration of Real and Complex Functions

We'll again let $(X, \mathcal{M}, \mu)$ be the fixed measure space for this section, except the codomain now will be either $\mathbb{R}$ or $\mathbb{C}$. For the case of $\mathbb{R}$, we can quickly extend our results from last section by splitting f into f^+ and f^- (which are both measurable if f is by corollary 2.1.12), and if at least one of f^+ or f^- is finite, then

$$\int f = \int f^+ - \int f^-$$

Notice that if they are both infinity, we have $\infty - \infty$ which is in general not well-defined. Thus, we will limit our attention to functions where $\int |f| < \infty$ since $|f| = f^+ + f^-$:

For the case of $\mathbb{C}$, using the triangle inequality we get that:

$$|f| = |\text{Re}f + \text{Im}f| \leq |\text{Re}f| + |\text{Im}f| \leq 2|f|$$

and so both $\text{Re}f$ and $\text{Im}f$ must be finite to get a well-defined result. In that case, we define

$$\int f := \int \text{Re}f + i \int \text{Im}f$$

We give a name for complex functions (the real functions being a special case) that satisfy this finiteness condition:

Definition 2.3.1: Integrable Function

Let $f : X \to \mathbb{C}$ be measurable. We say f is *integrable* if

$$\int |f| < \infty$$

Let L^1 represent the set of all integrable functions.

Like with L^+, we sometimes write $L^1(\mu)$, $L^1(X)$, or $L^1(X, \mu)$. If the codomain is $\mathbb{R}$, then we get that $\int |f^+| < \infty$ and $\int |f^-| < \infty$, as we claimed for our extension of the integral to $\mathbb{R}$. Furthermore, for any $E \in \mathcal{M}$, we will say that f is *integrable on E* if $\int_E |f| < \infty$.

The set of L^1 functions are well-behaved under addition and scalar multiplication in the sense that they form a vector-space:

Proposition 2.3.2: Vector Space on L^1

Let L^1 be the set of integrable functions. Then with $+$ and scalar multiplication, L^1 forms a vector-space (over $\mathbb{R}$ or $\mathbb{C}$ respective to the codomain). Furthermore, $\int$ is a linear functional on L^1 (i.e. $\int : L^1 \to \mathbb{R}$ is a linear functional)

***Proof*:**
the proof that $cf + dg$ is integrable comes from the triangle inequality ($|cf + dg| \leq |c||f| + |d||g| < \infty$). The fact that $cf \in L^1$ for any $c \in \mathbb{R}$ and $f \in L^1$ follows similarly.

$\int$ being a functional, we need to show linearity and $\int cf = c \int f$. For linearity, assume $h = f + g$

is integrable. Then $h^+ - h^- = f^+ - f^- + g^+ - g^-$, or, re-arranging, we get:

$$h^+ + f^- + g^- = h^- + f^+ + g^+$$

These are all functions in L^+, and so by linearity for of integrals for L^1 functions:

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+$$

re-arraning, we get:

$$\begin{aligned}
&= \int f + g \\
&= \int h = \int h^+ - \int h^- = \int f^+ - \int f^- + \int g^+ - \int g^- \\
&= \int f + \int g
\end{aligned}$$

showing linearity. Through a similar means, we can also show $\int cf = c \int f$, completing the proof.

2.3.3 Remark Since it is a vector-space, it has a basis and a dimension. What dimension is this vector-space? Since it is a real vector-space, it is isomorphic to $\mathbb{R}^N$ for appropriate cardinal N (N may be finite, countable, uncountable, etc). If $N \geq \aleph_0$, that means that it is rather difficult to define a measure on the space L^1[a]. Trying to find ways to study "measure" notions infinite dimensional vector-spaces (ex. boundedness, σ-finiteness, etc.) is accomplished by defining various different concepts like *seminorms*, *norms*, or *inner products*, which we shall do in chapter 4.

[a] For an illustration of the oddities of infinite dimensional vector-spaces, consider the [hyper]-volume of a unit ball S^n as $n \to \infty$; you should find that it goes to 0!

Next, we would like to establish some properties of integrable functions. One result is a including a useful inequality including absolute values. Another is how the results still work as long as things happen "a.e.". Finally, the set of all "interesting" points (points where $f(x) \neq 0$) are still σ-finite:

Proposition 2.3.4: Properties of Integrable Functions

Let $f, g \in L^1$. Then:

1. $\left|\int f\right| \leq \int |f|$
2. $\{x \mid f(x) \neq 0\}$ is σ-finite
3. $\int_E f = \int_E g$ for all $E \in \mathcal{M}$ if and only if $\int |f - g| = 0$ if and only if $f = g$ a.e.

***Proof*:**

1. The result is essentially immediate in the $\mathbb{R}$ case. In the $\mathbb{C}$ case, let α where $|\alpha| = 1$ be such that it "rotates" the integral so that it's in the real case, and then apply a similar strategy to that of the $\mathbb{R}$ case.

2. Notice that $\{x \mid f(x) \neq 0\} = \{x \mid f^+(x) > 0\} \cup \{x \mid f^-(x) < 0\}$, which we've shown in proposition 2.2.13 to be σ-finite

3. The fact that $\int |f - g| = 0$ if and only if $f = g$ a.e. comes from proposition 2.2.8, (since $|f-g| \in L^+$) so we will prove $\int_E f = \int_E g$ if and only if $\int |f-g| = 0$. Let's say $\int |f-g| = 0$. Then to show that $\int_E f = \int_E g$, it is equivalent to show that

$$\left|\int_E f - \int_E g\right| = 0$$

by the epsilon principle (i.e. $a = b$ if and only if $|a-b| < \epsilon$ for all ϵ). Using proposition 2.3.4, we get that:

$$\left|\int_E f - \int_E g\right| \leq \int \chi_E |f - g| \leq \int |f - g| = 0$$

completing the $\Rightarrow$ direction.

For the other direction, let's argue contra-positively and assume $\int |f - g| \neq 0$ a.e. Let

$$u = \mathrm{Re}(f - g) \qquad v = \mathrm{Im}(f - g)$$

so that u^+, u^-, v^+, v^- are the corresponding functions. Since $f \neq g$ a.e., at least one of these must have a set with positive measure. Without loss of generality, let's say $E = \{x \mid u^+(x) > 0\}$ and $\mu(E) > 0$. Then

$$\mathrm{Re}\left(\int_E f - g\right) = \mathrm{Re}\left(\int_E f - \int_e g\right) = \int_E u^+ + \int_E u^- = \mu(E) + 0 > 0$$

since $u^- = 0$ on E. Thus, the measure will be positive.

As a consequence of this, we usually think of integrable functions *up to a zero set*. If we wanted to, we can limit our scope to of integration to functions f that are measurable on E and E^c has zero-measure to functions where $f = 0$ on E^c. This also means that if we are working with an $\overline{\mathbb{R}}$-value function that is finite a.e., then it is equivalent to treat it as a $\mathbb{R}$-value function.

Proposition 2.3.4 also allows us to re-define L^1 in terms of functions that are equivalent up to a measure zero set. That is, we form an equivalence relation on L^1 if and only if $f = g$ a.e., and label the collection of equivalence classes as L^1. We abuse notation and write "$f \in L^1$" to mean we are selecting a function that is integrable (or an a.e.-defined integrable function). This is because we will usually be working with functions instead of the equivalence classes, and so this is a common abuse of notation.

The reason to introduce this new definition of L^1 is so that we can define a new metric on L^1:

Definition 2.3.5: L^1 metric

Let L^1 be the set of equivalence classes as defined above. Then

$$\rho(f, g) = \int |f - g|$$

is a metric on L^1

Symmetry and triangle inequality come immediately without even needing to use the fact that these are equivalence classes. Where the new construction of L^1 becomes important is to say $\int |f - g| = 0$ if and only if $f = g$. Since this is true a.e., then we need these two to be in an equivalence class. Since we defined a metric, we can define a notion of convergence, in particular, f_n converges to f in L^1 if and only if $\int |f - f_n| \to 0$, meaning $f_n \to f$ almost everywhere (however, L^1 convergence does not imply a.e. point-wise convergence, and a.e. point-wise convergence does not imply L^1 convergence, as we'll show in section 2.4).

We will soon explore more properties of this metric space, in particular, we would want this metric space to be complete so that the limit of a sequence of L^1 function's is L^1 and find some nice dense set of L^1 (so far, we only have the [point-wise] limit of measurable function is measurable, but no such result yet for integrable functions). To do so, we need more tools to analyze swapping limits and integrals. The next result we'll cover allows us to generalize the Monotone Conversely Theorem to the general case of $f_n \to f$ a.e., with the extra condition that each $|f_n|$ is bounded by a single $|g|$ with finite area (in this way, we are getting rid of the "escaping to infinity" cases in example 2.2.10)

Theorem 2.3.6: Dominated Convergence Theorem (DCT)

Let $\{f_n\}_{i=1}^{\infty} \subseteq L^1$ be a sequence of integrable functions such that

1. $f_n \to f$ a.e.
2. there exists a nonnegative $g \in L^1$ such that for all $n \in \mathbb{N}$, $|f_n| \leq g$ a.e.

Then

$$\int f = \lim_{n\to\infty} \int f_n$$

or equivalently:

$$\int \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int f_n$$

***Proof*:**
By proposition 2.1.19 and 2.1.20 , f is measurable (up to a null set, in which case we can appropriately redefine f on that null-set). Since $|f_n| < g$ for all $n \in \mathbb{N}$, $f < g$, and so $f \in L^1$.

Since all L^1 functions are broken down to real parts (complex functions into 2 real parts), it suffices to show the result for real value f_n and f. By assumption, we have $g + f_n \geq 0$ and $g - f_n \geq 0$. Thus, by Fatou's Lemma:

$$\int g + \int f \leq \liminf \int (g + f_n) = \int g + \liminf \int f_n$$

$$\int g - \int f \leq \liminf \int (g - f_n) = \int g - \limsup \int f_n$$

Therefore, $\limsup \int f_n \leq \int f \leq \liminf \int f_n$, and so the two value are actually equal and so

$$\int f = \lim_{n\to\infty} \int f_n$$

as we sought to show

Note that functions can still converge even if the hypothesis of the DCT fails:

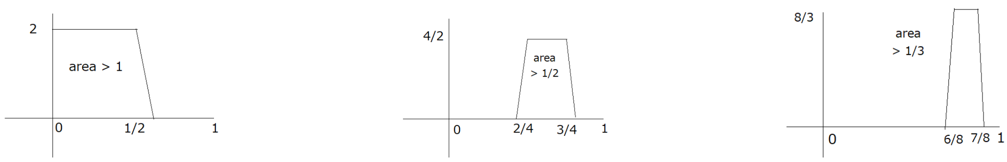
Example 2.3.7: Converges Without DCT

Construct a sequence of functions $\{f_n\}$ such that all the following hold:

1. $f_n : [0,1] \to \mathbb{R}$ is continuous
2. $f_n \geq 0$
3. $\lim_{n\to\infty} f_n(x) = 0$ for all $x \in [0,1]$
4. $\lim_{n\to\infty} \int_0^1 f_n(x)\, dm = 0$
5. $\sup_n f_n(x)$ is *not* integrable on $[0,1]$ with respect to the Lebesgue measure

(that is, just because the hypothesis of the dominated convergence theorem fails does not mean the conclusion does not hold: it is not an if and only if)

Proof. The proof essentially is a formalizing the following images:



(a) f_1 (b) f_2 (c) f_3

Figure 2.1: Visual intuition for question 3

We'll first define the indicator functions, then make these functions continuous. Let

$$g_1(x) = \begin{cases} 2^1/1 & x \in \left[0, \frac{1}{2}\right) \\ 0 & x \in \left[1 - \frac{1}{2}, 1\right] \end{cases} \quad g_2(x) = \begin{cases} 2^2/2 & x \in \left[\frac{1}{2}, \frac{1}{4}\right) \\ 0 & x \in \left[1 - \frac{1}{4}, 1\right] \end{cases} \quad g_3(x) = \begin{cases} 2^3/3 & x \in \left[\frac{3}{4}, \frac{7}{8}\right) \\ 0 & x \in \left[1 - \frac{7}{4}, 1\right] \end{cases}$$

and so on, defining:

$$g_n(x) = \begin{cases} 2^n/n & x \in [2^{n-1} - 2/2^{n-1}, 2^{n-1} - 1/2^{n-1}) \\ 0 & x \in [1 - 2^{n-1} - 1/2^{n-1}, 1] \end{cases}$$

Next, we can make each g_n continuous by simply adding a line to omit the jump:

$$f_1(x) = \begin{cases} 2^1/1 & x \in [0, 1/2) \\ -2^{10\cdot 1}\left(x - \left(\frac{1}{2} - 1/2^{10\cdot 1}\right)\right) & x \in (1/2, 1/2 + 1/2^{10\cdot 1}] \\ 0 & x \in [1 - (1/2 + 1/2^{10\cdot 1}), 1] \end{cases}$$

and defining each f_n similarly. Then it is clear by our construction that $f_n \to 0$: since the trapezoids are moving away, they will never cross our values again once they pass them (this is the opposite of the proof for question 2 where they constantly loop back, since the proof is similar the details were ommited for this proof). Furthermore,

$$\int f_n = 1/n + 1/2^{10n}$$

Thus, $\int f_n \xrightarrow{n\to\infty} 0$, showing that the integral also converges to 0. However,

$$\int \sup_n f_n \geq \int \sup_n g_n = \sum_{k=1}^{\infty} 1/k = \infty$$

showing that the $\sup_n f_n$ is *not* integrable, as we sought to show.

□

Therefore, the DCT is a sufficient, but not necessary condition for convergence. Sufficient and necessary conditions are a bit harder to come by, and we will not need to characterize limits commuting with integrals in these other ways, but for those who are curious:

https://journalofinequalitiesandapplications.springeropen.com/articles/10.1186/s13660-020-02502-w

This theorem will be key in finding the integral of many function constructed by limits, and will be used to prove completeness of L^1 (after defining Cauchy sequences in L^1 in section ref:HERE) A more immediate result is that the Dominated convergence theorem allows to define a form of countable linearity for L^1 functions:

Theorem 2.3.8: Linearity for Integrable Functions

Let $\{f_i\}_{i=1}^{\infty} \subseteq L^1$ be a sequence of functions, and $f = \sum_{i=1}^{\infty} f_i$. If $\sum_{i=1}^{\infty} |f_i| < \infty$, then $\sum_{i=1}^{\infty} f_i$ converges a.e. to a function in L^1, and

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

***Proof*:**
First, since $\sum_{i=1}^{\infty} |f_i| < \infty$, by proposition 2.2.7,

$$\int \sum_{i=1}^{\infty} |f_i| = \sum_{i=1}^{\infty} \int |f_i|$$

Thus, $g = |\sum_{i=1}^{\infty} f_i|$ is a function in L^1. The function g is finite a.e. by proposition 2.3.4[2], so by the dominated convergence theorem, any partial sum of functions will commute, and so taking the limit, we get:

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

as we sought to show

The dominated convergence theorem also gives us a dense set in L^1 to work with, namely the simple functions:

Theorem 2.3.9: Simple Functions Dense in L^1

Let $f \in L^1$. Then for all $\epsilon > 0$, there exists an integrable simple function $\varphi = \sum a_i \chi_{E_i}$ such that

$$\int |f - \varphi| d\mu < \epsilon$$

Furthermore, if μ is a Lebesgue-Stieljes measure on $\mathbb{R}$, the sets E_i can be taken to be finite union of open intervals, and better, there is a continuous function g that vanishes outside a bounded interval (i.e. $g = 0$ outside the bounded interval) such

$$\int |f - g| d\mu < \epsilon$$

***Proof*:**
p. 55 in the book. Also proven in fuller generality that simple functions are dense in L^p spaces, which includes L^1.

For the purposes of Ordinary Differential Equations (ODE's), we want to establish is a quick result to relate differentiable functions and integral of functions:

Theorem 2.3.10: Relating Derivative and Integral

Let $(X, \mathcal{M}, \mu)$ be a measure space, and suppose $f : X \times [a, b] \to \mathbb{C}$ for $-\infty < a < b < \infty$ and that $f(\cdot, t) : X \to \mathbb{C}$ is integrable for every $t \in [a, b]$, so we can define $F(t) = \int_X f(x, t)\, d\mu$.

1. Suppose that there exists a $g \in L^1$ such that $|f(x, t)| \leq g(x)$ for all x, t. Then if $\lim_{t \to t_0} f(x, t) = f(x, t_0)$ for every x, $\lim_{t \to t_0} F(t) = F(t_0)$; that is, if $f(x, \cdot)$ is continuous at every x, so is F.

2. Suppose $\frac{\partial f}{\partial t}$ exists and there is a $g \in L^1(\mu)$ such that $\left|\frac{\partial f}{\partial t}(x, t)\right| \leq g(x)$ for all x, t. Then F is differentiable and
$$F'(x) = \int \frac{\partial f}{\partial t}(x, t)\, d\mu(x)$$

The way I like to think about is that there is a system X that at different times t acts/behaves differently. This difference in how the system acts at any given time t is captured by $f(x, t)$. The function F capture the "total value" of this action/behavior (note that f is non-negative too, so all behavior has a "non-negative" output).

Now, if the behavior varies continuously, then the total output varies continuously, and if the behavior is smooth smooth enough so that it is differentiable (at any slice), then F is also differentiable, and it is in fact even easy to compute the value of F'.

***Proof*:**

1. Since $|f(x,t)|$ is dominated by g, simply apply the Dominated Convergence Theorem:

$$\lim_{t\to t_0} F(t) = \lim_{t\to t_0} \int f(x,t)\,d\mu(x) = \int \lim_{t\to t_0} f(x,t)\,d\mu(x) = \int f(x,t_0)\,d\mu(x) = F(t_0)$$

   where $\{t_n\}$ is any sequence converging to t (since any $f(x,t)$ satisfies the bound, no more restrictions are needed on the choice of sequence)

2. Since $\frac{\partial f}{\partial t}$ exists, the limit of the derivative exists, and so define

$$\frac{\partial f}{\partial t} = \lim_{t_n\to t_0} h_n(x) \qquad h_n(x) = \frac{f(x,t_n) - f(x,t_0)}{t_n - t_0}$$

   $\{t_n\}$ is any sequence converging to t. Since h_n difference and division of measurable functions, by proposition 2.1.10 it is measurable, and by the Dominated Convergence Theorem $\frac{\partial f}{\partial t}$ is measurable. Furthermore, by the Mean Value Theorem:

$$|h_n(x)| \leq \sup_{t\in[a,b]} \left|\frac{\partial f}{\partial t}(x,t)\right| \leq g(x)$$

   and so by the Dominated Convergence Theorem again:

$$\begin{aligned} F'(t_0) &= \lim_{t_n\to t} \left(\frac{F(t_n) - F(t_0)}{t_n - t_0}\right) \\ &= \lim_{t_n\to t} \left(\int h_n(x)\,d\mu(x)\right) \\ &\overset{\text{D.C.T.}}{=} \int \lim_{t_n\to t} h_n(x)\,d\mu(x) \\ &= \int \frac{\partial f}{\partial t}\,d\mu(x) \end{aligned}$$

   completing the proof

The final fact we shall present is a continuation of proposition 2.2.13 in the sense that we aim to further establish connections between some subsets of the domain and the resulting integral value in the co-domain, and see that if $\int$ is limited to some finite area of integration E ($\int_E < \infty$), does it actually respect continuity? In fact, it does! Even better, it is in a sense uniformally continuous with respect to the measure of the set that is being integrated over.

Proposition 2.3.11: $\int$ Uniform Continuity

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f \in L^1$. Then for all $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\mu(E) < \delta \ \Rightarrow \ \int_E |f| < \epsilon$$

***Proof*:**
Without loss of generality, let $f \geq 0$. Let $\epsilon > 0$, and choose φ such that $\int f < \int \varphi + \epsilon/2$, i.e. $\int f - \int \varphi < \epsilon/2$. Next, since φ is a simple function, we have that $\sup \varphi = M < \infty$ (being the finite sum of constant terms). Now, set $\delta = \frac{\epsilon}{2M}$ so that $\mu(E) < \delta$.

$$\int_E f - \varphi \leq \int f - \varphi$$

then re-arranging we get:

$$\int_E f \leq \int f - \varphi + \int_E \varphi < \frac{\epsilon}{2} + \frac{M\epsilon}{2M} = \epsilon$$

as we sought to show.

As a consequence of this, if we have a space of integrable functions $\mathcal{F}$ with the sup norm (convergence in the sup norm is equivalent to convergence), then for any $\mu(E) < \infty$, $\int_E$ is in fact *continuous*, that is, if $f_n \rightrightarrows f$, then:

$$\int_E \lim f_n = \lim \int_E f_n$$

let's fix some E where $\int_E 1 = M < \infty$. Take $f_n \rightrightarrows f$ so that $|f(x) - f_n(x)| < \epsilon$ for all $x \in E$ when $n > N$. Then:

$$\begin{aligned} \left| \int_E f(x)dx - \int_E f_n(x) \right| &= \left| \int_E f(x) - f_n(x) \right| \\ &\leq \int_E |f(x) - f_n(x)| \\ &\leq \int_E \epsilon \\ &= \epsilon M \end{aligned}$$

It quickly follows form here that the limit does indeed commute.

2.3.1 Relating Lebesgue and Riemann Integral

Let's consider $f \in L^1(m)$ to be a Lebesgue integrable function on $\mathbb{R}$. As a preliminary to this book, Riemann integration of real-valued functions have been studied. So far, we have done nothing to show that the definitions of the integral will produce the same result. They surely must, since the Riemann integral correctly captures the area under surfaces. In this section, we show how the Lebesgue integral is in fact an extension of the Riemann integral, meaning if we restrict to the set of all Riemann integrable functions (which will all be Lebesgue integral), then the Lebesgue and Riemann integral give the same result.

As a quick reminder of Riemann integration, choosing to follow Darboux' characterization, let $[a, b]$ be a closed interval (or more generally a compact subset of $\mathbb{R}^n$) Then for any finite partition $P = \{t_0, ..., t_n\}$ where $a = t_0 < t_1 < \cdots < t_n = b$, define:

$$U_P(f) = \sum_{i=1}^{n} M_i(t_i - t_{i-1}) \qquad L_P(f) = \sum_{i=1}^{n} m_i(t_i - t_{i-1})$$

where M_i and m_i are the supremum and infimum of f on $[t_{i-1}, t_i]$. Then take:

$$\overline{I}_a^b(f) = \inf_P U_P(f) \qquad \underline{I}_a^b = \sup_P L_p(f)$$

over all possible partitions P. If $\overline{I}_a^b(f) = \underline{I}_a^b$, then their common value is denoted as $\int_a^b f(x)\, dx$ and f is called Riemann integrable on $[a, b]$.

We'll now compare this to the Lebesgue integral:

Theorem 2.3.12: Lebesgue-Vitali Theorem

Let f be a bounded real-valued function on $[a, b]$.

1. If f is Riemann integrable, then f is Lebesgue measurable (and hence integrable on $[a, b]$ since f is bounded) and
$$\int_a^b f(x)\, dx = \int_{[a,b]} f\, dm$$
2. f is Riemann integrable if and only if $\{x \in [a, b] \mid f \text{ is discontinuous at } x\}$ has Lebesgue measure 0.

Note that $\chi_{\mathbb{Q}}$ on $[0, 1]$ is discontinuous everywhere, with discontinuities having Lebesgue measure 1, was shown to not be Riemann integrable (this was in fact the example of a non Riemann-integrable function presented in the preliminary chapter). However, it is certainly Lebesgue integrable!

***Proof*:**

1. Let f be Riemann integrable. Let M_i and m_i are defined as the supremum and infimum as defined above, and for each partition P, let:
$$G_P = \sum_{i=1}^{n} M_i \chi_{(t_{i-1}, t_i]} \qquad g_P = \sum_{i=1}^{n} m_i \chi_{(t_{i-1}, t_i]}$$
so $U_p(f) = \int G_P\, dm$ and $L_p(f) = \int g_P\, dm$. Now, there exists a sequence of partitions P_n such that $\max_i(t_i - t_{i-1})$ tends to zero and P_{n+1} is a refinement of P_n (meaning g_{P_n} increase in value while G_{P_n} decreases) such that $U_{P_n}(f)$ and $L_{P_n}(f)$ converge to $\int_a^b f(x)\, dx$. Let $G = \lim G_{P_n}$ and $g = \lim g_{P_n}$. Since $g \leq f \leq G$, by the dominated convergence theorem:
$$\int G\, dm = \int g\, dm = \int_a^b f(x)\, dx$$

Therefore, $\int G - g = 0$ and so $G = g$ a.e. by proposition 2.2.8, and so $G = f$ a.e.

Finally, since G is measurable (being the point-wise limit of measurable functions)and m is complete, f is measurable and

$$\int_[a, b] f \, dm = \int G \, dm = \int_a^b f(x) \, dx$$

2. This was exercise 23 in the textbook

The big advantage we have with this is that we can now use the computational techniques that were developed for Riemann integrals to compute particular Lebesgue integrals! In most cases in analysis, functions are locally Riemann integrable, and so we might what does the generality give us. The main thing it gives us is *completeness.* So though computing Lebesgue integrals might no be so easy (or if they are not an elementary Riemann integrable function), they are easier to work with in a theoretical framework, and help us define concepts like L^p space (which are important for Fourier Analysis).

(Note that in the infinite case, the Riemann and Lebesgue integral need not match. For example, $\sin(x)/x$ is Riemann integrable and produce a finite value, but it is not Lebesgue integral, producing an infinite value))

(A quick word on the Gamma function)

2.4 Modes of Convergence

So far, we have accumulated the following forms of convergence of functions:

1. uniform convergence $f_n \rightrightarrows f$
2. point-wise convergence $f_n \to f$
3. point-wise convergence a.e. $f_n \to f$ a.e.
4. L^1 convergence $f_n \to f$ if and only if $\int |f_n - f| \to 0$

The first 3 are weaker than then the next: uniform convergence $\implies$ point-wise convergence $\implies$ a.e. convergence, but the converse is false in each case. Furthermore, L^1 convergence implies none of those 3, and none of those 3 imply L^1 convergence. The examples to keep in mind to compare are:

1. $f_n = n^{-1}\chi_{[0,n]}$
2. $f_n = \chi_{[n,n+1]}$
3. $f_n = n\chi_{[0,1/n]}$
4. $f_1 = \chi_{[0,1]}$, $f_2 = \chi_{[0,1/2]}$, $f_3 = \chi_{[1/2,1]}$, $f_4 = \chi_{[0,1/4]}$, $f_5 = \chi_{[1/4,2/4]}$, and so on

We now introduce another way of thinking about convergence: instead of saying that the $\int f_n$ approaches that of f (i.e., evaluate the integral and look at the limiting behavior of the integral of the function), we will want there to exist an N such that $n \geq N$, the measure of the difference between f_n and f are epsilon away from the function we are converging to f. This is not the same thing as L^1 convergence, since this (1), (3) and (4) all converge to 0 in this type of convergence. Before any further analysis let's define our terms:

Definition 2.4.1: Cauchy in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *cauchy in measure* if:

$$\forall \epsilon > 0,\ \exists N \text{ such that } \forall n, m \geq N,\ \mu(\{x \mid |f_m(x) - f_n(x)| \geq \epsilon\}) < \epsilon$$

Equivalently, we can write the definition as $\forall \epsilon > 0$,

$$\mu(\{x \mid |f_n(x), f_m(x)| \geq \epsilon\}) \to 0 \text{ as } n, m \to \infty$$

Definition 2.4.2: Convergence in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *convergent in measure* or *converge in measure* if:

$$\forall \epsilon > 0,\ \lim_{n\to\infty} \mu(\{x \mid |f(x) - f_n(x)| \geq \epsilon\}) = 0$$

With these definition, notice that (2) is in fact *not* Cauchy in measure: the difference between the two functions.

Proposition 2.4.3: L^1 convergence Then Measure Convergence

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f_n \to f$ in L^1. Then $f_n \to f$ in measure

***Proof*:**
Let ϵ be given, and define

$$E_{\epsilon,n} = \mu(\{x \mid |f_n(x) - f(x)| \geq \epsilon\})$$

Then:

$$\int |f - f_n| \geq \int_{E_{\epsilon,n}} |f - f_n| \geq \epsilon\mu(E_{\epsilon,n})$$

Thus:

$$\mu(E_{n,\epsilon}) \leq \epsilon^{-1} \int |f - f_n| \to 0$$

showing that it converges in measure.

Note that convergence in measure does not imply convergence in L^1 as we saw in example (1) and (3).

With this established, we can go onto estalbish why we care about convergence in measure: it gives us another way of finding point-wise convergent functions by establishing a condition weaker than L^1 convergence:

Theorem 2.4.4: Cauchy in Measure Then Pointwise a.e.

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ Cauchy in measure. Then there exists a f such that $f_n \to f$ in measure, and furthermore there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \to f$ a.e.
If $f_n \to g$ in measure, then $f = g$ a.e.

***Proof*:**

We can choose a subsequence $\{g_j\} = \{f_{n_j}\}$ of $\{f_n\}$ such that if $E_j = \{x : |g_j(x) - g_{j+1}(x)| \geq 2^{-j}\}$, then $\mu(E_j) \leq 2^{-j}$. If $F_k = \bigcup_{j=k}^{\infty} E_j$, then $\mu(F_k) \leq \sum_k^{\infty} 2^{-j} = 2^{1-k}$, and if $x \notin F_k$, for $i \geq j \geq k$ we have

$$|g_j(x) - g_i(x)| \leq \sum_{l=j}^{i-1} |g_{l+1}(x) - g_l(x)| \leq \sum_{l=j}^{i-1} 2^{-l} \leq 2^{1-j}. \tag{2.31}$$

Thus $\{g_j\}$ is pointwise Cauchy on F_k^c. Let $F = \bigcap_1^{\infty} F_k = \limsup E_j$. Then $\mu(F) = 0$, and if we set $f(x) = \lim g_j(x)$ for $x \notin F$ and $f(x) = 0$ for $x \in F$, then f is measurable (see Exercises 3 and 5) and $g_j \to f$ a.e. Also, (2.31) shows that $|g_j(x) - f(x)| \leq 2^{1-j}$ for $x \notin F_k$ and $j \geq k$. Since $\mu(F_k) \to 0$ as $k \to \infty$, it follows that $g_j \to f$ in measure. But then $f_n \to f$ in measure, because

$$\{x : |f_n(x) - f(x)| \geq \epsilon\} \subset \{x : |f_n(x) - g_j(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |g_j(x) - f(x)| \geq \tfrac{1}{2}\epsilon\},$$

and the sets on the right both have small measure when n and j are large. Likewise, if $f_n \to g$ in measure,

$$\{x : |f(x) - g(x)| \geq \epsilon\} \subset \{x : |f(x) - f_n(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |f_n(x) - g(x)| \geq \tfrac{1}{2}\epsilon\}$$

for all n, hence $\mu(\{x : |f(x) - g(x)| \geq \epsilon\}) = 0$ for all ϵ. Letting ϵ tend to zero through some sequence of values, we conclude that $f = g$ a.e.

Corollary 2.4.5: L^1 convergence and Finding Subsequence

Let $f_n \to f$ in L^1. Then there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \to f$ a.e.

***Proof*:**
By proposition 2.4.3, $f_n \to f$ in measure, and by theorem 2.4.4, there exists a subsequence that converges to f a.e.

As we saw, $f_n \to f$ a.e. does not imply $f_n \to f$ in measure, as we saw in example (2). This is due to the "escape to infinity" that is happening, which is why we care about the dominated convergence theorem to help us stop such "sneaky infinity problem". However, if μ was a finite measure, then $f_n \to f$ a.e. does imply $f_n \to f$ in measure. In fact, since the area is bounded, we can some even stronger convergent conditions:

Theorem 2.4.6: Ergoff's Theorem

Let $(X, \mathcal{M}, \mu)$ be a measure space and suppose that $\mu(X) < \infty$. If $\{f_i\}$ are measurable complex-valued functions such that $f_n \to f$ a.e., then for every $\epsilon > 0$, there exists a $E \subseteq X$ such that $\mu(E) < \epsilon$ and $f_n \to f$ uniformally on $\mu(E^c)$

Proof:

Without loss of generality we may assume that $f_n \to f$ everywhere on X. For $k, n \in \mathbb{N}$ let

$$E_n(k) = \bigcup_{m=n}^{\infty} \{x : |f_m(x) - f(x)| \geq k^{-1}\}.$$

Then, for fixed k, $E_n(k)$ decreases as n increases, and $\bigcap_{n=1}^{\infty} E_n(k) = \varnothing$, so since $\mu(X) < \infty$ we conclude that $\mu(E_n(k)) \to 0$ as $n \to \infty$. Given $\epsilon > 0$ and $k \in \mathbb{N}$, choose n_k so large that $\mu(E_{n_k}(k)) < \epsilon 2^{-k}$ and let $E = \bigcup_{k=1}^{\infty} E_{n_k}(k)$. Then $\mu(E) < \epsilon$, and we have $|f_n(x) - f(x)| < k^{-1}$ for $n > n_k$ and $x \notin E$. Thus $f_n \to f$ uniformly on E^c.

The type of convergence in this theorem is sometimes called *almost uniform convergence.* Almost uniform convergence implies convergene a.e. and convergence in measure.

2.5 Product Measure

As we defined in definition 1.1.8, If $\mathcal{M}$ and $\mathcal{N}$ are σ-algebras, then we can define a new σ-algebra $\mathcal{M} \otimes \mathcal{N}$. If $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are measure spaces, we will show that we can define a new measure on $\mathcal{M} \otimes \mathcal{N}$ using μ and ν. We will use the tools we have done in constructing measure to construct this measure, and show that that there is a natural measure $\mu \times \nu$ with respect to the outer-measure given our space is σ-finite (this is just the uniqueness clause of theorem 1.3.7).

First, we'll give a name to sets in $\mathcal{M} \otimes \mathcal{N}$ that will be easier to measure (these will be our generating sets)

Definition 2.5.1: [Measurable] Rectangle

Let $A \in \mathcal{M}$ and $B \in \mathcal{N}$. Then $A \times B \in \mathcal{M} \otimes \mathcal{N}$ is called a *measurable rectangle* or just *rectangle* for short

Notice that if $A \times B$ and $C \times D$ are rectangles, they form an elementary family:

$$(A \times B) \cup (C \times D) = (A \cap C) \times (B \cap D) \qquad (A \times B)^c = (X \times B^c) \cup (A^c \times B)$$

and so by proposition 1.1.13, the collection of finite disjoint unions of rectangles forms an algebra $\mathcal{A}$, and they generate $\mathcal{M} \otimes \mathcal{N}$.

Furthermore, not all sets in $\mathcal{M} \otimes \mathcal{N}$ are measurable rectangles: if we take $(\mathbb{R}, \mathcal{L}(\mathbb{R}), m)$ and consider $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$, then the set of points $\Delta = \{(x, x) \mid x \in \mathbb{R}\}$ is in $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$ (being the countable intersection of ever smaller squares on the line), however it is very far from being the product of two sets in $\mathcal{L}(\mathbb{R})$.

We'll now define how to integrate simple functions on $X \times Y$ with the associated measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$: Let $A \times B$ be a rectangle that is the (finite or countable) disjoint union of rectangle $A_i \times B_i$. Then notice that:

$$\chi_{A \times B}(x, y) = \chi_A(x)\chi_B(y)$$

and so:

$$\chi_{A \times B}(x, y) = \sum_i \chi_{A_i \times B_i}(x, y) = \sum_i \chi_{A_i}(x)\chi_{B_i}(y)$$

If we integrate with respect to x, treating $\chi_{B_i}(y)$ as some constant, then by theorem 2.2.7

$$\int \chi_A(x)\chi_B(y)d\mu = \mu(A)\chi_B(y)d\mu = \mu(A)\chi_B(y) = \sum_i \mu(A_i)\chi_{B_i}(y)$$

It might be confusing which variable is associated to which measure, in which case it is common to write:

$$\int \chi_A(x)\chi_B(y)d\mu(x)$$

to emphasize the function with the x variable is the one being measured by μ. Integrating with respect to y yields:

$$\int \mu(A)\chi_B(y)d\nu = \mu(A)\nu(B) = \sum_i \mu(A_i)\nu(B_i)$$

Thus, we can define $\pi : \mathcal{A} \to \mathbb{R}$ where if $E \in \mathcal{A}$ is a finite disjoint union of rectangles $A_i \times B_i$, then:

$$\pi(E) = \sum_i \mu(A_i)\nu(B_i)$$

with the usual convention of $0 \times \infty = 0$ (i.e., if one of the dimensions is "0", then we are making the value 0). Notice that π is independent of representation, since any finite disjoint union of rectangles have a common refinement. Thus π is a pre-measure, and so by theorm 1.3.7, π generates an outer measure on $X \times Y$ whose restriction to $\mathcal{M} \otimes \mathcal{N}$ is a measure where restricted to $\mathcal{A}$ is π (or conversely, whose measure extends π). Such a measure is given a name:

Definition 2.5.2: Product Measure

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{M}, \nu)$ be measure spaces, and define π as in the previous construction. Then the induced outer-measure is called the *product measure* and is usually denoted $\mu \times \nu$.

Notice further that if μ and ν are σ-finite, so $\mu(X) = \sum_i^\infty \mu(A_i)$ and $\nu(Y) = \sum_i^\infty \nu(B_i)$, then $(\mu \times \nu)(X \times Y) = \sum_{i,j} \mu(A_i)\nu(B_j)$, and since all terms are finite, $\mu \times \nu$ is also σ-finite, in which case by the same theorem as invoked in defining $\mu \times \nu$, the product measure is the unique measure such that $\mu \times \nu(A \times B) = \mu(A)\nu(B)$.

Using induction, we can extend this construction to any finite product of measures, that is, for measure spaces $(X_i, \mathcal{M}_i, \mu_i)$, we can define $\mu_1 \times \cdots \mu_n$ on $\mathcal{M}_1 \otimes \mathcal{M}_2 \otimes \cdots \otimes \mathcal{M}_n$ such that

$$\mu_1 \times \mu_2 \times \cdots \mu_n(A_1 \times A_2 \times \cdots \times A_n) = \prod_i^n \mu_i(A_n)$$

and if every μ_i is σ-finite, then the result product measure is the unique extension from the premeasure on the rectangles of $\prod_i^n \mathcal{M}_i$.

Next, (should I prove associativity, it is exercise 45)

With what we've established so far, we can already measure sets using the outer-measure definition. However, this is very tedious, and ultimately comes down to some measure properties of the respective component measure. There is an easier way to use these component measure to get the resulting measure: notice in our construction of π that we integrated with respect to one variable first, and then the next. We will ostensibly generalizing this trick so that we can measure spaces, or more specifically integrate functions on product spaces, by fixing all variables except one, integrating on it, and then moving the result out (since it's a constant) and integrate the next variable, until we integrate over all variables.

Since we can generalise our results by induction, let's take $n = 2$ and consider the measure space $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$. The following definition tries to capture the idea of fixing everything except one variable:

Definition 2.5.3: n-section

Let $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$ be a product measure space, and consider $E \subseteq X \times Y$. Then for any $x \in X$, $y \in Y$, define the *x-section* E_x and *y-section* E^y to be:

$$E_x = \{y \in Y \mid (x, y) \in E\} \qquad E^y = \{x \in X \mid (x, y) \in E\}$$

If f is a function on $X \times Y$, we define the *x-section* f_x and *y-section* f^y to be:

$$f_x(y) = f(x, y) \qquad f^y(x) = f(x, y)$$

The most simple example of the section of a function would be:

$$(\chi_E)_x = \chi_{E_x} \qquad (\chi_E)^y = \chi_{E^y}$$

Proposition 2.5.4: Measure of Section

1. Let $E \in \mathcal{M} \otimes \mathcal{N}$. Then $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$ (i.e., all sets in a produce σ-algebra can be broken down into x-sections and y-section)
2. If f is $\mathcal{M} \otimes \mathcal{N}$-measurable, then f_x is $\mathcal{N}$-measurable and f^y is $\mathcal{M}$-measurable

***Proof*:**

1. The proof of this fact comes down to taking advantage of the fact that the pre-measure of rectangles generates $\mathcal{M} \otimes \mathcal{N}$. Let $\mathcal{R}$ be the collection of all subsets of $E \subseteq X \times Y$ such that $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$. This clearly contains the empty set, and the set must contain all rectangles, since if $A \neq 0$ and $x \in A$, then $(A \times B)_x = B$ (if $x \notin A$, then $(A \times B)_x = \emptyset$). Now, since

$$\left(\bigcup_i^\infty E_i\right)_x = \bigcup_i^\infty (E_i)_x \qquad (E^c)_x = (E_x)^c$$

by some basic set manipulation, we see that $\mathcal{R}$ is a σ-algebra. Since it contains the generating set of $\mathcal{M} \otimes \mathcal{N}$, we have that $\mathcal{R} \supseteq \mathcal{M} \otimes \mathcal{N}$. Thus, all the sets in $\mathcal{M} \otimes \mathcal{N}$ can be decompose in this way (note that we have not proved equality: we have not proved that if every slice is measurable then E is measurable)

2. By some set-theory manipulation, and part (1):

$$f_x^{-1}(B) = (f^{-1}(B))_x \qquad f^y{-1}(B) = (f^{-1}(B))^y$$

and so f_x and f^y must be $\mathcal{N}$ and $\mathcal{M}$ measurable (respectively)

With this, we are almost ready to start proving we can integrate $A \times B \in \mathcal{M} \otimes \mathcal{N}$ with respect to 1 measure at a time. We first need to prove a technical result for the next theorem which gives (again) another way of constructing σ-algebra. This one is particularly useful, since it will let us prove certain sets are σ-algebras by using the MCT and DCT:

Definition 2.5.5: Monotone Class

Let X be a set and $P(X)$ the powerset of X. Then the subset $\mathcal{C} \subseteq P(X)$ is called a *monotone class* if

1. it is cosed under countable unions of increasing sets, that is, if $E_1 \subseteq E_2 \subseteq \cdots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\cup_i E_i \in \mathcal{C}$
2. it is cosed under intersection unions of decreasing sets, that is, if $E_1 \supseteq E_2 \supseteq \cdots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\cap_i E_i \in \mathcal{C}$

Clearly, every σ-algebra is a monotone class (we've even used this property in a couple of proofs). Furthermore, the intersection of any family of monotone class is again a monotone class, and so we can define the unique smallest monotone class that contains the set $\mathcal{E} \subseteq P(X)$, where $\mathcal{E}$ is called the monotone class generated set, and the monotone class is called the monotone class generated by $\mathcal{E}$, and is dentoed $\mathcal{C}(\mathcal{E})$

Lemma 2.5.6: Monotone Class Lemma

Let X be a set and $\mathcal{A} \subseteq X$ be an algebra on some subsets of X. Then the monotone class $\mathcal{C}$ generated by $\mathcal{A}$ is equal to the σ-algebra $\mathcal{M}$ generated by $\mathcal{A}$, that is:

$$\mathcal{C}(\mathcal{A}) = \mathcal{M}(\mathcal{A})$$

***Proof*:**
Since $\mathcal{M}$ is itself a monotone class, we automatically have $\mathcal{E} \subseteq \mathcal{M}$, so we'll prove $\mathcal{E} \supseteq \mathcal{M}$.

This is just a lot of set-theory manipulation, I'll leave it for another time

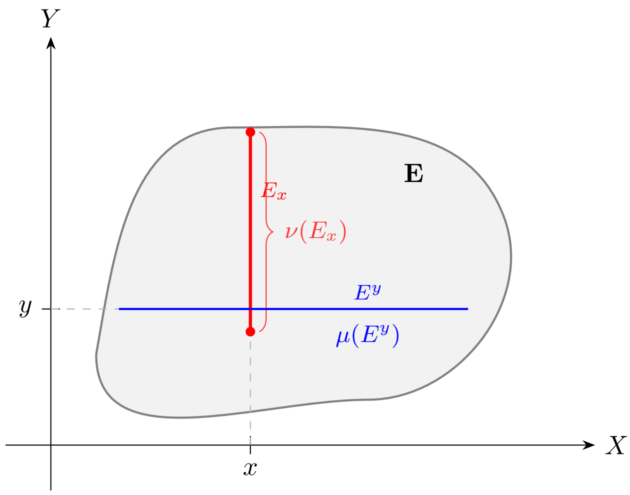
We now are able to start proving the main results:

Theorem 2.5.7: Fubini-Tonelli for Characteristic Functions

Let $(X, \mathcal{M}, \mu)$ and $(N, \mathcal{N}, \nu)$ be measure spaces with μ, ν being σ-finite. Then if $E \in \mathcal{M} \otimes \mathcal{N}$, the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable on X and Y respectively, and

$$(\mu \times \nu)(E) = \int \nu(E_x) d\mu = \int \mu(E^y) d\nu$$

If you are visual, you can think of $\mu(E_x)$ as measuring the area under the slice (inside the appropriate measure so that it doesn't have measure 0), and see that are saying there exists a function that maps x to every slice of the graph:



In particular, it maps x the measure of that slice, so we get a new function which at every slice captures the effect of the X component. After that, we integrate the function to get the effect from the Y component.

Furthermore, it will turn out that we can measure the slices with respect to X, or with respect to Y, and get the same result, hence the second statement of this theorem! After proving the theorem, we'll go over why the assumption in the statement of the theorem are important.

***Proof*:**
We'll first prove it for when μ and ν are finite, and then generalize for when they are σ-finite. The way we'll do it is by starting with a dense subset of $\mathcal{M} \otimes \mathcal{N}$ for which we know the conclusion holds, and then generalize the result using the MCT and DCT. In particular, we will be taking advantage that rectangles already satisfy the property of the theorem, and since rectangles are a generating set for $\mathcal{M} \otimes \mathcal{N}$, we will be able to extend this result any measurable set.

Let $\mathcal{C}$ be the subset of $\mathcal{M} \otimes \mathcal{N}$ for which the result of the theorem is true, that is, the two maps are measurable and we can integrate one way or another. Then if $A \in \mathcal{M}$ and $B \in \mathcal{N}$, we have that $A \times B \in \mathcal{C}$. To see this, first compute the values $\nu(E_x)$ and $\mu(E^y)$:

$$\nu(E_x) = \chi_A(x)\nu(B) \qquad \mu(E^y) = \mu(A)\chi_B(y)$$

Then the functions is measurable since it's a slice (theorem 2.5.4), and:

$$\int \nu(E_x)d\mu = \int \chi_A(x)\nu(B)d\mu = \mu(A)\nu(B) = \int \mu(A)\chi_B(y)d\nu = \int \mu(E^y)d\nu$$

Showing the two integrals are equivalent, and since $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$, all equality hold, and so $E = A \times B \in \mathcal{C}$. By additivity of the integral, it follows that all finite disjoint unions of rectangles are also in $\mathcal{C}$. Thus, by lemma 2.5.6, it suffices to show that $\mathcal{C}$ is a monotone class, for then it is a σ-algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, meaning $\mathcal{C} \supseteq \mathcal{M} \otimes \mathcal{N}$, and so all measurable sets can be broken down in this way.

First, let $\{E_n\}_{n=1}^{\infty} \subseteq \mathcal{C}$ be an increasing subset and let $E = \bigcup_n^{\infty} E_n$. We want to show that $E \in \mathcal{C}$. Since $E_n \in \mathcal{C}$ for all $n \in \mathbb{N}$, the sequence of y-section functions f_n, $f_n(y) = \mu((E_n)^y)$ are measurable and increase pointwise to $f(y) = \mu(E^y)$. Since the limit of measurable functions is measurable (corollary 2.1.13), f is measurable. Thus, we can find the value of $\int \mu(E^y)d\nu$ by applying the Monotone Convergence Theorem:

$$\begin{aligned}
\int \mu(E^y)d\nu &= \int \lim_{n\to\infty} \mu((E_n)^y)d\nu \\
&= \lim_{n\to\infty} \int \mu((E_n)^y)d\nu && \text{MCT} \\
&= \lim_{n\to\infty} (\mu \times \nu)(E_n) && E_n \in \mathcal{C} \\
&= (\mu \times \nu)(E) && \cup_i E_i = E
\end{aligned}$$

Doing the same trick with respect to x, we can see that:

$$\int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

which combined together shows us that:

$$\int \mu(E^y)d\nu = \int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

and so $E \in \mathcal{C}$.

Next, let $\{E_n\}_{n=1}^{\infty} \subseteq \mathcal{C}$ be a decreasing subset and let $\bigcap_n E_n = E$. Then since $E_1 \in \mathcal{C}$, the function f_1, $f_1(y) = \mu((E_1)^y)$ is measurable. In fact, it's in L^1, since:

$$\mu((E_1)^y) \leq \mu(X) < \infty \qquad \nu(Y) < \infty$$

and so the resulting integral must have finite measure. Furthermore, f_1 will dominate the sequence of functions f_n, and so we can apply the Dominated Convergence Theorem to do the same limit trick we have just done and conclude that $E \in \mathcal{C}$. Therefore, $\mathcal{C}$ is a monotone class, hence a σ-algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, and the result is true for all measurable sets.

Now, suppose that μ and ν are σ-finite. Then we can write $X \times Y$ to be the union of an increasing sequence of rectangles $\{X_i \times Y_i\}$ of finite measure. Thus, for any $E \in \mathcal{M} \otimes \mathcal{N}$, take $E \cap (X_i \times Y_i)$, then this set now has finite measure, and so what we have just proved applies giving us:

$$(\mu \times \nu(E \cap (X_i \times Y_i)) = \int \chi_{X_i}(x)\nu(E_x \cap Y_i)d\mu = \int \mu(E^y \cap X_i)\chi_{Y_i}(y)d\nu$$

and so, a final application of the Monotone Convergence Theorem gives us our desired result

The hypothesis that $\mu \times \nu$ is σ-finite is necessary:

Example 2.5.8: Necessity of σ-finite

To compute the iterated integrals, we start by finding the x and y section. Using the definition of the diagonal we get that: $(\chi_D)_x = \chi_{\{x\}}$ and $(\chi_D)_y = \chi_{\{y\}}$. Thus, we get:

$$\int \left[\int \chi_{\{y\}} d\mu(x)\right] d\nu(y) = \int 0\nu(y) = 0$$

and

$$\int \left[\int \chi_{\{x\}} d\nu(y)\right] d\mu(x) = \int 1 d\mu(x) = 1$$

showing that the iterated iterated iterated integrals are not equal. For the integral $\int \chi_D \, d(\mu \times \nu)$, since χ_D is a simple function, we have shown in class that the integral is the measure of D with respect to the product measure $\mu \times \nu$ (namely, since χ_D will be in the supremum set over which we are computing the integral), which is:

$$\int \chi_D d(\mu \times \nu) = \mu \times \nu(D) = \inf \left\{ \sum_i^\infty \mu(A_i)\nu(B_i) \;\middle|\; A_i \times B_i \in \mathcal{A},\ D \subseteq \bigcup_i A_i \times B_i \right\}$$

where $\mathcal{A}$ is the collection of all rectangles of $\mathcal{M} \otimes \mathcal{N}$. We'll show $\mu \times \nu(D) = \infty$.

Take any countable cover of D by rectangle $\mathcal{C}$ (so $\{A_i \times B_i\}_{i=1}^\infty = \mathcal{C}$ for appropriate $A_i \times B_i$). Without loss of generality, we can make these covers disjoint; we can refine the rectangles. If we can show that there must be a non-zero measure set A_i and an infinite measure set B_i such that $A_i \times B_i \in \mathcal{C}$ then we are done. For the sake of contradiction, let's say no such set exists in any countable cover $\mathcal{C}$. We'll show that $\mathcal{C}$ doesn't cover D.

Split $\mathcal{C}$ into $F \subseteq \mathcal{C}$ and $I \subseteq \mathcal{C}$ where F is the collection of all sets where A_i has ***finite*** non-zero measure (the sets have finite measure since if $A_i \subseteq X$, $\mu(A_i) \leq \mu(X) = 1$), and I is the set where B_i has ***infinite*** measure. By our contradiction assumption, these set's are disjoint, since the first component of the sets in I must have zero measure.

The set F can only cover countably many points of D: Since each B_i has finite measure, any set $A_i \times B_i \in F$ can cover at most finitely many points of D (given appropriate A_i), and so $\bigcup_{A_i \times B_i \in F} A_i \times B_i$ can at most cover countably many points of D (since the 2nd component of

D is covered by elements from B_i given appropriate A_i). Therefore, an uncountable number of points are not being covered by F.

The set I misses at least uncountably many points. Since each component A_i has zero-measure and the collection of x components of D (i.e. the set $[0,1]$) has measure one, the union of the sets A_i cannot cover all of the x-components (given appropriate B_i components) or else $0 = \mu(\bigcup_i A_i) \geq \mu([0,1]) = 1$, a contradiction. If the B_i component's don't line up correctly, then this set cover's even fewer points of D. Therefore, I can at most cover a measure zero amount of x-components points from D. Since any set two sets containing each other with a different measure (ex. $\alpha \subseteq \beta$, $0 = \mu(\alpha) \leq (\beta) = 1$) must have a difference with at least uncountable cardinality[a] ($|B \setminus A| \geq \aleph_1$), the set I must miss at least uncountably many points.

However, the cover F can only cover at most countably many points. Therefore, the cover $F \cup I = \mathcal{C}$ cannot cover all of D, a contradiction to the assumption that $\mathcal{C}$ is a cover for D. Hence, there is at least one set $A_i \times B_i$ with $\mu(A_i) > 0$ and $\mu(B_i) = \infty$ for any cover of D, meaning all covers of D must have a set with infinite measure, and so $\infty \leq \mu \times \nu(D)$, and hence

$$\int \chi_D \, d(\mu \times \nu) = \infty$$

and so the two iterated integrals, as well as the integral of χ_D has unequal measures, as we sought to show.

[a]If two sets containing each other had differed by countably many point, then they would have the same measure

Notice that the functions within the integral of the previous theorem are in fact the values of characteristic functions! Hence by the linearity of the integral, we have the result for simple functions. Thus, with the statement proven for simple functions on $\mathcal{M} \otimes \mathcal{N}$, we move on to proving it for integrable functions in general:

Theorem 2.5.9: Fubini-Tonelli Theorem

Let $(X, \mathcal{M}, \mu)$ and $(N, \mathcal{N}, \nu)$ be σ-finite measures. Then:

1. (Tonelli) If $f \in L^+(X \times Y)$, then the functions $g(x) = \int f_x d\nu$ and $h(y) \int f^y d\mu$ are in $L^+(X)$ and $L^+(Y)$ respectively, and

$$\int \left[\int f(x,y) d\nu(y) \right] d\mu(x) = \int f(x,y) d(\mu \times \nu) = \int \left[\int f(x,y) d\mu(x) \right] \nu(y)$$

2. (Fubini) If $f \in L^1(\mu \times \nu)$, then $f_x \in L^1(\nu)$ for a.e. $x \in X$ and $f^y \in L^1(\mu)$ for a.e. $y \in Y$. Furthermore, the a.e. defined function $g(x) = \int f_x d\nu$ and $h(x) = \int f^y d\mu$

The resulting integral is called the *iterated integral*

***Proof*:**

1. This essentially comes down to using MCT and the standard representation of measurable functions. By theorem 2.5.7, we have the simple functions satisfy the theorem. To extend it to integrable functions, take $f \in L^+(X \times Y)$, and let $\{f_n\}$ be some simple representation for it, that is, a sequence of simple functions converging upwards to f. From $\{f_n\}$, we

have a corresponding $\{g_n\}$ and $\{h_n\}$ converging upwards to g and h: $g_n = \int (f_n)_x$ and $h_n = \int (f_n)^y$. By theorem 2.5.7, these functions are measurable, and so h and g are measurable. With this, we can put together all our information to form the following chain of equalities:

$$\begin{aligned}
\int \left[\int f_x \, d\nu\right] d\mu = \int g d\mu = \int \lim g_n \, d\mu \\
&= \lim \int g_n \, d\mu \\
&= \lim \int f_n \, d(\mu \times \nu) && \text{theorem } 2.5.7 \\
&= \int f \, d(\mu \times \nu) && \text{MCT} \\
&= \lim \int f_n \, d(\mu \times \nu) \\
&= \lim \int h_n \, d\nu && \text{theorem } 2.5.7 \\
\int \left[\int f^y \, d\mu\right] d\nu = \int h d\nu = \int \lim h_n \, d\nu
\end{aligned}$$

Thus establishing the equality of Tonelli's Theorem. Furthermore, since $\int f \, d(\mu \times \nu) < \infty$, then $g < \infty$ a.e. and $h < \infty$ a.e., so $\int f_x < \infty$ and $\int f^y < \infty$, and therefore $f_x \in L^1(\nu)$ and $f^y \in L^1(\mu)$, which some of the preliminary results for Fubini's Theorem.

2. Let $f \in L^1(\mu \times \nu)$, so $f^+ \in L^+(\mu \times \nu)$ and $f^- \in L^+(\mu \times \nu)$ (or Ref and sgn$f$). Thus, the result applies to each component function, and so by linearity applies to f

Often, the parenthesis are dropped, and we combine the integrals into a double integral:

$$\int \left[\int f(x,y) \, d\mu(x)\right] d\nu(y) = \int \int f(x,y) \, d\mu(x) \, d\nu(y) = \int \int f \, d\mu \, d\nu$$

Also, you might wonder if instead of assuming $f \in L^+(X \times Y)$ or $f \in L^1(X \times Y)$, is it possible to instead assume that each $f_x \in L^+(\nu), f^y \in L^+(\mu)$ (L^1 respectively) and that the iterated integrals $\int \int f \, d\mu \, d\nu$ and $\int \int f \, d\nu \, d\mu$ existed, then $f \in L^+(X \times Y)$ (L^1 respectively), that is, is the converse true? The answer is actually no:

Example 2.5.10: Converse not True

Let $X = Y = \mathbb{N}$, $\mathcal{M} = \mathcal{N} = P(\mathbb{N})$, $\mu = \nu$ be the counting measures. Define:

$$f(m,n) = \begin{cases} 1 & m = n \\ -1 & m = n+1 \\ 0 & \text{otherwise} \end{cases}$$

Then $\int |f| d(\mu \times \nu) = \infty$, and $\int \int f d\mu d\nu$, $\int \int f d\nu d\mu$ exist and are unequal.

This is not quite a converse because the two iterated integrals are unequal, and I don't know of an example where they are equal, but $\int f d(\mu \times \nu)$ is not equal to it (i.e. infinity).

One practical application of Fubini-Tonelli's Theorem is to solve integrals by integrating over the simpler variable first. Usually, one first uses Tonelli's theorem to evaluate the integral $\int f\,d(\mu\times\nu)$ as an iterated integral $\int\int f\,d\mu\,d\nu$ and show that the result is finite, at which point Fubini's Theorem can be invoked to get $\int\int f\,d\mu\,d\nu = \int\int f\,d\nu\,d\mu$.

To close off the discussion on product measures, notice that if μ and ν are complete measure, $\mu\times\nu$ is almost never a complete measure. Let $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ be measure spaces, take $A\in\mathcal{M}$ is a non-empty set with $\mu(A)=0$, and suppose $\mathcal{N}\neq P(Y)$ (For example, choose $(\mathbb{R},\mathcal{L},m)$ for both measure spaces). If $E\in P(Y)\setminus\mathcal{N}$, then $A\times E\notin\mathcal{M}\times\mathcal{N}$ since not all slices of $A\times E$ (in particular, the slice E) are in the respective measure spaces (proposition 2.5.4). However, $A\times E\subseteq A\times Y$, and $(\mu\times\nu)(A\times Y)=0$ (since the measure is finite, any out-measure is equal to the usual outer-measure on these sets, which is $\mu(A)\nu(Y)=0\cdot n=0$, since we take on the convention that $0\cdot\infty=0$). Thus, $\mu\times\nu$ is not complete. If we want, we can apply theorem 1.2.8 to complete the measure, however we can no longer directly apply Fubini-Tonelli's Theorem. (Put Why Here!). We thus have to slightly amend Fubini-Tonelli's theorem for the completion:

Theorem 2.5.11: Fubini-Tonelli's Theorem for Complete Measures

Let $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ be complete σ-finite measures and let $(X\times Y,\mathcal{L},\lambda)$ be the completion of $(X\times Y,\mathcal{M}\otimes\mathcal{N},\mu\times\nu)$. If f is $\mathcal{L}$-measurable is either:

1. non-negative, that is $f\geq 0$
2. $f\in L^1(\lambda)$

Then f_x is $\mathcal{N}$-measurable for a.e. x and f^y is $\mathcal{M}$-measurable for a.e. y, and if (2) applies as well, then f_x and f^y are integrable for a.e. x and y. Furthermore, $x\mapsto\int f_x\,d\nu$ and $y\mapsto\int f^y\,d\mu$ are measurable (and in the case of (2), also integrable) and

$$\int f\,d\lambda = \int\int f(x,y)\,d\mu(x)\,d\nu(y) = \int\int f(x,y)\,d\nu(y)\,d\mu(x)$$

***Proof*:**
We'll do the following:

1. If $E\in\mathcal{M}\otimes\mathcal{N}$ and $\mu\times\nu(E)=0$, then $\nu(E_x)=\mu(E^y)=0$ for a.e. x and y
2. If f is $\mathcal{L}$-measurable and $f=0$ λ-a.e., then f_x and f^y are integrable for a.e. x and y, and $\int f_x d\nu=\int f^y d\mu=0$ for a.e. x and y (here, the completeness of μ and ν is needed)

lemma 1:

Proof. Note that $\chi_E\in L^+(X\times Y)$ and $(X,\mathcal{M},\mu)$ $(Y,\mathcal{N},\nu)$ are σ-finite, and so Tonelli's Theorem applies, meaning

$$\int\chi_E d(\mu\times\nu) = \int\left[\int(\chi_E)^y d\mu(x)\right]d\nu(y) = \int\left[\int(\chi_E)_x d\nu(y)\right]d\mu(x)$$

First, since χ_E is an indicator function, we see that $(\chi_E)_x=\chi_{E_x}$, since $\chi_{E_x}(y)=\chi_E(x,y)$.

Similarly, $(\chi_E)^y = \chi_{E^y}$. Therefore, we have:

$$\begin{aligned} 0 &= \int \chi_E d(\mu \times \nu) \\ &= \int \left[\int (\chi_E)^y d\mu(x) \right] d\nu(y) \\ &= \int \left[\int (\chi_E)_x d\nu(y) \right] d\mu(x) \end{aligned}$$

showing that the function $x \mapsto \int \chi_{E_x} d\nu$ and $y \mapsto \int \chi_{E^y} d\mu$ is 0 a.e., which implies $\nu(E_x) = 0$ and $\mu(E^y) = 0$ a.e. $\square$

lemma 2: Multiple parts of this proof take advantage of the fact that if μ is a measure and $\mu(A) = 0$, then $\int_A f d\mu = 0$ (since all simple functions φ approximating f with the property of $\varphi \leq f$ will have their integral be of the form $\sum_i^n c_i \chi_{A_i} = \sum_i^n c_i \cdot 0 = 0$ for $A_i \subseteq A$, and so the supremum of the integral of such simple functions must also be 0)

Proof. Let f be $\mathcal{L}$-measurable and $f = 0$ λ-a.e. We want to show that f_x and f^y is integrable for a.e. x and y and $\int f_x d\nu = \int f^y d\mu = 0$ for a.e. x and y. What we'll do is take advantage of Fubini-Tonelli and theorem 2.12 (which states that there exists a $\mathcal{M} \otimes \mathcal{N}$-measurable function g such that $f = g$ λ-a.e.)

First, since $f = g$ λ-a.e., $f - g = 0$ λ-a.e., meaning $(f - g)_x = f_x - g_x = 0$ for a.e. x (assuming completeness) and $f^y - g^y = 0$ for a.e. y (assuming completeness), and so $f_x = g_x$ and $f^y = g^y$ for a.e. x and y. This will be useful to us later.

Next, Consider $\int f d\lambda$. We want to take advantage of Fubini's theorem, which is true for g, and so we must re-write this expression in such a way that we get it equal to g. Let's say $S = \{(x, y) \in X \times Y \mid f(x, y) \neq g(x, y)\}$. We know that $\lambda(S) = 0$, and so:

$$0 = \int f \, d\lambda = \int_S f \, d\lambda + \int_{S^c} f \, d\lambda = \int_{S^c} f \, d\lambda \tag{2.1}$$

Since $S^c \in \mathcal{L}$, we have that $S^c = A \coprod N$ where $A \in \mathcal{M} \otimes \mathcal{N}$ and $N \subseteq B$ where $\mu \times \nu(B) = 0$. Thus, we get:

$$\int_{S^c} f d\lambda = \int_A f d\lambda + \int_N f d\lambda = \int_A f d\lambda$$

since $f = g$ on S^c, in particular $f = g$ on $A \in \mathcal{M} \otimes \mathcal{N}$, and λ is the extension of $\mu \times \nu$, we can re-write this as:

$$\int_A f d\lambda = \int_A g d(\mu \times \nu)$$

and so Fubini-Tonelli applies, giving us the iterated integrals are equal to 0 (carrying the 0 from equation (2.1)):

$$\int \left[\int_{A_x} g_x d\nu(y) \right] d\mu(x) = 0$$

and similarly for the y-section. Since $f = g$ on S^c, by what we've shown earlier, $f_x = g_x$ on A_x, (and similarly $f^y = g^y$ on A^y) and so:

$$\int_{A_x} f_x d\nu(y) = 0 \qquad \text{for a.e. } x$$

and similarly for the y-sections, which brings us close to our goal (it is possible that $A_x \subsetneq X$ or $A^y \subsetneq Y$). We now have to add back in points which we eliminated since the integral was zero on S and N:

$$\int_S f d\lambda = 0 \qquad \int_N f d\lambda = 0$$

Since they are measure zero sets in $\mathcal{L}$, there exists a $C, D \in \mathcal{M} \otimes \mathcal{N}$ such that $\mu \times \nu(C) = \mu \times \nu(D) = 0$ and $S \subseteq C$, $N \subseteq D$. Let $E = C \cup D$ so that $S \cup N \subseteq E$ and $\mu \times \nu(E) = 0$. Thus, we have:

$$\int_E f d\lambda = 0$$

since $\mu \times \nu(E) = 0$ and $E \in \mathcal{M} \otimes \mathcal{N}$, by the first lemma, $\mu(E^y) = \nu(E_x) = 0$ for a.e. x and y. But then, for a.e. x and y, $\int_{E_x} f_x d\nu(y) = \int_{E^y} f^y d\mu(x) = 0$. Since $X \times Y = S^c \cup S = A \cup E$ (by how we've broken the set down), then $X = A_x \cup E_x$ and $Y = A^y \cup E^y$, and so

$$0 \leq \int f_x d\nu(y) \leq \int_{A_x} f_x d\nu(y) + \int_{E_x} f_x d\nu(y) = 0$$

for a.e. x, implying $\int f_x d\nu(y) = 0$ for a.e. x (and similarly for y). Hence, they are integrable for a.e. x and y, completing the proof. ☐

With this, we can prove Fubini-Tonelli's Theorem for complete measures. First, let's build-up some result and notation. Let $f \in L^+(\lambda)$ (resp. $f \in L^1(\lambda)$). Then by theorem 2.12, there exists a g such that $f = g$ λ-a.e., meaning that $f - g = 0$ λ-a.e. Thus, by the 2nd lemma, $\int (f-g)_x = 0$ and $\int (f-g)^y = 0$ are integrable for a.e. x and y. Therefore, $\int f_x - g_x = 0$ and $\int f^y - g^y = 0$ for a.e. x and y, and so $\int f_x = \int g_x$ and $\int f^y = \int g^y$ for a.e. x and y.

We can now start concluding the proof. Since μ and ν are complete and $f_x = g_x$ ($f^y = g^y$) for a.e. x and y, by proposition 2.11, f_x is $\mathcal{N}$-measurable and f^y is $\mathcal{M}$-measurable for a.e. x and y, completing the 1st assertion of the theorem. In the $f \in L^1(\lambda)$ case, the proof we've just done in the previous paragraph shows that each f_x and f^y are integrable for a.e. x and y.

Next, since $G_x : x \mapsto \int g_x d\nu$ and $G^y : y \mapsto \int g^y d\mu$ is measurable (resp. integrable), by theorem 2.11, $F_x : x \mapsto \int f_x d\nu$ and $F^y : y \mapsto \int f^y d\mu$ is measurable (resp. integrable). Finally, since $F_x = G_x$ for a.e. x and $F^y = G^y$ for a.e. y, their integrals match, and since g is $\mathcal{M} \otimes \mathcal{N}$ measurable, Tonelli's (resp. Fubini's) theorem applies, and so by one more application of theorem 2.11 :

$$\int f \, d\lambda = \int \int f(x,y) \, d\mu(x) \, d\nu(y) = \int \int f(x,y) \, d\nu(y) \, d\mu(x)$$

Completing the proof

2.6 The n-dimensional Lebesgue Integral

In this section, we focus on $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ with the completed product measure $m \times m \times \cdots \times m = m^n$ on $\mathcal{L} \otimes \mathcal{L} \otimes \cdots \otimes \mathcal{L} = \mathcal{L}^n$. The domain $\mathcal{L}^n$ of m^n is called the set of *Lebesgue measurable sets in* $\mathbb{R}^n$: $(\mathbb{R}^n, \mathcal{L}^n, m^n)$ (sometimes, we will restrict m^n to have domain $(\mathcal{B}_{\mathbb{R}})^n = \mathcal{B}_{\mathbb{R}^n}$). If it is unambiguous, we will usually write $(\mathbb{R}^n, \mathcal{L}^n, m)$, dropping the power on the m, and for the $n = 1$ case, we will often write $\int f(x)\,dx := \int f\,dm$.

The following 3 theorems are simply generalizations of previous properties of the Lebesgue measure. In the following, let $E = \prod_{i=1}^{n} E_i$ where $E_i \subseteq \mathbb{R}$. We will call each E_i the *sides* of E

Theorem 2.6.1: Simplifying outer-measure

Suppose $E \in \mathcal{L}^n$

1. $m(E) = \inf\{m(U) \mid E \supseteq U,\ U \text{ is open}\} = \sup\{m(K) \mid K \subseteq E,\ K \text{ is compact}\}$
2. $E = A_1 \cup N_1 = A_2 \setminus N_2$ where $A_1 \in F_\sigma$, $A_2 \in G_\delta$, and $m(N_1) = m(N_2) = 0$
3. If $m(E) < \infty$, for all $\epsilon > 0$, there is a finite disjoint collection of rectangles $\{R_i\}_{i=1}^{n}$ whose sides are intervals such that $m(E\Delta \cup_i^n R_i) < \epsilon$

***Proof*:**
p. 70 in folland

Theorem 2.6.2: Completeness of $L^1(m)$

Let $f \in L^1(m)$ and let $\epsilon > 0$. Then there exists a simple function $\varphi = \sum_{i=1}^{n} a_i \chi_R$ where each R_i is a product of intervals such that

$$\int |f - \varphi| < \epsilon$$

and there is a continuous function that vanishes outside a bounded set such that

$$\int |f - g| < \epsilon$$

***Proof*:**
Just like in theorem 2.3.9, approximate f by a simple function, then apply proposition 2.6.1(3) to approximate φ appropriately. Finally, approximate φ by continuous functions by applying the natural generalization of the argument in theorem 2.3.9

Theorem 2.6.3: Translation Invariance in $\mathbb{R}^n$

Let $(\mathbb{R}^n, \mathcal{L}^n, m)$ be a measure space. Then m is translation invariant, that is, for any $a \in \mathbb{R}^n$, if we define $\tau_a : \mathbb{R}^n \to \mathbb{R}^n$, $\tau_a(x) = a + x$, then

1. If $E \in \mathcal{L}^n$, then $\tau_a(E) \in \mathcal{L}^n$ and $m(E) = m(\tau_a(E))$
2. If $f : \mathbb{R}^n \to \mathbb{C}$ is Lebesgue measurable, then so is $f \circ \tau_a$. Furthermore, if either $f \geq 0$ or $f \in L^1(m)$, then $\int (f \circ \tau_a)\,dm = \int f\,dm$

Notice that we did not mention scaling. This is because scaling is not abstracted to linear transformations which requires more work to show how the scaling of a linear transformation affects the value, and so we will prove that later in this section.

> ***Proof*:**
> Folland p.71

(here, some on cube approximation and Jordan content)

I'll for now simply state the effect of linear transformations and come back to this:

Theorem 2.6.4: Linear Transformation on Integrability

Suppose $T \in GL_n(\mathbb{R})$. Then:

1. If f is a Lebesgue measurable function on $\mathbb{R}^n$, so is $f \circ T$. If $f \geq 0$ or $f \in L^1(m)$, then:

$$\int f\, dm = |\det(T)| \int f \circ T\, dm$$

2. If $E \in \mathcal{L}^n$, then so is $T(E)$, and $m(T(E)) = |\det(T)|m(E)|$

> ***Proof*:**
> here

There is also the statement on how diffeomorphisms affect the integral.

3

Signed Measure and Differentiation

In this chapter, we will take a close look at how to define measure's with respect to integrals. In particular, given the appropriate integrable function f, we will want to define a measure μ such that

$$\mu(A) = \int_A f$$

After defining some terms in section 3.1, we will focus in section 3.2 and 3.3 on how to find a function f given a measure μ so that $\mu(A) = \int_A f$. The last 2 sections then focus on the special case where the function f that we found can actually be the *derivative*. In the simple case where $X = \mathbb{R}$, then we can define:

$$F(x) = \int_{(-\infty,x]} f dx = \int_{-\infty}^{x} f dx$$

and ask ourselves what conditions does f or F requires so that $f = F'$ (or, $f = F'$ a.e.). This line of reasoning will ultimately lead to a generalisation of the Fundamental Theorem of Calculus, which will tell us when can we recover the function f given we know $\int_A f$ for all A.

3.1 Signed Measures

Definition 3.1.1: Signed Measure

Let $(X, \mathcal{M})$ be a measurable space. A *signed measure* on $(X, \mathcal{M})$ is a function: $\nu : \mathcal{M} \to [-\infty, \infty]$ such that:

1. $\nu(\emptyset) = 0$
2. ν has at most one value of $\pm\infty$
3. If $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ is a disjoint collection, then

$$\nu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \nu(E_i)$$

   where if $\nu\left(\bigcup_{i=1}^{\infty} E_i\right) < \infty$, then $\sum_{i=1}^{\infty} \nu(E_i)$ converges absolutely.

Example 3.1.2: Signed Measures

Let $(X, \mathcal{M})$ be a measurable space.

1. Clearly, every measure on $\mathcal{M}$ is a signed measure. To distinguish between the two, we will often call measures *positive measures*.
2. Let μ_1 and μ_2 be two measure on $\mathcal{M}$ with at least one being finite. Then $\nu = \mu_1 - \mu_2$ is a signed measure.
3. If f is a $\mathcal{M}$-measurable function, then if at least one of $\int f^+$ or $\int f^-$ is finite,

$$\nu(E) = \int_E f$$

   is a signed measure. In this case, we call f the *extended μ-integrable function* (in contrast to $f \in L^1$ where both terms need to be finite)

The fact that it converges absolutely is to make sure that the order in which we sum the elements does not matter: If it did not converge absolutely, then it is possible to make the sum approach any value we want. Since $\mathcal{M}$ doesn't necessarily have an inheret order (in fact it almost never does), then having this dependence on the order of summing the elements is inconsistent with our notion of area (this is also why we only allow one of the direction to be infinite). It might seem that we must thus be careful on how we define our signed measure to make sure this condition holds, however we will soon show that every signed measrue can be decomposed into 2 measures. If the measure of all sets with positive measure and negative measure are both infinite ($\pm\infty$), then we have contradicted the definition of a signed measure. As just mentioned, all signed measures can be decomposed into two positive measures. We will spend this section proving that fact. In the next section, we will show that under appropriate conditions and choices of signed measure ν and positive meaure μ, we can find a measruable function f such that $\nu(E) = \int_E f d\mu$. This will essentially be the generalization of the derivative!

We start by building up to decomposing a signed measure into two positive meaures:

Proposition 3.1.3: Continuity from Above/Bellow for Signed Meaures

Let ν be a signed measure on $(X, \mathcal{M})$.

1. If $\{E_i\}$ is an increasing sequence in $\mathcal{M}$, then $\nu\left(\bigcup_i E_i\right) = \lim_{i\to\infty} \nu(E_i)$
2. If $\{E_i\}$ is an decreasing sequence in $\mathcal{M}$ where $\nu(E_1)$ is finite, then $\nu\left(\bigcap_i E_i\right) = \lim_{i\to\infty} \nu(E_i)$

***Proof*:**
Very similar to proposition 1.2.5

For the next proposition, we introduce a quick new definition: a set $P \in \mathcal{M}$ is called *positive* (resp. *negative*) if for all $S \subseteq P$ such that $S \in \mathcal{M}$, then $\nu(S) \geq 0$ (resp. $\nu(S) \leq 0$)

Lemma 3.1.4

Any countable union of positive sets is a positive set

***Proof*:**
Use proposition 3.1.3

Theorem 3.1.5: Hahn Decomposition Theorem

If ν is a signed measure on $(X, \mathcal{M})$, there exists a positive set P and a negative set N for ν $P \coprod N = X$ and $P \cap N = \emptyset$. If P' and N' is another such pair, then $P \Delta P$ and $N \Delta N'$ is a null set for ν

***Proof*:**
Let ν be a signed measure, so only ∞ or $-\infty$ can be achieved. If ∞ is achieved, take $-\nu$ so that ∞ is not achieved. Now, let m be the supremum of $\nu(E)$ where E ranges over all positive sets. Thus, there is a sequence of positive sets $\{P_i\}$ such that $\nu(P_i) \to m$. Let $P = \bigcup_i P_i$. Then by the previous lemma and proposition, P is a positive set and $\nu(P) = m$. Since ν does not achieve ∞, $m < \infty$. I claim that $N = X \setminus P$ is negative (thus completing the first assertion of the proof).

First, N cannot contain non-null positive sets. If $E \subseteq N$ was a positive set (i.e. $m(E) > 0$), then $E \cap P = \emptyset$ and $E \cup P$ is a positive set, so $m(E \cup P) = m(E) + m(P) > m$, which is a contradiction to m being the supremum.

Now, let's say there exists a $A \subseteq N$ such that $\nu(A) > 0$. We will reach a contradiction in using the following fact: since A cannot be positive, then there must exist a $B \subseteq A$ such that $\nu(B) < 0$. Then let $C = A \setminus B$ so that $\nu(C) > \nu(A)$ and $C \subseteq A$.

With this, we proceed in forming a contradiction: Define the following sequence $\{A_i\}$ and $\{n_i\}$ as follows:

1. Let n_1 be the smallest positive integer such that there exists a $A_1 \subseteq N$ where $\nu(A_1) > n_1^{-1}$ (this set exists since there is a $A \subseteq N$ where $\nu(A) > 0$

2. Let n_k be the smallest positive integer such that there exists a $A_k \subseteq A_{k-1}$ such that $\nu(A_{k-1}) > \nu(A_k) + n_k^{-1}$ (since A_{k-1} is positive, we use the earlier remark fact to find such a set)

From this sequence of positive decreasing sets, $A_1 \supseteq A_2 \supseteq \cdots$, let $A = \bigcap_i A_i$. Then by the fact that ν does not attain ∞ and the previous proposition:

$$\infty > \nu(A) = \lim \nu(A_i) > \sum_{i=1}^{\infty} n_i^{-1}$$

since the series converges, it must be that $n_i \to \infty$. Finally, once again take $B \subseteq A$ such that $\nu(A) > \nu(B) + n^{-1}$ for some positive integer n. Since $n_i \to \infty$, there exists an i sufficiently large so that $n < n_i$ and $B \subseteq A_i$, meaning we can replace n_i by n in the above construction. However, by construction, we picked the *minimum* integer every-time, and so this contradicts minimality! Therefore, it cannot be that there exists a set $A \subseteq N$ such that $\nu(A) > 0$, and so N is indeed negative.

If P' and N' where two other sets satisfying the decomposition condition, then we have that $P \setminus P' \subseteq P$ (since it's a subset) and $P \setminus P' \subseteq N'$ (since it is P minus all the positive set P') , so it is both positive and negative, which can only be possible if it is a null-set (similarly for $N \setminus N'$)

Using the Hahn decomposition, we can define a positive measure on P and N. Before proceeding to that theorem, we quickly give a name to the relation between the two measures that will be defined:

Definition 3.1.6: Mutually Singular Signed Measures

Let ν and μ be be two signed measure on $(X, \mathcal{M})$. Then they are called *mutually singular* or μ *is singular with respect to* ν if there exists a $E, F \in \mathcal{M}$ such that $E \cup F = X$, $E \cap F = \emptyset$, and:

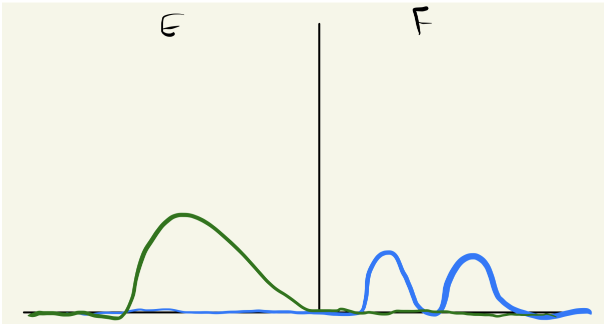
1. $\mu(E) = 0$

2. $\nu(F) = 0$

and we write mutually singular signed measures as:

$$\mu \perp \nu$$

In a sense, this means that these two sets are are meaningful on disjoint sets, in that sense they are "perpendicular" to each other. I like to keep an image like this in mind



Notice that it is possible that μ have zero values on F, and ν have zero values on E. They key is that they *must* have zero values on their respective sets.

Mutually singular measures can be applied more generally than in the context of decomposing a signed measure. For example, of δ_0 is the dirac-delta measure at 0 (i.e. $\delta(\{0\}) = 1$ and zero everywehre else), then $\delta_0 \perp \lambda$.

Theorem 3.1.7: Jordan Decomposition Theorem

If ν is a signed measure, there exists unique positive measure ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$

***Proof*:**
By theorem 3.1.5, let $X = P \cup N$ be the Hahn decomposition of X with respect to ν. Define $\nu^+(E) = \nu(E \cap P)$ and $\nu^-(E) = -\nu(E \cap N)$. Then by definition, $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$.

If there was another decomposition $\nu = \mu^+ - \mu^-$ where $\mu^+ \perp \mu^-$, then we have $E, F \in \mathcal{M}$ such that $E \coprod F = X$ and $E \cap F = \emptyset$, $\mu^+(F) = \mu^-(E) = 0$, so E, F is another Hahn decomposition for ν, and so $P \Delta E$ is null. Thus, for any $A \in \mathcal{M}$

$$\mu^+(A) = \mu^+(A \cap E) = \nu(A \cap E) = \nu(A \cap P) = \nu^+(A)$$

thus, $\mu^+ = \nu^+$, and similarly $\mu^- = \nu^-$, completing the proof.

Definition 3.1.8: Jordan Decomposition

Let ν be a signed measure on $(X, \mathcal{M})$. THen ν^+ and ν^- are caleld the *positive* and *negative variation* of ν, and $\nu = \nu^+ - \nu^-$ is called the *Jordan decomposition of* ν. The *total variation of* ν, denoted $|\nu|$ is defined as $|\nu| = \nu^+ + \nu^-$.

As an exercise, you should verify that $E \in \mathcal{M}$ is ν-null if and only if $|\nu|(E) = 0$, and $\nu \perp \mu$ if and only if $|\nu| \perp \mu$ if and only if $\nu^+ \perp \mu$ and $\nu^- \perp \mu$. With this definition, we will say that ν is finite (resp. σ-finite) if $|\nu|$ is finite (resp. σ-finite)

Finally, we can also get back to showing that any signed measure is of the form $\nu(E) = \int_E d\mu$ for appropriate μ, namely $\mu = |\nu|$ and $f = \chi_P - \chi_N$ for appropriate P and N from a Hahn decomposition. From this, we can define the integration with respect to a signed measure ν as follows:

Definition 3.1.9: Integration w.r.t. Signed Measures

Let ν be a signed measure, $\nu^+ - \nu^-$ be it's Hahn decomposition, and $f : X \to \mathbb{C}$ be a real function. Then we will say that f is *integrable with respect to the signed measure* ν if:

$$f \in L^1(\nu^+) \cap L^1(\nu^-) = L^1(\nu)$$

and define it's integral to be:

$$\int f d\nu = \int f d\nu^+ - \int f d\nu^- \qquad (f \in L^1(\nu))$$

3.2 Lebesgue-Radon-Nikodym Theorem

In this section, we show under what circumstances we can decompose a signed measure μ into $\nu + \rho$ where $\nu(A) = \int_A f$ and ρ is the "left-over" (also known as the singular part). We first define the "opposite" notion of mutually singular signed measures. The motivation for the vocabulary will come in section ref:HERE

Definition 3.2.1: Absolutely Continuous w.r.t. μ

Let ν be a signed measure on $(X, \mathcal{M})$ and μ is a positive measure[a] on $(X, \mathcal{M})$. We say that ν is *absolutely continuous with respect to* μ when if $\mu(E) = 0$ (given $E \in \mathcal{M}$) then $\nu(E) = 0$, and we write is as:

$$\nu \ll \mu$$

[a]this can be extended to a signed measure by taking $\nu \ll \mu$ if and only if $\nu \ll |\mu|$, but we don't need this for this document

I like to remember the direction of the arrow by remembering $\mu(E) = 0 \Rightarrow \nu(E) = 0$, so the arrow is pointing towards ν, so we write the arrow's pointing towards ν: $\nu \ll \mu$. The ν is absolutely continuous with respect to μ, and so it's ν "w.r.t." μ: ν is on the left, the μ on the right. If μ is any measure and 0 is the zero measure, it is always the case that $\mu \ll \mu$ and $0 \ll \mu$. The most important example of an absolutely continuous measures is given by meaures defined via integration:

Definition 3.2.2: Measures Defined By Densities

Let μ be a measure on x and let $f \in L^1(X)$. Then:

$$\nu(E) = \int_E f d\mu$$

is a measure with respect to f.

Clearly $\nu \ll \mu$, since if $\mu(E) = 0$, then $\int_E f d\mu = 0 = \nu(E)$.

It should be verified that $\nu \ll \mu$ if and only if $|\nu| \ll \mu$ if and only if $\nu^+ \ll \mu$ and $\mu^- \ll \mu$. Notice that absolute continuity is sort of the opposite of modular singularity in the sense that if $\nu \perp \mu$ and $\nu \ll \mu$, then it must be that $\nu = 0$. Note that $\mu \perp \nu$ does not imply $\mu(E) = 0 \Rightarrow \nu(E^c) = 0$: take

ν to be the Lebesgue signed measure on $\mathbb{R}$, so that the positive and negative sets are $[0, \infty)$ and $(-\infty, 0)$ and experiment.

The following theorem should give you a taste on why the term "continuity" comes around when ν is finite. In particular, it should give you a feeling of uniformally continuous. Later on, we'll show that absolute continuity is stronger than uniform continuity, and so this feeling should make sense:

Theorem 3.2.3: Epsilon-Delta Absolute Continuity

Let ν be a finite signed measure on $(X, \mathcal{M})$ and μ a positive measure on $(X, \mathcal{M})$. Then $\nu \ll \mu$ if and only if for every $\epsilon > 0$, there exists a $\delta > 0$ such that $\mu(E) < \delta \Rightarrow |\nu(E)| < \epsilon$.

***Proof*:**
Since $\nu \ll \mu$ if and only if $|\nu| \ll \mu$, and moreover $|\nu(E)| \leq |\nu|(E)$, it suffices to show it in the case where $\nu = |\nu|$ is positive.

($\Leftarrow$) Let the ϵ-δ definition hold so that for any $\epsilon > 0$, there exists a $\delta > 0$ such that for all $E \in \mathcal{M}$ that satisfy $\mu(E) < \delta$, then $\nu(E) < \epsilon$. If we pick all E such that $\mu(E) = 0$, then for any ϵ arbitrarily small, any δ will work, and so $\nu(E) < \epsilon$ for all ϵ, and so it must be that $\nu(E) = 0$

($\Rightarrow$) assume the contra-positive; that the ϵ-δ condition is not satisfied, meaning there exists a $\epsilon > 0$ such that for all $\delta > 0$, there is a E_δ such that $\mu(E_\delta) < \delta \Rightarrow \nu(E_\delta) \geq \epsilon$. In particular, for every $n \in \mathbb{N}$, $\mu(E_n) < 2^{-n} \Rightarrow \nu(E_n) \geq \epsilon$. let $F_k = \bigcup_{n=k}^{\infty} E_n$ and $F = \bigcap_k F_k$. Then

$$\mu(F_k) < \sum_{k=n}^{\infty} \frac{1}{2^n} = 2^{1-k}$$

and since $F \subseteq F_k$ for all F_k, it must be that $\mu(F) = 0$. However, by assumpiton $\nu(F_k) \geq \epsilon$ for all k. Since ν is finite, $\nu(F_1)$ is finite, so continuity from above implies

$$\nu(F) = \lim \nu(F_k) \geq \epsilon$$

and so $\nu \not\ll \mu$

An easy way to find a ν that is absolutey continuous with respect to μ is if we have a measure μ and a extended μ-integrable function f and define $\nu(E) = \int_E f \, d\mu$. If $f \in L^1$, then ν is finite, in which case we have the following corrollary:

Corollary 3.2.4: Absolute Continuity And L^1

If $f \in L^1(\mu)$, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\forall E \in \mathcal{M}, \ \mu(E) < \delta \ \Rightarrow \ \left| \int_E f \right| < \epsilon$$

***Proof*:**
apply the previous theorem on Ref and Imf

The definition of ν with respect to $\int_E f d\mu$ is common enough that we define the following notation for it:

Definition 3.2.5: Density Notation

$$d\nu = f d\mu$$

This should give you the feeling of change of variables, and indeed the two concepts will be related. I like to remember this notation by thinking that if we add an integral with respect to E, $\int_E$ to both sides, and imagine there is the constant function in-front of the $d\nu$ on the left hand side, we get back our previous notation of $\nu(E) = \int_E f d\mu$.

We are now in a position to fully classify signed measures given positive measures. The way we will generlize this is to start by remember that an extended μ integrable function f has one part that allow's for an infinite value. Thus, we can think of the domain of f as being split between a signed measure and a positive measure. To avoid pathologies, we will work over σ-finite measures (as usual). We first prove some technical lemma before continuing to the main result:

Lemma 3.2.6

Let μ and ν be finite measures on $(X, \mathcal{M})$. Then either $\mu \perp \nu$, or there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\nu \geq \epsilon\mu$ on E (that is, ν and μ have domain $M_E = \{E \cap F \mid F \in \mathcal{M}\}$, and E is a positive set for $\nu - \epsilon\mu$)

***Proof*:**
Let $X = P_n \coprod N_n$ be the Hahn decomposition for $\nu - n^{-1}\mu$, and let $P = \bigcup_n^\infty P_n$ and $N = \bigcap_n^\infty N_n = P^c$. Then N is a negative set on $\nu - n^{-1}\mu$ for all n a.e., and

$$0 \leq \nu(N) \leq n^{-1}\mu(N)$$

for all n, so $\nu(N) = 0$. If $\mu(P) = 0$, then $\nu \perp \mu$. If $\mu(P) > 0$, then $\mu(P_n) > 0$ for some n, and P_n is a positive set for $\nu - n^{-1}\mu$

Lemma 3.2.7: Sum and Absolute Continuity

Suppose $\{\nu_i\}$ is a sequence of positive meaures. If $\nu_i \perp \mu$ for all i, then $\sum_i^\infty \nu_i \perp \mu$ and if $\nu_i \ll \mu$ for all i then $\sum_i^\infty \nu_i \ll \mu$

***Proof*:**
Let $\nu_i \perp \mu$ for all i, so that $X = A_i \coprod B_i$ where $\nu_i(A_i) = 0$ and $\mu(B_i) = 0$ for all i. Since $\mu(B_i) = 0$ for all i, then $\mu\left(\bigcup_i B_i\right) = 0$. Similarly, $\sum_i^n \nu_i\left(\bigcap_i A_i\right) = 0$ since $\cap_i A_i \subseteq A_i$ for each i. By some basic set theory

$$X = \left(\bigcap_i A_i\right) \coprod \left(\bigcup_i B_i\right) \qquad \left(\bigcap_i A_i\right) \cap \left(\bigcup_i B_i\right) = 0$$

since each x is in either A_i or B_i and $A_i \coprod B_i = X$ for each i. Therefore, we have:

$$\sum_i \nu_i \perp \mu$$

For the second statement, let's say $\nu_i \ll \mu$ for all i. This implies that:

$$\begin{aligned} \mu(E) = 0 &\Rightarrow \nu_1(E) = 0 \\ &\Rightarrow \nu_2(E) = 0 \\ &\vdots \\ &\Rightarrow \nu_n(E) = 0 \\ &\vdots \end{aligned}$$

meaning if $\mu(E) = 0$, then each $\nu_i(E) = 0$. But then $\sum_i \nu_i(E) = 0$, and so

$$\sum_i^\infty \nu_i \ll \mu$$

as we sought to show.

Theorem 3.2.8: Lebesgue-Radon-Nikodym Theorem

Let ν be a signed σ-finite measure on $(X, \mathcal{M})$ and μ be a positive σ-finite measure on $(X, \mathcal{M})$. Then, there exists unique σ-finite measure λ and ρ on $(X, \mathcal{M})$ such that

$$\lambda \perp \mu, \qquad \rho \ll \mu \quad \text{and} \quad \nu = \lambda + \rho$$

Furthermore, there is an extended μ-integrable function f such that $d\rho = f d\mu$, and any two such functions are equal μ-a.e. and so we get:

$$\nu(A) = \int_A f d\mu + \lambda(A)$$

In other words, λ and ρ "split" the measure μ based on where μ is zero or not , and together they can be combined to make ν. Notice that there is no prior relation between the signed measure ν and the positive measure μ. This means that we can replace μ with μ', and find $\lambda' \perp \mu'$ and $\rho' \ll \mu'$, and still get $\nu = \lambda' + \rho'$, and so this decomposition of a signed measure with respect to a measure μ heavily depends on μ. This decomposition might make more sense if we conider the special case where $\nu \ll \mu$, in which case we ar left with $\nu = \rho$. The last statement of the theorem tell's us that:

$$d\rho = f d\mu \qquad \text{which is shorthand for} \qquad \rho(E) = \int_E f d\mu$$

Notice that this looks like the change of varaibles we were talking about earlier. Thus, in this case, this theorem is ultimately about a "change of variables" into the μ measure!

Finally, notice that this "change of variables" show's that the measure ν is sort of the "integral" of the measure of f, and so in a sense f is thus the "derivative" of ν since you "get back" ν by integrating. In other words, this should start giving you Fundamental Theorem of Calculus where $d/dx(\int f)) = f$. However, be weary that the function f is not [always] obtained by taking the limit of some function F, i.e. does does not always exist a function F such that $f(a) = F'(a) = \lim_{x\to a} \frac{F(x+a)-F(a)}{x-a}$. However, in the next section, we will see that in some special cases we *can* define a function F (which we obtain through ν) where $F' = f$. Furthermore, this does not imply that we can compute real measures by defining a distribution F like in the case of Lebesgue measures (which is the other result of the FTC). Thus, the function f can be thought of as a sort of generalization of the notion of derivative.

***Proof*:**
We will break down this proof into 3 cases: when ν and μ are finite positive measures, when ν and μ are σ-finite positive meaures, and finally when ν is a signed measure.

For the first case, let:

$$\mathcal{F} = \left\{ f : X \to [0,\infty] \;\middle|\; \int_E f\,d\mu \le \nu(E) = \int_E 1 d\nu \text{ for all } E \in \mathcal{M} \right\}$$

Remember that we can define a measure through the integral, so what we are asking for is if we can find all possible functions which can define us a measure that will be smaller than ν. We will end up taking the largest such function f, and labelling $d\rho = f d\mu$, and $d\lambda = d\nu - d\rho$. We can already see that $\rho \ll \mu$. What's left to show is that such an f exists, that $\lambda \perp \mu$ and that the result is unique.

Notice also how the construction can make you think of change of variables. Let's say for a moment that $\int_E f d\mu = \int_E 1 d\nu$. Then it is as though f is correcting the value so that the values are equal. This is actually what we are looking for, so we look for all f such that $\int_E f d\mu \le \int_E 1 d\nu$ and take the supremum.

First, $\mathcal{F}$ is nonempty since $0 \in \mathcal{F}$. Thus, let $a = \sup\left\{ \int f\,d\mu \;\middle|\; f \in \mathcal{F}\right\}$ (notice that $a \le \nu(X) < \infty$ since ν is finite), and choose a sequence $\{f_n\} \subseteq \mathcal{F}$ such that $\int f_n d\mu \to a$ and let $f = \sup_n f_n$. We would like to show that $a = \int f$. To that end, notice that if $f, g \in \mathcal{F}$, then $\max(f,g) \in \mathcal{F}$: if $A = \{x \mid f(x) > g(x)\}$, for any $E \in \mathcal{M}$:

$$\int_E \max(f,g) d\mu = \int_{E\cap A} f d\mu + \int_{E\setminus A} g d\mu \le \nu(E\cap A) + \nu(E \setminus A) = \nu(E)$$

Let $g_n = \max(f_1, f_2, ..., f_n)$. Then $g_n \in \mathcal{F}$, g_n increases pointwise to f, and $\int g_n\,d\mu \ge \int f_n\,d\mu$ by monotonicity. Thus, since a is the supremum, it must be that $\lim \int g_n\,d\mu = a$, and so by the MCT

$$a = \lim \int g_n d\mu = \int \lim g_n d\mu = \int f$$

and so $f \in \mathcal{F}$ (i.e. the supremum is "achieved") and $\int f\,d\mu = a$. In particular, $f < \infty$ a.e., so we may take f to be real-valued everywhere. Let $d\rho = f d\mu$ (we will soon show this $d\rho$ satisfies the conditions of the theorem).

We know define the measure λ which is singular with respect to μ. Let $d\lambda = d\nu - f d\mu$ (i.e. $\lambda(E) = \nu(E) - \int_E f d\mu$). Then $d\lambda$ is positive, since $f \in \mathcal{F}$, and is singular with respect to

μ. It it weren't, then by lemma 3.2.6, there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\lambda \geq \epsilon\mu$ on E. Then $\epsilon\chi_E d\mu \leq d\lambda = d\nu = fd\mu$, or rearranging $(f + \epsilon\chi_E)d\mu \leq d\nu$. Since $f \in \mathcal{F}$ and the $\epsilon\chi_E$ only effects the value on E, we just showed that $f + \epsilon\chi_E \in \mathcal{F}$. However, $\int f + \epsilon\chi_E d\mu = a + \epsilon\mu(E) > a$, contradicting the maximality of a. Hence it must be that $\lambda \perp \mu$.

Therfore, we have that $d\nu = fd\mu + d\lambda$ where $\lambda \perp \mu$. Recalling we set $d\rho = fd\mu$, we get that $d\nu = d\rho + d\lambda$, and by definition of ρ we have $\rho \ll \mu$ (see definition 3.2.2). Converting into our more usual notation, we see that $\nu(E) = \rho(E) + \lambda(E)$ for all E, and so $\nu = \rho + \lambda$ with $\lambda \perp \mu$ and $\rho \ll \mu$. It remains to show that this decomposition is unique. To that end, let's say $d\nu = d\lambda' + f'd\mu$. Then $d\lambda - d\lambda' = (f - f')d\mu$. Then (show that $d\lambda - d\lambda' \perp \mu$) and $(f' - f)d\mu \ll d\mu$, and since both sides must be equal, this forces that $d\lambda - d\lambda' = (f' - f)d\mu = 0$. Thus, $\lambda = \lambda'$, and by proposition 2.3.4, $f = f'$ μ-a.e. Thus, if ν and μ are of finite measure, the theorem is done.

Now let ν and μ be σ-finite measures so that X is a countable union of μ-finite sets and a countable union of ν-finite sets. By refining the two by taking intersections, we get a new collection $\{A_i\} \subseteq \mathcal{M}$ $\mu(A_i)$ and $\nu(A_i)$ is finite for all i and $X = \bigcup_i^\infty A_i$. We can define

$$\mu_i(E) = \mu(E \cap A_i) \qquad \nu_i(E) = \nu(E \cap A_i)$$

Then by what we've just shown, we have the decomposition:

$$d\nu_i = d\lambda_i - f_i d\mu$$

where $\lambda_i \perp \mu_i$. Now $\mu_i(A_i^c) = \nu_i(A_i^c) = 0$, so $\int_{A_i^c} fd\mu = 0$ and

$$\lambda_i(A_i^c) = \nu_i(A_i^c) - \int_{A_i^c} fd\mu = 0$$

for our construciton, let f_i be 0 on A_i^c. Now, let $\lambda = \sum_{i=1}^\infty \lambda_i$ and $f = \sum_{i=1}^\infty f_i$. Then by lemma 3.2.7:

$$d\nu = d\lambda - fd\mu$$

and $d\lambda$ and $fd\mu$ are σ-finite, as desired.

Finally, taking the case of ν being a signed measure, using the Jordan decomposition, apply the preceeding argument to ν^+ and $\nu-$, compelting the proof

Since the decomposition exists, it is given a name:

Definition 3.2.9: Lebesgue Decomposition

Let ν be a signed σ-finite measure and μ a positive σ-finite measure. Then the decomposition

$$\nu = \lambda + \rho \qquad \lambda \perp \mu \quad \rho \ll \mu$$

is called the *Lebesgue decomposition of ν with respect to μ*.

if $\nu \ll \mu$, then the Lebesgue-Radon-Nikodym theorem tell's us that $d\nu = fd\mu$. If the extra condition that $\nu \ll \mu$ is added, the theorem is often called *Radon-Nikodym's Theorem*, and f is given a special name:

Definition 3.2.10: Radon-Nikodym Derivative

Let $\nu \ll \mu$ and f be the function obtain from the Lebesgue Decomposition such that $d\nu = f d\mu$. Then f is called the *Radon-Nikodym derivative of ν with respect to μ*. We often denote f by:

$$\frac{d\nu}{d\mu} := f$$

As a consequence of this notation, we get the following representation of $\nu(E) = \rho(E)$ that looks like a fraction cancellation:

$$d\nu = \frac{d\nu}{d\mu} d\nu$$

Strictly speaking, $\frac{d\nu}{d\mu}$ should really be the class of function's that are equal to f μ-a.e., however just like with L^1, we often abuse notation and pick some function in this class when we use this notation). Most of the time, we will be working with the case of $\nu \ll \mu$. In Distribution theory, we will be studying the case where $\nu \not\ll \mu$.

Writing f in this form is very fitting, as in f respects the same properties as the derivative! Once we reach the special case of the measures being Lebesgue measures and the functions being differentiable functions, then f will in fact be the derivative of f.

Without yet doing concrete computations of f, if we take the Radon-Nikodym of $\nu_1 + \nu_2$ where $\nu_1 + \nu_1 \ll \mu$ then we have

$$\frac{d(\nu_1 + \nu_2)}{d\mu} = \frac{d\nu_1}{d\mu} + \frac{d\nu_1}{d\mu}$$

to see this, let $d(\nu_1 + \nu_2)/d\mu = f$. Since $\nu_1 + \nu_2 \ll \mu$, $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$. Then taking $d\nu_1 = f_1 d\mu$ and $d\nu_2 = f_2 d\mu$, then:

$$\begin{aligned}
&= \int \frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} d\mu \\
&= \int \frac{d\nu_1}{d\mu} d\mu + \int \frac{d\nu_2}{d\mu} d\mu \\
&= \int 1 d\nu_1 + \int 1 d\nu_2 \\
&= \nu_1(E) + \nu_2(E) \\
&= (\nu_1 + \nu_2)(E) \\
&= \int 1 d(\nu_1 + \nu_2) \\
&= \int_E \frac{d(\nu_1 + \nu_2)}{d\mu} d\mu
\end{aligned}$$

and since the two integrals are equal over every measurable set, we get:

$$\frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} = \frac{d(\nu_1 + \nu_2)}{d\mu}$$

which is the same property of differentiable functions. In particular, notice that similarly we have the following chain-rule property:

Proposition 3.2.11: Chain Rule For Measures

Let ν be a signed σ-finite measure on $(X, \mathcal{M})$, μ and λ positive σ-finite measures on $(X, \mathcal{M})$ such that $\nu \ll \mu$ and $\mu \ll \lambda$. Then:

1. If $g \in L^1(\nu)$, then $g(d\nu/d\mu) \in L^1(\mu)$ and

$$\int g d\nu = \int g(d\nu/d\mu) d\mu$$

2. Absolute continuity, $\ll$, is transitive: $\nu \ll \lambda$ and

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda}$$

***Proof*:**

1. This proof uses (what now should be) the standard measure-theory way to generalize results. First, we can consider ν^+ and ν^- seperately, so without loss of generality let's say $\nu \geq 0$. If we let $g = \chi_E$, then:

$$\int g\frac{d\nu}{d\mu} d\mu = \int \int \chi_E f d\mu = \int_E f d\mu$$

which is true by definition of f. Thus, it is true for any simple functions since the integral and the construction is linear, and so it is also true for non-negative function's by the MCT, and so it is true for functions in $L^1(\nu)$ by linearity again.

2. If we take ν to be μ and μ to be λ and set $g = \chi_E \frac{d\nu}{d\mu}$, we get:

$$\nu(E) = \int_E \frac{d\nu}{d\mu} d\mu = \int_E \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda} d\lambda$$

for all $E \in \mathcal{M}$, including X, and so by proposition 2.3.4

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda} \quad \lambda\text{-a.e.}$$

as we sought to show

As a consequence, if $\mu \ll \lambda$ and $\lambda \ll \mu$, then $(d\lambda/d\mu)(d\mu/d\lambda) = 1$ λ-a.e. and μ-a.e. Note that all of this is for the special case of $\nu \ll \mu$. If this is not the case (say $\nu \ll \mu$) we get a different notion of "$d\nu/d\mu$ (the example of $\delta_0 \perp \lambda$ has a "Radon-Nikodym derivative" of the dirac delta function).

Note that σ-finiteness is necessary:

Example 3.2.12: Withiout σ-Finiteness

Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$, m the Lebesgue measure and μ the counting measure on $\mathcal{M}$

1. $m \ll \mu$ but $dm \neq f d\mu$ for any f

Proof. First, if $\mu(E) = 0$, then $|E| = 0$, which we have shown that this implies $m(E) = 0$, and so $m \ll \mu$.

However, there does not exist an f such that $m(E) = \int_E f d\mu$. To see this, consider any $p \in [0, 1]$. Then

$$0 = m(\{p\}) = \int_{\{p\}} f d\mu = \int_{[0,1]} \chi_{\{p\}} f d\mu \stackrel{!}{=} f(p)\mu(\{p\}) = f(p)$$

where $\stackrel{!}{=}$ comes from the fact that $\chi_{\{p\}} f$ is in fact a simple funciton (being of the form $\sum_i c_i \chi_{E_i}$ with $i = 1$, $E_i = \{p\}$ and $c_i = f(p)$), and so matches the definition of the integral for simple functions (which we have shown in class is the value of the usual integl). Hence, we have that $f(p) = 0$ for all $p \in [0, 1]$. However, this implies that $f \equiv 0$. This leads to a contardiction, since:

$$1 = m([0, 1]) = \int_{[0,1]} f d\mu = \int_{[0,1]} 0 d\mu = 0$$

but $1 \neq 0$ □

Next, show that μ has no Lebesgue decomposition with respect to m

Proof. For the sake of contradiction, let $\mu = \lambda + \rho$ where $\lambda \perp m$ and $\rho \ll m$. Then for any $c \in [0, 1]$

$$\mu(\{c\}) = \lambda(\{c\}) = 1$$

since $m(\{c\}) = 0$ and $\rho \ll m$, $\rho(\{c\}) = 0$. But then, λ is never null on non-zero sets by monotonicity, implying that m must *always* be null. However, this is not true since m is the Lebesgue measure – a contradiction. Thus, μ does not have such a decomposition with respect to m. □

Before we move on to our last result of the section, I want to take a moment to talk about generalizations of the Radon-Nikodym theorem. In the theorem, we assumed that the measures must be σ-finite. However, in contrast to previous example where without σ-finiteness the theorem fails badly, we seem to have these assumption because the generalization requires more build-up. In particular, to fix the lack of control of σ-finiteness, we would have to define the notion of being *truly continuous*. If μ is σ-finite, being absolutely continuous and truly continuous are equivalent. The statement of the Radon-Nikodym can be reformulated in terms of truly continuous measures, and I found a stack-exchange post on this:

https://mathoverflow.net/questions/258936/radon-nikodym-theorem-for-non-sigma-finite-measures

Finally, we conclude with this final proposition we'll use later on:

Proposition 3.2.13

Let $\mu_1, ..., \mu_n$ be measures on $(X, \mathcal{M})$. Then there exists a measure μ (namely $\sum_i \mu_i$) such that $\mu_i \ll \mu$

***Proof*:**
exercise

3.3 Complex Measure

We know show how we can generalize the results when the codomain is $\mathbb{C}$. The reason to define complex measure is generalize the notion of the Radon-Nikodym derivative to the complex case so that we may define complex results. Most of the work we have done before translates one-to-one for the complex measures, with the exception of defining the absolute variation which we will spend most of this section covering.

Definition 3.3.1: Complex Measure

Let $(X, \mathcal{M})$ be a measure space. A function $\nu : \mathcal{M} \to \mathbb{C}$ is called a *complex measure* if

1. $\nu(\emptyset) = 0$
2. If $\{E_i\} \subseteq \mathcal{M}$ is a pair-wise disjoint sequence, then if $\sum_i^\infty \nu(E_i)$ converges absolutely, then
$$\nu\left(\bigcup_i^\infty E_i\right) = \sum_i^\infty \nu(E_i)$$

Just like for signed measures, the absolute convergence is so that the order in which we measure the sets does not matter.

Notice that the sequence $X, \emptyset, \emptyset, ...$ is a valid disjoint sequence, and so $\sum_i \nu(X_i) = \nu(X)$ must converge absolutely, and so ν must be a finite measure. Thus, a measure is a complex measure if it is finite. Another way of saying it is that if μ is a positive measure and $f \in L^1(\mu)$ (recall f is a complex function), then $f d\mu$ is also a complex measure. As is usually done with complex valued function, we define the signed measures $\nu_r(E) = \pi_r(\nu(E))$ and $\nu_i(E) = \frac{\pi_i(\nu(E))}{i}$ to take on the real and complex values of ν, so ν_i and ν_r are *finite* signed measures and $\nu = \nu_r + i\nu_i$. As a consequence, the range of ν is a bounded subset of $\mathbb{C}$.

Many of the previous concepts we've seen naturally generalize to the complex case:

1. We define $L^1(\nu)$ to be $L^1(\nu_r) \cap L^1(\nu_i)$
2. For $f \in L^1(\nu)$, set $\int f d\nu := \int f d\nu_r + i \int f d\nu_i$
3. If ν and μ are complex measures, then we that $\mu \perp \nu$ if $\nu_a \perp \nu_b$ for all $a, b \in \{r, i\}$.
4. If λ is a positive measure, we say $\nu \ll \lambda$ if $\nu_r \ll \lambda$ and $\nu_i \ll \lambda$
5. The Lebesgue-Radon-Nikodym Theroem generlizes to complex measures nicely; you would apply theorem 3.2.8 to the real and imaginary part seperately. To make sure there is no ambiguity in understanding the generalization:

   Let ν be a complex measure and μ a σ-finite measure on $(X, \mathcal{M})$. Then there exists a complex measure λ and a $f \in L^1(\mu)$ such that
$$d\nu = d\lambda + f d\mu \qquad \lambda \perp \mu$$

Furthermore, if there existed another λ' such that $\lambda' \perp \mu$ and $d\nu = d\lambda' + f'd\mu$, then $\lambda = \lambda'$ μ-a.e. and $f = f'$ μ-a.e.
If $\nu \ll \mu$, then we denote f a $\frac{d\nu}{d\mu}$

We next want to generalize the notion of *total variation* to the complex case. In the signed measure case, this was an easy task since $\mathbb{R}$ has an inherent order. In the complex case, this is not so straight forward: Given that a surface $S \subseteq \mathbb{C}$ is the image of some complex measure $\nu(E) = S$, what does it mean to capture the "total-variation" on that surface? There is more than one sensible answer to this question, namely since there is no "natural order" to $\mathbb{C}$, and so it would be nice to have succint way to capture it's *total variation*. There is in fact a nice and concise way of defining this which also generalizes the previous way of looking at total variation that captures the idea in 1 formula, and so we will build-up to it.

First, let ν be a complex measure and let $\mu = |\nu_r| + |\nu_i|$. Next, notice that $\nu \ll \mu$, since if $\mu(E) = 0$, then it must be that both $|\nu_r|$ and $|\nu_i|$ are zero, and so $\nu_r(E) = \nu_i(E) = 0$, and so $\nu(E) = 0$. Therefore, the Lebesgue decomposition of ν with respect to μ gives us a f such that $d\nu = fd\mu$. As an example of such an f, let's say that ν was simply a finite signed measure (which is a special case of a complex measure), so $\mu = |\nu|$. Then $f = (\chi_P - \chi_N)$ and:

$$d\nu = (\chi_P - \chi_N)d\mu = (\chi_P - \chi_N)d|\nu|$$

Which is the Jordan decomposition theorem we have shown earlier! Now, notice that $|\chi_P - \chi_N| = 1$ so that we can write:

$$d|\nu| = |\chi_P - \chi_N|d\mu \quad \text{which is shorthand for} \quad |\nu|(E) = \int_E |\chi_P - \chi_N|d\mu$$

If we re-write the last equation as $d|\nu| = |f|d\mu$, then we have that the measure $|\nu|$ is defiend by the absolute value of the measure defined by the derivative with respect to $|\nu|$. If we come back to the case of ν being a general complex measure, then we would get $\mu = |\nu_r| + |\nu_i|$, giving us a less obvious, but certainly existent decomposition $d|\nu| = |f|d\mu$ (i.e. $|\nu|(E) = \int_E |f|d\mu$). Thus, it seems that this property somehow "captures" the idea of total variation. We chose a particular μ for both decomposition, however it turns out that this property we have stated is independent of choice of μ up to an appropriate zero-measure set, and so we will define total-variation with respect to the following property:

Definition 3.3.2: Total Variation And Complex Differentiation

Let ν be a complex measure. Then $|\nu|$ is defined such that when $d\nu = fd\mu$ where μ is a positive measure, then $d|\nu| = |f|d\mu$

What we have shown with the μ we saw earlier is that $d\nu$ is always decomposeable into $d\nu = fd\mu$. We must now show that $|\nu|$ is in-fact independent of choice of μ (i.e. any choice will get the same result up zero an appropriate zero-measure set) given μ has the property of the definition. Let's say $d\nu = f_1 d\mu_1 = f_2 d\mu_2$ satisfy the total-variation property. Let $\rho = \mu_1 + \mu_2$. By proposition 3.2.11, we have that:

$$f_1 \frac{d\mu_1}{d\rho} d\rho = d\nu = f_2 \frac{d\mu_2}{d\rho} d\rho$$

Thus $f_1 \frac{d\mu_1}{d\rho} = f_2 \frac{d\mu_2}{d\rho}$ ρ-a.e. Since $\frac{d\mu_1}{d\rho}$ and $\frac{d\mu_2}{d\rho}$ are both non-negative, we have:

$$|f_1| \frac{d\mu_1}{d\rho} d\rho = \left| f_1 \frac{d\mu_1}{d\rho} d\rho \right| = \left| f_2 \frac{d\mu_2}{d\rho} d\rho \right| = |f_2| \frac{d\mu_2}{d\rho} d\rho \quad \rho\text{-a.e.}$$

Thus:

$$|f_1|d\mu_1 = |f_1|\frac{d\mu_1}{d\rho}d\rho = |f_2|\frac{d\mu_2}{d\rho}d\rho = |f_2|d\mu_2$$

and so, ν is independent of choice of μ. We next give some useful properties of total variation:

Proposition 3.3.3: Properties Of Total Variation

Let $(X, \mathcal{M}, \nu)$ be a complex measure space.

1. $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$
2. $\nu \ll |\nu|$ and $\frac{d\nu}{d|\nu|}$ has absolute value 1 $|\nu|$-a.e.
3. $L^1(\nu) = L^1(|\nu|)$ and if $f \in L^1(\nu)$, then $|\int f d\nu| \leq \int |f| d|\nu|$

***Proof*:**
Pick f such that $d\nu = f d\mu$ (as we've shown exists when defining $|\nu|$). Then:

$$|\nu(E)| = \left|\int_E f d\mu\right| \leq \int_E |f| d\mu = |\nu|(E)$$

showing the first claim, and also showing that $\nu \ll |\nu|$. For the next claim, let $g = \frac{d\nu}{d|\nu|}$ so

$$f d\mu = d\nu = g d|\nu|$$

so $g|f| = f$ μ-a.e. and so $|\nu|$-a.e. Since $|f| > 0$ $|\nu|$-a.e. as well, it must be that $g = 1$ $|\nu|$-a.e.
Next, (left as an exercise by Folland)

The next property we introduce shows that it interacts well with the triangle inequality

Proposition 3.3.4: Triangle Inequality Of Total Variation

Let ν_1 and ν_2 be complex measures on $(X, \mathcal{M})$. Then

$$|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$$

***Proof*:**
By proposition 3.2.13, we can write $\nu_1 = f_1 d\mu$ and $\nu_2 = f_2 d\mu$ for the same μ. Then:

$$d|\nu_1 + \nu_2| = |f_1 + f_2| d\mu \leq |f_1| d\mu + |f_2| d\mu = d|\nu_1| + d|\nu_2|$$

as we sought to show

Finally, though this characterization of total variation is a good definition, it can be hard to work with. The following gives an equivalent definition of total variation that is easier computationally:

Proposition 3.3.5: Equivalence Of Total Variation

Let ν be a complex measure on $(X, \mathcal{M})$. If $E \in \mathcal{M}$, define

1. $\mu_1(E) = \sup\{\sum_{i=1}^n |\nu(E_i)| \mid n \in \mathbb{N},\ E_1, ..., E_n \text{ disjiont},\ E = \coprod_{i=1}^n E_i\}$
2. $\mu_2(E) = \sup\{\sum_{i=1}^\infty |\nu(E_i)| \mid E_1, ... \text{ disjiont},\ E = \coprod_{i=1}^\infty E_i\}$
3. $\mu_3(E) = \sup\left\{\left|\int_E f d\nu\right| \mid |f| \leq 1\right\}$

Then $\mu_1 = \mu_2 = \mu_3 = |\nu|$

***Proof*:**
Throughout this proof, let:

$$S_1^E = \left\{\sum_{i=1}^n |\nu(E_i)| \,\middle|\, n \in \mathbb{N},\ E_1, ..., E_n \text{ disjiont},\ E = \coprod_{i=1}^n E_i\right\}$$

and

$$S_2^E = \left\{\sum_{i=1}^\infty |\nu(E_i)| \,\middle|\, E_1, ... \text{ disjiont},\ E = \coprod_{i=1}^\infty E_i\right\}$$

$(\mu_1 \leq \mu_2)$ Notice that $S_1^E \subseteq S_2^E$ (since for any finite disjoint sequence, we can extend it to an infinite disjoint sequence by letting $E_i = \emptyset$ for the rest of the terms), we get $\sup S_1^E \leq \sup S_2^E$, and since this is true for all E, $\mu_1 \leq \mu_2$

$(\mu_2 \leq \mu_3)$ In the process we will use the fact that there exists an f such that:

$$d\nu = f d|\nu|$$

By proposition 3.3.3, $\left|\frac{d\nu}{d|\nu|}\right| = |f| = 1$ $|\nu|$-a,e, and so $|f| \leq 1$ $|\nu|$-a.e. For the $|\nu|$-zero-measure points that are not less than or equal to 1, we can simply choose them to be equal to 0. This also implies that $|\overline{f}| \leq 1$. With this information we get that:

$$\overline{f} d\nu = \overline{f} f d|\nu| = |f|^2 d|\nu| = 1 d|\nu| = d|\nu| \quad |\nu|\text{-a.e.}$$

and so:

$$\begin{aligned}\sum_i^\infty |\nu(E_i)| &\leq \sum_i^\infty |\nu|(E_i) \\ &= \int_E d|\nu| \\ &= \int_E \overline{f} d\nu \\ &\leq \left|\int_E \overline{f} d\nu\right|\end{aligned}$$

and since $|f| \leq 1$, this is an element of $\left\{\left|\int_E f d\nu\right| \mid |f| \leq 1\right\}$, and since this was true for arbitrary E, we get: $\mu_2 \leq \mu_3$

($\mu_3 = |\nu|$) We'll first show $\mu_3 \leq |\nu|$. By the definition of the supremum, for all $\epsilon > 0$, there exists an f where $|f| \leq 1$ such that

$$\begin{aligned}
\mu_3 &\leq \left|\int_E f d\nu\right| + \epsilon \\
&\leq \int_E |f| d|\nu| + \epsilon && \text{prop. 3.13[c]} \\
&\leq \int_E d|\nu| + \epsilon && |f| \leq 1 \\
&\leq |\nu|(E) + \epsilon
\end{aligned}$$

and as $\epsilon \to 0$, we have that $\mu_3 \leq |\nu|(E)$. Going the other way, we can again define $\frac{d\nu}{d|\nu|}$, which by proposition 3.13 its absolute value is 1 $|\nu|$-a.e. Let $\overline{f} = \frac{d\nu}{d|\nu|}$ so that $1 = |f|^2 = f\overline{f}$ $|\nu|$-a.e. If f ever has value greater than 1, we can simply redefine them to be 0 since those value have measure 0. Then:

$$\begin{aligned}
|\nu|(E) &= \int_E d|\nu| \\
&= \int_E 1 d|\nu| \\
&= \int_E |f|^2 d|\nu| \\
&= \int_E |f| d|\nu| && \text{since } |f| = 1 \ |\nu|\text{-a.e.} \\
&= \int_E f\overline{f} d|\nu| \\
&= \int_E f \frac{d\nu}{d|\nu|} d|\nu| \\
&= \int_E f d\nu && \text{prop. 3.9} \\
&\leq \left|\int_E f d\nu\right|
\end{aligned}$$

and since $|f| \leq 1$,

$$\left|\int_E f d\nu|\right| \in \left\{\left|\int_E f d\nu\right| \;\middle|\; |f| \leq 1\right\}$$

and since this is true for arbitrary E:

$$|\nu| \leq \mu_3$$

and so, combining the two inequalities, we get $\mu_3 = |\nu|$

($\mu_3 \leq \mu_1$) We will do as the hint suggets and use simple functions to approximate f, in particular we will use the standard representation from theorem 2.10, let's label the increasing sequence $\{\varphi_k\}_{k=1}^{\infty}$. To take advantage of these functions, first break up ν into positive meaures, $\nu_r^+, \nu_i^+, \nu_r^-, \nu_i^-$ so that:

$$\left|\int_E f d\nu\right| = \left|\int_E f d\nu_r^+ - \int_E d\nu_r^- + i\left(\int_E f d\nu_i^+ - \int_E f d\nu_i^-\right)\right|$$

Then by the definition of integral, for all $\epsilon/4 > 0$, there must exists an $n \in \mathbb{N}$ such that

$$\begin{aligned}
&= \left|\int_E f d\nu_r^+ - \int_E f d\nu_r^- + i\left(\int_E f d\nu_i^+ - \int_e f d\nu_i^-\right)\right| \\
&\leq \left|\int_E \varphi_n d\nu_r^+ + \frac{\epsilon}{4} - \int_E \varphi_n d\nu_r^- - \frac{\epsilon}{4} + i\left(\int_E \varphi_n d\nu_i^+ + \frac{\epsilon}{4} - \int_e \varphi_n d\nu_i^- - \frac{\epsilon}{4}\right)\right| \\
&\leq \left|\int_E \varphi_n d\nu_r^+ - \int_E \varphi_n d\nu_r^- + i\left(\int_E \varphi_n d\nu_i^+ - \int_E \varphi_n d\nu_i^-\right)\right| + \epsilon \\
&= \left|\int_E \varphi_n d\nu\right| + \epsilon
\end{aligned}$$

Then by the definition of a simple function, $\varphi_n = \sum_{i=1}^n c_i \chi_{F_i}$ for appropriate F_i and c_i. Note that $c_i \leq 1$ since $|\varphi_n| \leq |f| \leq 1$. Also, note that $\chi_E \chi_{F_i} = \chi_{E \cap F_i}$. Then by definition of an integral:

$$\left|\int_E \varphi_n d\nu\right| + \epsilon = \left|\sum_{i=1}^n c_i \nu(F_i \cap E)\right| + \epsilon \leq \sum_{i=1}^n |c_i| \left|\nu(F_i \cap E)\right| + \epsilon$$

This is almost the result we need. The problem is that the collection $\{F_i \cap E\}$ is not guaranteed to cover E since there might be portions that are yet to be "precise" enough to bring it into the approximation. Since we only need to get bigger, we can solve this by adding $\bigcap_{i=1}^n F_i^c \cap E$, which is disjoint of every $F_i \cap E$, and so:

$$\begin{aligned}
\sum_{i=1}^n |c_i| \left|\nu(F_i \cap E)\right| + \epsilon &\leq \sum_{i=1}^n |c_i| \left|\nu(F_i \cap E)\right| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon \\
&\leq \sum_{i=1}^n \left|\nu(F_i \cap E)\right| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon
\end{aligned}$$

and so, combining all the inqualities, we get:

$$\left|\int_E f d\nu\right| \leq \sum_{i=1}^n \left|\nu(F_i \cap E)\right| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon$$

and so, by the definition of the supremum:

$$\left|\int_E f d\nu\right| \leq \sup S_1^E$$

and since this is true for arbitrary $\left|\int_E f d\nu\right|$:

$$\mu_3 \leq \mu_1$$

and so, we've established $\mu_1 \leq \mu_2 \leq \mu_3 [= |\nu|] \leq \mu_1$, and so they are all equal, as we sought to show

3.4 Differentiation of Euclidean Space

We now analyze the special case where $\mu = m$ is the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$. Since we are working over the Borel sets of $\mathbb{R}^n$, we have a metric available to us, and so we can define the pointwise derivative of ν with respect to m at x:

Definition 3.4.1: Pointwise Derivative

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, m)$ be our measure space, ν a signed or complex measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$, and $x \in \mathbb{R}^n$. Then if $B_r(x)$ is a ball of radius r around x, then if the following limit exists:

$$\lim_{r\to 0} \frac{\nu(B_r(x))}{m(B_r(x))} = F(x)$$

then we say that $F(x)$ is the *pointwise derivative of ν with respect to m at x*

We can replace balls of radius x with sets that shrink suitably quickly in some "regular" way. This will be advantageous to us later, since it will turn out we can make these "nicely shrinking" sets to be $(x, x+h)$ (or the appropriate generalisation in the $\mathbb{R}^n$ case) which will give us

$$\lim_{r\to 0} \frac{\nu(x+h)}{h}$$

notice that if we define ν in terms of a distribution as we've done for the outer measure in section 1.3, say G is our distribution, then our numerator becomes $G(x+h) - G(x)$, so in this case the definition *exactly* matches be the definition of the derivative in one dimension:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \overset{\text{F.t.C.}}{=} \frac{1}{h}\int_x^{x+h} f'(x)dx$$

We will get back to this important case after some more examination of the general case.

If $\nu \ll m$ so that $d\nu = f dm$, then $\nu(B_r(x))/m(B_r(x)) = \frac{1}{m(B_r(x))}\int_{B_r(x)} f d\mu$ is the average value of f on $B_r(x)$ (recall the 1-dimensional intuition of $\frac{1}{b-a}\int_a^b f(x)dx$). Since $r \to 0$, we can hope that $F = f$ m-a.e. (since the averaging is getting more precise with the smaller range), telling us that we can recover a density F with information about f. This will indeed be the case if $\nu(B_r(x))$ is finite for all r, x; this will be what we explore in this section. Essentially, from the point of view of f, what we're showing is that the derivative of the integral of f is f, i.e., the Fundamental Theorem of Calculus (FTC). For the rest of this section, the term integrable and a.e. is with respect to the Lebesgue measure m.

We start with an important technical lemma:

Lemma 3.4.2: build-up lemma I

Let $\mathcal{C}$ be a collection of open balls in $\mathbb{R}^n$ and let $U = \bigcup_{B\in\mathcal{C}} B$. Then for any $c \in \mathbb{R}$ such that $c < m(U)$, there exists a fininte disjoint subset of $\mathcal{C}$, $B_1, ..., B_k$, such that

$$\sum_{i=1}^k m(B_i) > \frac{c}{3^n}$$

In other words, for any value c less than $m(U)$, we can show that we only need to pick a finite number of balls from $\mathcal{C}$ and still keep the inequality, up to a correction factor of 3^{-n} where n is the dimension of our space, $\mathbb{R}^n$, and 3 comes form making sure the radius of the balls will be big enough to cover what we need.

***Proof*:**
We essentially sneakily get compactness by using theorem 2.6.1[1]. Using this theorem, we know that there exists a compact set $K \subseteq U$ such that $c < \mu(K)$, so there exists a finite subcover of balls from $\mathcal{C}$, say $A = \{A_1, ..., A_m\}$, that cover K. Since balls have finite radius and there are finitely many balls, let B_1 be the ball with largest radius from A. Then let B_2 be the largest ball from A that are disjoint from B_1. Continuing until we have exhausted all possibilites, we get a collection of balls $B = \{B_1, ..., B_k\}$. If there is an A_i that is not in B, then there must be a B_j such that $A_i \cap B_j \neq \emptyset$, and if j is the smallest integer with this property, then by our construction A_i can have radius that is at most that of B_j (or else we would have picked A_i instead of B_j in our construction). Thus, we can take a ball B_j^* whose radius is 3 times that of B_j and have the same center as B_i (i.e., they are concentric) so that $A_i \subseteq B_j^*$. Then by this construction, $K \subseteq \bigcup_j^k B_j^*$, and so

$$c < \mu(K) \leq \sum_j^k m(B_j^*) = 3^n \sum_j^k m(B_j)$$

and so $\frac{c}{c^n} \leq \sum_j^k m(B_j)$, as we sought to show

With this lemma ready to be referenced, we define another important concept that we will use

Definition 3.4.3: Locally Integrable

Let $f : \mathbb{R}^n \to \mathbb{C}$ be a function. Then we say that f is *locally integrable* (with respect to the Lebesgue measure) if:

$$\forall K \in \mathcal{B}_{\mathbb{R}^n}, m(K) < \infty, \int_K |f(x)|dx < \infty$$

and is denoted L^1_{loc}

Definition 3.4.4: Average

Let $f \in L^1_{\text{loc}}$. Then we define the average $A_r f(x)$ on $B_r(x)$ to be

$$A_r f(x) = \frac{1}{B_r(x)} \int_{B_r(x)} f(y)dy$$

This is simply the observation we made in the previous paragraph but made into a definition. We next prove that varying r or x by a bit will only change $A_r f(x)$ by a bit, i.e., $A_r f(x)$ is continuous:

Lemma 3.4.5: Build-Up Lemma II

Let $f \in L^1_{\text{loc}}$. Then $A_r f(x)$ is continuous on both r ($r > 0$) and x ($x \in \mathbb{R}^n$)

Notice that f certainly need not be continuous, while $1/m(B_r(x))$ is clearly continuous. It is the $\int$ operation that is forcing this result to be continuous.

***Proof*:**
First, taking advantage that we're working over the Lebesgue measure in $\mathbb{R}^n$, so that if $c = B_1(0)$, then $cr^n = B_r(x)$ and $m(\partial B_r(x)) = 0$ (where $\partial B_r(x) = \{y \mid \|x - y\| = r\}$). This will come back soon.

To actually prove continuity, we'll show that $A_{\lim_{r\to r_0}} f(x) = \lim_{r\to r_0} A_r f(x)$ and $A_r f(\lim_{x\to x_0}) = \lim_{x\to x_0} A_r(x)$. To that end, take a sequene $r \to r_0$ and $x \to x_0$, and notice that the $\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)}$ pointwise on $\mathbb{R}^n \setminus \partial B_{r_0}(x_0)$ (imagine $x = x_0$ is fixed and the radius goes to r_0 by oscilating around r_0 so that the pointwise limit doesn't exist for the border). Thus, $\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)}$ a.e., and so we have that $\chi_{B_{r_0+1}(x_0)}$ dominates the $\chi_{B_r(x)}$ if $r < r_0 + 1/2$ and $|x - x_0| < 1/2$. Thus, by the Dominated Converting Theorem:

$$\begin{aligned}
A_{\lim\limits_{r\to r_0} r} f(\lim_{x_0\to x} x) &= \int_{B_{\lim\limits_{r\to r_0} r}(\lim\limits_{x\to x_0} x)} f(y)dy \\
&= \int_{B_r(x)} \lim_{r,x} f(y)dy \\
&= \lim_{r,x} \int_{B_r(x)} f(y)dy && \text{DCT}
\end{aligned}$$

and so $\int_{B_r(x)} f(y)dy$ is continuous. Since $B_r(x) = cr^n$ and r^n is a continuous funciton, and the product of two continuous function is continuous, we have that

$$A_r f(x) = \frac{1}{cr^n} \int_{B_r(x)} f(y)dy$$

is also continuous, as we sought to show

Since $A_r f$ is continuous, it is measurable. Next, we can define the largest possible average given a point x:

Definition 3.4.6: Hardy-Littlewood Maximal Function

Let $f \in L^1_{\text{loc}}$. Then the *Hardy-Littlewood Maximal function* is defined as

$$Hf(x) := \sup_{r>0} A_r|f|(x) = \sup_{r>0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y)|dy$$

Note that Hf is measurable since $(Hf)^{-1} = \bigcup_{r>0}(A - r|f|)^{-1}((a, \infty))$ is open for any $a \in \mathbb{R}$ by the previous lemma. What the next theorem shows us is that if we fix some number $\alpha \in \mathbb{R}_{>0}$, and try to take the measure of all points where $Hf(x) > \alpha$ (i.e., we are measuring the area of the largest possible average over an area of points with that larger average), then this value is in fact bounded by $\int |f(x)|$ provided an appropraite constant C/α:

Theorem 3.4.7: The Maximal Theorem

Let $(\mathbb{R}^n, \mathcal{L}, m)$ be a measure space. There exists a constant C such that for all $f \in L^1$ and all $\alpha > 0$

$$m(\{x \mid Hf(x) > \alpha\}) \leq \frac{C}{\alpha}\int |f(x)|dx$$

***Proof*:**
Let $E_\alpha = \{x \mid Hf(x) > \alpha\}$. By suprmum property of $Hf(x)$, there exists a $r_x > 0$ such that

$$\frac{1}{m(B_{r_x}(x))}\int_{B_{r_x}(x)} |f(y)|dy = A_{r_x}|f|(x) > \alpha \quad \Leftrightarrow \quad m(B_{r_x}(x)) < \frac{1}{\alpha}\int_{B_{r_x}(x)} |f(y)|dy \tag{3.1}$$

Clearly, the balls $B_{r_x}(x)$ cover E_α. So for any $c < m(E_\alpha)$, by lemma 3.4.2, there exists a finite collection $r_{x_1}, ..., r_{x_k} \in E_\alpha$ such that

$$\sum_{i=1}^{n} m(B_{r_{x_j}}(x_j)) > \frac{c}{3^n}$$

and so, combining the inequalities we have and moving values around, we get:

$$c < 3^n \sum_{i=1}^{n} m(B_{r_{x_j}}(x_j)) \overset{\text{eq. } 3.1}{\leq} \frac{3^n}{\alpha}\int_{B_{r_{x_j}}} |f(y)|dy \leq \frac{3^n}{\alpha}\int_{\mathbb{R}^n} |f(y)dy|$$

Since this is true for all c, letting $c \to m(E_\alpha)$, we get the desired result, as we sought to show.

With all this build-up, we are ready to show the result we stated at the beginning, that $A_r f(x) \to f(x)$ a.e.

Before stating the theorem, it is useful to define the notion of limit superior for real-valued functions:

Definition 3.4.8: Limit Superior

Let $\varphi : \mathbb{R} \to \mathbb{R}$. Then:

$$\limsup_{r \to R} \varphi(r) := \lim_{\epsilon \to 0} \sup_{0 < |r - R| < \epsilon} \varphi(r) = \inf_{\epsilon > 0} \sup_{0 |r - R| < \epsilon} \varphi(r)$$

As an exercise, show that:

$$\lim_{r \to R} \varphi(r) = c \quad \text{iff} \quad \limsup_{r \to R} |\varphi(r) - c| = 0$$

Theorem 3.4.9: Almost Lebesgue Differentiation Thoerem

If $f \in L^1_{\text{loc}}$. Then
$$\lim_{r \to 0} A_r f(x) = f(x)$$
for a.e. $x \in \mathbb{R}^n$. Phrased another way:
$$\lim_{r \to 0} \frac{1}{mB_r(x)} \int_{B_r(x)} f(x) - f(y) dy = 0$$
for a.e. $x \in \mathbb{R}^n$.

***Proof*:**
First, let's show we can limit our attention to function in L^1. In particular, If $A_r f(x) \to f(x)$ for a.e. x, for any $N \in \mathbb{N}$, we can bound what x's we choose by $|x| \leq N$. Since for any $r \leq 1$, the values $A_r f(x)$ depend only on $f(y)$ for $|y| \leq N+1$, we can replace f with $f\chi_{B_{N+1}(0)}$, so $f \in L^1$

By theorem 2.6.2, for every $\epsilon > 0$, there exists a continuous integrable function g such that $\int |g(y) - f(y)| dy < \epsilon$. Since g is continuous, for every $x \in \mathbb{R}^n$, for all $\delta > 0$, there exists an $r > 0$ such that $|y - x| < r$ implies $|g(y) - g(x)| < \delta$. From this, we get the result to work for g:

$$\begin{aligned}
|A_r g(x) - g(x)| &= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y) dy - g(x) \right| \\
&= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y) dy - \frac{m(B_r(x))}{m(B_r(x))} g(x) \right| \\
&= \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} g(y) - g(x) dy \right| \\
&< \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} \delta dy \right| \\
&= \frac{1}{m(B_r(x))} |m(B_r(x))\delta| \\
&= \delta
\end{aligned}$$

and so, since δ can be anything, we hve that $A_r g(x) \to g(x)$ for a.e. x.

Now we use this result to prove it for f. We will use the Maximal theorem to bound the result. Let:

$$\begin{aligned}
\limsup_{r \to 0} |A_r f(x) - f(x)| &= \limsup_{r \to 0} |A_r f(x) - A_r g(x) + A_r g(x) - g(x) + g(x) - f(x)| \\
&= \limsup_{r \to 0} |A_r (f - g)(x) + (A_r g - g)(x) + (g - f)(x)| \\
&\leq H(f - g)(x) + 0 + |f - g|(x)
\end{aligned}$$

Or, in terms of sets, if

$$E_\alpha = \left\{ x \;\middle|\; \limsup_{r \to 0} |A_r f(x) - f(x)| > \alpha \right\} \qquad F_\alpha = \{x \mid |f - g| > \alpha\}$$

then

$$E_\alpha \subseteq F_{\alpha/2} \cup \{x \mid H(f-g)(x) > \alpha/2\}$$

Furthermore, we have that

$$(\alpha/2)m(F_{\alpha/2}) \leq \int_{F_{\alpha/2}} |f(x) - g(x)|dx < \epsilon$$

Thus, by the maximal theroem:

$$m(E_\alpha) \leq \frac{2\epsilon}{\alpha} + \frac{2C\epsilon}{\alpha}$$

and since ϵ is arbitrary, $m(E_\alpha) = 0$ for all $\alpha > 0$. However, $\lim_{r\to 0} A_r f(x)$ for all $x \notin \bigcup_{i=1}^{\infty} E_{1/n}$, finishing the proof.

In the theorem, we proved it for $f(x)$; it can in fact be shown that it can be proven for $|f|$. To show this, we define the following concept for reference:

Definition 3.4.10: Lebesgue Set

Let f be a measurable function. Then the *Lebesgue set* is :

$$L_f := \left\{ x \;\middle|\; \lim_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(x) - f(y)|dy = 0 \right\}$$

The following theorem shows that the points on which the Lebesgue set fails on is measure 0, and so the result is true for L_f as well

Theorem 3.4.11: Upgrade To Lebesgue Set

Let $f \in L^1_{\text{loc}}$. Then $m((L_f)^c) = 0$

***Proof*:**
This proof will take advantage of the zero measure set in theorem 3.4.9, and that we can form a dense set of $\mathbb{C}$ which will correspond the zero-measure set from this previous theorem, meaning we can make a limit argument that will work for the proof.

First, for each $c \in \mathbb{C}$, we can apply theorem 3.4.9 to $g_c = |f(x) - c|$ to get that:

$$\lim_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - c|dy = |f(x) - c|$$

up to some Lebesgue zero measure set, say N_c. Now, Let D be any countable dense subset of $\mathbb{C}$ and let $N = \bigcup_{c\in D} N_c$ (so $m(N) = 0$). If $x \notin N$, then for any $\epsilon > 0$, we can choose some $c \in D$ such that $|f(x) - c| < \epsilon$. Thus:

$$|f(y) - f(x)| < |f(y) - c| + \epsilon$$

Thus, we have that:

$$\limsup_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - f(x)|dy \leq |f(x) - c| + \epsilon \leq 2\epsilon$$

and since ϵ can be arbitrary small, we have that $\limsup_{r\to 0} \frac{1}{m(B_r(x))}\int_{B_r(x)} |f(y)-f(x)|dy = 0$, and so if $x \notin N$, the theorem applies. But then $0 = m(N) = m(L_f)^c)$, as we sought to show.

We have essentially proven the main results of this section; the last thing we want to do is generalize the types of sets that shrink to x. So far, we have been using balls, however that is not necessary. Thus, we will generalize the types of sets and state the most general version of this "averaging" theorem

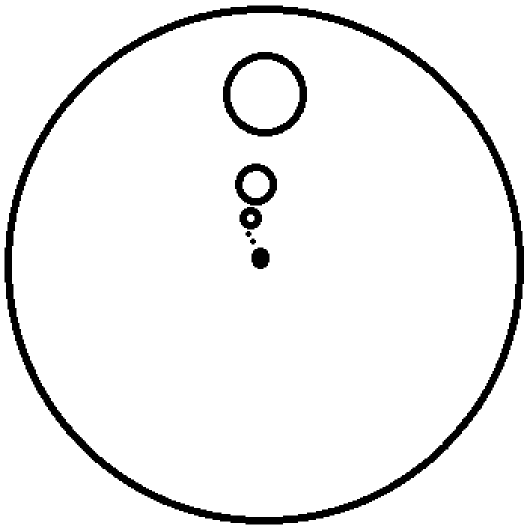
Definition 3.4.12: Sets That Shrink Nicely

A family of set $\{E_r\}_{r>0}$ of Borel subsets of $\mathbb{R}^n$ is said to *shrink nicely* if :

1. $E_r \subseteq B_r(x)$ for each r (so $m(E_r) \leq m(B_r(x))$)
2. There is a constant $\alpha > 0$ that is indepenent of r such that $m(E_r) > \alpha m(B_r(x))$

Example 3.4.13: Nicely Shrinking Sets

1. Let $U \subseteq B_1(0)$ be any borel subset. such that $m(U) > 0$, and consider $E_r = \{x + ry \mid y \in U\}$. You can imagine that U is some circle of $B_1(0)$ and see that, if $x = 0$:



   Then we see that E_r shrinks to to x (even if $x \notin B_1(0)$)
2. in $\mathbb{R}$, the set $E_r = (x, x + r]$ also shrink nicely.

With this, we can finally state the result we've building up to in full generality:

Theorem 3.4.14: The Lebesgue Differentiation Theorem

Let $f \in L^1_{\text{loc}}$. Then for every x in the Lebesgue set, that is almost every x:

$$\lim_{r\to 0} \frac{1}{m(E_r)} \int_{E_r} |f(y) - f(x)| dy = 0$$

and

$$\lim_{r\to 0} \frac{1}{m(E_r)} \int_{E_r} f(y) dy = f(x)$$

for every family of nicely shrinking sets $\{E_r\}_{r>0}$ that shrink nicely to x

Written in terms of measure's this tells us that if $d\nu = f d\mu$, then:

$$\lim_{r\to 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

***Proof*:**
To prove the first equality, we simply use the definition of nicely shrinking sets:

$$\begin{aligned}
\frac{1}{m(E_r)} \int_{E_r} |f(y) - f(x)| dy &\leq \frac{1}{m(E_r)} \int_{B_r(x)} |f(y) - f(x)| dy \\
&\leq \frac{1}{\alpha m(B_r(x))} \int_{E_r} |f(y) - f(x)| dy
\end{aligned}$$

so we can use theorem 3.4.11.

For the second equality, intead of putting absolute value, can start without them, in which case we would do the same think and then use theorem 3.4.9, and move the $f(x)$ to the other side.

3.4.1 Regular Measures

We will apply these results to a special class of measures which, as is common with such definitions, we take some aspects that we like from the Lebesgue measure on $\mathbb{R}^n$, and want to study measure's more generally (in this case, borel measures on $\mathbb{R}^n$) that have the same property:

Definition 3.4.15: Regular Measure

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, \nu)$ be a borel measure space. Then ν is called *regular* if:

1. $\nu(K) < \infty$ for all compact subset $K \subseteq \mathbb{R}^n$
2. $\nu(E) = \inf\{\nu(U) \mid U \text{ open}, E \subseteq U\}$ for every $E \in \mathcal{B}_{\mathbb{R}^n}$

It is in fact provable that the first condition implies the second. We have shown this for $\mathbb{R}$ (i.e. $n = 1$) in theorem 1.4.2 and 1.4.4. In the study of Radon measures, this proof can be done for arbitrary n (Folland chapter 7.2).

Notice that a regular measure is automatically σ-finite, since $\mathbb{R}^n$ is σ-compact. As usual, a signed or complex measure ν is regular if $|\nu|$ is regular.

If $f \in L^+(\mathbb{R}^n)$, then the measure fdm is regular if and only if $f \in L^1_{\text{loc}}$. This means that we can generalize the Lebesgue differentiation theorem to regular measures in general instead of measures defined on densities which are locally integrable. Let's first check that that assertion is indeed the case. First, it should be clear that $f \in L^1_{\text{loc}}$ is equivalent to $\nu(K) < \infty$ for all compact sets. To show 2, we'll first show it works for bounded borel sets E. By theorem 2.6.1, for any $\delta > 0$, there exists an open set $U \supseteq E$ such that $\mu(U) < \mu(E) + \delta$, thus $\mu(U \setminus E) < \delta$. By theorem 3.2.3, for all $\epsilon > 0$, there exists a $U \supseteq E$ such that $\int_{U\setminus E} fdm < \epsilon$, so $\int_U fdm < \int_E fdm + \epsilon$, thus $\nu(U) < \nu(E) + \epsilon$. This completes the requirement of the infimum. For the unbounded case, take $E = \cup_i E_i$ where each E_i is finite, finding it in each case while choosing $\int_{U_i\setminus E_i} fdm < \epsilon 2^{-i}$.

With this equivalence, it becomes worthwile exploring Lebesgue Differentiation Theorem in the case of regular measures:

Theorem 3.4.16: Lebesgue Differentiation With Regular Measure

Let ν be a regular signed or complex Borel measure on $\mathbb{R}^n$, and let $d\nu = d\lambda + fdm$ be its Lebesgue decomposition. Then for m-almost every $x \in \mathbb{R}^n$:

$$\lim_{r\to 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

for every nicely shrinking family $\{E_r\}_{r>0}$ that nicely shrinks to x

***Proof*:**
We have essentially already done this proof, except now we have to consider we are adding $d\lambda$, $d\nu = d\lambda + fdm$, and so we would like to show that $\frac{\lambda(E_r)}{m(E_r)} \to 0$.

First, notice that $d|\nu| = d|\lambda| + |f|d\mu$ (or $|f|dm$?). Thus, regularity in ν implies regularity of λ and fdm (this should be proven, though it is a bit tedious[a]). Since $f \in L^1_{\text{loc}}$, it is enough to show that if $\lambda \perp m$, then for all x, $\lambda(E_r)/m(E_r) \to 0$ m-a.e as $r \to 0$. Without loss of generality, we can let $E_r = B_r(x)$ and assume λ is positive since:

$$\left|\frac{\lambda(E_r)}{m(E_r)}\right| \le \frac{|\lambda|(E_r)}{m(E_r)} \le \frac{|\lambda|(B_r(x))}{m(E_r)} \le \frac{|\lambda|(B_r(x))}{\alpha m(B_r(x))}$$

for some $\alpha > 0$. Now, let's say $\lambda \ge 0$ (if $\lambda = 0$, then we'd be done), and take $A \in \mathcal{B}_{\mathbb{R}^n}$ such that $\lambda(A) = m(A^c) = 0$ (considering that $\lambda \perp m$). We will do the usual trick when it comes to showing that something has measure zero by assuming it doesn't, havign some sets define define in terms of a " $1/k$ sequence", so to speak, and show we reach a contradiction. Consider:

$$F_k = \left\{ x \in A \;\middle|\; \limsup_{r\to 0} \frac{\lambda(B_r(x))}{m(B_r(x))} > \frac{1}{k} \right\}$$

We will show that $m(F_k) = 0$ for all k, meaning the set of points for which $\lambda(E_r)/m(E_r) \to 0$ is 0 m-a.e. First, since λ is regular, for all $\epsilon > 0$, there exists a $U_\epsilon \supseteq A$ such that $\lambda(U_\epsilon) < \epsilon$. Since U_ϵ is open, for each point $x \in U_\epsilon$, there exists a ball such that $B(x) \subseteq U_\epsilon$. By definition of F_k, $\lambda(B(x)) > \frac{m(B(x))}{k}$. or $k\lambda(B(x)) > m(B(x))$. Next, take $V_\delta = \cup_x B(x)$. Then by construction

$F_k \subseteq V_\epsilon$. We'll show that $m(V_\epsilon) \leq 3^n k\epsilon$ for all ϵ, and so $m(F_k) = 0$ or all k. To do this, notice that by assumption there is a constant c such that $c < m(V_\epsilon)$. Thus, by lemma 3.4.2, there exist disjoint balls around points $x_1, ..., x_n$ such that

$$c < 3^n \sum_1^n m(B_{x_i}) \leq 3^n k \sum_1^n \lambda(B_{x_i}) \leq 3^n k\lambda(V_\epsilon) \leq 3^n k\lambda(U_\epsilon) \leq 3^n k\epsilon$$

showing that $m(F_k) \leq m(V_\epsilon) < 3^n k\epsilon$, and since ϵ is arbitrary, we have that $m(F_k) = 0$ for all k, as we sought to show.

[a]It was homework 8, q26

3.5 Functions of Bounded Variation

We will now work towards the Fundamental Theorem of Calculus in its generalized form. This generalization is not in the direction of higher dimensions, as is the case with Stokes Theorem which is a generalization into higher dimensions, but requires much nicer conditions like working over smooth manifolds and differential forms. Rather, we want to stay in $\mathbb{R}$ (or $\mathbb{C}$) and see if we can weaken the conditions (i.e. Weaker than differentiable) and see when we can still apply FTC. Essentially, we want to see if we can define a *density* from probability using the closest notion of a derivative of a function.

In the case of $\mathbb{R}$, we have alraedy explored in section 1.4 that Borel measures defined on $\mathbb{R}$ are defined through an associated distribution F, which is a right-continuous increasing function. In particular, we defined μ_F based on a increasing right-continuous function F (i.e. a distribution) to be $\mu([a, b]) = F(b) - F(a)$. We will use the tools we build up in the last section to finally prove the Fundamental theorem of Calculus in full generality. We will then explore when we can define distributions when we have a complex function, and substitute the notion of increasing function with a function of *bounded variation*.

The first thing we will do is show how increasing functions are a.e. differentiable (which to me is sort of crazy)!!

Proposition 3.5.1: Increasing Then a.e. Differentiable

Let $F : \mathbb{R} \to \mathbb{R}$ be an increasing function. Then the set of points where F is discontinuous is countable.
Furthermore, if $G(x) = \lim_{y\to x^+} F(y) = F(x+)$, then F and G are differentiable a.e. and $F' = G'$ a.e.

***Proof*:**
First, if x is continuous at x, then $F(x+) = F(x-)$, so the interval $(F(x-), F(x+))$ is empty. If it is not empty, then since F is increasing, each $(F(x-), F(x+))$ must be disjoint. So, for any $N \in \mathbb{N}$, for all $|x| \leq N$, we have that $(F(x-), F(x+)) \subseteq (F(-N), F(N))$. Now, here is the trick of this proof: notice that

$$\sum_{|x|<N} [F(x+) - F(x-)] \leq F(N) - F(-N) < \infty$$

since the $F(x-)$ is equal to $F(x_+)$ at appropriate locations, and so the sum cancels out (note that this sum is absolutely continuous since $F(x+) \geq F(x-)$). To see more clearly what's going on ,if we label all the points in $|x| < N$ through an indexing set A, and let $x_\alpha = (F(x_\alpha -) - F(x_\alpha -))$, then:

$$\sum_{\alpha < A} x_\alpha < \infty$$

From here, we use a the fact that this the sum of uncountably many positive x_α, and use the fact that the sum of uncountably many positive elements must be infinite if more than countably elements are non-zero, but since the sum *must* be finite $\{x \in (-N, N) \mid f(x+) \neq F(x-)\}$ is countable. Since this is true for all N, the set of discontinuities of F must be countable, proving the first assertion.

For the second assertion, first note that G is increasing and right continuous and $G = F$ at all points excepts where F is discontinuous. To show G is differentaible a.e, by proposition 1.4.2

$$G(x+h) - G(x) = \begin{cases} \mu_G((x, x+h]) & \text{if } h > 0 \\ \mu_G((x+h, x]) & \text{if } h < 0 \end{cases}$$

Notice that $\{(x-r, x]\}$ and $\{(x, x+r]\}$ shrink nicely to x as $r = |h| \to 0$. Furhtermore μ_G is regual by theorem 1.4.2. Thus, by theorem 3.4.16, we have that:

$$G'(x) = \lim_{r \to 0} \frac{G(x+h) - G(x)}{h} = \lim_{r \to 0} \frac{\mu_G(E_r)}{m(E_r)}$$

exists for a.e. x. The final part of this proof is to show that if $H = G - F$, then H' exists and is equal to zero a.e.

Let $\{x_i\}$ be an enumeration of the points where $H \neq 0$. Since $H = G - F$, $H(x_i) > 0$. Let's say $\delta_i : \mathbb{R} \to \mathbb{R}$ is a function where $\delta_i(x_i) = 1$ and zero eveywehre else. Then for any $N \in \mathbb{N}$, since H is increasing we have $\sum_{\{i \ \mid \ |x_i| < N\}} H(x_i) < \infty$. Thus, we can define a new measure:

$$\mu = \sum_i H(x_i)\delta_i$$

By what we've shown, μ is finite on all compact sets by the previous lines reasoning, and so by the theorem's we've been using μ is regular. Furthermore, $\mu \perp m$ since $m(E) = \mu(E^c) = 0$ where $E = \{x_i\}_{i=1}^\infty$. Thus:

$$\left| \frac{H(x+h) - H(x)}{h} \right| \leq \frac{H(x+h) + H(x)}{|h|} \overset{!}{\leq} \frac{\mu\,(x - 2|h|, x + 2|h|)}{|h|}$$

where $\overset{!}{\leq}$ comes from trying to ingulf $H(x+h)$ and $H(x)$ into an interval.

Since all measure are regular, by theorem 3.4.16 this limit exists and $H' = 0$ a.e., completing the proof.

With this, we now have the tools to try and explore how to think about "increasing function" of the form $F : \mathbb{R} \to \mathbb{C}$. Since $\mathbb{C}$ has no inheret order, we must be more precise on how we define "increasing" and how do we make sure it doesn't explode (just like how when defining the borel measures through a distribution)

To understand the motivation behind the following definitions, imagine the following scenario: let F be the function representing the movement of a particle as it travels accross in time with respect to t (the graph being the position of the particle relative to, say, where it started). The *total variation* from time a to time b of the particle would be the total distance the particle has travelled in that time frame. if F were differentiable, this would simply be:

$$\int_a^b |F'(x)|dx$$

However, what if F is not differentiable? In that case, we will take the following approach: partition $[a, b]$ into $[a, t_1], [t_1, t_2], ..., [t_{n-1}, b]$. Then, linearly approximate F by a straight line from $(t_{i-1}, F(t_{i-1}))$ to $(t_i, F(t_i))$. Then, take the limit over all partitions (in particular, as partitions get finer). If you are worried that such a limit is not well-defined, remember that polynomials can arbitrarily well approximate continuous functions on $\mathbb{R}$, and in most cases F is right-continuous. This heuristic might be more rigorous, but it at least make me feel comfortable with doing this idea

We will do a similar idea for the complex case, except we will take the norm (which we'll also call absolute value) of the distances:

Definition 3.5.2: Total Variation

Let $f : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$. Then define the *total variation of F* to be:

$$T_F(x) = \sup\left\{\sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| \;\middle|\; n \in \mathbb{N},\ -\infty < x_0 < x_1 < \cdots < x_n = x\right\}$$

I should be clear by construction that T_F is an increasing function. Since we put absolute values inside the sum, then as we refine the partition, the sum will get bigger. Therefore, If we have $a < b$ and consider $T_F(b)$, it is harmless to assume a is one of the subdivision points (since we can refine and get a partition with a being part of it). It follows that we have:

$$T_F(b) - T_F(a) = \sup\left\{\sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| \;\middle|\; n \in \mathbb{N},\ a = x_0 < x_1 < \cdots < x_n = x\right\}$$

Using this fact, we can define the following:

Definition 3.5.3: Bounded Variation

Let T_F be the total variation of the function F. Then if:

$$\lim_{t\to\infty} T_F(t) = T_F(\infty) < \infty$$

Then F is said to be of *bounded variation* (on $\mathbb{R}$). We denote the set of of all functions of bounded variation as BV. If we are concerned with all function of bounded variation on an interval $[a, b]$, we will denote this as $BV([a, b])$.

Note that if $F \in BV$, then $F \in BV([a, b])$, since it's total variation on $[a, b]$ is $T_F(b) - T_F(a)$ which is less thatn the total vriation of T_F. If $F \in BV([a, b])$, then we can extend F to be in BV by letting $F(x) = F(a)$ for all $x \leq a$ and $F(b) = F(x)$ for all $b \leq x$. Further note that we rarely actually compute $T_F(b)$, but we use the fact that a function is in BV to get many theoretical results:

Example 3.5.4: Bounded Variation

1. If $F : \mathbb{R} \to \mathbb{R}$ is bounded and increasing, then $F \in BV$ $(T_f(x) = F(x) - F(-\infty))$
2. If $F, G \in BV$ and $a, b \in \mathbb{C}$ then $aF + bG \in BV$
3. If F is differentiable on $\mathbb{R}$ and F' is bounded, then $F \in BV([a, b])$ for $-\infty < a < b < \infty$ (by the mean value theorem)
4. If $F(x) = \sin(x)$, then $F \in BV([a, b])$ for all $-\infty < a < b < \infty$, but $F \in BV$
5. if $F = x \sin\left(\frac{1}{x}\right)$ for $x \neq 0$ and $F(0) = 0$, then $F \notin BV([a, b])$ for $a \leq 0 < b$ and $a < 0 \leq b$

Before we prove stuff about BV functions, let's first prove a lemma:

Lemma 3.5.5: Sum Is Increasing

Let $F \in BV$ be a real valued function. Then $T_F + F$ and $T_F - F$ are increasing functions.

***Proof*:**
Take $x < y$ and $\epsilon > 0$. Since T_F is a supremum, take appropriat $x_0 < \cdots x_n = x$ such that

$$\sum_i^n |F(x_i) - F(x_{i-1})| \geq T_F(x) - \epsilon$$

Next, notice that we can make this into an element of the supremum set of $T_f(y)$ (also known as an approximating sum) by doing

$$\sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)|$$

With this sum, take $F(y) = [F(y) - F(x)] + F(x)$. With some manipulation, we get:

$$T_f(y) \pm F(y) \geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x)$$

Since F is a real valued function, we cna simply take cases to notice that:

+	F(x) - F(y)	+	F(x) - F(y)	2|F(y) - F(x)|
+	F(x) - F(y)	-	F(x) - F(y)	0
-	F(y) - F(x)	+	F(x) - F(y)	0
-	F(y) - F(x)	-	F(x) - F(y)	2|F(y) - F(x)|

$$\begin{aligned} T_f(y) \pm F(y) &\geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x) \\ &\geq T_f(x) - \epsilon \pm F(x) \end{aligned}$$

letting $\epsilon \to 0$, we see that:

$$T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$$

and since x and y was arbitrary, we have establish that $T_F + F$ and $T_F - F$ are both increasing functions, we as sought to show

Theorem 3.5.6: Properties Of BV

1. $T_F \in BV$ is an increasing function
2. $F \in BV$ if and only if $\text{Re}F \in BV$ and $ImF \in BV$
3. If $F : \mathbb{R} \to \mathbb{R}$ then $F \in BV$ if and only if F is the difference of two bounded increasing functions; for $F \in BV$ these functions may be taken to be $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$
4. If $F \in BV$, then $F(x+) = \lim_{y\to x^-} F(y)$ and $F(x-) = \lim_{y\to x^+} F(y)$ exist for all $x \in \mathbb{R}$, as do $F(\pm\infty) = \lim_{y\to\pm\infty} F(y)$
5. If $F \in BV$, the set of points at which F is discontinuous is countable
6. If $F \in BV$ and $G(x) = F(x+)$, then F' and G' are exist and are equal a.e.

***Proof*:**

1. This essentially comes from the definition
2. This is essentially immediate, and so is left as exam review
3. Look at example 3.5.4 to see how the $\Rightarrow$ direction goes. For the $\Leftarrow$ direction, by lemma 3.5.5, the equation $F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$ express F as the difference of two increasing functions. To show boundedness, we showed that $T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$ for $y > x$. Re-arranging and taking advantage of the fact that $F \in BV$, we get:

$$|F(y) - F(x)| \leq T_F(y) - T_F(x) \leq T_F(\infty) - T_F(-\infty) < \infty$$

   showing that F (and hence T_F) is bounded

4,5,6 follow from 1,2,3 along with proposition 3.5.1.

Note that (6) is particularly powerful since we now have that a function simply needs to be of bounded variation to have a derivative a.e. (not necessarily increasing anymore). The converse is false: being differentiable a.e. (even everywhere) does not imply BV (think of $\sin(1/x)$ on $(0, 1]$).

The equation in part 2 are important enough to be given a name:

Definition 3.5.7: Jordan Decomposition

Let $F \in BV$ be a real-valued function. Then the *jordan decomposition* of F is:

$$F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$$

where $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$ are called the *positive variation of F* and *negative variation of F* respectively.

Since $x^+ = \max(x, 0) = \frac{1}{2}(|x| + x)$ and $x^- = \max(-x, 0) = \frac{1}{2}(|x| - x)$ for $x \in \mathbb{R}$, we get that:

$$\frac{1}{2}(T_F \pm F)(x) = \sum \left\{ \sum_i^n [F(x_i)F(x_{i-1})]^\pm \;\middle|\; x_0 < \cdots < x_n = x \right\} \pm \frac{1}{2}F(-\infty)$$

We are almost at a stage to connect borel measures to BV functions. There is one more peice of the puzzle that we need:

Definition 3.5.8: Normalized Bounded Variation (NBV)

Let $F \in BV$ be a function of bounded variation. Then if F is right-continuous and $F(-\infty) = 0$, then F is said to be a function of *normalized bounded variation.* We will denote the collection of such functions as:

$$NBV := \{F \in BV \mid F \text{ is right continuous and } F(-\infty) = 0\}$$

Example 3.5.9: NBV

Let $F \in BV$. Define $G = F(x+) - F(-\infty)$. Then $G \in NBV$, and $F' = G'$ a.e.

First, to show that $G \in BV$, by theorem 3.5.6[2,3], if F is real and $F = F_1 - F_2$ where F_1, F_2 are increasing, then $G(x) = F_1(x+) - [F_2(x+) - F(-\infty)]$, which is the difference of two increasing functions. That $G(-\infty) = 0$ comes by definition. Hence $G \in NBV$.

In fact, every $F \in BV$ almost defines an NBV function through T_F:

Lemma 3.5.10: BV, then T_F almost NBV

If $F \in BV$, then $T_F(-\infty) = 0$. If F is also right continuous, then T_F is also right-continuous (making $T_F \in NBV$)

***Proof*:**
Let $\epsilon > 0$ so that for any x and $x_0 < \cdots < x$ we ahve:

$$\sum F(x_{i+1}) - F(x_i) = T_f(x) - T_f(x_0) > T_F(x) - \epsilon$$

So $T_f(x_0) < \epsilon$ or more generally for any $y \leq x_0$ $T_f(y) < \epsilon$. Since epislon is arbitray, as we let $y \to 0$ we get $T_f(-\infty) = 0$

For the next assertion, suppose F is right-continuous. I'll finish this proof later:

Now suppose that F is right continuous. Given $x \in \mathbb{R}$ and $\epsilon > 0$, let $\alpha = T_F(x+) - T_F(x)$, and choose $\delta > 0$ so that $|F(x+h) - F(x)| < \epsilon$ and $T_F(x+h) - T_F(x+) < \epsilon$ whenever $0 < h < \delta$. For any such h, by (3.24) there exist $x_0 < \cdots < x_n = x + h$ such that

$$\sum_1^n |F(x_j) - F(x_{j-1})| \geq \tfrac{3}{4}[T_F(x+h) - T_F(x)] \geq \tfrac{3}{4}\alpha,$$

and hence

$$\sum_2^n |F(x_j) - F(x_{j-1})| \geq \tfrac{3}{4}\alpha - |F(x_1) - F(x_0)| \geq \tfrac{3}{4}\alpha - \epsilon.$$

Likewise, there exist $x = t_0 < \cdots < t_m = x_1$ such that $\sum_1^n |F(t_j) - F(t_{j-1})| \geq \frac{3}{4}\alpha$, and hence

$$\begin{aligned}\alpha + \epsilon &> T_F(x+h) - T_F(x) \\ &\geq \sum_1^m |F(t_j) - F(t_{j-1})| + \sum_2^n |F(x_j) - F(x_{j-1})| \\ &\geq \tfrac{3}{2}\alpha - \epsilon.\end{aligned}$$

Thus $\alpha < 4\epsilon$, and since ϵ is arbitrary, $\alpha = 0$. ∎

We finally get to establish the connection between complex meaures and functions in BV:

Theorem 3.5.11: Complex Meaures And BV

If μ is a complex Borel meaure on $\mathbb{R}$ (i.e. a finite signed measure) and $F(x) = \mu((-\infty, x])$, then $F \in NBV$. Conversely, if $F \in NBV$, then there is a unique complex Borel measure μ_F such that $F(x) = \mu_F((-\infty, x])$

Proof:

Let μ a complex measure. Then we can decompose it into $\mu = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$ where $\mu_i^\pm$ are positive finite measures. If $F_i^\pm(x) = \mu_i^\pm((-\infty, x])$, then $F_i^\pm$ is increasing, right continuous, $F_i^\pm(-\infty) = 0$, and $F_i^\pm(\infty) = \mu_i^\pm(\mathbb{R}) < \infty$. Thus, by theorem 3.5.6[2,3], the function

$$F = F_1 + -F_1^- + i(F_2^+ - F_2^-) \in NBV \tag{3.2}$$

Conversely, by theorem 3.5.6 and lemma 3.5.10, any $F \in NBV$ can be written in the from of equation (3.2) with the $F_i^\pm$ increasing and in NBV. By Theorem 1.4.2, each $F_i^\pm$ gives a measure $\mu_i^\pm$, so $F(x) = \mu_F((-\infty, x])$ where $\mu_F = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$.

Finally, to show $|\mu_F| = \mu_{T_F}$ (this was left as an outlined exercise 28)

Corollary 3.5.12: Total Variation Of μ_F

Let $F \in NBV$, μ_F be the associated complex measure on $\mathbb{R}$, and $G(x) = |\mu_F|((-\infty, x])$. Then:

$$|\mu_F| = \mu_{T_F}$$

***Proof*:**
We break down the proof in the following steps:

1. From the definition of T_F, $T_F \leq G$

   Proof. We need to show $T_F(x) \leq G(x)$ In the previous homework, we showed that:

   $$|\mu_F|((-\infty, x]) = \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; \coprod_i E_i = (-\infty, x]\right\}$$

   In particular, we can choose are disjoint sets to be h-intervals and get:

   $$\begin{aligned}|\mu_F|((-\infty, x]) &= \sup\left\{\sum_i^\infty |\mu_F((x_i, x_{i+1}])| \;\middle|\; \coprod_i (x_i, x_{i+1}] = (-\infty, x]\right\} \\ &= \sup\left\{\sum_i^\infty |F(x_{i+1}) - F(x_i)| \;\middle|\; \coprod_i (x_i, x_{i+1}] = (-\infty, x]\right\}\end{aligned}$$

   and by definition of T_F

   $$T_F(x) = \sup\left\{\sum_i^N |F(x_{i+1}) - F(x_i)| \;\middle|\; N \in \mathbb{N},\ x_0 < \cdots < x_N = x\right\}$$

   Notice that the set for $|\mu_F|$ contains that of $T_F(x)$ since we can set the rest of the infinite value for T_F to 0, and so $T_F \leq G = |\mu_F|$ □

2. $|\mu_F(E)| \leq \mu_{T_F}(E)$ when E is an interval, and hence E is a borel set.

   Proof. Without loss of generality, we will work with intervals of the form $(a, b]$. We only require these type of intervals for the σ-algebra that will be generatedi n part (c).

   First, since $F \in NBV$, then it is right-continuous, and so by lemma 3.28 T_F is right-continuous, and so by lemma 3.28 again $T_F \in NBV$. Thus, by theorem 3.29, there is an assocaited complex borel meaure such that $T_F(x) = \mu_{T_F}((-\infty, x])$. So $\mu_{T_F}((a, b]) = T_F(b) - T_F(a)$. With that, we proceed with the proof.

   First, the emptyset (the trivial interval) is immediate since $|\mu_F(\emptyset)| = 0 = \mu_{T_F}(\emptyset)$. Next, to show the inequality is true for bounded intervals, notice that:

   $$|\mu_F((a, b])| = |\mu_F((-\infty, b]) + \mu_F((-\infty, a]) = |F(b) - F(a)|$$

Next, recall that:

$$\mu_{T_F}((a,b]) = T_F(b) - T_F(a) = \sup\left\{ \sum_i^n |F(x_{i+1}) - F(x_i)| \;\middle|\; n \in \mathbb{N}\ a = x_0 < \cdots < x_n = b \right\}$$

So, $|F(b) - F(a)|$ is inside the supremum set, and so:

$$= |F(b) - F(a)| \leq T_F(b) - T_F(a) = \mu_{T_F}((a,b])$$

Next, for rays, notice we can split a ray into the countable union of h-intervals:

$$|\mu_F((-\infty, b])| = \left| \sum_{i=0}^{\infty} \mu_F((b-i-1, i-k]) \right| \leq \sum_{i=0}^{\infty} |\mu_F((b-i-1, i-k])|$$

where we can now apply the results for bounded intervals to get that:

$$\leq \sum_{i=0}^{\infty} |\mu_F((b-i-1, i-k])| \leq \sum_{i=0}^{\infty} \mu_{T_F}((a,b]) = \mu_{T_F}((-\infty, b])$$

Thus $|\mu_F((-\infty, b])| \leq \mu_{T_F}((-\infty, b])$, and similarly for rays of the form (a, ∞).

To show it's true for all Borel sets, first recall that h-intervals form an algebra (their disjoint union forms an elementary family, and hence by proposition 1.7). Next, let $\mathcal{C} = \{E \in \mathcal{B}_{\mathbb{R}} \mid |\mu_F(E)| \leq \mu_{T_F}(E)\}$. This set clearly contains all h-intervals by what we've just shown. We'll show that this set is in fact a monotone class, and hence by the monotone class lemma contains all Borel sets.

This is simply a matter of using the appropraite definitions. Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{C}$ be an increasing sequence. Then:

$$\begin{aligned}
\left| \mu_F\left(\bigcup_i^{\infty} E_i \right) \right| &= \left| \lim_{i\to\infty} \mu_F\left(\bigcup_i^{\infty} E_i \right) \right| && \text{definition} \\
&= \lim_{i\to\infty} \left| \mu_F\left(\bigcup_i^{\infty} E_i \right) \right| && \text{limit property} \\
&\leq \lim_{i\to\infty} \mu_{T_F}(E_i) && \text{all } E_i \in \mathcal{C} \\
&= \mu_{T_F}\left(\bigcup_i^{\infty} E_i \right) && \text{definition}
\end{aligned}$$

and hence $\mathcal{C}$ is closed under increasing sequences. A similar argument works for decreasing sequences (in particular since μ_{T_F} is finite, it will hold true for any decreasing sequence). Hence, by the monotone class lemma, $\mathcal{C}$ contains the Borel set, showing that $|\mu_F(E)| \leq \mu_{T_F}(E)$ for all $E \in \mathcal{B}_{\mathbb{R}}$, completing the proof.

□

3. $|\mu_F| \leq \mu_{T_F}$, and hence $G \leq T_F$ (use exercise 21.)

Proof. Recall that

$$|\mu_F|(E) = \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; E_i \in \mathcal{B}_\mathbb{R}, \coprod_i E_i = E\right\}$$

where I modified what's inside the parenthesis to make it more explicit that the sets are inside $\mathcal{B}_\mathbb{R}$. Using what we have just proven:

$$\begin{aligned}
|\mu_F|(E) &= \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; E_i \in \mathcal{B}_\mathbb{R}, \coprod_i E_i = E\right\} \\
&\leq \sup\left\{\sum_i^\infty \mu_{T_F}(E_i) \;\middle|\; E_i \in \mathcal{B}_\mathbb{R}, \coprod_i E_i = E\right\} \\
&\leq \sup\left\{\mu_{T_F}(\coprod_i E_i) \;\middle|\; E_i \in \mathcal{B}_\mathbb{R}, \coprod_i E_i = E\right\} && \text{disjoint} \\
&\leq \sup\left\{\sum_i^\infty \mu_{T_F}(E) \;\middle|\; E_i \in \mathcal{B}_\mathbb{R}, \coprod_i E_i = E\right\} \\
&= \mu_{T_F}(E)
\end{aligned}$$

Thus, $|\mu_F|(E) \leq \mu_{T_F}(E)$. If we take $E = (-\infty, x]$, then we get $|\mu_F|(E) = G(E) \leq \mu_{T_F}(E)F$, completing the proof.

In particular, since we've shown the other direction, we have that $G = \mu_{T_F}$. ☐

With the connection established for between NBV functions and complex measures, it is natural to ask which function in NBV correspond to measure μ such that $\mu \perp m$ or $\mu \ll m$? The following proposition answers that:

Proposition 3.5.13: NBV And Radon-Nikodym with respect to m

If $F \in NBV$, then $F' \in L^1(m)$. Moreover:

1. $\mu_F \perp m$ if and only if $F' = 0$ a.e.
2. $\mu_F \ll m$ if and only if $F(x) = \int_{-\infty}^x F'(t)dt$

***Proof*:**
First, note that μ_F is regular by theorem 1.4.4. Next, observe that:

$$F'(x) = \lim_{r \to 0} \frac{\mu_F(E_r)}{m(E_r)}$$

where $E_r = (x, x+r]$ or $(x-r, x]$ by applying theorem 3.4.16

Since we have now characterized μ_F in terms of F, we can characterize $\mu_F \ll m$ in terms of F:

Definition 3.5.14: Absolute Continuity w.r.t. F

A function $F : \mathbb{R} \to \mathbb{C}$ is said to be *absolutely continuous* if

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that}$$

$$\text{for any set of disjoint intervals } (a_1, b_1), ..., (a_N, b_N),$$

$$\sum_1^N (b_i - a_i) < \delta \Rightarrow \sum_i^N |F(b_i) - F(a_i)| < \epsilon$$

More generally, F is absolutely continuous on $[a, b]$ if all the open intervals lie in $[a, b]$.

Notice that if F is absolutely continuous, then F is automaticaly uniform (pick $N = 1$). If F is everywhere differentiable and F' is bounded, then F is absolutely continuous, since $|F(b_i) - F(a_i)| \leq (\max |F'|)(b_i - a_i)$ by the Mean Value Theorem. More generally, on a compact set, we have that:

$$\text{absolutely continuous } \subseteq \text{ uniformally continuous } \subseteq \text{ continuous}$$

and on a compact interval:

$$\begin{gathered} \text{continosly differentiable} \\ \subseteq \\ \text{Lipchitz continuous} \\ \subseteq \\ \text{absolutely continuous} \\ \subseteq \\ \text{Bounded Variation} \\ \subseteq \\ \text{differentiable a.e.} \end{gathered}$$

where Lipchitz continuous means for all $x, y \in \mathbb{R}$, there exists an M such that $|f(x) - f(y)| \leq M|x - y|$. After going over the FTC for Lebesgue integrals, you can prove as an exercise that Lipchitz continuous implies abslutely continuous. Usually, Lipchitz continuity is used as a simple way of finding absolutely continuous functions.

To make sure this new notion links with the absolute continuity of the measure:

Proposition 3.5.15: Linking Absolute Continuity Of F And Measures

Let $F \in NBV$. Then F is absolutely continuous if and only if $\mu_F \ll m$

***Proof*:**

($\Leftarrow$) Let $\mu_F \ll m$. Then since μ_F is finite, we can apply theorem 3.2.3 to the sets $E = \bigcup_i^N (a_i, b_i]$, and get that F is absolutely continuous

($\Rightarrow$) Let E be a borel set such that $m(E) = 0$. Let $\epsilon > 0$ so that we can choose a δ that satisfies the definition of absolute continuity of F. By theorem 1.4.4, we can find open sets $U \supseteq U_1 \cdots \supseteq E$ such that $m(U_1) < \delta$ (which also implies $m(U_i) < \delta$ by monotonicity) and $\mu_F(U_i) \to \mu_F(E)$. Since each U_i is the countable union of disjoint intervals $(a_{i,j}, b_{i,j})$, for all $N \in \mathbb{N}$:

$$\sum_k^N |\mu_F((a_{i,j}, b_{i,j}))| \leq \sum_k^N |F(b_{i,j}) - F(a_{i,j})| < \epsilon$$

by absolute continuity. Letting $N \to \infty$, we get that $|\mu_F(U_i)| < \epsilon$, and so $|\mu_F(E)| < \epsilon$. Since this is true for all epsilon, it must be that $|\mu_F(E)| = 0$, showing that $\mu_F \ll m$, as we sought to show

With this, we can relate L^1 functions to NBV functions:

Corollary 3.5.16: Relating L^1 And NBV Functions

Let $f \in L^1$. Then the function $F(x) = \int_{-\infty}^x f(t)dt$ is in NBV, is absolutely continuous, and $f = F'$ a.e. Conversely, if $F \in NBV$, is absolutely continuous, then $F' \in L^1(m)$ and $F(x) = \int_{-\infty}^x F'(t)dt$

***Proof*:**
This follows from proposition 3.5.13 and 3.5.15.

If the function is bounded, then we get a slightly stronger result:

Lemma 3.5.17

If F is absolutely continuous on $[a, b]$, then $F \in BV([a, b])$

***Proof*:**
We will find an upper bound for the total variation of F, that is, we'll find an N such that $T_f(b) < N$. Choose $\epsilon = 1$ and let δ be the appropriate choice to satisfy absolute continuity: for any finite set of disjoint intervals (a_i, b_i):

$$\sum(b_i - a_i) < \delta \Rightarrow \sum |F(b_i) - F(a_i)| < \epsilon$$

Choose N to be the greatest integer that is smaller than $\delta^{-1}(b-a)+1$. Then, for any subdivision $a = x_0 < x_1 < \cdots < x_n = b$, we can collect the subintervals (x_i, x_{i+1}) into at most N groups such that the sum of each group is less than δ (adding subdivision's if necessary). Then the sum $\sum |F(x_{i+1}) - F(x_i)|$ over each group is 1, and so the total variation of F on $[a, b]$ is at most N

Is the converse true: If $F \in BV$, then does it imply that F is aboslutely continuous? The answer is no. Even better, F can even be NBV, uniformally continuous, and increasing (so differentiable a.e.), but still not absolutely continuous:

Example 3.5.18: NBV, Continuous, But Not Absolutely Continuous
Let F be the cantor function. Note that F is increasing, $F(-\infty) = 0$, and $T_F(\infty) = 1/2$ (since it's continuous, it is integrable, and integarting give $1/2$). Thus μ_F is defined. However, it is not absolutely continuous with respect to m. However, F is not absolutely continuous (you should prove this).

Theorem 3.5.19: Fundamental Theorem Of Calculus For Lebesgue Integrals

Let $-\infty < a < b < \infty$ and $f : [a, b] \to \mathbb{C}$. Then the following are equivalent:

(a) F is absolutely continuous

(b) $F(x) - F(a) = \int_a^x f(t)dt$ for some $f \in L^1([a, b], m)$

(c) F is differentiable a.e. on $[a, b]$, $F' \in L^1([a, b], m)$ and $F(x) - F(a) = \int_a^x F'(t)dt$

***Proof*:**
First, to show $(a) \Rightarrow (c)$, we can subtract some constant from F so that $F(a) = 0$. Doing the usual extension of setting $F(x) = 0$ for $x < a$ and $F(x) = F(b)$ for $x > b$, then $F \in NBV$. Thus, by lemma 3.5.17 using corrollary 3.5.16. Then, that $(c) \Rightarrow (b)$ comes immetidaly since (c) is a stronger condition than (b). For $(b) \Rightarrow (a)$, if we set $f(t) = 0$ for $t \notin [a, b]$ then $f \in L^1(m)$ and so by corrollary 3.5.16 we get the final result.

If $F \in NBV$, then we usually write an integral of a function g with respect to the associated μ_F as $\int gdF$ or $\int g(x)dF(x)$. These types of integrals are called *Lebesgue-Stieltjes integrals*. I'm not sure why they were introduced, but this theorem was presented for generalizing integration by parts to these types of integrals

Theorem 3.5.20: Integration By Parts For Lebesgue-Stieltjes Integrals

Let $F, G \in NBV$ where at least one of them is continuous. Then for $-\infty < a < b < \infty$

$$\int_{(a,b]} FdG + \int_{(a,b]} GdF = F(b)G(b) - F(a)G(a)$$

***Proof*:**
F and G are linear combinations of increasing functions in NBV (by Theorem 3.27(a,b) in Folland), so a simple calculation shows that it suffices to assume F and G increasing. Suppose for the sake of definiteness that G is continuous, and let $\Omega = \{(x, y) : a < x \leq y \leq b\}$. We use Fubini's theorem to compute $\mu_F \times \mu_G(\Omega)$ in two ways:

$$\begin{aligned}\mu_F \times \mu_G(\Omega) &= \int_{(a,b]} \int_{(a,y]} dF(x)\, dG(x) = \int_{(a,b]} [F(y) - F(a)]\, dG(y)\\ &= \int_{(a,b]} F\, dG - F(a)[G(b) - G(a)],\end{aligned}$$

and since $G(x) = G(x-)$,

$$\mu_F \times \mu_G(\Omega) = \int_{(a,b]} \int_{[x,b]} dG(y)\, dF(x) = \int_{(a,b]} [G(b) - G(x)]\, dF(x)$$
$$= G(b)[F(b) - F(a)] - \int_{(a,b]} G\, dF.$$

Subtracting these two equations, we get the desired result.

Finally, we end this chapter with a quick discussion about different terminology when it comes down to decomposing measures.

Definition 3.5.21: Discrete And Continuous Measures

Let μ be a complex Borel measure on $\mathbb{R}^n$

1. μ is called *discrete* if if there is a coutnable subset $\{x_i\}_i \subseteq \mathbb{R}^n$ such that $\sum_i c_i < \infty$ and $\mu = \sum c_i \delta_{x_i}$ where $\delta_{x_i}(x_i) = 1$ and is zero everywhere else (i.e. the pointmass at x).
2. μ is called *continuous* if for all $x \in \mathbb{R}^n$, $\mu(\{x\}) = 0$

It is easy to see that any measure can be broken down into $\mu = \mu_d + \mu_c$ where μ_d is discrete and μ_c is continuous. The main thing is to insure that μ_d is over a countable set: let $E = \{x \mid \mu(\{x\}) \neq 0\}$. Taking any countable subset $F \subseteq E$, then $\sum_{x \in F} \mu(\{x\})$ converges absolutely to $\mu(F)$ so $\left\{x \in E \;\middle|\; \mu(\{x\}) > k^{-1}\right\}$ is finite, and so it must be that E is countable, so $\mu_d(A) = \mu(A \cap E)$ is discrete and $\mu_c(A) = \mu(A \setminus E)$ is continuous.

If μ is discrete, then $\mu \perp m$, and if μ is continuous then $\mu \ll m$. Thus, we can further break down the decomposition into:

$$\mu = \mu_d + \mu_{ac} + \mu_{sc}$$

where μ_d is the discrete part, μ_{ac} is the absolutely continuous part, and μ_{sc} is the *singularly continuous* part. So far, we have mainly worked with measures with zero singularly continuous parts. There do exist measures with nonzero singularly continuous parts (Folland says the surface measure of the sphere for $n > 1$ is an example). An example for $n = 1$ is rarer, but they do exist. By proposition 3.5.13, we know that we are looking for a corresponding $F \in NBV$ such that $F' = 0$ a.e. In fact, the cantor function is such an example (when extended in the way we keep doing to make sure functions are in NBV). In fact, using the cantor function, we can construct an increasing function F that is continuous, $F' = 0$ a.e., and that is *strictly increasing*:

Example 3.5.22: Interesting Singularly Continuous Function

1. If $\{F_i\}_{i=1}^{\infty}$ is a sequence on nonegative increasing functions on $[a, b]$ such that $F(x) = \sum_1^{\infty} F_i(x) < \infty$ for all $x \in [a, b]$, then $F'(x) = \sum_1^{\infty} F_i'(x)$ for a.e. $x \in [a, b]$ (It suffices to assume $F_i \in NBV$. Consider the measure μ_{F_i})

***Proof*:**
First, notice that since the F_i are nonnegative, by letting $F(x) = 0$ for $x < a$ and $F(b) = F(y)$ for $b < y$, we have that $F_i(-\infty) = 0$. Furthermore, the the function $G(x) = F(x+)$ and $G_i(x) = F_i(x+)$ is equal a.e. with the same derivative wherever it is defined (which it is a.e.), and so if G and G_i has a particular derivative, so do F and F_i. Thus, Without loss of generality, we'll assume $F, F_i \in NBV$.

Since they are in NBV, consider the induced complex borel measure μ_F and μ_{F_i} given by theorem 3.29. Since these measures are finite, they are σ-finite, and so admit a Radon-Nykodym derivative:

$$d\mu_F = f dm + d\lambda \qquad d\mu_{F_i} = f_i dm + d\lambda_i$$

where $f dm \ll m$, $f_i dm \ll m$ and $d\lambda \perp m$, $d\lambda_i \perp m$. We will show that $f dm = \sum f_i dm$. First, we need to show that $\mu_f = \sum \mu_{F_i}$, and then use the previous homework exercise (question 8) to conclude the equality.

By definition of F, $\mu_F((-\infty, b]) = \sum_i^\infty \mu_{F_i}((-\infty, b])$. Since the two measures agree on a generating set (closed rays form a generated set), they agree everywehre, meaning $\mu_F = \sum_i^\infty \mu_{F_i}$. Thus, we have that (using absolute convergence to re-arrange the sum)

$$f dm + d\lambda = d\mu_F = \sum_{i=1}^{\infty} d\mu_{F_i} = \sum f_i dm + \sum \lambda_i$$

Since $(\sum d\lambda_i) \perp m$ and $(\sum f_i dm) \ll m$, by question 8:

$$d\lambda = \sum d\lambda_i \qquad f dm = \sum f_i dm$$

Finally, we can invoke theorem 3.22 to connect the information about f and f_i to derivative information. By letting $E_r = (x, x + r]$ (since the F and F_i's are always positive), we get:

$$\begin{aligned}
F'(x) &= \lim_{r\to 0} \frac{F(x+r) - F(x)}{r} \\
&= \lim_{r\to 0} \frac{\mu_F(E_r)}{m(E_r)} \\
&= f(x) && \text{thm. 3.22} \\
&= \sum f_i(x) && m\text{-a.e.} \\
&= \sum \left(\lim_{r\to 0} \frac{\mu_{F_i}(E_r)}{m(E_r)} \right) && \text{thm 3.22} \\
&= \sum \left(\lim_{r\to 0} \frac{F_i(x+r) - F_i(x)}{r} \right) \\
&= \sum F'(x)
\end{aligned}$$

as we sought to show.

2. Let F denote the cantor function on $[0, 1]$ (see 1.5), and set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Let $\{[a_n, b_n]\}$ be an enumeration of the closed subintervals of $[0, 1]$ with rational endpoints, and let $F_n(x) = F\left(\frac{x-a_n}{b_n-a_n}\right)$. Then $G = \sum_1^\infty 2^{-n} F_n$ is continuous and strictly

increasing on $[0,1]$ and $G' = 0$ a.e. (use exercise 39)

> ***Proof*:**
> We break this proof down in many steps.
>
> First, F is differentiable a.e. Notice that F is increasing and bounded, and so by theorem 3.27(e) F is a.e. differentiable. Notice that we can extend F as $F(x) = F(0)$ $x \leq 0$ and $F(1) = F(y)$ $1 \leq y$. In this way, $F \in NBV$.
>
> Next, $F' = 0$ a.e. To see this, notice that by definition of F, the function is constant on C^c where C is the cantor set. Since the Cantor set is closed, C^c is open. On any open interval of C^c, F is constant, and so $F' = 0$ on C^c.
>
> As a consequence, every F_n has zero derivative almost everywhere. At any point where F is differentiable:
> $$F_n' = F'\left(\frac{x-a_n}{b_n-a_n}\right)\left(\frac{x-a_n}{b_n-a_n}\right)' = 0\left(\frac{x-a_n}{b_n-a_n}\right)' = 0$$
> Showing that F_n' is a.e. 0.
>
> Using this result, we can apply exercise 39 to get $G' = \sum_n 2^{-n} F_n$. Since each $2^{-n}F_n$ is a non-negative increasing function has derivative 0 a.e.:
> $$G' = \sum_{n=1}^{\infty} 2^{-n} F_n' = 0 \qquad \text{a.e.}$$
> Which gives us our first result.
>
> Next, let's prove G is continuous. To see this, let $G_n = \sum_{k=1}^{n} 2^{-k} F_k$. Notice that $2^{-k}F_k$ is continuous being the product of two continuous functions (F_k is continuous being the composition of two continuous functions). By definition of F_k, the largest value if 1, and so $\left|\sum_1^n 2^{-k} F_n\right| \leq \sum_1^n 2^{-k} = 1 - 2^{-n}$. Then we can combine these two pieces of information to show that G_n uniformally converges to G. In particular, for all $\epsilon > 0$, chosoe $n > \frac{1}{2^\epsilon}$ so that $|G - G_n| < \epsilon$ for all $x \in [0,1]$. Thus, G is continuous.
>
> Finally, to show G is strictly increasing, choose $x, y \in [a,b]$ such that $x < y$. Since x is strictly less than y, there exists an ϵ such that $x \notin [y-\epsilon, y]$. Since the rational's are dense, we can choose $[a_n, b_n]$ from our enumeration such that $x < a_n < b_n < y$ so that $x, y \notin [a_n, b_n]$. As a consequence, notice that $G(x)$ must be at least 2^{-n} away from $G(y)$, since:
> $$\sum_n^{\infty} 2^{-n} F_n(x) < \sum_n^{\infty} 2^{-n} F_n(y)$$
> and so G is strictly increasing.

I also want to give a quick word on convexity. I'd like to write more about convex functions, like their definition and some properties, but for now I'll just write down in bullet form what I found interesting:

1. Convex functions on compact intervals are absolutely continuous
2. As a consequence, FTC applies. In fact convex functions can be characterized like so : f is convex if and only if f' is strictly increasing. If f is twice differentiable, this leads to the

common notion of $f''(x) \geq 0$

3. In a sense, linear functions are a sepcial type of convex function. Convex functions are closed under addition and sclara multiplication. Thus, linear function being both convex and concave mean they are closd under subraction, which is why they form a vector-space, but convex functions do not. The inequalities of linear function are also all "trivial", preciesly because of linearity, while those of convex functions are more complex.

Part II

Functional Analysis

The transition from Calculus and Linear Algebra to Functional Analysis marks a fundamental shift in mathematical perspective. In classical analysis, we study individual functions, their derivatives, and their integrals. In linear algebra, we study vectors in finite dimensions. *Functional Analysis* we treat the function f itself as a point and look at transformations between spaces of these functions.

Note the term functional historically meant a function F take takes a function returns a number. We shall show that most linear functional are integrating against a function as we'll show in section 6.2 and chapter 7), and many important exceptions will be shown to still integrating against distributions.

The Historical Imperative: Taming the Infinite

Historically, the field grew out of a crisis in physics and differential equations. By the early 20th century, physicists dealing with Quantum Mechanics and classical wave equations realized that their problems could not be solved by looking at individual coordinates.

- In Quantum Mechanics, the state of a particle is not a position in $\mathbb{R}^3$, but a wave function ψ. The superposition principle—the fact that if ψ_1 and ψ_2 are states, then $\alpha\psi_1 + \beta\psi_2$ is a state—suggests that these wave functions live in a linear vector space.
- In Partial Differential Equations, viewing the heat or wave equation as an operation on an infinite-dimensional space allows us to decouple the "geometry" of the operator from the specific details of the function.

However, unlike the finite-dimensional world of $\mathbb{R}^n$, where linear algebra is rigid and predictable, infinite-dimensional spaces are wild. The intuitions of Euclidean geometry breaks down: bounded sets are no longer compact, linear maps are not necessarily continuous, and the notion of "convergence" splinters into multiple, distinct topologies.

Functional Analysis is the rigorous attempt to tame this infinite wilderness. It builds a hierarchy of structures to recover the tools of calculus and algebra in the infinite setting.

The Three Pillars of the Theory

The narrative of this course is driven by the tension between three mathematical forces:

1. The Algebraic Structure (Linearity): At its core, we are doing linear algebra. We want to solve equations like $Lx = y$. If L is a matrix, we invert it. If L is a differential operator, we want to do the same. The dream of Functional Analysis is to treat differential operators as "infinite matrices" and apply the Spectral Theorem to diagonalize them. When this works (as in Hilbert Spaces), we can solve complex PDEs with the same ease as a 2×2 matrix equation.
2. The Topological Structure (Continuity): Algebra alone is insufficient in infinite dimensions. In $\mathbb{R}^n$, all norms are equivalent, and continuity is automatic for linear maps. In infinite dimensions, this rigidity evaporates. We must introduce *Topology* via norms and metrics to distinguish between operators that are "well-behaved" (bounded) and those that are explosive (unbounded, like differentiation). The interplay between the algebraic dual space (all linear functionals) and the topological dual space (continuous functionals) becomes a central theme of the theory.

3. The Geometric Structure (Distance and Angle): This is where the "Zoo" of spaces emerges.
   (a) *Hilbert Spaces (L^2):* These are "correct" generalization of finite-dimensional euclidean space. They possess an inner product, allowing us to measure angles and define orthogonality.
   (b) *Banach Spaces ($L^p, C(K)$):* They have a notion of size (norm) but lack a notion of angle (i.e. a linear notion of relatedness). The geometry here can be strange – unit balls can be "pointy" or "flat," and the shortest path to a subspace might not be unique.
   (c) *Fréchet and Topological Vector Spaces:* necessary for distributions and generalized functions, where the concept of a "norm" is abandoned for families of semi-norms. They are equivalent to invariant metric spaces

There are more general spaces, but they are beyond an introductory text.

Ultimately, Functional Analysis is the study of *Dualities*. It is the duality between a space and the (linear) functions acting on it. We will see that by lifting concrete problems into these abstract spaces, we gain a new vantage point. The goal of this Part is to map this universe, understanding how the shape of the space dictates the physics of the operators within it.

4

Metric Function Spaces

(This entire article is so enlightening: Intuition on Function Spaces)

Recall in section 2.3 that the space L^1 is a vector-space. As an exercise, it should be shown that it is an infinite dimensional vector-space[1]. Since it is a real vector-space, it is isomorphic to $\mathbb{R}^N$ for appropriate cardinality N. However, N is usually uncountable, finding such N is equivalent to the axiom of choice, and the notion of measure on such a space is not well-defined. Instead, to extract information from a vector space that is an infinite dimensional function space, we assign functionals to extract information. The type of space we work with determines which type of functionals we can assign (inner products, norms, semi-norms, metrics, etc.)

The term *Function Analysis* or *Functional Analysis* is the traditional name for the study of infinite-dimensional vector-spaces over $\mathbb{R}$ or $\mathbb{C}$ and the linear maps between them. When in finite dimension, we have a lot of very nice algebraic results such as: Let V be a finite dimensional vector space,

1. If $W \subseteq V$ and $f \in W^*$, then f can be extended to be a linear functional on V
2. Bijective homomorphisms are isomorphisms
3. Others

What makes functional analysis different form linear algebra over finite dimensional vector space is that in finite dimension, there is only one "resonable" topology that makes linear functions automatically continuous. This is not the case for ininite dimensional spaces (recall the amount of interpretations we saw of how a sequence $\{f_n\}$ of function on $\mathbb{R}$ can convege $f_n \to f$). This chapter will be an introduction to this world.

(Why do we want maps to be continuous? Is it that we saw that many spaces are Banach spaces, and we want to thus apply algebraic results, but they are infinite dimensional meaning many finite-

[1] answer: the functions $\chi_{[n,n+1)}$ are linearly independent for all $n \in \mathbb{N}$

dimensional results don't hold, and so we limit our attention to "continuous" continuous functions to fix this?)

(When does a metric induce a norm: https://math.stackexchange.com/questions/166380/not-every-metric-is-induced-from-a-norm)

(I don't konw if this is the right place to put this, but it is really really really cool. It is bassically the classical duality between algebra and geoemtry. Given a topological space X, we can define an algebra by taking the set of all continuous functions $C(X)$ with the operations being pointwise. Now, given just the information $C(X)$, does there exist a X such that $C(X)$ is isomorphic to A? In other words, can we recover X? The answer is yes if X is a locally compact topological space! This is the algebra/topology duality. The same is true if X is a Riemann manifold, though you need more information, something called the *spectral triple*. Later, Grothendick generlized this and we got to create form any algebra A a topological space Spec(A)!!! Given the appropriate topological properties on Spec(A), we can recover A!! In other words, there is this way to go back and forth between algebra and geometry. This same exists in analysis too: the dual of $C_c(X)$ for a LCH X is intimately related to Radon measure, and the dual of $C_c(X)$ with the appropriate conditions let's us generlize the notion of derivatives to locally integrable functions!!!)

One of the key problems a norm tries to fix is making sure there is still a reasonable "finiteness" to your space. Though a normed vector-space doesn't have an ordering, there is something simlar to it: in the same way the distance between any two real numbers if a real numbers (i.e. finite) the *norm* of any element in a Banach space is *finite*. Thus, though a space might be infinite like L^1, there is still some notion of finiteness allowing us to bring in many of the concepts we have worked with before and avoid the pathologies of the infinite.

(wouldn't mind mentioning how one of the key take-aways for Banach spaces is the Hahn-Banach Theorem, Open mapping theorem, and uniform boundedness principle.)

4.1 Basic Function Spaces

Example 4.1.1: pointwise convergence

1. here

2. Note that it's possible that a poinwise converging sequence of functions "miss" points. Take

$$f_n(x) : (a, b) \to \mathbb{R} \qquad f_n(x) = x^n$$

then $f_n \to 0$, the zero-function, since every point is indeed eventually convering to 0. However, it seems like there will be a point which is sort of "up there". We need a strong notion to fix that

Definition 4.1.2: Uniform Convergence

Let M and N be arbitrary metric spaces. A sequence of function $f_n : M \to N$ *converges uniformally* to the limit $f : M \to N$ if for each $\epsilon > 0$ there is an N such that for all $n \geq N$, and all $x \in M$,

$$|f_n(x) - f(x)| < \epsilon$$

The function f is the *uniform limit* of the sequence (f_n) and we write

$$f_n \rightrightarrows f \qquad \text{or} \qquad \text{unif} \lim_{n\to\infty} f_n = f$$

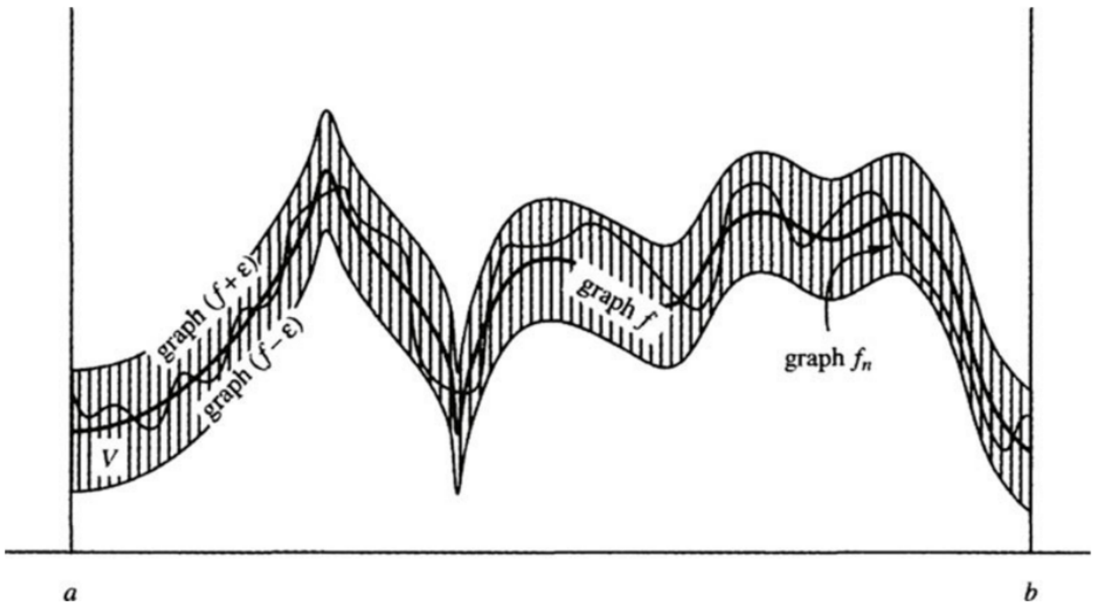


Figure 4.1: uniform convergence visually

Theorem 4.1.3: Uniform Convergence Preserve's Continuity

if $f_n \rightrightarrows f$ and eachf_n is continuous at x_0 then f is continuous at x_0

***Proof*:**
Key is to use the triangle inequality. Details are omitted to be field. Once you choose any $\epsilon > 0$, you know for some $n \geq N$,

$$|f_n(x) - f(x)| < \frac{\epsilon}{3}$$

Then you know for f_N at x_0 there is a $\delta > 0$ st

$$|f_N(x) - f_N(x_0)| < \frac{\epsilon}{3}$$

and then you do the triangle inequality twice, to get split the first equation into 3 equations with f_N being part of it at least once, then use they're all less than $\frac{\epsilon}{3}$, and done

So, with $f_n(x) = x^n$, these function do *not* uniformally converge to 0, but to

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

Note that the converse is not true: if at every point x_0, f_n is continuous and f is continuous at x_0, this does not imply (f_n) is uniformally convergent:

Example 4.1.4: growing steeple

This is called the growing steeple:

$$f_n(x) = \begin{cases} n^2 x & \text{if } 0 \leq x \leq \frac{1}{n} \\ 2n - n^2 x & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{if } \frac{2}{n} \leq x \leq 1 \end{cases}$$

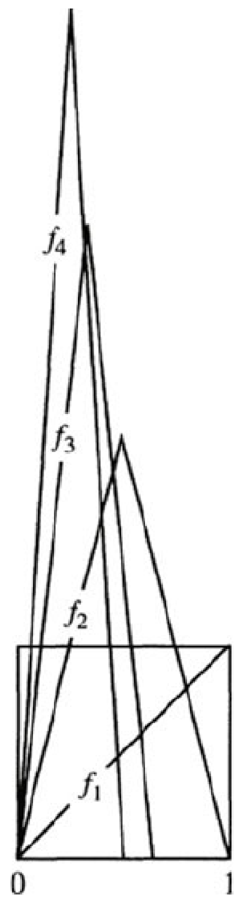


Figure 4.2: growing steeple function

and if you don't like that it goes to infinity, multiply it by $\frac{1}{n}$

Application to function spaces

Definition 4.1.5: Space Of Bounded Functions

Let $C_b([a,b], \mathbb{R})$ represent the set of all bounded functions from $f : [a,b] \to \mathbb{R}$

Then we can define the following norm on this space:

Definition 4.1.6: Sup Norm

Let $f \in C_b$. Then:

$$\|f\| = \sup\{|f(x)||x \in [a,b]\}$$

you should check that this is indeed a norm, and that it will induce a metric as well, and hence a metric topology, and hence the idea of convergence.

In this metric space under this metric, the definition is the same as uniform convergence

Theorem 4.1.7: sup norm convergence if and only if uniformally convergence

Convergence with respect to the sup metric d is equivalent to uniform convergence

***Proof*:**
The two definition's are in fact identical.

Since $d(f_n, f) \to 0$, then for every $x \in [a,b]$ the distnce is becomming smaller, so $|f(x) - f_n(x)|\epsilon$, and so $f_n \rightrightarrows f$.

Theorem 4.1.8: C_b Is A Complete Metric Space

C_b is a complete metric space

***Proof*:**
p.216. If you take any cauchy sequence of functions, then at every point, it's a cuachy sequence in $\mathbb{R}$. use the face that $\mathbb{R}$ is complete while using the sup-norm to complete the proof.

With this, we can show that the space of continuous bounded functions is complete too.

Definition 4.1.9: Continuous Bounded Funcitons

Let $C^0([a,b], \mathbb{R})$ denote the set of continuous functions from $f : [a,b] \to \mathbb{R}$. Note that since the domain is compact, the image is compact, and in $\mathbb{R}$ that implies it's bounded, so $C^0 \subset C_b$.

Corollary 4.1.10: C^0 Is A Compelte Metric Space

C^0 Is A Compelte Metric Space

***Proof*:**
By theorem 4.1.3, every convergent sequence in C^0 lies in C^0, thus C^0 contains it's limits, and thus is closed. so it's a closed subset of a compelte metric space. Thus, by HERE, it's complete

4.2 Application to other concepts

We'll now explore how uniform convergence works with derivatives, integrals, and series

Let

$$F_n = \sum_{i=1}^{n} f_i(x)$$

Then we can say that (F_n) conveges uniformally just like with (f_n) since it's also a sequence of functions.

If we have

$$F_n = \sum_{i=1}^{\infty} f_i(x)$$

Then we can say it converges uniformally if the sequence of partial sums converges uniformally.

Also, we can say the series of absolutes values $\sum |f_k(x)|$ converges then the series $\sum f_k$ *converges absolutely*

Theorem 4.2.1: Weierstrass M-Test

If $\sum M_k$ is a convergetn sereis of constants and if $f_k \in C_b$ satisfies $\|f_k\| \leq M_k$ for all k then $\sum f_k$ converges uniformly and absolutely

***Proof*:**
For n large enough, we can use the triangle enquality on the partial sums to expand out our series:

$$\begin{aligned} d(F_n, F_m) &\leq d(F_n, F_{n-1}) + \cdots + d(F_{m+1}, F_m) \\ &= \sum k = m + 1^n \|f_k\| \leq \sum k = m + 1^n M_k \end{aligned}$$

and thus, since $\sum M_k$ converges, then we can always use it's epsilon for large enough m, n. This shows that (F_n) is cauchy, and so since C_b is compelte, it converges uniformly.

Theorem 4.2.2: Uniform Limits Through Integrals

The uniform limit of Riemann integrable functiosn is Riemann integrable, and hte limit of hte integrals is the integral of hte limit

$$\lim_{n\to\infty} \int_a^b f_n(x)dx = \int_a^b \text{uniflim}_{n\to\infty} f_n(x)dx$$

***Proof*:**
p.218

Note that since all bounded contiuous function are integrable, but not all integrable bounded functions are continuous, this means that the subset of C_b of functions for which this theorem applies is between

C_b and C_0 (since not all bounded functions are integrable. If $\mathcal{R}$ represent all integrable funcitons:

$$C_b \supset \mathcal{R} \supset C^0$$

Now, using the fact that a limit can pass through integrable functions, we can show that infnite sums of of integrated functions is the sme as the integral of the infinite sum of functions:

Theorem 4.2.3: Term By Term Integration Theorem

A uniformly convergetn series of integrable functions $\sum f - k$ can be integrated term-by-term in the sense that

$$\int_a^b \sum_{k=0}^{\infty} f_k(x)dx = \sum_{k=0}^{\infty} \int_a^b f_k(x)dx$$

***Proof*:**
Use the definition, then apply theorem 6.

Now, we move on to differentiable functions

Theorem 4.2.4: Uniform Limit Of Differentiable Functions Is A Differentiable Function

The uniform limit of a sequecne of idfferentiable functiosn is a differentiable function provided that the sequence of derivaties also converges uniformally

***Proof*:**
p.219-220

Assuming f' is continuous makes the proof easy enough using the fundamental theorem of calculus by splitting up $f_n(x) = f_n(a) + \int_a^x f_n'(t)dt$.

For f' not continuous, more to check. Check the book, it's not too bad to follow

This also looks cool: here

Note that this can fail if the derivative doesn't converge uniformally

Example 4.2.5: derivative not converge uniformally
Take

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}$$

Then this series converges to $f(x) = |x|$, which is not differentiable! The derivatiev converges point-wise, but not uniformally.

Note that in complex analysis, this will not be the case! the uniform limit of a complex differentiable function is complex differentiable, and the sequence of derivatives will automatically be unifomrally convergent to a limit!

LIke before, infinite sums work well

Corollary 4.2.6: Series Of Infinite Sum And Uniform Convergence

A uniformly convergent sereis of differentaible functiosn can be differentiated term-by-term, provided that the derivative series converges uniformly,

$$\left(\sum_{k=0}^{\infty} f_k(x)\right)' = \sum_{k=0}^{\infty} f_k'(x)$$

***Proof*:**
Using the definition of convergence for infinite series, apply theroem 4.2.4 on the partial sums.

4.3 Compactness and equicontinuity

We now want to apply the Heine-Borel theorem to C^0, because checking something is closed and bounded is relatively easy, and for some reason, we want to able to easily find convergent-subsequence of convergent sequences (will come back here once I figure out why TBD)

In C^0, closed and bounded sets are usually not compact. Take

$$B = \left\{ f \in C^0([0,1], \mathbb{R}) | \, \|f\| \leq 1 \right\}$$

that is, teh set of all functions who's supremum over $[0,1]$ is 1. This set is not closed: it contains convergent sequences that do not lie inside B, however the value it converges to is *not* continuous

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}$$

This comes down to the fact that $f(x)$ is infinite dimensional: Heine-Borel will applies to finite-dimensional spaces!

To show that C^0 is an infinite dimensional vector-space over $\mathbb{R}$, I'll show that it has an infinite dimensional linearly independent set (I'll not be showing it's a basis here, but since a set exists, the basis must at least have this cardinatliy) Take the set of "smooth bump" funtions, where f_0 start it's bump at 0 and ends at 1. Then $\{f_n | n \in \mathbb{Z}\}$ is a linearly independent set of continuous (even smooth) functions. Thus, the space must at least have a countably infinite dimension.

Now back to finding the appropriate conditions for the Heine-Borel property. To do this, we need a new condition on our functions:

Definition 4.3.1: [Uniform] Equicontinuous

A sequence of functions (f_n) in C^0 is **equicontinuous** if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s - t| < \delta, \forall n \in \mathbb{N} \Rightarrow |f_n(s) - f_n(t)| < \epsilon$$

The functions f_n are *equally continuous.*

Note that the δ depends on teh ϵ, but *not* on the n I put [uniform] in the definition because there are definition of equicontinuity which are not "uniform", that is, you pick a point for which the conditions applies:

Definition 4.3.2: Point-Wise Equicontinuous

A sequence (f_n) in C^0 is *pointwise equicontintuous* if

$$\forall \epsilon > 0 \forall x \in [a,b], \exists \delta > 0 \text{ such that } |x-t| < \delta, n \in \mathbb{N}, |f_n(x) - f_n(t)| < \epsilon$$

From here on out, we will say equicontinuity instead of uniform equicontinuity.

Example 4.3.3: Example

1. The set of function $f : \mathbb{R} \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$ are equicontinuous. However, they are not uniformally continuous.
2. Take $f : [0,1] \to \mathbb{R}$, $f_n(x) = n$. That is, a constant function. Then $\{f_n\}$ is an equicontinuous sequence. However $\{f_n\}$ is not a convergent sequence (and so not uniformally convergent).
3. let $\{f\}$ be a equicontinuous set with a single function. Then f is uniformally continuous by definition. In this sense, equicontinuity is intimately related to uniform continuity.

<++>

This definition can also be used to define an equicontinuous set:

Definition 4.3.4: Equicontinuous Set

The set $\mathcal{E} \subset C^0$ is equicontinuous if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s-t| < \delta \text{ and } f \in \mathcal{E} \Rightarrow |f_(s) - f(t)| < \epsilon$$

Essentially, if a function is continuous, than for all $\epsilon > 0$, there exists an open neighborhood U such that $d(f(x), f(y)) < \epsilon$ for all $y \in U$. Equicontinuity means that a single neighborhood U can be chosen that will work for all the functions f in the collection $\mathcal{E}$.

You might recall BolzanoWeierstrass theorem:

Theorem 4.3.5: BolzanoWeierstrass Theorem

Let $M = \mathbb{R}^n$. Then every bounded sequence has a convergent subsequence.

***Proof*:**
Here is the sketch of the proof.

Since it's bounded, put it in a ball, and take the closure of the ball. Then it's closed. So it's compact. Take the cover: $B_1 = \{B_1(x) | x \in \mathbb{R}^n\}$. Then it has a finite sub-cover B_1'. Then one of the ball must have infinitely many points. Let's say $b_1' \in B_1'$. Then cover this ball with $B_{1/2}(x)$ balls, and repeat the same process. Keep doing this for every n. Then you can pick a sub-sequence from each $B_{1/n}(x)$ which will converge.

We'll be proving the analogous of this with equicontinuous sequences in C^0:

Theorem 4.3.6: Arzela-Ascoli Theorem

Let M be a compact metric space. Every uniformally bounded equicontinuous sequence of functions in $C^0(M, \mathbb{R})$ has a uniformly convergent subsequence

***Proof*:**
p.226. Fun to follow!

Notice how this relates to compactness: if every subsequence of f_n converges, then the closure of $\{f_n | n \in \mathbb{N}\}$ is compact!

The last part of the proof also shows that you can weaken the condition a bit, and say that if you converge on a dense subset, then you're uniformally convergent.

Corollary 4.3.7: Arzela-Ascoli Propagation Theorem

Pointwise convergence of an equicontinuous sequence of functions on a dense subset of the domain *propagates* to uniform convergence on the whole domain

***Proof*:**
last part of the proof for Ascoli

Another thing that I don't konw where to put, but I think is neat to consider:

1. Let's say that you have a sequence of function where $f_n \to f$ pointwise. you also have sequences of functions $f_{n,m} \to f_n$, that is, every f_n has a sequnce of function's converging to it. Then is there a subseuqence of $f_{n,m}$ (for varaible n, m) that converges to f?

<++>

If you're derivative is bounded, we cna actually prove this with a much weaker claim: only 1 point must be convergent for the whole thing to be uniformally continuous!!

Corollary 4.3.8: Differentiable Convergent Subsequence

Assume that $f_n : [a, b] \to \mathbb{R}$ is a sequence of differentiable functions whose derivatives are uniformly bounded. If for one point x_0, the sequence $(f_n(x_0))$ is bounded as $n \to \infty$, then the sequence (f_n) has a subsequence that converges uniformly on the whole interval $[a, b]$.

***Proof*:**
Let M be the bound of $|f_n'(x)|$ for all $n \in \mathbb{N}$ and $x \in [a, b]$. Equicontinuity of (f_n) then follows from the MVT:

$$|s - t| < \delta \Rightarrow |f_n(s) - f_n(t) = |f_n'(\alpha)||s - t| \leq M\delta$$

and then set $\delta = \frac{\epsilon}{(M+1)}$. THis shows that (f_n) is equicontinuous!

Now to show that the sequence is uniformally bounded, let C be a bound for $|f_n(x_0)|$ for all $n \in \mathbb{N}$. THen

$$\begin{aligned} |f_n(x) \leq |f_n(x) - f_n(x_0)| + f_n(x_0)| &\leq M|x - x_0| + C \\ &\leq M|b - a| + C \end{aligned}$$

showing that the sequence (f_n) is bouded in C^0! Since it's uniformally bounded and equicontinuous, by Arzela-Ascoli's theorem (4.3.6), $f(f_n)$ has a uniformally convergent subsequence!

As a quick asside, this theorem has two uses in other domain we will not discuss;

1. A sequecne of solutions to a continuous ordinary differential equation in $\mathbb{R}^M$ has a subsequence that converges to a limit, and that limit is also a solutionof the ODE
2. A sequence of complex analytic functions that converges pointwise, converges uniformly (on compact subsets of the domain of definition) and the limit is complex analytic.

However, we'll be using Arzela-Ascoli for Heine-Borel:

Theorem 4.3.9: Heine-Borel Theroem In A Function Space

A subset $\mathcal{E} \subseteq C^0$ is compact if and only if it is closed, bounded, and equicontinuous

***Proof*:**

($\Rightarrow$) Since C^0 is a complete metric space, it is totally bounded (it is automatically closed since C^0 is clopen). This means for some $\epsilon > 0$, ew can cover $\mathcal{E}$ with finitely many balls of size $\epsilon/3$

$$\mathcal{C} = \left\{ B_{\epsilon/3}(f_k) | \text{appropriate } f_k \right\}$$

Note that each f_k is uniformally continuous since the domain is compact. Thus, there exists a $\delta > 0$ st

$$|s - t| < \delta \Rightarrow |f_k(s) f_k(t)| < \frac{\epsilon}{3}$$

Now, if $f \in \mathcal{E}$, the for some k we hvae $f \in B_{\epsilon/3}(f_k)$, and $|s - t| < \delta$ will give us

$$|f(s) - f(t)| \leq |f(s) - f_k(s)| + (f_k(s) - f_k(t)| + |f_k(t) - f(t)| < 3\frac{\epsilon}{3} = \epsilon$$

showing $\mathcal{E}$ is equicontinuous.

($\Leftarrow$) Assume $\mathcal{E}$ is closed, bounded, and equicontinuous. Then if (f_n) is a sequence in $\mathcal{E}$, by Arzela-Ascoli's theorem, some subsequence converges uniformly to a limit. This limit lies in $\mathcal{E}$ since $\mathcal{E}$ is clsoed. Thus, by ref:HERE, $\mathcal{E}$ is compact

4.4 Uniform Approximation in C^0

We now move onto a really interesting part of analysis: Given a continuous but nondifferentiable function f (so, any function in C^0), we we can "arbitrarily" approximate f with a smooth function

g! Even better, this smooth function will be a polynomial!!

In order for this to be possible, it would mean that for any $f \in C^0$, there is a polynomial close to it. That is what we're about to prove:

Theorem 4.4.1: Weierstrass Approximation Theorem

The set of polynomials is dense in $C^0([a, b], \mathbb{R})$

Note that since the function is continuous on a compact set, it is uniformally continuous (a fact we'll use in the 2nd proof).

Decomposing what density means for a moment: for each $f \in C^0$, and each $\epsilon > 0$, there is a polynomial function $p(x)$ such that for all $x \in [a, b]$

$$|f(x) - p(x)| < \epsilon$$

meaning that we can arbitrarily approximate $f(x)$!!

***Proof*:**
#1 The key idea to building $p(x)$ from f is to sample values of f and recombining them in a clever way.

Without loss of generality, let's say $[a, b] = [0, 1]$. For each $n \in \mathbb{N}$, consider

$$p_n(x) = \sum_{k=0}^{n} \binom{n}{k} c_k x^k (1 - x)^{n-k}$$

where $c = f\left(\frac{k}{n}\right)$ and $\binom{n}{k}$ is the binomial coefficient. This polynomial is called the **Bernstein polynomial**. We'll show that (p_n) converges uniformally to f as $n \to \infty$, i.e., $(p_n) \rightrightarrows f$.

***Proof*:**
#2 This proof uses convolution. to link the nice properties of polynomials with the less nice propertries of general continuous functions.

Without loss of generality, let's let $[a, b] = [0, 1]$ (they are homeomorphic). Let $f \in C^0([0, 1], \mathbb{R})$. What we'll do is manipulate this functions to a nicer homeomorphic equivalent, and approximate that function with a polynomial.

Let $g(x) = f(x) - (mx + b)$, where

$$m = \frac{f(1) - f(0)}{1} = f(1) - f(0) \qquad \text{and} \qquad b = f(0)$$

Since we're adding a polynomial to $f(x)$, $g \in C^0$. Notice that

$$g(0) = f(0) - (f(1) - f(0))\,0 - f(0) = 0 \qquad g(1) = f(1) - f(1) + f(0) - f(0) = 0$$

so $g(0) = 0 = g(1)$. This shows that given any f, it is equivalent to consider f in the "form" of g (the two being homeomorphic), so let $f(0) = 0 = f(1)$. In our construction, we'll sometimes need the domain to extend beyond $[0, 1]$ for technical reasons, so let's say $f(x) = 0$ for $x \in \mathbb{R} - [0, 1]$ (this is a smooth extension with zero measure, so nothing bad can come from this).

Next, we will construct the nice function we'll convolute with. Let

$$\beta_n(t) = b_n(1-t^2)^n \qquad -1 \leq t \leq 1$$

where the constant b_n is chosen so that $\int_{-1}^{1} b_n(t)dt = 1$. For $0 \leq x \leq 1$, set

$$P_n(x) = \int_{-1}^{1} f(x+t)\beta_n(t)dt$$

This is a weighted average of the value of f using the weight function β_n. This is where we use convlution to get the nice properties of $\beta_n(t)$ to f. We claim that P_n is a polynomial, and $P_n(x) \rightrightarrows f(x)$ as $n \to \infty$.

To check that P_n is a polynomial, we use a change of variables: $u = x + t$. Then for $0 \leq u \leq 1$ we have

$$P_n(x) = \int_{x-1}^{x+1} f(u)\beta_n(u-x))du = \int_{0}^{1} f(u)\beta_n(u-x)du$$

where the last integral is because $f = 0$ outside $[0,1]$. The function $\beta_n(u-x) = b_n(1-(u-x^2))^n$ is a polynomial in x whose coefficients are polynomials in u. We can pull out all the x terms, since the integral is with respect to x, and since the integral is definite with concrete endpoints, it is a constant, and so we have a polynomial!! In other words, we managed to "eliminate" the complexity of f.

To check that $P_n \rightrightarrows f$ as $n \to \infty$, we need to estimate $\beta_n(t)$. We claim that if $\delta > 0$ then

$$\beta_n(t) \rightrightarrows 0 \text{ as } n \to \infty \text{ and } \delta \leq |t| \leq 1$$

Notice that $\delta > 0$, so $0 \leq |t| \leq 1$, and so we are ignoring the fact that $x = 0$ diverges.

You can prove this directly, but I'll skip it for now (p.233)

Now, we'll deduce $P_n \rightrightarrows f$. Let $\epsilon > 0$. Since f is uniformally continuous, let $\delta > 0$ such that $|t| < \delta$ imples $f(x+t) - f(x)| < \frac{\epsilon}{2}$. Since β_n has integral 1 on $[-1,1]$, we can slip in β_n in the following equality:

$$\begin{aligned}
|P_n(x) - f(x)| &= \left| \int_{-1}^{1} (f(x+t) - f(x))\, \beta_n(t)dt \right| \\
&\leq \int_{-1}^{1} |f(x+t) - f(x)|\beta_n(t)dt \\
&= \int_{|t|<\delta} |f(x+t) - f(x)|\beta_n(t)dt + \int_{|t|\geq\delta} |f(x+t) - f(x)|\beta_n(t)dt \\
&< \epsilon/2 + 2M \int_{|t|\geq\delta} B_n(t)dt
\end{aligned}$$

where the 1st integral is less than $\frac{\epsilon}{2}$ becuase HERE, and the 2nd because HERE. Since δ is positive, and $\beta_n \rightrightarrows 0$, then the 2nd integral becomes $\frac{\epsilon}{2}$ as well.

Thus, $P_n \rightrightarrows f$ – as we claimed

With this, we can prove the general Stone-Weierstrass theroem on compact metric spaces: that is, change $[0,1]$ into a general comapct metric space.

To do this, we'll first define an algebra on the functions. Let $A \subseteq C^0(M, \mathbb{R})$ such that if $f, g \in A$, $c \in \mathbb{R}$, then

$$f + g \in C^0(M, \mathbb{R}),\ cf \in C^0(M, \mathbb{R}),\ f \cdot g \in C^0(M, \mathbb{R})$$

where each operation is defined pointwise (ex. $(f + g)(x) = f(x) + g(x)$). For examples, polynoimals form an algebra.

The function algbera *vanishes at point* p if $f(p) = 0$ for all $f \in A$. FOr example, A with polynoimals with zero constatn term vanishes at $x = 0$.

THe function algebra *seperates poitns* if for each pari of distinct points $p_1, p_2 \in M$, ther eis a function $f \in A$ such that $f(p_1) \neq f(p_2)$. FOr example, the function algebra of all tirgonometric polynoimals seperates the points $[0, 2\pi)$ and vanishes nowhere

Theorem 4.4.2: Stone-Weierstrass Theorem

If M is a compact metric sapce and A is a functino-algebra in $C^0(M, \mathbb{R})$ that vanishes nowhere and seperates points, then A is dense in $C^0(M, \mathbb{R})$

Note that the Weierstrass theorem is a specia lcase of this one, but it will be used to prove this one

***Proof*:**
We first prove 2 lemmas to help us

If A vanisehs nowhere and seperates points then there exists $f \in A$ with specified values at any pari of disitnct ponts

Proof. □

1. power series: definition, integration, differentiation
2. define analytic
3. Weierstrass aproximation theroem(set of polynomials is dense in $C^0([a, b], \mathbb{R})$)
4. Stone-Weierstrass Theorem
5. Banach Contraction Principle?
6. Picard's Theorem?

<++>

4.5 Ordinary Differential Equations

Definition 4.5.1: Fixed Point

Let $f : M \to M$. a fixed point is a point $p \in M$ such that $f(p) = p$

Definition 4.5.2: Contraction

Let M be a metric space. A *contraction* of M is a mapping $f : M \to M$ such that for some $k < 1$ and all $x, y \in M$ $d(f(x), f(y)) \leq kd(x, y)$

Theorem 4.5.3: Banach Contraction Principle

Supose that $f : M \to M$ is a contraction and the metric space M is complete. Then$f$ has a unique fixed point p and for any $x \in M$ the iterate $f^n(x) = f \circ f \circ \cdots \circ f(x)$ converges to p as $n \to \infty$

***Proof*:**
p.240

Theorem 4.5.4: Brouwer Fixed-Point Theroem

Suppose that $f : B^m \to B^m$ is continuous where B^m is a closed unit ball in $\mathbb{R}^m$. Then f has a fixed point $p \in B^m$

***Proof*:**
see topology notes

Let $U \subseteq \mathbb{R}^m$. A *vector ODE* on U is given as m simultaneous scalar equations

$$\begin{aligned}
x_1' &= f_1(x_1, x_2, ..., x_m) \\
x_2' &= f_2(x_1, x_2, ..., x_m) \\
&... \\
x_m' &= f_m(x_1, x_2, ..., x_m)
\end{aligned}$$

where each f_i is a function from U to $\mathbb{R}$. One seeks m real-valued functions $x_1(t)$, ..., $x_m(t)$ such that

$$\begin{aligned}
\frac{dx_1(t)}{dt} &= f_1(x_1(t), x_2(t), ..., x_m(t)) \\
\frac{dx_2(t)}{dt} &= f_1 2x_1(t), x_2(t), ..., x_m(t)) \\
&... \\
\frac{dx_m(t)}{dt} &= f_m(x_1(t), x_2(t), ..., x_m(t))
\end{aligned}$$

hold identically and simultaneously. Notice that $x_1, ..., x_n$ are all inside the $f_i(t)$. The functions $x_1(t), ..., x_m(t)$ add said to solve the ode with the initial condition

$$(x_1(0), x_2(0), ..., x_m(0))$$

the ode can be thought geometrically if we cake

$$f(x) = (f_1(x), ..., f_m(x))$$

where $x = (x_1, ..., x_m)$, making f a vector field. thus, what we seek is a path γ such that

$$\gamma'(t) = f(\gamma(t))$$

some more build-up

The way of thinking about ODE's is a vector-fields. When you have an ODE, you'll get

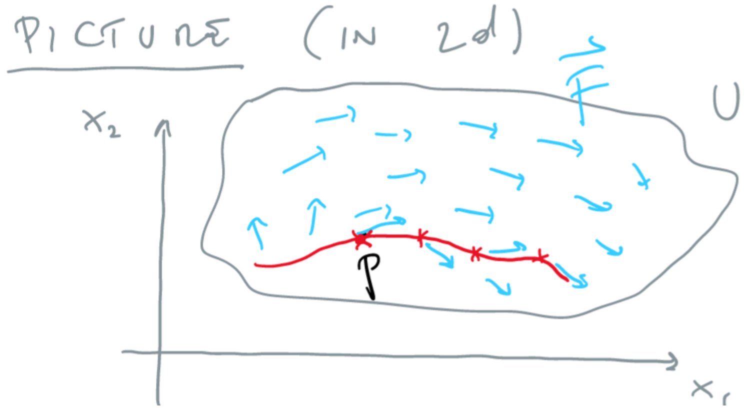


Figure 4.3: ODE as a Vector Field

Theorem 4.5.5: picard's theorem

given $p \in u$ there exists an F-trajectory $\gamma(t)$ in u through p. this means that $\gamma : (a, b) \to u$ solves

$$\gamma'(t) = F(\gamma(t))$$

locally, γ is unique

The locallness is very important. If it wasn't local, you can have functions with grow infinitely fast, or curl around to itself. It is the localness that will let us use contractions.

Proof. p.245 Pugh

First, since F is continuous, there exists a compact neighborhood $N = \overline{N}_r(p) \subseteq U$ and a constant M such that $|F(x)| \leq M$. Choose a t small enough so that

$$tM \leq r \qquad tL < 1$$

so that that tM is within the epsilon ball and tL will satisfy the contraction condition.

Considder teh set $\mathcal{C}$ of all continuous functions $\gamma, \sigma : [-t, t] \to N$ with respect to the metric

$$d(\gamma, \sigma) = \sup \{|\gamma(t) - \sigma(t)||t \in [-t, t]\}$$

the set $\mathcal{C}$ is a complete metric space. Given $\gamma \in \mathcal{C}$, define a new curve $\Phi(\gamma)$ as

$$\Phi(\gamma)(t) = p + \int_0^t F(\gamma(s))ds$$

Solving for this is a continuous curve γ that will also be differentiable (TBD). We just need a fixed point of Φ

To do this, we just check that Φ is a cntraction of $\mathcal{C}$. First, does Φ send $\mathcal{C}$ to itself? Well, given $\gamma \in \mathcal{C}$, we see that $\Phi(\gamma)(t)$ is a continuous (differentiable) vector-valued function of t and that

$$|\Phi(\gamma)(t) - p| = \left| \int_0^t F(\gamma(s))ds \right| \leq tM \leq r$$

implying Φ does send $\mathcal{C}$ to itself. Next, Φ contracts $\mathcal{C}$ since

$$\begin{aligned} d(\Phi(\gamma), \Phi(\sigma)) &= \sup_t \left| \int_0^t F(\gamma(s)) - F(\sigma(s))ds \right| \\ &= t \sup_s |F(\gamma(s)) - F(\sigma(s))| \\ &= t \sup_s L|\gamma(S) - \sigma(S)| \leq tLd(\gamma, \sigma) \end{aligned}$$

and $tL < 1$! Therefore Φ has a fixed point γ, and $\Phi(\gamma) = \gamma$, and therefore, $\gamma(t)$ sovles oru integral, which implies γ is differentiable and solve (20) (TBD) $\square$

4.6 Nowhere Differentiable Continuous Functions

Theorem 4.6.1: Nowhere Differentiable Continuous Functions

There exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ that has a derivative at no point whatsoever

***Proof*:**
We'll construct a function by taking an infinite series of sawtooth functions wich will sum to this:

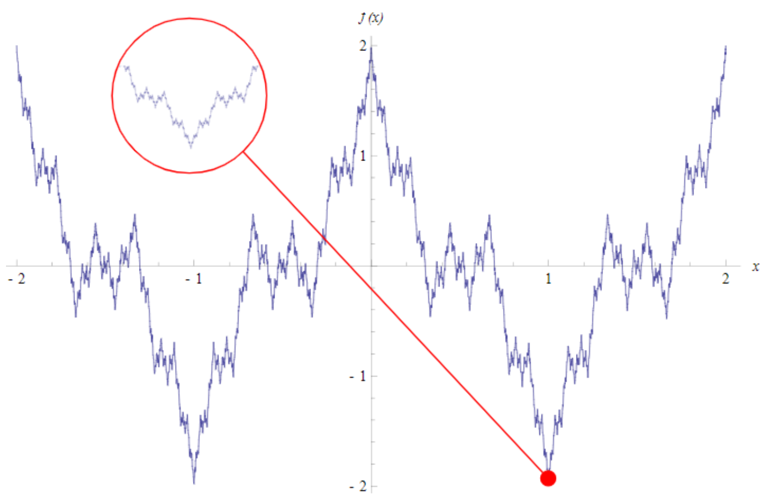


Figure 4.4: Nowehere Differential, yet Continuous Functions

Let the letters k, m, n dentoes integers. Start with the sawtooth function $\sigma_0 : \mathbb{R} \to \mathbb{R}$ defined as

$$\sigma_0(x) = \begin{cases} x - 2n & \text{if } 2n \leq x \leq 2n+1 \\ 2n+2-x & \text{if } 2n+1 \leq x \leq 2n+2 \end{cases}$$

Note that σ_0 is periodic with period 2

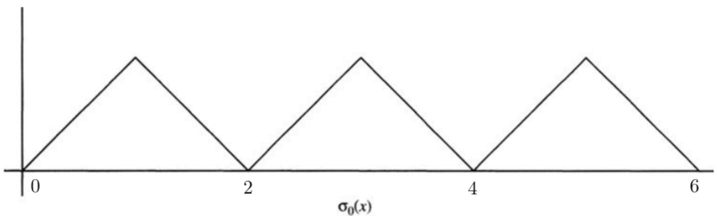


Figure 4.5: σ_0

Next, defined the next sawtooth functions by compression σ_0:

$$\sigma_k(x) = \left(\frac{3}{4}\right)^k \sigma_0(4^k x)$$

the period will be $\frac{2}{4^k}$, since $\sigma(x) = \sigma(x + m\left(\frac{2}{4^k}\right))$ for any $m \in \mathbb{Z}$.

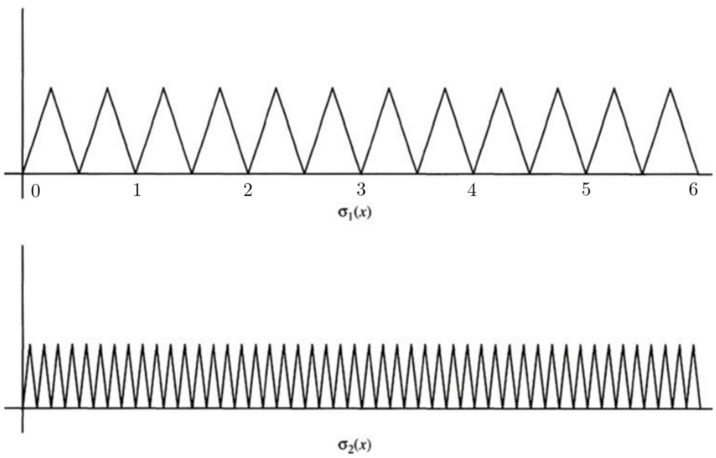


Figure 4.6: two examples of the itteration

By the M-test, $\sum \sigma_k(x0)$ converges uniformly to a limit f since $|\sigma_k| \leq \left(\frac{3}{4}\right)^k$ and

$$f(x) = \sum_{k=0}^{\infty} \sigma_k(x)$$

is continuous. We'll show that f is nowhere differentiable, that is

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

does not exist. Since this must work no matter how $h \to 0$, pick $\delta_m = \frac{1}{2\cdot 4^m}$. Notice that $\delta_m \to 0$ as $m \to \infty$. We'll show that this limit does not exist:

$$\lim_{m\to 0}\frac{f(x\pm\delta_m - f(x)}{\delta_m}$$

where the $\pm$ is because we want this limit to reprent approach from either the left or right.

Using proposition 4.2.6, we get the derivative is

$$\sum_{k=0}^{\infty}\frac{\sigma_k(x\pm\delta_m)-\sigma(x)}{\delta_m}$$

We'll split this sum into 3 cases: $k > m$, $k = m$, $k < m$

($k > m$) Since $\delta_m = \frac{2}{4^k}$

$$\delta_m = \frac{1}{2}\frac{1}{4^m} = m\sigma_k \qquad m \in \mathbb{Z}$$

and thus

$$\sigma_k(x\pm\delta_m)-\sigma_k(x) = \sigma_k(x)-\sigma_k(x) = 0$$

meaning these terms do not contribute to the sum. We can essentially re-write our sum to

$$\sum_{k=0}^{n-1}\frac{\sigma_k(x\pm\delta_m)-\sigma(x)}{\delta_m}$$

($k = m$) σ_m will be monotone on either $[x, x+\delta_m]$ or $x-\delta-m, x]$:

Figure 4.7: Demonstration of monotonicity

and therefore

$$|frac\sigma_m(x+\delta_m)-\sigma_m(x)\delta_m| = 3^m$$

$(k < m)$ using what we've just learn from the previous exercice, we can crudely estimate each σ_k:

$$\left|\frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m}\right| \leq 3^k$$

giving us

$$\sum k < m \left|\frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m}\right| \leq \sum_{k<m} 3^k = \frac{3^m - 1}{3 - 1}$$

Thus, this gives us

$$\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{m\to\infty} \sum_{k=0}^{\infty} \frac{\sigma_k(x \pm \delta_m) - \sigma(x)}{\delta_m} = \lim_{m\to\infty} 0 \pm 3^m + (3) \geq 3^m - \frac{3^m - 1}{2} = \frac{3^m}{2} + \frac{1}{2}$$

which diverges as $m \to \infty$! Therefore, this function is *not* differentiable at any point

4.7 Baire Theory

This construction might look like a particular case we need to construct to find, but it turns out that almost all continuous functions are nowhere differentiable! If you define your notion of probability right, then the chance of picking a continuous differentiable function form the set of continuous function is 0%! The way we'll make this rigorous is by taking a detour into Baire spaces.

A set D is dense in M if $\overline{D} = M$. Equivalently, it is dense in M if D for any open $W \subseteq M$, $W \cap M \neq \emptyset$. The intersection of dense sense need not be dense (ex. $\mathbb{Q}$ and $\mathbb{Q}^c$ in $\mathbb{R}$). However, if U, V are *open*-dense sets in M, then $U \cap V$ will also be open-dense in M. If $W \subseteq M$ is an arbitrary open subset. Then $U \cap W$ is also open, and so by denseness of V, $V \cap (U \cap W)$ is also non-empty! re-arranging using

4.8 Cardinality quick

I'm not sure where to add this yet, but I thought it is a good exercise to appreciate the oddity of $\aleph_1$

Pove that C^0 and $\mathbb{R}$ have the same cardinality

It is clear that C^0 has at least as many function as real numbers since every constant function is continuous, so it comes down to show there are no more. This is exercise 6 in Pugh's book, and its marked with a double star.

Found this too

https://math.stackexchange.com/questions/477/cardinality-of-set-of-real-continuous-functions

4.9 Condition for Smooth to be Analytic

Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth function. Then we can always take its' tailor approximations at a point $T_a f$, and ask whether there exists an open interval U around a such that $T_a f|_U = f|_U$, namely:

$$f(h) = \sum_{k=1}^{\infty} \frac{a_k (h-a)^k}{k!}$$

Not all smooth functions are analytic, as the following typical example demonstrates:

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

which can be verified by taking it's Taylor series which is

$$T_0(f) = \sum_{k=1}^{\infty} 0x^k = 0$$

However, we can precisely understand when a smooth is analytic by looking at the growth rate of the derivative, as we shall demonstrate in this chapter.

We can take a partial Taylor approximation and add on to it the remainder:

$$f(t) = \underbrace{\sum_{j=0}^{k-1} \frac{f^{(j)}(x)}{j!}(t-x)^j}_{\text{Polynomial}} + \underbrace{R_{k-1}(t)}_{\text{Remainder}}$$

Let us estimate $R_k(t)$.

Theorem 4.9.1: Lagrange Remainder Theorem

Let f be a smooth function, or at least $C^{k-1}(\mathbb{R})$ and let U be some interval. Let $P_{k-1}(t)$ be the partial Taylor series around a. Then for fixed t:

$$f(t) = P_{k-1}(t) + \frac{f^{(k)}(\xi_t)}{k!}(t-a)^k$$

***Proof*:**
Given the partial Taylor series $P_{k-1}(t)$ around a, fix a t_0 (without loss of generality $t_0 > a$). Then there always exists a constant C such that:

$$f(t_0) = P_{k-1}(t_0) + \frac{Q}{k!}(t_0 - a)^k$$

Define the function:

$$F(z) = f(t_0) - \sum_{j=0}^{k-1} \frac{f^{(j)}(z)}{j!}(t_0 - z)^j - \frac{Q}{k!}(t_0 - z)^k$$

Then $F(a) = F(t_0) = 0$ and F is differentiable, hence by Roles theorem there exists a ξ_t such that:

$$F'(\xi_t) = 0$$

Taking the actual derivative we get:

$$\frac{d}{dz}\left(\frac{f^{(j)}(z)}{j!}(t-z)^j\right) = \frac{f^{(j+1)}(z)}{j!}(t-z)^j - \frac{f^{(j)}(z)}{j!}j(t-z)^{j-1}$$

this creates a telescoping sum gives:

$$F'(z) = -\frac{f^{(k)}(z)}{(k-1)!}(t-z)^{k-1} + Q\frac{k(t-z)^{k-1}}{k!} = \left(-f^{(k)}(z) + Q\right)\frac{(t-z)^{k-1}}{(k-1)!}$$

where the equation is 0 at ξ_t. Now, as $a \neq t_0$, $(t-\xi_t)^{k-1} \neq 0$, hence:

$$(-f^{(k)}(\xi_t) + Q) = 0 \qquad \Rightarrow \qquad f^{(k)}(\xi_t) = Q$$

as we sought to show.

Using this, we can bound the remainder term. Given a interval $(r-a, r+a) = U$, for any $t \in U$, $|a-t| \leq r$. Then, for any choice of t, we have $f^{(k)}(\xi_t)$, and we can define:

$$M_k = \sup_{t \in U} |f^{(k)}(t)|$$

Hence:

$$|R_{k-1}(t)| = \left|\frac{f^{(k)}(\xi)}{k!}(t-a)^k\right| \leq \frac{M_k}{k!}r^k$$

This is the worst the function can grow. Then by the root test (which compares the growth rate of of function to $\sum_k \frac{1}{k^n}$[2]) if the quantity:

$$L = \limsup_{k\to\infty} \sqrt[k]{\frac{M_k}{k!}r^k} = r \limsup_{k\to\infty} \sqrt[k]{\frac{M_k}{k!}}$$

is strictly less than 1, it converges, if it is strictly greater than 1 it diverges, and if $L = 1$ then it is indeterminate. Note how R_{k-1} is dependent on the radius r.

Definition 4.9.2: Derivative Growth Rate

Let $I = (x-r, x+r)$ be an interval. Then the *bounded derivative growth rate* of a smooth function f on I is:

$$\alpha_r = \limsup_{k\to\infty} \sqrt[k]{\frac{M_k}{k!}}$$

Using definition 4.9.2, we have:

$$L = r \cdot \alpha_r$$

[2] $L < \rho, 1$, taking nth powers we see by the comparison test that L converges

Note that α_r is dependent on the radius r as we take the max over an interval of radius r. Now, since $|f^{(k)}(a)| \leq M_k$, we have that

$$R_a = \frac{1}{\limsup_{k\to\infty} \sqrt[k]{\frac{|f^{(k)}(a)|}{k!}}} \leq \frac{1}{\alpha_r}$$

Hence:

$$\frac{1}{\alpha_r} \leq R_a$$

Theorem 4.9.3: Taylor Convergence

Let I be an interval. If $\alpha \cdot r < 1$ then the Taylor series converges uniformly to f on the interval I.

***Proof*:**
Choose $\delta > 0$ such that $(\alpha + \delta)r < 1$. The Taylor remainder formula from Chapter 3, applied to the $(r-1)$st-order remainder, gives

$$f(x+h) - \sum_{k=0}^{r-1} \frac{f^{(k)}(x)}{k!}h^k = \frac{f^{(r)}(\theta)}{r!}h^r$$

for some θ between x and $x+h$. Thus, for r large we have

$$\left| f(x+h) - \sum_{k=0}^{r-1} \frac{f^{(k)}(x)}{k!}h^k \right| \leq \frac{M_r}{r!}r^r = \left(\left(\frac{M_r}{r!} \right)^{1/r} r \right)^r \leq ((\alpha+\delta)r)^r.$$

Since $(\alpha + \delta)r < 1$, the Taylor series converges uniformly to $f(x+h)$ on I.

Theorem 4.9.4: Analytic then bounded derivative growth

If f is expressed as a convergent power series $f(x+h) = \sum c_k h^k$ with radius of convergence $R > \sigma$ then f has bounded derivative growth rate on I.

***Proof*:**
The proof of theorem 4.9.4 uses two estimates about the growth rate of factorials. If you know Stirling's formula they are easy, but we prove them directly.

$$\lim_{r\to\infty} \frac{r}{\sqrt[r]{r!}} = e \tag{4.1}$$

$$0 < \lambda < 1 \implies \limsup_{r\to\infty} \sqrt[r]{\sum_{k=r}^{\infty} \binom{k}{r} \lambda^k} < \infty. \tag{4.2}$$

Taking logarithms, applying the integral test, and ignoring terms that tend to zero as $r \to \infty$

gives

$$
\begin{aligned}
\frac{1}{r}(\log r! - \log r!) &= \log r - \frac{1}{r}(\log r + \log(r-1) + \cdots + \log 1) \\
&\sim \log r - \frac{1}{r}\int_1^r \log x \, dx = \log r - \frac{1}{r}(x \log x - x)\Big|_1^r \\
&= 1 - \frac{1}{r},
\end{aligned}
$$

which tends to 1 as $r \to \infty$. This proves (4.1).

To prove (4.2) we write $\lambda = e^{-\mu}$ for $\mu > 0$, and reason similarly:

$$
\begin{aligned}
\sum_{k=r}^{\infty} \binom{k}{r} \lambda^k &= \sum_{k=r}^{\infty} \frac{k(k-1)\dots(k-r+1)}{r!} e^{-k\mu} \\
&\leq \frac{1}{r!} \sum_{k=r}^{\infty} k^r e^{-k\mu} \approx \frac{1}{r!} \int_r^{\infty} x^r e^{-\mu x} \, dx \\
&= \frac{-1}{r!} e^{-\mu x} \left(\frac{x^r}{\mu} + \frac{r x^{r-1}}{\mu^2} + \frac{r(r-1)x^{r-2}}{\mu^3} + \cdots + \frac{r!}{\mu^{r+1}} \right) \Big|_r^{\infty} \\
&\leq \frac{1}{r!} e^{-\mu r} (r+1) r^r \frac{1}{\min(1,\mu)^{r+1}}.
\end{aligned}
$$

According to (4.1) the rth root of this quantity tends to $e^{1-\mu} / \min(1,\mu)$ as $r \to \infty$, completing the proof of (4.2).

By assumption the power series $\sum c_k h^k$ has radius of convergence R and $\sigma < R$. Since $1/R$ is the $\limsup$ of $\sqrt[k]{|c_k|}$ as $k \to \infty$, there is a number $\lambda < 1$ such that for all large k we have $|c_k \sigma^k| \leq \lambda^k$. Differentiating the series term by term with $|h| \leq \sigma$ gives

$$
\begin{aligned}
|f^{(r)}(x+h)| &\leq \sum_{k=r}^{\infty} k(k-1)(k-2)\dots(k-r+1)|c_k h^{k-r}| \\
&\leq \frac{r!}{\sigma^r} \sum_{k=r}^{\infty} \binom{k}{r} |c_k \sigma^k| \leq \frac{r!}{\sigma^r} \sum_{k=r}^{\infty} \binom{k}{r} \lambda^k.
\end{aligned}
$$

for r large. Thus,

$$
M_r = \sup_{|h| \leq \sigma} |f^{(r)}(x+h)| \leq \frac{r!}{\sigma^r} \sum_{k=r}^{\infty} \binom{k}{r} \lambda^k.
$$

According to (4.2),

$$
\alpha = \limsup_{r \to \infty} \sqrt[r]{\frac{M_r}{r!}} \leq \frac{1}{\sigma} \limsup_{r \to \infty} \sqrt[r]{\sum_{k=r}^{\infty} \binom{k}{r} \lambda^k} < \infty,
$$

and f has bounded derivative growth rate on I.

Theorem 4.9.5: Analyticity Theorem

A smooth function is analytic if and only if it has locally bounded derivative growth rate.

***Proof*:**
Assume that $f : (a, b) \to \mathbb{R}$ is smooth and has locally bounded derivative growth rate. Then $x \in (a, b)$ has a neighborhood N on which the derivative growth rate α is finite. Choose $\sigma > 0$ such that $I = [x - \sigma, x + \sigma] \subset N$ and $\alpha\sigma < 1$. We infer from theorem 4.9.3 that the Taylor series for f at x converges uniformly to f on I. Hence f is analytic.

Conversely, assume that f is analytic and let $x \in (a, b)$ be given. There is a power series $\sum c_k h^k$ that converges to $f(x + h)$ for all h in some interval $(-R, R)$ with $R > 0$. Choose σ with $0 < \sigma < R$. We infer from theorem 4.9.4 that f has bounded derivative growth rate on I.

Thus, while smooth functions have no bounds for the derivative:

$$|f^{(k)}(x)| \leq \text{anything, no bound}$$

analytic functions satisfy:

$$|f^{(k)}(x)| \leq C \cdot \frac{k!}{R^k}$$

5

Banach and Fréchet Spaces

(these are all spaces key spaces to consider with normed

(alternative title: Fréchet, Banach, and Hilbert Spaces)

5.1 Banach Spaces

Let k be either $\mathbb{R}$ or $\mathbb{C}$, let V or X denote a vector-space over k, with 0 representing the zero vector. Let kx denote the subspace of X spanned by x. If $M, N \subseteq X$, then $M + N = \{m + n \mid m \in M, n \in N\}$.

Definition 5.1.1: Semi-Norm And Norm

Let V be a vector-space. A *semi-norm* on V is a function $x \mapsto \|x$
from V to $[0, \infty)$ satisfying the propeties:

1. $\|x + y\| \leq \|x\| + \|y\|$ (the triangle inequality)
2. $\|\lambda x\| = |\lambda|\|x\|$ (homogeneity of scalars)

If furthermore:

(3) $\|x\| = 0$ if and only if $x = 0$

then the function is called a *norm*

Definition 5.1.2: Normed Vector-Space

Let V be a vector space. Then if there exists a function $\|\cdot\|$ that is defined on V, then $(V, \|\cdot\|)$ is called a *normed vector-space*.

If V is a normed vector-space, we can induce a metric $\rho(x, y) = \|x - y\|$, namely since:

$$\|x - z \leq \|x - y\| + \|y - z\| \qquad \|x - y\| = \|(-1)(x - y)\| = \|y - x\|$$

Since any metric induces a topology, the induced topology by the metric induced by a norm $\|\cdot\|$ is called the *norm topology* on V. A nice consequence of the induced norm topology is that addition and scalar multiplication is continuous. From the axioms of a semi-norm, $\|0\| = 0$. There are norms for which $x \neq 0$ but $\|x\| = 0$, as we'll soon show. Note that the norm $\|\cdot\| : V \to \mathbb{F}$ is evidently continuous in the topology it induces[1]. If $x_n \to x$ in X, then for all $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that for all $n \geq N$, $\|x_n - x\| < \epsilon$. By the reverse triangle inequality:

$$|\|x_n\| - \|x\|| \leq \|x_n - x\| < \epsilon$$

showing that $|\|x_n\| - \|x\||$ get's arbitrarily close, and so $\|x_n\| \to \|x\|$, i.e. $\|\cdot\|$ is continuous. This means that the pre-image of $\{0\}$ of $\|\cdot\|$ is closed. Furthermore, it is a vector-space. Denoting

$$K = \{x \in V \mid \|x\| = 0\}$$

Then if $x, y \in K$, $0 \leq \|x+y\| \leq \|x\|+\|y\| = 0$, and $\|cx\| = |c|\|x\| = |c|0 = 0$. Hence, we may consider V/K. What norm to put on V/K that is natural with respect to the projection map $\pi : V \to V/K$ will be addressed soon.

Two norms $\|\cdot\|_1$ and $\|\cdot\|_1$ are said to be *equivalent* if there exits some $C_1, C_2 > 0$ st

$$C_1\|\cdot\|_1 \leq \|\cdot\|_2 \leq C_2\|\cdot\|_1$$

Equivalent norms define equivalent metrics and the same topology, and so the same Cauchy sequences. Recall that in finite dimensions, all norms are equivalent, meaning there is only one notion of distance. As we'll see, this is not the case in infinite dimensions.

Since we have a metric, we have a notion of completeness. As is the case when working with analysis, we will require our spaces to be complete:

Definition 5.1.3: Banach Space

Let V be a normed vector space. Then V is a *Banach space* if it is complete with respect to it's induced metric.

Example 5.1.4: Banach Spaces

1. Every finite dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ is a Banach space. In this space, all linear map are continuous.

2. Let K be a compact space, and take the set of all continuous function, $C_0(K)$. Then $C(K)$ is a Banach space with the supremum norm $\|f\|_\infty = \sup_{x \in K} |f(x)|$ (also called uniform

[1] Note that in the metric space topology, continuous if and only if sequentially continuous

norm). We will later show (BanachMazur theorem) that all real separable Banach spaces are isometric (the isomorphisms in the category of Banach spaces) to a subspace of $C([0,1])$, giving us a theoretical universal way of thinking about Banach spaces. We shall later see that $C([0,1])$ has the Approximation property and has a Schauder basis, but there exists Banach spaces without these properties. This shows that for Banach spaces, it is possible to have nice total spaces with "bad" subspaces. In particular, if X is an isometric embedding into $C([0,1])$ there are not always nl nice projection $P : C([0,1]) \to X$.

If the space is not separable, if the smallest dense set D of a Banach space X has density of cardinality α, then X is isometric to $C_0([0,1]^\alpha, \mathbb{R})$[a]

3. If X is a topological space, then $B(X,k)$ and $BC(X,k)$ (all bounded functionals and bounded continuous functionals) form a Banach space with the uniform norm $\|f\|_u = \sup_{x\in X} |f(x)|$.
4. For p where $1 \leq p \leq \infty$, the space $L^p(X)$ for some measurable space X is a Banach space with the L^p-norm (recall that $L^p(\mu)$ is the set of equivalence classes functions that agree a.e.; if $L^p(\mu)$ represented only functions, then $\|\cdot\|_p$ would be a semi-norm)
5. Two other common examples of Banach spaces we will encounter later will be special subspaces spaces of differentiable functions: Banach Algebras, and Hilbert Spaces.

There are also examples of spaces that are not Banach Spaces:

1. Let X be a locally compact Hausdorff space and consider $C_c(X)$ with the ∞-norm. Then it is not a Banach space, not being complete. It is easy to see that $C_c(X)$, the set of continuous bounded function, is a Banach space and contains $C_c(X)$, and so it suffices to find the closure of $C_c(X)$ in $C_b(X)$. The closure turns out to be $C_0(X)$.
2. Let $U \subseteq \mathbb{R}^n$ be open and take $C_0(U, \mathbb{C})$. Then This space is *not* a Banach space
3. The space $H(U, \mathbb{C})$ of all holomorphic functions from U to $\mathbb{C}$ is *not* a Banach Space
4. The space $C_c^\infty(\mathbb{R}^n)$ is *not* a Banach space
5. (If you know some PDE's) The space of distributions $\mathcal{D}'$ is *not* a Banach Space

These are all examples of a weaker structure known as a *topological vector space* which we shall cover in section 5.4.

[a] An interesting result using Banach-Mazur's theorem by Luis Rodríguez-Piazza shows that the space of smooth functions with the uniform norm is isometric to the space of nowhere differentiable functions

There is a nice way to check whether a vector space V is a Banach space which requires the following definition (this essentially follows from the fact that continuity if and only if sequential continuity in a metric space and that $\|\cdot\|$ is continuous)

Definition 5.1.5: Absolutely Convergent

Let V be a normed vector space. and $\{x_k\}_k^\infty \subseteq V$ a sequence. Then the sequence is said to *converge absolutely* if

$$\sum_k^\infty \|x_k\| < \infty$$

Lemma 5.1.6: Cauchy and Absolute Convergence

Let V be a normed vector space. Then V is complete if and only if every absolutely convergent series in V converges.

The idea is that $\mathbb{R}$ has an order, making it easier to produce many arguments about sequences.

***Proof*:**
Let's first say that X is complete and that $\sum_{n=1}^\infty \|x\| < \infty$. To show it converges, we take advantage of completeness and show there is a Cauchy sequence. In particular, let $S_n = \sum_{k=1}^n x_k$ be the partial sum. Then since the series is absolutely convergent, there for all $n > m$,

$$\|S_n - S_m\| \leq \sum_{m+1}^n \|x_k\| \to 0 \quad \text{as } m, n \to \infty$$

so the sequence $\{S_n\}$ is Cauchy, and so converges, so $\{x_k\}$ converges.

Conversely, let's say an absolutely convergent series converges, and choose any Cauchy sequence $\{x_k\}$. Since it's a Cauchy sequence, choose a subsequence such that

$$\left\|x_{k_{i+1}} - x_{k_i}\right\| < 2^{-i}$$

Take $y_1 = x_{k_1}$ and $y_i = x_{i+1} - x_i$ for $i > 1$ so that $\sum_{i=1}^n y_i = x_{k_n}$. Then notice that $\{y_i\}$ is in fact absolutely convergent:

$$\sum_{i=1}^\infty \|y_i\| \leq \|y_1\| + \sum_{i=1}^\infty 2^{-i} = \|y_1\| + 1 < \infty$$

So $\sum_{i=1}^\infty y_i = \lim x_{k_n}$ exists. Since the subsequence of every Cauchy sequence approaches the same "point" (just like convergent sequences), we see that $\{x_n\}$ also converges to the same point, showing that Cauchy sequences are in fact convergent sequences, as we sought to show.

Next, we would want to study which linear functions between two vector-spaces are *continuous*. It will *not* be the case that all linear functions will be continuous. The following builds up to the classification of continuous linear functions:

Definition 5.1.7: Bounded Linear Map

Let $T : X \to Y$ be a linear map between two normed vector space. Then if there exists some C such that for all $x \in X$

$$\|T(x)\| \leq C\|x\|$$

Then T is said to be *bounded*, or is a *bounded linear operator*.

Note how this is different then boundedness of a set (ex. the range is bounded if $\|T(x)\| \leq C$). This definition is would be useless to us since no nonzero linear map would satisfy it. All linear map over finite dimensional vector spaces clearly bounded (can you find the C?), so this concept gets interesting in infinite dimesion

Proposition 5.1.8: Bounded $\Leftrightarrow$ Continuous

Let $T : X \to Y$ be a linear map between two normed vector spaces. Then the following are equivalent:

1. T is continuous
2. T is continuous at 0
3. T is bounded

***Proof*:**
If T is the zero map, then we're done, so assume $T \neq 0$

(1) implies (2) by definition. For (2) implies (3), Since T is continuous at 0, for for all $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|x - 0\| < \|x\| < \delta$, then $\|T(x) - T(0)\| = \|T(x) - 0\| = \|T(x)\| < \epsilon$. Choosing $\epsilon = 1$, we get that there exists a δ_1 such that

$$\|x\| < \delta_1 \to \|T(x)\| < 1$$

we know want to show there exists a C such that $\|T(y)\| \leq C\|y\|$ for all y in the domain. We will do this by taking advantage of linearity and shrinking element so that we can take advantage of continuity.
Define $x(y) = \frac{\delta_1}{2\|y\|}y$ so that

$$\|x(y)\| = \left\|\frac{\delta_1}{2\|y\|}y\right\| = \frac{\delta_1}{2}\frac{1}{\|y\|}\|y\| = \frac{\delta_1}{2} < \delta_1$$

Therefore, by continuity and linearity:

$$\|T(x(y))\| = \left\|T\left(\frac{\delta_1}{2\|y\|}y\right)\right\| = \frac{\delta_1}{2\|y\|}\|T(y)\| < 1$$

since this is true for all y, we if we re-arrange we have:

$$\|T(y)\| < \frac{2}{\delta_1}\|y\|$$

showing that T is indeed bounded.

Finally, if $\|T(x)\| \leq C\|x\|$, then for all $\epsilon > 0$, choose $\delta = C^{-1}\epsilon$ so that

$$\|T(y) - T(x)\| = \|T(y - x)\| \leq \epsilon$$

completing the proof.

Note that for $T \in B(X,Y)$ to be an isomorphism, we require that T^{-1} also be bounded[2]. If $\|T(x)\| = \|x\|$ for all x, then T is called an *isometry*.

Since the topology is that of norms, then continuous linear functions between normed vector-spaces are the homomorphisms, meaning they preserve the norm (to the degree of boundedness mentioned in the above proposition) and the vector-space structure (to the degree the map is injective). We will denote the set of bounded linear functions (which we will call linear operators) as $B(X,Y)$, $\mathcal{L}(X,Y)$, or $\mathrm{Hom}_k(X,Y)$, where we understand that since X and Y are normed spaces $\mathrm{Hom}_k(X,Y)$ refers to continuous lienar maps.

Like with most fields of mathematics, we are more often more interested in the functions between spaces rather than the spaces itself, for it is function that ultimatly give us information about the structure we are working with. Thus, if we can use the technics we will be developing on $B(X,Y)$, we may gain many interesting results. For this to be the case, we require to introduce a norm on $B(X,Y)$. Naturally, we want that the norm on $B(X,Y)$ to be compatible with X and Y in some way, especially the codomain Y (as we usually investigate the structure of the image of a map). Sicne we are working with bounded operators, we have that $\|T(x)\| \leq C\|x\|$. Furthermore, since we are working with linear operators, by some scaling we see that we only need to consider when $\|x\| = 1$. This moivates the following norm on $B(X,Y)$:

Definition 5.1.9: Operator Norm

$B(X,Y)$ is a normed vector space with norm[a]

$$\begin{aligned}\|T\| &= \sup_{\|x\|=1} \|T(x)\| \\ &= \inf\{C \mid \|T(x)\| \leq C\|x\| \text{ for all } x\} \\ &= \sup_{\|x\|\neq 0} \frac{\|T(x)\|}{\|x\|}\end{aligned}$$

If $\|T(x)\| = \|x\|$, then T is called an *isometry*

[a]the third equality holds if $V \neq \{0\}$

The operator norm can be thought of as the smallest consant that will satisfy $\|T(x)\| \leq C\|x\|$ when T is bounded, that is $\|T(x)\| \leq \|T\|\|x\|$. In the spacial case where we have $B(X,k)$, we write V^* and call it the dual space with the *norm topology*. It is useful to establish when is $B(X,Y)$ a Banach space

[2]We will show using the open mapping theorem that it suffices to show T is continuous and bijective

Proposition 5.1.10: Completness Of $B(X, Y)$

Let X, Y be normed vector-space and $B(X, Y)$ the normed vector-space of all bounded linear functions from X to Y. If Y is complete, so is $B(X, Y)$.

***Proof*:**
Let (T_n) be a cauchy sequence in $B(X, Y)$. For each $x \in X$, we have a cauchy sequence $(T_n(x)) \subseteq Y$, and thus a limit vector which we will denote $T(x)$. This defines a function $T : X \to Y$ which is clearly linear. The inequality

$$\begin{aligned}\|T(x) - T_n(x)\| &= \lim_m \|T_m(x) - T_n(x)\| \\ &\leq (\limsup_m \|T_m - T_n\|)\|x\|\end{aligned}$$

shows that T is bounded and that $T_n \to T$.

Hence, V^* is always a banach space, since $\mathbb{R}$ and $\mathbb{C}$ are complete. Next, we see what happens with common vector-space constructions like products and quotients.

Subspaces, Product spaces, Quotient Spaces

There are 3 equivalent norms we can put on the product of two normed vector-spaces:

1. $\|(x, y)\| = \max(\|x\|, \|y\|)$
2. $\|(x, y)\| = \|x\| + \|y\|$
3. $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$

The key to these being equivalent is that we are only multiplying by a finite number of spaces, a different strategy would have to be implemented for norms on infinite dimensional vector spaces (recall the different ℓ^p-norms put on the set of all sequences). For subspace, it is clear that if $W \subseteq V$ is a subspace of a normed vector space V, then W has an induced norm with respect to V. What is more interesting is if V is a banach space and we want W to be a Banach space. We then require all sequences to converge. Since V is Hausdorff (even metrizable since there is a norm on V), we see that W is complete if W is a *closed subspace* with respect to the topology of V. For quotients V/W, we would like to define a new norm on V/W so that the natural projeciton map $\pi : V \to V/W$ is continuous. This leads us to the following definition:

Proposition 5.1.11: Banach Quotient Space

Let X be a normed vector space and $Y \subseteq X$ a subspace of X. Then $\pi : X \to X/Y$ is the quotient map. This induces the topology on X/Y by defining:

$$\|\pi(x)\| = \|x + Y\| = \inf_{y \in Y} \|x - y\|$$

which is a *seminorm* on X/Y. If Y is closed, then this is in fact a norm, and if furthermer X is Banach, then X/Y is a banach space.

The intuition in my mind for this definition of quotient norm is to "eliminate" any effect of the quotienting space Y on any element of $x + Y$. Take for example $\mathbb{R}^2/\mathbb{R}$. Then the the norm of $\|(x, y) + \mathbb{R}\|$ should really be $|x|$. Furthermore, π is naturally a continuous map since it is norm-decreasing, meaning $\|\pi(x)\| \leq \|x\|$ (simply choose $C = 1$).

***Proof*:**
For subadditivity, we take advantage of the definition of infimum: for every $x_1, x_2 \in X$, there exists an $\epsilon > 0$ and elements $y_1, y_2 \in Y$ such that

$$\begin{aligned}\|\pi(x_1)\| + \|\pi(x_2)\| + \epsilon &\geq \|x_1 - y_2\| + \|x_2 - y_2\| \\ &\geq \|x_1 + x_2 - (y_1 + y_2)\| \\ &\geq \|\pi(x_1 + x_2)\|\end{aligned}$$

Since ϵ was arbitrary, we have subaditivity. Homogeneity is immediate since π is linear, and hence we have a seminorm.

If $\|\pi(x)\| = \|x + Y\| = 0$, there is a sequence $(y_n) \subseteq Y$ such that $\|x - y_n\| \to 0$. If Y is closed, $\lim_n(y_n) = x \in Y$, meaning $\pi(x) = 0$. It follows that X/Y is a normed space if and only if Y is closed in X.

Now, say $(z_n) \subseteq X/Y$ is a cauchy sequence. Then we can find a subsequence (z'_n) were

$$\|z'_{n+1} - z'_n\| < 2^{-n}$$

By induction we can choose $x_n \in X$ such that $\pi(x_n) = z'_n$ and $\|x_{n+1} - x_n\| < 2^{-n}$. If X is banach, (x_n) converges to an element $x \in X$, and since π is "norm decreasing" (the norm the same or smaller), then (z'_n) converges to $\pi(x)$. Hence, (z_n) converges to $\pi(x)$, and so X/Y is complete, as we sought to show.

Proposition 5.1.12: Universal Property of Quotient Maps

If $T \in B(X, Y)$ where X and Y are normed spaces, and if $W \subseteq X$ is a closed subspace of X containing $\ker(T)$, then then there exists an operator $\tilde{T} \in B(X/W, Y)$ defined pointwise as $\tilde{T}(\pi(x)) := T(x)$ such that

$$\|\tilde{T}\| = \|T\|$$

in particular, the following diagram commutes in **Ban**:

$$\begin{tikzcd} X \arrow[r, "T"] \arrow[d, "\pi"'] & Y \\ X/W \arrow[ur, "\tilde{T}"'] & \end{tikzcd}$$

***Proof*:**
First, by elementary linear algebra we have that $\tilde{T}$ is an operator on X/W, which is a normed space by the earlier proposition. Since

$$\|\tilde{T}(\pi(x))\| = \|T(x)\| = \|T(x - z)\| \leq \|T\|\|x - z\|$$

for every $z \in W$, it follows form the definition of the quotient norm that

$$\|\tilde{T}(\pi(x)) \leq \|T\|\|\pi(x)\|$$

hence, $\|\tilde{T}\| \leq \|T\|$. The reverse inequality $\|\tilde{T}\| \geq \|T\|$ is clear since Q is norm decreasing.

Proposition 5.1.13: Extending Banach Spaces

Let X be a normed space and Y a closed subspace. Then if Y and X/Y are Banach spaces, then X is a Banach space.

***Proof*:**
Let $(x_n) \subseteq X$ be a cauchy sequence. Since $\pi : X \to X/Y$ is norm-decreasing, $(\pi(x_n)) = (y_n) \subseteq X/Y$ is a cauchy sequence. Since X/Y is complete, it converges to a point $\overline{x} \in X/Y$. Next, by the definition of the quotient norm, for every x_n, there exists an n st

$$\|(x - x_n) - y_n\| \leq \frac{1}{n} + \|\pi(x - x_n)\|$$

Then we get:

$$\begin{aligned}\|y_n - y_m\| &= \|(y_n + x - x_n) - (y_m + x - x_m) + x_n - x_m\| \\ &\geq \frac{1}{n}\|\pi(x - x_n)\| + \frac{1}{m}\|\pi(x - x_m)\| + \|x_n + x_m\|\end{aligned}$$

which is clearly cauchy, and hence since Y is banach it is complete and so this sequence converges to y. Finally, it is easy to see that

$$\|x_n - (x + y)\| \leq \|x_n - (x - y_n)\| + \|y - y_n\|$$

showing that $x - n \to x + y$, as we sought to show.

Finite space naturally have none of the problems introduced in infinite dimensions and preserve all of their nice properties even when subspace of infinite dimensinoal vectorspace:

Proposition 5.1.14: Finite Dimensional Normed Space

Every finite-dimensional subspace Y of a normed space X is a Banach space, and consequently is closed in X. Moreover, if $\dim(Y) = n$, every linear isomrophism of $\mathbb{F}^n$ onto Y is a homeomorphism

***Proof*:**
p. 46 Analysis now.

Working with infinite dimensional space dense subsets become of key interest. As a simple example, the smooth bump functions form a dense subset of L^p for all $1 \leq p < \infty$. The following simplifies for us how we may define functions from spaces given a known dense set:

Proposition 5.1.15: Defining Operator using Dense Sets

Let X and Y be banach spaces and X_0 is a dense subspace of X. Then every operator $T_0 \in B(X_0, Y)$ has a unique extension to an operator $T \in B(X, Y)$ and $\|T\| = \|T_0\|$

***Proof*:**
For each $x \in X$, there is a sequence $(x_0) \subseteq X_0$ that converges to x: $x_n \to x$. Since T_0 is a bounded operator ($\|T_0(x_n)\| \leq C\|x_n\|$), it is clear that $(T_0(x_0))$ is a convergent sequence in Y; let's say it converges to $T(x)$. It should be clear that if we choose another sequence converging to x and do this procedure, then we will get the same value of $T(x)$. Thus, the map $x \to T(x)$ is a operator that is linear, extends T_0, and by construciton

$$\|T\| = \|T_0\|$$

an excellent example would be polynomial over a compact set are dense over continuous functions over compact sets, which themselves are dense over all L^p spaces. Hence in the compact case, for real or complex functions, it suffices to define our function over a the polynomials. For the case of locally compact spaces, Note that smooth bump functions are dense in L^p, and since they have compact support a polynomial sequnce uniformally converges to them. Hence, we may define a lot of operators on different real/complex functions spaces over locally compact spaces using polynomals.

For future reference, it will be important to remember that all normed spaces can be completed and that the norm between a normed space X and its completion $\overline{X}$ is isometric:

Proposition 5.1.16: Unique Extension To Banach Space

For each normed space X, there is a Banach space $\overline{X}$, uniquely determiend up to isometric isomoprhism, such that $\overline{X}$ contains X as a dense subspace

***Proof*:**
p. 47 Analysis now

An important concept in any infinite dimensional space is to figure out which sets are compact. For example, you may recall in real analysis that we proved the Stone-Weistrass theorem, or even simpler that in $\mathbb{R}^n$ a set is compact if it is closed and bounded. The following shows that subsets that are “almost finite-dimensional” are compact.

Proposition 5.1.17: Compact Sets In Banach Spaces

Let $K \subseteq V$ be a subset of a Banach Space V. Then the following are equivalent:

1. K is compact
2. K is sequentially compact
3. K is closed and bounded, and for every $\epsilon > 0$, K lies in the ϵ-neighborhood

$$\{x \in V \mid \|x - y\| < \epsilon \text{ for some } y \in W\}$$

   of a finite dimensional subspace W of V.

Suppose further that there is a nested sequence $V_1 \subseteq V_2 \subseteq \cdots$ of finite dimensional subspaces of V such that $\bigcup_{n=1}^{\infty} V_n$ is dense. Then the following statement is equivalent to the first three:

4. K is closed and bounded, and for every $\epsilon > 0$, there exits an n such that K lies in the ϵ-neighborhood of V_n

***Proof*:**
exercise

Example 5.1.18: Compact Sets In Banach Spaces

Let $1 \leq p < \infty$ and take a set $K \subseteq \ell^p(\mathbb{N})$. In order for it to be compact in the norm topology, it needs to be closed, bounded, and also uniformly p^{th}-power integrable at spatial infinity, that is for every $\epsilon > 0$, there exists an $n > 0$ such that

$$\left(\sum_{m>n} |f(m)|^p \right)^{1/p} \leq \epsilon$$

for all $f \in K$. This shows that the moving bump set is not uniformly p^{th} power integrale and thus not a compact subset, even though it's closed and bounded.

This condition doesn't translate to the continuous equivalent $L^p(\mathbb{R})$. We also need some uniform integrability at a "finte scale", which would be described using harmonic analysis tools like Fourier transforms, and hence we simply mention it here without developing it further.

Given the Banach space $\text{End}_k(V)$, it is natural have a second binary operation of composition and ask if it is also continuous (hence bounded). More generally, if $T \in B(X, Y)$, and $S \in L(Y, Z)$, then

$$\|ST(x)\| \leq \|S\|\|T(x)\| \leq \|S\|\|T\|\|x\|$$

and so $ST \in B(X, Z)$. In particular, that makes $B(X, X) = \text{End}_k(X)$ into an algebra, and when X is complete, $B(X, X)$ is a *Banach Algebra*. More generally:

Definition 5.1.19: Banach Algebra

Let X be a normed k-algebra where addition, multiplication, and scalar multiplication are continuous with respect to the induced topology. Then if X is complete, X is a *Banach Algebra*.

For the multiplication to be continuous, it suffices to check that $\|xy\| \leq \|x\|\|y\|$ (can you see why?). If X has a multiplicative unit, it is called *unital*. Not all Banach algebra's have a multiplicative unit, $C_0(X)$ where X is a locally compact (hausdroff) space and $C_0(X)$ is the set of functions from X to $\mathbb{R}$ (or $\mathbb{C}$) that vanish at infinity is *not* unital. On the otherhand, $C_b(X)$, the set of bounded functions with pointwise operations with the supremum norm *is* unital. We will work more with Banach Algebras later.

The Local Geometry of Banach Spaces

While infinite-dimensional Banach spaces can be geometrically wild (as we will see with L^1 and L^∞), their finite-dimensional subspaces exhibit a surprising regularity. We previously noted that all norms on a finite-dimensional space are equivalent. The following theorem, a cornerstone of the "Local Theory of Banach Spaces," makes a much stronger claim: Euclidean geometry is "locally universal".

Theorem 5.1.20: Dvoretzky's Theorem

Let $(X, \|\cdot\|)$ be an infinite-dimensional Banach space. For every integer $n \geq 1$ and every $\epsilon > 0$, there exists a subspace $E_n \subset X$ with $\dim(E_n) = n$ such that E_n is $(1+\epsilon)$-isomorphic to the Euclidean space ℓ_2^n.
In terms of the Banach-Mazur distance, this means:

$$d(E_n, \ell_2^n) = \inf \|T\|\|T^{-1}\| \leq 1 + \epsilon$$

where the infimum is taken over all isomorphism $T : E_n \to \ell_2^n$.

Geometrically, this is often described as the "slicing problem." If you visualize the unit ball of a Banach space X as a high-dimensional convex body K, Dvoretzky's theorem states that there is always a slice through the center of K (a subspace) that is nearly spherical.

This theorem highlights a tension between local and global structure:

1. *Globally*, X might be very far from a Hilbert space (like ℓ_∞ or the Enflo space).
2. *Locally*, X contains "copies" of Hilbert spaces of all finite dimensions.

Thus, while the "global" geometry of Banach spaces is a vast zoo of structures, the "atomic" local geometry is always Euclidean.

Exercise 5.1.1

1. Let $A, B \subseteq V$ are two closed subspaces. Show that $A + B$ need not be a closed subspace! However, if one of A or B is finite dimensional, then $A + B$ is closed.

2. Show that the open unit ball in X maps onto (by π) the open unit ball of X/Y, but a similar statement about closed unit balls holds only in special cases.

3. For the categorically inclined, it should be verified that the topology on X/Y is the quotient topology corresponding to the equivalent relation $\sim$ where $x_1 \sim x_2$ if and only if $x_1 - x_2 \in Y$.

5.2 Baire Category Theorem

In this section, we introduce some of the basic tools that allow us to work with functions between Banach spaces. Of keen interest to us is the *open mapping theorem*, the *closed graph theorem*, and the *uniform boundedness theorem*. The open mapping theorem make's it sufficient that a surjective continuous linear operator is open, which in particular means that if it is also bijective it is an isomorphism. The closed graph theorem allows us to show that a linear operator $T : X \to Y$ is continuous if the graph $\Gamma(T)$ is closed, i.e. if all sequences in the graph converge. This will lead to great simplifications in showing a function is continuous. Finally, the uniform boundedness theorem will give a powerful criterion for a family of operators that converge pointwise to converge uniformally.

The underpinning results we need are about dense sets. Let's say we have a countable family of dense sets, which can be thought of as our "generalized basis". Then it is natural that their intersection may be empty, for example taking linearly independent transcendental elements $r_1, r_2, ... \in \mathbb{R}$, then $\bigcap_r \mathbb{Q} + r = \emptyset$. What we need is to insure that the intersection is non-empty. For that, we require that the limit of an infinite intersection exists (ex. our space is LCH or a complete metric space), and that our sets have a nice property that insures non-emptyness. The following deals with the case of the space being a complete metric space:

Proposition 5.2.1: Baire Category Theorem I

Let (A_n) be a sequence of open dense subset of a complete metric space (X, d). Then $\bigcap_n A_n$ is dense in X.

***Proof*:**
We want to show that $\bigcap_n A_n$ is dense, so for every ball $B_0 \subseteq X$, $B_0 \cap \bigcap_n A_n \neq \emptyset$.

Let B_0 be any closed ball in X with radius $r > 0$. Then since A_1 is dense in X and is open, the set $A_1 \cap \operatorname{int}(B_0)$ contains a closed ball B_1 with radius $2^{-n_1} r$ for n_1 sufficiently large. Since A_2 is an open set and B_1 is a closed ball, we can repeat this with $A_2 \cap \operatorname{int}(B_1)$ and find a closed ball B_2 with radius $2^{-n_2} r$ with $n_2 > n_1$ sufficinetly large. By induction, we get a sequence of closed balls (B_n) in X such that $B_n \subseteq A_n \cap \operatorname{int}(B_{n-1})$, each with radius $< 2^{-n_k} r$ for every k. Since X is complete (and Hausdorff, since metric topological are normal), there exists a unique point $\{x\}$ such that

$$\{x\} = \bigcap_n B_n \subseteq B_0 \cap \left(\bigcap A_n\right)$$

This shows that $\bigcap A_n$ intersects every nontrivial ball X, which implies density, completing the proof.

Note that this is a fully topological proof, and hence is invariant under homeomorphism. In particular, completeness is *not* a topological property (it is a uniform property), however $(0, 1) \cong_{\mathbf{Top}} \mathbb{R}$.

Furthermore, since each U_n is open, it's compliment must be nowhere dense. For if $O \subseteq \overline{U_n^c} = U_n^c$ (by closedness) for some non-empty open set O, then

$$X \supsetneq O^c = \overline{O^c} \supset \overline{U_n} = X$$

a contradiction. This leads to the following very useful corrollary:

Corollary 5.2.2: Countable union of Empty Interior Sets

The countable union of closed sets with empty interior has empty interior. If X is a complete metric space, $X \neq \bigcup_i \overline{X_i}$ for nowhere dense sets X_i.

As a consequence, if $X = \bigcup_n X_n$, then there exists some n_0 such that X_{n_0} has nonempty interior.

***Proof*:**
If (A_n) is a sequence of open dense sets, (A_n^c) is a sequence of nowhere dense sets (the interior of their closure is empty). By the above, their union is a nonempty set with empty interior.

It follows that since nowhere dense sets have closure of nowhere dense sets have empty interior, but a complete metric space has non-empty open sets, X cannot be the union of such sets.

This consequence is very interesting, since it shows that through a countable process, we require a set with nonempty interior to form our entire space. This motivates the following definition:

Definition 5.2.3: Meager Set

Let X be a topological space. Then $A \subseteq X$ is said to be *meager* (or of first-category) if it is a countable union of nowhere dense sets. The compliment of a measure set is a *residual* (or co-meager, or of second-category).

Example 5.2.4: Interesting Meager Sets

1. The Cantor set is meager in $\mathbb{R}$
2. Take $C([0,1])$ with the uniform convergence topology. Let E be the set of all continuous functions that are differentiable somewhere. Then E is in fact a *meager* set in $C([0,1])$. To see this, take:

$$E_n = \{f \mid \exists x_0 \in [0,1],\ f(x) - f(x_0)| < n|x - x_0|\ \forall x \in [0,1]\}$$

   Then all differentiable functions are contained in $\bigcup_n E_n$, and it should be shown that this is a closed set and that there exists functions g such that $\|f - g\|_u < \epsilon$ for any $f \in E_n$ but $g \notin E_n$. Hence, we satisfied the conditions of Bare-Category theorem. As a consequence, the set of continuous nowhere differentiable functions is risidual! Thus more generally, smooth and analytic functions are meager.

Proposition 5.2.5: Baire Category Theorem II

Let $T : V \to W$ be a bounded linear operator between two Banach spaces, $T \in B(V, W)$. Then if the image of the unit ball $B_1(0)$ is dense in some ball $B_r(0)$ with $r > 0$, $\overline{T(B_1(0))} = \overline{B_r(0)}$, then

$$B_{(1-\epsilon)r}(0) \subseteq T(B_1(0))$$

for every $\epsilon > 0$

In a sense, if $T(B_1(0))$ is dense in some ball $B_r(0)$, then it is the "supremum" of $B_r(0)$

***Proof*:**
Let $U = T(B_1(0))$. Since $T(B_1(0))$ is dense in $B_r(0)$, for any $y \in B_r(0)$ and $\epsilon > 0$, there is a $y_1 \in U$ such that $\|y - y_1\| < \epsilon r$. Consider now ϵU. This set is dense in $B_{\epsilon r}(0)$. Hence, there exists a $y_2 \in \epsilon U$ such that $\|(y - y_1) - y_2\| < \epsilon^2 r$. Continuing by inductiong, we get a sequence (y_n) where $y_n \in \epsilon^{n-1} U$ and

$$\left\| y - \sum_{k=1}^{n} y_k \right\| < \epsilon^n r$$

Choose now $x_n \in B_1(0)$ such that $T(x_n) = y_n$ and $\|x_n\| \leq \epsilon^{n-1}$. Then $\sum x_N = x \in X$ converges, and by definition $T(x) = y$. Since $\|x\| \leq \sum \epsilon^{n-1} = (1-\epsilon)^{-1}$ [a] we have

$$(1-\epsilon)^{-1} U \supseteq B_r(0)$$

completing the proof.

[a] recall that $1 - b^n = (1-b)(1 + b + b^2 + \cdots + b^{n-1})$

Theorem 5.2.6: Open Mapping Theorem

Let X and Y be Banach spaces and $T \in B(X, Y)$ be a bounded linear operator with $T(X) = Y$. Then T is an open map (i.e. it is a surjective bicontinuous map)

***Proof*:**
Since T is surjective:

$$Y = T(X) = \bigcup_n \overline{T(_n(0))}$$

Then by corollary 5.2.2, not all closed sets $\overline{TB_n(0)}$ have an empty interior. Thus, there is an n and a ball $B_\epsilon(y)$ such that

$$B_\epsilon(y) \subseteq \overline{T(B_n(0))}$$

Thus, $T(B_1(0))$ is dense in $B_{n^{-1}\epsilon}(y)$. Since

$$2B_{n^{-1}\epsilon}(0) \subseteq B_{n^{-1}\epsilon}(y) - B_{n^{-1}\epsilon}(y)(???)$$

$T(B_1(0))$ is also dense in $B_{n^{-1}\epsilon}(0)$. By Bare Category Theorem II, $B_\delta(0)$ is contained in $T(B_1(0))$ for every $\delta < n^{-1}\epsilon$. Since every open set $U \subseteq X$ is a union of balls, we conclude from lineartiy of T that $T(U)$ conain a neighborhood[a] around each of its points, thus $T(U)$ is open, completing the proof.

[a]in fact closed balls

Corollary 5.2.7: Bijective And Continuous

Let $T : X \to Y$ be a bijective bounded operator. Then it has a bounded inverse, that is T is in fact a Banach-space isomorphism

The following also let's us recover a result similar to the one showing us that there is only one norm on a finite dimensional vector space up to equivalence, namely no (non-equivalent) norm on a Banach space can dominate another:

Corollary 5.2.8: Equivalent Banach Spaces

Let X be a vector-space, and a Banach space under two norms $\|\cdot\|_a$ and $\|\cdot\|_b$. Then if $\|\cdot\|_a \leq \alpha\|\cdot\|_b$ for some $\alpha > 0$, then there is a $\beta > 0$ such that $\|\cdot\|_b \leq \beta\|\cdot\|_a$

***Proof*:**
Take $\mathrm{id} : (X, \|\cdot\|_b) \to (X, \|\cdot\|_a)$. This is certainly bijective, continuous by the boundedness assumption, and open by the open mapping theorem. Hence it is a homeomorphism, meaning it has a bounded inverse, giving the desired result.

The next powerful result we get is a quick way of proving a function is bounded:

Theorem 5.2.9: Closed Graph Theorem

Let $T : X \to Y$ be an operator between Banach spaces. Then if the graph

$$\Gamma(T) = \{(x, y) \in X \times Y \mid T(x) = y\}$$

is closed in $X \times Y$,then T is bounded

***Proof*:**
Using the ∞-norm on $X \times Y$, assume that $\Gamma(T)$ is closed in $X \times Y$. Then the projection map $\pi_1 : \Gamma(T) \to X$ $\pi(x, T(x)) = x$ is a bijective norm decreasing bijective map, and hence has a bounded inverse π_1^{-1}. The other projection $\pi_2 : \Gamma(T) \to D$ given by $\pi(x, T(x)) = T(x)$ is also a bijective norm decreasing. Hence, we have:

$$T = \pi_2 \circ \pi^{-1}$$

showing that T is bounded, completing the proof.

The key take-away is that if we are verifying the continuity of a linear map $T : X \to Y$ (so $x_n \to x$ implies $T(x_n) \to T(x)$), we just need to verify the closedness of the graph. The closedness of the graph implies that if $x_n \to x$ and $T(x_n) \to y$, then $T(x) = y$. Notice how this simplifies proving a function is continuous, since we did not need to show that $\lim_n T(x_n) = T(x)$, simply that it conveges to *something*, namely if $T(x_n) \to y$, then $T(x) = y$. Note how completeness is crucial in both these theorems (exercise 29-31 folland for counter-examples).

The converse of the closed graph theorem, that if $T : X \to Y$ is bounded between Banach spaces, then $\Gamma(T)$ is closed, is elementary: let's say T is bounded and that $T(x_n) \to y$. Since $x_n \to x$, $\lim_n T(x_n) = T(x)$. Then:

$$\begin{aligned} \|T(x) - y\|_Y &\leq \|T(x) - y + T(x_n) - T(x_n)\|_Y \\ &\leq \|T(x - x_n)\|_Y + \|T(x_n) - y\|_Y \\ &\leq C\|x - x_n\|_X + \|T(x_n) - y\| \xrightarrow{n\to\infty} 0 \end{aligned}$$

hence, $\|T(x) - y\| = 0$, implying $T(x) = y$ and showing $\Gamma(T)$ is closed.

Finally, the following result is a way to pass from pointwise boundedness to uniform boundedness, only requiring the completness of our domain! It can be thought of as one of the "pillars" of functional analysis:

Theorem 5.2.10: Principle Of Uniform Boundedness

Let X, Y be normed vector spaces and $A \subseteq L(X, Y)$. Then if X is a Banach space and $\sup_{T\in A} \|T(x)\| < \infty$ for all $x \in X$, then $\sup_{T\in A} \|T\| < \infty$

This theorem is also called by some the BanachSteinhaus Theorem.

***Proof*:**
If A is a constant function, then we're done, so assume A contains nonconstant functions. We want to upgrade the result applies uniformally for all x inside a ball so that $\sup_{T\in A} \|T\| < m$ for some $m < \infty$. Let

$$E_n = \left\{ x \in X \;\middle|\; \sup_{T\in A} \|T(x)\| \leq n \right\} = \bigcap_{T\in A} \{x \in X \mid \|T(x)\| \leq n\}$$

Then certainly the E_n's are closed and $X = \bigcup_n E_n$. Then by the Baire Category Theorem, there must exist some $n \in N$ such that E_n has nonempty interior. Hence, for some $x_0 \in E_n$ and $r > 0$, $B_r(x_0) \subseteq E_n$. Since E_n is closed, $\overline{B_r}(x_0) \subseteq E_n$.

Now, consider $u \in X$ such that $\|u\| \leq 1$ and take any $T \in A$. Then:

$$\begin{aligned} \|T(u)\|_Y &= \epsilon^{-1}\|T(x_0 + \epsilon u)\| - T(x_0)\|_Y \\ &\leq \epsilon^{-1}(\|T(x_0 + \epsilon u)\|_Y + \|T(x_0)\|_Y) \\ &\leq \epsilon^{-1}(n + n) \end{aligned}$$

Hence, taking the supremum over u in the unit ball of X and over $T \in F<$ we get:

$$\sup_{T\in F} \|T\|_{B(X,Y)} \leq 2\epsilon^{-1} n < \infty$$

completing the proof.

If X is simply a normed vector-space, we may keep this theorem by simply taking the set suprum of the set of nonmeager points. The key realization is that $\sup_{T\in A} \|T(x)\|$ is a different supremum for each x, but is finite forall x! That is, we are not working with $\sup_{\substack{T\in A \\ x\in X}} \|T(x)\|$ since that would make the result trivial:

$$\sup_{T\in A}\|T\| = \sup_{\substack{T\in A\\ \|x\|=1}} \|T(x)\| \leq \sup_{\substack{T\in A\\ x\in X}} \|T(x)\|$$

This is what makes the Uniform boundedness principle so strong, we got a *uniform* bound by having a *local* bound for all x!

Corollary 5.2.11: Net Convergence

Let $(T_\lambda)_{\lambda\in\Lambda} \subseteq B(X,Y)$ be a net set $(T_\lambda(x))_{\lambda\in\Lambda}$ is bounded and converges in Y or every $x \in X$. Then there is a $T \in B(X,Y)$ such that $T_\lambda(x) \to T(x)$ for every $x \in X$.

***Proof*:**
Since $T_n(x) \to T(x)$, $\sup_n \|T_n(x)\| < \infty$ for all $x \in X$. Hence, by the uniform boundedness principle, $\sup_n \|T_n\| < \infty$, that is for all n:

$$\sup_{\|x\|\neq 0} \frac{\|T_n(x)\|}{\|x\|} \leq \sup \|T_n\| = C$$

Hence, $\|T_n(x)\| \leq C\|x\|$ where C is independent of n. Taking $n \to \infty$ we get:

$$\|T(x)\| \leq C\|x\|$$

But this is the condition for $T \in L(X,Y)$, as we sought to show.

If our net is a sequence, then convergence of eac hsequence $(T_n(x))$ for all $x \in X$ automatically implies boundedness. It also suffices in the principle of uniform boundedness to know that each net is bounded and that the net is convergent for a dense set of elements.

5.3 Dual Spaces

Let V be a Banach space over k (k being either $\mathbb{R}$ or $\mathbb{C}$). Since we are in the category of Banach spaces, we'll consider $V^* := \mathrm{Hom}(V,k)$ to the collection of all continuous linear functionals (i.e. bounded linear functionals). If V is finite dimensional, then it is clear what the structure of V^*, namely since $V \cong V^*$. In infinite dimensions, it is in fact hard to identify when a funcitonal is continuous and bounded. Famously, d/dx is an unbounded operator, and $\int$ is bounded given an appropriate restriction of the domain. It would be good if we can find some easier type of function that we know can be extended to a functional. We start by doing just that, in particular we start by proving the Hahn-Banach extension theorem, which shows that there exists "enough" linear functions in X^* to make their study intersting (this theorem is the equivalent of the Uryohn's Theorem from topology).

The key "simpler" function is the following:

Definition 5.3.1: Linear Subfunctional

Let $m : V \to \mathbb{R}$ be a subatitive function $m(x+y) \leq m(x)+m(y)$ that is positive homogeneous $m(tx) = tm(x)$ for all $t \geq 0$ in $\mathbb{R}$. Then m is called a *linear subfunctional.*

Note that it is not a semi-norm since homogeneity only needs to hold for positive scalars. On the otherhand, every semi-norm (and hence norms) are example of minkowski functionals. Note that the result works just as well over real or complex fields:

Proposition 5.3.2: Converting Complex And Real Linear Functionals

Let X be a vector space over $\mathbb{C}$. If f is a complex linear functional on X, and $u = \mathrm{Re} f$, the nu is a real linear functional, and $f(x) = u(x) - iu(ix)$ for all $x \in X$. Conversely, if u is a real lienar functional on X and $f : X \to \mathbb{C}$ is defiend by $f(x) = u(x) - iu(ix)$, then f is a complex linear functional. In this case, if X is normed, $\|u\| = \|f\|$

***Proof*:**
If f is complex linear then $u = \mathrm{Re} f$ is clealry real linear and $\mathrm{Im} f(x) = -\mathrm{Re}[if(x)] = -u(ix)$, so $f(x) = u(x) - iu(ix)$. On ther other hiand, if u is real linea, then $f(x) = u(x) - iu(ix)$ is clearly R liner and

$$f(ix) = u(ix) - iu(-x) = u(ix) + iu(x) = if(x)$$

thus f is $\mathbb{C}$-linaer. Finally, if X is norme,d since $|u(x)| = |\mathrm{Re} f(x)| \leq |f(x)|$, $\|u\| \leq \|f$
. On the other ahdn, if $f(x) \neq 0$, let $\alpha = \overline{\mathrm{sgn}(f(x))}$. Then

$$|f(x)| = \alpha f(x) = f(\alpha x) = u(\alpha x)$$

since $f(\alpha x)$ is real. Hence $|f(x)| \leq \|U\|\|\alpha x\| = \|u\|\|x\|$, thus $\|f\| \leq \|u$
, hence

$$\|f\| = \|u\|$$

completing the proof.

Theorem 5.3.3: Hahn-Banach Extension theorem

Let X be a real vector space, p a sublinear functional on X, $M \subseteq X$ a subspace of X, and f a linear funtional on M such that $f(x) \leq p(x)$ for all $x \in M$. Then there exitss a lienar functional F on X such that $F(x) \leq p(x)$ for all $x \in X$ and $F|_M = f$

***Proof*:**
We shall start with a "base case" and then use Zorn's lemma (in this context, think of it as transfinite induction). Let $x \in X \setminus M$. We shall extend f to a functional g on $M + \mathbb{R}x$ that satisfies $g(y) \leq p(y)$ on $M + \mathbb{R}x$. First, first any $y_1, y_1 \in M$, we have

$$f(y_1) + f(y_2) = f(y_1 + y_2) \leq p(y_1 + y_2) \leq p(y_1 - x) + p(x + y_2)$$

where x is our above chosen $x \in X$. Re-arranging, we get:

$$f(y_1) - p(y_1 - x) \leq p(x + y_2) - f(y_2)$$

Since this is true for all $y_1, y_2 \in M$, we have:

$$\sup_{y\in M}\{f(y) - p(y-x)\} \leq \inf_{y\in M}\{p(x+y) - f(y)\}$$

Thus, we can find an α satisfying:

$$\sup_{y\in M}\{f(y) - p(y-x)\} \leq \alpha \leq \inf_{y\in M}\{p(x+y) - f(y)\}$$

Define

$$g : M + \mathbb{R}x \to \mathbb{R} \qquad g(y + \lambda x) = f(y) + \lambda\alpha$$

Clearly g is linear, $g|_M = f$, and on M $g(y) \leq p(y)$ (equivalently when $\lambda = 0$). When $\lambda > 0$, then for any $y \in M$:

$$g(y + \lambda x) = \lambda[f(y/\lambda) + \alpha] \leq \lambda[f(y/\lambda) + p(x + (y/\lambda)) - f(y/\lambda)] = p(y + \lambda)$$

and when $\lambda = -\mu < 0$:

$$g(y + \lambda x) = \mu[f(y/\mu) - \alpha] \leq \mu[f(y/\mu) - f(y/\mu) + p((y/\mu) - x)] = p(y + \lambda x)$$

Thus, $g(z) \leq p(z)$ for all $z \in M + \mathbb{R}x$.

But now, notice that g satisfies the same condition as in the hypothesis: g is a linaer functional on a subspace $M + \mathbb{R}x \subseteq X$ such that $g(z) \leq p(z)$ for all $z \in M + \mathbb{R}x$. We can thus finish off by noticing that we can create a partial order given by inclusion of the domains (since the domains resrct appropraitely), and we clearly see that this partial is closed under unions of chains. Hence, it has a maximal, which through an easy proof by contradiction can be shown to be X, completing the proof.

Before moving on to some of the fascinating consequence, we first insure the we understand how to move this result to complex functionals. For order to be well-defined, we must insure that we are using seminorms instead of general lienar subfunctionals, that is if p is a seminorm and $f : X \to \mathbb{R}$ is lienra, then the inequality $f \leq p$ is equivalent to $|f| \leq p$ since$|f(x)| = \pm(x) = f(\pm x)$ and $p(-x) = p(x)$.

Proposition 5.3.4: Complex Hahn-Banach Theorem

Let X be a vetor space over $\mathbb{C}$, p a seminorm on X, $M \subseteq X$ a subspace of X and f a compelx lienar functional on M such that $|f(x)| \leq p(x)$ for all $x \in M$. Then there exists a compelx linear functional F on X such that $|F(x)| \leq p(x)$ for all $x \in X$ and $F|_M = f$

***Proof*:**
Let $u = \mathrm{Re}f$. By The Hahn-Banach Theorem there exists a linear extension of u to X such that $|U(x)| \leq p(x)$ for all $x \in X$. Let $F(x) = U(x) - iU(ix)$. Then F is a complex linear extension of f, and by proposition 5.3.2, if $\alpha = \overline{\mathrm{sgn}(f(x))}$ we have $|f(x)| = \alpha F(x) = F(\alpha x) = U(\alpha x) \leq p(\alpha x) = p(x)$, completing the proof.

Proposition 5.3.5: Hahn-Banach Consequences

Let X be a normed vector-spae. Then:

1. If $M \subseteq X$ is a closed subpsace and $x \in X \setminus M$, there exists a $f \in X^*$ such that $f(x) \neq 0$ and $f|_M = 0$. In particulr, if $\delta = \inf_{y \in M} \|x - y\|$, there exists an f such that
$$\|f\| = 1 \qquad f(x) = \delta$$
2. If $0 \neq x \in X$, there exists $f \in X^*$ such that $\|f\| = 1$ and $f(x) = \|x\|$
3. The bounded lienar functionals on X seperate points
4. if $x \in X$, the double dual $\hat{x} : X^* \to \mathbb{C}$ given by $\hat{x}(f) = f(x)$ is alienar isometry from X to X^{**}

***Proof*:**

1. $M \subseteq X$ be a closed subspace and $\delta = \inf_{y \in M} \|x - y\|$. Define
$$f : M + \mathbb{C}x \qquad f(y + \lambda x) = \lambda\delta$$
Then $f(x) = \delta$, $f|_M = 0$, and for $\lambda \neq 0$ by the infinum proeprty:
$$|f(y + \lambda x)| = |\lambda|\delta \leq |\lambda|\|\lambda^{-1}y + x\| = \|y + \lambda x\|$$
Hence, taking $p(x) = \|x\|$ to be the norm (which is a lienar subfunctional defined on X), by the Hahn-Banach theorem F on X where $F(x) = \delta$ and $F|_M = 0$. Finally, if $\|y + \lambda x\| = 1$, then certainly the supremum of all such values when plugged into F must be 1, hence $\|F\| = 1$
Alternatively, apply the result from part (a) to X/Y, then note that functionals in $(X/Y)^*$ may be regarded as elements in X^* that annihilate Y
2. In the above, let $M = \{0\}$ be the trivial vector-space and define $f(x) = \|x\|$.
3. If $x \neq y$, by (b) there exists a $f \in X^*$ such that $f(x - y) \neq 0$, hence $f(x) \neq f(y)$
4. Certainly $\hat{x}$ is a lienar functioanl onX^* and $x \mapsto \hat{x}$ is linaer. Furthermore
$$|\hat{x}(f)| = |f(x)| \leq \|f\|\|x\|$$
hence $\|\hat{x}\| \leq \|x\|$. On the other hand, by (b), we get $\|\hat{x}\| \geq \|x\|$, hence we have an isometry

The first (and second) point can be thought of as a generalization of Uryshon's lemma, and also inspires the following definition:

Definition 5.3.6: Annihilator

For a subspace $Y \subseteq X$ of a normed space X, the *annihilator* of Y is defined as:

$$Y^{\perp} = \{\varphi \in X^* \mid \varphi(y) = 0, \ \forall y \in Y\}$$

Similarly, we define the annihilator of a subspace $Y^* \subseteq X^*$ to be

$$(Y^*)^{\perp} = \{x \in X \mid \varphi(x) = 0, \ \forall \varphi \in Y^*\}$$

It is immedaite that $Y \subseteq (Y^{\perp})^{\perp}$, and it follows that in fact $Y = (Y^{\perp})^{\perp}$. for every norm closed subspace $Y^* \subseteq X^*$.

The third point is useful if we have setup a senario where for all $f \in X^*$:

$$f(T(x)) = f(y) \qquad f \in X^*$$

Since the f separate points, it must be that

$$T(x) = y$$

in other words, separation of points can be thought of as as sort of "injectivity". As an exercise, show that if $T : X \to Y$ is a linear operator and for eac $f \in Y^*$, $f \circ T \in T^*$, then T is bounded (hint: used closed graph theorem and the above remark).

The 4th point inspires a quick look at double duals. If we let $\hat{X} = \{\hat{x} \mid x \in X\}$ and let X^{**} be the double dual, then since X^{**} is always complete, the closure of $\hat{X}$ in X^{**}, $\overline{\hat{X}}$ is banach space, and $x \mapsto \hat{x}$ embeds X into $\overline{\hat{X}}$ as a dense subspace. The sapce $\overline{\hat{X}}$ is called the *completion* of X, and if X Is itself Banah then $\overline{\hat{X}} = \hat{X}$. If X is finite dimensional, then $\hat{X} = X^{**}$, since they have the same dimension. This is not guaranteed in infinite dimension, hecne if $\hat{X} = X^{**}$, we shall call X a *reflexive space*. For example, $C_c^{\infty}([0,1])$ will be shown to not be reflexive. We shall see later that L^p spaces are reflexive.

5.4 Weak Topologies and TVS's

(This post seems like an excellent post to follow! link here)

There are many topologies on which the operations are continuous (i.e. + and scalar multiplication is continuous) that do not arrise from norms. The topology on a vector-space induced by a norm is in fact quite strong. In general, such a space (in infinite dimensions) doesn't have many compact set, and so we loose many of the nice results of working on comapct or pre-compact sets. This is naturally unfavourable in analysis, since many arguments rely on properties of sequences converging. Looking at functional analysis as the study of sequences, norms induce a particular definition of convergence: $\|x - x_n\|_V \to 0$. However, there are many other notions of convergence that are weaker that are quite useful, for example:

1. pointwise convergence
2. convergence in measure

3. smooth convergence
4. convergence on compact sets

None of these types of convergence are induced by a norm. We will instead be induced new types of topologies on our vector-spaces. The two types of topologies we will usually impose on V will be known as the *weak topology* and the *weak* topology*. We will define analogous topologies on $\mathrm{Hom}_k(V, W)$ which somewhat confusingly will be named *strong operator topology* and *weak operator topology*.

Definition 5.4.1: Topological Vector Space (TVS)

Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$. Then V is a *topological vector space* (TVS) if the operation of addition and scalar multiplication are continuous. A topological vector-space is said to be *locally convex* if there is a base for the topology consisting of convex sets.

It should be verified that the translation map $x \mapsto x_0 + x$ and the dilation map $x \mapsto \lambda x$ are both homeomorphissm for any scalar λ and vector x_0.

Example 5.4.2: Topological Vector Spaces

1. Let V be a normed vector-space with the topology induced by the norm. Then $\|c \cdot x\| \leq |c|\|x\|$ (in fact $=$) and $\|+(x,y)\| \overset{!}{\leq} 2(\|x\| + \|y\|)$ where $\overset{!}{\leq}$ comes from the triangle inequality and the fact that $\|(x,y)\| = \max(\|x\|, \|y\|)$. Hence, V is a topologicla vector space. If $\|-\|$ was a quasi-norm so that we replace the triangle inequality with $\|x+y\| \leq K(\|x\| + \|y\|)$ then a quasi-normed space V is still a topological vector space.
2. Let V be a semi-normed space so that $\|x\| = 0$ does not necessarily imply $x = 0$. Then V is a topological vector space with the topology induced by the semi-norm. V is Hausdroff if and only if the semi-norm is a norm.
3. Any subbspace $W \subseteq V$ of a topological vector space V is a topological vector space with the subspace topology. Similarly for products and quotients.
4. Let V be a vector-space and $(\mathcal{T}_\alpha)_{\alpha \in A}$ be a collection of topologies on V that makes V a topological vector space. Let $\mathcal{T} = \langle \mathcal{T}_\alpha \rangle_{\alpha \in A}$ be the topology generated by $\bigcup_{\alpha \in A} \mathcal{T}_\alpha$ (this is the weakest topology that contains each $\mathcal{T}_\alpha$). Then $(V, \mathcal{T})$ is also a topological vector space. Importantly, a sequence (resp. net) $x_n \to x$ in $\mathcal{T}$ if and only if $x_n \to x$ in $\mathcal{T}_\alpha$ for all $\alpha \in A$ (recall that $x_n \to x$ in a general topology space if for any neighborhood U of x, there exists an N such that for all $n \geq N$, $x_n \in U$). Commonly, we will have $(\mathcal{T}_\alpha)_{i=1}^\infty$ be generated by countable collection of semi-norms. A complete space generated by a countable collection of semi-norms is called a *Fréchet space*.
5. (Pointwise Convergence) Let X be a set and $\mathbb{C}^X$ the collection of all complex valued functions $f : X \to \mathbb{C}$. Then $\mathbb{C}^X$ is a complex-vector-space. For each $x \in X$, define the seminorm:

$$\|f\|_x := |f(x)|$$

Then the topology generated by all of these seminorms is the *topology of pointwise convergence* on $\mathbb{C}^X$ (it is also the product topology on this space). A sequence $(f_n) \subseteq \mathbb{C}^X$ converges to

f in this topology if and only if it converges pointwise. Note that if X has more than one point, then none of the semi-norms individually generate a Hausdroff topology, but taken all together the induced topology is Haudsorff.

6. (Uniform Convergence) Let X be a topological space and $C(X)$ the set of complex-valued continuous function $f : X \to \mathbb{C}$. If X is not compact, then in general we would not expect functions in $C(X)$ to be bounded, so the sup-norm will not necessarily make $C(X)$ into a normed vector-space. Nonetheless, we can still define "balls" $B_r(f) \subseteq C(X)$ by:

$$B_r(f) := \left\{ g \in C(X) \;\middle|\; \sup_{x \in X} |f(x) - g(x)| \leq r \right\}$$

these do form a topological structure on $C(X)$, but not a topological vector space structure since scalar multiplication is no longer continuous. Nonetheless, we have that $(f_n) \subseteq C(X)$ converges in this topology to a limit $f \in C(X)$ if and only if f_n converge uniformaly to f. Thus $\sup_{x \in X} |f_n(x) - f(x)|$ is finite for sufficinetly large n and converges to 0 as $n \to \infty$.

7. (Uniform Convergence on Compact sets) Let X and $C(X)$ be as in the above example. For every compact subset $K \subseteq X$, define a seminorm:

$$\|f\|_{C(K)} = \sup_{x \in K} |f(x)|$$

The topology generated by all these seminorms as k ranges over all compact subsets of X is called the *topology of uniform convergence on compact sets*. It is stronger than the topology of pointwise convergence but weaker than the topology of uniform convergence, and is frequently used in complex analysis. A sequence $(f_n) \subseteq C(X)$ converges to $f \in C(X)$ in this toplogy if and only if f_n converges uniformly to f on each compact set. Is the resulting space a topological vector space?

8. (Smooth Convergence) Let $C^\infty([0,1])$ be the space of smooth functions $f : [0,1] \to \mathbb{C}$. Then we can define the C^k-norm on this space for any non-negative integer k by the formula:

$$\|f\|_{C^k} := \sum_{i=1}^{k} \sup_{x \in [0,1]} |f^{(i)}(x)|$$

where $f^{(i)}$ is the ith derivative of f. The topology generated by all C^k norms $k = 0, 1, 2, ...,$ is the *smooth topology.* A sequence f_n converges in this topology to a limit f if $f_n^{(i)}$ converges uniformally to $f^{(i)}$ for each $i \geq 0$. Is this a topological vector space?

9. (Convergence in measure) Let $(X, \mathcal{M}, \mu)$ be a measure space and $L(X)$ the set of measurable functions $f : X \to \mathbb{C}$. Then the sets

$$B_{r,\epsilon}(f) := \{g \in L(X) \mid \mu(\{x \mid |f(x) - g(x)| \geq r\} < \epsilon)\}$$

for $f \in L(X)$, $\epsilon, r > 0$ forms the basis for a topology that makes $L(X)$ into a topological vector space. A sequence $(f_n) \subseteq L(X)$ converges to f in this topology if and only if it converges in measure.

10. Let $[0,1]$ be given the usual Lebesgue measure. Then show that $L^\infty([0,1])$ cannot be given a topological vector space structure in which a sequence $(f_n) \subseteq L^\infty([0,1])$ converges to f in this topology if and only if it converges a.e.

11. Let V be an $\mathbb{R}$-vector space. a subset $U \subseteq V$ is called *algebraically open* if
$$\{t \in \mathbb{R} \mid x + tv \in U\}$$
is open for all $x, v \in V$. Then any open set in a topological vector space is algebraically open. Note that there are algebraically open sets that are not open (take any dense subset of S^1, say D, and $\mathbb{R} \setminus \{D\}$.) This larger set of algebraically open sets forms a topology, but this topology does not give the $\mathbb{R}^2$ structure of a topological vector space.

12. Let X be a LCH space. On $\mathbb{C}^X$, the topology of uniform convergence on compact sets is defined by the seminorms $p_K(f) = \sup_{x \in K} |f(x)|$ as K rangers over compact subsets of X. If X is σ-compact and $\{U_n\}$ are as in proposition ref;HERE (folland p. 168), this topology is defined by the seminorms $p_n(f) = \sup_{x \in \overline{U_n}} |f(x)|$. In this case, $\mathbb{C}^X$ is easly seen to be complete, so it is a Fréchet space. By proposition ref:HERE, so is $C(X)$.

13. The space $L^1_{\text{loc}}(\mathbb{R}^n)$ is a Fréchet space with the topology defined by the seminorms $p_k(f) = \int_{|x| \le k} |f(x)| dx$ (completeness follows from completness of L^1).

14. Here is another useful place where TVS are defined. Let's say we want to study a space in which d/dx is continuous (and hence bounded) on $C^\infty([0,1])$. However, this is essentially impossible to define the topology through a norm. Indeed, if $f_\lambda(x) = e^{\lambda x}$, then $(d/dx) f_\lambda = \lambda f_\lambda$, and so $\|d/dx\| \ge \lambda$ for all λ, no matter what norm is used on $C^\infty([0,1])$. There are a few ways to rectify this:
    (a) One can consider differentiation as an unbounded operator form V to W where W is a suitable banace space and X is a dense subpsac eof W (exercise 30 in Folland)
    (b) One can consider differentiation as a bounded linear map from one Banach space V to a different one W such that $V = C^k([0, 10])$ and $W = C^{k-1}([0,1])$ (execise 9)
    (c) One can consider differentiation as a continuous operator on a locally convex space V whose topology is not given by a norm. It is easy to construct families of seminorms on the space of smooth functions st differentiation becomes continuous almost by definition. For example, the seminorms:
    $$p_k(f) = \sup_{0 \le x \le 1} |f^{(k)}(x)|$$
    where $k \in \{0, 1, 2, ...\}$ makes $C^\infty([0,1])$ into a Fréchet space (the completeness is proved in exercise 9), and d/dx is continuous by proposition 5.15, since $p_k(f') = p_{k+1}(f)$. Other examples are in exer exercsie 45 in chatper 9 of Folland.

15. The space $\mathbb{R}^\infty = \varinjlim \mathbb{R}^n$ is a Fréchet space; can you see the norms?

Proposition 5.4.3: Topological Vector Space And Hausdorff

Let V be a topological vector space. Then V is Hausdroff if and only if $\{0\}$ is closed

***Proof*:**
exercise

We may ask what properties of Banach spaces extend to topological vector space's. Some nice properties of Banach spaces include the uniform boundedness principle and the open mapping theorem. These can be extended to the slighty larger class of *Fréchet spaces* that we mentioned in the examples above. We will single them out and talk about them in the next section.

If V is a topological vector space, we can take V^* to be the set of all continuous functionals from V to the base field $k \in \{\mathbb{R}, \mathbb{C}\}$. Since V has no norm, V^* has no norm topology. We can however still define interesting topologies on it. This topology is dual to a weaker topology we can put on V:

Definition 5.4.4: Weak And Weak* Topologies

Let V be a topological vector space and V^* its dual. Then

1. the *weak* topology* on V^* is the topology generated by the seminorms $\|\lambda\|_x := |\lambda(x)|$ for all $x \in V$
2. the *weak topology* on V is the topology generated by the seminorms $\|x\|_\lambda = |\lambda(x)|$ for all $\lambda \in V^*$

If V is also a normed vector-space, then the weak topology on V is weaker than the norm topology. If V is a normed space, then so is V^* and $(V^*)^*$. the weak* topology on V^* makes V^* into a topological vector space, and is a topology weaker than the weak topology on V^* (which is defined using the double dual $(V^*)^*$). When V is reflexive ($V \cong V^{**}$), then the weak and weak* topologies on V^* are in fact equivalent.

Definition 5.4.5: Weak Convergence

Let V a vector-space endowed with the weak topology. Then a sequence $(x_n) \subseteq V$ that converges in V with the weak topology is called *weakly convergent* and is said to *weakly converge* to a point x. A sequence weakly converges $x_n \to x$ if and only if

$$\lambda(x_n) \to \lambda(x) \qquad \forall \lambda \in V^*$$

and is usually denoted as $x_n \rightharpoonup x$. Similarly, $\lambda_n \rightharpoonup \lambda$ if $\lambda_n(x) \to \lambda(x)$ for all $x \in V$ (λ_n, as a function of V, converges pointwise to λ)

Proposition 5.4.6: Normed Vector Spaces Weak Topology

Let V be a normed vector-space. Then the weak topology on V and the weak* topology on V^* are both Hausdorff. In particular, we conclude that in the weak and weak* limits, when they exist they are unique

***Proof*:**
Use the Hahn-Banach Theorem

Example 5.4.7: Norm, Weak, Weak* Convergence
Let $V = c_0(\mathbb{N})$, $V^* = \ell^1(\mathbb{N})$, and $V^{**} = \ell^\infty(\mathbb{N})$. Let $e_1, e_2, ...$ be the standard basis of any of these vetor-spaces.

1. The sequence (e_i) converges weakly in V to zero, but not not converge in norm in V
2. (easy to think up some examples)

A natural generalization of the weak* topology on X^* can be imposed $B(X, Y)$ for some TVS Y. Just like for X^*, we can impose a weak topology on it, hewer there is another natural topology we may induce on $B(X, Y)$:

Definition 5.4.8: Strong And Weak Operator Topology

Let $B(X, Y)$ be a all continuous operators between Fréchet spaces. Then

1. the *weak operator topology* is the topology induced by the seminorm $T \mapsto |\lambda(T(x))|$ for all $x \in X$ and $\lambda \in Y^*$
2. If furthermore Y is a Banach space, then the *strong operator topology* (or *operator norm topolgoy*) is the topology induced by the seminorms $T \mapsto \|T(x)\|_Y$ for all $x \in X$

Hence, a sequence $(T_n) \subseteq B(X, Y)$ converges in the strong operator topology to a limit T if and only if $T_n(x) \to T(x)$ in the norm topology on Y, and T_n converges to T in the weak operator topology if $T_n(x) \to T(x)$ in e weak topology on x. Hence, we see that the weaker operator topology is weaker than the strong operator topology, which in turn is (somewaht confusingly) weaker than the operator norm topology.

A final result we shall point out is hat we can reprhase the uniform boundedness principle for convergence now as follows:

Proposition 5.4.9: Uniform Boundedness Principle Reformulation

Let $(T_N) \subseteq B(X, Y)$ be a sequence of bounded linear operators from a Banach space X to a normed space Y, and let $T \in B(X, Y)$ be another bounded lienar operator, and let D be a dense subspace of X. Then the following are equivalent:

1. T_n converges in the strong operator topology fo $B(X, Y)$ to T
2. T_n is bounded in the operator norm (i.e. $||T_n||_{op}$ is bounded) and the restriction of T_n to D converges in the strong operator topology of $B(D, Y)$ to the restriction of T to D

5.5 Fréchet Spaces

When constructing a space, we often cannot define a space simply through a norm; not all elements may be "finite". However, we can often construct a space using countably many normed spaces, or usually semi-normed space $\|x\| = 0$ does not imply $x = 0$, for example if f is an integrable function). Due to the countable nature of this construction, most types of convergences are accomplished by a family of semi-norms. For example, convergence on compact sets, pointwise convergnce, uniform convergence, and smooth convergence are all convergence in Fréchet spaces. In analysis, there will be many spaces that will Fréchet spaces, notably when you encounter distributions and require to define a topology on the dual space. Hence, a moment to get a better undersatnding of them is worth it.

Theorem 5.5.1: Semi-Norms Generate TVS

Let $\{p_\alpha\}_{\alpha\in A}$ be a family of norms on a vector space V. If $x \in X$, $\alpha \in A$, and $\epsilon > 0$, let

$$U_{x,\alpha,\epsilon} = \{y \in V \mid p_\alpha(y-x) < \epsilon\}$$

and let $\mathcal{T}$ be the topology generated by the sets $U_{x,\alpha,\epsilon}$. Then:

1. For each $x \in X<$ the inite intersections of the sets $U_{x,\alpha,\epsilon}$ $(\alpha \in A, \epsilon > 0)$ form a neigborhood base at x
2. if $\langle x_i\rangle_{i\in I}$ is a net in V ,then $x_i \to x$ if and only if $p_\alpha(x_i - x) \to 0$ for all $\alpha \in A$.
3. $(V, \mathcal{T})$ is a locally convex topological vector-space

***Proof*:**

1. If $x \in \bigcap_1^k U_{x_j,\alpha_j,\epsilon_j}$, let $\delta_j = \epsilon_j - p_\alpha(x - x_j)$. By the triangle inequality

$$x \in \cap_1^k U_{x,\alpha_j,\delta_j} \subseteq \cap_1^k U_{x-j,\alpha_j,\epsilon_j}$$

Thus, the assertion now follows from the fact that the topology generated consists of $\emptyset$, X, and all unions of finite intersections.

2. By part (1), it suffices to see that $p_\alpha(x_i - x) \to 0$ if and only if $\langle x_i\rangle$ is eventually in $U_{x,\alpha,\epsilon}$ for all $\epsilon > 0$.
3. Continuity is easy from part (b), for if $x_i \to x$ nd $y_i \to y$, then

$$p_\alpha((x_i + y_i) - (x + y)) \leq p_\alpha(x_i - x) + p_\alpha(y_i - y) \to 0$$

so $x_i + y_i \to x + y$. If furthermore $\lambda_i \to \lambda$, then eventually $\lambda_i \leq C + |\lambda| + 1$, so

$$p + \alpha(\lambda_i x_i - \lambda x) \leq p_\alpha(\lambda_i(x_i - x)) + p_\alpha((\lambda_i - \lambda)x) \leq CP_\alpha(x_i - x) + |\lambda_i - \lambda| p_\alpha(x)$$

Hence, $\lambda_i x_i \to \lambda x$. Furthermore, the sets $U_{x,\alpha,\epsilon}$ are convex, for if $y, z \in U_{x,\alpha,\epsilon}$, then

$$p_\alpha(x - [t + (1-t)z]) \leq p_\alpha(tx - ty) + p_\alpha((1-t)x + (1-t)z) < t\epsilon + (1-t)\epsilon = \epsilon$$

Then local convexity follows from (1)

Definition 5.5.2: Fréchet Space

A complete Hausdorff topological vector-space whose topology is defined by a countable family of seminroms is called a *Fréchet space.*

Proposition 5.5.3: Continuity in Fréchet Space

Let $T : V \to W$ be a linear map between two vector spaces. Let V be a topology induced by a family of semi-norms $(\| - \|_{V_\alpha})_{\alpha \in A}$ and W the topology induced by a family of semi-norm $(\| - \|_{W_\beta})_{\beta \in B}$. Then T is continuous if and only if for each $\beta \in B$, there exists a finite subsets $A_\beta \subseteq A$ and a constant C_β such that

$$\|T(x)\|_{W_\beta} \leq C_\beta \sum_{\alpha \in A_\beta} \|x\|_{V_\alpha} \tag{5.1}$$

for all $x \in V$.

***Proof*:**
Let equation 5.1 hold and consider a convergent net $\langle x_i \rangle \subseteq X$. Then by theorem 5.5.1 $p_\alpha(x_i - x) \to 0$ for all α, thus by the inequality $\|T(x_i) - T(x)\|_{S_\beta} \to 0$ for all β, which implies $T(x_i) \to T(x)$. Since T preserves convergent nets, T is continuous.

Conversely, if T is continuous for every $\beta \in B$ choose a neighbor U_β of 0 in V such that $\|T(x)\|_{W_\beta} < 1$ for all $x \in U_\beta$. By theorem 5.5.1, we may assume that $U_\beta = \cap_{1k} U_{x\alpha_{\epsilon_j}}$. Let $\epsilon = \min(\epsilon_1, \epsilon_2, ..., \epsilon_k)$ so that $\|T(x)\|_{W_\beta} < 1$ whenever $p_{\alpha_j}(x) < \epsilon$ for all j. Now, for $x \in X$, either:

1. $p_{\alpha_j})(x) > 0$ for some j. Then let $y = \epsilon x / (\sum_1^k p_{\alpha_j}(x))$ so tht $p_{\alpha_j}(y) < \epsilon$. Then:

$$\|T(x)\|_{W_\beta} = \sum_1^k \epsilon^{-1} p_{\alpha_j}(x) \|T(y)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

2. If $p_{\alpha_j}(x) = 0$ for all j, then $p_{\alpha_j}(rx) = 0$ for all j and all $r > 0$. Thus $r\|T(x)\|_{W_\beta} = \|T(rx)\|_{W_\beta} < 1$ for all $r > 0$, and os $\|T(x)\|_{W_\beta}$. Then certainly

$$\|T(x)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

covering both cases, we are done.

The following proposition is an interesting characteriztion of Fréchtet spaces:

Proposition 5.5.4: Hausdorff and invariant Metric

Let X be Frćhet space given by $\{p_\alpha\}_{\alpha \in A}$. Then

1. X is Haudorff if and only if for each $x \neq 0$, there exists $\alpha \in A$ such that $p_\alpha(x) \neq 0$
2. If X is Hausdorff and A is countable, then X is metrizable with a trnslation invariant metric (i.e. $\rho(x, y) = \rho(x + z, y + z)$ for all $x, y, z \in X$)

***Proof*:**
exercise 43 Folland chatper 5

Note that topologically, all these spaces are the same (as the bellow theorem will sketch). So what really matters is what extra information these semi-norms (or this invariant metric) provides:

Theorem 5.5.5: Anderson-Kadec Theorem

Every infinite-dimensional separable Fréchet space is homeomorphic to the product space $\mathbb{R}^{\mathbb{N}}$ (the countable product of real lines equipped with the product topology).

Note that L^∞ is *inseparable* and hence is not included in this theorem.

***Proof*:**
(Sketch). The proof is historically a combination of results by Kadec (1966) and Anderson (1966), establishing a chain of homeomorphisms. We outline the two main steps:

1. The Banach Case: Kadec proved that every infinite-dimensional separable Banach space is homeomorphic to the Hilbert space ℓ_2. The construction typically involves re-norming the space with a strictly convex equivalent norm (often called a Kadec norm) and defining a homeomorphism via radial projection and coordinate mapping.
2. The Product Case: Anderson proved that ℓ_2 is homeomorphic to $\mathbb{R}^{\mathbb{N}}$. This is often shown using "separable absorption" theorems or the "notback-and-forth" method to manage the coordinate systems.

Since every Fréchet space is a projective limit of Banach spaces (and using the assumption of separability), one can extend the homeomorphism to the general case. Consequently, despite their distinct linear and geometric properties, spaces such as the Schwartz space $\mathscr{S}(\mathbb{R})$, the space of holomorphic functions $H(\mathbb{C})$, and $C^\infty[0,1]$ are all topologically indistinguishable from $\mathbb{R}^{\mathbb{N}}$.

Proposition 5.5.6: dense convergence, then strong convergence

Let $(T_n)_1^\infty \subseteq L(V,W)$ where $\sup_n \|T_n\| < \infty$ and $T \in L(V,W)$. If $\|T_n(x) - T(x)\| \to 0$ for all x in a dense subset D of X, then $T_n \to T$ strongly.

***Proof*:**
We want to show that $\|T_n(x) - T(x)\| \to 0$ for all x. Let $C = \sup\{\|T\|, \|T_1\|, \|T_2\|, ...\}$. Then given $x \in X$ and $\epsilon > 0$, choose $x' \in D$ such that $\|x - x'\| < \epsilon/3C$. Since $T_n(x') \to T(x)$, choose n large enough so that $\|T_n(x') - T(x')\| < \epsilon/3$. Then

$$\begin{aligned}
\|T_n(x) - T(x)\| &\leq \|T_n(x) - T_n(x')\| + \|T_n(x') - T(x')\| + \|T(x') - T(x)\| \\
&\leq 2C\|x - x'\| + \frac{1}{3}\epsilon \\
&< \epsilon
\end{aligned}$$

but then $T_n(x) \to T(x)$ strongly, as we sought to show.

One important result is that in the weak operator topology, unit balls are always compact:

Theorem 5.5.7: Alaoglu's Theorem

Let X be a normed space, and B^* the closed unit ball of X^*, $B* = \{f \in X^* \mid \|f\| \leq 1\}$. Then B^* is compact in the weak*-topology.

***Proof*:**
We shall show that B^* is a closed subspace of a compact space. For each $x \in X$, define

$$D_x = \{z \in \mathbb{C} \mid \|z\| \leq \|x\|\}$$

and take $D = \prod_{x \in X} D_x$. Then by Tychnoff's Theorem, D is compact. Now, we may exactly identify the elements of D the complex-valued function $\varphi : X \to \mathbb{C}$ such that $|\varphi(x)| \leq \|x\|$ for all $x \in X$. Then B^* would conssit of the elements of D that are linear. Hence, we must just show that B^* is closed. The induced topology B^* from the product topology on D and the weak* topology on X^* are both the topology of pointwise convergence (for D by the product construction), and hence checking for closedness in X is equivalence to checking closedness of D. Take a net $\langle f_\alpha \rangle$ in B^* that converges to $f \in D$ for any $x, y \in X$ and $a, b \in \mathbb{C}$. Then:

$$f(ax + by) = \lim f_\alpha(ax + by) = \lim[af_\alpha(x) + bf_\alpha(y)] = af(x) + bf(y)$$

showing that f is linear, and certainly $\|f\| \leq 1$, hence $f \in B^*$, making it compact and completing the proof.

Note: Usually, since we are working with a vector space we would want to conclude that X^* is locally compact (the "it suffices to show the result at 0" trick). However this is *not* the case (exercise 49(b) folland).

(The following was copied from nLab, and is very informative!)

Every complete locally convex topological vector space T is the cofiltered projective limit of Banach spaces in the category of locally convex spaces. (Note that Fréchet spaces are additionally required to be metrisable, so this is more general.)

(see this here) Now given that a Fréchet space admits a decreasing sequence of convex balanced and absorbing neighborhoods, it follows immediately that: Every Fréchet space is a sequential projective limit of Banach spaces.

(this is also really cool on homomorphisms of Fréchet space here)

6

L^p space

Look at this blog!! https://terrytao.wordpress.com/2009/01/09/245b-notes-3-lp-spaces/
In this chapter, we will be studying Banach spaces which are defined through the integral: L^p spaces. These spaces are a generalization of L^1 spaces, where if $f \in L^p$, then it must decay "faster" at infinity, and f doesn't blow up too fast at any finite point. In other words, L^p spaces capture a type of "speed of growth". As we have seen in the previous chapters, limits of integrals are central to many constructions. For some constructions, being in L^1 was enough, but for others (like Fourier series) we will require that function be more than finite in integral (i.e. more than $f \in L^1$). L^p spaces will give a nice set of examples of Banach spaces that are relatively simple to define.

My current intuition is that they are treated as the intermediate to being able to bound a function: it is sometimes nontrivial to immediately say a function is bounded, but we can build-up to it by saying it's L^3, which will imply L^5, which will imply L^7, and so on allowing us to conclude it's L^∞, which means a functions is bounded up to a zero-measure set. This is a common phenomena in analysis where we may not be able to bet a function to fit exactly in one space or another (in our case L^1 or L^∞ which are bounded a.e. functions), but we can put the function in intermediary buckets.

6.1 Basics

So far, we have limited our attention to measurable functions f such that $\int |f| < \infty$ and have labelled them L^1. Often, what we are looking are function with more precise "convergent properties", in particular:

$$\int |f|^p < \infty$$

for some $p \geq 1$, including ∞ (which we will define shortly). We start by labelling the colletion of all these functions:

Definition 6.1.1: L^p Space

Let f be measurable. for $p \in (1, \infty)$[a], define the *p-norm* to be:

$$\|f\|_p = \left[\int |f|^p \, d\mu\right]^{1/p}$$

From this norm, define:

$$L^p(X, \mathcal{M}, \mu) := \left\{ f : X \to \mathbb{C} \;\middle|\; f \text{ is measurable and } \|f\|_p < \infty \right\}$$

[a] we will define L^∞ later

As with L^1, we often abbreviate to $L^p(\mu)$, $L^p(X)$, or just L^p. As with L^1, we'll let L^p be the equivalence class of the set of functions such that $|f|^p$ is integrable where $f \sim g$ if and only if $\int |f - g|^p = 0$. Just like L^1, two functions will be equal in L^p (i.e. belong in the same equivalence class) if they are equal almost everywhere. Furthermore, we claimed that $\|\cdot\|_p$ is a norm. That $\|f\|_p = 0$ if and only if $f = 0$ a.e. comes almost immediately, since

$$\left(\int |f|^p\right)^{1/p} = 0 \quad \text{iff} \quad \int |f|^p = 0 \quad \text{iff} \quad \int |f| = 0 \quad \text{iff} \quad f = 0 \text{ a.e.}$$

For $\|cf\|_p = |c| \, \|f\|_p$, we take advantage of our normalisation:

$$\|cf\|_p = \left(\int |cf|^p\right)^{1/p} = \left(\int |c|^p |f|^p\right)^{1/p} = (|c|^p)^{1/p} \left(\int |f|^p\right)^{1/p} = |c| \, \|f\|_p$$

However, the triangle inequality is not so trivial. We will in fact take good part of this section to show that the triangle inequality is satisfied. For now, we take a moment to explore an example of L^p spaces and L^p functions to gain an intuition:

Example 6.1.2: L^p spaces

1. Let $f : [0, \infty) \to \mathbb{R}$ $f(x) = \frac{1}{x^{1/p}}$. Then $f \in L^q$ for $q > p$. This comes down to the fact that:

   $$x > x^{1/p} \;\Leftrightarrow\; \frac{1}{x} < \frac{1}{x^{1/p}}$$

   and by the fact that $\int \left|\frac{1}{x^n}\right| < \infty$ if $n > 1$. We have therefore found a class of functions that fits into all possible $p \in [1, \infty)$. In fact, we can find functions that fit into any connected subinterval of $[1, \infty)$! In particular, let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \to \mathbb{C}$ a complex measurable function on X. Let

   $$E := \left\{ p \in [1, \infty) | \;\middle|\; \int_X |f|^p < \infty \right\}$$

   (a) Then E is connected, that is, if $p, q \in E$ such that $p < r < q$ for some r, then $r \in E$

   Proof. Clearly, if $E = \emptyset$ or $|E| = 1$, then E is connected because the only non-trivial clopen set is $\emptyset$ and E, so let $|E| > 1$.

Pick $p, q \in E$ such that $p < q$, and consider a point r such that $p < r < q$. We'll show that $\int_X |f|^r < \infty$ using the following trick:

$$|f| = \chi_{|f|\geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f|\geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^q = \chi_{|f|\geq 1}|f|^q + \chi_{|f|<1}|f|^q$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f|\geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^q = \int \chi_{|f|\geq 1}|f|^q + \int \chi_{|f|<1}|f|^q$$

It thus suffice show that $\int \chi_{|f|\geq 1}|f|^r < \infty$ and $\int \chi_{|f|<1}|f|^r < \infty$. This is essentially immediate:

$$\chi_{|f|\geq 1}|f|^r \leq \chi_{|f|\geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^p < \infty$$

Therefore

$$\int |f|^r < \infty$$

showing that $r \in E$, as we sought to show. ☐

(b) Show that we can pick a measure and an f such that $E = (a, b), [a, b), [a, b]$, or $\{a\}$ for $1 \leq a \leq b \leq \infty$. In other words, all connected subsets of $[1, \infty)$ are possible for E (You may freely use the fact that any Riemann integrable function is also integrable with respect to the Lebesgue measure and its Riemann integral is equal to the Lebesgue integral)
Hint: Think about integrability properties of functions like x^{-a} and $\frac{-1}{\log^2(x)x} = \frac{d}{dx}\frac{1}{\log(x)}$

Proof. Throughout this hole proof we'll be using Riemann integration as a substitute for Lebesgue integration (we showed earlier that this is allowed). We'll be using these function's to construct our intervals:

i. $f(x) = x^{-1/a}$ on $(1, \infty)$: Notice that

$$\int_1^\infty x^{-1/a}dx = \lim_{x\to\infty} \frac{x^{-1/a+1}}{-1/a+1} - \frac{1}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in (a, \infty]$.

ii. $f(x) = x^{-1/a}$ on $(0, 1)$: Notice that

$$\int_0^1 x^{-1/a}dx = \frac{1}{-1/a+1} - \lim_{x\to 0} \frac{x^{-1/a+1}}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in [1, a)$.

iii. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(1, \infty)$: This integral doesn't have an elementary solution, however we can still ascertain some basic convergence properties. In particular, if $p = a$, then:

$$\int_1^\infty |f(x)|^a dx = \lim_{x \to \infty} \left[-\frac{1}{\ln(x)} \right] - \lim_{x \to 1} \left[-\frac{1}{\ln(x)} \right]$$

so $\|f\|_p < \infty$ if and only if $p \in [a, \infty]$

iv. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(0, 1)$: This result is similar to the other 3 cases we've proven. The result for this integral shows that $\|f\|_p < \infty$ if and only if $p \in [1, a]$.

Thus, we have proven all possible "sub-rays" of $[1, \infty]$ can exist. To show all possible sub-intervals (closed, open, or half), we simply add the appropriate function's to produce the result. For example, if we want to have the interval $[a, b)$, simply take:

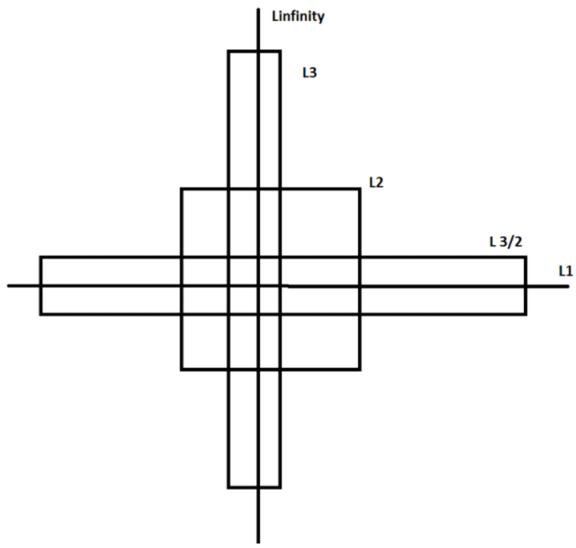
$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}} \chi_{(1,\infty)} + x^{-1/b} \chi_{(0,1)}$$

which clearly has interval of convergence of $[a, b)$. Similarly, to get a singleton $\{a\}$, take:

$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}} \chi_{(1,\infty)} + \frac{1}{(\ln^2(x)x)^{1/a}} \chi_{(0,1)}$$

Thus, we have all possible connected subintervals of $[1, \infty]$. □

So if $f \in L^p$, it does not imply that $f \in L^q$ for $p > q$ or $p < q$! Later on, we will show some of the possible relations between L^p spaces. A more accurate intuition is the following [a]: If $f \in L^p$ for any p, then f must decays fast enough at infinity, and f must not blow up too fast at any finite point. As p increases, the first requirement gets less strict, and the second gets more strict. If X is bounded, the first requirement is vacuous, and if X is discrete, the second is vacuous. Sometimes this visual is used to represent these relations:



2. A particular case that will be common for analyzing sequences is when μ is the counting measure on A. In that case, we usually denote $L^p(\mu)$ by $\ell^p(A)$. If $A = \mathbb{N}$, we often simply

abbreviate notation to ℓ^p. If we write $f : \mathbb{N} \to \mathbb{R},\ f(k) = a_k$, then the sequence f (i.e. $\{a_k\}$ is in ℓ^p if:

$$\sum_{k=1}^{\infty} |a_k|^p < \infty$$

Thus, ℓ^1 is the set of convergence series.

[a] credit to this post on stack-exchange: https://mathoverflow.net/questions/314050/intuition-about-lp-spaces

Notice that we are "normalizing" each integral by a factor of $1/p$. This normalization is done to make sure that the constant can be brought out of the norm. It might seem tempted to say that:

$$\left(\int |f|^p\right)^{1/p} \text{ “=” } \int |f|$$

However, this is most certainly not true (simply take $f(x) = x$). As mentioned before, this makes it non-trivial to show that $\|\cdot\|_p$ is a norm on L^p. However, it is easy to see that L^p is a vector-space. If $f, g \in L^p$, then:

$$|f+g|^p \leq [2\max(|f|,|g|)]^p \leq 2^p(|f|^p + |g|^p)$$

which is finite by assumption, and so $f + g \in L^p$. It is also immediate that $cf \in L^p$. Thus, we may freely add and multiply by L^p functions without worrying about leaving L^p. Because of this, it makes it meaningful to ask whether

$$\|f_1 + f_2\|_p \leq \|f_1\|_p + \|f_2\|_p$$

What we will now do is show that $\|\cdot\|_p$ is does indeed satisfy the triangle inequality for $p \geq 1$. If $p < 1$, then the triangle inequality fails in general:

Example 6.1.3: $p < 1$ then no Triangle Inequality

Take $a, b > 0$ and $0 < p < 1$. Then for any $t \in \mathbb{R}$, we have that:

$$t^{p-1} > (a+t)^{p-1}$$

Integrating both sides from 0 to b, we get:

$$\begin{aligned}
\int_0^b t^{p-1} &> \int_0^b (a+t)^{p-1} \\
\frac{b^p}{p} &> \frac{(a+b)^p}{p} \\
b^p &> (a+b)^p \\
a^p + b^p &> (a+b)^p \\
(a^p + b^p)^{1/p} &> a + b
\end{aligned}$$

Now, if E and F are disjoint union with positive finite measures in X such that $\mu(E)^{1/p} = a$ and $\mu(E)^{1/p} = b$, then $\chi_E, \chi_F \in L^p$. Taking advantage of the fact that $\chi_E\chi_F = 0$ (i.e. the zero function) since $E \cap F = \emptyset$, then:

$$\|\chi_E + \chi_F\|_p = (a^p + b^p)^{1/p} > a + b = \|\chi_E\|_p + \|\chi_F\|_p$$

Thus, the $\|\cdot\|_p$ for $0 < p < 1$ cannot be a norm.

Thus, we stick to showing the case where $p \geq 1$ First, we will state a simple inequality fact similar Jensen's inequality. This is in fact the key idea behind the entire proof:

Lemma 6.1.4: Young's Inequality

If $a, b \geq 0$ and $0 < \lambda < 1$, then:

$$a^\lambda b^{1-\lambda} \leq \lambda a + (1-\lambda)b$$

with equality if and only if $a = b$. In particular, notice that if $\frac{1}{p} + \frac{1}{q} = 1$, then if $\lambda = p^{-1}$, then $(1-\lambda) = q^{-1}$ and so:

$$a^{1/p}b^{1/q} \leq \frac{1}{p}a + \frac{1}{q}b$$

***Proof*:**
This proof is trying to capture what this image is doing in formal language:

https://www.desmos.com/calculator/jdznjzqcju

If $a = b$, equality clearly holds. If $a = 0$ or $b = 0$, inequality clearly holds. If $a, b \neq 0$, then dividing by b and letting $t = a/b$, we get

$$a^\lambda b^{-\lambda} \leq \lambda(a/b) + (1-\lambda) \quad \Leftrightarrow \quad t^\lambda \leq \lambda t + (1-\lambda) \quad \Leftrightarrow \quad t^\lambda - \lambda t \leq (1-\lambda) \tag{6.1}$$

From here, we use some calculus: we see that $t^\lambda - \lambda t$ is strictly increasing for $0 < t < 1$ (since $a, b \geq 0$, $t \geq 0$) and strictly decreasing for $t > 1$, so the maximal value occurs at $t = 1$, and in fact we get that the maximal value is $1 - \lambda$:

$$1^\lambda - \lambda 1 = 1 - \lambda \leq 1 - \lambda$$

and so the inequalities in equation (6.1) hold! Equality happens when $t = a/b = 1$ which implies $a = b$, completing the proof.

Usually, Young's inequality is stated as

$$a^\alpha b^\beta \leq \alpha a + \beta b \qquad 0 \leq \alpha, \beta \leq 1, \quad \alpha + \beta = 1$$

which is exactly what we have, since if we fix some α, then it must be that $\beta = 1 - \alpha$. Another formulation of Young's inequality is:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with $p^{-1} + q^{-1} = 1$. This proof is a "Jensen equality" like proof that more clearly shows the role of concavity and so I'll dedicate a moment to prove this one too:

***Proof*:**
If $a = 0$ or $b = 0$, the result is immediate, so assume $a, b > 0$. Then notice that since log is monotone:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad \Leftrightarrow \quad \log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right)$$

which is great, since we know log is concave and so Jensen's inequality applies. Since $p^{-1}+q^{-1}=1$, we get $t=1/p$ and $(1-t)=1/q$, and so by Jensen's inequality:

$$\log(ta^p+(1-t)b^q) \geq t\log(a^p)+(1-t)\log(b^q) = \frac{p}{p}\log(a)+\frac{q}{q}\log(b) = \log(ab)$$

completing the proof.

The second interpretation of $p^{-1}+q^{-1}=1$ in Young's inequality is a nice way to write the straight line (or sub-straight line) equation without the $1-\lambda$ term, but just two numbers. It furthermore let's us relate two numbers that can be much larger. For example, if I choose $p=100$, then it must be that $q=100/99$ so that

$$p^{-1}+q^{-1}=1/100+99/100=1$$

which, if we consider p and q to be the exponents associated to L^p and L^q, we see that this gives us a way to relate different L^p space!

To see this more succinctly, we use this convexity property given by Young's inequality in the "build-up" lemma towards the triangle inequality. This lemma is important later on in functional analysis since it let's us bound the operation of multiplication of tow functions fg, (namely, $\|fg\|_1$ will be less than a value on the right hand side that seperates f and g). For our current purposess, it is important for the triangle inequality:

Theorem 6.1.5: Hölder's Inequality

Let $1 \leq p, q \leq \infty$ satisfy:

$$\frac{1}{p}+\frac{1}{q}=1$$

or equivalently $q=p/(p-1)$, where we will interpret $\frac{1}{\infty}$ as 0. Then if f and g are measurable functions on X then:

$$\|fg\|_1 \leq \|f\|_p \|g\|_q$$

In particular, if $f \in L^p$ and $g \in L^q$ then $fg \in L^1$. Equality holds if and only if $\alpha|f|^p=\beta|g|^q$ a.e. for some constants α, β with $\alpha\beta \neq 0$

Notice that when $p=q=2$ (if $p=2$, it must be that $q=2$ to satisfy the equation), then this inequality is called the *Cauchy-Schwartz inequality*[1]. Notice too that this is the only time when $p=q$; this will become important when we analyze the $p=2$ case later.

I also found this intuition on a stack-exchange: the slower f decays, the faster g needs to decay, and the faster f blows up, the more we need g to not blow up. The previous sentence is equally true if we switch f and g, which convinces us that L^p spaces should be reflexive, and by considering $f=g$, we see that the restrictions on f and g should balance at exactly $p=2$, so L^2 should be self-dual.

Notice too that since we are taking the absolute value of the function on the inside, the integral can be thought of as a sort of "concave function", where the input would be, say in the special case of $\mathbb{R}$ as the domain, some interval. This makes it easier to see why this inequality holds (notice that convex functions must preserve monotonicity, and so does the integral).

[1]Technically we need to define an inner product on L^2 to satisfy this inequality. However, as well see in chapter 8, L^2 is indeed a Hilbert space

***Proof*:**
If $\|f\|_p = 0$ or $\|g\|_p = 0$, then $f = 0$ or $g = 0$ a.e., and so equality holds. Similarly if $\|f\|_p = \infty$ or $\|g\|_p = \infty$, inequality immediately holds. Furthermore, notice that if we find a particular f and g for which the result holds, then the result holds for af and bg since

$$\begin{aligned} \Leftrightarrow \ \|afbg\|_1 &\leq |ab| \, \|af\|_p \, \|bg\|_q \\ \Leftrightarrow \ |ab| \, \|fg\|_1 &\leq |ab| \, \|f\|_p \, \|g\|_q \\ \Leftrightarrow \ \|fg\|_1 &\leq \|f\|_p \, \|g\|_q \end{aligned}$$

So, it suffices to show the result when $\|f\|_p = \|g\|_q = 1$, and equality holds of $|f|^p = |g|^q = 1$. To this end, we use the previous lemma and set $a = |f(x)|^p$ and $b = |g(x)|^q$ and $\lambda = p^{-1}$ (so that $1 - \lambda = 1 - p^{-1} = q^{-1}$). Then by the previous lemma we have:

$$|f(x)|^{p \cdot p^{-1}} |g(x)|^{q \cdot q^{-1}} \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \quad \Leftrightarrow \quad |f(x)g(x)| \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \tag{6.2}$$

Now, integrating both sides, we get:

$$\|fg\|_1 \leq p^{-1} \int |f|^p + q^{-1} \int |f|^q = p^{-1} + q^{-1} = 1 = \|f\|_p \, \|g\|_q$$

where equality holds when equality holds in equation (6.2) a.e., which holds only when $|f|^p = |g|^q$ a.e., as we sought to show.

For a visual conformation, you can check out:

https://www.desmos.com/calculator/gwqvfv4tie

When p and q satisfy the equality in the equation, we call them *conjugate exponents*. As mentioned before, the conjugate exponents is simply another way or writing the equation of a straight line to apply Young's inequality. From this, we can take advantage of this convexity property yet again to prove the triangle inequality of $\|\cdot\|_p$, which has a different name since for the special case of the p-norm due to the fact it is used on other places and needs to be referenced:

Theorem 6.1.6: Minkowski's Inequality

If $1 \leq p < \infty$ and $f, g \in L^p$. Then:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Before proceeding with the proof, a quick word on notation to avoid confusion:

$$\|f\|_p^p = \left(\int |f|^p \right)^{p/p} = \int |f|^p$$

We often work with $\|f\|_p^p = \int |f|^p$ to get rid of the $1/p$ exponent to be able to work with the linearity of the integral, and re-introduce it at the end to get the desried result. We did not have to do this for Hölder's inequality since we reduced to a very nice special case.

***Proof*:**
If $p = 1$, then the result immediately follows by the regular triangle inequality and linearity of the integral. If $f + g = 0$ a.e., then the result also follows, so assume that $f + g \neq 0$ a.e., which using the triangle inequality:

$$|f+g|^p \leq (|f| + |g|)|f+g|^{p-1} = |f||f+g|^{p-1} + |g||f+g|^{p-1}$$

So, we will apply Hölder's inequality using the fact that $q = \frac{p}{p-1}$ and integrating:

$$\begin{aligned}
\|f+g\|_p^p &= \int |f+g|^p d\mu \\
&= \int |f+g| \cdot |f+g|^{p-1} d\mu \\
&\leq \int (|f| + |g|)|f+g|^{p-1} d\mu \\
&= \int |f||f+g|^{p-1} d\mu + \int |g||f+g|^{p-1} d\mu \\
&= \left\| |f|(|f+g|^{p-1}) \right\|_1 + \left\| |g|(|f+g|^{p-1}) \right\|_1 \\
&\leq \||f|\|_p \|f+g\|_q^{p-1} + \|g\|_p \|f+g\|_q^{p-1} && \text{Hölder's Inequality} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^{(p-1)\left(\frac{p}{p-1}\right)} d\mu \right)^{\frac{p-1}{p}} && \text{recall } q = \frac{p}{p-1} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^p d\mu \right)^{\frac{p-1}{p}} \\
&= (\|f\|_p + \|g\|_p) \frac{\|f+g\|_p^p}{\|f+g\|_p}
\end{aligned}$$

Thus, multiplying both sides by $\frac{\|f+g\|_p}{\|f+g\|_p^p}$, we get

$$\|f+g\|_p \leq \|f\|_p + \|g\|_p$$

As we sought to show

Thus, we get that $\|\cdot\|_p$ is indeed a norm! Thus L^p is a normed vector space. In fact, just like L^1, it is a complete normed vector space, i.e., a Banach space. We first recall a lemma to simply our task. To understand the lemma, notice that every norm defines a metric:

$$\rho(x, y) = \|x - y\|$$

and so defines a topology, in particular a normal space, that has the notion of sequences well-defined. Next, recall that if $\{x_n\}$ is a sequence in V, it's series is said to converge if there exists an $x \in V$ such that $\sum_{n=1}^{\infty} x_n = x$, and is said to be absolutely convergence if $\sum_{n=1}^{\infty} \|x_n\| < \infty$ (this is lemma 5.1.6). If you did not cover this result yet, it is important to go see it now to understand the following result:

Theorem 6.1.7: L^p is a Banach Space

For $1 \leq p < \infty$, L^p is a complete normed vector-space, i.e., a Banach space. Moreover, if $f_n \to f$ in L^p, then there exists a subsequence that converges pointwise to f a.e.

***Proof*:**
Let $1 \leq p < \infty$. By the previous lemma, it suffices to show that every absolutely convergent series converges:

$$\sum_{k=1}^{\infty} \|f\|_p = B < \infty \quad \Rightarrow \quad \exists f \text{ s.t. } \sum_{k=1}^{\infty} f_k = f \in L^p$$

Let $G_n = \sum_{k=1}^{n} |f_k|$ be the partial sums of the series and $G = \sum_{k=1}^{\infty} |f_k|$ be the series. Then by Minkowski's inequality, we get:

$$\|G_n\|_p \leq \sum_{k=1}^{n} \|f_k\|_p \leq B \quad \forall n \in \mathbb{N}$$

and so $\|G\|_p = \left(\int |G|^p\right)^{1/p} < \infty$ and so $\left(\int |G|^p\right) < \infty$. Since $G_n \leq G_{n+1}$, and $G_n : X \to [0, \infty)$, by the Monotonce Convergence Theorem:

$$\int \lim_{n\to\infty} G_n^p = \lim_{n\to\infty} \int G_n^p \leq B^p < \infty$$

and so $G \in L^p$. By proposition 2.3.4, $G(x) < \infty$ a.e., so the series $\sum_{k=1}^{\infty} f_k$ converges a.e. If we let $F = \sum_{k=1}^{\infty} f_k$, we have that $|F| \leq G$ a.e., and so $F \in L^p$. From this, we can see that:

$$\left\| F - \sum_{k=1}^{\infty} f_k \right\|_p^p = \int \left| F - \sum_{k=1}^{\infty} f_k \right|^p \to 0$$

Thus, the series $\sum_{k=1}^{\infty} f$ converges (to F) in the p-norm, and so by lemma 5.1.6, L^p is complete, as we sought to show.

This is the proof the prof was doing in class, but I got stuck so found a different proof (the above proof) (Commented it out for now)

This also finally showed that L^1 is a complete space under $\|\cdot\|_1$! Note the importance of taking a subsequence: since $f_n \to f$ in the p-norm, i.e. the "sum-averaging" of the function converges, then there can be lots of wiggle room for hwo the functions f_n approach f:

Example 6.1.8: Taking A Subsequence
Construct a sequence of functions such that all the following hold:

1. $f_n : [0, 1] \to \mathbb{R}$ is continuous
2. $0 \leq f_n(x) \leq 1$ for every x
3. $\lim_{n\to\infty} \int_0^1 f_n dm = 0$
4. For every $x \in [0, 1]$ we have that $\lim_{n\to\infty} f_n(x)$ does not exist

(that is, it is definitely important to take a subsequence in order to deduce pointwise convergence from converge in L^1)

Proof. Essentially, we'll be taking advantage that $\sum_n \frac{1}{n}$ does not converge. Let $f_1(x) = 1$, that is, that constant function with constant 1.

For g_2, we split the interval into 2 equal parts. Define

$$g_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ 0 & x \in (1/2, 1] \end{cases}$$

This function is not continuous. We will make it continuous by simply climbing to it (and for other values, we'll be descending back down too). At the point 1/2, slope down at a speed of 2^2 until we reach 0:

$$f_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ -2^2 \left(x - \left(1 - \frac{1}{4}\right)\right) & x \in [1/2, 3/4] \\ 0 & x \in (3/4, 1] \end{cases}$$

From here, we will define our function in a sort of "recursive" sense. Starting form the inteval $[0, 1/2)$, we keep the right end point, and add an interval of length 1/3 to it: $[1/2, 5/6)$. Now, define f_3 to be 1 on $[1/2, 5/6]$, and zero everywhere else. Next, to define f_4, repeat the process by adding 1/4 to 5/6, giving us the interveal $[5/6, 13/12]$. Note that 13/12 goes beyond 1. We thus wrap it around back to 0: $[0, 1/12] \cup [5/6, 1]$. Now, define a function f_3 to be 1 on this set, and zero everywhere else.

In each of these functions, we extend them to be continuous by attaching them a slope which has 2^n as a factor. In this way, the slope shrinks in size quicker than the size of the rectangles, and so we can morally imagine the problem to be one of indicator functions.

We can continue doing this, defining every f_k by appending $1/k$ and wrapping around when we go past 1 and then making the function continuous. Clearly, $\int f_k \to 0$ as $k \to \infty$ since each inteval will be of total length $1/k + 1/2^k$. However, f_k does not converge anywhere: for any $x \in [0, 1]$, we can choose a subsequence from $f_k(x)$ (say $f_{k_l}(x)$) such that $f_{k_l}(x) = 1$. or $f_{k_l}(x) = 0$

To construct such a sequence, it is easier to imagine our construction without the wrapping around, that is, we keep adding intervals of length $1/k$ as though our co-domain was $[0, \infty)$. Then, our first value of l will be the value for which $x + 1$ is in some interval of length $1/k$ for appropriate k. This interval will exist, since $\sum_k 1/k$ diverges. Similarly, our next value of l for which $x + 2$ will be in some interval of length $1/k$ for appropraite k. We can continue doing this indefinitely, showing that there exists a subsequence of f_k that doesn't converge to 0, meaning f_k cannot converge to 0. On the other hand, it is easy to see that we can choose a subsequence that converges to 0 (since f_k is zero after the trapezoid passes by our point).

Since x was arbitrary, this will work any $x \in [0, 1]$. Thus, f does not converge anywhere, as we sought to show. □

Retruning to analyzing L^p spaces, we can now extend theorem 2.3.9 to show that simple function are dense in every L^p!

Theorem 6.1.9: Simple Functions Dense in L^p

For $1 \leq p < \infty$, the set of simple functions $f = \sum_{i=1}^n a_i \chi_{E_i}$ where $\mu(E_i) < \infty$ for all i is dense in L^p

***Proof*:**
First of all, the set of simple functions is in L^p for every $1 \leq p < \infty$. Next, choosing any $f \in L^p$, choose some standard representation $\{f_n\}$ where $f_n \to f$ a.e[a]. Then since $|f_n| \leq |f|$, we get that $|f_n| \in L^p$. Furthermore

$$|f - f_n|^p \leq |2f|^p = 2^p|f|^p \in L^1$$

and so by the dominated convergence theorem, we get:

$$\int |f - f_n|^p \to 0 \quad \Rightarrow \quad \left(\int |f - f_n|^p\right)^{1/p} = \|f - f_n\|_p \to 0$$

Moreover, if $f_n = \sum_{k=1}^\infty a_k \chi_{E_k}$ was a simple function where all E_i are pairwise disjoint and a_i are nonzero, it must be that $\mu(E_i) < \infty$ since

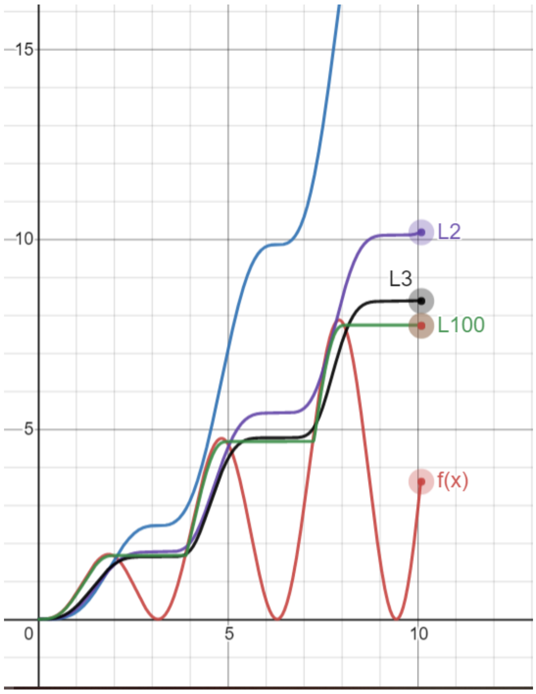
$$\sum_{k=1}^\infty |a_k|^p \mu(E_k) = \int |f_n|^p < \infty$$

completing the proof

[a] Recall that the any measurable function has a standard representation, not just integrable functions

Some further remarks about which sets are dense can be enlightening. For $1 \leq p, q < \infty$, L^p is dense in L^q, since each simple set of the form we descriped are subsets of L^p for all $p \geq 1$. If we modify the characteristic function to make them continuous (simply draw a connecting line as we've done in example 6.1.8), then we see that $C_0(\mathbb{R})$ (continuous functions that vanish at infinity) are dense in L^p for every $1 \leq p < \infty$. If we restrict to closed intervals, then $C([a,b])$ is dense in $L^p([a,b])$ since the standard representation becomes uniformaly convergent on bounded domains (see theorem 2.1.18).

So far, we have defined L^p spaces where $0 < p < \infty$ and showed that for $1 \leq p < \infty$, L^p is a Banach space. Interestingly, if we take a look at a graph of $\|f\|_p$ as $p \to \infty$, we'll find that there is a striking pattern that appears:



Notice that as $p \to \infty$, the resulting $\|f\|_p$ value got closer to a max function. In fact, this is almost the case! We might be tempted to then say that $\|f\|_\infty = \sup f$. However, the problem with this is that we might have a measure 0 set that diverges, while the supremum of the rest of the function is attained by a value $k < \infty$. The idea of $\sup f$ on all but a zero measure set will be called the *essential supremum*, and all functions that have a finite essential supremum will be denoted as L^∞. In this way, L^∞ is simply a natural extension of our definition of L^p:

To see the intuition behind this, let us simply consider the p-norm of $\|\vec{v}\|_p = (|3|^p + |10|^p)^{1/p}$

1. Case $p = 1$: $(3^1 + 10^1)^{1/1} = 13$. (The 3 contributes about 23% to the total)

2. Case $p = 2$ (Euclidean): $\sqrt{3^2 + 10^2} = \sqrt{9 + 100} = \sqrt{109} \approx 10.44$. (The 3 matters less now. The 10 is dominating)

3. Case $p = 10$: $(3^{10} + 10^{10})^{1/10} = (59,049 + 10,000,000,000)^{1/10}$. Notice that 3^{10} is negligible dust compared to 10^{10}. $\approx (10^{10})^{1/10} = 10.00000\textit{something}$

4. Case $p = 100$: The 3^{100} is mathematically invisible next to 10^{100}. The sum is basically just 10^{100}. When you take the $1/100$ root, you are just returning the base number: 10.

Definition 6.1.10: L^∞ Space

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f : X \to [0, \infty]$ be a measurable function. Let

$$S = \left\{ r \in \mathbb{R} \;\middle|\; \mu(f^{-1}((r, \infty])) = 0 \right\}$$

Then the *essential supremum* of f, denoted $\|f\|_\infty$ or $\operatorname{ess\,sup} f$ is :

$$\|f\|_\infty := \begin{cases} \infty & \text{if } S = \emptyset \\ \inf S & \text{if } S \neq \emptyset \end{cases}$$

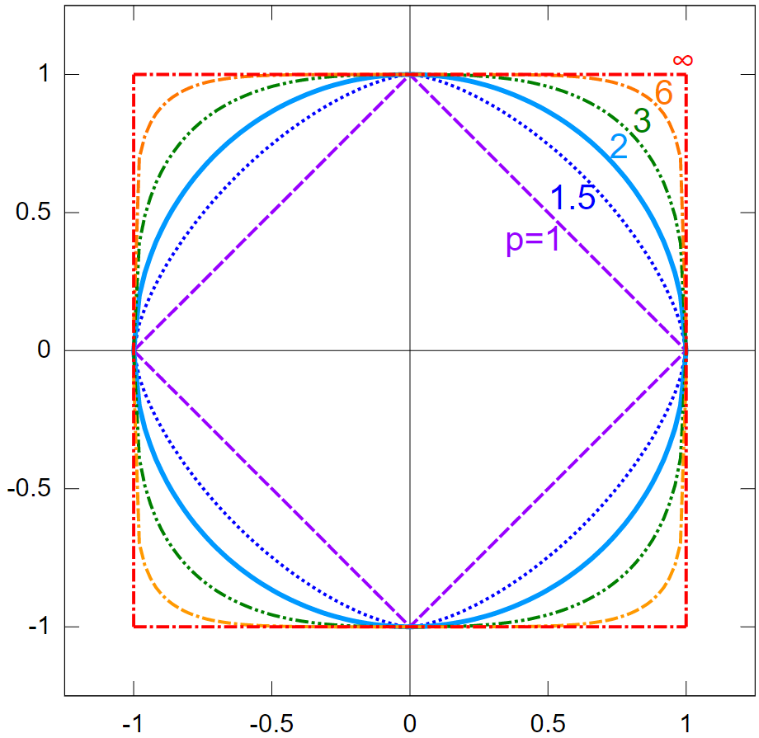
or, equivalently:

$$\|f\|_\infty = \inf_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} \left(\sup_{x \in E} |f(x)| \right)$$

If f is a complex measurable, function, then $\|f\|_\infty = \| \, |f| \, \|_\infty$

Essentially, $\|f\|_\infty$ is the least upper bound of f, ignoring measure zero sets worth of points. Using this definition, we can define L^∞ to be the equivalence classes of functions f such that $\|f\|_\infty < \infty$.

Another quick way of visualizing the relation between all p-norms, $1 \leq p \leq \infty$ is by taking the ℓ^p norm on $\mathbb{R}^2$, taking the function $x \mapsto \|x\|_p = \sqrt[p]{x^p + y^p}$, and considering when $\|x\| = 1$, or in terms of set notation $S_p = \left\{ x \in \mathbb{R}^2 \;\middle|\; \|x\|_p = 1 \right\}$. We'd get:



A couple more quick remarks: $L^\infty(\mu)$ depends on μ only insofar as μ determines which sets have measure zero. If μ and ν are mutually absolutely continuous ($\mu \ll \nu$ and $\nu \ll \mu$[2]) then $L^\infty(\mu) = L^\infty(\nu)$. Secondly, if μ is not semi-finite, it is useful to adpot a slightly useful definition of L^∞ (Folland left this as exercsie 23-25 in chatper 6).

Most properties of L^p for $1 \le p < \infty$ generalises naturally to the L^∞ case:

Proposition 6.1.11: Properties of L^∞

Let $(X, \mathcal{M}, \mu)$ be a measure and L^∞ the set of measurable function satisfying $\|\cdot\|_\infty < \infty$. Then:

1. If f, g are measurable functions on X, then
$$\|fg\|_1 \le \|f\|_1 \|g\|_\infty$$
If $f \in L^1$ and $g \in L^\infty$, then $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ if and only if $|g(x)| = \|g\|_\infty$ a.e. on the set where $f(x) \neq 0$
2. $\|\cdot\|_\infty$ is a norm on L^∞
3. $\|f_n - f\|_\infty \to 0$ if and only if there exists $E \in \mathcal{M}$ such that $\mu(E^c) = 0$ and $f_n \to f$ uniformally on E
4. L^∞ is a Banach space
5. The simple functions are dense in L^∞

As a consequence of the well-definedness of L^∞, $1^{-1} + \infty^{-1} = 1$ is indeed a well-defined notion (so $\infty^{-1} = 1/\infty = 0$). Thus, 1 and ∞ are conjugate exponents!

Proof:

We will prove (1), from which the rest of the results follow.

1. We'll be using the $\|f\|_\infty = \inf\limits_{\substack{E\in\mathcal{M}\\ \mu(X\setminus E)=0}} (\sup_{x\in E}|f(x)|)$ definition of the essential supremum to take advantage of the supremum within the definition. Our goal is to show that
$$\int |fg| = \|fg\|_1 \le \|f\|_1 \|g\|_\infty$$
First, we will limit our attention of $\int |fg|$ to some $E \in \mathcal{M}$ such that $\mu(X \setminus E) = \mu(E^c) = 0$ (i.e. ignoring a zero-measure set). Then:
$$\int |fg| = \int_E |fg| + \int_{E^c} |fg| = \int_E |fg|$$
Next, we take advantage the fact that we are working with an infimum to see that:
$$\|f\|_\infty \le \sup_{x\in A} |f(x)| < \|f\|_\infty + \epsilon$$

[2] see definition 3.2.1

Therefore, we can take:

$$\int_E |fg| \leq \int_E \left| f \cdot \sup_{x\in E} |g(x)| \right| = \int_E |f| \cdot \left| \sup_{x\in E} |g(x)| \right| = \left| \sup_{x\in E} |g(x)| \right| \int_E |f|$$

where the last equality comes from the fact that the supremum is a constant. Finally, by the property of the infimum, we have:

$$\left| \sup_{x\in E} |g(x)| \right| \int_E |f| \leq (\|g\|_\infty + \epsilon) \int |f|$$

Notice that this inequality does not depend on ϵ, and so:

$$\|fg\|_1 = \int_E |fg| \leq \|f\|_1 \|g\|_\infty$$

as we sought to show

Notice how similar $\|\cdot\|_\infty$ is to the uniform metric $\|f\|_u = \sup_{x\in X} f(x)$. If we are dealing with Borel measures that assign a positive value to all open sets, then if f is continuous we in fact have $\|f\|_\infty = \|f\|_u$ (since $\{x \mid |f(x)| > a\}$ is open). In this way, we can consider the space of bounded continuous functions as a (closed!) subspace of L^∞.

We conclude with some final remarks about the density of simple functions in L^∞. Notice that for L^p where $p < \infty$, simple functions where $\mu(E_i) < \infty$ where dense. This condition is no longer present for density in L^∞; we in fact require simple functions where $\mu(E) = \infty$ now since the function $1(x) = 1$ is in $L^\infty(\mathbb{R})$. This also means that $C_0(\mathbb{R})$ cannot be dense in L^∞. In the case of $L^\infty([a, b])$ $C([a, b])$, not all funcions can be converged to uniformally dense: consider $f : [0, 1] \to \mathbb{R}$, $f(x) = 0$ for $x \in [0, 1/2]$ and $f(x) = 1$ for $x \in [1/2, 1]$. The no continuous function uniformally converges to f in the ∞-norm.

6.1.1 Relating L^p spaces: Interpolation

In this section, we will show how we can relate L^p spaces with each other. Fundamentally, the way we get any order with L^p spaces is through Hölder's inequality. We take full advantage of it in the following proofs. The end-goal is to show that if $p < q < r$ where we allow $r = \infty$, then:

$$L^p \cap L^r \subseteq L^q \subseteq L^p + L^r$$

This should reflect on the example that we have demonstrated in example 6.1.2. In particular, the larger p is, the less f has decay at infinity, but the more f must not explode at any point. In other words p gets bigger, we can let p decay slower, but it must behave better at points where it can explod3. If our space X is bounded, then we don't have to care about decay (this will give us that $L^p \supseteq L^q$ where $p < q$). If X is discrete, we don't have to care about exploding at any point (this will give us that $\ell^p \subseteq \ell^q$ if $p < q$). We thus explore these relation's in this section. These types of relations bewteen space via addition of functions or intersection of spaces is called *interpolation*.

Proposition 6.1.12: L^p **subset**

Let $0 < p < q < r \leq \infty$. Then

$$L^q \subseteq L^p + L^r$$

In particular, each $f \in L^q$ is the sum of a function in L^p and a function in L^q

***Proof*:**
This proof is really similar to that of example 6.1.2[1]. Let $f \in L^q$. Then consider:

$$|f| = \chi_{|f|\geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f|\geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^r = \chi_{|f|\geq 1}|f|^r + \chi_{|f|<1}|f|^r$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f|\geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^r = \int \chi_{|f|\geq 1}|f|^r + \int \chi_{|f|<1}|f|^r$$

Then we get:

$$\chi_{|f|\geq 1}|f|^p \leq \chi_{|f|\geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^q < \infty$$

Therefore, $\chi_{|f|\geq 1}f \in L^p$ and $\chi_{|f|<1}f \in L^r$. If $r = \infty$, then clearly $\|\chi_{|f|\leq 1}f\|_\infty \leq 1$, showing the result still holds, as we sought to show.

Proposition 6.1.13: L^p **superset**

Let $0 < p < q < r \leq \infty$. Then

$$L^p \cap L^r \subseteq L^q$$

In particular, if if we find that f is bounded in growth from above and bellow by p and r, $\|f\|_p < \infty$ and $\|f\|_r < \infty$, then f is bounded in growth by q.

***Proof*:**
Let $\|f\|_r < \infty$ and $\|f\|_q < \infty$. We need to show that $\|f\|_p < \infty$. Notice that it is equivalent to show that $\|f\|_r^n < \infty$ and $\|f\|_q^m < \infty$ then $\|f\|_p^\ell < \infty$ for $n, m, \ell \in \mathbb{R}_{\geq 0}$ (since if it's finite, then all powers will be finite). We'll prove it in the case where $r = \infty$ and $r < \infty$. In the infinite case, we simply take advantage of bounding properties of the essential supremum, and in the other case we use Hölder's inequality.

Starting with the $r = \infty$ case, notice that

$$|f|^q = |f|^{q-p+p} = |f|^q|f|^{q-p} \leq \|f\|_\infty^{q-p}|f|^p$$

and so, taking the integral and the qth root, we get:

$$\begin{aligned}
\|f\|_q &= \left(\int |f|^q\right)^{1/q} \\
&\leq \left(\int \|f\|_\infty^{q-p}|f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p\right)^{p/pq} \\
&= \|f\|_\infty^{1-\frac{p}{q}} \|f\|_p^{p/q} < \infty
\end{aligned}$$

The fact that the last inequality is less than infinity comes from the that $\|f\|_p < \infty$ and $\|f\|_\infty < \infty$, so any power is less than infinity, and so $\|f\|_q < \infty$. Furthermore, notice that the sum of the power's is 1; this fact will come back for another intuition of $\|f\|_\infty$.

For the $r < \infty$ case the key is to find a way to integrate p and r using conjugate exponents. Since

$$p < q < r$$

then

$$\frac{1}{p} > \frac{1}{q} > \frac{1}{r}$$

therefore, define the continuous function $\lambda \mapsto \frac{1-\lambda}{p} + \frac{\lambda}{r}$. Since $(0,1)$ is open, and $1/p > 1/r$, there exists a λ_0 such that

$$\frac{1-\lambda_0}{p} + \frac{\lambda_0}{r} = \frac{1}{q}$$

manipulating the equation around to get conjugate exponents, notice that:

$$\frac{(1-\lambda_0)q}{p} + \frac{\lambda_0 q}{r} = 1 \quad \Leftrightarrow \quad \frac{1}{\frac{p}{(1-\lambda_0)q}} + \frac{1}{\frac{r}{\lambda_0 q}} = 1$$

Thus, we have that $\frac{p}{(1-\lambda_0)q}$ and $\frac{r}{\lambda_0 p}$. With conjugate exponents established, we proceed with trying to bound $\|f\|_q$. As we commented, it is equivalent to bound $\|f\|_q^m$, so let $m = q$ (this will be a common trick to apply Hölder's inequality). Also, since $1 - \lambda_0 + \lambda_0 = 1$, we have that

$q = \lambda_0 q + (1-\lambda_0)q$, giving us the ability to split up our integral:

$$\begin{aligned}
\|f\|_q^q &= \int |f|^q \\
&= \int |f|^{\lambda_0 q + (1-\lambda_0)q} \\
&= \int |f|^{\lambda_0 q}|f|^{(1-\lambda_0)q} \\
&= \left\| |f|^{\lambda_0 q}|f|^{(1-\lambda_0)q} \right\|_1 \\
&\leq \left\| |f|^{\lambda_0 q} \right\|_{\frac{p}{(1-\lambda_0)q}} \left\| |f|^{(1-\lambda_0)q} \right\|_{\frac{r}{\lambda_0 q}} && \text{Hölder's inequality} \\
&= \left(\int |f|^p \right)^{\frac{(1-\lambda_0)q}{p}} \left(\int |f|^r \right)^{\frac{\lambda_0 q}{r}} \\
&= \|f\|_p^{(1-\lambda_0)q} \|f\|_r^{\lambda_0 q} < \infty
\end{aligned}$$

Since the last two terms are finite, any power of them are also finite, and so we have bounded $\|f\|_q^q < \infty$, and so $\|f\|_q < \infty$, as we sought to show.

Proposition 6.1.14: ℓ^p Covariant Inclusion

If $0 < p < q \leq \infty$, then $\ell^p(A) \subseteq \ell^q(A)$

***Proof*:**
They key part is to take advantage of the properties of the counting measure. Since the only set of measure 0 in the counting measure is the empty-set, we have that the essential supremum is actually the supremum!

Let's say $f \in \ell^p$, so that $\sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$. We'll start by showing that $\|f\|_\infty < \infty$. Since the $\|f\|_\infty$ is in fact the supremum, we have that:

$$\|f\|_\infty^p = \sup_{a\in A} |f(a)|^p \leq \sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$$

Therefore, $\|f\|_\infty < \infty$.

For the $q < \infty$ case, using proposition 6.1.13 with $\lambda = p/q$, we get:

$$\|f\|_q \leq \|f\|_p^\lambda \|f\|_\infty^{1-\lambda} \leq \|f\|_p$$

therefore, $\|f\|_q < \infty$, completing the proof.

Proposition 6.1.15: L^p Contravariant Inclusion in finite measure

Let $(X, \mathcal{M}, \mu)$ be a measure space where $\mu(X) < \infty$. If $0 < p < q \leq \infty$, then $L^p \supseteq L^q$

***Proof*:**
We need to show that if $f \in L^q$ then $f \in L^p$. Since $f \in L^q$, $\|f\|_q < \infty$, so $\|f\|_q^n < \infty$ for all $n \in \mathbb{N}$. If we can show that $\|f\|_p^m \leq \|f\|_q^n$ for some $m, n \in \mathbb{N}$, then the proof is complete.

We'll apply Hölder's inequality in the caes of $q = \infty$ and $q < \infty$. If $q = \infty$, then:

$$\|f\|_p^p = \int |f|^p = \int |f|^p \cdot 1 \overset{\text{H.}}{\leq} \|f\|_\infty^p \int 1 = \|f\|_\infty^p \mu(X) < \infty$$

since $\int 1 = \mu(X)$. If $q < \infty$, then we do a similar argument. First, since $p < q$, notice that:

$$\frac{1}{\frac{q}{p}} + \frac{1}{\frac{q}{q-p}} = \frac{p}{q} + \frac{q-p}{q} = \frac{q}{q} = 1$$

and so $\frac{q}{p}$ and $\frac{q}{p-q}$ are conjugate exponents. Therefore, by Hölder's inequality:

$$\begin{aligned}
\|f\|_p^p &= \int |f|^p \\
&= \int |f|^p \cdot 1 \\
&= \||f|^p \cdot 1\|_1 \\
&\leq \||f|^p\|_{\frac{q}{p}} \|1\|_{\frac{q}{p-q}} && \text{Hölder's Inequality} \\
&= \left(\int |f|^{p\frac{q}{p}}\right)^{\frac{p}{q}} \left(\int 1\right)^{\frac{q-p}{p}} \\
&= \|f\|_q^p \, \mu(X)^{\frac{q-p}{p}} < \infty
\end{aligned}$$

therefore, $\|f\|_p^p < \infty$, so $\|f\|_p < \infty$, showing that $f \in L^p$, as we sought to show.

Proposition 6.1.16: Relating L^p and L^∞

Let $p \in [1, \infty)$ and assume $\|f\|_p < \infty$. Show that:

$$\lim_{p \to \infty} \|f\|_p = \|f\|_\infty$$

***Proof*:**
For ease of typing, we simplify notation: $\|f\|_p := \|f\|_{L^p(\mu)}$

We'll show that $\lim_{p\to\infty} \|f\|_p \leq \|f\|_\infty$ and $\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$. For the $\leq$ direction, note that $|f| \leq \|f\|_\infty$ a.e., and since x^n is a monotone function on $[0, \infty)$, we have $|f|^n \leq \|f\|_\infty^n$. Then for some fixed λ_0:

$$\begin{aligned}\|f\|_p &= \left(\int |f|^p\right)^{1/p} \\ &= \left(\int |f|^{p+\lambda_0-\lambda_0}\right)^{1/p} \\ &= \left(\int |f|^{p-\lambda_0}|f|^{\lambda_0}\right)^{1/p} \\ &\leq \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}}\end{aligned}$$

where the last inequality is the same trick as part (a) of this question. Since λ_0 is fixed and this inequality holds true for all p, as $p \to \infty$:

$$\lim_{p\to\infty} \|f\|_p \leq \lim_{p\to\infty} \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}} = \|f\|_\infty$$

For the $\geq$ direction, we have to bound $\|f\|_p$ from bellow by something that includes $\|f\|_\infty$, but ultimately the inequality will be independent of the choices we make. In particular, let $\epsilon > 0$. Take all points which are epsilon close to the essential supremum (which exists by the infimum property). Let M_ϵ be our set, in particular $M_\epsilon = \{x \mid |f(x)| \geq \|f\|_\infty - \epsilon\}$. Since $\|f\|_\infty = k$, then $\mu(M_\epsilon) > 0$. Thus:

$$\|f\|_p = \left(\int |f|^p\right)^{1/p} \geq \left(\int_{M_\epsilon} (\|f\|_\infty - \epsilon)^p\right)^{1/p} \overset{!}{=} (\|f\|_\infty - \epsilon)\mu(M_\epsilon)^{1/p}$$

where $\overset{!}{=}$ comes from the fact that we are integrating over a constant region. Since $\mu(M_\epsilon) > 0$, we have that $\mu(M_\epsilon)^{1/p} > 0$ for all p, and $\mu(M_\epsilon)^{1/p} \xrightarrow{p\to\infty} 1$. Furthermore, notice our inequalities are independent of choice of ϵ, and so since we can let ϵ be as small as we want:

$$\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$$

and since we've shown the $\leq$ direction, we have:

$$\lim_{p\to\infty} \|f\|_p = \|f\|_\infty$$

as we sought to show

Now that we have examined L^p spaces in more detail, we finish off this seciton with some comments on the usefulness these spaces:

1. L^1 is natural for integration; we will essentially work with variation's of L^1 when we integrate. From an algebraic perspective, it can be thought of as a generalizatgion of the notion of "counting" all points in a space, and working in spaces where the count is finite.

2. L^∞ is useful as a broadening of the space of uniform-bound functions. It is a generlization of the idea of working with functions that are bounded.

3. L^p interpolates in between L^1 and L^∞. Many times, L^1 or L^∞ spaces have pathological behaviors (examples of which I yet do not know). L^p spaces behave more nicely, and can sometimes be used to go from L^1 to L^∞ step by step using the inequalities we constructed (ex. we got a function is L^1, we want to show it's L^∞, and we use the proceeding inequalities to build our way up)

4. L^2 is particularly nice since it is a Hilbert space, and so many euclidean intuitions work on L^2 and help us construct many nice properties. A lot of Fourier analysis happens on L^2, though sometimes we do use L^p.

   Another way of thinking of L^2 is that it is a generlization of the notion of a space that *can* have an inner product (this will be made rigorous with a particular universal property, see theorem 8.1.24). In ℓ^2, we would simply get a natural generalization of an inner product on a vector-space, while in L^2, we get the continuous equivalent.

6.2 The dual of Lebesgue Space

By Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. We can thus define a function:

$$\varphi_g : L^p \to \mathbb{R} \qquad \varphi_g(f) = \int fg$$

and the operator norm of φ_g is always at most $\|g\|_q$[3]. With the opeartor norm, the mapping $g \mapsto \varphi_g$ will almost always be an isometry:

Proposition 6.2.1: Operator Norm And Isometry

Let p and q be conjugate exponents, $1 \leq q < \infty$. If $g \in l^q$, then:

$$\|g\|_q = \|\varphi_g\| = \left\{ \left| \int fg \right| \;\middle|\; \|f\|_p = 1 \right\}$$

if μ is semifinite, this results alos holds for $q = \infty$

***Proof*:**
Folland p. 188

Conversely, if $f \mapsto \int fg$ is a bounded linaer functional on L^p, then $q \in L^q$ in almost all cases:

[3]if $p = 2$, it is usually defined as mapping to $\int f\overline{g}$. This can also be done to $p \neq 2$, there it is just easier to work by conjugating for complex numbers

Theorem 6.2.2: L^p Linear Functional Implies G In L^q

Let p and q be conjugate exponents. Suppose that g is a measurable function on X such that $fg \in l^1$ for all f inteh space Σ of simple functions that vanish outside a set of finite measure, and the quanitty

$$M_q(g) = \{|fg| \mid f \in \Sigma,\ \|f\|_p = 1\}$$

is finite. Also, suppose either that $S_g = \{x \mid g(x) \neq 0\}$ is σ-finite or tha tμ is semi-finite. Then $g \in L^q$ and $M_q(g) = \|g\|_q$.

***Proof*:**
Folland p. 189

Finally, perhaps the most stricking thing is that the maps $g \mapsto \int fg$ are almost always a surjection of $(L^p)^*$!

Theorem 6.2.3: Description Of Dual Of Lp

Let p and q be conjugate exponents. If $1 < p < \infty$, for each $\varphi \in (L^p)^*$ there exits $g \in L^q$ such that $\varphi(f) = \int fg$ for all $f \in L^p$, and hence L^q is isometrically isomorphic to $(L^p)^*$. The same conclusion holds for $p = 1$ provided μ is $1s$-finite

***Proof*:**
Folland p. 190

Corollary 6.2.4: Lp Is Reflective

If $1 < p < \infty$, L^p is reflective, that is $(L^p)^{**} \cong L^p$

(Some comment in the case of $p = 1$ and $p = \infty$)

6.3 Some Useful Inequalities

In this section, we explore more inequalties, since they are at the heart of working with L^p spaces. We first present a common inequality that shows up often in statistics:

Theorem 6.3.1: Chebychev's Inequality

Let $f \in L^p$ $(0 < p < \infty)$. Then for any $\alpha > 0$

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

***Proof*:**
Let $E_\alpha = m(\{x \mid |f(x)| > \alpha\})$. Then:

$$\|f\|_p^p = \int_\Omega |f|^p \geq \int_{E_\alpha} |f|^p \geq \int_{E_\alpha} \alpha^p = \alpha^p m(E_\alpha)$$

re-arranging gives us:

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

as we sought to show

The next inequality gives us a general statement on being abel to get a function from $L^p(\mu)$ "transformed" into a function of $L^p(\nu)$:

Theorem 6.3.2: Transitioning L^p Spaces

Let$(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measures spaces, K be a $M \otimes N$-measurable function on $X \times Y$, and let's say there exists a $C > 0$ such that

$$\int |K(x,y)| d\mu(x) \leq C \qquad \int |K(x,y)| d\nu(y) \leq C$$

for a.e. x and y. Then if $f \in L^p(\nu)$ $(1 \leq p \leq \infty)$, the integral:

$$Tf(x) = \int K(x,y) f(y) d\nu(y)$$

converges absolutely for a.e. $x \in X$, $Tf \in L^p(\mu)$, and $\|Tf\|_p \leq C\|f\|_p$

***Proof*:**
Let $p \in [1, \infty]$, and let q be it's conjugate exponent. Then:

$$|K(x,y)f(y)| = |K(x,y)||f(y)| = |K(x,y)|^{\frac{1}{p}+\frac{1}{q}}|f(y)| = (|K(x,y)|^{1/p})(|K(x,y)|^{1/q}f(y)|)$$

and applying Hölder to the functions in parenthesis, we get:

$$\begin{aligned}\int |K(x,y)f(y)| d\nu(y) &\leq \left(\int |K(x,y)| d\nu(y)\right)^{1/q} \left(\int |K(x,y)||f(y)|^p d\nu(y)\right)^{1/p} \\ &\leq C^{1/q} \left(\int |K(x,y)||f(y)|^p d\nu(y)\right)^{1/p}\end{aligned}$$

for a.e. $x \in X$. Hence, taking the integral over $X \times Y$, using Tonelli's theorem, and substituting the result we just found, we get:

$$\begin{aligned}\int \left[\int |K(x,y)f(y)d\nu(y)\right]^p d\mu(x) &\leq \int\int C^{p/q}|K(x,y)||f(y)|^p d\nu(y)d\mu(x) \\ &= C^{p/q}\int\int |K(x,y)||f(y)|^p d\mu(x)d\nu(y) \qquad \text{Tonelli} \\ &\leq C^{p/q}\int |f(y)|^p \int K(x,u)d\mu(x)d\nu(y) \\ &= C^{p/q+1}\int |f(y)|^p d\nu(y)\end{aligned}$$

Since $f \in L^p(\nu)$, the last integral is finite. Since this is true for a.e. x, $K(x,\cdot)f \in L^1(\nu)$ for a.e. x. Thus, Tf is well-defined a.e., meaning we can integrate it, and by the inequality we just showed:

$$\int |Tf(x)|^p d\mu(x) = \|Tf\|_p^p \leq C^{p/q+1}\|f\|_p^p$$

taking the pth root of each side get's us the desired inequality

The next result we work towards is a generalization of Minkowski: instead of summing we will be taking integrals. You can also interpret part (a) as you can move the $(|\cdot|^p)^{1/p}$ "down" a layer of integrals, and part (b) as the generalization of monotonicity ($|\int f| \leq \int |f|$).

Theorem 6.3.3: Minkowski's Inequality For Integrals

Let $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ be σ-finite meaure spaces and let f be a [real?] $M\otimes N$-measurable function on $X\times Y$. Then:

1. if $f \geq 0$ and $1 \leq p \leq \infty$, then:

$$\left[\int\left(\int f(x,y)d\nu(y)\right)^p d\nu(x)\right]^{1/p} \leq \int\left[\int f(x,y)^p d\mu(x)\right]^{1/p} d\nu(y)$$

2. If $1 \leq p \leq \infty$, $f(\cdot,y) \in L^p(\mu)$ for a.e. y and the function $y \mapsto \|f(\cdot,y)\|_p$ is in $L^1(\nu)$, then $f(x,\cdot) \in L^1(\nu)$ for a.e. x, the function $x \mapsto \int f(x,y)d\nu(y)$ is in $L^p(\mu)$ and

$$\left\|\int f(\cdot,y)d\nu(y)\right\|_p \leq \int \|f(\cdot,y)\|_p \, d\nu(y)$$

***Proof*:**
If $p=1$, then this becomes Tonelli's Theorem, so consider $1 < p < \infty$ (the case for $p=\infty$ comes later). Let q be it's conjugate exponent so that $1/p+1/q=1$. Suppose $\mu,\nu<\infty$ and $\|f\|_\infty < \infty$ (i.e. it's essential supremum is finite). For notational simplicity, define $H(x) = \int_Y f(x,y)d\nu(y)$.

Since the measure is finite and f is esentially bounded, we have that $\|H\|_p < \infty$. Then:

$$\begin{aligned}
\|H\|_p^p &= \int_X \left(\int_Y f(x,y)d\nu(y)\right)^p d\nu(x) \\
&= \int_X \left(\int_Y f(x,y)d\nu(y)\right)(H(x))^{p-1}d\mu(x) && x \text{ indep. of } y \\
&= \int_X \int_Y f(x,y)H(x)^{p-1}d\mu(x)d\nu(y) && \text{Fubini} \\
&\overset{!}{\leq} \int_X \|f(x,\cdot)\|_p \|H\|_p^{p-1} d\nu(y) && \text{Hölder, } q = p/(p-1) \\
&\leq \|H\|_p^{p-1} \int_X \|f(x,\cdot)\|_p d\nu(y)
\end{aligned}$$

where $\overset{!}{\leq}$ I'm not sure how Hölder's inequality was used. Finally, dividng by $\|H(x)\|_p^{p-1}$ get's us:

$$\|H\|_p \leq \int_X \|f(x,\cdot)\|_p d\nu(y)$$

For the general case, Yakov left it as an exercise to do $X = \bigcup_n X_n$, $Y = \bigcup Y_n$, $E_n = \{x \mid f(x) < n\}$, and

$$\tilde{X}_n = X_n \cap E_n \qquad \tilde{Y}_n = Y_n \cap E_n$$

and apply the result to each of these, and the monotone convergence theorem at the end (feels similar to how it was done for Fubini Tonnelli).

For $p = \infty$, This comes down to monotonicity of integrals:

$$\left\|\int f(x,y)d\nu(y)\right\|_\infty \leq \left|\int f(x,y)d\nu(y)\right| \leq \int |f(x,y)d\nu(y) = \int \|f(x,y)\|_\infty d\nu(y)$$

where the last inequality comes from the fact that the integral ignores zero-measure sets.

There are a few more results in the book, but Yakov decided to prove the generalized Hölder's inequality instead for now:

Theorem 6.3.4: Generalized Hölder's Inequality

Let $1 \leq p_i \leq \infty$ for $1 \leq i \leq n$ and $\sum_i^n p_i^{-1} = r^{-1} \leq 1$. If $f_i \in L^{p_i}$, then $\prod_i^n f_i \in L^r$ and

$$\left\|\prod_i^n f_i\right\|_r \leq \prod_i^n \|f_i\|_{p_i}$$

***Proof*:**
If $r = \infty$, then $r^{-1} = 0$, so all the $p_i^{-1} = 0$, so all the $p_i = \infty$, in which case we simply take:

$$\left|\prod_i^n f_i\right| \leq \prod_i^n |f_i|$$

For $r < \infty$, just like for Hölder's inequality, it suffice to assume $r = 1$, since we can do:

$$\sum \frac{1}{\frac{p_i}{r}} = 1$$

In that case, notice that:

$$\begin{aligned} \left\| \prod_i^n f_i \right\|_r^r &= \int |f_1 f_2 \cdots f_n|^r \\ &= \int f_1^r f_2^r \cdots f_n^r \\ &\leq \|f_1\|_{p_1}^r \|f_2\|_{p_2}^r \cdots \|f_n\|_{p_n}^r \end{aligned}$$

Notice that each f_i is raised to the power of p_i/r. (TBD)

Now, we do induction on i. Let the induction hypothesis be:

$$\left(\sum_i^{n-1} p_i^{-1} \right) + p_n^{-1} = r^{-1} + p_n^{-1} = 1$$

Then by Hölder's inequality:

$$\int f_1 \cdots f_n| \leq \|f_1 \cdots f_{n-1}\|_r \|f\|_{p_n} \overset{\text{I.H.}}{\leq} \|f_1\|_{p_1} \cdots \|f_n\|_{p_n}$$

giving our desired result

The final property we'll present a way to represent functions in terms of the integral:

Proposition 6.3.5: Layer Cake Representation

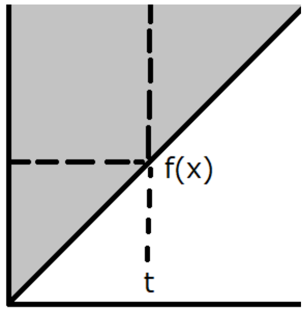
Let $\varphi(t) : [0, \infty] \to [0, \infty]$ be an increasing function which is C^1 on $(0, \infty)$, satisfies $\varphi(0) = 0$ and satisfies $\lim_{t\to\infty} \varphi(t) = \varphi(\infty)$. Let (X, M, μ) be a σ-finite measure space and $f : X \to [0, \infty]$ be a non-negative measurable function. Then:

$$\int_X \varphi(f(x))\, d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) > t\})\varphi'(t) dt$$

One consequence of this equation is that with $\varphi(t) = t^p$ for $1 < p < \infty$ is the following formula for the L^p norm of f:

$$\|f\|_p = \left(p \int_0^\infty \mu(\{x \mid f(x) > t\}\, t^{p-1} \right)^{1/p} dt$$

***Proof*:**
The proof can be captured pictorially as integrating over the following area:



Let $S \subseteq X \times [0,1]$, $S = \{(x,t) \mid f(x) \geq t\}$. With this set, define $F : X \times [0,\infty] \to [0,\infty]$, $F(x,t) = \varphi'(t)\chi_S$. That is, we are going to integrate the "upper triangle". Since φ is an increasing C^1 function and $\varphi'(t) > 0$, we have $F \in L^+(X,[0,\infty])$. Therefore, we can apply Tonneli's Theorem and see that it's equivalent to integrate over the iterated integrals to compute integrals.

To compute it, we need to be clever with the bounds we choose which depend on which iterated integral we choose. The different direction of the iterated integral will ultimately lead to the desired result. If we integrate with respect to $d\mu(x)dt$, then we take slices across t: $(\varphi'(t)\chi_S)_t = \varphi'(t)\chi_{S_t}$, where $(\chi_S)_t = \chi_S(x,t) = \chi_{S_t}$, with $S_t = \{x \mid f(x) \geq t\}$. And so we get:

$$\int_0^\infty \int_X \chi_{S_t}\varphi'(t) = \int_0^\infty \int_{\{x \mid f(x) \geq t\}} \varphi'(t)d\mu(x)dt = \int_0^\infty \mu(\{x \mid f(x) \geq t\})\varphi'(t)dt$$

Now, integrating with respect to $dtd\mu(x)$, we take x slices: $(\varphi'(t)\chi_S)_x = \varphi'(t)\chi_{S_x}$ where $(\chi_S)_x = \chi_S(x,t) = \chi_{S_x}$, with $S_x = \{x \mid 0 \leq f(x)\}$. And so we get:

$$\int_X \int_0^\infty \chi_{S_x}\varphi'(t)dtd\mu = \int_X \int_0^{f(x)} \varphi'(t)dtd\mu(x)$$

and since $[0, f(x)]$ is bounded and $\varphi(0) = 0$, the Lebesgue integral matches the Riemann integral, and so we get that:

$$\int_X \varphi(f(x))d\mu(x)$$

and by Tonnelli's theorem:

$$\int_X \varphi(f(x))d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\})\varphi'(t)dt$$

As we sought to show. Using this, if we let $\varphi(t) = t^p$, then we see that φ is C^1 on $(0,\infty)$ and $\varphi(0) = 0$ for $1 \leq p < \infty$. Thus, the value of $\|f\|_p$ (where we take f^p instead of $|f|^p$ since $f \geq 0$) is:

$$\begin{aligned} \|f\|_{L^p} = \left(\int |f|^p\right)^{1/p} = \left(\int f^p\right)^{1/p} &= \left(\int_X \varphi(f(x))d\mu(x)\right)^{1/p} \\ &= \left(p\int_0^\infty \mu(\{x \mid f(x) > t)\}\, t^{p-1}\right)^{1/p} dt \end{aligned}$$

Another use of this formula is by taking $\varphi(t) = t$. Then we will get that:

$$\int f(x)d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\})dt$$

This can be useful if we have information of what is happening to the growth of f as f get's larger, in particular if we know that somehow, the amount of points such that $\mu(\{x \mid f(x) \geq t\})$ is shrinking rapidly enough say that $\mu(\{x \mid f(x) \geq t\}) \leq \frac{C}{t^r}$ for some constant C).

6.4 Interpolation: The Riesz-Thorin Theorem

As we saw in the previous section, if a function is in L^p and L^r, it is also in L^q for all $p < q < r$. This is a type of "interpolation" of spaces. A natural question arises: if we have a linear operator T that is bounded on L^p and bounded on L^r, is it also bounded on L^q? The answer is yes, and the Riesz-Thorin theorem provides the precise quantitative relationship between the operator norms.

The proof relies on a powerful complex analysis technique known as the Phragmén-Lindelöf principle, specifically a version called the Three Lines Lemma.

Lemma 6.4.1: Three Lines Lemma

Let φ be a bounded continuous function on the strip $S = \{z \in \mathbb{C} \mid 0 \leq \text{Re}(z) \leq 1\}$ that is holomorphic on the interior of S. If $|\varphi(z)| \leq M_0$ for $\text{Re}(z) = 0$ and $|\varphi(z)| \leq M_1$ for $\text{Re}(z) = 1$, then for all $t \in [0, 1]$ and z with $\text{Re}(z) = t$:

$$|\varphi(z)| \leq M_0^{1-t} M_1^t$$

***Proof*:**

For $\epsilon > 0$, define $\varphi_\epsilon(z) = \varphi(z) M_0^{z-1} M_1^{-z} e^{\epsilon(z^2-1)}$. The term $e^{\epsilon(z^2-1)}$ helps handle the boundedness at infinity within the strip. By the maximum modulus principle applied to rectangles in the strip as the height goes to infinity, one can show $|\varphi_\epsilon(z)| \leq 1$ in the strip. Letting $\epsilon \to 0$ yields the result.

Theorem 6.4.2: Riesz-Thorin Interpolation Theorem

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces. Let $p_0, p_1, q_0, q_1 \in [1, \infty]$. Define p_t and q_t for $t \in (0, 1)$ by:

$$\frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1}, \qquad \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1}$$

Suppose T is a linear operator defined on the set of simple functions Σ such that for all $f \in \Sigma$:

$$\|Tf\|_{q_0} \leq M_0 \|f\|_{p_0} \quad \text{and} \quad \|Tf\|_{q_1} \leq M_1 \|f\|_{p_1}$$

Then for all $t \in (0, 1)$, T extends to a bounded operator from $L^{p_t}(\mu)$ to $L^{q_t}(\nu)$ satisfying:

$$\|Tf\|_{q_t} \leq M_0^{1-t} M_1^t \|f\|_{p_t}$$

***Proof*:**
Let $t \in (0,1)$ be fixed. By duality (Proposition 6.2.1), it suffices to estimate $|\int (Tf)g\, d\nu|$ where f and g are simple functions with $\|f\|_{p_t} = 1$ and $\|g\|_{q_t'} = 1$ (where q_t' is the conjugate exponent of q_t).

We construct a family of functions to apply the Three Lines Lemma. Let $f = |f|e^{i\theta}$ and $g = |g|e^{i\psi}$ be the polar decompositions. For any complex number z in the strip $0 \le \mathrm{Re}(z) \le 1$, define:

$$\alpha(z) = (1-z)\frac{1}{p_0} + z\frac{1}{p_1}, \qquad \beta(z) = (1-z)\frac{1}{q_0'} + z\frac{1}{q_1'}$$

Note that for $z = t$, $\alpha(t) = 1/p_t$ and $\beta(t) = 1/q_t'$. Now define deformations of f and g:

$$f_z = |f|^{p_t \alpha(z)} e^{i\theta}, \qquad g_z = |g|^{q_t' \beta(z)} e^{i\psi}$$

(If $f = 0$ or $g = 0$, set the function to 0). Note that $f_t = f$ and $g_t = g$.

Define the function $\Phi(z)$ on the strip $0 \le \mathrm{Re}(z) \le 1$:

$$\Phi(z) = \int_Y (Tf_z) g_z \, d\nu$$

Since f and g are simple, f_z is a finite linear combination of terms like $c_k \alpha^z$, making $\Phi(z)$ holomorphic and bounded.

We check the boundary conditions. For $\mathrm{Re}(z) = 0$, $f_z \in L^{p_0}$ and $g_z \in L^{q_0'}$. Specifically:

$$\|f_{iy}\|_{p_0}^{p_0} = \int ||f|^{p_t(\frac{1}{p_0} + i\mathrm{Im}(\alpha))}|^{p_0} = \int |f|^{p_t} = \|f\|_{p_t}^{p_t} = 1$$

Similarly $\|g_{iy}\|_{q_0'} = 1$. Thus by Hölder's inequality and the hypothesis on T:

$$|\Phi(iy)| = \left| \int T(f_{iy}) g_{iy} \right| \le \|Tf_{iy}\|_{q_0} \|g_{iy}\|_{q_0'} \le M_0 \|f_{iy}\|_{p_0} \cdot 1 = M_0$$

Similarly, for $\mathrm{Re}(z) = 1$, we find $|\Phi(1+iy)| \le M_1$.

By the Three Lines Lemma, $|\Phi(t)| \le M_0^{1-t} M_1^t$. Since $\Phi(t) = \int (Tf)g$, and this holds for all unit simple functions f, g, we conclude:

$$\|T\|_{L^{p_t} \to L^{q_t}} \le M_0^{1-t} M_1^t$$

Counter-Example: The Linear Transport Equation

Consider the simple advection equation in one dimension:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

where $c \neq 0$ is a real constant velocity.

1. **The Operator:** The solution is a translation of the initial state $f(x)$:

$$u(x,t) = f(x - ct)$$

Therefore, the time-shift operator S_t acts as:

$$(S_t f)(x) = f(x - ct)$$

2. **The Adjoint:** To check if S_t is self-adjoint, we compute its adjoint S_t^* using the standard L^2 inner product $\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\,dx$.

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(x - ct)\overline{g(x)}\,dx$$

Perform a change of variables $y = x - ct$, so $x = y + ct$ and $dx = dy$:

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(y)\overline{g(y + ct)}\,dy$$

$$\langle S_t f, g \rangle = \langle f, S_{-t} g \rangle$$

This shows that the adjoint is the inverse shift:

$$S_t^* = S_{-t}$$

3. **Conclusion:** For S_t to be self-adjoint, we require $S_t = S_t^*$. This would imply $S_t = S_{-t}$, or $f(x - ct) = f(x + ct)$ for all functions f. This is clearly false for $c \neq 0$ and $t \neq 0$. Thus, S_t **is not self-adjoint.**

General Context

- **Unitary vs. Self-Adjoint:** In conservative systems (like Quantum Mechanics or the Wave Equation), the evolution operator S_t is typically **unitary**, meaning $S_t^* = S_t^{-1} = S_{-t}$. A unitary operator is rarely self-adjoint (unless $S_t^2 = I$, which is not true for general time evolution).

- **When is it Self-Adjoint?** S_t *is* self-adjoint in specific dissipative systems, such as the **Heat Equation** ($u_t = \Delta u$). In this case, $S_t = e^{t\Delta}$. Since the Laplacian Δ is a self-adjoint operator (with appropriate boundary conditions), the operator exponential $e^{t\Delta}$ is also self-adjoint for real t.

Zak Katz: I see. Does that mean that S_t is not always "diagonalizable"?

Raycast AI: No, that is a common misconception! **"Not self-adjoint" does not mean "not diagonalizable."**

The time-shift operator S_t for the Transport Equation (which I used as a counter-example previously) **is** diagonalizable, even though it is not self-adjoint.

Here is the breakdown of why this works and when it fails.

1. The Key Concept: Normal Operators

In linear algebra and functional analysis, a deeper property governs diagonalizability: **Normality**. An operator A is **Normal** if it commutes with its adjoint:

$$AA^* = A^*A$$

The **Spectral Theorem** states that **any Normal operator is diagonalizable** (specifically, unitarily diagonalizable).

- **Self-Adjoint Operators** are Normal ($A = A^*$, so $AA = AA$).
- **Unitary Operators** are Normal ($A^* = A^{-1}$, so $AA^{-1} = A^{-1}A = I$).

2. Revisiting the Transport Equation (The "Yes" Case)

Let's look at the Transport Equation again ($u_t + u_x = 0$), where S_t shifts the function: $f(x) \to f(x-t)$.

- We found that $S_t^* = S_{-t}$ (the inverse shift).
- Does it commute?
$$S_t S_t^* = S_t S_{-t} = S_0 = I$$
$$S_t^* S_t = S_{-t} S_t = S_0 = I$$
- Because $S_t S_t^* = S_t^* S_t$, the operator is **Normal**.

How do we "diagonalize" it? We use the **Fourier Transform**. In the frequency domain (Fourier space), the shift operator becomes multiplication:

$$\mathcal{F}(S_t f)(k) = e^{-ikt} \hat{f}(k)$$

Multiplication by a scalar (or function) is the infinite-dimensional equivalent of a diagonal matrix. The Fourier basis functions e^{ikx} are the "eigenvectors" of the time-shift operator.

Conclusion: The Transport operator is not self-adjoint, but it is normal, so it **is** diagonalizable.

---

3. When is S_t NOT Diagonalizable? (The "No" Case)

For S_t to fail to be diagonalizable (to be "defective"), the system must be **Non-Normal** in a specific way that creates "Jordan Blocks." This happens when eigenvectors coalesce and you lose a complete basis (this is called an **Exceptional Point** in physics).

Example: A "Defective" System of PDEs Consider a coupled system of two variables $u(x,t)$ and $v(x,t)$:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

Here, the change in u depends on v, but v is constant in time.

1. **Solve it:**
   - $v(x,t) = v(x,0)$ (constant)
   - $u_t = v(x,0) \implies u(x,t) = u(x,0) + t \cdot v(x,0)$

2. **The Time-Evolution Operator S_t:** We can write S_t as a matrix acting on the vector $\begin{pmatrix} f \\ g \end{pmatrix}$:

$$S_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Is it Diagonalizable?** Check the matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - The eigenvalues are $\lambda = 1$ (with algebraic multiplicity 2).
   - However, there is only **one** independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we are missing a second eigenvector, this operator is **not diagonalizable** for any $t \neq 0$.

Summary Table

Type of PDE	**Operator Property**	**Self-Adjoint?**	**Diagonalizable?**
Heat Eq. ($u_t = \Delta u$)	Self-Adjoint	**Yes**	**Yes** (Real eigenvalues)
Transport / Wave	Unitary	**No**	**Yes** (Complex eigenvalues)
Defective System	Non-Normal	**No**	**No** (Jordan Block)

Bottom Line: Most standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because they are typically governed by Normal operators. You usually have to construct specific coupled systems or non-Hermitian boundary conditions to break diagonalizability.

Zak Katz: Let me be clear: S_t shifts *in time* not in space. So to be clear: let a PDE $F(x_1, \ldots, x_n, t) = 0$ PDE w.r.t time t, if $u(x,t)$ is a solution and $f_t(x) = u(x,t)$ is a "state" at time t, and Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a. (NOT the space shift operator $f(x) \to f(x+t)$). Is S_t always diagonalizable?

Raycast AI: Thank you for the clarification. I understand the distinction you are making: you are asking about the **time-evolution operator** (often called the Propagator, $U(t)$ or S_t) that maps the initial state $u(\cdot, 0)$ to the state at time t, $u(\cdot, t)$.

The answer remains: **No, S_t is not always diagonalizable.**

While many standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because their generators are Normal (Self-Adjoint or Skew-Adjoint), it is very easy to construct physical systems where S_t is **defective** (non-diagonalizable).

This happens specifically when the system exhibits **Linear Instability** or **Critical Damping** (often associated with "Exceptional Points" in modern physics).

The Mathematical Reason: Jordan Blocks

If a PDE can be written in the form $\frac{\partial}{\partial t}\psi = L\psi$, the time-evolution operator is formally $S_t = e^{Lt}$.

For S_t to be diagonalizable, the generator L must possess a complete basis of eigenfunctions. If L has a **Jordan Block** (i.e., it has generalized eigenvectors but not enough true eigenvectors), then the exponential $S_t = e^{Lt}$ will also possess a Jordan block structure for any $t \neq 0$, making it non-diagonalizable.

Here are two concrete counter-examples.

Counter-Example 1: The "Resonant" System (Algebraic Growth)

Consider a system of two fields, $u(x,t)$ and $v(x,t)$, on a domain Ω. The PDE is:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

where λ is a constant. This is a standard linear PDE system.

1. **Find the Solution:** This is a coupled system.
   - $v_t = \lambda v \implies v(x,t) = e^{\lambda t}v(x,0)$.
   - $u_t = \lambda u + v$. Substituting the solution for v: $u_t = \lambda u + e^{\lambda t}v(x,0)$.
   - The solution for u involves a secular term (a linear t):

   $$u(x,t) = e^{\lambda t}u(x,0) + \mathbf{te}^{\boldsymbol{\lambda}\mathbf{t}}v(x,0)$$

2. **The Time-Evolution Operator S_t:** Writing this as an operator matrix acting on the initial vector $\psi_0 = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$:

   $$S_t = e^{\lambda t}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Why it is NOT Diagonalizable:** Look at the matrix part $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - **Eigenvalues:** It has a single eigenvalue $\mu = 1$ (repeated).
   - **Eigenvectors:** It has only **one** linearly independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we need two independent eigenvectors to span the 2D space of fields at every point x, and we only have one, S_t is **not diagonalizable** for any $t \neq 0$.

Physical Interpretation: The term $te^{\lambda t}$ represents "algebraic growth" (or transient growth). Whenever you see a solution of the form $t \times$ (exponential), it is the signature of a non-diagonalizable operator. A diagonalizable operator would only have pure sums of exponentials.

Counter-Example 2: Critical Damping (Classical Mechanics / PDE)

Consider the Damped Harmonic Oscillator equation for a field $u(x,t)$, which is a second-order PDE in time:

$$u_{tt} + 2\gamma u_t + \gamma^2 u = 0$$

(Here $\gamma > 0$ is the damping coefficient).

To analyze S_t, we convert this to a first-order system (state-space form) by defining $v = u_t$:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

The Critical Case: The eigenvalues of the matrix $L = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}$ are found by $\det(L - \omega I) = 0$.

$$\omega^2 + 2\gamma\omega + \gamma^2 = (\omega + \gamma)^2 = 0$$

We have a repeated eigenvalue $\omega = -\gamma$.

Because the matrix L is not the identity matrix but has a repeated eigenvalue, it is a non-trivial Jordan block. The solution is:

$$u(t) = C_1 e^{-\gamma t} + C_2 \mathbf{te}^{-\gamma \mathbf{t}}$$

The presence of the $te^{-\gamma t}$ term confirms that the propagator S_t cannot be diagonalized. If it were diagonalizable, we could write the solution purely as $u(t) = Ae^{\omega_1 t} + Be^{\omega_2 t}$, which implies strictly exponential decay without the linear polynomial pre-factor.

Summary

- **Is S_t always diagonalizable? No.**
- **When does it fail?** It fails at **Exceptional Points**, where two or more eigenmodes of the system coalesce (become parallel), making the set of eigenvectors incomplete. This results in solutions that grow or decay with polynomial factors (like t, t^2) multiplying the exponentials.

Zak Katz: Okay. Now, when it *is* diagonalizable, is it *always* the case that we can use Fourier transforms to find an eigenbasis?

Raycast AI: No. Absolutely not.

Just because an operator is diagonalizable does not mean the Fourier Transform provides the basis.

The **Fourier Transform** is a very specific tool that only diagonalizes operators that are **Translation Invariant** (i.e., operators with **Constant Coefficients** defined on the entire space $\mathbb{R}^n$).

If your PDE has **variable coefficients** or is defined on a **bounded domain**, the Fourier basis (e^{ikx}) will generally fail to diagonalize the operator, yet a different eigenbasis might still exist perfectly fine.

Here is the hierarchy of logic:

1. **Diagonalizable:** Means *there exists* a basis $\{\varphi_n(x)\}$ such that the operator acts as multiplication on these functions.
2. **Fourier Diagonalizable:** Means that specific basis is $\{\varphi_k(x) = e^{ikx}\}$.

When does Fourier work?

Fourier analysis works **only** if your operator L commutes with the translation operator T_a (where $T_a f(x) = f(x+a)$).

- **Example:** The Heat Equation $u_t = u_{xx}$.
  - The second derivative behaves the same at $x = 0$ as it does at $x = 100$. Space is homogeneous.
  - Therefore, the eigenfunctions are plane waves e^{ikx}.

When does Fourier fail (but the operator is still diagonalizable)?

Fourier fails whenever **spatial symmetry is broken**. This happens in three common scenarios:

1. Variable Coefficients (Space is not homogeneous) Consider the **Quantum Harmonic Oscillator** (one of the most famous diagonalizable systems in physics).

$$i\frac{\partial\psi}{\partial t} = -\frac{\partial^2\psi}{\partial x^2} + x^2\psi$$

- **Is it Diagonalizable?** Yes. The Hamiltonian is Self-Adjoint. It has a discrete, perfect set of eigenfunctions.
- **Are they Fourier Modes? No.** The term x^2 breaks translation invariance (physics looks different at $x = 0$ vs $x = 100$).
- **The Actual Basis:** The eigenfunctions are **Hermite Functions** (a Gaussian multiplied by a polynomial):
  $$\varphi_n(x) = H_n(x)e^{-x^2/2}$$
  If you tried to apply a Fourier Transform to this equation, the x^2 multiplication term turns into a Laplacian in k-space ($-\partial_k^2$), so you effectively just swap the PDE for an identical PDE in frequency space without solving it.

2. Bounded Domains (The "Boundary Condition" Problem) Consider the heat equation $u_t = u_{xx}$ on a finite interval $x \in [0, L]$ with Dirichlet boundary conditions $u(0) = u(L) = 0$.

- **Is it Diagonalizable?** Yes.
- **Is it the Fourier Transform?** No. The Continuous Fourier Transform uses e^{ikx} for any real k.
- **The Actual Basis:** The basis is strictly **Sine functions**: $\sin(\frac{n\pi x}{L})$.

– While related to Fourier *Series*, the Continuous Fourier *Transform* basis (e^{ikx}) is invalid here because e^{ikx} does not satisfy the boundary conditions (it doesn't vanish at 0 and L). You are forced onto a discrete subset of the frequency spectrum.

3. Different Geometries If your PDE is on a sphere (like weather patterns on Earth):

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

- **Is it Diagonalizable?** Yes.
- **The Actual Basis:** The eigenfunctions are **Spherical Harmonics** $Y_l^m(\theta, \varphi)$.
- Standard planar Fourier modes (e^{ikx}) make no sense here.

Summary Table

PDE Feature	**Is S_t Diag.?**	**Fourier?**	**The Actual Eigenbasis**
Constant Coeffs on $\mathbb{R}$ ($u_t = u_{xx}$)	Yes	**Yes**	Plane Waves (e^{ikx})
Variable Coeffs ($u_t = xu_x$)	Yes	**No**	Power functions (x^λ)
Potential Well ($u_t = i\Delta u + iV(x)u$)	Yes	**No**	Hermite / Legendre / Bessel
Bounded Domain ($x \in [0, 1]$)	Yes	**No**	Fourier *Series* (Sines/Cosines)

Conclusion: Diagonalizability guarantees that *some* coordinate system exists where the operator is simple. It does not guarantee that the coordinate system is the one given by Fourier.

Zak Katz: Okay. Is it a necesssary and sufficient condition that S_t is diagonalizable by the Fourier basis iff the diagonalizable operator is translation invariant? Just to make sure I understand what translation invariant means here, we are talking about *space* translation, and we mean that if S_t is our operator and τ_a is our space-translation, then:

$$S_t(\tau_a(f)) = \tau_a(S_t(f))$$

Raycast AI: Yes, precisely.

For a linear operator S_t acting on functions defined on the entire space $\mathbb{R}^n$ (or a periodic domain), the condition is **necessary and sufficient**:

> **Theorem:** A bounded linear operator S_t is diagonalized by the Fourier Transform if and only if it is **Translation Invariant**.

Your definition of translation invariance is exactly correct:

$$[S_t, \tau_a] = S_t\tau_a - \tau_a S_t = 0 \quad \text{for all } a \in \mathbb{R}^n$$

where τ_a is the spatial shift operator.

Here is the logic for why this is an "if and only if" relationship.

1. Why is it Sufficient? (Translation Invariant $\implies$ Fourier Diagonal)

This relies on a fundamental principle of linear algebra: **Commuting operators share a common eigenbasis.**

1. Consider the family of translation operators $\{\tau_a\}_{a\in\mathbb{R}}$.
2. What are the eigenfunctions of a translation? Let's verify that the plane waves e^{ikx} are eigenfunctions of τ_a:
$$\tau_a(e^{ikx}) = e^{ik(x+a)} = e^{ika} \cdot e^{ikx}$$
The eigenvalue is the constant number e^{ika}, and the eigenfunction is e^{ikx}.
3. Since your time-evolution operator S_t commutes with **every** translation operator τ_a (for all shifts a), it must share the same set of eigenfunctions.
4. Therefore, the eigenfunctions of S_t must also be e^{ikx}.
5. If the eigenfunctions are e^{ikx}, then by definition, the operator is diagonalized by the Fourier Transform.

2. Why is it Necessary? (Fourier Diagonal $\implies$ Translation Invariant)

If an operator is diagonal in Fourier space, it acts as a simple multiplication. Let's call the Fourier transform $\mathcal{F}$.

1. Assume S_t is a Fourier Multiplier: $\widehat{S_t u}(k) = m(k)\widehat{u}(k)$ for some multiplier function $m(k)$.
2. The translation operator τ_a in Fourier space is also just a multiplication (by the phase factor e^{ika}).
3. Since regular multiplication of scalars is commutative ($a \cdot b = b \cdot a$), the multiplication by $m(k)$ commutes with the multiplication by e^{ika}.
4. Therefore, S_t and τ_a commute.

Summary

This concept is the bedrock of signal processing and linear systems theory (where these are called **Linear Time-Invariant (LTI)** systems, or in your PDE context, "Linear Space-Invariant" systems).

- **Does S_t commute with space shifts?**
  - **Yes:** You can use Fourier Transforms (e.g., Heat Eq, Wave Eq, Schrödinger with constant potential).

  - **No:** You cannot use Fourier Transforms to find the eigenbasis (e.g., Schrödinger with Harmonic Oscillator potential, variable coefficient Wave Eq).

Zak Katz: Right. Is ther a "theory" for diagonalizable operators S_t?

Raycast AI: Yes, there is a vast and rich "theory" for this, but it doesn't have a single name like "The Theory of Diagonalizable Operators." Instead, it is split into different fields of **Functional Analysis** depending on *why* the operator is diagonalizable.

If you are looking for the generalized version of "Fourier Analysis" for operators that are **not** translation invariant (i.e., not solvable by Fourier Transform), here is the map of the theory.

1. The "Gold Standard": The Spectral Theorem

This is the master theory. It applies if your operator S_t (or its generator L) is **Normal** ($LL^* = L^*L$).

- **The Theory: Spectral Theory of Normal Operators**.
- **What it says:** Any normal operator on a Hilbert space is unitarily equivalent to a multiplication operator.
- **Physical Meaning:** There *always* exists an orthonormal basis of eigenfunctions (or a "continuous basis" / spectral measure) that diagonalizes the operator.
- **How to find the basis:** This is the hard part. It usually requires solving the **Sturm-Liouville Problem** (for 1D) or using **Scattering Theory** (for unbounded domains).

2. The "Variable Coefficient" Theory: Sturm-Liouville Theory

If your PDE has variable coefficients but is in 1D (or separable), this is the standard replacement for Fourier analysis.

- **The Setup:** Your generator looks like $L = \frac{d}{dx}[p(x)\frac{d}{dx}] + q(x)$.
- **The Result:** Just like Fourier series uses $\sin(nx)$ and $\cos(nx)$, Sturm-Liouville theory guarantees a set of orthogonal eigenfunctions $\varphi_n(x)$ weighted by a density function.
- **Example:** Legendre polynomials, Bessel functions, Hermite polynomials (Quantum Harmonic Oscillator). These are the "custom Fourier bases" for specific geometries or potentials.

3. The "Abstract" Theory: Semigroup Theory (C_0-Semigroups)

This is the modern mathematical framework for time-evolution operators S_t.

- **The Setup:** You treat S_t as a "one-parameter semigroup" $T(t)$ acting on a Banach space.
- **The Theorem:** The **Hille-Yosida Theorem**.

- **What it does:** It characterizes exactly when a differential operator L generates a stable time-evolution S_t. While Hille-Yosida doesn't guarantee *diagonalizability* (it only guarantees existence and uniqueness of the flow), it is the starting point for analyzing the spectrum of S_t.
- **Spectral Mapping Theorem:** This theorem relates the eigenvalues of the generator L to the eigenvalues of the propagator S_t (basically $\sigma(S_t) = e^{t\sigma(L)}$). If you know the spectrum of L, you know the spectrum of S_t.

4. The "Non-Normal" Theory: Riesz Bases and Spectral Operators

What if S_t is diagonalizable but the eigenvectors are **not orthogonal**? (This happens in non-conservative systems, like convection-diffusion with non-standard boundary conditions).

- **The Theory: Non-Self-Adjoint Spectral Theory** (or the theory of **Riesz Bases**).
- **What it says:** You can still expand solutions in a basis of eigenfunctions, but because the basis is not orthogonal ($\langle \varphi_n, \varphi_m \rangle \neq \delta_{nm}$), you need a "dual basis" $\{\psi_n\}$ to find the coefficients.
  - Expansion: $f = \sum a_n \varphi_n$
  - Coefficients: $a_n = \langle f, \psi_n \rangle$ (instead of $\langle f, \varphi_n \rangle$).
- **Why it matters:** This is currently a very hot topic in fluid dynamics and laser physics, where systems are often "almost" normal but mathematically non-self-adjoint.

Summary: Which "Theory" do you need?

If your operator S_t is...	**The Theory is called...**	**The Basis is...**
Translation Invariant	Harmonic Analysis	Fourier Transform
Self-Adjoint / Normal	Spectral Theory	Orthonormal Eigenbasis
1D Variable Coeffs	Sturm-Liouville Theory	Special Functions
Not Normal (but diag.)	Non-Self-Adjoint Theory	Riesz Basis (Biorthogonal)
Not Diagonalizable	Jordan Form	Generalized Eigenvectors

7

Radon Measures and Dual Theory

(I really like the results of this chapter and am adamant to write it down here, especially the duality results. This chapter is a place-holder for now)

(intuition here once I have my own)

7.0.1 Convention In this chapter X will be a *locally compact Hausdorff space* (LCH space).

7.1 Positive Linear Functionals on Compact Support Continuous Functions

(I think there is a reason we start with $C_c(X)$ which I will eventually write down either here or in the chapter intro)

Definition 7.1.1: Positive Functional

Let $C_c(X)$ denote the real continuous functions with compact support. A linear functional $I : C_c(X) \to \mathbb{R}$ is called *positive* if $I(f) \geq 0$ whenever $f \geq 0$.

A positive functional was not defined to necessarily be continuous, however the positivity condition gives a sort of continuity that with $C_c(X)$ "having" compact-open topology, however this topology is in fact too coarse since as we'll see we shall want $\text{supp}(f) \subseteq K$ for compact $K \subseteq X$ which is not necessarily the case for the compact-open topology. Instead, we can form a Banach space $C_K(X)$ with the sup-norm $\|f\|_{u,K} = \sup_{x\ inK} |f(x)|$ and then take:

$$C_c(X) = \cup_{K \subseteq X} C_K(X)$$

and put the usual *filtered colimit topology* on it (in classical literature known as inductive limit topology). By the universal property of the topology, a functional is continuous if it is continuous on each $C_K(X)$.

Proposition 7.1.2: Continuity of Positivity

Let I be a positive linear functional on $C_c(X)$. Then for each compact $K \subseteq X$ there exists a C_K such that

$$|I(f)| \leq C_K \|f\|_u \qquad \forall f \in C_c(X) \text{ where } \mathrm{supp} f \subseteq K$$

***Proof*:**
Let $f \geq 0$. Given $K \subseteq X$, by Urysohn's lemma choose $\varphi \in C - c(X, [0,1]0$ such that $\varphi = 1$ on K. Then if $\mathrm{supp}(f) \subseteq K$,

$$|f(x)| \leq \|f\|_u \varphi(x) \qquad \text{i.e.} \qquad |f| \leq \|f\|_u \varphi$$

Hence $\|f\|_u \varphi - f \geq 0$. As f is positive $\|f\|_u \varphi + f \geq 0$. Thus by positivity of I:

$$\|f\|_u I(\varphi) - I(f) \geq 0 \qquad \text{and} \qquad \|f\|_u I(\varphi) + I(f) \geq 0$$

Hence:

$$\|I(f)| \leq I(\varphi)\|f\|_u$$

as we sought to show.

Now, given μ is a Bore measure on X such that $\mu(K) < \infty$ for all compact $K \subseteq X$, then clearly $C_c(X) \subseteq L^1(\mu)$ So the map:

$$f \mapsto \int f d\mu$$

is a positive linear functional on $C_c(X)$. What we shall show is that *every* positive linear functional on $C_c(X)$ is of this form, and with a little more regularity μ can be made unique in some way feels like this should be said earlier.

Definition 7.1.3: Regular Borel Measure

Let μ be a Borel measure on X and $E \subseteq X$ a Borel subset. Then μ is *outer-regular* on E if:

$$\mu(E = \inf \{\mu(U) \mid U \supseteq E, U \text{ open}\}$$

inner regular E if:

$$\mu(E = \inf \sup \mu(K) K \subseteq E, K \text{ compact}$$

and regular on E if E is inner- and outer-regular. The Borel μ is called outer-regular, inner-regular, and regular if all Borel sets are outer-regular, inner-regular, and regular respectively.

In most spaces, a regular measure shall work. However, we shall have some edge-cases of spaces that are not σ-compact on which we cannot define a regular measure (see ref:HERE for where this comes up)

Example 7.1.4: Too Big For Regular Measure
(you can make one on the LCH space $X = [0, \omega_1)$ which is the collection of all countable ordinals, which is not σ-compact, and take the Borel σ-algebra given the Dieudonné Measure)

Definition 7.1.5: Radon Measure

Let μ be a measure on X. Then if μ is a Borel measure that is:

1. finite on all compact sets
2. outer-regular on all Borel sets
3. inner-regular on all open sets

then μ is called a *Radon Measure*.

Proposition 7.1.8 shall show that all Radon measure are also inner-regular on all σ-finite sets.

Now, for the following theorem define the following notation: if $U \subseteq X$ is open and $f \in C_c(X)$ then:

$$f \prec U \qquad \text{iff} \qquad 0 \leq f \leq 1 \text{ and } \operatorname{supp}(f) \subseteq U$$

Note this is note quite the same as $0 \leq f \leq \chi_U$ as we this would imply $\operatorname{supp}(f) \subseteq \overline{U}$.

Theorem 7.1.6: Riesz Representation Theorem

If I is a positive linear functional on $C_c(X)$, there is a unique Radon measure μ on X such that $I(f) = \int f d\mu$ for all $f \in C_c(X)$. Moreover, μ satisfies

$$\mu(U) = \sup \{I(f) \mid f \in C_c(X), f \prec U\} \text{ for all open } U \subseteq X \tag{7.1}$$

and

$$\mu(K) = \inf \{I(f) \mid f \in C_c(X), f \geq \chi_K\} \text{ for all compact } K \subseteq X \tag{7.2}$$

***Proof*:**
We will first tackle the uniqueness of such a measure, which will naturally guide us toward the correct construction for existence.

Suppose such a Radon measure μ exists. Then for any open $U \subseteq X$, if $f \prec U$ (meaning $f \in C_c(X)$, $0 \leq f \leq 1$, and $\operatorname{supp}(f) \subseteq U$), we have:

$$I(f) = \int f d\mu \leq \int \chi_U d\mu = \mu(U)$$

Conversely, for any compact $K \subseteq U$, Urysohn's Lemma guarantees a function $f \in C_c(X)$ such that $f \prec U$ and $f = 1$ on K. Thus:

$$\mu(K) = \int \chi_K d\mu \leq \int f d\mu = I(f)$$

Since μ is a Radon measure, it is inner regular on open sets, meaning $\mu(U) = \sup_{K \subseteq U} \mu(K)$. Combining this with the inequalities above shows that $\mu(U)$ must be exactly the supremum of

$I(f)$ for $f \prec U$, satisfying (7.1). Since a Radon measure is determined by its values on open sets (due to outer regularity), μ is unique.

To prove existence, we simply turn this uniqueness property into a definition. For any open U, define:

$$\mu(U) = \sup\{I(f) \mid f \in C_c(X),\ f \prec U\}$$

and for an arbitrary set $E \subseteq X$, we define the outer measure μ^* standardly as:

$$\mu^*(E) = \inf\{\mu(U) \mid U \supseteq E,\ U \text{ open}\}$$

Note that for open sets, $\mu^*(U) = \mu(U)$ since $U \subseteq V \implies \mu(U) \leq \mu(V)$. To complete the theorem, we proceed in four steps:

1. Show μ^* is an outer measure.
2. Show every open set is μ^*-measurable.

Once these two are established, Carathéodory's Theorem (theorem 1.3.4) implies that all Borel sets are measurable and $\mu = \mu^*|_{\mathcal{B}_X}$ is a Borel measure. We then must show:

3. μ satisfies the inner regularity condition (7.2).
4. The functional is recovered by the integral: $I(f) = \int f d\mu$.

1. *μ^* is an outer measure:* The only non-trivial property to check is countable subadditivity on open sets (since the extension to arbitrary sets follows from proposition 1.3.2). Let $U = \bigcup_{j=1}^{\infty} U_j$. We want to show $\mu(U) \leq \sum \mu(U_j)$. Pick any $f \prec U$. Then $K = \text{supp}(f)$ is a compact subset of U, so it is covered by finitely many U_j, say N. Using a Partition of Unity ref:HERE (Proposition 4.41 in folland), we can find $g_1, \ldots, g_N \in C_c(X)$ such that $g_j \prec U_j$ and $\sum g_j = 1$ on K. Then $f = \sum_{j=1}^{N} f g_j$, and since each $f g_j \prec U_j$:

$$I(f) = \sum_{j=1}^{N} I(f g_j) \leq \sum_{j=1}^{N} \mu(U_j) \leq \sum_{j=1}^{\infty} \mu(U_j)$$

   Taking the supremum over all such f, we get $\mu(U) \leq \sum \mu(U_j)$, as desired.

2. *Open sets are measurable:* we need to show that for any open U and any E with finite outer measure, $\mu^*(E) \geq \mu^*(E \cap U) + \mu^*(E \setminus U)$. First, assume E is open. Given $\epsilon > 0$, we can choose an $f \prec E \cap U$ such that $I(f)$ is close to $\mu(E \cap U)$, specifically $I(f) > \mu(E \cap U) - \epsilon$. Similarly, since $E \setminus \text{supp}(f)$ is open, we choose $g \prec E \setminus \text{supp}(f)$ with $I(g) > \mu(E \setminus \text{supp}(f)) - \epsilon$. Since f and g have disjoint supports, $f + g \prec E$. Thus:

$$\begin{aligned} \mu(E) \geq I(f+g) &= I(f) + I(g) \\ &> \mu(E \cap U) + \mu(E \setminus \text{supp}(f)) - 2\epsilon \\ &\geq \mu^*(E \cap U) + \mu^*(E \setminus U) - 2\epsilon \end{aligned}$$

   Letting $\epsilon \to 0$ gives the result for open E. For a general E, we approximate it from the outside by an open $V \supset E$ such that $\mu(V) < \mu^*(E) + \epsilon$ and apply the previous logic to V.

3. *Inner Regularity:* We verified outer regularity by construction; now we check (7.2) for compact K. If $f \in C_c(X)$ and $f \geq \chi_K$, let $U_\epsilon = \{x \mid f(x) > 1 - \epsilon\}$. This is an open set containing K. If we take any $g \prec U_\epsilon$, then $f \geq (1-\epsilon)g$, so $I(g) \leq (1-\epsilon)^{-1}I(f)$. Taking the supremum over g, we get $\mu(K) \leq \mu(U_\epsilon) \leq (1-\epsilon)^{-1}I(f)$. As $\epsilon \to 0$, $\mu(K) \leq I(f)$. Conversely, for any open $U \supset K$, Urysohn's Lemma gives us an f where $\chi_K \leq f \prec U$, so $I(f) \leq \mu(U)$. Taking the infimum over U confirms the equality.

4. *Recovering the Integral:* it suffices to prove $I(f) = \int f d\mu$ for functions with range $[0,1]$ (by linearity). The strategy is to slice the range of f into N strips of height $1/N$, similar to constructing a Lebesgue integral. Let $K_j = \{x \mid f(x) \geq j/N\}$ for $j = 0, \ldots, N$. We define a "layer" function f_j that lives between K_j and K_{j-1}:

$$f_j = \min\left\{\max\left\{f - \frac{j-1}{N}, 0\right\}, \frac{1}{N}\right\}$$

Notice that $f = \sum_{j=1}^{N} f_j$. We can bound f_j by characteristic functions: $\frac{1}{N}\chi_{K_j} \leq f_j \leq \frac{1}{N}\chi_{K_{j-1}}$. Integrating this gives:

$$\frac{1}{N}\mu(K_j) \leq \int f_j d\mu \leq \frac{1}{N}\mu(K_{j-1})$$

Applying the functional I (and using the fact that $Nf_j \prec U$ for any open $U \supset K_{j-1}$) yields the same bounds:

$$\frac{1}{N}\mu(K_j) \leq I(f_j) \leq \frac{1}{N}\mu(K_{j-1})$$

Summing these up for all j, both $\int f d\mu$ and $I(f)$ are trapped in the same interval of size $\frac{1}{N}(\mu(K_0) - \mu(K_N))$. Since $\mu(K_0)$ is finite (it's the support of f) and N is arbitrary, the difference must be zero, so $I(f) = \int f d\mu$, completing the proof.

It is worth noting that the construction in this proof actually yields something stronger than just the statement of the theorem. We obtain not just a Borel measure μ, but an extension $\overline{\mu}$ of μ to the σ-algebra of all μ^*-measurable sets. Recall that μ is outer regular. Because of this, for any $E \subseteq X$, the outer measure is effectively approximated by Borel sets:

$$\mu^*(E) = \inf\{\mu(B) \mid B \in \mathcal{B}_X,\ B \supseteq E\}$$

This means that μ^* corresponds exactly to the outer measure induced by μ. Consequently, the extension $\overline{\mu}$ relates to the standard completion of a measure: if μ is σ-finite, $\overline{\mu}$ is precisely the completion of μ. In the general case, it is the saturation of the completion.

Finally, while we have focused on $\mathcal{B}_X$, some authors prefer to work with a slightly smaller σ-algebra, specifically the one generated by the continuous functions themselves. For ease of indexing, we box the definition:

Definition 7.1.7: Baire Sets

The σ-algebra generated by $C_c(X)$ (i.e., the smallest σ-algebra with respect to which every $f \in C_c(X)$ is measurable) is denoted by $\mathcal{B}_X^0$. The elements of $\mathcal{B}_X^0$ are called *Baire sets.*

Proposition 7.1.8: Inner Regularity on σ-finite Sets

Every Radon measure is inner regular on all of its σ-finite sets.

***Proof*:**
Let μ be a Radon measure and E be a σ-finite set. We proceed by cases.

1. Assume $\mu(E) < \infty$. The strategy here is to approximate E from the outside with an open set, and then approximate that open set from the inside with a compact set, ensuring we stay "mostly" inside E. For any $\epsilon > 0$, since μ is outer regular on Borel sets, we can choose an open $U \supseteq E$ such that $\mu(U) < \mu(E) + \epsilon$. Since μ is inner regular on open sets, we can find a compact $F \subseteq U$ such that $\mu(F) > \mu(U) - \epsilon$.

   However, F is not necessarily a subset of E. The part of F that "spills over" is contained in $U \setminus E$. Notice that:
   $$\mu(U \setminus E) = \mu(U) - \mu(E) < \epsilon$$
   Since this "error set" $U \setminus E$ is Borel, we can use outer regularity again to cover it with an open set $V \supseteq U \setminus E$ such that $\mu(V) < \epsilon$. Now, define $K = F \setminus V$. Since V is open, K is a closed subset of the compact set F, and thus K is compact. Furthermore, since we removed the part covering $U \setminus E$, we have $K \subseteq E$. Finally, we check the measure:
   $$\mu(K) = \mu(F) - \mu(F \cap V) > (\mu(E) - \epsilon) - \mu(V) > \mu(E) - 2\epsilon$$
   Since ϵ is arbitrary, μ is inner regular on E.

2. Assume $\mu(E) = \infty$. Since E is σ-finite, it is an increasing union of sets E_j with $\mu(E_j) < \infty$ and $\mu(E_j) \to \infty$. For any large number N, there exists some j such that $\mu(E_j) > N$. Since E_j has finite measure, by the argument in Case 1, we can find a compact $K \subseteq E_j \subseteq E$ such that $\mu(K)$ is arbitrarily close to $\mu(E_j)$, and specifically $\mu(K) > N$. Thus, the supremum of the measures of compact subsets is infinite, satisfying inner regularity.

Corollary 7.1.9: Regularity of Radon Measures

Every σ-finite Radon measure is regular. Furthermore, if X is σ-compact, every Radon measure on X is regular.

7.2 Dual of $C_0(X)$

8

Hilbert Spaces

There are many reasons why Hilbert spaces are interesting. We shall start with a motivation stemming from looking for the ideal space where the product of two functions is still integrable, and then take a moment to elaborate on some more advanced intuitions of Hilbert spaces for readers with more background.

We are naturally interested in integrating products of functions:

$$\int_X fg$$

The problem we may encounter when working over L^1 is that $\int_X fg$ need not be finite. We thus may say "let us thus work work the functions in which this *is* well-defined". If $L'(X)$ is the space of all functions measurable functions where for any choice $f, g \in L'(X)$:

$$\int_X f\overline{g} < \infty \qquad \text{i.e.} \qquad f\overline{g} \in L^1$$

then we have just made a circular-reasoning mistake, for we are quantifying existence of elements in $L'(X)$ using elements of $L'(X)$. Instead, we may ask: given a set S, what is the subset of measurable functions T such that:

$$T = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X f\overline{g} < \infty,\ \forall g \in S \right\}$$

The two extremes of this are if $S = \{0\}$, then T is all measurable functions, and if S is all measurable functions, then $T = \{0\}$ only contains the zero function. The ideal is that there is a middle ground choice of S where $S = T$ so that we don't need to keep track of the two spaces we are choosing our functions from, namely we would not like to have $(S, T)(X)$. Fortunately, as will be proven in section 6.2, this is *exactly* the space L^2. Namely, if $S = L^2$, then:

$$T = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X f\overline{g} < \infty \ \forall g \in L^2 \right\} = L^2$$

Though we can't prove this here, the high level is that if $f, g \in L^2$, then by the Cauchy-Schwartz inequality[1]

$$\int_X |fg| \leq \left(\int_X |f|^2\right)^{1/2} \left(\int_X |g|^2\right)^{1/2} = \|f\|_2 \|g\|_2$$

showing that $L^2 \subseteq T$. Conversely, by Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. If $p = 2$, then $q = 2$, showing that all functionals $(L^2)^*$ are characterized by L^2 functions:

$$\varphi_g : L^p \to \mathbb{R} \qquad \varphi_g(f) = \int fg$$

showing that $T \subseteq L^2$, hence $T = L^2$. From this perspective, L^2 is the ideal space in which to multiply functions and they remain integrable (i.e. $f, g \in L^2$, $fg \in L^1$)! We shall call such multiplication the *inner product*:

$$\langle f, g \rangle = \int_X f\overline{g}$$

The conjugation in the definition is used so that $f\overline{f} = |f|^2$. Noe that we are not looking for the product of two functions to remain within L^1. In section 10.1.1, we shall see that *convolution* will have the desired property so that $f, g \in L^1(G)$ then $f * g \in L^1(G)$ for G a locally compact group, in particular that shall be a *group algebra*.

8.0.1 post-Hoc Motivation A lot of the following motivation came *a-posteriori*, that is, I felt much more comfortable with Hilbert spaces after leaning about L^p spaces, and generalizing some result from Riemannian Geometry to Banach spaces to notice that L^2 is the only space with flat geometry, hence really is the "correct" generalization of euclidean space/geometry in infinite dimensions.

Banach spaces provide a robust framework for measuring the magnitude of elements, allowing us to quantify convergence and bound operations via the triangle and Minkowski inequalities. However, a fundamental deficiency in general Banach spaces is the lack of a linear measurement theory to quantify *similarity*. While we can measure the total size of a sum $\|f + g\|$, we often lack the granular information required to isolate the contribution of f from g. For instance, determining how much a vector projects onto a specific basis element is straightforward in Euclidean space but algebraically obstructed in general norms.

To make this deficiency precise, consider how the inner product arises in Euclidean geometry. Define the *energy* of a vector as $E(x) = \frac{1}{2}\|x\|_2^2$. Then the directional derivative of this energy at x in the direction of a perturbation y is:

$$\frac{d}{dt} E(x + ty)\Big|_{t=0} = \frac{d}{dt} \frac{1}{2} \langle x + ty, x + ty \rangle\Big|_{t=0} = \langle x, y \rangle$$

Thus, the inner product $\langle x, y \rangle$ is not just an algebraic artifact; it physically represents the *sensitivity* of the energy of x to a perturbation by y. In particular, the energy satisfies the clean decomposition:

$$E(x + y) = E(x) + E(y) + \langle x, y \rangle$$

[1]which shall be proven in this section, but the reader most likely has already encountered it and hence it is used here to give intuition

The interaction term $\langle x, y\rangle$ is *bilinear*—it decomposes the interference between two vectors according to the simple rules of superposition. In general Banach spaces, however, such a clean, linear decomposition of the interaction is impossible due to the non-flat geometry of the unit ball.

The natural question, then, is: *what replaces the inner product in L^p?* The correct generalization is to replace the quadratic energy with the *p-energy*:

$$F(f) = \frac{1}{p}\|f\|_p^p = \frac{1}{p}\int |f|^p$$

and ask: how does $F(f)$ change if we perturb f by a function g? To answer this, we must compute the gradient ∇F, which requires first understanding the natural duality of L^p spaces.

The Duality Map. What does it mean to go in the "same direction" in a function space? (For simplicity we work throughout with L^p spaces for $1 < p < \infty$.[2]) Rigorously, we are asking for the functional φ_f of unit norm that *maximizes* the evaluation at f. We can interpret this φ_f as the *optimal filter* or sensor for f—it captures the full "energy" of f without loss ($\varphi_f(f) = \|f\|_p$ and $\|\varphi_f\|_p = 1$).

By theorem 6.2.3 (and the equality condition of Hölder's inequality), this functional corresponds to $g \in L^q$ such that:

$$g(x) = |f(x)|^{p-1}\mathrm{sgn}(f(x))$$

Note that $g \in L^q$ since:

$$|g|^q = |f|^{q(p-1)} = |f|^{qp-q} \overset{!}{=} |f|^p$$

where $\overset{!}{=}$ holds since $1/p + 1/q = 1$ iff $p + q = pq$ iff $p = q(p-1)$. Hence as $f \in L^p$:

$$\int |g|^q = \int |f|^p < \infty$$

Thus, we have a map $f \mapsto \varphi_f$. We define the *duality map*:

$$J_p : L^p \to L^q, \qquad f \mapsto \varphi_f$$

As $(L^q)^* \cong L^p$, we can equivalently view this as a map $J_p : L^p \to (L^p)^*$.

The Central Geometric Fact. The duality map J_p is not an algebraic accident—it is the *geometric normal* to the level sets of the p-energy. To see this, consider the function:

$$F(x) = \frac{1}{p}\|x\|_p^p = \frac{1}{p}\sum |x_i|^p$$

(where the ℓ^p norm is used for visualization, but the logic holds for integrals). From multivariable calculus, the unit sphere is the level set $F(x) = \frac{1}{p}$, and the gradient $\nabla F(x)$ is the vector strictly perpendicular (normal) to the level set at point x. Computing the gradient (the Fréchet derivative):

$$\frac{\partial}{\partial x_i}\left(\frac{1}{p}|x_i|^p\right) = |x_i|^{p-1}\mathrm{sgn}(x_i)$$

[2]The reasoning can extend to nuclear Fréchet spaces, which admit descriptions as projective limits of Hilbert spaces. For general metric spaces, a different approach is needed via CAT(k) geometry; see [8] for details.

So, the gradient vector is:

$$\nabla F(x) = \big(|x_1|^{p-1}\mathrm{sgn}(x_1), \ldots, |x_n|^{p-1}\mathrm{sgn}(x_n)\big)$$

This is exactly our duality map:

$$\boxed{\nabla F(x) = J_p(x)}$$

This identification is the key to the entire theory. It tells us that the optimal functional for f—the element of $(L^p)^*$ that best "measures" f—is precisely the gradient of the energy at f: the direction of steepest increase in p-energy. The duality map is a *geometric* object, not merely an algebraic one.

The p-Pairing and Its Interpretation. With the gradient in hand, we can now answer our original question: how does the p-energy $F(f)$ change under a perturbation by g? The first-order variation is[3]:

$$dF(f; g) = \langle \nabla F(f), g\rangle_{\text{dual}} = \int \nabla F(f) \cdot g$$

Substituting $\nabla F = J_p(f)$, this defines a natural pairing:

$$\langle f, g\rangle_p := \int J_p(f) \cdot g$$

Note that it is important to pass through $J_p(f)$ since $\int fg$ need not be finite in general. However, by Hölder's inequality:

$$\int |J_p(f) \cdot g| < \infty$$

This pairing has a precise geometric interpretation. The gradient $J_p(f)$ is a cotangent vector—a 1-form $dF|_f \in T_f^* L^p$—and the perturbation g is a tangent vector $g \in T_f L^p$. Thus the pairing is the evaluation of a 1-form on a tangent vector:

$$\langle f, g\rangle_p = dF|_f(g) \;\in\; T_f^* L^p \times T_f L^p \to \mathbb{R}$$

This canonical evaluation pairing between the tangent and cotangent spaces exists in *every* geometric setting without any choice of metric. What J_p adds is the ability to *produce* a specific cotangent vector from a tangent vector, turning the pairing from a bilinear form on $T_f L^p \times T_f^* L^p$ into a (generally nonlinear) form on $T_f L^p \times T_f L^p$—that is, internalizing the measurement within the space itself.

The Infinitesimal Meaning of V^*. We are now in a position to make a precise statement about the geometric content of the dual space. In a bare vector space without additional structure, the dual V^* is merely an algebraic object—a collection of linear maps with no geometric significance beyond probing linearly. However, in L^p equipped with its norm, the identity $J_p = \nabla F$ reveals that *every element of $(L^p)^*$ arises as the gradient of the energy functional*—a differential $dF|_f$. This endows $(L^p)^*$ with genuine infinitesimal meaning: its elements are not just linear maps, but *cotangent vectors* that encode the first-order sensitivity of the geometry.

Crucially, cotangent vectors are a different species of infinitesimal object than tangent vectors. A tangent vector $g \in T_f L^p \cong L^p$ represents an *infinitesimal direction of motion*—it answers "which way?". A cotangent vector $\varphi \in T_f^* L^p \cong L^q$ represents an *infinitesimal measurement*—it answers "how fast is this quantity changing?". These are related by the evaluation pairing $\varphi(g) = dF|_f(g)$, but they

[3]The dF here can be thought of as the Gâteaux derivative, commonly defined for functions between locally convex topological vector spaces.

transform in opposite ways: tangent vectors push forward under maps, while cotangent vectors pull back. The duality map $J_p : T_f L^p \to T_f^* L^p$ provides a (nonlinear) identification between these two types of infinitesimal data, allowing us to transfer the directional interpretation onto the dual space. However, this transfer *depends essentially on the norm geometry*—it is canonical only for $p = 2$.

Geometrically, if we view the identity map $f \mapsto f$ as the radial vector field (a *section* of the tangent bundle TL^p), then the duality map J_p transforms this into a covector field (a *section* of the cotangent bundle T^*L^p). In the language of Riemannian geometry, this transformation is the *musical isomorphism* $\flat$ (flat), which maps a vector field X to a 1-form $X^\flat$ by lowering indices via the metric tensor. The four corollaries that follow are all consequences of this single geometric fact.

Corollary 1: Nonlinearity of the Musical Isomorphism. The duality map is *not* linear:

$$J_p(2f) = |2f|^{p-1}\mathrm{sgn}(2f) = 2^{p-1}|f|^{p-1}\mathrm{sgn}(f) = 2^{p-1}J_p(f)$$

This is only linear in the special case of $p = 2$, since:

$$J_2(nf) = |nf|^{2-1}\mathrm{sgn}(nf) = nJ_2(f)$$

This distinction is profound. In a general Banach space ($p \neq 2$), the musical isomorphism is *nonlinear*: the operation of lowering an index depends on the magnitude of the vector itself, meaning the “metric” changes as we move through the space. However, when $p = 2$, this map is linear. This implies a linear bundle isomorphism $TL^2 \cong T^*L^2$, effectively trivializing the distinction between vectors and covectors. Only in L^2 is the musical isomorphism a linear operator, meaning the “force” required to correct an error is linearly proportional to the error itself.

Corollary 2: Non-Symmetry of the Pairing. The pairing $\langle f, g\rangle_p$ is *not* symmetric for $p \neq 2$. This is an immediate consequence of the nonlinearity of J_p: the 1-form $dF|_f$ evaluated on g has no reason to equal the 1-form $dF|_g$ evaluated on f, since the gradient depends nonlinearly on its base point. If however $p = 2$, then $J_p(x)$ maps identically to itself and we recover the symmetric form $\int fg$. This asymmetry is the first signal that the geometry of L^p is *not flat* for $p \neq 2$.

Corollary 3: Failure of Rotational Invariance. This lack of symmetry in the pairing $\langle f, g\rangle_p$ is directly linked to the fact that L^2 is the unique L^p space that is *rotationally invariant.* Geometrically, rotational invariance means that the properties of a vector depend only on its length, not on its alignment with the coordinate axes. Mathematically, this requires that the duality map commutes with rotations. If R is a rotation operator (an isometry), we expect the “normal vector” of the rotated point to be the rotated “normal vector” of the original point:

$$J_p(Rf) \stackrel{?}{=} R(J_p(f))$$

In L^2, since J_2 is linear (essentially the identity), this holds trivially: $J_2(Rf) = Rf = R(J_2(f))$. However, for $p \neq 2$, the map $f \mapsto |f|^{p-1}\mathrm{sgn}(f)$ applies a non-linear distortion component-wise. Rotating the vector mixes the components, and applying the power law *after* mixing is not the same as applying it *before.*

To see this concretely, consider ℓ^3 ($p = 3$) in $\mathbb{R}^2$ and let $f = (1, 0)$. If we first compute the functional and then rotate by $\pi/4$ (45°), we get:

$$R(J_3(f)) = R(1^2, 0^2) = R(1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

However, if we rotate the vector first, the components become $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Applying the duality map J_3 (squaring components) yields:

$$J_3(Rf) = J_3\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Since $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \neq (\frac{1}{2}, \frac{1}{2})$, we have $J_p(Rf) \neq R(J_p(f))$. This confirms that for $p \neq 2$, the geometry is *anisotropic*: the shape of the unit ball looks different depending on the angle from which you view it.

Corollary 4: Birkhoff-James Orthogonality. With this understanding of the position-dependent metric, we can clarify the notion of orthogonality. In Euclidean geometry, orthogonality is symmetric ($x \perp y \iff y \perp x$) and additive. In general Banach spaces, without a global bilinear form, we must rely on *Birkhoff-James orthogonality*: we say $x \perp y$ if moving in the direction of y does not decrease the length of x ($\|x + \lambda y\| \geq \|x\|$). Geometrically, this means the line passing through x in the direction of y is tangent to the unit sphere at x.

Analytically, utilizing the duality map J_p, this condition is equivalent to requiring that the tangent vector y be annihilated by the cotangent vector generated by x:

$$\langle J_p(x), y \rangle = 0$$

This is precisely the statement that y lies in the kernel of the 1-form $dF|_x$—a natural geometric condition. However, it reveals the fundamental asymmetry in non-Hilbert geometry. In L^2, because J_2 is linear, the condition is symmetric. However, in L^p ($p \neq 2$), the non-linearity of J_p breaks reciprocity. The set of vectors orthogonal to x (kernel of $J_p(x)$) does not align symmetrically with the set of vectors orthogonal to y. Consequently, it is common to have $x \perp y$ but $y \not\perp x$. This lack of symmetry illustrates why Hilbert spaces are the unique setting where Euclidean geometric intuition remains valid.

The Riemannian Perspective. The corollaries above all concern the first derivative of the energy (the gradient J_p). The full geometry of the space, however, is determined by how this gradient *changes*—the second derivative, which acts as the metric tensor. In this framework, the metric tensor $g_{ij}(x)$ at a point x dictates how to measure the length of tangent vectors. In our context, this metric is generated by the Hessian of the energy function $F(x) = \frac{1}{p}\|x\|_p^p$. For L^2, this Hessian is proportional to the identity operator everywhere; the metric is *constant* (flat). However, for L^p ($p \neq 2$), the second derivative depends on position x, specifically scaling as $|x|^{p-2}$. This implies that the "metric" changes as we move around the space.

It is worth noting why the theory breaks down at the boundary cases. For L^1, the unit ball is a diamond with corners; at these corners, the gradient is not unique (it is a set-valued subgradient). For L^∞, the unit ball has flat faces; on these faces, the gradient is non-unique or vanishes in certain directions. This lack of smoothness (L^1) or rotundity (L^∞) prevents a well-defined duality map, leaving L^p ($1 < p < \infty$) as the sweet spot for smooth geometry.

Topological vs. Geometric Structure. To visualize the practical consequences of this non-flat geometry, it is helpful to consider the "shape" of the unit ball in infinite dimensions. In finite dimensions, all norms are equivalent, meaning the unit ball of one norm can always be fit inside a scaled unit ball of another. In infinite dimensions, this equivalence breaks. The curvature of the norm dictates how "energy" is allowed to concentrate without bound. The L^∞ unit ball acts like a strict box, while the L^1 unit ball allows for infinitely long thin spikes.

Ideally, we would like to know if these distortions are topological or "merely" geometric. This is answered by the *Mazur Map* $M_{p,q} : L^p(B) \to L^q(B)$ where B is the unit ball. The map:

$$M_{p,q}(f) = \mathrm{sgn}(f) \cdot |f|^{p/q}$$

shows that all $L^p(B)$ spaces are *uniformly homeomorphic* to each other (not just homeomorphic as theorem 5.5.5 shows). This means that we can smoothly deform the "spikes" of L^1 into the "sphere" of L^2. However, while the Mazur map preserves the topology, it destroys the linear structure (additivity). The Hilbert space L^2 is the unique fixed point where this topological structure aligns perfectly with the linear algebraic structure (where the Mazur map would be the identity).

This geometric disparity explains why convergence depends on the norm. Convergence $f_n \to 0$ is not about movement, but about entering arbitrarily small ϵ-balls. Consider the sequence $f_n(x) = x^n$ on $[0, 1]$. In the L^1 geometry, this sequence travels down one of the "spikes" of the unit ball: the functions get thinner and thinner, their area shrinking to zero despite their height remaining constant. In the L^∞ geometry, however, these functions are trapped on the surface of the unit sphere; they cannot enter the ϵ-ball because the L^∞ "box" does not extend down the "spikes" of concentration.

Consequences for Hilbert Spaces. The unique geometry of Hilbert spaces—specifically self-duality, isotropy, and the inner product—unlocks powerful results unavailable in general Banach spaces. The preceding analysis explains *why*: in L^2, the duality map J_2 is the identity, making the musical isomorphism linear, the pairing symmetric, and the geometry flat. The following are key consequences we will explore:

1. *The Projection Theorem (Best Approximation):* In general Banach spaces, finding the element in a subspace M closest to a point x is difficult (the closest point might not be unique, e.g., the flat edge of L^1). In Hilbert spaces, the strict convexity and "roundness" of the unit ball guarantee that for any closed subspace M, there exists a *unique* element $P_M x$ minimizing the distance $\|x - m\|$. Furthermore, the error vector $x - P_M x$ is orthogonal to the subspace.

2. *The Riesz Representation Theorem (Self-Duality):* The duality map $J : H \to H^*$ is linear and bijective. This means we can completely identify linear functionals with vectors. Every continuous linear functional φ acts exactly like an inner product: $\varphi(x) = \langle x, y \rangle$ for a unique y. This collapses the study of H^* to the study of H.

3. *Orthonormal Bases and Fourier Series:* In L^p, we may have a Schauder basis[4], but determining the coefficients is non-trivial. In Hilbert spaces, orthogonality allows us to decouple coefficients. This is key for the Fourier transform, where any element is reconstructed by projection: $x = \sum \langle x, e_n \rangle e_n$. Additionally, the *Parseval Identity* allows us to measure size by summing squared coefficients: $\|x\|^2 = \sum |\langle x, e_n \rangle|^2$.

4. *Existence of Adjoints:* In Banach spaces, the adjoint of $T : X \to Y$ maps $Y^* \to X^*$. Because Hilbert spaces are self-dual, we can pull this operator back to the original space. Thus, for every bounded operator T on a Hilbert space, there is a unique adjoint T^* on the *same space* satisfying $\langle Tx, y \rangle = \langle x, T^* y \rangle$. This is foundational for spectral theory.

In summary, while Banach spaces provide the necessary topological framework (completeness), Hilbert spaces add the rigid geometric structure required to generalize linear algebra to infinite dimensions.

[4] A sequence $\{b_n\}$ where every $v \in V$ has a unique expansion $v = \sum \alpha_n b_n$.

The duality map J_p, understood as the gradient of the p-energy, reveals the precise mechanism: it is the *cotangent structure* of the space—the infinitesimal measurement theory—that determines the geometry. The canonical evaluation pairing between tangent and cotangent vectors is the universal source of "derivative-like" phenomena; J_p is the geometric data that internalizes this pairing within the space itself. In L^2 alone, this internalization is linear, making vectors and covectors—directions and measurements—two aspects of a single object.

8.1 Basic definition and properties

Definition 8.1.1: Inner Product Space

Let H be a complex vector space with an inner product $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{C}$ where

1. $\langle x, y \rangle = \overline{\langle y, x \rangle}$
2. $\langle x + y, z \rangle = \langle x, z \rangle + \langle x, z \rangle$
3. $\langle ax, y \rangle = a\langle x, y \rangle$, $a \in \mathbb{C}$
4. $\langle x, x \rangle \in (0, \infty)$ for all nonzero $x \in H$
5. $\langle x, x \rangle = 0$ if and only if $x = 0$

We'll define $\|x\|_2 = \sqrt{\langle, x, x \rangle}$ where the 2 subscript hints that this will become the L^2-norm and is added to not hold ambiguity with the L_1 norm. This is naturally a norm since:

1. $\|x\|_2 = \sqrt{\langle x, x \rangle} = 0$ if and only if $x = 0$
2. $\|ax\|_2 = \sqrt{\langle ax, ax \rangle} = \sqrt{a\overline{a}\langle x, x \rangle} = \sqrt{a^2\langle x, x \rangle} = |a|\sqrt{\langle x, x \rangle} = |a|\|x\|_2$

Proving the triangle inequality is a bit more difficult: IF we square both sides, we it is equivalent to show:

$$\|x + y\|_2^2 \leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2$$

expanding out the left hand side, we get:

$$\|x + y\|_2^2 = \|x\|_2^2 + 2\text{Re}\langle x, y \rangle + \|y\|_2^2$$

so we have to show that $\text{Re}\langle x, y \rangle \leq \|x\|_2\|y\|_2$. In fact, a bit more is true: $|\langle x, y \rangle| \leq \|x\|_2\|y\|_2$. Though we are using this inequality to prove the triangle inequality, notice that this in fact gives us a general bound on the value of the inner product! This inequality is one of the central tools used in the study of Hilbert spaces:

Theorem 8.1.2: Cauchy-Schwartz Inequality

$$|\langle x, y \rangle| \leq \|x\|_2\|y\|_2$$

***Proof*:**
If $\|y\|_2 = 0$ then $y = 0$, and then $|\langle x, 0\rangle| = 0$, so assume $\|y\|_2 \neq 0$. Then for all $t \in \mathbb{R}$

$$\begin{aligned} 0 &\leq \langle x - ty, x - ty\rangle \\ &= \|x\|_2^2 + 2t\mathrm{Re}\langle x, y\rangle + t^2\|y\|_2^2 \\ &= \|x\|_2^2 + \frac{2\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} + \frac{\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} && t = \frac{\mathrm{Re}\langle x, y\rangle}{\|y\|_2^2} \\ \Rightarrow |\mathrm{Re}\langle x, y\rangle| &\leq \|x\|_2\|y\|_2 \end{aligned}$$

To finish the proof, pick $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ such that

$$|\mathrm{Re}(\alpha\langle x, y\rangle)| = |\langle x, y\rangle|$$

Notice this does not break the inequality, and so we can repeat the entire process but with α taken into account with t, and so:

$$|\langle x, y\rangle \leq \|x\|_2\|y\|_2$$

as we sought to show

With this, the rest of the proof that $\| - \|_2$ induced by $\langle -, -\rangle$ is a norm follows

Theorem 8.1.3: Triangle Inequality For Induced Norm

$$\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$$

***Proof*:**

$$\begin{aligned} \|x + y\|_2^2 &= \langle x + y, x + y\rangle \\ &\leq \|x\|_2^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|_2^2 \\ &\leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2 \end{aligned}$$

The Cauchy-Shwartz inequality is the tool we use to bring in geometry as we know it into Hilbert space. For example, in real inner product spaces, we will define the angle between two vectors to be θ where θ satisfies:

$$\langle u, v\rangle = \cos(\theta)\|u\|_2\|v\|_2$$

Many other cool properties of the Cauchy-Shwartz inequality can be found here: (commented)

Now, notice we can define a metric by doing $\rho(x, y) = \|x - y\|_2$:

1. $\rho(x, x) = \|x - x\|_2 = 0$ if and only if $x - x = 0$ so $x = x$
2. $\rho(x, y) = \|x - y\|_2 \geq 0$
3. $\rho(x, y) = \|x - y\|_2 = \|x - y + y - z\|_2 \leq \|x - y\|_2 + \|y - z\|_2 = \rho(x, y) = \rho(y, z)$

Thus, we can define a natural topology on any pre-Hilbert space. If that topology is complete, we give it a name:

Definition 8.1.4: Hilbert Space

Let H be an inner product space. Then H is a Hilbert space if it is *complete* with respect to the induced metric.

Example 8.1.5: Hilbert Spaces

1. $\mathbb{R}^n$ is a (real) Hilbert space with the inner product being:

$$\langle x, y\rangle = \sum_i x_i y_i$$

verifying this is a Hilbert space is left as an exercise.

2. $L^2_\mu(X)$ for a complex measure μ is a Hilbert space with the inner product being:

$$\langle f, g\rangle_{L^2} = \int_X f(x)\overline{g(x)}d\mu(x)$$

To show it's well-defined, recall that if $a, b \geq 0$:

$$2|ab| = 2|a||b| \leq |a|^2 + |b|^2$$

Thus:

$$|f(x)\overline{g(x)}| = |f(x)||g(x)| \leq |f(x)|^2 + |g(x)|^2$$

As integrating keeps inequality when integrating positive functions:

$$\|f\overline{g}\|_1 \leq \int_X |f(x)|^2 + \int_X |g(x)|^2 < \infty$$

showing it is well-defined. Alternatively, using Cauchy-Schwartz:

$$\|f\overline{g}\|_1 \leq \|f\|_2\|g\|_2 < \infty$$

3. As an exercise, show that $\ell^2(\mathbb{Z})$ is a Hilbert Space.

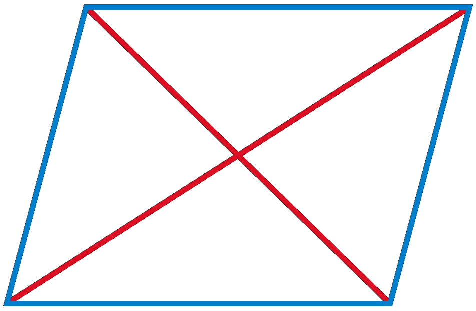
Hilbert spaces are very euclidean in nature, for example:

Theorem 8.1.6: Parallelogram Law

Let $\|-\|$ be a norm. Then $\|-\|$ is induced by an inner product if and only if:

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Visually, this means that the "area" of a parallelogram is can be measured in two ways:



***Proof*:**

$$\begin{aligned}\|x+y\|^2+\|x-y\|^2 &= \langle x+y,x+y\rangle+\langle x-y,x-y\rangle\\ &= 2\langle x,x\rangle+\langle x,y\rangle-\langle x,y\rangle+\langle y,x\rangle-\langle y,x\rangle+2\langle y,y\rangle\\ &= 2\langle x,x\rangle+2\langle y,y\rangle\\ &= 2\|x\|^2+2\|y\|^2\end{aligned}$$

It turns out that a norm satisfying the parallelogram law will in fact define an inner product. This shows how norms define geometries that are more general than euclidean:

Example 8.1.7: Exploring Parallelogram Law

1. (The polarization Identity)$\mathcal{H}$ be a Hilbert space. For any $x, y \in \mathcal{H}$,

$$\langle x,y\rangle=\frac{1}{4}(\|x+y\|^2-\|x-y\|^2+i\|x+iy\|^2-i\|x-iy\|^2)$$

Proof. This is simply a matter of computation:

$$\begin{aligned}&\frac{1}{4}(\|x+y\|^2-\|x-y\|^2+i\|x+iy\|^2-i\|x-iy\|^2)\\ &=\frac{1}{4}(\langle x+y,x+y\rangle-\langle x-y,x-y\rangle+i\langle x+iy\rangle-i\langle x-iy,x-iy\rangle)\\ &=\frac{1}{4}(\langle x,x\rangle+\langle x,y\rangle+\langle y,x\rangle+\langle y,y\rangle-(\langle x,x\rangle-\langle x,y\rangle-\langle y,x\rangle+\langle y,y\rangle)\\ &\quad+i(\langle x,x\rangle+\langle x,iy\rangle+\langle iy,x\rangle+\langle iy,iy\rangle-(\langle x,x\rangle-\langle x,iy\rangle-\langle iy,x\rangle+\langle iy,iy\rangle))\\ &=\frac{1}{4}(2\langle x,y\rangle+2\langle y,x\rangle+i(2\langle x,iy\rangle+2\langle iy,x\rangle))\\ &=\frac{1}{4}(2\langle x,y\rangle+2\langle y,x\rangle+i(-i)2\langle x,y\rangle+2ii\langle y,x\rangle)\\ &=\frac{1}{4}(2\langle x,y\rangle+2\langle y,x\rangle+2\langle x,y\rangle-2\langle y,x\rangle)\\ &=\frac{1}{4}4\langle x,y\rangle\\ &=\langle x,y\rangle\end{aligned}$$

□

2. If $\mathcal{H}'$ is another Hilbert space, a linear map from $\mathcal{H}$ to $\mathcal{H}'$ is unitary if and only if it is isometric and surjective

Proof. Let U be unitary so that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$ for any $x, y \in \mathcal{H}$. Letting $x = y$ gives us $\|Ux\|_2 = \|x\|_1$, giving us that it's an isometry. Since U is invertible, it is surjective, and so U is a surjective isometry.

Conversely, let U be a surjective isometry. We need to show its invertible (it suffices to show it's injective), an isometry, and that U and U^{-1} are bounded. Clearly, U is bounded since $\|U(x)\| \leq 1\|x\|$. For injectivity, let $Ux = Uy$, so $Ux - Uy = 0$. Taking the norm on both sides, we get:

$$0 = \|0\|_2 = \|Ux - Uy\|_2 = \|U(x-y)\|_2 = \|x - y\|_1$$

where the last equality is due to U being an isometry. Since $\|z\| = 0$ if and only if $z = 0$, we have $x - y = 0$, i.e. $x = y$, proving that U is injective. Along with surjectivity, that makes U invertible. With this, we can also show that U^{-1} is bounded: for any $y \in \mathcal{H}'$, there exists an $x \in \mathcal{H}$ such that $U(x) = y$. Thus:

$$\|U^{-1}(y)\| = \|x\| \overset{!}{=} \|U(x)\| = \|y\|$$

where $\overset{!}{=}$ comes from U being an isometry, showing that $\|U^{-1}(y)\| \leq 1 \cdot \|y\|$, making it also bounded.

Finally, to show that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$, we will use part (a):

$$\begin{aligned}
\langle x, y\rangle &= \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2) \\
&\overset{!}{=} \frac{1}{4}(\|U(x+y)\|^2 - \|U(x-y)\|^2 + i\|U(x+iy)\|^2 - i\|U(x-iy)\|^2) \\
&= \frac{1}{4}(\|U(x)+U(y)\|^2 - \|U(x)-U(y)\|^2 + i\|U(x)+iU(y))\|^2 - i\|U(x)-iU(y)\|^2) \\
&= \langle U(x), U(y)\rangle
\end{aligned}$$

where $\overset{!}{=}$ comes from U being an isometry. Thus:

$$\langle Ux, Uy\rangle = \langle x, y\rangle$$

As we sought to show. □

Proposition 8.1.8: Inner Product is Continuous

Let V be a Hilbert space. Then $\langle -, -\rangle : V \times V \to \mathbb{R}$ (or $\mathbb{C}$) is continuous

***Proof*:**
As $V \times V$ has the product topology, it is enough to show that if $x_n \to x$ and $y_n \to y$ in V then

$$\langle x_n, y_n\rangle \to \langle x, y\rangle$$

Take:

$$\begin{aligned} |\langle x,y\rangle - \langle x_n,y_n\rangle| &= |\langle x,y\rangle - \langle x_n,y\rangle + \langle x_n,y\rangle - \langle x_n,y_n\rangle| \\ &= |\langle x-x_n,y\rangle + \langle x_n,y-y_n\rangle| \\ &\leq |\langle x-x_n,y\rangle| + |\langle x_n,y-y_n\rangle| & \leq \|x-x_n\|\|y\| + \|x_n\|\|y-y_n\| \\ &\to 0 \end{aligned}$$

where the last line comes since $x_n \to x$ and $y_n \to y$, completing the proof.

The next concept we consider is a way of defining two vectors being somehow “unrelated” or going in some notion of “direction” that doesn't give information for the other vector:

Definition 8.1.9: Orthogonal

We say that x is *orthogonal* to y if $\langle x,y\rangle = 0$. We will write $x \perp y$

If two vectors are orthogonal, the triangle inequality becomes instant:

Theorem 8.1.10: Pythagoras

$$x \perp y \Rightarrow \|x+y\|^2 = \|x\|^2 + \|y\|^2$$

***Proof*:**

$$\|x+y\|^2 = \langle x+y, x+y\rangle = \|x\|^2 + \|y\|^2$$

We might wonder if we take $E \subseteq H$, can we find all vectors in H that are somehow “dual” to E? Define it's *orthogonal compliment* to be:

$$E^\perp = \{x \in H \mid \langle x,y\rangle = 0 \ \forall y \in E\}$$

Notice that E is *always* a closed subspace:

Proof. The fact it's a subspace follows from linearity of the second component.t

for closed, suppose there is a sequecne $\{x_n\} \subseteq E^\perp$ where $x_n \to x \in H$. Then:

$$|\langle x,y\rangle| \leq |\langle x-x_n,y\rangle| \leq \|x-x_n\|\|y\| \to 0 \quad \text{as } n \to \infty$$

□

We can also find closed (and not closed) subspace of hilbert spaces:

Example 8.1.11: Subspaces Examples

1. If $\mathcal{H}$ is a finite-dimensional Hilbert space, then all subspace are closed
2. If $\mathcal{H}$ is infinite-dimensional, consider $\mathcal{H} = L^1([0,1])$ with the L^2 inner product. and take $P([0,1]) \subseteq L^1([0,1])$ to be the subspace of all polynomial functions. This is is certainly not equal to $L^1([0,1])$, however polynomial functions are dense in $L^1([0,1])$ (they can arbitraily

approximate continuous bump characteristic functions, and hence simple functions). However, their closure is $L^1([0,1])$, showing that $P([0,1])$ cannot be closed.

3. Assume $E \subseteq \mathcal{H}$ is an open subspace. Then since $0 \in E$, there exists a $r > 0$ such that
$$B_r(0) \subseteq E$$
However, for any nonzero element $x \in \mathcal{H}$,
$$\frac{r}{2\|x\|}x \in B_r(0) \subseteq E$$
Since E is closed under scalar multiplication, this shows that $x \in E$, so $\mathcal{H} \subseteq E$, so $\mathcal{H} = E$, showing that the only open subspace of $\mathcal{H}$ is $\mathcal{H}$ itself.

Thus, we are interested in closed subspace since are closed under limits, meaning we can use our tools from analysis to get the following decomposition result:

Theorem 8.1.12: Hilbert Space Decomposition

Let $E \subseteq \mathcal{H}$ be a closed subspace. Then:
$$\mathcal{H} = E \oplus E^\perp$$
In particular, for any $x \in H$, there is a unique $y \in E$ and $z \in E^\perp$ such that $x = y + z$.

***Proof*:**
Let $x \in \mathcal{H}$, and let $\delta = \inf\{\|x - y\| = \rho(x, y) \mid y \in E\}$ (which exists by completeness of $\mathbb{R}^n$ and $\mathcal{H}$). let $\{y_n\} \subseteq E$ be a sequence such that $\|x - y_n\| \to \delta$. We will show that $\{y_n\}$ is a cauchy sequence, and then find a limit point for it. By the parralelogram law:
$$2(\|x - y_n\| + \|x - y_m\|) = \|y_n - y_m\|^2 + \|(y_n + y_n) - 2x\|^2$$
Since $y_n + y_m \in E$, for sufficinetly large N so that $n, m > N$:
$$\begin{aligned}\|y_n - y_m\|^2 &= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\|\frac{1}{2}(y_n + y_n) - x\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\delta^2\end{aligned}$$
and as $n, m \to \infty$, this sequence goes to $4\delta^2 - 4\delta^2 = 0$, and so $\|y_n - y_m\|\rho(y_n, y_m) \to 0$, meaning $\{y_i\}_i$ is a cauchy sequence. Take $y \lim y_n$. Then since E is closed, $y \in E$. By construction $\|x - y\| = \delta$. Now let $z = x - y$.

I claim that $z \in E^\perp$, that is $\langle z, u\rangle = 0$ for all $u \in E$. So, consider $\langle u, z\rangle$ for any $u \in E$. If $\langle u, z\rangle$ is complex, multiply it by a [complex] unit-length scalar to make it real ($\langle u, z\rangle = 0$ if and only if $a\langle u, z\rangle = 0$). With the fact that it's real, we will do a sneaky trick where we take advantage of optimization of differentiable functions. In particular, let
$$f : \mathbb{R} \to \mathbb{R} \quad f(t) = \|z + tu\|^2 = \|z\|^2 + 2t\langle z, u\rangle + \|u\|^2$$
f is certainly a differentiable function, being a polynomial with a square-root on $\mathbb{R}$. Furthermore, $z = x - y$ is the minimal value of $\|x - y\|$, and so the minimal value of $\|x - y\|^2$. It follows

that $z + ut$ is not minimal, and so this equation is has a minimum when $t = 0$. Thus, we get $0 = f'(0) = 2\langle u, z\rangle$, implying that $\langle u, z\rangle = 0$. Since u was arbitrary, we get that $z \in E^\perp$.

To show that z is the minimal distance from x, let's say there was another $z' \in E^\perp$ that satisfied the equation. Then by the Pythagorean theorem:

$$\begin{aligned}
\|x - z'\| &= \|(x - z) + (z - z')\|^2 \\
&= \|x - z\|^2 + \|z - z'\|^2 && x - y \in E, z - z' \in E^\perp \\
&\geq \|x - z\|^2
\end{aligned}$$

So equality holds if and only if $z = z'$, meaning z is of minimal distance to x. A similar reasoning shows the same for y.

To show uniqueness, let's say $y + z = x = y' + z'$. Then $y - y' = z - z' \in E \cap E^\perp = \{0\}$, so $y - y' = z - z' = 0$, which can be used to show that $y = y'$ and $z = z'$, completing the proof.

We said that orthogonal vector are somehow dual to each other. However, there is the more classical notion of "dual" vector spaces. Let $\mathcal{H}^*$ be set of all linear functions from $\mathcal{H}$ to the underlying field (usually $\mathbb{C}$, but sometimes $\mathbb{R}$); such linear functions are called functionals. It turns out that all linear functionals can be written in the form $f_y(x) = \langle x, y\rangle$, and we can define a norm on $\mathbb{H}^*$ such that $\|f_y\| = \|y\|$. Thus, $y \mapsto f_y$ will define a conjugate-linear isometry from $\mathcal{H}$ to $\mathcal{H}^*$. The following theorem shows tha this map is surjective:

Theorem 8.1.13: Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)

Let $f \in \mathcal{H}^*$. Then there exists a $y \in \mathcal{H}$ such that $f(x) = \langle x, y\rangle$ for all $x \in X$.

***Proof*:**

It easy easy to check uniqueness: if $\langle x, y\rangle = \langle x, y'\rangle$ for all x, then let $x = y - y'$. Manipulating the equation, we get that $\|y - y'\| = 0$, so $y = y'$.

If f is the zero functional, then take $y = 0$. If otherwise, then let $\mathcal{M} = \{x \mid f(x) = 0\}$ be the kernel of f. It is clearly a subspace, and any squence will also necesarily converge within $\mathcal{M}$, and so it is also closed. Furthermore, $\mathcal{M}$ is a proper subspace, since $f \neq 0$. Thus, $\mathcal{M}^\perp \neq \{0\}$ by the previous proof.

Now, pick any $z \in \mathcal{M}^\perp$ with $\|z\| = 1$. Then if we set $u = f(x)z - f(z)x$, we have that $u \in \mathcal{M}$. Hence:

$$0 = \langle u, z\rangle = f(x)\|z\|^2 - f(z)\langle x, z\rangle = f(x) - \langle x, \overline{f(z)}z\rangle$$

re-arrangine and setting $y = \overline{f(z)}z$, we get

$$f(x) = \langle y, z\rangle$$

as we sought to show.

The uniqueness makes this map to be injective, and by linearity of f and the first component of the inner product makes it is an isomorphism. Furthermore, since $\|f_y\| = \|y\|$, limits are also preserved, and so this in fact defines an isomorphism between $\mathcal{H} \cong \mathcal{H}^*$. Notice how this differs from vector-spaces where there is no natural isomorphism between V and V^* (there is one between V and V^{**}).

Since we are working over a vector-space, we should be able to define a basis. Since we are studying infinite dimensional vector-spaces, we often care more about "countable linear combination" rather than "finite linear combination" as we have for vector-spaces. To distinguish the two, we will call *Hamel basis* a set of linearly independent elements for which all *finite linear combinations* can represent every element of the vector space. We will call a *Schauder basis* a set of linearly independent elements for which all *countable inear combinations* can represent every element of the vector space. If every element in a Schauder basis has norm 1 and is orthogonal, we call it an *orthonormal basis*. Since we have an inner-product that is complete, it would be nicest if this vector-space has an orthonormal basis. We strive to this end in the following build-up:

Theorem 8.1.14: Bessel's Inequality

If $\{u_\alpha\}_{\alpha\in A}$ is an orthonormal set in $\mathcal{H}$, then for any $x \in \mathcal{H}$:

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 \leq \|x\|^2$$

As a consequence, $\{\alpha \mid \langle x, u_\alpha\rangle \neq 0\}$ is countable.

***Proof*:**
It suffices to show that $\sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 \leq \|x\|^2$ for any finite subset $F \subseteq A$, since if it fails for some countable amount, it will fail for some finite amount. Then we see that:

$$\begin{aligned}
0 &\leq \left\| x - \sum_{\alpha\in F} \langle x, u_\alpha\rangle u_\alpha \right\| \\
&= \|x\|^2 - 2\mathrm{Re}\left\langle x, \sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha \right\rangle + \left\|\sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha\right\| \\
&= \|x\|^2 - 2\sum_{\alpha\in F}|\langle x, u_\alpha\rangle|^2 + \sum_{\alpha\in F}|\langle x, u_\alpha\rangle|^2 && \text{Pythagorean Theorem} \\
&= \|x\|^2 - \sum_{\alpha\in F}|\langle x, u_\alpha\rangle|^2
\end{aligned}$$

and manipulating the final equation we get:

$$\sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 \leq \|x\|^2$$

as we sought to show

This in particular is important for uncountable sets. In the following we establish when equality holds, and thus when an orthonormal set is an orthonormal basis

Theorem 8.1.15: Orthonormal Basis Condition

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set in $\mathcal{H}$. Then the following are equivalent:

1. Finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$.
2. (Completeness) If $\langle x, u_\alpha \rangle = 0$ for all α, then $x = 0$
3. For each $x \in \mathcal{H}$, $x = \sum_{\alpha \in A} \langle x, u_\alpha \rangle u_\alpha$ where the right hand side sum has only countably many nonzero terms and converges in the norm topology no matter the order of the terms
4. (Parseval's Identity) $\|x\|^2 = \sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2$ for all $x \in \mathcal{H}$

***Proof*:**

(1) $\Rightarrow$ (2) Assume $\langle f, u_\alpha \rangle = 0$. As finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$, for any $f \in \mathcal{H}$ there is a sequence g_n such that $\|f - g_n\| \xrightarrow{n \to \infty} 0$. As $\langle f, u_\alpha \rangle = 0$, then $\langle f, -g_n \rangle = 0$. By bi-linearity and Cauchy-Schwartz:

$$\|f\|^2 = \langle f, f \rangle = \langle f, f - g_n \rangle \leq \|f\| \|f - g_n\| \xrightarrow{x \to \infty} 0 \tag{8.1}$$

hence $f = 0$

(2) $\Rightarrow$ (3) As the nonzero portion of the sum is countable, we can reduce to the countable case. For any $f \in \mathcal{H}$, let:

$$S_n(f) = \sum_{k=1}^{n} a_k u_k \qquad a_k = \langle f, u_k \rangle$$

Let's first show $S_n(f) \to g$ for some $g \in \mathcal{H}$, then show $g = f$. First, we show an alternative proof of Bessel's identity with a satisfying trick. Notice that certainly $S_n(f) \perp (f - S_n(f))$. Thus:

$$\|f\|^2 = \|S_n(f)\|^2 + \|f - S_n(f)\|^2 = \sum_{k=1}^{n} a_k u_k + \|f\|^2$$

Taking $n \to \infty$ we get:

$$\|f\| \geq \sum_{k=1}^{\infty} a_k u_k$$

Importantly, $\sum_{k=1}^{\infty}$ *converges*, and hence $S_n(f)$ forms a Cauchy sequence in $\mathcal{H}$, so by completeness $S_n(f) \to g$. To show $f = g$, notice that for any k, for sufficiently large n:

$$\langle f - S_n(f), u_k \rangle = a_k - a_k = 0$$

Thus:

$$\langle f - g, u_k \rangle = 0 \qquad \forall k$$

so by (2), $f - g = 0$, so $f = g$.

(3) $\Rightarrow$ (4) Assuming (3) so that $= \sum_k^\infty \langle f, u_k \rangle$, then $\|f - S_n(f)\| \to 0$ so plugging this into equation (8.1):

$$\|f\|^2 = \sum_k^\infty |a_k|^2$$

(4) $\Rightarrow$ (1) Notice that $S_n(f)$ is a finite linear combinations elements u_k, hence using equation (8.1) again we get the desired implication.

Any orthonormal set satisfying these conditions is given a name:

Definition 8.1.16: Orthonormal Basis

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set. Then if the set satisfies any of the 3 conditions in theorem 8.1.15, it is called an *orthonormal basis*.

An argument similar to the one for vector-spaces gives that every Hilbert spaces has an orthonormal basis, essentially an orthonormal basis is an orthonormal set that is maximal:

Theorem 8.1.17: Hilbert Space Basis

Every Hilbert space has an orthonormal basis.

***Proof*:**
Application of Zorn's lemma: The collection of ordered orthonormal sets ordered by inclusion has a maximal element, and maximality is equivalent to the first property of theorem 8.1.15.

Example 8.1.18: Orthonormal Basis

Given a Hilbert space $L^2(G)$ where $G = \mathbb{R}^n$ or $\mathbb{T}^n$, we would want to find an orthonormal basis to get decomposition of functions. go over all of these!

1. Legendre basis (can be thought of as "global" Taylor approximation) (must add integrate image. Also a good example of orthonormal basis that is not the Fourier one)

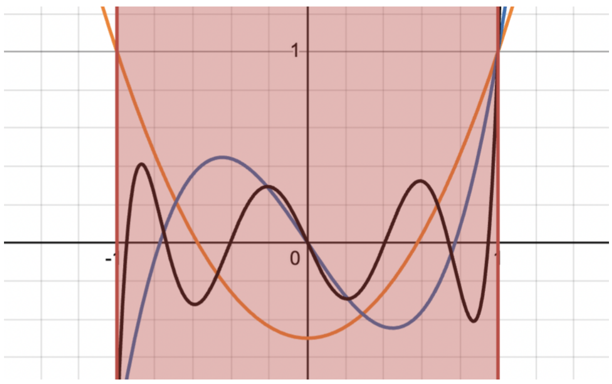


Figure 8.1: Legendre polynomials integrate to 0

2. Hermite basis
3. Jacobi Basis
4. Chebyshev

8.1.19 Basis On Banach Space Without an inner product, we cannot define an orthonormal basis on L^p. We can ask if we can at least define a *Schauder basis*, i..e any $f \in L^p$ is a (potentially) linear of a *single* sequence $\{b_n\}$ with different weights:

$$f = \sum a_n b_n$$

This was actually a very difficult problem and is related to the following question: is every compact operator on a Banach space the limit of finite operators. It isn't too hard to check this is the case in Hilbert spaces (exercise ref:HERE). However, this was proven in the negative by Per Enflo [7] (for a refinement of the proof see [3]). A consequence of this is that not every Banach space has a Schauder basis [2]. Another consequence of this paper for Banach spaces is that not all Banach space embed into $\ell^p(\mathbb{Z})$, $L^p(\mathbb{R}^d)$, or $L^p(K)$ for a compact set K; the family of Banach spaces is in fact larger (ex. $C(K)$ Is a Banach space not belonging to any of these spaces) .

Now that we have established a basis for Hilbert spaces, we will separate them into 3 categories: Those with finite, countable, and uncountable basis. In most cases, we will be working with countable basis due to the following property:

Proposition 8.1.20: Separable Hilbert Spaces

A Hilbert space $\mathcal{H}$ is separable if and only if it has a countable orthonormal basis, in which case every orthonormal basis for $\mathcal{H}$ is countable.

***Proof*:**
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Though all separable Hilbert spaces are "equivalent" (isometric), it is often the specific manifestation of the Hilbert space that matters as it frames how operators on the space transform functions (ex. the choice of basis, the choice of inner product).

Theorem 8.1.21: Enflo's Rigidity Theorem

A Banach space is uniformly homeomorphic to a Hilbert space if and only if it is linearly isomorphic to a Hilbert space.

***Proof*:**
This was a seminal result by Per Enflo (1969) [6]. It settled the question of whether the "uniform" structure of a space determines its linear structure in the context of Hilbert spaces. Since L^p (for $p \neq 2$) is not isomorphic to L^2, this theorem implies they are not uniformly homeomorphic either.

8.1.22 Hierarchy of Sameness It is useful to summarize how "same" two infinite-dimensional separable spaces can be. Let X and Y be infinite-dimensional separable Banach spaces (e.g., L^p and L^2).

1. Topological Homeomorphism : They are homeomorphic by the *Anderson-Kadec Theorem*, i.e. all such spaces are homeomorphic to $\mathbb{R}^{\mathbb{N}}$. They are topologically indistinguishable.
2. Uniform Homeomorphism: Generally no. By *Enflo's Rigidity Theorem*, if X is a Hilbert space and Y is not linearly isomorphic to it (like $L^p, p \neq 2$), they are *not* uniformly homeomorphic. *Note:* Their unit balls *are* uniformly homeomorphic via the **Mazur Map**, but this does not extend to the whole space.
3. Linear Isomorphism (The "Algebraic" level): Is there a linear bijection? Depends. L^p is not isomorphic to L^q for $p \neq q$.
4. Isometric Isomorphism (The "Geometry" level): They don't they preserve distance. The geometry of the unit ball (round vs. diamond vs. box) is strictly preserved here.

We next talk about the isomorphisms in the category of Hilbert spaces:

Definition 8.1.23: Unitary Map

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$, and let $U : \mathcal{H}_1 \to \mathcal{H}_2$. Then U is called an *isometry* if if it is a linear map that preserves inner products:

$$\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$$

If it is also bijective, it is a *unitary map*.

Note that by taking $x = y$, we get that $\|x\|_1 = \|Ux\|_2$, i.e. it is an isometry. In fact, the converse is true too: every surjective isometry is unitary. With the definition of an isomorphism, we can see that all Hilbert spaces are isomorphic to $\ell^2(A)$ given basis A:

Theorem 8.1.24: Classifying Hilbert Spaces

Let $\{e_\alpha\}_{\alpha \in A}$ be an orthonormal basis for $\mathcal{H}$. Then the correspondence

$$x \mapsto \hat{x} \qquad \hat{x}(\alpha) = \langle x, u_\alpha \rangle$$

is a unitary map from $\mathcal{H}$ to $\ell^2(A)$.

***Proof*:**
We'll show it's a surjective isometry, which is equivalent to it being unitary.

First off, the map $x \to \hat{x}$ is linear since the inner product is linear in each component, and it is an isometry by Parseval identity: $\|x\|^2 = \sum |\hat{x}(x)|^2$.

For surjectivity, let $f \in \ell^2(A)$, so $\sum_{\alpha \in A} |f(\alpha)^2| < \infty$, so the Pythagorean shows that all partial sumns of the series $\sum f(\alpha) u_\alpha$ are Cauchy [a], so $x = \sum_\alpha f(\alpha) u_\alpha$ exists in $\mathcal{H}$ and $\hat{x} = f$.

[a] Note that only countably many terms are nonzero

8.2 Operators on Hilbert Spaces

In this section, we try to generalize many notion's we've seen before, including:

1. adjoints, self-adjoint, and normal operators

2. sign and absolute value and when can we decompose a function into this form

3. diagonalizable

4. polar decomposition (think of how for every $z \in \mathbb{C}$, $z = \arg(z)|z|$)

Adjoints

Hilbert spaces have very simple dual structure, since $\mathbb{H} \cong \mathbb{H}^*$ in a very natural way with the inner product of $\mathbb{H}$. Thus, the problem of finding the structure of dual spaces is fully finished for hilbert spaces. Due to this very simple dual structure, we will be able to properly define an "opposite" operator given any operator $T : \mathbb{H}_1 \to \mathbb{H}_2$. In essense, there is an interesting symmetry in the structure of any map: Given a map $T : \mathbb{H}_1 \to \mathbb{H}_2$ between two hilbert spaces, there is a natural dual map $T^* : \mathbb{H}_2^* \to \mathbb{H}_1^*$. Hence, we have

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2^*, \mathbb{H}_1^*)$$

and since in Hilbert spaces $\mathbb{H}_i \cong \mathbb{H}_i^*$, we may envision this map as "flipping" the direction of arrows in a natural way:

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2, \mathbb{H}_1)$$

Not only does this map exist, but it is an isometric isomrophism (in **Ban**)! We will start with the special case of $\mathrm{Hom}(\mathbb{H}, k) = \mathbb{H}^*$. We will find that $(-)^*$ is a map

$$(-)^* : \mathrm{Hom}(\mathbb{H}, \mathbb{H}) \to \mathrm{Hom}(\mathbb{H}, \mathbb{H})$$

The following shows how we can define such a dual and some consequences:

Lemma 8.2.1: Endomorphism and And Inner Products

Let $\mathbb{H}$ be a complex hilbert space. Then there is a bijective isometric correspondence between $\mathrm{End}_k(\mathbb{H})$ and bounded sesquilinear forms on $\mathbb{H}$ given by tne map $T \to B_T$ where

$$B_T(x, y) = \langle x, T(y) \rangle$$

***Proof*:**
Let $T \in \mathrm{End}_k(\mathbb{H})$. Then certainly B_T is a sequilinear form on $\mathbb{H}$ bounded by $\|T\|$ since

$$\begin{aligned}\|B_T\| &= \sup\{|B_T(x,y)| \mid \|x\| \leq 1,\ \|y\| \leq 1\} \\ &= \sup\{|\langle x, T(y)\rangle| \mid \|x\| \leq 1, \|y\| \leq 1\} \\ &\leq \|T\|\end{aligned}$$

On the other hand

$$\begin{aligned}\|T(x)\|^2 &= \langle T(x), T(x)\rangle \\ &= B_T(T(x), x) \\ &\leq \|B_T\|\|T(x)\|\|x\| \\ &\leq \|B_T\|\|T\|\|x\|^2\end{aligned}$$

showing that $\|T\| \leq \|B_T\|$, and so $\|T\| = \|B_T\|$.

Theorem 8.2.2: Existence Of Adjoint

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then there exists a unique T^* in $\mathrm{End}_k(\mathbb{H})$ such that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

for all $x, y \in \mathbb{H}$. The map $T \mapsto T^*$ is conjugate linear, contravariant with respect to multiplication, an isometry of $\mathrm{End}_k(\mathbb{H})$ onto itself, an involution, and satisfies

$$\|T^*T\| = \|T\|^2$$

***Proof*:**
Fix $T \in \mathrm{End}_k(\mathbb{H})$, and let $B_T(x,y)$ be a continuous sesquilinear form defined as $(x,y) \mapsto \langle T(x), y\rangle$. Then by lemma 8.2.1, there exists a $T^* \in \mathrm{End}_k(\mathbb{H})$ such that $T^* \to B_T$. In particular

$$B_T(x,y) = \langle x, T^*(y)\rangle$$

Thus,

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

giving us existence of such an operator. We now establish all the properties named in the theorem. By conjugate symmetry

$$\begin{aligned}\langle x, T^{**}(y)\rangle &= \langle T^*(x), y\rangle \\ &= \overline{\langle y, T^*(x)\rangle} \\ &= \overline{\langle T(y), x\rangle} \\ &= \langle x, T(y)\rangle\end{aligned}$$

Thus, T^{**} and T correspond to the same sesquilienar form, thus $T^{**} = T$. This also shows that $T \mapsto T^*$ is an involution. Furthermore:

$$\langle x (ST)^* y\rangle = \langle S(T(x)), y\rangle = \langle T(x), S^*(y)\rangle = \langle x, T^*(S^*(y))\rangle$$

thus $(S \circ T)^* = T^* \circ S^*$ showing $(-)^*$ is a contravariant endofunctor. Since the operator norm is submiltiplicative, we have $\|T^*T\| \leq \|T\|\|T^*\|$. On the other hand, applying $\angle T(x), y\rangle = \langle x, T^*(y)\rangle$ to T^* we get

$$\|T(x)\|^2 = \langle T(x), T(x)\rangle = \langle T^*(T(x)), x\rangle = \|T^*T\|\|x\|^2$$

showing that $\|T\|^2 \leq \|T^*T$
. Combining these we get $\|T\| \leq \|T^*\|$, hence $\|T\| = \|T^*\|$ (since $T^{**} = T$). This finishes the proof.

Note that if $\mathbb{H}$ is a finite dimensional complex Hilbert space, then each operator T has a matrix representation, say $(\alpha_{ij)})$. Then $T^* = (\overline{\alpha_{ji}})$, i.e. $T^* = \overline{T^t}$ where T^t is the transpose given the matrix representation.

Definition 8.2.3: Adjoint And Self-Adjoint

Let $T \in B(\mathbb{H})$ be a continuous linear operator between Hilbert spaces. Then T^* is called the *adjoint* of T. If $T = T^*$, then T is said to be *self adjoint* (or *Hermitian*).

Proposition 8.2.4: Adjoint Properties I

Let $T \in \text{End}_k(\mathbb{H})$. Then
$$\ker(T^*) = (T(\mathbb{H}))^\perp$$

Proof:
Since $\langle T(x), y\rangle = \langle x, T^*(y)\rangle$ we have $y \in \ker(T^*)$ when $y \in (T(\mathbb{H}))^\perp$. Conversely, if $y \in (T(\mathbb{H}))^\perp$, then $T^*(y) \in \mathbb{H}^\perp = \{0\}$.

For ease of reference, we shall list some of the results we have proven in the theorem:

Proposition 8.2.5: Adjoint Properties II

Let $T, S \in B(\mathbb{H})$. Then:

1. $(T + S)^* = T^* + S^*$
2. $(cT)^* = \overline{c}T^*$
3. $(TS)^* = S^*T^*$
4. $(T^*)^* = T$
5. $I^* = I$

Proof:
If you're not convinced of any of these, double check the theorem or try proving it again.

Proposition 8.2.6: Adjoint Properties III

Let $T \in \text{End}_k(\mathbb{H})$. Then the following are equivalent:

1. T is invertible (so $T^{-1} \in \text{End}_k(\mathbb{H})$)
2. T^* is invertible
3. T and T^* are bounded away from zero.
4. T and T^* are injective, and $T(\mathbb{H})$ is closed
5. T is injective and $T(\mathbb{H}) = \mathbb{H}$

***Proof*:**
Since T is invertible, $T^{-1}T = TT^{-1} = I$, and so $(T^{-1})^*$ is the inverse of T^*, giving $(1) \Leftrightarrow (2)$.

For $(1) \Rightarrow (3)$, we say that an operator F is bounded away from zero if

$$\|x\| \leq C\|F(x)\|$$

for some constant C. In our case:

$$\|x\| = \|T^{-1} \circ T(x)\| \leq \|T^{-1}\|\|T(x)\|$$

Since $(1) \Leftrightarrow (2)$, the same holds for T^*.

For $(3) \Rightarrow (4)$, We have $\|T(x)\| \geq \epsilon\|x\|$ and $\|T^*(x)\| \geq \epsilon\|x\|$ for some $\epsilon > 0$ and all $x \in \mathbb{H}$. Thus certainly both T and T^* are injective: if $T(x) = T(y)$ then

$$0 = \|T(x) - T(y)\| = \|T(x - y)\| \geq \epsilon\|x - y\|$$

hence $\|x - y\| = 0$, so $x = y$. Furthermore, $\|T(x) - T(y)\| \geq \|x - y\|$ imples that $T(\mathbb{H})$ is complete, hence a closed subspace of $\mathbb{H}$.

For $(4) \Rightarrow (5)$, by corollary **??** ($\overline{Y} = (Y^{\perp})^{\perp}$) and proposition 8.2.4 we get

$$\overline{T(\mathbb{H})} = (T(\mathbb{H})^{\perp})^{\perp} = (\ker(T^*))^{\perp} = \{0\}^{\perp} = \mathbb{H}$$

where the last equality come from T^* bein injective.

Finally, for $(5) \rightarrow (1)$, we use the open mapping theorem.

This allows us to generlize this result to the followin

Proposition 8.2.7: Inversing Maps

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be hilbert spaces and consider an operator $T \in \mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$. Then there exists a unique operator $T^* \in B(\mathbb{H}_2, \mathbb{H}_1)$ such that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

show that the map $T \to T^*$ is conjugate linear isometry of $\mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$ to $\mathrm{Hom}_k(\mathbb{H}_2, \mathbb{H}_1)$ and satisfies

$$\|T^*T\| = \|T^2 = \|TT^*\|$$

***Proof*:**
Let

$$\hat{T} = \begin{pmatrix} 0 & 0 \\ T & 0 \end{pmatrix} \in \mathrm{End}_k(\mathbb{H}_1 \oplus \mathbb{H}_2)$$

By theorem 8.2.2, there is a unique function $\hat{T}^*$ such that

$$\langle \widehat{T}(x_1, x_2), (y_1, y_2)\rangle = \langle (x_1, x_2), \widehat{T}^*(y_1, y_2)\rangle$$

By definition of $\widehat{T}$, we see

$$\widehat{T}^* = \begin{pmatrix} 0 & T^* \\ 0 & 0 \end{pmatrix}$$

Hence we have that T^* is a function with domain and codomain $T^* : \mathbb{H}_2 \to \mathbb{H}_1$. It follows that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

and by theorem 8.2.2, T and T^* respect all the desired properties, completing the proof.

For the rest of this seciton, we will focus on $T \in B(\mathbb{H})$ insead of $T \in B(\mathbb{H}_1, \mathbb{H}_2)$. The reason for this is to find suitable decompositino theorems of T generlizing the notion of eigenvalues.

Recall that over finite dimensional vector spaces, $TT^* = T^*T$ implied T was diagonalizable, with the usual special case of $T = T^*$ (self-adjoint) being looked into. We would like to know if this notion of diagonalizability for normal operators remains true for infinite dimensional Hilbert spaces. We shall see that this is not the case, some more conditions is required (for ex. a special case of digaonalizable operators will be comapct normal operators). Before we get into the technicalities, I want to put a word on how I think about $TT^* = T^*T$. In my mind, when two functions commute, it means that one function is somehow “invariant” with respect to the other. Think of PDE’s with equations that commute with the differential operator that makes it invariant to the Lorenzian metric. In this case, I like to think of T being somehow “symmetric”. Since $TT^*(x) = T^*T(x)$ for all $x \in \mathbb{H}$, and T^* is in some ways the dual (think of ref:HERE, the generalization), we see that T is “invariant” to the effect of the dual T^*. We shall see in a moment that the invariant to the dual is translated to T and T^* being metrically identical. In the simpleset case of $T = T^*$, it is evident why this is the case. If $T = T^*$, we can deepen our intuition with complex numbers: if we consider $T : \mathbb{C} \to \mathbb{C}$ where $T(z) = z$, then $T^*(z) = \overline{z}$. Hence, $T = T^*$ if and only if $T(\mathbb{C}) \subseteq \mathbb{R}$, since only the real number are invariant under conjugation.

Definition 8.2.8: Normal Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *normal* if it commutes with its adjoint, i.e. if $T^*T = TT^*$

As an immedaite consequence, we have

$$\|T(x)\| = \langle T^* \circ T(x), x\rangle^{1/2} = \langle T \circ T^*(x), x\rangle^{1/2} = \|T^* x\|$$

showing that T and T^* are metrically identical. In fact, by the palalalogram law (also known as polarization identity) if $T \in \text{End}_k(\mathbb{H})$ such that T and T^* ar emetrically identical, we can show $\langle T^* \circ T(x), y\rangle = \langle T \circ T^*(x), y\rangle$ for all x, y and so T is normal by lemma 8.2.1. By proposition 8.2.6, an operator is normal if and only if it is bounded away from zero.

Positive Operators

Definition 8.2.9: Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *positive* if $T = T^*$ and $\langle T(x), x\rangle \geq 0$ for all $x \in \mathbb{H}$. If $k = \mathbb{C}$, the positivey condition implies self-adjointness by the polarization identity.

Certainly, the sum of two positive operators is positive (geometrically, positive operatros form a cone in $\text{End}_k(\mathbb{H})$). However the product is not positive, and need not be self-adjoint.

Example 8.2.10: Positive And Non-Positive Operators

here

However, if the opeartors commute (if the product is self-adjoint), we see that there is a product that is positive since ($ST = S^{1/2}TS^{1/2} \geq 0$ by what we'll show in the next proposition)

Let $B(\mathbb{H})_{sa} \subseteq \text{End}_{\mathbb{R}}(\mathbb{H})$ be the collection of self-adjoint operators. Then this is a closed subspace, this subspace intersected with the conte of positive operators defines an ordder $S \leq T$ if $T - S \geq 0$. By taking the case of 2 dimensional real subspaces and $\text{End}_{\mathbb{R}}(\mathbb{R}^2)$, this order is not total, not even a lattice order.

We require the following lemma to gain some important results (which will be an immedate consequence of the spectral theorem in ref:HERE)

Lemma 8.2.11: Square Root Lemma

If $S \leq T$ in $B(\mathbb{H})_{sa}$, then $A^*SA \leq A^*TA$ for every $A \in \text{End}_k(\mathbb{H})$. If morever $) \leq S$, then $\|S\| \leq \|T\|$

***Proof*:**
Since $S \leq T$, we know that $\langle (T - S)(y), y\rangle \geq 0$ for evey $y \in \mathbb{H}$. Replacing y with $A(x)$ for appropraite $x \in \mathbb{H}$, it follows that $A^*(T - S)A \geq 0$, so $A^*SA \leq A^*TA$.

If $0 \leq S \leq T$, by the Cauchy-Schwarz inequality on the positive sesquilinear form $\langle S(\cdot), \cdot\rangle$, we get that for every pair of unit vectors $x, y \in \mathbb{H}$

$$|\langle S(x), y\rangle|^2 \leq \langle S(x), x\rangle\langle S(y), y\rangle \leq \langle T(x), x\rangle\langle T(y), y\rangle \leq \|T\|^2$$

Hence, $\|S\|^2 \leq \|T\|^2$ by lemma 8.2.1.

Lemma 8.2.12: Approximation Lemma

There is a sequence $(p_n) \subseteq \mathbb{R}[x]$ of polynomials with positive coefficients such that $\sum p_n$ converges uniformally on $[0,1]$ to

$$t \mapsto 1 - (1-t)^{1/2}$$

***Proof*:**
This is immediate from real analysis, but we can also construct it explicitly.If $q_0 = 0$ and $q_n(t) = \frac{1}{2}(t + q_{n-1}(t)^2)$, then we can see that each coefficent of q_n is positive and that $q_n(t) \leq 1$ for every $t \in [0,1]$. Morever, since

$$2(q_{n+1} - q_n) = q_n^2 - q_{n-1}^2 = (q_n - q_{n-1})(q_n + q_{n-1})$$

it follows by induction that each polynomial $p_n = q_n - q_{n-1}]$ has positive coefficients. Taking the q_n as elements of $C([0,1])$, we see they converge pointwise to $[0,1]$ to a function q such that $2q(t) = t + q(t)^2$. Thus, $q(t) = 1 - (1-t)^{1/2}$. Sicne $q \in C([0,1])$ and $[0,1]$ is compact, by the monotone ocnvergence $q_n \nearrow q$ si in fact uniform by Dini's lemma, hence

$$\sum p_n = \lim q_n = q$$

completing the proof.

Proposition 8.2.13: Square-Rooting Positive Operator

Let $T \in \mathrm{End}_k(\mathbb{H})$ be a positive operator. Then there is a unique positive operator $T^{1/2}$ such that $(T^{1/2})^2 = T$. Moreover, $T^{1/2}$ commutes with every operator commuting with T.

***Proof*:**
Analysis Now p.92

Proposition 8.2.14: Invertible Positive Operator

A positive operator T is invertible in $\mathrm{End}_k(\mathbb{H})$ if and only if $T \geq \epsilon I$ for some $\epsilon > 0$. I that case, $T^-1 \geq$ and $T^{1/2}$ is invertib,e and $(T^{-1})^{1/2} = (T^{1/2})^{-1}$. If $T \leq S$, then $S^{-1} \leq T^{-1}$

***Proof*:**
Analayis now p.94

Projection

Let $W \subseteq \mathbb{H}$ be a closed subspace of a hilbert space H. Then we know that $\mathbb{H} = W + W^{\perp}$. In particular, for eah $y \in \mathbb{H}$, we have $y = w + w^{\perp}$, $w \in W$ and $w^{\perp} \in W^{\perp}$. Since

$$\alpha y_1 + \beta y_2 = (\alpha x_1 + \beta x_2) + (\alpha x_1^{\perp} + \beta x_2^{\perp})$$

is the decomposition of a linear combination, it follows that $P : \mathbb{H} \to \mathbb{H}$ given by $P(y) = x$ is an operator in $\text{End}_k(\mathbb{H})$ with $\|P\| \leq 1$. Note P is idempotent ($P^2 = P$), self-adjoint, and positive since

$$\langle P(y_1), y_2\rangle = \langle x_1 < x_2 + x_2^{\perp}\rangle = \langle x_1, x_2\rangle = \langle x_1 + x_1^{\perp}, x_2\rangle = \langle y_1, P(y_2)\rangle$$

and

$$\langle P(y), y\rangle = \langle x, x + x^{\perp}\rangle = \|x\|^2 \geq 0$$

Definition 8.2.15: (Orthogonal) Projection

The map P in the above is called the *(orthogonal) projection.*

If P is a self-adjoint and idempotent in $\text{End}_k(\mathbb{H})$, then

$$W = \{x \in \mathbb{H} \mid P(x) = x\} = P(\mathbb{H})$$

is a closed subsapce of $\mathbb{H}$. If $w^{\perp} \in W^{\perp}$, we have (since $P = P^*$) that

$$\|P(w^{\perp})\|^2 = \langle w^{\perp}, p^2(w^{\perp})\rangle = 0$$

Thus, P is an orthogonal projection of $\mathbb{H}$ onto W. Note that $I - P$ is the projection on $W^{\perp}$.

Projections are some of the east operators we may construct, and can be thought as the "characteristic function" from real analysis. The analogue of simple function is then obtained by decomposing $\mathbb{H}$ into a finite orthogonal sum

$$\mathbb{H} = W_1 \oplus W_2 \oplus \cdots \oplus W_n$$

corresponding to the projections $P_1, ..., P_n$ which are pairwise orthogonal ($P_iP_j = 0$ for $i \neq j$ satisfying $\sum P_i = I$). We may then consider

$$T = \sum \lambda_n P_n$$

for scalars $\lambda_1, \lambda_2, ..., \lambda_n \in k$. This may look familiar to the notion of *diagonalization.* Indeed, we may generalize the concept to infintie dimensions like so:

Definition 8.2.16: Diagonalizable

Let $T \in \text{End}_k(\mathbb{H})$ be an operator. Then T is said to be *diagonalizable* if there is an orthonormal basis $\{e_i \mid i \in I\}$ for $\mathbb{H}$ and a bounded set $\{\lambda_i \mid i \in I\} \subseteq k$ such that

$$T(x) = \sum \lambda_i \langle x, e_i\rangle e_i$$

The number $\langle x, e_i\rangle$ are the coordiantes for x in the basis $\{e_i\}$ and each λ_i is an *eigenvalue* for T corresponding to the eignevector e_i> Denotign P_i to be the projecti nofo $\mathbb{H}$ onto ke_i, we have $P_i(x) = \langle x, e_i\rangle e_i$, so we may write the operator as:

$$T = \sum \lambda_i P_i$$

Note that the above sum is not necessarily convergent in the norm topology on $\mathrm{End}_k(\mathbb{H})$, but only pointwise convergent (i.e. in convergent in the strong operator topology). If T is diagonalizable, then T^* is diagonalizable with

$$T^*(x) = \sum \overline{\lambda_i}\langle x, e_i\rangle e_i$$

with eigenvalues $\{\overline{\lambda_i}\}$. Furthermore, if T is diagonalizable, $T^*T = TT^*$ (with eigenvalues $|\lambda_i|^2$ with basis $\{e_i\}$), and hence T is normal. T is self-adjoint if and only if all λ_i are real and T is positive if and only if $\lambda_i \geq 0$ for all i. If $\mathbb{H}$ is finite-dimensional, every normal operator is diagonalizable. This is not generally true for infinite dimensions. In general, most "interesting" (normal) operators on infinite-dimensional hilbert sapces are nondiagonalizable, and must be handled with a continuous analogue of the concept of a basis (this will be the spectrum, see section 8.3 and chapter 9).

Unitary Operators

Recall $T : \mathbb{H} \to \mathbb{H}$ is unitary if T is an isometric isomorphism from $\mathbb{H}$ onto itself (if $k = \mathbb{R}$ we say it is an *orthogonal operator*). By the polarization identity, we see that $U^*U = UU^* = I$, so that U is normal and invertible with $U^{-1} = U^*$. Conversely, if $U \in \mathrm{End}_k(\mathbb{H})$ such that $U^{-1} = u^*$ then U is unitary since

$$\|U(x)\|^2 = \langle U^*U(x), x\rangle = \|x\|^2$$

So U is an isometry and is surjective since U is invertible. If $\{e_i \mid i \in I\}$ is an orthonormal basis for $\mathbb{H}$ and U is unitary, then $\{Ue_i \mid i \in I\}$ is a basis for $\mathbb{H}$. Conversely, we saw that every transition between orthonormal bases is given by a unitary operator. For future reference the denote the group of unitary operators $U(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ (it should be evident it forms a group).

Definition 8.2.17: Unitary Equivalent

Let $S, T \in \mathrm{End}_k(\mathbb{H})$. Then we say that S and T are *unitarily equivalnet* of $S = UTU^*$ for some unitary operator U.

It should be clear that unitary equivalent preserves norm, self-adjointness, normality, diagonalizability, and unitarity.

Definition 8.2.18: Partial Isometry

Let $U \in \mathrm{End}_k(\mathbb{H})$. Then U is said to be a *partial isometry* if there is a closed subpspace $X \subseteq \mathbb{H}$ such that $U|_X$ is isometric and $U|_{X^\perp} = 0$

This immediately implies that $X^\perp = \ker(U)$, hence $X = \bar{(}U^*(\mathbb{H}))$. Setting $P = U^*U$, then

$$\langle P(x), x\rangle = \|U(x)^2 = \|x\|^2$$

for every $x \in X$, hence $P(x) = x$ by the Cauchy-Schwarz inequality. Furthermore, $P(x^\perp) = 0$ for every $x^\perp \in X^\perp$, so P is a projetion of $\mathbb{H}$ onto X. Conversely, if $U \in \mathrm{End}_k(\mathbb{H})$ such that $U^*U = P$ for some projeciotn P, taking $X = P(\mathbb{H})$ and see from the equation $\|U(x)$
$^2 = \langle p(x), x\rangle$ that U is isometric on X and 0 on $X^\perp$, and so U is a partial isometry. Using this, we can also show that U^* is a partial isometry. Since $U(I - P) = 0$,

$$(UU^*)^2 = UU^*UU^* = UPU^* = UU^*$$

so UU^* is self-adjoint idempodent, i.e. a projection. Note that U^* is the projectio nof $U(\mathbb{H})$ back onto $U^*(\mathbb{H})$ so that U^* is a *partial inverse* of U.

Be weary that there exist isometries that are not unitary (since they need not be surjective). For example, if $\mathbb{H}$ is a separable and has countable basis $\{e_n \mid n \in \mathbb{N}\}$, we have

$$S\left(\sum \alpha_n e_n\right) = \sum \alpha_n e_{n+1}$$

Then S (the unilateral shift operator) is an isometry of $\mathbb{H}$ ont othe subspace $\{e_1\}^{|}perp$ and consequently S^* is a patial isometry of $\{e_1\}^\perp$ onto $\mathbb{H}$. In particular, $S^*S = I$, wheas SS^* is the projection onto $\{e_1\}^\perp$.

Theorem 8.2.19: Polar Decomposition of T

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then there is a unique positive operator $|T| \in \mathrm{End}_k(\mathbb{H})$ such that

$$\|T(x)\| = \||T|(x)\|$$

and $|T| = (T^*T)^{1/2}$. Moreover, there is a unique partial isometry U with kernel $\ker(U) = \ker(T)$ and $U \circ |T| = T$. In particular

$$U^*U|T| = |T| \qquad U^*T = |T| \qquad UU^*T = T$$

***Proof*:**
Analysis Now p.96

We may think of U as the generalized "sign" of T and $|T|$ as the generalized "absolute value" of T. By the formula

$$T^* = |T|U^* = U^*(U|T|U^*)$$

and the unique of polar decomposition, we see that U^* is the sign of T^*, whereas $U|T|U^*$ is the absolute value fo T^*. In general, the absolute value of a sum or product of noncommuting operators doens't have much relation to the sum or product of the absolute values. Similarly, the sign (in the finite dimensional case), can't always be chosen to be unitary (the unilateral shift is an immedaite counter example). For T to have the form $U|T|$ for some unitary operator U, it is necessary and sufficient that the cloesd subspaces $\ker(T)$ and $\ker(T^*)$ have the same dimension (finite or infinite). Easy cases of this general theorem are estalblisehd in the next result

Proposition 8.2.20: Polar Decomposition For Isomorphisms

Let $T \in \mathrm{End}_k(\mathbb{H})$ be invertible. Then the partial isometry in its polar decomposition is unitary

***Proof*:**
Analysis Now p.97

Proposition 8.2.21: Polar Decomposition For Normal Operators

Let $T \in \text{End}_k(\mathbb{H})$ be normal. Then there is a unitary operator W commuting with T, T^*, and $|T|$ such that $T = W|T|$

***Proof*:**
Analysis Now p.97

I want to add one more thing, I just don't konw where: for any $T \in B(\mathbb{H})$, $T = T_1 + iT_2$ for self-adjoint operators T_1, T_2. This lets us generalize the spectral theorem for noraml operators, since we will konw it for self-adjoint operator.

Lemma 8.2.22: lemma I

If $T = T^*$ and $\|T\| \leq 1$, the operator $U = T + i(I - T^2)^{1/2}$ is unitary and $T = \frac{1}{2}(U + U^*)$

***Proof*:**
Analysis Now p.97

Lemma 8.2.23: Lemma II

Let $S \in \text{End}_k(\mathbb{H})$ with $\|S\| < 1$. Then for every unitary U, there are unitaries U_1, V_1 such that

$$S + U = U_1 + V_1$$

***Proof*:**
Analysis Now p.98

Theorem 8.2.24: Russo-Dye-Gardner Theorem

Let $T \in \text{End}_k(\mathbb{H})$ with

$$\|T\| < 1 - \frac{2}{n} \qquad n > 2$$

Then there are unitary operators $U_1, U_2, ..., U_n$ such that

$$T = \frac{1}{n}(U_1 + U_2 + \cdots + U_n)$$

***Proof*:**
Analysis Now p.98

Since the self-adjoint elements span $\text{End}_k(\mathbb{H})$, we see from lemma 8.2.22 that every $T \in \text{End}_k(\mathbb{H})$ is the linear combination of (at least 4) unitary operators The estimate in the Russo-Dye-Gardner Theorem is much more precise with respect to the norm, and shows that every element in the open unit ball of $\text{End}_k(\mathbb{H})$ is a convex combination (indeed, a mean) of unitary operators. On the surface of the ball this need not be the case, in fact the unilateral shift cannot be expressed as a convex combination of unitaries (exercise 3.2.8 in Gert K. Pederson Analysis Now).

Proposition 8.2.25: Numerical Radius Of Operator

Let $\mathbb{H}$ be a complex hilbert space and $T \in \mathrm{End}_{\mathbb{C}}(\mathbb{H})$. Then we deifn eth *numerical radius* of T as

$$|||T||| = \sup\{\langle T(x), x\rangle \mid x \in \mathbb{H}, \|x\| \leq 1\}$$

Then $\|\cdot\|$ is a norm on $\mathrm{End}_{\mathbb{C}}(\mathbb{H})$ satisfying $\frac{1}{2}\|\cdot\| \leq |||\cdot||| \leq \|\cdot\|$. More over, $|||T^2||| \leq |||T|||^2$ for every T and $|||T||| = \|T$
if T is normal

***Proof*:**
Analysis Now p.99

8.3 Compact Operators

Throughout this section, let $\mathbb{H}$ be a hilbert space over $k \in \{\mathbb{R}, \mathbb{C}\}$ unless otherwise specified.

Definition 8.3.1: Finite Rank Operator

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then T is said to have *finite rank* if $T(\mathbb{H})$ is finite-dimensional subspace of $\mathbb{H}$. Denote by $B_f(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ the set of all finite rank opeartors.

Clearly, $B_f(\mathbb{H})$ is a subspace, even a subalgebra, even an ideal of $\mathrm{End}_k(\mathbb{H})$. If $T \in B_f(\mathbb{H})$, by proposition 8.2.4, we get an orthogonal decomposition $\mathbb{H} = T(\mathbb{H}) \oplus \ker(T^*)$, which shows that $T^*(\mathbb{H}) - T^*T(\mathbb{H})$, so T^* has finite rank. Thus, $B_f(\mathbb{H})$ is *self-adjoint* ideal in $\mathrm{End}_k(\mathbb{H})$ $((B_f(\mathbb{H}))^* = B_f(\mathbb{H})$ as a set). The subset $B_f(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ is similar in relation to $C_c(X) \subseteq C_b(X)$ when X is a locally compact Hausdorff space. In particular, these classes describe the local phenomena on $\mathbb{H}$ and on X. Passing to a limit in norm may destroy the "locality". but enogh structure is preservd to make these "quasilocal" operators and functions very useful and interesting. We shall stud the closure of $B_f(\mathbb{H})$ in this section as a noncommutative analogue of $C_0(X)$ in function theory.

Lemma 8.3.2: Net And Compact Operators

There is a net $(P_\lambda)_{\lambda\in\Lambda}$ of projection in $B_f(\mathbb{H})$ such that $\|P_\lambda(x) - x\| \to 0$ for each $x \in \mathbb{H}$.

that is, the projection are "dense" in $B_f(\mathbb{H})$.

***Proof*:**
We want to find a next P_n such that $\|P_n(x) - x\| \to 0$. Let $\{e_i \mid i \in I\}$ be an orthonormal basis for $\mathbb{H}$ and let N be a net of finite subsets of I ordered by inclusion. For each $n \in N$, let P_n denote the projection of $\mathbb{H}$ onto $\mathrm{span}\{e_i \mid i \in n\}$ so that $(P_n)_{n\in N}$ is a net in $B_f(\mathbb{H})$.

Now, if $x \in \mathbb{H}$, then $x = \sum a_i e_i$, thus

$$\|P_n(x) - x\|^2 = \sum_{i \notin n} |a_i|^2$$

By Parseval's identity, this tends to zero, completing the proof.

Theorem 8.3.3: Closed Unit Balls In Hilbert Spaces

Let $B \subseteq \mathbb{H}$ be a closed unit ball in a hilbert space $\mathbb{H}$. Then the following conditions on an operator $T \in \text{End}_k(\mathbb{H})$ are equivalent

1. $T \in \overline{B_f(\mathbb{H})}$
2. $T|_B$ is a weak-norm continuous function from B into $\mathbb{H}$
3. $T(B)$ is compact in $\mathbb{H}$
4. $\overline{T(B)}$ is compact in $\mathbb{H}$
5. each net in B has a subnet whose iamge under T converges in $\mathbb{H}$

***Proof*:**
Analysis Now p.105

Definition 8.3.4: Compact Operators

Let $T \in B(\mathbb{H})$. Then T is compact if $T(B)$ is compact. The set of compact operators is denoted by $B_0(\mathbb{H})$ to show that these operators "vanish at infinity".

Lemma 8.3.5: Diagonalizable Operator Is Compact

Let $T \in \text{End}_k(\mathbb{H})$. Then T is comapct if and only if its eigenvalues $\{\lambda_i \mid i \in I\}$ belongs to $c_0(I)$

***Proof*:**
Let $T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$. If $T \in B_0(\mathbb{H})$, then for any $\epsilon > 0$, define

$$I_\epsilon = \{i \in I \mid |\lambda_i| \geq \epsilon\}$$

If I_ϵ is infinite, the net $(e_i)_{i \in I_\epsilon}$ converges weakly to zero for any well-ordering of I_ϵ since $\langle e_i, x \rangle \to 0$ by Parsaval's identity. Sicne $\|T(e_i)\| = |\lambda_i| \geq \epsilon$ for $i \in I_\epsilon <$ this contradicts theorem 8.3.3(3). Thus, I_ϵ must be finite for each $\epsilon > 0$, hence λ_i must vanish at infinity.

Conversely, if I_ϵ is finite for all $\epsilon > 0$. et

$$T_\epsilon = \sum_{i \in I_\epsilon} \lambda_i \langle \cdot, e_i \rangle e_i$$

Then T_ϵ has finite rank, and

$$\begin{aligned}\|(T - T_\epsilon)(x)\|^2 &= \left\|\sum_{i\notin I_\epsilon} \lambda_i \langle x, e_i\rangle e_i\right\|^2 \\ &= \sum_{i\notin I_\epsilon} |\lambda_i|^2 |\langle x, e_i\rangle|^2 \\ &\le \epsilon^2 \|x\|^2\end{aligned}$$

hence, $\|T - T_\epsilon\| \le \epsilon$, thus $T \in B_0(\mathbb{H})$ by theorem 8.3.3(1), completing the proof.

Lemma 8.3.6: Eigen Vectors Between Adjoint

Let x be an eigenvector for a normal operator $T \in \mathrm{End}_k(\mathbb{H})$ corresponding to an eigenvalue λ. Then x is an eigenvector for T^* corresponding to the eigenvalue $\overline{\lambda}$. Eigenvectors for T correspondign to different eigenvalues are orthogonal.

***Proof*:**
Note that the operator $T - \lambda I$ is normal and it's adjoitn is $T^* - \overline{\lambda} I$. COnversely,

$$\|(T^* - \overline{\lambda} I)(x)\| = \|(T - \lambda I)(x)\| = 0$$

If $T(y) = \gamma y$ where $\gamma \neq \lambda$, then we may certainly assume $\lambda \neq 0$ so that

$$\begin{aligned}\langle x, y\rangle &= \lambda^{-1}\langle T(x), y\rangle \\ &= \lambda^{-1}\langle x, T^*(y)\rangle \\ &= \lambda^{-1}\langle x, \overline{\gamma} y\rangle \\ &= \lambda^{-1}\gamma\langle x, y\rangle\end{aligned}$$

Hence, $\langle x, y\rangle = 0$, showing they are different eigenvectors, completing the proof.

Lemma 8.3.7: Normal Compact Operator, then Eivenvalue Exists

Every normal, compact operator T on a complex Hilbert space $\mathbb{H}$ has an eigenvalue λ with $|\lambda| = \|T\|$

***Proof*:**
Analysis now p. 107

Theorem 8.3.8: Normal Compact, Then Diagonalizable

Every normal compact operator $T \in \mathrm{End}_k(\mathbb{H})$ is diagonalizable and its eigenvalues (counted-with multiplicity) vanish at infinite. Conversely, each such operator is normal and compact.

***Proof*:**
Analysis now p.108

It will be convenient for future sections to introduce a new notation: Let $x \odot y$ represent the rank one operator in $\text{End}_k(\mathbb{H})$ determined by the vectors $x, y \in \mathbb{H}$ and the formula

$$(x \odot y)(z) = \langle z, y \rangle x$$

The map $(x, y) \mapsto x \odot y$ is sesquilinear map of $\mathbb{H} \times \mathbb{H}$ into $B_f(\mathbb{H})$. If $\|e\| = 1$ then $e \odot e$ is the one-dimensional projection of $\mathbb{H}$ on $\mathbb{C}e$. Every normal compact operator $\mathbb{H}$ can now be written by the above theorem in the form

$$T = \sum \lambda_i e_i \odot e_j$$

For suitable orthonormal basis $\{e_i \mid i \in I\}$. The sum will converge in norm, since either the set $I_0 = \{i \in I \mid \lambda_i \neq 0\}$ is finite (so that $T \in B_f(\mathbb{H})$) or else countably infinite in which case the sequence $\{\lambda_i \mid i \in I_0\}$ converges to zero.

Definition 8.3.9: Spectrum

We say that the compact set

$$^{(}T) = \{\lambda_i \mid i \in I_0\} \cup \{0\}$$

is the *spectrum* of T.

For every continuous function f on $^{(}T)$, define

$$f(T) = \sum f(\lambda_i) e_i \odot e_i$$

Then $f(T)$ is compact if and only if $f(0) = 0$ and the map $f \mapsto f(T)$ Is an isometric $*$-preserving homomorhpism of $C(^{(}T))$ into $\text{End}_k(\mathbb{H})$. If $f(z) = \sum \alpha_{nm} z^n \overline{z}^m$ is a polynomial of two commuting variables z and $\overline{z}$, then $f(T) = \sum \alpha_{nm} T^n (T^*)^m$. This result is the *spectrial (mapping) theorem* for normal compact operators. In the next chapter, we shall generalize this to any normal operator.

Calkin Algebra and Index

Since $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$ is a closed ideal, we may define $B(\mathbb{H})/B_0(\mathbb{H})$ with the quotient norm[5]. This quotient is given a name

Definition 8.3.10: Calkin Algebra

Let $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$. Then the algebra

$$B(\mathbb{H})/B_0(\mathbb{H})$$

is called the *Calkin Algebra* of $\mathbb{H}$

We shall see that many properties of $B(\mathbb{H})$ are conveniently expressed in terms of the Calkin algebra.

[5]This is even a C^*-algebra

Definition 8.3.11: Compact Pertubation

If S and T are elements in $B(\mathbb{H})$ and $S - T \in B_0(\mathbb{H})$, we say that S is a *compact pertubation* of T.

This is antoher way of saying that S and T hav the same image in the Calkin algebra. Such "local" pertubation happen frequently in application and properties of an operator that are invariatn under compact pertubations are highly used. The best known of these we shall study know is known as the *index*

Theorem 8.3.12: Atkinson's Theorem

Let $T \in B(\mathbb{H})$. Then the following are equivalent:

1. There is a unique operator $S \in B(\mathbb{H})$ such that ST and TS are the projections on $(\ker(T))^\perp$ and $\ker(T^*)^\perp$, respectively, and both projections have finite co-rank
2. For some operator $S \in B(\mathbb{H})$ both operators $ST - I$ and $TS - I$ are compact
3. the image of T is invertible in the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$
4. Both $\ker(T)$ and $\ker(T^*)$ are finite-dimensional subsapces and $T(\mathbb{H})$ is closed

***Proof*:**
Analysis now p.109

In essence, Fredholm operators are operators that are are almost invertible up to some finite-dimension beign ignored, in particular they are invertible modulo compact operators.

Definition 8.3.13: Fredholm Operator

The operator satisfying the condition's in Atkinson's theorem are called *Fredholm operators*, and the class of them is dneoted $F(\mathbb{H})$.

Definition 8.3.14: Index

For each $T \in F(\mathbb{H})$, define the *index* of T to be

$$\text{index}(T) = \dim(\ker(T)) - \dim(\ker(T^*))$$

If we shooe S for T as in (1) of Atkinson's theorem and write $suchthat = I - P$ and $TS = I - Q$ with P and Q projection of rank, then another way of writing this would be

$$\text{index}(T) = \text{rank}(P) - \text{rank}(Q)$$

By (3) of Atkinson's Theorem, the product of Fredholm operators is again a Fredholm operator. In particular, every product $RT \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ and R is invertible in $B(\mathbb{H})$, and in this case

$$\text{index}(RT) = \text{index}(TR) = \text{index}(T)$$

since R is bijective. It should also be clear that $T^* \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ with index $\operatorname{index}(T^*) = -\operatorname{index}(T)$. Finally, the partial inverse S to T in (1) of Atkinson's theorem is a Fredholm operator with index $S) = -T)$. F

Definition 8.3.15: Indexed Fredholm Operators

For each $n \in \mathbb{Z}$ define

$$F_n(\mathbb{H}) = \{T \in F(\mathbb{H}) \mid T) = n\}$$

All of these subsets are nonemtpy, since if S is the unilateral shift, then $S^n \in F_{-n}(\mathbb{H})$ and $(S^*)^n \in F_n(\mathbb{H})$ for every $n > 0$.

Lemma 8.3.16: Finite Rank And 0 Index

If $A \in B_f(\mathbb{H})$, then $I + A \in F_0(\mathbb{H})$

***Proof*:**
Analysis now p.110

Corollary 8.3.17: Fredholm Alternative

If $A \in B_0(\mathbb{H})$ and $\lambda \in \mathbb{C} \setminus \{9\}$, then either $\lambda I - A$ is invertible in $B(\mathbb{H})$ or λ is an eigenvalue of A with finite multiplicity, in which case $\overline{\lambda}$ is an eigenvalue for A^* with the same multiplicity

***Proof*:**
Let $T + I - \lambda^{-1}A$. Since $T \in F_0(\mathbb{H})$, by lemma 8.3.16 we know that $T(\mathbb{H})$ is closed and that

$$\dim(\ker(T)) = \dim(\ker(T^*)) < \infty$$

If these dimension are 0, then $T(\mathbb{H}) = (\ker(T^*))^\perp = \mathbb{H}$ and $\ker(T) = \{0\}$, hence T is invertible by proposition 8.2.6, completing the proof.

This lemma implies that the spectrum of a compact operator (i.e. those λ such that $\lambda I - T$ is not invertible, as we'll see in the next chapter) consists of $\{0\}$ and the eigenvalues for T. IN particualr, it is a countable subset of $\mathbb{C}$ with 0 as the only possible accumulatpion point.

Theorem 8.3.18: Index Invariant Under Compact Operator

Let $T \in F(\mathbb{H})$ and A be a compact operator. Then

$$T + A) = T)$$

***Proof*:**
Analysis now p.111

Proposition 8.3.19: Fredholm Class Is Open

Each Fredholm class $F_n(\mathbb{H})$ for each $n \in \mathbb{H}$ is open in $B(\mathbb{H})$

***Proof*:**
analysis now p.112

Proposition 8.3.20: Grading Structure Of Freidholm Operators

If $T_1 \in F_n(\mathbb{H})$ and $T_2 \in F_m(\mathbb{H})$ then $T_1T_2 \in F_{n+m}(\mathbb{H})$

***Proof*:**
Analysis now p. 112

We close with an interesting topological fact that brings some light to the index .Let G denote the group of invertible elements of the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$. and G_0 is the connected component of the identity (which is the group generated by the elements of the form $\exp(A)$ for some $A \in B(\mathbb{H})/B_0(\mathbb{H})$). Then G_0 is an open and closed subgroup of G and G/G_0 is a discrete group that labels the connected components in G. In this case, G/G_0 is isomorphic to $\mathbb{Z}$, and the index of a Fredholm operator T is the image of T under the compsoed quotinet maps from $F(\mathbb{H})$ to G and from G to G/G_0 $(= \mathbb{Z})$

8.4 Trace

8.5 *Intuition to add

This is a conversation I had with Gemini that I really want to incorporate somewhere as it gave some good insights. I will save it here for now:

Zak Katz: Let $F(x_1, \ldots, x_n, t)$ represent a PDE w.r.t time. Let's say $u(x,t)$ represent a solution and $f_t(x) = u(x,t)$ is a "state" at time 0. Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a.

My question is: Is S_t always self-adjoint? If yes, can you prove it, and if not, can you give a counter-example?

Raycast AI:

`Tool: web_search`

No, the time-shift operator S_t is not always self-adjoint.

In fact, for many common PDEs (like the wave equation, transport equation, or Schrödinger equation), S_t is almost never self-adjoint. It is usually unitary (energy-conserving) or a contraction semi-group (dissipative).

9

Spectral Theory

In short, Spectral theory generalizes the notion of eigen vectors and values to infinite dimensional operators. In the previous chapter, we saw how there is a notion of diagonalizability for compact normal operators. In this chapter, we extend this result to normal operators. We also along the way learn some important dualities such as the Gelfand duality which has some useful applications, and is an intersting motivator for classical algebraic geometry.

9.1 Banach Algebras

Recall the definition of a banach algebra

Definition 9.1.1: Banach Algebra Revisited

A *Banach Alagebra* is an algebra A with a submultiplicative norm (so that it is continuous) st the unerlying vector space is a banach space.

All the same applies for Banach Algebras as we expect. Note that for A/I to be a Banach algebra for an ideal I, I has to be closed, just like for subspaces. If A is unital and $A \neq \{0\}$, then it is certainly unique and $\|I\| \geq 1$ (why not =?? Analysis now p.128) If A has no units, we may try to isometrically embed it into a larger Banach Algebra $\widehat{A}$ such that $A \subseteq \widehat{A}$ becomes an ideal such that every nonzero ideal of $\widehat{A}$ has a nonzero intersection with A (such an ideal is called an *essential ideal*). This can be thought as the algebraic counterpart of the notion of compactification of a topological space. For classical Banach algebra, there is usually a natural way of *adjoining a unit*, but there is always a general method of doing so that is the counter-part of one-point compactification. Take the direct sum $A \oplus k = \widehat{A}$ along with the product

$$(x, \alpha)(y, \beta) = (xy + \alpha y + \beta x, \alpha\beta)$$

and the submultiplicative 1-norm. Then $A \oplus k$ is a unital Banach algebra with $I(0,1)$ and contains A as an ideal of co-dimension 1.

Definition 9.1.2: Regular Representation

Let A be a Banch Algebra. Then the (left) *regular representation* $\rho : A \to B(A)$ by $\rho(x)(y) = xy$ for $x, y \in A$.

It should be verified that ρ is a norm decreasing algebra homomorphism, and if A is unital, the inequalities

$$\|A\| \le \|\rho(A)\|\|I\| \le \|A\|\|I\|$$

show that ρ is a homeomrophism. Renorming A by using the equivalent norm from $B(A)$, we may assume that $\|I\| = 1$ (if $A \ne \{0\}$), which we do.

Definition 9.1.3: Approximate Unit

A net $(E_\lambda)_{\lambda \in \Lambda}$ in the unit ball of a Banah algebra A is an *approximate unit* if

$$\lim E_\lambda x = \lim x E_\lambda = x$$

for all $x \in A$.

Example 9.1.4: Approximate Units

Can you think of approximate units for the non-unital Banach algebra $C_0(X)$, $B_0(\mathbb{H})$ and $L^1(\mathbb{R}^n)$? Show that the existence of an approximate unit guarantees that the regular representation ρ is an isometry

We shall denote $A^\times = \mathrm{GL}(A)$ the set of all invertible elements of A (in algebra, we'd usually call these elements unital, but the term unit in functional analysis is more commonly reserved for the identity)

Lemma 9.1.5: Almost Invertible Elements

Let $x \in A$ where A is a unital banach algebra with $\|A\| < 1$. Then $I - A \in \mathrm{GL}(A)$ and

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

***Proof*:**
Since the norm is submultiplicative $\|x^n\| \le \|x\|^n$, thus the series $\sum x^n$ converges in x to an element y. Sicne $xy = yx = y - I$ by manipulating the above power series, it follwos that $y = (I - x)^{-1}$, completing the proof.

Proposition 9.1.6: Properties Of $\mathrm{GL}(A)$

Let A be a unital algebra and $\mathrm{GL}(A)$ be it's multiplicative group. Then $\mathrm{GL}(A)$ is an open subset of A, an the map $x \mapsto x^{-1}$ is a homeomorphism of $\mathrm{GL}(A)$

***Proof*:**
Let $x \in \mathrm{GL}(A)$ and $y \in A$, Then by lemma 9.1.5

$$y = x - (x - y) = x(I - x^{-1}(x - y)) \in \mathrm{GL}(A)$$

provided that $\|A^{-1}(A - B)\| < 1$. In particular, the ball $B_\epsilon(x)$ is contained in $\mathrm{GL}(A)$ for $\epsilon < \|x^{-1}\|^{-1}$, which proves that $\mathrm{GL}(A)$ is open. If $y \to a$, we see from the above equation and lemma 9.1.5 that $y^{-1} \mapsto a^{-1}$, thus inverting is a continuous process and therefore a homeomorphism since $(A^{-1})^{-1} = A$, completing the proof.

From ehre on out, all Banach algebra will be over $\mathbb{C}$, i.e. all Banach algebras will be *complex*. See exercise ref:HERE on the complexicication of general real Banach algebras (or E.2.1.13 in Analysis Now).

Definition 9.1.7: Spectrum

Let A be a complex unital Banach algebra. Then for every $x \in A$, define the *spectrum* of x t obe the set

$$^{(}x) = \{\lambda \in \mathbb{C} \mid \lambda I - x \notin \mathrm{GL}(A)\}$$

The comliment of $^{(}x)$ is the *resolvent set*.

On the resolvent set, we can define the resolvent function

$$R(x, \lambda) = (\lambda I - A)^{-1}$$

Definition 9.1.8: Spectral Radius

Let A be a complex unital Banach algera, $x \in A$, and $^{(}x)$ be its spectrum. Then the smalleest $r \geq 0$ such that $^{(}x) \subseteq B_r(0)$ is called the *spectral radius* of x, and is denoted $r(x)$. THus

$$r(x) = \sup \left\{ |\lambda| \;\middle|\; \lambda \in^{(} x) \right\}$$

Lemma 9.1.9: Build-Up Lemma I

Let $x \in A$ and $f(z) = \sum \alpha_n z^n$ be a holomorphic function in a region contined in the closed disk $B_r(0)$ with $\|x\| \leq r$. then we can define $f(x) = \sum \alpha_n x^n$ in A. Morever, if $\lambda \in^{(} x)$ and $|\lambda| \leq r$, then $f(\lambda) \in^{(} f(x))$

***Proof*:**
The series $f(x)$ is absolutely convergence ($\sum |\alpha_n| \|x\|^n < \infty$), so certainly $f(x) \in A$. Moerover

$$\begin{aligned} f(\lambda)I - f(x) &= \sum \alpha_n(\lambda^n I - x^n) \\ &= (\lambda I - x)\sum \alpha P_{n-1}(\lambda, x) \\ &= (\lambda I - x)y \end{aligned}$$

where $P_{n-1}(\lambda, x) = \sum_{k=0}^{n-1} \lambda^k A^{n-k-1}$ so that $\|P_{n-1}(\lambda, x)\| \leq nr^{n-1}$, which implies the series of polynomials converges in A to an element y commuting with x. We see form this computation that if $f(\lambda)I - f(x) \in \mathrm{GL}(A)$ with inverse v, then yv will be the inverse of $\lambda I - x$, so tha $\lambda I - x \in \mathrm{GL}(A)$

Lemma 9.1.10: Build-Up Lemma II

For every $x\ inA$, we have
$$r(x) \leq \inf \|A^n\|^{1/n}$$

***Proof*:**
If $|\lambda| > \|A\|$, thn by lemma 9.1.5

$$(\lambda I - x)^{-1} = \lambda^{-1}(I - \lambda^{-1}x)^{-1} = \sum \lambda^{-n-1}x^n$$

which shows that $r(x) \leq \|x\|$. Now if $\lambda \in^(x)$, then $\lambda^n \in^(x^n)$ by lemma 9.1.9. Thus $|\lambda^n| \leq \|x^n\|$ from what we've just shown. It follows that $r(x) \leq \|x^n\|^{1/n}$ for every n, completing the proof.

Theorem 9.1.11: Spectrum Is Compact

Let $x \in A$ be an element of a complex unital Banach algebra A. Then the spectrum of x, $^(x)$ is a compact nonempty subset of $\mathbb{C}$, and the spectral radius of x is the limit of the convergence sequence $\|A^n\|^{1/n}$

***Proof*:**
Analysis now p.131, really cool proof that I should take the time to write down!

A cool consequence of this is that we get a classical result of Representation theory known as Artin-Wedderburn for algebraically closed algebras:

Corollary 9.1.12: Algebraically Closed Division Algebra $\mathbb{C}$

Let A be a unital Banach algebra over $\mathbb{C}$. Then if A is a division ring, $A = \mathbb{C}$

***Proof*:**
For each $x \in A$, there is some $\lambda \in sp(x)$ by the above theorem. Thus, $\lambda I - x \notin \mathrm{GL}(A)$, hence $x = \lambda I$.

9.2 Continuous Functional Calculus

Let $T \in B(\mathbb{H})$ and consider $\sigma(T)$. Consider $B(\sigma(T))$, which is a $*$-algebra where $*$ is complex onjugation. We may consider this algebra with the supremum norm. In this section, we'll be looking itno various sub $*$-algebra of $B(\sigma(T))$, for example the polynomials functions on $\sigma(T)$ or the continuous functions on $\sigma(T)$

Definition 9.2.1: Functional Calculus

Let $T \in B(\mathbb{H})$. Then a *functinoal calculus* of T is a continuous $*$-homomorphism from $*$-algebra of functions $(\sigma(T) \to \mathbb{C})$ to $B(\mathbb{H})$ which sents the constant function 1 to the idneitty operator,a and sends the identity function $x \mapsto x$ to T

Since it's a $*$-homomorphism, we have
$$\varphi(\overline{f}) = \varphi(f)^*$$
since we have the supremum norm and the $*$-homomorphism is continuous, $f_n \to f$ in the supremum norm then $f_n(T) \to f(T)$ in the norm topology of $B(\mathbb{H})$.

(more here, this is really cool!)

Part III

Harmonic Analysis

(maybe make this a part. There is a problem tho that distribution theory is a more general theory. May move that up to the Function Analysis part and make a chapter on Tempered distributions instead)

10

Fourier Analysis

In the 19th century, Joseph Fourier was trying to find out how to model the movement of heat through solid bodies, in particular model how heat seems to dissipate and homogenise within its medium. For simplicity, let's say this body was a rode, and we can represent any point on the rode with the interval $[a, b]$. If $u : [0, t] \times [a, b] \to \mathbb{R}$ is a function that at any given point $c \in [a, b]$ and at any given time $t_0 \in [0, t]$ give you the temperature, then Fourier devised that the function u must respect the following differential equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

which can be interpreted that in any movement in time for some point $c \in [a, b]$, the temperature at c will equal to the *average* of its neighbor (see 3Blue1Brown's video on Fourier analysis for an excellent visualization of this). After choosing the value at $u|_{t_0=0}$ which represent the initial distribution of heat in the rode $[a, b]$, as t_0 changes $u|_{t_0}$ represents the temperature of the rod at every point at time t_0 . Working with this equation, Fourier saw that trigonometric identities are a solution for u given the restraint (with suitable scaling[1]). The next insight Fourier had was that a huge amount of functions can be interpreted as an infinite sum of trigonometric functions, and even better, he found how to find the appropriate coefficients for the trigonometric functions! In particular, this family of equations is *dense* in the space of all L^2 functions on a compact interval (or even better, they form an orthonormal basis), and since the PDE in question is *linear*, we can use the Dominated convergence theorem to show that the resulting sequence can mean that $u(0, \cdot)$ can be any L^2-integrable function (or really, simply integrable since the domain is compact). This is known as the *Fourier Series*, and going form a function to its Fourier series is called the *Fourier Transform*.

This was the beginning of Fourier Analysis, also known as Harmonic Analysis[2]. Since then, it was

[1]In particular, some care must be given to the boundary conditions, but this is something that would be covered in more detail in a proper class in PDE's

[2]Sometimes, Harmonic Analysis is used as a generalization of Fourier Analysis in the case of non-abelian groups, something which will be made more clear in this chapter

asked what spaces of functions admit a Fourier transform, can we go back and forth between the Fourier series and the original function, and can we generalize the domain of the functions (this will be pushed to be compact abelian group, and more generally locally compact groups). The great advantage of being able to find the space of functions that have a Fourier transform is that (under suitable conditions) derivatives work really well with infinite series, trigonometric functions have some of the simple derivatives (they are periodic of order 4), and the Fourier Transform allows us to transfer our problem into a potentially nice space (similarly to how in Representation Theory, it is much easier to find the representation of the monster group to study it rather than directly study it).

These days, Harmonic Analysis has found a large range of applications, from solving my different types of PDE's, to having connections to algebraic number theory (the pursuit of this connection is known as *Langlands Program*). In this chapter, we will cover the elementary theory required to understand the theory of Fourier analysis.

10.1 Basic Definitions for Fourier Analysis

In this chapter, we will always work with $\mathbb{R}^n$ with the Lebesgue measure, denoted m. If $E \subseteq \mathbb{R}^n$ is measurable, $L^p(E)$ (or $L^p(E, m)$) is the set of all L^p functions from E to $\mathbb{R}$, $C^k(E)$ is the set of all k-times continuously differentiable functions from E to $\mathbb{R}$, $C^\infty(E)$ (the set of all smooth functions form E to $\mathbb{R}$) is equal to $\cap_{k=1}^\infty C^k(E)$, and $C_c^\infty(E)$ is the set of all smooth function whose compact support is in E. If $E = \mathbb{R}^n$, we will often drop the domain of the function space, so:

$$L^p = L(\mathbb{R}^n), \qquad C^\infty = C^\infty(\mathbb{R}^n) \qquad C_c^\infty = C_c^\infty(\mathbb{R}^n)$$

The inner product and norm on $\mathbb{R}^n$ is the usual dot product and euclidean norm:

$$x \cdot y = \sum_i^n x_i y_i \qquad \|x\| = |x| = \sqrt{x \cdot x}$$

Derivative play a prominent role in this chapter, and so we establish important notation. We will assume the standard basis and write:

$$\partial_j = \frac{\partial}{\partial x^j}$$

and for higher order mixed derivatives, we will use a multi-index, which is an ordered n-tuple $\alpha = (\alpha_1, ..., \alpha_n)$. As usual:

$$|\alpha| = \sum_1^n \alpha_i \quad \alpha! = \prod_1^n \alpha_i! \quad \partial^\alpha = \left(\frac{\partial}{\partial x_i}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n}$$

and for any $x = (x_1, x_2, ..., x_n) \in \mathbb{R}^n$:

$$x^\alpha = \prod_1^n x_i^{\alpha_i}$$

Be aware that $|\alpha|$ and $|x|$ do not means the same thing; the meaning should be clear from context. For reference, the product rule on a multi-index expands to:

$$\partial^\alpha(fg) = \sum_{\beta+\gamma=\alpha} \frac{\alpha!}{\beta!\gamma!}(\partial^\beta f)(\partial^\gamma g)$$

A common function we'll use a lot in C_c^∞ is:

$$\psi(x) = \eta(1 - |x|^2) = \begin{cases} \exp((|x|^2 - 1)^{-1})) & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

where $\eta(t) = e^{1/t}\chi_{(0,\infty)}(t)$. Another common subset of C^∞ that will be of use is the following which ensures that a function and all its derivatives decay faster than polynomial time:

Definition 10.1.1: Schwartz Space

The set $\mathcal{S} \subseteq C^\infty$ is the set of all smooth functions and all their derivatives which vanish faster than any power of $|x|$, that is, for any $n \in \mathbb{N}$ and any multi-index α, if we define:

$$\|f\|_{(N,\alpha)} = \sup_{x\in\mathbb{R}^n} (1 + |x|)^n |\partial^\alpha f(x)|$$

Then:

$$\mathcal{S} = \left\{ f \in C^\infty \;\middle|\; \|f\|_{(N,\alpha)} < \infty \text{ for all } N, \alpha \right\}$$

Example 10.1.2: Schwartz Functions

1. Naturally, $C_c^\infty \subseteq \mathcal{S}$
2. $f_\alpha(x) = x^\alpha e^{-|x|^2}$ for any multi-index α is in $\mathcal{S}$

Note that if $f \in \mathcal{S}$, then $\alpha f \in L^p$ for all α and all $p \in [1, \infty]$ since $|\partial^\alpha f(x)| \leq C_N(1 + |x|)^{-N}$ for all $N \in \mathbb{N}$, and $(1 + |x|)^{-N} \in L^p$, for all $N > n/p$

Proposition 10.1.3: $\mathcal{S}$ is a Fréchet Space

The Schwartz space $\mathcal{S}$ is a Fréchet space with the topology defined by the norms $\|\cdot\|_{(N,\alpha)}$

***Proof*:**
All is easy to prove with the possible exception of completeness. Let $\{f_k\}$ be a cauchy sequence in $\mathcal{S}$ so that $\|f_i - f_j\|_{(N,\alpha)} \to 0$ for all N, α. In particular, for each α the sequence $\{\partial^\alpha f_k\}$ converges uniformly to some g_α. Letting e_i denote the standard ith basis vector, we get:

$$f_k(x + te_i) - f_k(x) = \int_0^t \partial_i f_k(x + se_i)ds$$

Letting $k \to \infty$, we get:

$$g_0(x + te_i) - g_0(x) = \int_0^t g_{e_i}(x + se_i)ds$$

and the fundamental theorem of calculus tells us that $g_{e_i} = \partial_i g_0$, and so by induction we get $g_\alpha = \partial^\alpha g_0$ for all α. From this, it is now easy to verify that $\|f_k - g_0\|_{(N,\alpha)} \to 0$ for all α and $N \in \mathbb{N}$.

A useful characterization of $\mathcal{S}$ is also the following:

Proposition 10.1.4: Characterizing Schwartz Functions

If $f \in C^\infty$, Then $f \in \mathcal{S}$ if and only if $x^\beta \partial^\alpha f$ is bounded for all multi-indices α, β, if and only if $\partial^\alpha(x^\beta f)$ is bounded for all multi-indices α, β.

***Proof*:**
If $f \in \mathcal{S}$, then naturally $|x^\beta| \le (1+|x|)^N$ for $\beta \le N$. Conversely, notice that $\sum_1^n |x_i|^N$ is strictly positive on the unit sphere $|x| = 1$ (which is compact), and hence has a positive minimum, say δ. It follows that $\sum_1^n |x_i|^N \ge \delta |x|^N$ for all x since both sides are homogeneous of degree N, and hence:

$$(1+|x|)^N \le 2^N(1+|x|^N) \le 2^N \left[1 + \delta^{-1} \sum_1^n |x_i^N|\right] \le 2^N \delta^{-1} \sum_{|\beta| \le N} |x^\beta|$$

Giving the first equivalence. For the second equivalence, notice that for every α, $\partial^\alpha(x^\beta f)$ is a linear combination of terms of the form $x^\gamma \partial^\delta f$ and vice versa, by the product rule, completing the proof.

Next we define a periodic function to be a function for which $f(x+k) = f(x)$ for all $x \in \mathbb{R}^n$ and $k \in \mathbb{Z}^n$. Without loss of generality the period of the functions will be of size 1. Thus, we can consider the space on which we define periodic functions to be $\mathbb{R}^n/\mathbb{Z}^n \cong (\mathbb{R}/\mathbb{Z})^n \equiv \mathbb{T}^n = S^1 \times S^1 \times \cdots \times S^1$, i.e. we can think of defining period functions on an n-torus (note that $\mathbb{T}^1 = S^1$, but $\mathbb{T}^2 \ne S^2$). Naturally, $\mathbb{T}^n$ is a compact Hausdorff space. It can be identified with all points $z = (z_1, z_2, ..., z_n) \in (\mathbb{C}^n)^*$ with $|z_i| = 1$ via the map

$$(x_1, x_2, ..., x_n) \mapsto (e^{2\pi i x_1}, ..., e^{2\pi i x_n})$$

For measure-theoretic purposes, we will sometimes identify $\mathbb{T}^n$ with the unit cube:

$$Q = \left[\frac{-1}{2}, \frac{1}{2}\right)^n$$

and when we talk about the Lebesgue measure on $\mathbb{T}^n$, we will mean the induced measure from Q. This means that $m(\mathbb{T}^n) = 1$. We will think of functions on $\mathbb{T}^n$ as either periodic functions on $\mathbb{R}^n$ or as functions on Q (the context will make it clear which we pick).

Exercise 10.1.1

1. Is the sum of two periodic functions periodic?

10.1.1 Convolution

In this section, we introduce a new way of multiplying two functions that will be very useful and very natural. The naturality comes from the discrete case, and so we will start with it: recall that if we have two polynomials $p(x)$ and $q(x)$ (which can be thought of as formal sums, bringing in the naturality), then the multiplication formula which groups the terms of the same degree is:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{i+j=k} p_i q_j\right) x^k = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{l=0}^{k} p_l q_{k-l}\right) x^k$$

If we set p_k or q_{l-k} to zero if k or $l-k$ is negative, then we can re-write the sum as:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i\right) \cdot \left(\sum_{i=0}^{\deg(q)} q_j x^j\right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{-\infty}^{\infty} p_l q_{k-l}\right) x^k$$

If we let $p[n]$ and $q[n]$ represent the coefficient of the nth degree term, and $h = pq$, then:

$$h[n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

We can write this as:

$$h[n] = [p * q][n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

where $p*q$ is a compact notation the right hand side. The summation can be thought of as repersenting the discrete measure. Generalized to the continuous case, this is called *convolution*:

Definition 10.1.5: Convolution

Let f, g be measurable on $\mathbb{R}^n$. Then the *convolution* of f and g is a function $f * g : \mathbb{R}^n \to \mathbb{R}$ defined by:

$$f * g(x) := \int f(x-y)g(y)dy$$

for all x such that the integral exists

To see this in action, Let $f = \chi_{[0,1]}$ and $g = \chi_{[0,1]}$, and you'll find that $f * g$ is a a triangle. You can imagine sliding the graph of f and finding the area of the intersection of f and g, and plotting it. There are many ways to make sure that $f * g$ is a.e. defined. For example, if f was bounded and compactly supported, g can be locally integrable. Note that in many cases, we will need that $K(x, y) = f(x-y)$ be measurable on $\mathbb{R}^n \times \mathbb{R}^n$. If we let $s(x, y) = x - y$, then since s is continuous, if f is Borel measurable, so is K. (Folland makes a comment here saying we can "assume" that this is the case, which I'm guessing is f being Borel measurable in the definition of convolution, because of his theorem 2.12, which is my proposition 2.1.20, and he says the same can happen for Lebesgue measurability). Usually, we will use convolution to somehow combine the property of two functions f and g. We will soon see that convolution often preserves the "nice" properties the "nicer" function, so that we would convolution, say, a smooth function f with an integrable function g to get a smooth function $f * g$.

I would also be remiss if I don't mention that convolution is like multiplying two functions in a way the respects operation the *domain*. If f, g are real functions, then $f \cdot g$ respects the multiplication in the *codomain* as it multiplies the outputs. As $f * g$ is the continuous case of polynomial multiplication which multiplies, it respects the multiplication happening in the *domain*. To motivate why this is worth pointing out, consider a finite group G, $f \in \mathrm{Hom}_{\mathbf{Set}}(G, \mathbb{R})$, and interpret it as an element of $\mathbb{C}[G]$ so that $f = \sum_i a_i g_i$. Then we can always make $r \in \mathbb{R}$ act on f via:

$$r \cdot f(g_i) = rf(g_i)$$

However, we can also make $g \in G$ act on f:

$$g \cdot f(g_i) = f(g^{-1}g_i)$$

where we inverse for associativity:

$$\begin{aligned}(g \cdot (h \cdot f))(x) &= (h \cdot f)(g^{-1}x) \\ &= f(h^{-1}(g^{-1}x)) \\ &= f((h^{-1}g^{-1})x) \\ &= f((gh)^{-1}x) \\ &= ((gh) \cdot f)(x)\end{aligned}$$

This is the same thing as defining $g \cdot f$ given $f \in \mathbb{C}[G]$ since:

$$g\dot{f} = g \cdot \sum_{g_i \in G} a_i g_i = \sum_{g_i \in G} a_i (gg_i) = \sum_{g^{-1}g_i \in G} a_i g_i$$

Thus, we may think of the translation operation as an action from the structure of the *domain*, when the domain has a group structure. It shall turn out that the way of characterizing the orthonormal basis for L^2 that gives rise to the Fourier transform is to choose the one that's invariant under τ_y.

Definition 10.1.6: Translation Operator

If $f : \mathbb{R}^n \to \mathbb{R}$, then

$$y * f(x) = \tau_y f(x) = f(x - y)$$

Notice that:

$$\|\tau_y f\|_p = \|f\|_p \qquad \|\tau_y f\|_u = \|f\|_u$$

Using this, we can give another definition of uniformally continuous: f is uniformally continuous if:

$$\|\tau_y f - f\|_u \to 0$$

Lemma 10.1.7: Compact Support Functions And UC

If $f \in C_c(\mathbb{R}^n)$, then f is uniformally continuous

***Proof*:**
should be easy proof (Folland p.238)

Proposition 10.1.8: Translation Is Continuous In Lp

If $1 \leq p < \infty$, then translation is continuous in the L^p norm, in particular if $f \in L^p$ and $z \in \mathbb{R}^n$, then

$$\lim_{y \to 0} \|\tau_{y+z} f - \tau_z f\|_p = 0$$

Note this proposition is false when $p = \infty$

> ***Proof***:
> Folland p. 238

The following results all work over $\mathbb{R}^n$ or $\mathbb{T}^n$ (interpreted as the unit cube)

Proposition 10.1.9: Algebraic Properties Of Convolution

Assume all the following integrals are defined and converge absolutely. Then:

1. $f * g = g * f$
2. $(f * g) * h = f * (g * h)$ (this result in particular requires absolute convergence)
3. $f * (g + h) = f * g + f * h$
4. $a(f * g) = (af) * g$ for all $a \in \mathbb{C}$
5. for all $z \in \mathbb{R}^n$, $\tau_z(f * g) = (\tau_z f) * g = f * (\tau_z g)$
6. If A is the closure of $\{x + y \mid x \in \text{supp}(f),\ y \in \text{supp}(g)\}$, then $\text{supp}(f * g) \subseteq A$

Proof:

1. $$\begin{aligned} f * g(x) &= \int f(x-y)g(y)dy \\ &= \int f(z)g(x-z)dz && z = x - y,\ y = x - z \\ &= \int g(x-z)f(z)dz \\ &= g * f(x) \end{aligned}$$
2. Follows from (1) from Fubini's theorem
3. Follows from direct computation
4. Follows from direct computation
5. Follows from direct computation and (1)
6. If $x \notin A$, then for any $y \in \text{supp}(g)$, we have $x - y \notin \text{supp}(f)$, thus $f(x-y)g(y) = 0$, i.e. $f * g(x) = 0$

One of the first properties important to establish is the relation between convolution of a function and p-norms:

Theorem 10.1.10: Young's Inequality

Let $1 \leq p, q, r \leq \infty$ be such that $p^{-1} + q^{-1} = r^{-1} + 1$, and let $f \in L^p$ and $g \in L^q$. Then $f * g(x)$ exists for a.e. x, $f * g \in L^r$ and

$$\|f * g\|_r \leq \|f\|_p \|g\|_q$$

Notice in the statement that we have something a little different than the usual conjugate exponents. To see why, there is another thing worth pointing out, which is that this condition is stable under scaling. In particular, if $f_\lambda(x) = f(\lambda x)$ and $g_\lambda(x) = g(\lambda x)$. Then:

$$\begin{aligned} \|f_\lambda\|_p &= \left(\int_{\mathbb{R}^n} f(\lambda x)^p \right)^{1/p} \\ &= \left(\int_{\mathbb{R}^n} \lambda^{-n} f(y)^p \right)^{1/p} && y = \lambda x,\ dy = \lambda^n dx \\ &= \lambda^{-n/p} \|f\|_p \end{aligned}$$

where λ^n is the determinant of the Jacobian matrix. So:

$$\begin{aligned} (f_\lambda * g_\lambda)(x) &= \int_{\mathbb{R}^n} f(\lambda x - \lambda y) g(\lambda y) dy \\ &= \lambda^{-n} (f * g)(\lambda x) \end{aligned}$$

Thus we have:

$$\begin{aligned} \|f_\lambda * g_\lambda\|_r &= \lambda^{-n} \|(f * g)_\lambda\|_r \\ &= \lambda^{-n/r - n} \|f * g\|_r \\ &\leq \lambda^{-n/p - n/q} \|f\|_p \|g\|_q \end{aligned}$$

Notice that this inequality must hold for *all* $\lambda > 0$. If the exponents of λ on the left and right sides were different, we could force a contradiction by taking $\lambda \to 0$ or $\lambda \to \infty$ (one side would vanish while the other explodes). Thus, the exponents must be equal:

$$-n\left(1 + \frac{1}{r}\right) = -n\left(\frac{1}{p} + \frac{1}{q}\right)$$

Dividing out the $-n$, we recover the necessary condition:

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$$

Intuitively, the "extra" $+1$ comes from the integration dy in the definition of convolution, which adds n dimensions of volume (unlike pointwise multiplication which preserves the exponent balance of Hölder's inequality).

***Proof*:**
First, consider $r = \infty$, meaning $r^{-1} = 0$, $p^{-1} + q^{-1} = 1$. So, we want to show $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$. So:

$$\begin{aligned}
\|f * g\|_\infty &= \sup_x \left| \int f(x-y) g(y) dy \right| \\
&\leq \|f(x - \cdot)\|_p \|g\|_q \\
&\stackrel{!}{=} \|f\|_p \|g\|_q
\end{aligned}$$

where $\stackrel{!}{=}$ comes from $\int |h(x-y)| dy = \int |h(y)| dy$ where $z = x - y$, $dz = (-1)^n dz$ so the $(-1)^n$ dissapears in the absolute value. (Note: I corrected the L^1 typo to L^∞ here to match the case $r = \infty$).

Next, if p or q is 1 (say $p = 1$), then $1 + q^{-1} = r^{-1} + 1$, so $q^{-1} = r^{-1}$. Then we would have:

$$\|f * g\|_q \leq \|f\|_1 \|g\|_q$$

To see that this happens, we use theorem 6.3.2. In particular, let $K(x, y) = f(x - y)$. Then if $\|g\|_q$ exists, $\|K(x, \cdot)\|_1 = \|f\|_1$ and $\|K(\cdot, y)\| = \|f\|_1$, giving us the result.

Now, assume $1 < p < r$, $1 < q < r$. First, we manipulate it to be able to use Hölder's inequality:

$$|f(y)g(x-y)| = \left(|f(y)|^p |g(x-y)|^q\right)^{1/r} |f(y)|^{1-p/r} |g(x-y)|^{1-q/r}$$

Next, notice that:

$$\frac{1}{r} + \frac{r-q}{rq} + \frac{r-p}{rp} = \frac{1}{r}\left(1 + \frac{r}{q} - 1 + \frac{r}{p} - 1\right) = 1$$

So, we can apply general Hölder's inequality since $r - p, r - q \neq 0$ which gives us:

$$\begin{aligned}
|(f * g)(x)| &\leq \left(\int_{\mathbb{R}^n} |f(y)^p| |g(x-y)|^q dy\right)^{1/r} \cdot \|f\|_p^{(r-p)/r} \|g\|_q^{(r-q)/r} \\
\Rightarrow \quad |(f * g)(x)|^r &\leq \left(\int_{\mathbb{R}^n} |f(y)^p| |g(x-y)|^q dy\right) \cdot \|f\|_p^{r-p} \|g\|_q^{r-q}
\end{aligned}$$

And now, we can integrate the left hand side with respect to x and use Tonnelli's theorem:

$$\Rightarrow \quad \|(f * g)\|_r^r \leq \|f\|_p^p \|f\|_p^{r-p} \|g\|_q^q \|g\|_q^{r-q} = \|f\|_p^r \|g\|_q^r$$

completing the proof.

If we focus on the special case of $p = q = r = 1$, then $f * g \in L^1(\mathbb{R}^n)$. Then proposition 10.1.9 and Young's inequality (theorem 10.1.10) turns $L^1(\mathbb{R}^n)$ into a "commutative associative algebra". Note that this algebra has no multiplicative identity (i.e. We are working over a *rng*). It does have a notion of an identity in a limit, which we will shortly explore. In fact, if we slightly correctly loosen our notion of function to *distribution*, we shall naturally recover an identity (explored in chapter 11). Note that if $f, g \in L^2(\mathbb{R}^n)$ then $f * g \in L^\infty(\mathbb{R}^n)$, namely convoluting forces the functions the result to be bounded.

Before exploring a notion of identity, it must be shown that convoluting two functions where one nicer than the other usually keeps the properties of the nicer function:

Lemma 10.1.11: Convolution Preserves C_c^k

Let $f \in L^1$ and $g \in C_c^k$ (more generally, $\partial^\alpha g$ is bounded for $|\alpha| \leq k$). Then

$$f * g \in C^k$$

and

$$\partial^\alpha (f * g) = (f * \partial^\alpha g)$$

***Proof*:**
In Folland textbook, he says it's clear from theorem 2.3.10.

This is a matter of computation. We'll do the case of $k = 1$, since the rest is a matter of induction. First, write what you know about $f * g$:

$$\begin{aligned} f * g(x) &= \int_{\mathbb{R}^n} f(x-y)g(y)dy \\ &= \int_{\mathbb{R}^n} f(y)g(x-y)dy \end{aligned}$$

So consider

$$\lim_{h \to 0} \frac{f * g(x+h) - f * g(x)}{h} = \lim_{h \to 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy$$

Notice that:

$$\left| \frac{g(x+h-y) - g(x-y)}{h} \right| \leq \frac{|g'(c)| \cdot h}{h}$$

meaning we have a dominating function, and so we can apply dominated convergence theorem to move the limit in!

$$\begin{aligned} \lim_{h \to 0} \frac{f * g(x+h) - f * g(x)}{h} &= \lim_{h \to 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \lim_{h \to 0} \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \end{aligned}$$

and since $g \in C_c^1$ (in particular, g is differnetiable), this limit exists, and since g has comapct support, the integral is still well-defined and is C^1 (see the remarks we made at the beginning).

From here we can apply the induction, and the proof is done.

Proposition 10.1.12: Smooth Function Dense in L^p

Compact smooth functions (and hence Schwartz functions) are dense in L^p.

***Proof*:**
hint: simple functions are dense in L^p. Convolute them with a smooth bump function.

Proposition 10.1.13: Convolution in Schwartz Space

Let $f, g \in \mathcal{S}$. Then $f * g \in \mathcal{S}$

***Proof*:**
First, $f * g \in C^\infty$ by lemma 10.1.11. Next, we establish the following for the coming chain of inequalities:

$$1 + |x| \overset{\triangle\text{ineq.}}{\leq} 1 + |x - y| + |y| \leq (1 + |x - y|)(1 + |y|)$$

and so:

$$\begin{aligned}(1 + |x|)^N |\partial^\alpha (f * g)(x)| &\leq \int (1 + |x - y|)^N |\partial^\alpha f(x - y)| (1 + |y|)^N |g(y)| dy \\ &= \|f\|_{(N,\alpha)} \|g\|_{(N+n+1,\alpha)} \int (1 + |y|)^{-n-1} dy\end{aligned}$$

which is certainly finite

With this information, we are going to define a notion of "multiplicative idenity"

Definition 10.1.14: Good Kernels

Let $\{\varphi_t\}_{t>0}$ be a set of functions of the form $\varphi_t : \mathbb{R}^n \to \mathbb{C}$. Then they are called a *good kernel* if $t \in (0, \epsilon_0)$ and:

1. $\sup_{t>0} \|\varphi_t\|_1 < \infty$
2. $\int_{\mathbb{R}^n} \varphi_t(x) dx = 1$
3. $\lim_{t \to 0} \int_{|x|>\delta} |\varphi_t(x)| dx = 0$ for all $\delta > 0$

Theorem 10.1.15: Good Kernels Like Mult. Identity

Let $f \in L^p$ for $1 \leq p < \infty$ (note it is $<$, not $\leq$), then:

$$f * \varphi_t \to f$$

in L^p as $t \to \infty$. Furthermore, if f is continuous at some x_0, then:

$$f * \varphi_t(x_0) \to f(x_0)$$

in L^p

***Proof*:**
It is equivalent to show $f * \varphi_t - f \to 0$, so for all x:

$$\begin{aligned}
f * \varphi_t(x) - f(x) &= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - f(x)\cdot 1 \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - f(x)\cdot \int_{\mathbb{R}^n} \varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y)dy - \int_{\mathbb{R}^n} f(x)\varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} f(x-y)\varphi_t(y) - f(x)\varphi_t(y)dy \\
&= \int_{\mathbb{R}^n} [f(x-y) - f(x)]\varphi_t(y)dy \\
&= \int_{|y|\le\delta} [f(x-y) - f(x)]\varphi_t(y)dy + \int_{|y|>\delta} [f(x-y) - f(x)]\varphi_t(y)dy
\end{aligned}$$

and so, taking the p-norm we get:

$$\|f * \varphi_t - f\|_p \le \int_{|y|\le\delta} \|f(\cdot - y) - f(\cdot)\|_p|\varphi_t(y)|dy + \int_{|y|>\delta} 2\|f\|_p|\varphi_t(y)|dy$$

at this point, we are almost done. If we let $c = \sup_{t>0}\int_{\mathbb{R}^n}|\varphi_t(y)|dy$, then we get:

$$\limsup_{t\to 0}\|f * \varphi_t - f\|_p \le c \limsup_{\delta\to 0,\ |y|\le\delta} \|f(\cdot - y) - f(\cdot)\|_p = 0$$

where the last inequality comes from the shrinking delta.

It was left as an exercise to do the point-convergence in the continuous case.

Note that if f is uniformally continuous and bouned, then $f * \varphi_t \to f$ uniformally, and if $\int \varphi = a$, then $f * \varphi_t \to af$.

Example 10.1.16: Good Kernels
Let $X(x) \in C_c^\infty(\mathbb{R}^n)$ where $\int X = 1$. Then:

$$\varphi_t(x) = t^{-n}X(x/t)$$

is a good kernel (ex. let $X(x) = \frac{1}{\sqrt{\pi}}e^{-x^2}$). This is just a matter of computation. Note that $\varphi_t = 0$ if $\frac{x}{t} \ge c_0$ for some c_0, and so $t \le \frac{x}{c_0}$. It [somehow] follows that C_c^∞ is *dense* in L^p $(1 \le p < \infty)$ (note the $<$ and not $\le$)

For another Another example of a family of good kernel, take:

$$K_\delta(x) = \delta^{-1/2}e^{\frac{-\pi x^2}{\delta}}$$

These can be visualized here:

https://www.desmos.com/calculator/e8xc37r3ck

Notice that since the function $X \in C_c^\infty(\mathbb{R})$ is smooth on compact supporet then by lemma 10.1.11, $f * X$ is also smooth, and will remain smooth for every φ_n. Since $f * \varphi_n \to f$, we see that a sequence of smooth functions converges to an L^p function, meaning smooth functions are *dense* in L^p!

I found this interesting stack-exchange post on the importance of good Kernels:

https://math.stackexchange.com/questions/3294744/good-kernels-that-exist-in-the-real-world

Using the fact that convolution preserves the "nicer" structure, we strengthen's Urysohn's lemma (or, if you know some differential geometry, generlizes partitions of unity)

Theorem 10.1.17: Smooth Urysohn's Lemma

Let $K \subseteq \mathbb{R}^n$ be compact and $U \subseteq \mathbb{R}^n$ be open and contain K ($U \supseteq K$). Then there exists a $f \in C_c^\infty$ such that $0 \leq f \leq 1$ and $f \equiv 1$, K and $\operatorname{supp}(K) \subseteq f$.

***Proof*:**
Since K is compact, then $\delta = \rho(K, U^c)$ (the shortest distance from K and U^c) is non-zero. We are going to do a similar trick to that of lemma 3.4.2: let $V = \left\{x \mid \rho(x, K) < \frac{\delta}{3}\right\}$. Choose some φC_c^∞ such that $\int \varphi = 1$ and $\varphi(x) = 0$ for $|x| \geq \delta/3$ (as an example, $\{\psi_t\}$ to be a good kernel and take $\varphi = \left(\int \psi\right)^{-1} \psi_{\delta/3}$).

Now, let $f = \chi_V * \varphi$. Then as we've shown before, $f \in C_c^\infty$, and it is easy to see that $0 \leq f \leq 1$, $f \equiv 1$ on K, and $\operatorname{supp}(f) \subseteq \{x \mid \rho(x, K) \leq 2\delta/3\} \subseteq U$

10.2 Fourier Transform

Consider again the heat equation on a compact interval:

$$\frac{du}{dt} = \frac{\partial^2 u}{\partial x^2} \tag{10.1}$$

where u is a solution. If u represents a rod that is at time $t = 0$ half hot on the left and half cold on the right, with the two halves in contact at the midpoint, then as we let time move forward, the temperature will equalize due to diffusion. The shape goes from a step function, to a smoothed profile resembling the error function $\operatorname{erf}(x)$, and as time $\to \infty$ it converges to a constant function representing the average temperature. These are all functions with distinctly different behaviours.

Now, assume that the heat of the rod at $t = 0$ has the shape of $\sin(x)$ or $\cos(x)$. Then as we let time propagate these functions dampen uniformly, notably *they are always the same function*, simply scaled. This is essentially the behaviour of *eigenvectors*, namely as time moves forward we are scaling by a factor. To have eigenvectors, we need a linear transformation between vector spaces. Let S_t be the *time evolution operator* that takes in a state f_{t_0} (a function on the domain that is a t_0-slice of a solution $u(x, t_0)$) and outputs f_{t_0+t}, in particular $S_t : \mathscr{P}(X) \to \mathscr{P}(X)$ where $\mathscr{P}(X)$ is the state space over X and S_t is the evolve by t operator:

$$S_t(u(-, t_0)) = u(-, t_0 + t)$$

Or, viewed as an (abelian) monoid action[3]:

$$S : [0,\infty) \times \mathscr{P}(X) \to \mathscr{P}(X) \qquad t \cdot f_{t_0} = f_{t_0+t}$$

Note that it isn't "obvious" what type of space $\mathscr{P}(X)$ is, we will assume that this is a Hilbert space; see [4] for the justification of this assumption. At minimum it is certainly the case that any (smooth) L^2-integrable function is a valid starting state that can then evolve according to the heat equation. If the reader can accept that there is a reasonable sense in which integrable functions can be differentiated as shall be demonstrated in chapter 11, the reader can let $\mathscr{P}(X) = L^2([0,1])$.

The choice of L^2 is not arbitrary. As we saw in the preceding section, L^2 is the unique L^p space where the duality map J_p is linear, the geometry is flat, and vectors and covectors are canonically identified. The Fourier transform exploits this: it is an *isometry* of L^2 (Plancherel's theorem), which is possible precisely because the "rotation" from position coordinates to frequency coordinates preserves the inner product. On L^p for $p \neq 2$, the Fourier transform does not stay within the same space; instead, the Hausdorff-Young inequality gives

$$\|\hat{f}\|_{L^q} \leq C\|f\|_{L^p}, \qquad \tfrac{1}{p} + \tfrac{1}{q} = 1, \quad 1 \leq p \leq 2$$

mapping L^p into its dual L^q—reflecting the nonlinear duality of those spaces. Only at $p = 2$ does the transform remain an endomorphism, because only there is $V \cong V^*$ linearly.

Now, certainly S_t is linear as the heat equation is linear and homogeneous[4]. We can thus ask if it has eigenvectors, that is functions $f_r \in \mathscr{P}(X)$ where

$$S_t f_r(x) = \lambda_t f_r(x)$$

If we let t vary, we can define a function $\lambda(t)$ dependent on the time. Given a compact domain $X = [a,b]$, by direct computations we can check that $\sin(kx)$ is an eigenvector, where this is indeed a state as it is a t-slice of:

$$u(x,t) = e^{-k^2 t}\sin(kx)$$

For example at $t = 0$:

$$S_t(\sin(kx)) = e^{-k^2 t}\sin(kx)$$

With the assumption that $\mathscr{P}$ is a Hilbert space, if these eigenfunctions form an orthonormal basis, we can decompose a function into a "weighted sum" of these well-behaved functions with respect to time. For the rest of this section we shall take for granted $\{\sin(kx)\}_{k\in\mathbb{N}}$ form an orthonormal basis for S_t (proven in theorem 10.2.4) after appropriate normalization depending on the domain.

It must be pointed out that the same argument works for $\cos(kx)$:

$$S_t(\cos(kx)) = e^{-k^2 t}\cos(kx)$$

and these *also* form an orthonormal basis that *also* diagonalizes S_t. These two sets are individually complete orthogonal bases for $L^2([0,1])$ (with appropriate boundary conditions), so their union is *overcomplete*—it contains "twice as many" basis elements as needed. Moreover, the combined set is not even orthogonal:

$$\int_0^1 \sin(\pi x)\cos(2\pi x)dx = \frac{-2}{3\pi} \neq 0$$

[3]The heat equation is well-posed forward in time but ill-posed backward: reversing diffusion amplifies noise catastrophically (failing Hadamard's conditions for well-posedness; see [4]). Hence S_t forms a monoid $[0,\infty)$ rather than a group—time evolution is irreversible.

[4]It is also continuous and normal, but for that we would need to choose whether we consider $\mathscr{P}$ as a Banach space or a distribution space; intuitively the heat equation dissipates energy so $\|S_t f\| \leq \|f\|$

More fundamentally, as we shall see, neither set alone diagonalizes the *translation* operator—they only block-diagonalize it—and the passage to complex exponentials will be forced by the requirement of full diagonalization.

Thus, the question shifts into one about finding a *natural* eigenbasis for S_t. A-priori, it is not necessarily the case that these are the only eigenbases. What we need is for the eigenbasis to "respect" some inherent geometry of the space. Phrased differently, the eigenbasis needs to be *invariant* with respect to some operator(s), where the operators represent some symmetry of the space. To get a better grasp of invariance of operators, we may start looking at other PDEs and considering diagonalizing the operator S_t with respect to the solutions of those PDEs.

For example, if we solve the Heat equation on a sphere:

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

the operator can still be diagonalized (see [4]). What we shall want is for an eigenbasis that respects the spherical geometry, namely for it to be invariant to an $SO(3)$ action. Such eigenfunctions do exist, they are the *Spherical Harmonics* $Y_l^m(\theta, \varphi)$, and they are invariant to rotation[5]

$$\Delta(f \circ R) = (\Delta f) \circ R$$

To give another example that is famous in physics, consider the Quantum Harmonic Oscillator (QHO):

$$i\frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + x^2 \psi$$

This PDE is diagonalizable (see [4]), but it is *not* diagonalizable by the Fourier basis e^{ikx}. The reason, in the language of our framework, is that the potential $V(x) = x^2$ *breaks translation invariance*: since $x^2 \neq (x+a)^2$, the QHO Hamiltonian $H = -\partial_x^2 + x^2$ does not commute with the translation operator τ_a. By the general principle we are developing—different symmetries produce different eigenbases—the translation eigenbasis cannot diagonalize the QHO.

Instead it was shown to be diagonalized (uniquely) by the *Hermite Functions* (a product of the Hermite polynomials H_n with a Gaussian):

$$\varphi_n(x) = H_n(x) e^{-x^2/2}$$

The relevant symmetry is the *Heisenberg-Weyl algebra* generated by the ladder operators $a = \frac{1}{\sqrt{2}}(x + \partial_x)$ and $a^\dagger = \frac{1}{\sqrt{2}}(x - \partial_x)$, whose eigenbasis is precisely the Hermite functions.

A revealing computation shows *why* the Fourier basis fails here. If we attempt to apply the Fourier transform, the $-\partial_x^2$ term becomes multiplication by k^2 in frequency space, but the x^2 multiplication term becomes a Laplacian $-\partial_k^2$ in k-space. The result is the *identical* PDE in frequency space:

$$i\frac{\partial \hat{\psi}}{\partial t} = k^2 \hat{\psi} - \frac{\partial^2 \hat{\psi}}{\partial k^2}$$

The Fourier transform merely swaps the PDE for a copy of itself without solving it. In physics, this reflects the fact that the QHO Hamiltonian is *invariant* under the Fourier transform—it treats

[5]Important technicality: $SO(3)$ is not abelian, hence it cannot be that *all* rotation operators simultaneously commute. The best solution is to pick a "maximal set" of commuting symmetries: usually the Total Angular Momentum (L^2, representing the amount of rotation invariance) and Azimuthal Symmetry (L_z, rotation around the z-axis). This choice of a maximal commuting family is the analytic shadow of selecting a *maximal torus* (Cartan subalgebra) in the Lie algebra $\mathfrak{so}(3)$.

position and momentum symmetrically—creating a fascinating link between abstract harmonic analysis and the concrete geometry of quantum mechanics, a connection to be explored in another book.

So what is the invariance we want to impose for the operator S_t with respect to the heat equation? In the example of the sphere, we had the group $SO(3)$. A compact interval has no inherent invariance simply because it is not a group. However, by gluing the end points of the interval we get a 1-*torus* (a circle)[6]. We now have the rotation group operation on $\mathbb{T}^1$, or equivalently the translation operation on $\mathbb{R}/\mathbb{Z}$[7]. This does shift the geometry and hence the inherent dynamics of our PDEs; the problem of solving a PDE on a compact interval will be a subset of solving PDEs on a torus as shall be addressed shortly. Let us first simply consider S_t on $L^2(\mathbb{T}^1)$ and see how it interacts with the translation operator. Since we are working on the torus, we now have that $\sin(2\pi kx)$ and $\cos(2\pi lx)$ *are* orthogonal (where the 2π-scaling was to ensure a full period of each lies within the torus), and $\{\sin(kx), \cos(lx)\}_{k,l\in\mathbb{N}}$ is an orthonormal basis (after appropriate scaling and stretching). Now, note that sin and cos bases are *not* translation invariant, for example:

$$\tau_a(\sin(kx)) = \sin(k(x+a)) = \sin(kx)\cos(ka) + \cos(kx)\sin(ka)$$

and so they are not eigenvectors of τ_a. However, they still *block-diagonalize* the translation operator τ_a with blocks of size 2:

$$\begin{pmatrix} \cos(ka) & \sin(ka) \\ -\sin(ka) & \cos(ka) \end{pmatrix}$$

If we look at these blocks, we shall see that they are all rotation matrices by an angle ka; in other words the imposition is that we did not have enough eigenvalues. By extending the codomain to $\mathbb{C}$, we can readily verify that:

$$\tau_a(e^{iz}) = e^{ia} \cdot e^{iz}$$

and so $\{e^{ikz}\}_{k\in\mathbb{Z}}$ is our desired eigenbasis for τ_a. Now, the question remains whether translations of these functions commute with S_t, namely:

$$S_t(\tau_a(f)) = \tau_a(S_t(f)) \qquad \forall a \in \mathbb{T}^1$$

and indeed:

$$\begin{aligned}
\tau_a(S_t(e^{ikx})) &= \tau_a(e^{-k^2t}e^{ikx}) \\
&= e^{-k^2t}e^{ik(x+a)} \\
&= e^{-k^2t}e^{ika}e^{ikx} \\
&= e^{ika}S_t(e^{ikx}) \\
&= S_t(e^{ika}e^{ikx}) && S_t \text{ is linear} \\
&= S_t(\tau_a(e^{ikx}))
\end{aligned}$$

What this means physically is that the heat equation dissipates the same way no matter where you are on a ring-shaped rod. If the rod was half wood and half metal, then this translation invariance shall fail.

[6]More generally, any rectangular domain is homeomorphic to $[0,1]^n$ which then can be glued to become $\mathbb{T}^n = (S^1)^n$, and so this same translation structure appears in higher dimensions

[7]Really $\mathbb{C}/\mathbb{Z}$ as we shall justify shortly

This principle is the spectral-theoretic shadow of *Noether's theorem*: continuous symmetries of a physical system correspond to conservation laws. Translation invariance corresponds to conservation of *momentum*, and the Fourier decomposition $f = \sum \hat{f}(k)e^{ikx}$ is precisely the decomposition into states of definite momentum k. Each Fourier mode evolves independently—its amplitude changes but its frequency is preserved—because momentum is conserved by any translation-invariant dynamics.

Now, notice that as $\tau_a \circ S_t = S_t \circ \tau_a$ for all $a \in \mathbb{T}^1$ (or in commutator bracket notation $[\tau_a, S_t] = 0$ for all a, t), the eigenvectors of τ_a are the same eigenvectors as S_t. The eigenvectors of S_t are in general hard to compute, but the eigenvectors of τ_a are *precisely* the functions e^{ikx}. In fact, any (linear) operator L where $\tau_a \circ L = L \circ \tau_a$ will have the same eigenvectors. Thus, we are actually finding eigenvectors of the translation operators τ_a and then finding transformations that commute with translation. Note that any Linear Differential Operator with Constant Coefficients

$$L = a_n \frac{d^n}{dx^n} + \cdots + a_1 \frac{d}{dx} + a_0$$

and any convolution operator:

$$L(f) = f * g = \int f(y)g(x-y)dy$$

will commute with translation. This is also why the Heat Equation, Wave Equation, and Schrödinger Equation can all be solved (or transformed) using the Fourier transform.

From Local to Global: The Exponential Map

We have established that the Fourier basis functions e^{ikx} are the eigenfunctions of the "global" spatial translation operator τ_a. However, the reader likely recognizes these functions primarily as the solutions to the differential equation $f' = \lambda f$. This connection between the *global* translation τ_a and the *local* derivative $D = \frac{d}{dx}$ is not a coincidence—it is a central theme of Lie Theory, and it provides the structural explanation for *why* exponentials are the natural Fourier basis.

In chapter 11, it shall be shown that it makes sense to infinitely differentiate integrable functions. For now, we rely on the intuition that any continuous function can be approximated by a smooth function (and locally by a polynomial).

Consider the Taylor expansion of a function f around a point x:

$$f(z) = f(x) + f'(x)(z-x) + \frac{f''(x)}{2!}(z-x)^2 + \dots$$

If we wish to evaluate f at a shifted point $x + y$, the Taylor series gives:

$$f(x+y) = f(x) + yf'(x) + \frac{y^2}{2!}f''(x) + \dots$$

Notice that we can re-write this using the differentiation operator $D = \frac{d}{dx}$:

$$f(x+y) = \left(I + yD + \frac{y^2 D^2}{2!} + \dots \right) f(x)$$

Recognizing the series inside the parenthesis as the definition of the matrix exponential, we arrive at the compact operator identity:

$$\tau_y = e^{yD}$$

This equation states that *differentiation generates translation.*

Now, consider the spectral consequences. If an operator A has an eigenvector v with eigenvalue λ ($Av = \lambda v$), then the operator exponential e^A acts on v as:

$$e^A v = \left(I + A + \frac{A^2}{2!} + \dots\right) v = \left(1 + \lambda + \frac{\lambda^2}{2!} + \dots\right) v = e^\lambda v$$

Applying this to our spatial operators: since $f(x) = e^{ikx}$ is an eigenfunction of the derivative D with eigenvalue ik (i.e., $De^{ikx} = ike^{ikx}$), it must automatically be an eigenfunction of the translation operator $\tau_y = e^{yD}$ with eigenvalue e^{iyk}.

This confirms our earlier finding: the Fourier basis e^{ikx} simultaneously diagonalizes the derivative (and by extension the Laplacian $\Delta = D^2$) and the translation group.

Theorem 10.2.1: Eigenvector Mapping Lemma

Any eigenvector with eigenvalue λ of an operator A is an eigenvector of the operator e^A with eigenvalue e^λ.

***Proof*:**
Follows from the power series definition of the matrix exponential shown above.

Remark. The full *Spectral Mapping Theorem* is the stronger statement $\sigma(e^A) = e^{\sigma(A)}$ for the entire spectrum, not just eigenvalues. The eigenvector case stated above is the special case sufficient for our purposes.

To see this explicitly, observe the action of these operators on the basis function $f_k(x) = e^{ikx}$:

- *Derivative:* $Df_k = (ik)f_k$ (Eigenvalue ik)
- *Laplacian:* $\Delta f_k = D^2 f_k = (ik)^2 f_k = -k^2 f_k$ (Eigenvalue $-k^2$)
- *Translation:* $\tau_a f_k = e^{aD} f_k = e^{a(ik)} f_k$ (Eigenvalue e^{ika})

In the language of representation theory, the differentiation operator is the *Lie Algebra* generator $\mathfrak{g}$, and the translation operator is the *Lie Group* element $e^{\mathfrak{g}}$. The Fourier basis provides the unique coordinate system where the group action and its generator are both simple scalings.

The Fourier Variable as a Cotangent Coordinate. The eigenvalue list above reveals a deeper geometric structure that connects directly to the duality theory presented in chapter 8. Observe the pattern: the derivative $D = \frac{d}{dx}$ is a *tangent* operation (it acts on the spatial/position variable x), yet its eigenvalue ik is a scalar that labels the *frequency* k. In physics, x is position and k is momentum—and these are canonically dual variables, related as tangent and cotangent coordinates.

The Fourier transform makes this duality explicit. Under the transform:

- The derivative $\frac{d}{dx}$ (a tangent operation) becomes *multiplication by ik* (a cotangent scalar).
- Multiplication by x (a tangent/position operation) becomes $-i\frac{d}{dk}$ (a cotangent derivative).

The Fourier transform *exchanges tangent and cotangent roles.* In the language of symplectic geometry, it is a canonical transformation swapping position and momentum coordinates on the phase space $T^*\mathbb{R}$. Where the duality map J_p from chapter 8 provides a *pointwise, infinitesimal* passage from tangent to cotangent data (via the gradient of the energy), the Fourier transform achieves this passage *globally* via integration against the characters e^{ikx}.

This perspective also explains the QHO's Fourier invariance noted above. The QHO Hamiltonian $H = -\partial_x^2 + x^2$ maps under the Fourier transform to $k^2 - \partial_k^2 = -\partial_k^2 + k^2$: the identical operator with x replaced by k. This is precisely because H treats position and momentum *symmetrically*—it sits on the diagonal of the tangent-cotangent duality, which is why the Fourier transform (which swaps the two) leaves it invariant.

Boundary Problem

Having understood why the Fourier basis is structurally natural, we now examine two important practical consequences: how boundary conditions on an interval arise from the symmetry of the torus, and how the Fourier transform decouples partial differential equations.

The immediate point that must be clarified when switching to $\mathbb{T}^1$ is that this shifts the dynamics of any PDE we are solving, as now points at the boundary are affected by their neighbourhood. However, for PDEs that will respect the group structure of $\mathbb{T}^1$, the effects at the boundary can be "cancelled out" in two separate ways that shall recover the original two bases $\{\sin(kx)\}_{k\in\mathbb{Z}}$ and $\{\cos(kx)\}_{k\in\mathbb{Z}}$ we saw before (again, appropriately normalized). The key insight is that the interval $[0, 1]$ can be viewed as a *sub-domain* of the torus $\mathbb{T}^1$ (identified with $[-1, 1]$) subject to a symmetry constraint. While the torus possesses full translation invariance, the interval possesses *reflection invariance*. By imposing a symmetry on the torus, we create "virtual boundaries" at the fixed points of the reflection ($x = 0$ and $x = 1$). This gives rise to two distinct orthonormal bases for $L^2([0, 1])$, each corresponding to a different physical boundary condition:

- *Odd Symmetry (Dirichlet Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *odd*, satisfying $u(-x) = -u(x)$. At the fixed point $x = 0$, this symmetry forces $u(0) = -u(0)$, implying $u(0) = 0$. Physically, the heat flowing from the right is perfectly cancelled by the "negative heat" flowing from the left, creating a virtual cold wall. The eigenfunctions respecting this symmetry are precisely the sine functions:

$$\sin(n\pi x) = \frac{e^{in\pi x} - e^{-in\pi x}}{2i}$$

  Restricting these functions to $[0, 1]$ yields the *Sine Basis*, which diagonalizes the Laplacian subject to $u(0) = u(1) = 0$.

- *Even Symmetry (Neumann Conditions):* Consider the subspace of functions on $\mathbb{T}^1$ that are *even*, satisfying $u(-x) = u(x)$. At the fixed point, the derivative must be odd, forcing $u'(0) = 0$. Physically, this represents an insulated boundary where heat reflects back without loss. The eigenfunctions respecting this symmetry are the cosine functions:

$$\cos(n\pi x) = \frac{e^{in\pi x} + e^{-in\pi x}}{2}$$

  Restricting these to $[0, 1]$ yields the *Cosine Basis*, which diagonalizes the Laplacian subject to $u'(0) = u'(1) = 0$.

This leads to a result in Harmonic Analysis known as *half-range expansions*: while sines and cosines *together* form a basis for the full torus, *either* set alone forms a complete orthonormal basis for the interval $[0, 1]$.

Decoupling PDE

To understand the consequences better, take a solution $u : \mathbb{T}^1 \times [0, \infty) \to \mathbb{C}$. Then, think of each (space) neighborhood in the usual heat equation as influencing its infinitesimal neighbourhood, namely the $\frac{\partial^2 u}{\partial x^2}$ term determines how each neighbouring point affects each other as time propagates. We can think of this as being uncountably many dependent equations.

Let us now change the "basis" of our state space from the position basis $\{\delta_x\}_{x\in\mathbb{T}^1}$ to the eigenbasis of the spatial operator $\{e^{ikx}\}_{k\in\mathbb{Z}}$. By taking the Fourier transform, i.e. transforming to the translation-invariant orthonormal basis, we change a solution to:

$$u(x, t) : \mathbb{T}^1 \times [0, \infty) \to \mathbb{C} \qquad \xmapsto{\mathcal{F}_x} \qquad \hat{u}(k, t) : \mathbb{Z} \times [0, \infty) \to \mathbb{C}$$

where the $\mathbb{Z}$ is the index for the eigenbasis $\{f_k\}_{k\in\mathbb{Z}}$. Then if $\hat{u}_k$ is the function whose input is the index of the eigenvector and output is its coefficient, we see that as $\hat{u}_k$ propagates through time, $\hat{u}_k(t)$ is only dependent on the original function up to scaling. In fact we shall show that the heat equation PDE decomposes into the equations:

$$\frac{d\hat{u}_k}{dt} = -k^2 \hat{u}_k \qquad \forall k \in \mathbb{Z}$$

where now we have *countably many decoupled* equations.

Non-Compact Domain

Notice that $\sin, \cos, e^z$ are not L^2-integrable on $\mathbb{R}^n$ (resp. $\mathbb{C}^n$), hence these cannot form an orthonormal basis on $L^2(\mathbb{R}^n)$. Thus, the initial foray into Fourier will focus on functions with domain $\mathbb{T}^n$. In section 10.2.2 we shall show how to generalize the tools to the non-compact case, more specifically, the locally compact case.

The structural reason for this shift is illuminated by *Pontryagin duality* (section 12.2). The characters of a group G themselves form a group $\hat{G}$ (the *Pontryagin dual*) under pointwise multiplication, and the Fourier transform is a map $L^2(G) \to L^2(\hat{G})$. For the compact group $G = \mathbb{T}^1$, the dual group is discrete: $\hat{G} = \mathbb{Z}$ (indexed by frequency k), and the Fourier transform produces a *discrete* sequence of coefficients $\hat{f}(k)$. Conversely, for the discrete group $G = \mathbb{Z}$, the dual is compact: $\hat{G} = \mathbb{T}^1$. And remarkably, $G = \mathbb{R}$ is *self-dual*: $\hat{\mathbb{R}} \cong \mathbb{R}$, so the Fourier transform maps functions of a continuous variable to functions of a continuous variable, replacing the discrete sum $\sum \hat{f}(k)e^{ikx}$ with the continuous integral $\int \hat{f}(\xi)e^{2\pi i\xi x}\, d\xi$. Pontryagin duality ($\hat{\hat{G}} \cong G$) is then the abstract statement that the Fourier transform is involutive.

Generalizing using Representation Theory

In the preceding discussion, our motivation was to solve a specific time-dependent PDE (the heat equation). We discovered that the key to solving it lay not in the time evolution S_t itself, but in the underlying symmetry of the spatial domain (translation invariance). The basis functions e^{ikx} worked because they diagonalized the translation operator τ_a, which we then understood via the exponential map $\tau_a = e^{aD}$ as the passage from the Lie algebra (differentiation) to the Lie group (translation).

We now abstract this principle. We no longer need a specific "time" equation; instead, we look for the basis that diagonalizes the action of the symmetry group itself. By finding the irreducible representations of the symmetry group, we construct a "universal" eigenbasis that will automatically diagonalize *any* linear operator respecting that symmetry (be it the Laplacian, the Heat Operator, or others).

To make this concrete, let us recall the representation theory of finite groups. If G is a finite group, we can represent G on a vector space V (over a field k) by taking a homomorphism $\rho : G \to \mathrm{GL}(V)$.

$$G \curvearrowright V \qquad \rho(g)v = g \cdot v$$

As we are not interested in any arithmetic limitations, we take $k = \mathbb{C}$. Then by Maschke's theorem (see [9, chapter 24]) the induced $\mathbb{C}[G]$-module structure on V is semi-simple, meaning:

$$V \cong_G \bigoplus_i V_i$$

where V_i are irreducible representations. Given the basis for the above isomorphism, any $g \in G$ will give a matrix representation $\rho(g)$ which respects this block-diagonal decomposition:

$$\rho(g) = \begin{bmatrix} \mathbf{M}_1(g) & 0 & 0 & 0 \\ 0 & \mathbf{M}_2(g) & 0 & 0 \\ 0 & 0 & \mathbf{M}_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

where $\mathbf{M}_i(g)$ represents the matrix block acting on the subspace V_i. Crucially, if G is *abelian*, Schur's Lemma implies that *all* irreducible representations are 1-dimensional. Hence, the matrices become scalars:

$$\rho(g) = \begin{bmatrix} \lambda_1(g) & 0 & 0 & 0 \\ 0 & \lambda_2(g) & 0 & 0 \\ 0 & 0 & \lambda_3(g) & 0 \\ 0 & 0 & 0 & \dots \end{bmatrix}$$

Since G is abelian, the operators $\{\rho(g)\}_{g \in G}$ commute, and the construction above shows they are *simultaneously diagonalizable* by a single basis.

Now, we generalize this to function spaces. While for a finite group the vector space of functions $\mathbb{C}[G]$ is finite-dimensional, equipping it with an inner product structure is essential to define unitarity and orthogonality. A finite group G carries a natural unique measure (the counting measure), which we can normalize to be a probability measure (the Haar measure $\mu(g) = 1/|G|$). This allows us to define the Hilbert space $L^2(G)$ with inner product:

$$\langle f, h \rangle = \frac{1}{|G|} \sum_{x \in G} f(x)\overline{h(x)}$$

We define the *Right Regular Representation*[8] of G on this space:

$$G \curvearrowright L^2(G) \qquad (R_g \cdot f)(x) = f(xg)$$

[8] or, Left Regular Representations $(L_g \cdot f)(x) = f(g^{-1}x)$

We verify that this action is unitary (and hence can be diagonalized with an orthonormal basis). Since right-multiplication by g is a bijection on the group G (it permutes the elements), the sum over x is identical to the sum over $y = xg$:

$$\begin{aligned}\langle R_g f, R_g h\rangle &= \frac{1}{|G|}\sum_{x\in G} f(xg)\overline{h(xg)} \\ &= \frac{1}{|G|}\sum_{y\in G} f(y)\overline{h(y)} \\ &= \langle f, h\rangle\end{aligned}$$

Thus, the operators R_g are unitary. Since G is abelian, the operators $\{R_g\}_{g\in G}$ form a commuting family of unitary operators. By the spectral theorem for unitary operators, there exists an orthonormal basis $\mathcal{B}$ of $L^2(G)$ that simultaneously diagonalizes the action. If $\varphi \in \mathcal{B}$ is such a basis vector, then:

$$R_g(\varphi) = \chi_\varphi(g)\varphi$$

where $\chi_\varphi(g) \in \mathbb{C}$ is the eigenvalue. Since the representation is a homomorphism ($R_{gh} = R_g R_h$), the eigenvalues must satisfy the multiplicative property $\chi_\varphi(gh) = \chi_\varphi(g)\chi_\varphi(h)$. Moreover, since R_g is unitary, $|\chi_\varphi(g)| = 1$ and hence the codomain is S^1. These homomorphisms $\chi : G \to S^1$ are precisely the *characters* of the group. Thus,

> the group-theoretic eigenbasis for $L^2(G)$ is simply the set of characters!

We shall explore this more in section 12.2.

10.2.2 Advanced: Operators Commuting with Translations We focused on finding the basis that diagonalizes translations (R_g). But what are the operators that commute with all translations?

$$T \circ R_g = R_g \circ T \qquad \forall g \in G$$

A fundamental result in Harmonic Analysis states that any such linear operator T is a *Convolution Operator*. That is, there exists a function $k \in L^1(G)$ (the kernel) such that $Tf = f * k$.
The deeper algebraic structure is this: the algebra of all translation-invariant operators on $L^2(G)$ is isomorphic to the *group algebra* $L^1(G)$ acting by convolution. The Fourier transform simultaneously diagonalizes this entire algebra—it is the *Gelfand transform* of the commutative Banach algebra $L^1(G)$, mapping the group algebra to its spectrum (the space of characters). This is the structural reason why the Fourier transform converts convolution to pointwise multiplication: it maps the group algebra to its maximal ideal space, where every element acts as a scalar.
This explains why the Heat Operator, the Laplacian, and other physical operators can all be written as convolutions: they are simply the most general linear maps that respect the symmetry of the space.

The Infinite Compact Case

In the case where the domain is infinite (like the torus $\mathbb{T}^n$), we must be more careful. First, let us stick to compact domains so that we generalize the finite case directly. Functions defined on a rectangular domain with periodic boundary conditions are equivalent to functions on the torus $\mathbb{T}^n = (S^1)^n$. Our state space is the Hilbert space $L^2(\mathbb{T}^n, \mathbb{C})$.

We want to act by translation by $\mathbb{R}^n$, but since the domain is periodic, this descends to a faithful action of the compact group $\mathbb{R}^n/\mathbb{Z}^n \cong \mathbb{T}^n$. It is easy to check that translation is unitary ($\langle \tau_y f, \tau_y g\rangle = \langle f, g\rangle$) using the translation invariance of the Lebesgue measure.

We now invoke the *Peter-Weyl Theorem*, which is the generalization of Maschke's theorem to compact topological groups (see [5]). It states that the unitary representation of a compact group on $L^2(G)$ decomposes into a Hilbert direct sum of finite-dimensional irreducible representations:

$$L^2(G) \cong_G \widehat{\bigoplus_i} V_i$$

Since $\mathbb{T}^n$ is abelian, all irreducible representations V_i are *one-dimensional*. Being one-dimensional and unitary, they are given by characters $\chi : \mathbb{T}^n \to S^1$.

Thus, the Peter-Weyl theorem guarantees that the characters of $\mathbb{T}^n$ form a complete orthonormal basis for $L^2(\mathbb{T}^n)$. We have recovered our Fourier basis purely from the algebraic structure of the domain, without reference to any specific differential equation!

10.2.1 Fourier Transform on n-Torus

Finally, we look at the details and formally define the Fourier transform. Let us first formalize the eigenvectors of the translation operator:

Theorem 10.2.3: Eigenvector of Translation Functions

Let φ be a measurable function on $\mathbb{R}^n$ (resp. $\mathbb{T}^n$) such that $\varphi(x+y) = \varphi(x)\varphi(y)$ and $|\varphi(x)| = 1$. Then there exists a $\xi \in \mathbb{R}^n$ (resp. $\xi \in \mathbb{T}^n$) such that $\varphi(x) = e^{2\pi i \xi \cdot x}$ where $\xi \cdot x$ is the dot product.

***Proof*:**
We prove it for $\mathbb{R}$, the result naturally generalizing for $\mathbb{R}^n$ and automatically working for $\mathbb{T}^n$. Let $a \in \mathbb{R}$ such that $\int_0^a \varphi(t)dt \neq 0$. Such an a exists, or else by the Lebesgue Differentiation Theorem $\varphi = 0$ a.e. Setting $A = \left(\int_0^a \varphi(t)dt\right)^{-1}$, we get:

$$\varphi(x) = \varphi(x)AA^{-1} = A\int_0^a \varphi(x)\varphi(t)dx = A\int_0^a \varphi(x+t)dt = A\int_x^{x+a} \varphi(t)dt$$

Thus, φ must be continuous, being the integral of a locally integrable function. But that makes the function inside the integral continuous, and so φ is in fact C^1 (in fact C^∞, but we will not need this fact). Now, notice that:

$$\varphi'(x) = A[\varphi(x+a) - \varphi(x)] = B\varphi(x)$$

where $B = A(\varphi(a)-1)$. This is a simple ODE, which can be solved to show we get an exponential (and is unique by the uniqueness of a solution to an ODE). If you don't know about ODEs, then notice that

$$\frac{d}{dx}(e^{-Bx}\varphi(x)) = 0$$

so $e^{-Bx}\varphi(x)$ is constant. Since $\varphi(0) = 1$, we have $\varphi(x) = e^{Bx}$, and since $|\varphi(x)| = 1$, B must be purely imaginary meaning $B = 2\pi i\xi$ for some $\xi \in \mathbb{R}$, completing the proof for $\mathbb{R}$. For $\mathbb{T}$, we would consider φ to be periodic with period 1 and $e^{2\pi i\xi} = 1$ if and only if $\xi \in \mathbb{Z}$

For the n-dimensional case, let $e_1, e_2, ..., e_n$ be the standard basis for $\mathbb{R}^n$ and define the coordiante functions $\varphi_i(t) = \varphi(te_i)$. Naturally, each φ_i satisifes the property $\varphi_i(t+s) = \varphi_i(t)\varphi_i(s)$ on $\mathbb{R}$, and so $\varphi_i(t) = e^{2\pi i\xi_i t}$, and so:

$$\varphi(x) = \varphi\left(\sum_1^n x_i e_i\right) = \prod_1^n \varphi_i(x_i) = e^{2\pi i\xi\cdot x}$$

completing the proof.

Hence, a function satisfying this translation invariance is of the form $f(x) = ce^{2\pi i\xi\cdot x}$ for some constant $c \in \mathbb{R}$. With the classification of these functions, we will want to use these to decompose reasonably arbitrary functions in $\mathbb{R}^n$ and $\mathbb{T}^n$ (i.e. we will show they form a dense set). In the case of L^2 with $\mathbb{T}^n$, we get a very precise result:

Theorem 10.2.4: Orthonormality Of Translation Functions

Let $E_k(x) = e^{2\pi ik\cdot x}$. Then the set $\{E_k \mid k \in \mathbb{Z}^n\}$ is an orthonormal basis for $L^2(\mathbb{T}^n)$

***Proof*:**
For orthonormality, notice that:

$$\int_0^1 e^{2\pi ikt}dt = \begin{cases} 1 & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

using this, notice that if $n \neq m$, then:

$$\begin{aligned} \langle E_a, E_b\rangle &= \int_0^1 e^{2\pi iat}\overline{e^{2\pi ibt}}dt \\ &= \vdots \\ &= \frac{1}{2\pi i(a-b)}(e^{2\pi i(a-b)} - e^0) \\ &= \frac{1}{2\pi i(a-b)}(1-1) \\ &= 0 \end{aligned}$$

and if $n = m$, then we get

$$\int_0^1 e^{2\pi ikt}e^{2\pi i-kt}dt = \int_0^1 dt = 1$$

For density, since $E_xE_y = E_{x+y}$, the set of finte linear combinations of the E_k's form an algebra which clearly seperates points on $\mathbb{T}^n$, and $E_0 = 1$, $\overline{E_k} = E_{-k}$. Since $\mathbb{T}^n$ compact, by the Stone-Weiestrass theorem, this algebra is dense in $C(\mathbb{T}^n)$ in the uniform norm and hence in the L^2 norm, and $C(\mathbb{T}^n)$ is dense in $L^2(\mathbb{T}^n)$, and so $\{E_k\}$ is a orthonormal basis for $L^2(\mathbb{T}^n)$, as we sought to show.

Using this, we can now define the Fourier transform:

Definition 10.2.5: Fourier Transform And Fourier Series on Torus

Let $f \in L^2(\mathbb{T}^n)$, $\langle -, - \rangle$ be the induced inner product, and E_k the orthonormal basis from theorem 10.2.4. Then the *Fourier Transform* $\hat{f} : \mathbb{Z}^n \to \mathbb{C}$ is the function defined as:

$$\hat{f}(k) = \langle f, E_k \rangle = \int_{\mathbb{T}^n} f(x) e^{-2\pi i k \cdot x} dx$$

The map $f \mapsto \hat{f}$ is called the *Fourier transform*. The representation

$$f = \sum_{k \in \mathbb{Z}^n} \hat{f}(k) E_k$$

is called the *Fourier series* of f.

Example 10.2.6: Discrete Spectrum Examples (Torus)

1. *The Pure Tone (Finite Spectrum):* Let $f(x) = \sin(2\pi m x)$ for some integer m. Using Euler's identity:
$$f(x) = \frac{e^{2\pi i m x} - e^{-2\pi i m x}}{2i}$$
The Fourier transform is a pair of spikes: $\hat{f}(m) = \frac{1}{2i}$, $\hat{f}(-m) = -\frac{1}{2i}$, and $\hat{f}(k) = 0$ everywhere else. A smooth, oscillating function on the torus has a *finite* number of non-zero coefficients.

2. *The Square Wave (Algebraic Decay):* Consider the periodic function equal to $+1$ on $[0, 1/2)$ and -1 on $[1/2, 1)$.
$$\hat{f}(k) = \frac{1 - (-1)^k}{\pi i k} \quad \text{for } k \neq 0$$
This gives non-zero values only for odd k, decaying as $1/k$. Discontinuities in the function lead to coefficients that decay slowly (algebraically). This slow decay causes the partial sums to oscillate violently near the jump (Gibbs Phenomenon).

3. *The Periodized Gaussian (Jacobi Theta Function):* Instead of a single Gaussian (which doesn't live on a torus), we sum infinitely many copies:
$$f(x) = \sum_{n \in \mathbb{Z}} e^{-\pi (x-n)^2}$$
Remarkably, the Fourier series coefficients are sampled from the Gaussian itself:
$$\hat{f}(k) = e^{-\pi k^2}$$
This connects local geometry to global geometry. Even though the function wraps around the circle, its coefficients still decay super-fast (exponentially), reflecting the smoothness of the function.

4. *The Dirac Comb (Flat Spectrum):* Let $f = \delta_0$ be the Dirac delta distribution at the origin (periodized).
$$\hat{f}(k) = \int_{\mathbb{T}} \delta_0(x) e^{-2\pi i k x} dx = 1$$
Extreme localization in space (a single point) leads to extreme delocalization in frequency (all frequencies contribute equally).

By the definition of an orthonormal basis, the Fourier series of f converges to f in the L^2-norm. In the next sections, we will address point-wise convergence. As this is simply a conversion of basis:

Theorem 10.2.7: Parseval's Identity On $\mathbb{T}^n$

$$\|\hat{f}\|_2 = \|f\|_2$$

***Proof*:**
Theorem 10.2.4 shows that $L^2(\mathbb{T}^n)$ surjects onto $\ell^2(\mathbb{Z}^n)$ and forms an orthonormal basis, hence by theorem 8.1.15
$$\|\hat{f}\|_2 = \|f\|_2$$

Let us take some time concretely interpreting $\hat{f}(k)$. First, $\hat{f}(0)$ represents the "average total energy" in the system:
$$\hat{f}(0) = \int_{\mathbb{T}^n} f(x) dx$$
where as $m(\mathbb{T}^n) = 1$ the normalization is hidden. If we visualized this, we can think of it as the center of mass of the graph of a compact function wrapped around the circle. Then $\hat{f}(n)$ for $n \in \mathbb{Z}^k$ is that same graph wrapped around in each cardinal direction n_i number of times, and then taking its average:
$$\hat{f}(n) = \int_{\mathbb{T}^n} f(x) e^{2\pi i x \cdot n} dx$$

Now, let's say that instead of trying to take the formal approach we've done, we motivate the coefficient formula geometrically. Consider the inner product as a projection that will project onto one of the axis given by one of the terms in the series:
$$f = \cdots + c_{-1} e^{-1 \cdot 2\pi i t} + c_0 e^{0 \cdot 2\pi i t} + c_1 e^{1 \cdot 2\pi i t} + c_2 e^{2 \cdot 2\pi i t} + \cdots$$

How do we find the coefficients? First, by the DCT:
$$\begin{aligned}\int f(t) dt &= \int \left(\cdots + c_{-1} e^{-1 \cdot 2\pi i t} + c_0 e^{0 \cdot 2\pi i t} + c_1 e^{1 \cdot 2\pi i t} + c_2 e^{2 \cdot 2\pi i t}\right) dt \\ &= \cdots + \int c_{-1} e^{-1 \cdot 2\pi i t} dt + \int c_0 e^{0 \cdot 2\pi i t} dt + \int c_1 e^{1 \cdot 2\pi i t} dt + \int c_2 e^{2 \cdot 2\pi i t} dt + \cdots\end{aligned}$$

Now, notice that all terms except the one containing c_0 are disappear, since the rotation of by the exponential will always guarantee to make the integrals average to zero! Thus, c_0 can be thought of as the average mass point of f. To find the other terms, we simply have to multiply f at the beginning by $e^{k 2\pi i t}$
$$\int f(t) e^{-k 2\pi i t} dt$$

and then we have shifted over all of our terms except the kth term, which we can now find the integral of! In fact, notice we have just recovered:

$$\hat{f}(k) = \int f(t)e^{-k2\pi it}dt$$

giving us a geometric intuition for this formula!

Now, this gives the Fourier transform on $L^2(\mathbb{T}^n, \mathbb{C})$, but we often want to work with different L^p spaces. First, the definition of the Fourier transform $\hat{f}(k)$ makes sense if $f \in L^1(\mathbb{T}^n)$ since:

$$|\hat{f}(\xi)| = \left|\int f(x)e^{2\pi ix\cdot\xi}dx\right| \leq \int \left|f(x)e^{2\pi ix\cdot\xi}\right| dx = \int |f(x)| = \|f\|_1$$

Hence for each $\xi \in \mathbb{Z}^n$, if $f \in L^1(\mathbb{T}^n)$ then $|\hat{f}(\xi)| \leq \|f\|_1$ showing us that $\hat{f}$ must be essentially bounded:

$$\|\hat{f}\|_\infty$$

so the Fourier series would then extend to a norm-decreasing map from $L^1(\mathbb{T}^n)$ to $\ell^\infty(\mathbb{Z}^n)$. In particular, this shows us that there is a sort of duality between the speed of decay of a function f and its Fourier transform $\hat{f}$. The nature of this duality is made explicit in the following theorem:

Theorem 10.2.8: Hausdorff-Young Inequality over $\mathbb{T}^n$

Let $1 \leq p \leq 2$ and q be the conjugate exponent of p so that $\frac{1}{p} + \frac{1}{q} = 1$. If $f \in L^p(\mathbb{T}^n)$, then

$$\hat{f} \in \ell^q(\mathbb{Z}^n) \qquad \text{and} \qquad \|\hat{f}\|_q \leq \|f\|_p$$

***Proof*:**

Since $\|\hat{f}\|_\infty \leq \|f\|_1$ and $\|\hat{f}\|_2 = \|f\|_2$ for $f \in L^1$ and $f \in L^2$, this follows from the Riesz-Thorin interpolation theorem (theorem 6.4.2)

Example 10.2.9: Advanced: The Theta Function and Modularity

1. *The Heat Kernel Trace:* Recall example 10.2.6.3, where we periodized the Gaussian to get $K_t(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}e^{2\pi inx}$. If we evaluate this at the identity ($x = 0$), we obtain the classical *Theta Function* $\theta(it)$:
$$\theta(it) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$$
In the language of Number Theory, this is a modular form (specifically of weight 1/2). The fact that the Fourier transform of a Gaussian is a Gaussian translates directly into the *functional equation* of the Theta function via the Poisson Summation Formula:
$$\theta(t^{-1}) = \sqrt{t}\theta(t)$$
This identity is the "analytic engine" that proves the analytic continuation of the Riemann Zeta function $\zeta(s)$.

2. *Gauss Sums (Arithmetic Fourier Transform):* If we replace our field $\mathbb{C}$ with a finite field $\mathbb{F}_p$ and our torus with the additive group of the field, the Fourier Transform becomes the *Gauss*

Sum:

$$g(a) = \sum_{x \in \mathbb{F}_p} \chi(x) e^{2\pi i a x/p}$$

Just as $\int |f|^2 = \int |\hat{f}|^2$ (Plancherel), we have $|g(a)| = \sqrt{p}$. The "square root cancellation" of Fourier coefficients in analytic number theory is the discrete analogue of the decay rates we study in analysis.

10.2.2 Fourier Transform on Euclidean Space

We often want to work with $L^2(\mathbb{R}^n)$. The problem is that $e^{2\pi i k}$ is *not* an element of $L^2(\mathbb{R}^n)$ as its integral is infinite. To generalize the domain to $\mathbb{R}^n$, we need to shift perspectives. Recall that all unitary representation of $\mathbb{T}^n$ are given by functions in:

$$\text{Hom}(\mathbb{T}^n, S^1) \cong \mathbb{Z}^n$$

This is the *Pontryagin dual* of $\mathbb{T}^n$ and represents all the characters. We can generalize this and choose a locally compact (topological) groups and consider the Pontryagin dual $\text{Hom}(G, S^1)$. As shall be shown in chapter 12, there shall always be a unique invariant Borel measure defined on any locally compact group up to scaling (ex. on $\mathbb{R}$ it is the Lebesgue measure and on $\mathbb{Z}$ it is the counting measure). It can be shown that $\text{Hom}(G, S^1)$ is also a locally compact group and hence has a unique measure. When G is compact, $\text{Hom}(G, S^1)$ is *discrete* and the natural Haar measure is the counting measure. If G is not compact but locally compact, $\text{Hom}(G, S^1)$ is *continuous*, most importantly:

$$\text{Hom}(\mathbb{R}^d, S^1) \cong \mathbb{R}^d \qquad \text{given by} \qquad (\xi \mapsto e^{2\pi i x \cdot \xi}) \mapsto x$$

Notice that if G is compact and abelian, $\text{Hom}(G, S^1)$ represents all the characters. Hence if G is *locally compact*, then $\hat{G} = \text{Hom}(G, S^1)$, can interpreted as the generalization of characters.

10.2.10 Advanced: The Notion of Points If you have seen Grothendieck's notion of the *Functor of Points*, then the "right" way to see points is not as elements of a set, but as *morphisms* from a test object. In algebraic geometry, a point of a scheme X is determined by the morphisms $T \to X$. In harmonic analysis, this philosophy manifests as *Tannaka-Krein Duality*. We can reconstruct a group G solely from its category of representations $\text{Rep}(G)$. A "geometric point" $g \in G$ corresponds precisely to a tensor automorphism of the fiber functor (the forgetful functor to vector spaces). Thus, the Fourier Transform is simply the mechanism that translates between the geometric perspective (points as elements) and the spectral perspective (points as representations), proving they are dual aspects of the same underlying structure.

We can thus generalize the orthonormal basis interpretation of the Fourier transform to

$$\mathcal{F} : L^2(G, \mathbb{C}) \to L^2(\hat{G}, \mathbb{C})$$

Of course we have not proven it is well-defined, namely since $\hat{G}$ need not be an orthonormal basis anymore. For example if $f \in L^2(\mathbb{R}^d)$, it is not necessarily the case that $f \in L^1(\mathbb{R}^d)$ however as we've defined it so far we integrate against $e^{2\pi i x \cdot \xi}$:

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{2\pi i x \cdot \xi} dx$$

Simply take $(1+|x|)^{-1} \in L^2(\mathbb{R})$ which not in $L^1(\mathbb{R})$. However, $\mathcal{F}$ shall still be a well-defined map between L^2 spaces as shall be shown in upcoming sections. Note that in the case of $G = \mathbb{T}^n$ this is exactly what we had before:

$$\mathcal{F} : L^2(\mathbb{T}^n, \mathbb{C}) \to \ell^2(\mathbb{Z}^n, \mathbb{C})$$

There is a lot of details that are brushed off for now; this is all covered in chapter 12.

Now, if $f \in L^1$, $\hat{f}(\xi)$ will certainly be bounded (by $\|f\|_1$), and hence it is essentially bounded:

$$\|\mathcal{F}(f)\|_\infty < \infty$$

We thus have our first definition of Fourier transform on $\mathbb{R}^d$:

Definition 10.2.11: Fourier Transform on L^1

Let $f \in L^1(\mathbb{R}^n)$. Then the *Fourier Transform* $\hat{f} : \mathbb{R}^n \to \mathbb{C}$ is the function defined as:

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x} dx$$

The map $f \mapsto \hat{f}$ is called the *Fourier transform.*

As it turns out, $\mathcal{F}(f)$ is also continuous, hence it is not only essentially bounded (in L^∞) but *actually* bounded (in BC):

Theorem 10.2.12: Riemann-Lebesgue Lemma

$$\mathcal{F}(L^1(\mathbb{R}^n)) \subseteq \mathrm{BC}(\mathbb{R}^n)$$

***Proof*:**

Let $f \in L^1(\mathbb{R}^n)$ and consider $\{\xi_k\}_{k=1}^\infty \subset \mathbb{R}^n$ such that $\xi_k \to \xi$. It suffices to show $\lim_{k\to\infty} \hat{f}(\xi_k) = \hat{f}(\xi)$.

By the definition of the Fourier transform,

$$\hat{f}(\xi_k) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi_k \cdot x}\, dx.$$

Let $F_k(x) = f(x)e^{-2\pi i \xi_k \cdot x}$. Since the complex exponential is continuous, we have pointwise convergence for every $x \in \mathbb{R}^n$:

$$\lim_{k\to\infty} F_k(x) = f(x) \lim_{k\to\infty} e^{-2\pi i \xi_k \cdot x} = f(x) e^{-2\pi i \xi \cdot x}.$$

Next, for all k and x, the integrand is bounded by the magnitude of f:

$$|F_k(x)| = \left| f(x) e^{-2\pi i \xi_k \cdot x} \right| = |f(x)| \cdot 1 = |f(x)|.$$

Since $f \in L^1(\mathbb{R}^n)$, $|f|$ dominates it. Thus, by the DCT:

$$\lim_{k\to\infty} \int_{\mathbb{R}^n} F_k(x)\, dx = \int_{\mathbb{R}^n} \lim_{k\to\infty} F_k(x)\, dx = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x}\, dx = \hat{f}(\xi).$$

showing $\hat{f}$ is continuous, as we sought to show.

Proposition 10.2.13: Properties Of Fourier Transform

Let $f, g \in L^1(\mathbb{R}^n)$. Then:

1. $\widehat{(\tau_y f)}(\xi) = e^{-2\pi i\xi\cdot y}\hat{f}(\xi)$ and $\tau_\eta(\hat{f}) = \widehat{h}$ where $h(x) = e^{2\pi i\eta\cdot x}f(x)$
2. If T is an invertible linear transformation on $\mathbb{R}^n$ and $S = (T^*)^{-1}$ is its inverse transpose, Then $\widehat{(f \circ T)} = |\det(T)|^{-1}\hat{f} \circ S$. In particular, if T is a rotation, then $\widehat{(f \circ T)} = \hat{f} \circ T$. Furthermore, if $T(x) = t^{-1}x$ (for $t > 0$), then $\widehat{(f \circ T)}(\xi) = t^n\hat{f}(t\xi)$.

   Note: This implies the Heisenberg Uncertainty Principle qualitatively: dilating (widening) f results in contracting (narrowing) $\hat{f}$.
3. Convolution becomes multiplication after the Fourier Transform:
$$\widehat{(f * g)} = \hat{f}\hat{g}$$
4. (*Conjugate Symmetry*) if f is real-valued, then $\hat{f}$ (which is still complex-valued) satisfies the Hermitian symmetry condition:
$$\hat{f}(-\xi) = \overline{\hat{f}(\xi)}$$
   This implies the information in the codomain is 'half' redundant.
5. If $x^\alpha f \in L^1$ for $|\alpha| \leq k$ then $\hat{f} \in C^k$ and
$$\partial^\alpha\hat{f}(\xi) = [\widehat{(-2\pi ix)^\alpha f}](\xi)$$
6. if $f \in C^k$, $\partial^\alpha f \in L^1$ for $|\alpha| \leq k$, and $\partial^\alpha f \in C_0$, for $|\alpha| \leq k-1$, then
$$\widehat{(\partial^\alpha f)}(\xi) = (2\pi i\xi)^\alpha\hat{f}(\xi)$$
7. upgraded Riemann-Lebesgue lemma:
$$\mathcal{F}(L^1(\mathbb{R}^n)) \subseteq C_0(\mathbb{R}^n)$$

***Proof*:**

1. Computing directly:
$$\widehat{(\tau_y f)}(\xi) = \int f(x-y)e^{-2\pi i\xi\cdot x}dx$$
   Let $z = x - y$, then $dx = dz$ and $x = z + y$:
$$= \int f(z)e^{-2\pi i\xi\cdot(z+y)}dz = e^{-2\pi i\xi\cdot y}\int f(z)e^{-2\pi i\xi\cdot z}dz = e^{-2\pi i\xi\cdot y}\hat{f}(\xi)$$
   For the second part, let $h(x) = e^{2\pi i\eta\cdot x}f(x)$.
$$\hat{h}(\xi) = \int e^{2\pi i\eta\cdot x}f(x)e^{-2\pi i\xi\cdot x}dx = \int f(x)e^{-2\pi i(\xi-\eta)\cdot x}dx = \hat{f}(\xi - \eta) = \tau_\eta(\hat{f})(\xi)$$

2. Apply theorem 2.6.4 (Change of Variables). Let $y = Tx$, then $x = T^{-1}y$ and $dx = |\det T|^{-1}dy$:

$$\widehat{(f \circ T)}(\xi) = \int f(Tx)\exp(-2\pi i\xi \cdot x)dx$$
$$= |\det T|^{-1}\int f(y)\exp(-2\pi i\xi \cdot T^{-1}y)dy$$

   Using the definition of the adjoint (transpose), $\xi \cdot T^{-1}y = (T^{-1})^*\xi \cdot y = S\xi \cdot y$:

$$= |\det T|^{-1}\int f(y)\exp(-2\pi i S\xi \cdot y)dy$$
$$= |\det T|^{-1}\hat{f}(S\xi)$$

3. By Fubini's Theorem (justified since $f, g \in L^1$):

$$\widehat{(f * g)}(\xi) = \int \left(\int f(x-y)g(y)dy\right)e^{-2\pi i\xi \cdot x}dx$$
$$= \int g(y)\left(\int f(x-y)e^{-2\pi i\xi \cdot x}dx\right)dy$$

   Applying the translation property (proven in 1) to the inner integral where f is translated by y:

$$= \int g(y)\left(e^{-2\pi i\xi \cdot y}\hat{f}(\xi)\right)dy$$
$$= \hat{f}(\xi)\int g(y)e^{-2\pi i\xi \cdot y}dy$$
$$= \hat{f}(\xi)\hat{g}(\xi)$$

4. exercise.

5. By Differentiating under the integral sign (justified by the Dominated Convergence Theorem and the assumption $x^\alpha f \in L^1$):

$$\partial^\alpha \hat{f}(\xi) = \partial_\xi^\alpha \int f(x)e^{-2\pi i\xi \cdot x}dx = \int f(x)\partial_\xi^\alpha(e^{-2\pi i\xi \cdot x})dx = \int f(x)(-2\pi i x)^\alpha e^{-2\pi i\xi \cdot x}dx$$

6. Assuming first that $n = 1$ and $k = 1$. Since $f \in C_0$, f vanishes at boundaries. We integrate by parts:

$$\widehat{f'}(\xi) = \int_{-\infty}^{\infty} f'(x)e^{-2\pi i\xi x}dx$$

   Let $u = e^{-2\pi i\xi x}$ and $dv = f'(x)dx$. Then $du = -2\pi i\xi e^{-2\pi i\xi x}dx$ and $v = f(x)$.

$$= \left[f(x)e^{-2\pi i\xi x}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x)(-2\pi i\xi)e^{-2\pi i\xi x}dx$$

The boundary term is 0 because $f \in C_0$.

$$= 2\pi i\xi \int_{-\infty}^{\infty} f(x)e^{-2\pi i\xi x}dx = 2\pi i\xi \hat{f}(\xi)$$

For $n > 1$, we compute $\widehat{(\partial_j f)}$ by integrating by parts with respect to the j-th variable x_j. The boundary terms vanish because $f \in C_0(\mathbb{R}^n)$. The general case follows by induction on $|\alpha|$.

7. By the previous part, if $f \in C_c^1$ (Compactly supported C^1 functions), then $\widehat{\partial_j f}(\xi) = 2\pi i\xi_j \hat{f}(\xi)$. Since $\widehat{\partial_j f}$ is bounded (as the Fourier transform of an L^1 function), $|\xi_j \hat{f}(\xi)|$ is bounded. This implies $\hat{f}(\xi)$ decays as $|\xi| \to \infty$, so $\hat{f} \in C_0$.

   By a similar argument to proposition 10.1.12, C_c^1 is dense in L^1. Let $f \in L^1$ and $f_n \in C_c^1$ such that $f_n \to f$ in L^1. The Fourier transform is a bounded linear map from L^1 to L^∞ (specifically $\|\hat{g}\|_\infty \le \|g\|_1$), so:

$$\|\hat{f}_n - \hat{f}\|_\infty \le \|f_n - f\|_1 \to 0$$

   Thus $\hat{f}_n \to \hat{f}$ uniformly. Since C_0 is closed under the uniform norm (a uniform limit of functions vanishing at infinity also vanishes at infinity), we conclude $\hat{f} \in C_0$.

Notice that (5) and (6) give an interesting relation between the smoothness of f and the decay of $\hat{f}$: the smoother f is, the faster $\hat{f}$ decays (and vice-versa).

Proposition 10.2.14: Regularity and Decay Duality

The smoothness of a function determines the decay of its Fourier transform, and the decay of a function determines the smoothness of its Fourier transform. In particular:

1. *Decay $\Rightarrow$ Smoothness:* If $(1 + |x|)^k f(x) \in L^1(\mathbb{R}^n)$ for some integer $k \ge 1$, then $\hat{f} \in C^k(\mathbb{R}^n)$. Specifically, differentiation converts to multiplication by a polynomial in the spatial domain:

$$\partial^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^n} (-2\pi i x)^\alpha f(x)e^{-2\pi i\xi\cdot x}\, dx$$

   for all multi-indices $|\alpha| \le k$.

2. *Smoothness $\Rightarrow$ Decay:* If $f \in C^k(\mathbb{R}^n)$ and $\partial^\alpha f \in L^1(\mathbb{R}^n)$ for all $|\alpha| \le k$ (and f vanishes at infinity), then $\hat{f}$ decays rapidly:

$$|\hat{f}(\xi)| \le \frac{C}{(1 + |\xi|)^k}$$

   for some constant C.

***Proof*:**

1. **(Decay $\Rightarrow$ Smoothness):** Formally differentiating:

$$\frac{\partial}{\partial \xi_j}\hat{f}(\xi) = \frac{\partial}{\partial \xi_j}\int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}\,dx = \int_{\mathbb{R}^n} f(x)\frac{\partial}{\partial \xi_j}\left(e^{-2\pi i\xi\cdot x}\right)\,dx.$$

Calculating the derivative of the exponential gives $(-2\pi i x_j)e^{-2\pi i\xi\cdot x}$. Thus the integrand becomes $(-2\pi i x_j)f(x)e^{-2\pi i\xi\cdot x}$. To justify this operation rigorously, we need a dominating function. Observe that:

$$\left|(-2\pi i x_j)f(x)e^{-2\pi i\xi\cdot x}\right| = 2\pi|x_j||f(x)| \le 2\pi|x||f(x)|.$$

By our hypothesis, $(1+|x|)^k f \in L^1$, which implies $|x|f \in L^1$. Thus, the integrand is dominated by an integrable function, and the Dominated Convergence Theorem allows differentiation. Repeating this for higher derivatives up to order k yields the result.

2. **(Smoothness $\Rightarrow$ Decay):** We use the property that $\widehat{\partial^\alpha f}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$. Since $\partial^\alpha f \in L^1$, by the standard Riemann-Lebesgue Lemma (or simply the boundedness of the Fourier transform), we know that its transform is bounded:

$$\|\widehat{\partial^\alpha f}\|_\infty \le \|\partial^\alpha f\|_1 < \infty.$$

Substituting the identity, we have:

$$|(2\pi i\xi)^\alpha \hat{f}(\xi)| \le \|\partial^\alpha f\|_1.$$

This implies that for any multi-index α with $|\alpha| \le k$, the product $|\xi^\alpha \hat{f}(\xi)|$ is bounded. Since a function dominated by $1/|\xi|^k$ is equivalent to being bounded when multiplied by $|\xi|^k$, we conclude:

$$|\hat{f}(\xi)| \le C(1+|\xi|)^{-k}.$$

Corollary 10.2.15: Paley-Wiener theorem

If $f \in C_c^\infty(\mathbb{R}^n)$ has compact support (i.e., there exists $R > 0$ such that $f(x) = 0$ for $|x| > R$), then its Fourier transform $\hat{f}(\xi)$ extends to a holomorphic (complex analytic) function on $\mathbb{C}^n$.

***Proof*:**

Let $z \in \mathbb{C}^n$. Define the complex extension of the Fourier transform as:

$$F(z) = \int_{|x|\le R} f(x)e^{-2\pi i x\cdot z}\,dx$$

Since f has compact support, the integration range is effectively bounded by the ball $B_R(0)$. To show $F(z)$ is holomorphic, it suffices to show it is complex differentiable with respect to each component z_j. We compute the partial derivative by differentiating under the integral sign:

$$\frac{\partial}{\partial z_j}F(z) = \int_{|x|\le R} f(x)\frac{\partial}{\partial z_j}\left(e^{-2\pi i x\cdot z}\right)\,dx = \int_{|x|\le R} f(x)(-2\pi i x_j)e^{-2\pi i x\cdot z}\,dx$$

To justify integral converges, let $z = \xi + i\eta$. The magnitude of the exponential term is:

$$\left|e^{-2\pi i x\cdot(\xi+i\eta)}\right| = \left|e^{-2\pi i x\cdot\xi}\right|\left|e^{2\pi x\cdot\eta}\right| = e^{2\pi x\cdot\eta}$$

Since the domain of integration is bounded ($|x| \le R$), this term is bounded by $e^{2\pi R|\eta|}$. Furthermore, f and x_j are bounded on this domain. Thus, the integrand is bounded for any fixed z, and the integral is well-defined.

Since the derivative exists at every point $z \in \mathbb{C}^n$, the function $F(z)$ is entire (holomorphic everywhere). Consequently, the Fourier transform $\hat{f}(\xi)$ is simply the restriction of this entire function to the real subspace $\mathbb{R}^n$, implying that $\hat{f}$ is real-analytic.

Notice that $C_c^\infty(\mathbb{R}^n)$ does not map into compact spaces, since an analytic function cannot be non-zero and bounded. The closest equivalent is Schwartz space:

Corollary 10.2.16: Fourier Transform On Schwartz Space

The map $\mathcal{F}$ maps the Schwartz space $\mathcal{S}$ into itself

***Proof*:**
If $f \in \mathcal{S}$, then $x^\alpha \partial^\beta f \in L^1 \cap C_0$ for all α, β, so by proposition 10.2.13, $\hat{f}$ is C^∞ and

$$\widehat{(x^\alpha \partial^\beta f)} = (-1)^{|\alpha|}(2\pi i)^{|\beta|-|\alpha|}\partial^\alpha(\xi^\beta \hat{f}).$$

Thus $\partial^\alpha(\xi^\beta \hat{f})$ is bounded for all α, β, whence $\hat{f} \in \mathcal{S}$. Moreover, since $\int(1+|x|)^{-n-1}dx < \infty$,

$$\|\widehat{(x^\alpha \partial^\beta f)}\|_u \le \|x^\alpha \partial^\beta f\|_1 \le C\|(1+|x|)^{n+1}x^\alpha \partial^\beta f\|_u.$$

It then follows that $\|\hat{f}\|_{(N,\beta)} \le C_{N,\beta}\sum_{|\gamma|\le|\beta|}\|f\|_{(N+n+1,\gamma)}$ by proposition 10.1.4, so the Fourier transform is continuous on $\mathcal{S}$.

10.2.17 Distribution Theory and Compactly Supported In distribution theory (chapter 11), it shall be shown that $C(\mathbb{R}^n)$ can be recovered by integrating against smooth functions with compact support. Thus, C_c^∞ would be ideal space for the Fourier transform *if* they mapped back to themselves. However, by the boundedness argument above it is not the case. *However*, in over totally disconnected topological groups, such as those present in the representation theory of absolute galois groups, the equivalent of Schwartz space *is* C_c^∞; see [1] for more details.

Example 10.2.18: Continuous Spectrum Examples (Euclidean)

1. *The Box Function (The Sinc Function):* Let $f(x) = \mathbb{1}_{[-1/2,1/2]}(x)$ (The indicator function of width 1).
$$\hat{f}(\xi) = \int_{-1/2}^{1/2} 1\cdot e^{-2\pi i\xi x}dx = \frac{e^{-\pi i\xi} - e^{\pi i\xi}}{-2\pi i\xi} = \frac{\sin(\pi\xi)}{\pi\xi} \equiv \operatorname{sinc}(\xi)$$
Unlike the Square Wave on the torus (which had discrete coefficients $1/k$), this has a *continuous* spectrum that oscillates and decays as $1/\xi$. The compact support in space ($f = 0$

outside the box) forces the transform to extend infinitely (it cannot be compactly supported).

2. *The Two-Sided Exponential (Lorentzian):* Let $f(x) = e^{-2\pi|x|}$. This is a continuous L^1 function with a "cusp" at $x = 0$.
$$\hat{f}(\xi) = \frac{1}{\pi}\frac{1}{1+\xi^2}$$
This is the Cauchy (or Lorentzian) distribution. Notice the duality: Exponential decay in space leads to Polynomial ($1/\xi^2$) decay in frequency. The lack of smoothness at $x = 0$ (the cusp) prevents the frequency decay from being exponential.

3. *The Gaussian (Self-Dual):* Let $f(x) = e^{-\pi x^2}$. As proven in theorem 10.3.1:
$$\hat{f}(\xi) = e^{-\pi\xi^2}$$
The Gaussian is the unique eigenfunction of the Fourier Transform. It is the perfect balance point of the Uncertainty Principleit is equally localized in both space and frequency.

4. *The Shifted Function (Modulation):* Let $f(x)$ be a standard Gaussian centered at $x = 100$ rather than 0: $g(x) = f(x - 100)$.
$$\hat{g}(\xi) = e^{-2\pi i(100)\xi}\hat{f}(\xi) = e^{-2\pi i 100\xi}e^{-\pi\xi^2}$$
In Euclidean space, shifting a function far away does not change the "energy envelope" of its spectrum (it's still a Gaussian), but it causes the complex phase to wind rapidly.

Example 10.2.19: Advanced uses of Fourier Transform

1. *The Poincaré Bundle:* In our definition of the Fourier Transform $\hat{f}(\xi) = \int f(x)e^{-2\pi i\xi\cdot x}dx$, the kernel $K(x, \xi) = e^{-2\pi i\xi\cdot x}$ plays the central role. In Algebraic Geometry, if A is an abelian variety (say a complex torus V/Λ), its dual $\hat{A}$ parametrizes line bundles on A. The universal line bundle $\mathcal{P}$ on $A \times \hat{A}$ (the Poincaré bundle) is the geometric geometrization of the kernel $e^{-2\pi i\xi\cdot x}$. The Fourier transform generalizes to the *Fourier-Mukai Transform*, an equivalence of derived categories of coherent sheaves:
$$\Phi_{\mathcal{P}}(\mathcal{F}^\bullet) = \mathbf{R}\pi_{2,*}(\pi_1^*\mathcal{F}^\bullet \otimes \mathcal{P})$$
Here, the integration $\int dx$ is replaced by the pushforward $\mathbf{R}\pi_{2,*}$, and the product is the tensor product.

2. *The Spectrum of the Primes (Riemann's Explicit Formula):* You may have heard that the zeros of the Riemann Zeta function constitute the "spectrum" of the prime numbers. This is literal, not metaphorical. Consider the Chebyshev function $\psi(x) = \sum_{p^k\leq x}\log p$, which counts primes with a logarithmic weight. Riemann's Explicit Formula links this step function to the zeros $\rho = \frac{1}{2} + i\gamma$ of $\zeta(s)$:
$$\psi(x) = x - \sum_{\rho}\frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2})$$

If we make the logarithmic change of variables $x = e^u$, the sum over zeros becomes:

$$\sum_\rho \frac{e^{u(\frac{1}{2}+i\gamma)}}{\frac{1}{2}+i\gamma} = e^{u/2} \sum_\gamma \left(\frac{1}{\frac{1}{2}+i\gamma} \right) e^{i\gamma u}$$

This is a Fourier series (or integral) in the variable u!

- The "time" Domain signal is the distribution of primes (spikes at $u = \log p$).
- The "Frequency" Domain is the set of imaginary parts of zeta zeros (γ).

Just as a musical chord is "recovered" by summing its constituent frequencies, the location of prime numbers is recovered by summing the waves defined by the Riemann Zeros. This duality is an instance of Fourier Duality on the ideles.

3. *Pontryagin Duality as a Moduli Problem:* We defined $\hat{G} = \mathrm{Hom}(G, S^1)$. For a number theorist, we can view the frequency domain $\hat{G}$ as a *moduli space* of rank-1 unitary local systems on G.

   - If $G = \mathbb{R}$, the moduli space is $\mathbb{R}$ (Euclidean Fourier Transform).
   - If $G = S^1$, the moduli space is $\mathbb{Z}$ (Fourier Series).
   - If $G = \mathbb{Z}$, the moduli space is S^1 (Discrete Time Fourier Transform).

   The "Duality" implies that the moduli space of the moduli space recovers the original space.

4. *The Circle Method (Diophantine Equations):* How many integer solutions are there to $n_1^k + \cdots + n_s^k = N$? (Waring's Problem). This is a combinatorial problem that Fourier Analysis turns into an analytic one. Construct the generating function (a trigonometric polynomial):

   $$f(\alpha) = \sum_{m=1}^{N^{1/k}} e^{2\pi i m^k \alpha}$$

   The number of solutions $R(N)$ is exactly the N-th Fourier coefficient of the function $f(\alpha)^s$:

   $$R(N) = \int_0^1 (f(\alpha))^s e^{-2\pi i N\alpha} d\alpha$$

   The Key insight of Hardy and Littlewood was to split this integral (the torus) into "Major Arcs" (near rational numbers with small denominators, where the sum behaves predictably) and "Minor Arcs" (where the sum oscillates and cancels out, similar to the Riemann-Lebesgue lemma).

5. *Spectral Geometry (Can You Hear the Shape of a Drum?):* The set of eigenvalues $\{\lambda_n\}$ of the Laplacian Δ on a compact Riemannian manifold (M, g) is called the *spectrum*. This is the set of "pure frequencies" the manifold can support. The Fourier Transform allows us to recover geometric data from this spectral data via the trace of the Heat Kernel:

   $$\mathrm{Tr}(e^{t\Delta}) = \sum_{n=0}^{\infty} e^{-\lambda_n t} \sim \frac{1}{(4\pi t)^{d/2}} \left(\mathrm{Vol}(M) + \frac{t}{6} \int_M \mathrm{Scal}_g + O(t^2) \right)$$

   As $t \to 0$ (high frequencies), the asymptotics of the heat kernel allow us to "hear" the dimension, the volume, and the total scalar curvature of the manifold.

Note: Milnor gave examples of isospectral tori (16-dimensional) that are not isometric, proving that you cannot hear the *entire* shape, but Fourier analysis recovers the local invariants.

10.3 Fourier Integral and Series

Let us now show that up to measure zero the Fourier transform is injective. Equivalently, let us now look into finding the inverse Fourier transform. Let $f \in L^1$. Then define:

$$f^\vee(x) = \hat{f}(-x) = \int f(\xi)e^{2\pi i\xi\cdot x}d\xi$$

We'll show that if $f, \hat{f} \in L^1$, then $(\hat{f})^\vee = f$. This cannot be done by appealing to Fubini's theorem as the integral:

$$(\hat{f})^\vee(x) = \int\int f(y)e^{2\pi i\cdot x}e^{2\pi i\cdot y}dyd\xi$$

is not in $L^1(\mathbb{R}^n \times \mathbb{R}^n)$ (it will a.e. diverge). Instead, we do the usual trick of sequences that converge to the desired values. The solution is the usual solution of taking a limit of functions that satisfy the property and insuring that the desired properties are preserved in the limit. First, a technical lemma that also has an amazing tangential property that simply must be mentioned:

Theorem 10.3.1: Fourier Transform Fixed Point

Let $f(x) = e^{-\pi a|x|^2}$ where $a > 0$. Then

$$\hat{f}(\xi) = a^{-n/2}e^{-\pi|\xi|^2/a}$$

In particular if $a = 1$ so that $f(x) = e^{-\pi|x|^2}$, then

$$\hat{f} = f$$

***Proof*:**

First consider the case $n = 1$. Since the derivative of $e^{-\pi ax^2}$ is $-2\pi ae^{-\pi ax^2}$, by proposition 10.2.13(e) we have

$$(\hat{f})'(\xi) = (-2\pi ixe^{-\pi ax^2})^\wedge(\xi) = \frac{i}{a}(f')^\wedge(\xi) = \frac{i}{a}(2\pi i\xi)\hat{f}(\xi) = -\frac{2\pi}{a}\xi\hat{f}(\xi).$$

Then:

$$\begin{aligned}
\frac{d}{d\xi}\left(e^{\pi\xi^2/a}\cdot\hat{f}(\xi)\right) &= \left[\frac{d}{d\xi}\left(e^{\pi\xi^2/a}\right)\right]\cdot\hat{f}(\xi) + e^{\pi\xi^2/a}\cdot\hat{f}'(\xi) \\
&= e^{\frac{\pi}{a}\xi^2}\cdot\left(\frac{2\pi}{a}\xi\right)\cdot\hat{f}(\xi) + e^{\pi\xi^2/a}\cdot\hat{f}'(\xi) \\
&= e^{\pi\xi^2/a}\left[\frac{2\pi}{a}\xi\hat{f}(\xi) + \hat{f}'(\xi)\right] \\
&= e^{\pi\xi^2/a}\left[\frac{2\pi}{a}\xi\hat{f}(\xi) + \left(-\frac{2\pi}{a}\xi\hat{f}(\xi)\right)\right] \\
&= e^{\pi\xi^2/a}[0] = 0
\end{aligned}$$

It follows that $(d/d\xi)(e^{\pi\xi^2/a}\hat{f}(\xi)) = 0$, so that $e^{\pi\xi^2/a}\hat{f}(\xi)$ is constant. To evaluate the constant, set $\xi = 0$. Then computing the integral:

$$\hat{f}(0) = \int e^{-\pi a x^2}dx = a^{-1/2}.$$

The n-dimensional case follows by Fubini's theorem, since $|x|^2 = \sum_1^n x_j^2$:

$$\begin{aligned}
\hat{f}(\xi) &= \prod_1^n \int \exp(-\pi a x_j^2 - 2\pi i \xi_j x_j)dx_j \\
&= \prod_1^n [a^{-1/2}\exp(-\pi\xi_j^2/a)] = a^{-n/2}\exp(-\pi|\xi|^2/a).
\end{aligned}$$

completing the proof.

Theorem 10.3.2: The Fourier Inversion Theorem

Let $f \in L^1$ and $\hat{f} \in L^1$. Then f agrees a.e. with a continuous function f_0, and

$$(\hat{f})^\vee = (f^\vee)^\wedge = f_0$$

***Proof*:**
Given $t > 0$ and fixed $x \in \mathbb{R}^n$, set

$$\varphi(\xi) = \exp(2\pi i\xi\cdot x - \pi t^2|\xi|^2).$$

We use the standard Gaussian $g(x) = e^{-\pi|x|^2}$ and the scaling notation $g_t(y) = t^{-n}g(y/t)$. Using the translation and dilation properties of the Fourier transform on the Gaussian, we have:

$$\hat{\varphi}(y) = t^{-n}\exp(-\pi|x-y|^2/t^2) = g_t(x-y).$$

Since $f, \varphi \in L^1$, Fubini's theorem justifies the multiplication formula $\int \hat{f}\varphi = \int f\hat{\varphi}$:

$$\int e^{-\pi t^2|\xi|^2}e^{2\pi i\xi\cdot x}\hat{f}(\xi)d\xi = \int \hat{f}(\xi)\varphi(\xi)d\xi = \int f(y)\hat{\varphi}(y)dy = f * g_t(x).$$

Now we take the limit as $t \to 0$:

1. Right hand side: since $\int g(y)dy = 1$, $\{g_t\}$ is an approximate identity. Thus $f * g_t \to f$ in the L^1 norm.
2. Left hand side: since $\hat{f} \in L^1$, the function is dominated by $|\hat{f}|$. By the Dominated Convergence Theorem, for every x:

$$\lim_{t\to 0} \int e^{-\pi t^2|\xi|^2} e^{2\pi i\xi\cdot x}\hat{f}(\xi)d\xi = \int e^{2\pi i\xi\cdot x}\hat{f}(\xi)d\xi = (\hat{f})^\vee(x).$$

Since the LHS converges pointwise to $(\hat{f})^\vee$ and the RHS converges to f in L^1 (which implies pointwise convergence of a subsequence a.e.), the limits must be equal almost everywhere.

It follows that $f = (\hat{f})^\vee$ a.e. Since $(\hat{f})^\vee$ is the Fourier transform of an L^1 function $(\hat{f})$, it is continuous, completing the proof.

Corollary 10.3.3

If $f \in L^1$ and $\hat{f} = 0$, then $f = 0$ a.e.

Corollary 10.3.4: Fourier isomorphism on Schwartz Space

$\mathcal{F}$ is an isomorphism of $\mathcal{S}$ onto itself.

***Proof*:**
By corollary 10.2.16, $\mathcal{F}$ maps S continuously into itself, and hence so does $f \mapsto f^\vee$, since $f^\vee(x) = \hat{f}(-x)$. By the Fourier inversion theorem, these maps are inverse to each other.

Corollary 10.3.5: Fourier Transform Has Order 4

Let $\mathfrak{F}$ represent the Fourier transform. Then:

$$\mathfrak{F}^4 = I$$

and 4 is the smallest such number

***Proof*:**
First:

$$\begin{aligned} \mathfrak{F} \circ \mathfrak{F}(f)(x) &= \mathfrak{F}(\mathfrak{F}(f)) \\ &= \int_{\mathbb{R}^d} \hat{f}(\xi)e^{2\pi i\xi\cdot x}d\xi \\ &= \int_{\mathbb{R}^d} \hat{f}(\xi)e^{2\pi i\xi\cdot(-x)}d\xi \\ &= f(-x) && \text{Fourier inversion formula} \end{aligned}$$

Hence applying it it again gives the desired result.

We shall give a name to the representation of $f(x)$ in terms of its inversion:

Definition 10.3.6: Fourier Integral

Let $f \in L^1$ and $\hat{f} \in L^1$. Then the *Fourier integral* at x is:

$$f(x) = \int \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi$$

An important result using the Fourier inversion theorem is the dual of proposition 10.2.14:

Corollary 10.3.7: Inverse Regularity and Decay

1. **Decay $\Rightarrow$ Smoothness:** If $(1+|\xi|)^k \hat{f}(\xi) \in L^1(\mathbb{R}^n)$ for some integer $k \geq 1$, then $f \in C^k(\mathbb{R}^n)$. The derivative is given by:

$$\partial^\alpha f(x) = \int_{\mathbb{R}^n} (2\pi i \xi)^\alpha \hat{f}(\xi) e^{2\pi i x \cdot \xi} \, d\xi$$

   for all multi-indices $|\alpha| \leq k$.

2. **Smoothness $\Rightarrow$ Decay:** If $\hat{f} \in C^k(\mathbb{R}^n)$ and $\partial^\alpha \hat{f} \in L^1(\mathbb{R}^n)$ for all $|\alpha| \leq k$, then f decays rapidly:

$$|f(x)| \leq \frac{C}{(1+|x|)^k}$$

   for some constant C.

***Proof*:**
The Inverse Fourier Transform is defined by:

$$f(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i x \cdot \xi} \, d\xi.$$

This operator differs from the forward Fourier transform only by the sign of the exponential ($+i$ instead of $-i$). Since $|e^{i\theta}| = |e^{-i\theta}| = 1$, the absolute values used in the norm estimates and convergence arguments of **??** remain unchanged.

Thus, the proof follows immediately by applying **??** to the function $\hat{f}$ (with a trivial sign change in the derivative formula).

At last we are in a position to derive the analogue of the orthonormal basis as we have done in theorem 10.2.4:

Theorem 10.3.8: The Plancherel Theorem

If $f \in L^1 \cap L^2$, then $\hat{f} \in L^2$; and $\mathcal{F}|_{L^1 \cap L^2}$ extends uniquely to a unitary isomorphism on L^2.

***Proof*:**
Let $X = \{f \in L^1 : \hat{f} \in L^1\}$. Since $\hat{f} \in L^1$ implies $f \in L^\infty$, we have $X \subset L^2$ by proposition 6.1.13, and X is dense in L^2 because $\mathcal{S} \subset X$ and $\mathcal{S}$ is dense in L^2 by proposition 10.1.12.

Given $f, g \in X$, let $h = \overline{\hat{g}}$. By the properties of the Fourier transform under complex conjugation and inversion, we have $\hat{h} = \bar{g}$. Now, using the multiplication formula (Parseval's relation for L^1):

$$\int f\bar{g} = \int f\hat{h} = \int \hat{f}h = \int \hat{f}\overline{\hat{g}}.$$

Thus $\mathcal{F}|_X$ preserves the L^2 inner product. In particular, by taking $g = f$, we obtain $\|\hat{f}\|_2 = \|f\|_2$. Since $\mathcal{F}(X) = X$ by the inversion theorem, $\mathcal{F}|_X$ extends by continuity to a unitary isomorphism on L^2.

It remains only to show that this extension agrees with $\mathcal{F}$ on all of $L^1 \cap L^2$. But if $f \in L^1 \cap L^2$ and $g(x) = e^{-\pi|x|^2}$ as in the proof of the inversion theorem, we have $f * g_t \in L^1$ by Young's inequality and $(f * g_t)^\wedge \in L^1$ because $(f * g_t)^\wedge(\xi) = e^{-\pi t^2|\xi|^2}\hat{f}(\xi)$ and $\hat{f}$ is bounded. Hence $f * g_t \in X$; moreover, by theorem 10.1.15, $f * g_t \to f$ in both the L^1 and L^2 norms. Therefore $(f * g_t)^\wedge \to \hat{f}$ both uniformly and in the L^2 norm, completing the proof.

From this, we upgrade The Hausdorff-Young Inequality over $\mathbb{R}^n$:

Theorem 10.3.9: The Hausdorff-Young Inequality

Suppose that $1 \leq p \leq 2$ and q is the conjugate exponent to p. If $f \in L^p(\mathbb{R}^n)$, then

$$\hat{f} \in L^q(\mathbb{R}^n) \qquad \text{and} \qquad \|\hat{f}\|_q \leq \|f\|_p$$

Folland goes on an interesting point about trying to make an $\mathbb{R}^n$ input function periodic which leads to the Poisson Summation Formula; I'm skipping this for now

(here is a theorem I want to add since it related to the analytic theorem)

Theorem 10.3.10: Fourier Decay and Analyticity

Let f be a smooth function such that its Fourier transform $\hat{f}$ satisfies the exponential decay estimate

$$|\hat{f}(\xi)| \leq Ce^{-R2\pi|\xi|}$$

for some constants $C, R > 0$. Then f has a global derivative growth rate α satisfying

$$\alpha \leq \frac{1}{R}.$$

Consequently, by theorem 4.9.5, f is analytic with radius of convergence at least R.

***Proof*:**
Recall that the Fourier transform diagonalizes the differentiation operator. Specifically, the Fourier transform of the k-th derivative is given by $\widehat{f^{(k)}}(\xi) = (2\pi i\xi)^k\hat{f}(\xi)$. By the Fourier

inversion formula, we can express the k-th derivative as:

$$f^{(k)}(x) = \int_{-\infty}^{\infty} \widehat{f^{(k)}}(\xi) e^{2\pi i x\xi}\, d\xi = \int_{-\infty}^{\infty} (2\pi i\xi)^k \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

We wish to bound the magnitude $M_k = \sup_x |f^{(k)}(x)|$. Applying the triangle inequality and the assumed decay bound on $\hat{f}$:

$$\begin{aligned} |f^{(k)}(x)| &\leq \int_{-\infty}^{\infty} |2\pi\xi|^k |\hat{f}(\xi)|\, d\xi \\ &\leq \int_{-\infty}^{\infty} (2\pi|\xi|)^k C e^{-R|\xi|}\, d\xi \\ &= 2C(2\pi)^k \int_0^{\infty} \xi^k e^{-R\xi}\, d\xi. \end{aligned}$$

The integral on the right is a standard form related to the Gamma function (specifically, $\int_0^\infty t^k e^{-t} dt = k!$). Making the substitution $u = R\xi$, we have $\xi = u/R$ and $d\xi = du/R$:

$$\int_0^{\infty} \xi^k e^{-R\xi}\, d\xi = \frac{1}{R^{k+1}} \int_0^{\infty} u^k e^{-u}\, du = \frac{k!}{R^{k+1}}.$$

Substituting this back into our estimate for M_k:

$$M_k \leq 2C(2\pi)^k \frac{k!}{R^{k+1}} = \left(\frac{2C}{R}\right) k! \left(\frac{2\pi}{R}\right)^k.$$

Now we compute the derivative growth rate α:

$$\begin{aligned} \alpha &= \limsup_{k\to\infty} \sqrt[k]{\frac{M_k}{k!}} \\ &\leq \limsup_{k\to\infty} \sqrt[k]{\frac{2C}{R}\left(\frac{2\pi}{R}\right)^k} \\ &= \frac{2\pi}{R}. \end{aligned}$$

(Note: The factor of 2π appears due to the chosen convention of the Fourier transform. If the decay rate R is defined relative to angular frequency, the factor vanishes). In either case, α is finite and inversely proportional to the decay constant R. Thus, the derivative growth rate is bounded, and f is analytic.

I don't know if I wanna continue this section for now, about Cecaro sums and etc

10.4 Point-wise Convergence of Fourier Series

We have established that for $f \in L^2(\mathbb{T}^n)$, the Fourier series converges to f in the L^2-norm. However, norm convergence does not imply point-wise convergence. We might ask: if we pick a specific point

x_0, does the partial sum sequence $S_N(f)(x_0)$ converge to $f(x_0)$?

$$S_N(f)(x) = \sum_{|k|\leq N} \hat{f}(k)e^{2\pi ik\cdot x}$$

Surprisingly, for general integrable functions, the answer is often no. The mechanism governing this convergence is the *Dirichlet Kernel*.

Recall that convolution in the spatial domain corresponds to multiplication in the frequency domain. The partial sum $S_N(f)$ is the convolution of f with the inverse Fourier transform of the characteristic function of the range $[-N, N]$.

Definition 10.4.1: Dirichlet Kernel

The *Dirichlet Kernel* D_N on $\mathbb{T}^1$ is defined as:

$$D_N(x) = \sum_{k=-N}^{N} e^{2\pi ikx} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}$$

Consequently, the partial sums of the Fourier series are given by:

$$S_N(f)(x) = (f * D_N)(x) = \int_{\mathbb{T}} f(x-t)D_N(t)dt$$

The equality in the above equation is a simple application of the geometric series with ratio $r = e^{2\pi ix}$:

$$\begin{aligned}
\sum_{k=-N}^{N} r^k = r^{-N}\frac{1-r^{2N+1}}{1-r} &= \frac{r^{-N}-r^{N+1}}{1-r} \\
&= \frac{r^{-(N+1/2)}-r^{N+1/2}}{r^{-1/2}-r^{1/2}} \quad \text{(multiplying top and bottom by } r^{-1/2}\text{)} \\
&= \frac{e^{-2\pi i(N+1/2)x}-e^{2\pi i(N+1/2)x}}{e^{-\pi ix}-e^{\pi ix}} \\
&= \frac{-2i\sin((2N+1)\pi x)}{-2i\sin(\pi x)} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}
\end{aligned}$$

The convergence of $S_N(f)$ depends on whether D_N is a "good kernel" (an approximate identity). While $\int D_N = 1$, it fails the absolute integrability condition: $\int |D_N| \to \infty$ as $N \to \infty$. Specifically, $\|D_N\|_1 \sim \frac{4}{\pi^2}\log N$. This logarithmic growth allows for pathological behavior.

Theorem 10.4.2: Failure of Convergence

1. There exists an $f \in L^1(\mathbb{T})$ such that the Fourier series diverges *everywhere* (Kolmogorov, 1923).
2. There exists an $f \in C(\mathbb{T})$ (continuous!) such that the Fourier series diverges at a dense set of points (Du Bois-Reymond).

***Proof*:**
We will not construct the specific function f (which requires the Uniform Boundedness Principle), but we will prove the underlying reason: the divergence of the Lebesgue constants $L_N = \|D_N\|_1$.

$$L_N = \int_{-1/2}^{1/2} \left| \frac{\sin((2N+1)\pi x)}{\sin(\pi x)} \right| dx$$

Since $|\sin(\pi x)| \leq \pi |x|$, we have a lower bound:

$$L_N \geq \int_{-1/2}^{1/2} \frac{|\sin((2N+1)\pi x)|}{\pi |x|} dx = 2 \int_0^{1/2} \frac{|\sin((2N+1)\pi x)|}{\pi x} dx$$

Let $u = (2N+1)\pi x$. The integral becomes:

$$\frac{2}{\pi} \int_0^{(N+1/2)\pi} \frac{|\sin u|}{u} du \geq \frac{2}{\pi} \sum_{k=1}^{N} \int_{(k-1)\pi}^{k\pi} \frac{|\sin u|}{u} du$$

In the interval $[(k-1)\pi, k\pi]$, $1/u \geq 1/k\pi$. Also $\int_0^\pi |\sin u| du = 2$.

$$L_N \geq \frac{2}{\pi} \sum_{k=1}^{N} \frac{1}{k\pi} \cdot 2 = \frac{4}{\pi^2} \sum_{k=1}^{N} \frac{1}{k} \sim \frac{4}{\pi^2} \log N$$

Since the norms of the operators $S_N : C(\mathbb{T}) \to \mathbb{C}$ are unbounded, the Banach-Steinhaus theorem implies there exists a dense set of functions in $C(\mathbb{T})$ for which the series diverges.

This tells us that continuity alone is *not* strong enough to guarantee that we can recover the function point-wise from its frequencies using the standard summation. To guarantee convergence, we need slightly more regularity than continuity. We look for conditions that cancel out the oscillations of the Dirichlet kernel.

Theorem 10.4.3: Pointwise Convergence Tests

Let $f \in L^1(\mathbb{T})$.

1. *Dini's Criterion (Local):* If f satisfies a local Hölder condition at x (or more generally, $\int_{-1/2}^{1/2} |\frac{f(x+t)-f(x)}{t}| dt < \infty$), then $S_N(f)(x) \to f(x)$.
2. *Dirichlet-Jordan Test (Global):* If f is of *Bounded Variation* on $\mathbb{T}$, then for every x:

$$\lim_{N\to\infty} S_N(f)(x) = \frac{f(x^+) + f(x^-)}{2}$$

***Proof*:**
We prove Dini's Criterion. We wish to show $S_N(f)(x) - f(x) \to 0$. Using the fact that

$\int D_N(t)dt = 1$, we write:

$$S_N(f)(x) - f(x) = \int_{-1/2}^{1/2} (f(x-t) - f(x))D_N(t)dt$$

Substitute the explicit formula for $D_N(t)$:

$$= \int_{-1/2}^{1/2} \frac{f(x-t) - f(x)}{\sin(\pi t)} \sin((2N+1)\pi t)dt$$

Define $g(t) = \frac{f(x-t)-f(x)}{t} \cdot \frac{t}{\sin(\pi t)}$. The term $\frac{t}{\sin(\pi t)}$ is bounded and smooth near $t = 0$. By the Dini condition, $\frac{f(x-t)-f(x)}{t}$ is integrable near 0. Thus $g(t) \in L^1(\mathbb{T})$. The expression becomes:

$$\int_{-1/2}^{1/2} g(t) \sin((2N+1)\pi t)dt$$

This is simply the imaginary part of the $(2N+1)$-th Fourier coefficient of $g(t)$. By the Riemann-Lebesgue Lemma, $\hat{g}(k) \to 0$ as $k \to \infty$. Thus the integral vanishes as $N \to \infty$.

If we cannot guarantee convergence for continuous functions, we can modify our summation method. Instead of partial sums, we take the average of partial sums (Cesàro means).

Theorem 10.4.4: Fejér's Theorem

Let $\sigma_N(f) = \frac{1}{N+1}\sum_{n=0}^{N} S_n(f)$ be the Cesàro means. Then $\sigma_N(f) = f * K_N$, where K_N is the *Fejér Kernel*:

$$K_N(x) = \frac{1}{N+1}\left(\frac{\sin((N+1)\pi x)}{\sin(\pi x)}\right)^2$$

Unlike the Dirichlet kernel, $K_N(x) \geq 0$ and $\int K_N = 1$. Consequently, if $f \in C(\mathbb{T})$, then $\sigma_N(f) \to f$ *uniformly.*

***Proof*:**
The formula for K_N follows from summing the geometric series for D_n. The key properties are: 1. $K_N(x) \geq 0$. 2. $\int_{-1/2}^{1/2} K_N(x)dx = 1$. 3. For any $\delta > 0$, $\int_{|x|\geq\delta} K_N(x)dx \to 0$ as $N \to \infty$ (Mass concentration).

Let f be continuous on $\mathbb{T}$. Since $\mathbb{T}$ is compact, f is uniformly continuous. Given $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x-t) - f(x)| < \epsilon/2$ whenever $|t| < \delta$.

$$\begin{aligned} |\sigma_N(f)(x) - f(x)| &= \left|\int_{-1/2}^{1/2} (f(x-t) - f(x))K_N(t)dt\right| \\ &\leq \int_{-1/2}^{1/2} |f(x-t) - f(x)|K_N(t)dt \end{aligned}$$

Split the integral into region $A = \{|t| < \delta\}$ and $B = \{|t| \geq \delta\}$. On A: $|f(x-t) - f(x)| < \epsilon/2$. Since $\int K_N = 1$, the integral over A is $\leq \epsilon/2$. On B: $|f(x-t) - f(x)| \leq 2\|f\|_\infty$. The integral is bounded by $2\|f\|_\infty \int_{|t|\geq\delta} K_N(t)dt$. Using the explicit formula, for $|t| \geq \delta$, $K_N(t) \leq \frac{1}{N+1}\frac{1}{\sin^2(\pi\delta)}$. As $N \to \infty$, this goes to 0. Thus for large N, the second term is $< \epsilon/2$. The convergence is uniform because δ depended only on the uniform continuity of f, not on x.

10.4.5 Historical Note For a long time, it was unknown if $f \in L^2(\mathbb{T})$ converged pointwise. In 1966, Lennart Carleson proved that if $f \in L^2(\mathbb{T})$, then $S_N(f)(x) \to f(x)$ for *almost every* x. This was later extended by Hunt to all L^p for $p > 1$. This is a deep and difficult result, requiring a careful decomposition of the operator to avoid the logarithmic divergence of L^1.

10.5 *From Harmonic Analysis to Number Theory

In the preceding section, we developed a powerful principle: the natural eigenbasis for a function space is determined by the symmetry group of the domain, and the Fourier transform is the change of basis to this eigenbasis. We saw three instances:

Domain	**Symmetry**	**Eigenbasis**	**Transform**
$\mathbb{T}^n$	Translation ($\mathbb{R}^n/\mathbb{Z}^n$)	$e^{2\pi i k\cdot x}$	Fourier series
S^2	Rotation $SO(3)$	Spherical harmonics Y_l^m	Spherical transform
$\mathbb{R}$ (QHO)	Heisenberg-Weyl	Hermite functions φ_n	Hermite transform

The entire Langlands program can be understood as the systematic application of this principle to the domains that arise in number theory. In this section, we trace this path from concrete to abstract, beginning with the “missing” symmetry from the preceding section.

The Mellin Transform: Fourier Analysis for Multiplication

Recall from the preceding section that we commented on *dilation invariance* without developing it. We now fill this gap, as it is the gateway to number theory.

The Fourier transform arises from translation invariance: the Haar measure dx on the additive group $(\mathbb{R}, +)$ satisfies $d(x+a) = dx$, and the characters of this group are $x \mapsto e^{2\pi i\xi x}$, giving:

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i\xi x}\, dx$$

Now consider the *multiplicative* group $(\mathbb{R}_{>0}, \times)$. Its Haar measure is $\frac{dx}{x}$, which is dilation-invariant: $\frac{d(ax)}{ax} = \frac{dx}{x}$. The characters of $(\mathbb{R}_{>0}, \times)$ are the power functions $x \mapsto x^s$ for $s \in \mathbb{C}$ (since $x \mapsto x^s$ is the unique continuous homomorphism $(\mathbb{R}_{>0}, \times) \to (\mathbb{C}^*, \times)$). This gives the *Mellin transform*:

$$\mathcal{M}f(s) = \int_0^{\infty} f(x)\, x^s\, \frac{dx}{x}$$

The Mellin transform is thus the Fourier transform on the multiplicative group, in exactly the same sense that the ordinary Fourier transform is the Fourier transform on the additive group. This can be made precise: the substitution $x = e^t$ converts the Mellin transform into a two-sided Laplace transform, and restricting to the critical line $s = \frac{1}{2} + i\xi$ recovers a Fourier transform in the variable $t = \log x$.

The following table, which should be compared to the eigenvalue table from the Lie theory section, summarizes the analogy:

	Additive (Fourier)	**Multiplicative (Mellin)**
Group	$(\mathbb{R}, +)$	$(\mathbb{R}_{>0}, \times)$
Haar measure	dx	dx/x
Characters	$e^{2\pi i\xi x}$	x^s
Invariance	Translation: $f(x) \mapsto f(x+a)$	Dilation: $f(x) \mapsto f(ax)$
Generator (Lie algebra)	$D = \frac{d}{dx}$	$\Theta = x\frac{d}{dx}$
Eigenvalue of generator	$De^{i\xi x} = i\xi \cdot e^{i\xi x}$	$\Theta(x^s) = s \cdot x^s$
Convolution	$(f * g)(x) = \int f(y)g(x-y)\,dy$	$(f *_\times g)(x) = \int f(y)g(x/y)\,\frac{dy}{y}$

The operator $\Theta = x\frac{d}{dx}$ is the *Euler operator* (or multiplicative derivative); it generates dilations just as D generates translations, via the exponential map $e^{a\Theta}f(x) = f(e^a x)$. This is Lie theory applied to the multiplicative group.

10.5.1 Cotangent Interpretation of the Mellin Variable In the language of the preceding sections, the Fourier variable ξ is a cotangent (momentum) coordinate dual to the position variable x. Similarly, the Mellin variable s is a cotangent coordinate dual to $\log x$—it measures the "multiplicative momentum" or *scaling rate* of a signal. Where the Fourier transform decomposes a function into oscillatory modes $e^{i\xi x}$ of definite frequency, the Mellin transform decomposes a function into power-law modes x^s of definite *scaling exponent*. The Mellin transform is the natural tool whenever the geometry is multiplicative rather than additive—and this is precisely the situation in number theory, where the fundamental operations are multiplication of integers and primes.

The Prototype: Riemann's Zeta Function

We now demonstrate how the Mellin transform connects harmonic analysis to number theory through the simplest and most important example. Define the Riemann zeta function by:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \mathrm{Re}(s) > 1$$

The key observation is that this sum is a *discrete Mellin transform*. In the same way that a Fourier series $\sum \hat{f}(k)e^{ikx}$ samples the Fourier eigenbasis at integer frequencies, the zeta function samples the Mellin eigenbasis $n \mapsto n^{-s}$ at positive integers. But we can also write $\zeta(s)$ as a *continuous* Mellin transform by introducing the *theta function*:

$$\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad t > 0$$

This is the heat kernel on the circle $\mathbb{R}/\mathbb{Z}$ evaluated at time $t/(4\pi)$—precisely the object from our opening section! Now, the completed zeta function is the Mellin transform of (a modification of) the theta function:

$$\Lambda(s) := \pi^{-s/2}\,\Gamma(s/2)\,\zeta(s) = \frac{1}{2}\int_0^\infty \big(\theta(t)-1\big)\,t^{s/2}\,\frac{dt}{t}, \qquad \mathrm{Re}(s) > 1$$

The factor $\pi^{-s/2}\Gamma(s/2)$ is not a nuisance—it is the *archimedean local factor*, arising from the Mellin transform of the Gaussian $e^{-\pi x^2}$ (itself the eigenvector of the Fourier transform, just as in our QHO discussion!). Now the *functional equation* of $\zeta(s)$ follows from the *modularity* of θ, which in turn follows from *Poisson summation*:

$$\text{Poisson Summation} \Longrightarrow \theta(1/t) = t^{1/2}\theta(t) \xRightarrow{\text{Mellin}} \Lambda(s) = \Lambda(1-s)$$

This chain reveals the deep structure: the functional equation of the zeta function is a *consequence of Fourier duality* (Poisson summation), transmitted to the multiplicative world via the Mellin transform. The symmetry $s \leftrightarrow 1-s$ reflects the self-duality of the lattice $\mathbb{Z} \subset \mathbb{R}$ under the Fourier transform.

Let us see how this works. The theta function is a sum of Gaussians $e^{-\pi n^2 t}$ over the integer lattice $\mathbb{Z}$. Poisson summation (the Fourier-analytic identity $\sum_{n\in\mathbb{Z}} f(n) = \sum_{k\in\mathbb{Z}} \hat{f}(k)$, which is itself a consequence of Peter-Weyl for the group $\mathbb{T}^1 = \mathbb{R}/\mathbb{Z}$) applied with $f(x) = e^{-\pi x^2 t}$ gives:

$$\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t} = \frac{1}{\sqrt{t}} \sum_{k\in\mathbb{Z}} e^{-\pi k^2/t} = \frac{1}{\sqrt{t}}\,\theta(1/t)$$

Now, to pass from the additive world (theta, Poisson summation) to the multiplicative world (zeta, functional equation), we apply the Mellin transform. Splitting the integral at $t = 1$ and using $\theta(1/t) = \sqrt{t}\,\theta(t)$ to fold the integral onto $[1,\infty)$ where $\theta(t)-1$ decays exponentially, one obtains the meromorphic continuation and the symmetry $\Lambda(s) = \Lambda(1-s)$. The proof is a direct computation; the reader is referred to [4] for full details.

Dirichlet Characters: Representation Theory Meets Arithmetic

We now connect the character theory developed in the preceding section to classical number theory. Consider the finite abelian group $(\mathbb{Z}/N\mathbb{Z})^\times$—the group of units modulo N. By our general theory, its characters $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \to S^1$ form a group $\widehat{(\mathbb{Z}/N\mathbb{Z})}^\times$ that is (non-canonically) isomorphic to $(\mathbb{Z}/N\mathbb{Z})^\times$ itself.

These characters are exactly the *Dirichlet characters* modulo N: extending χ to all of $\mathbb{Z}$ by setting $\chi(n) = 0$ when $\gcd(n,N) > 1$, we obtain the arithmetic functions that Dirichlet used in 1837 to prove his theorem on primes in arithmetic progressions. The key tool is the *orthogonality of characters* (section 12.2):

$$\frac{1}{\varphi(N)} \sum_{\chi \bmod N} \chi(a)\overline{\chi(b)} = \begin{cases} 1 & \text{if } a \equiv b \pmod{N} \\ 0 & \text{otherwise} \end{cases}$$

This is precisely the "Fourier inversion formula" for the finite group $(\mathbb{Z}/N\mathbb{Z})^\times$. Dirichlet used it to isolate primes in a single residue class: the sum over *all* primes is controlled by the pole of $\zeta(s)$, and by multiplying by characters and using orthogonality, one projects onto primes $p \equiv a \pmod{N}$.

To each Dirichlet character χ, we associate the *Dirichlet L-function*:

$$L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1-\chi(p)p^{-s}}, \qquad \mathrm{Re}(s) > 1$$

The Euler product reflects the fact that χ is *multiplicative*: $\chi(mn) = \chi(m)\chi(n)$, which holds because χ is a group homomorphism. This Euler product is the multiplicative analogue of the fact that convolution operators (translation-invariant operators) become multiplication operators under the Fourier transform: the Dirichlet series *is* a multiplicative convolution, and the Euler product is its diagonalization.

Each $L(s,\chi)$ admits a completed version $\Lambda(s,\chi)$ with a functional equation $\Lambda(s,\chi) = \epsilon(\chi)\Lambda(1-s,\overline{\chi})$, proved by the same theta function method: replace $\theta(t)$ by a twisted theta function $\theta_\chi(t) = \sum \chi(n)e^{-\pi n^2 t/N}$, apply Poisson summation, and take the Mellin transform. The Gauss sum $G(\chi) = \sum_{a \bmod N} \chi(a)e^{2\pi i a/N}$ that appears is exactly the *finite Fourier transform* of χ—harmonic analysis on the finite group $(\mathbb{Z}/N\mathbb{Z})^\times$ evaluated at a specific point.

Tate's Thesis: Fourier Analysis on the Adèles (the GL_1 Case)

The proofs sketched above—theta function, Poisson summation, Mellin transform—work but involve many case-by-case computations depending on the number field and the character. In 1950, John Tate, following a suggestion of his advisor Emil Artin, found a remarkable reformulation that makes the entire argument *canonical*: perform Fourier analysis not on $\mathbb{R}$ or $\mathbb{Q}_p$ individually, but on all completions of $\mathbb{Q}$ *simultaneously*, via the ring of *adèles*.

The key idea is the local-global principle in harmonic analysis. A number field K has many completions: one for each prime $\mathfrak{p}$ (giving a p-adic field $K_\mathfrak{p}$) and one for each archimedean place (giving $\mathbb{R}$ or $\mathbb{C}$). Each completion is a locally compact field, so we can do harmonic analysis on each one separately. The *adèle ring* $\mathbb{A}_K$ is the restricted product of all these local fields:

$$\mathbb{A}_K = \prod_v{}' K_v$$

This is a locally compact topological ring, and it contains K as a discrete co-compact subgroup (just as $\mathbb{Z}$ sits discretely inside $\mathbb{R}$). This is the crucial structural analogy:

Classical Fourier Analysis	**Adelic Fourier Analysis**
$\mathbb{Z} \hookrightarrow \mathbb{R}$ (discrete, co-compact)	$K \hookrightarrow \mathbb{A}_K$ (discrete, co-compact)
Poisson summation on $\mathbb{R}/\mathbb{Z}$	Poisson summation on $\mathbb{A}_K/K$
$\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$	$\theta_f = \sum_{\alpha\in K} f(\alpha t)$, $\quad f$ Schwartz-Bruhat
Mellin transform $\to \Lambda(s)$	Adelic zeta integral $\to \Lambda(s,\chi)$
$\Lambda(s) = \Lambda(1-s)$	$\zeta(f,\chi,s) = \zeta(\hat{f},\chi^{-1},1-s)$

Tate defines an *adelic zeta integral*:

$$\zeta(f,\chi,s) = \int_{\mathbb{A}_K^\times} f(x)\,\chi(x)\,|x|^s\,d^\times x$$

where f is a Schwartz-Bruhat function on $\mathbb{A}_K$, χ is a character of the *idèle class group* $\mathbb{A}_K^\times/K^\times$ (a Hecke character—the adelic generalization of a Dirichlet character), and $|x|$ is the adelic norm. The fundamental result is:

All degree-one L-functions can be realized as adelic zeta integrals with suitably chosen f and χ.

The functional equation then follows from adelic Poisson summation in a single, uniform argument that works for all number fields simultaneously. Crucially, the adelic zeta integral *factors* as a product of local integrals:

$$\zeta(f,\chi,s) = \prod_v \int_{K_v^\times} f_v(x_v)\,\chi_v(x_v)\,|x_v|_v^s\,d^\times x_v$$

Each local factor is a piece of harmonic analysis on a single local field K_v. At unramified non-archimedean places, the local integral evaluates to $(1-\chi(\mathfrak{p})|\mathfrak{p}|^s)^{-1}$—the Euler factor. At the archimedean place $v=\infty$, taking $f_v(x)=e^{-\pi x^2}$, the local integral is $\pi^{-s/2}\Gamma(s/2)$—the gamma factor we saw earlier.

This is the representation-theoretic meaning of the gamma factor and the Euler product: they are the contributions of individual local symmetries (harmonic analysis at each prime and at ∞), assembled into a global object by the adelic framework.

From the perspective of the Langlands program, Tate's thesis is the GL_1 case: the Hecke characters $\chi : \mathbb{A}_K^\times/K^\times \to \mathbb{C}^\times$ are exactly the *automorphic representations of* GL_1, and their L-functions are the degree-one L-functions. Class field theory then provides the bridge to the Galois side: the Artin reciprocity map gives a canonical isomorphism between Hecke characters and one-dimensional representations of $\mathrm{Gal}(\overline{K}/K)^{\mathrm{ab}}$, so that:

$$\boxed{\left\{\begin{array}{c}\text{1-dimensional representations}\\ \text{of } \mathrm{Gal}(\overline{K}/K)\end{array}\right\} \xleftrightarrow{\text{Class Field Theory}} \left\{\begin{array}{c}\text{Automorphic representations}\\ \text{of } \mathrm{GL}_1(\mathbb{A}_K)\end{array}\right\}}$$

This is the GL_1 Langlands correspondence. It is a *theorem* (class field theory). The Langlands program conjectures that this correspondence generalizes to all n.

Hecke Operators: The Number-Theoretic Translation Operators

We now climb from GL_1 to GL_2. The key insight is that Hecke operators play *exactly the same structural role* for modular forms that translation operators τ_a play for the Fourier basis.

Recall our framework: we found the natural eigenbasis by identifying a commuting family of symmetry operators (translations τ_a), then simultaneously diagonalizing them (Peter-Weyl / spectral theorem). For modular forms, the role of the translation operators is played by the *Hecke operators* T_p (one for each prime p). These are linear operators on the space of modular forms $M_k(\mathrm{SL}_2(\mathbb{Z}))$ defined by averaging over certain lattice correspondences.

The crucial properties, paralleling our earlier analysis, are:

1. *The Hecke operators commute:* $T_p \circ T_q = T_q \circ T_p$ for all primes p, q. This is the analogue of $\tau_a \circ \tau_b = \tau_b \circ \tau_a$.
2. *The Hecke operators are normal (self-adjoint with respect to the Petersson inner product):* This is the analogue of τ_a being unitary on L^2.
3. *The spectral theorem applies:* There exists an orthonormal basis of $M_k(\mathrm{SL}_2(\mathbb{Z}))$ consisting of *simultaneous eigenfunctions* for all T_p. These are the *Hecke eigenforms*, and they are the "Fourier basis" of number theory.

Now comes the remarkable connection. If $f(z) = \sum_{n=0}^{\infty} a_n q^n$ (where $q = e^{2\pi i z}$) is a normalized Hecke eigenform, then:

$$T_p(f) = a_p \cdot f$$

That is, the *Fourier coefficient* a_p is simultaneously the *Hecke eigenvalue* λ_p. The Fourier expansion of a modular form and its spectral decomposition under Hecke operators are one and the same thing! This is analogous to how the Fourier coefficient $\hat{f}(k) = \langle f, e^{ikx} \rangle$ is simultaneously the "projection onto the k-th eigenspace of translation".

Furthermore, because Hecke eigenvalues are multiplicative ($a_{mn} = a_m a_n$ when $\gcd(m, n) = 1$), the L-function of a Hecke eigenform admits an *Euler product*:

$$L(f, s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s} = \prod_p \frac{1}{1 - a_p p^{-s} + p^{k-1-2s}}$$

This Euler product is the GL_2 analogue of the Euler product for Dirichlet L-functions. Each local factor $(1 - a_p p^{-s} + p^{k-1-2s})^{-1}$ is the inverse of a quadratic polynomial in p^{-s}—the *Hecke polynomial*—and its roots encode the *Satake parameters* α_p, β_p at the prime p. The Mellin transform of the eigenform gives the completed L-function, and modularity of f (i.e., the transformation law under $\mathrm{SL}_2(\mathbb{Z})$) implies the functional equation, by the same theta-function mechanism as before.

10.5.2 The Ramanujan Conjecture as a Spectral Bound The Ramanujan conjecture (proved by Deligne in 1974 for holomorphic modular forms) states that the Hecke eigenvalues of a cuspidal eigenform of weight k satisfy:

$$|a_p| \leq 2p^{(k-1)/2}$$

In our framework, this is a *spectral bound*: it constrains the eigenvalues of the Hecke operators T_p, analogous to how unitarity constrains the eigenvalues of τ_a to lie on S^1. In the language of automorphic representations, the Ramanujan conjecture asserts that local components of cuspidal automorphic representations are *tempered*—their matrix coefficients lie in $L^{2+\epsilon}$ but not in L^2, which is a condition on how "spread out" the representation is in the unitary dual. The general Ramanujan conjecture for GL_n over number fields remains one of the deepest open problems in the Langlands program.

The Langlands Program: Symmetry → Eigenbasis → L-function

We can now state the pattern that unifies everything. At each level of complexity, the same three-step procedure applies:

1. **Identify the symmetry group** acting on the function space.

2. **Decompose** L^2 into irreducible representations (the "eigenbasis").

3. **Attach an L-function** to each irreducible component.

Level	**Symmetry Group**	**Eigenbasis**	**L-function**
$\mathbb{T}^1$	$\mathbb{R}/\mathbb{Z}$ (translation)	$\{e^{2\pi ikx}\}_{k\in\mathbb{Z}}$	—
GL_1	$\mathbb{A}_K^\times/K^\times$	Hecke characters χ	$L(s,\chi)$
GL_2	$\mathrm{GL}_2(\mathbb{A})/\mathrm{GL}_2(K)$	Hecke eigenforms	$L(f,s)$
GL_n	$\mathrm{GL}_n(\mathbb{A})/\mathrm{GL}_n(K)$	Automorphic repr. π	$L(s,\pi)$
Gen. G	$G(\mathbb{A})/G(K)$	Automorphic repr. π	$L(s,\pi,r)$

In each case, the space $L^2(G(K)\backslash G(\mathbb{A}))$ plays the role of the "state space" $\mathscr{P}(X)$, and the spectral decomposition is the generalization of the Peter-Weyl theorem. For $G = \mathrm{GL}_1$, this recovers Tate's thesis. For $G = \mathrm{GL}_2$, the cuspidal part of the decomposition recovers the Hecke eigenforms. For general reductive G, the decomposition is the content of the *theory of Eisenstein series* (due to Langlands), which decomposes L^2 into a discrete (cuspidal) part and a continuous part:

$$L^2(G(K)\backslash G(\mathbb{A})) = L^2_{\mathrm{disc}} \oplus L^2_{\mathrm{cont}} = \bigoplus_{\pi \text{ cusp}} m(\pi)\cdot\pi \quad\oplus\quad \int^{\oplus} (\text{Eisenstein series})$$

This is exactly the analogue of the discrete spectrum (eigenfunctions) versus continuous spectrum (generalized eigenfunctions) that arises in the spectral theory of the Laplacian on non-compact manifolds.

The *Langlands reciprocity conjecture* then asserts a correspondence between the spectral side (automorphic representations) and the arithmetic side (Galois representations):

$$\boxed{\left\{\begin{matrix} n\text{-dimensional representations} \\ \text{of } \mathrm{Gal}(\overline{K}/K) \end{matrix}\right\} \xleftrightarrow{\text{Langlands}} \left\{\begin{matrix} \text{Automorphic representations} \\ \text{of } \mathrm{GL}_n(\mathbb{A}_K) \end{matrix}\right\}}$$

preserving L-functions on both sides. For $n = 1$, this is class field theory (a theorem). For $n = 2$, special cases include the modularity theorem (Wiles et al., 1995), which proved that every elliptic curve over $\mathbb{Q}$ is modular.

For a general reductive group G (not just GL_n), Langlands introduced the *L-group* (or Langlands dual group) ${}^LG$: a complex reductive group whose root datum is dual to that of G. The correspondence then relates:

$$\left\{\begin{matrix} \text{Homomorphisms} \\ \mathrm{Gal}(\overline{K}/K) \to {}^LG(\mathbb{C}) \end{matrix}\right\} \xleftrightarrow{\text{Functoriality}} \left\{\begin{matrix} \text{Automorphic representations} \\ \text{of } G(\mathbb{A}_K) \end{matrix}\right\}$$

The idea of *functoriality* is that a homomorphism ${}^LH \to {}^LG$ between L-groups should induce a *transfer* of automorphic representations from H to G, compatible with L-functions. This is the "master conjecture" of the Langlands program, from which essentially all other conjectures follow.

The Selberg Trace Formula: Poisson Summation for Reductive Groups

The reader may wonder: where is the analogue of *Poisson summation* in this general framework? The answer is the *Selberg trace formula* (and its generalization by Arthur). Recall that our proof of the functional equation for $\zeta(s)$ had the structure:

$$\text{Poisson summation (Fourier duality)} \xrightarrow{\text{Mellin}} \text{Functional equation}$$

The Selberg trace formula generalizes this to a *spectral-geometric duality* for the space $\Gamma\backslash G/K$, where G is a Lie group, K a maximal compact subgroup, and Γ a discrete subgroup. It equates two ways of computing the trace of a convolution operator on $L^2(\Gamma\backslash G)$:

$$\underbrace{\sum_{\pi} m(\pi)\,\widehat{h}(\pi)}_{\text{Spectral side}} \;=\; \underbrace{\sum_{\{\gamma\}} \operatorname{vol}(\Gamma_\gamma\backslash G_\gamma)\,\mathcal{O}_\gamma(h)}_{\text{Geometric side}}$$

The *spectral side* sums over automorphic representations π, weighted by their multiplicities $m(\pi)$ and the Fourier transform $\hat{h}(\pi)$ of a test function h. The *geometric side* sums over conjugacy classes $\{\gamma\}$ in Γ, weighted by volumes and *orbital integrals* $\mathcal{O}_\gamma(h)$.

For a compact hyperbolic surface $M = \Gamma\backslash\mathbb{H}$, this becomes:

$$\sum_n h(r_n) = \frac{\operatorname{Area}(M)}{4\pi}\int_{-\infty}^{\infty} r\tanh(\pi r)\,h(r)\,dr + \sum_{\{\gamma\}_{\mathrm{hyp}}} \frac{\ell(\gamma_0)}{2\sinh(\ell(\gamma)/2)}\,\hat{h}(\ell(\gamma))$$

where r_n are the spectral parameters of the Laplacian eigenvalues $\lambda_n = \frac{1}{4} + r_n^2$, the sum on the right runs over hyperbolic conjugacy classes (closed geodesics), and $\ell(\gamma)$ is the geodesic length. This is strikingly analogous to the explicit formulas relating zeros of $\zeta(s)$ to prime numbers, with:

Zeta function	**Selberg trace formula**
Zeros of $\zeta(s)$	Eigenvalues of Δ on M
Prime numbers p	Primitive closed geodesics on M
$\log p$	Length of geodesic $\ell(\gamma_0)$
Explicit formula	Trace formula

Arthur's generalization of the Selberg trace formula to arbitrary reductive groups over number fields is one of the most powerful tools in the Langlands program. The "comparison of trace formulas" between two groups G and H—matching orbital integrals on the geometric side to transfer automorphic representations on the spectral side—is the primary technique for proving cases of functoriality. The proof of the fundamental lemma by Ngô Bao Châu (Fields Medal, 2010) was a key breakthrough in making this comparison rigorous.

The Satake Isomorphism: Peter-Weyl at Each Prime

There is one more connection to our harmonic analysis framework that deserves emphasis. At each unramified prime p, the *spherical Hecke algebra* $\mathcal{H}(G(\mathbb{Q}_p), K_p)$ consists of K_p-bi-invariant compactly supported functions on $G(\mathbb{Q}_p)$, where $K_p = G(\mathbb{Z}_p)$ is a maximal compact subgroup. This algebra acts on the K_p-fixed vectors of any smooth representation—exactly paralleling how the convolution algebra acts on $L^2(G)$ in our Peter-Weyl discussion.

The *Satake isomorphism* states:

$$\mathcal{H}(G(\mathbb{Q}_p), K_p) \;\cong\; \mathbb{C}[X_*(T)]^W \;\cong\; \mathcal{O}(\hat{T}/W)$$

where T is a maximal torus, W is the Weyl group, $X_*(T)$ is the cocharacter lattice, and $\hat{T}$ is a torus in the dual group ${}^LG$. This is a deep generalization of the statement that the characters of a compact abelian group form its dual group: the Satake isomorphism identifies the “spectrum” of the Hecke algebra at p with semisimple conjugacy classes in ${}^LG(\mathbb{C})$.

This is precisely how the Langlands dual group ${}^LG$ *emerges* from harmonic analysis: it is the group whose representation ring classifies the spherical representations at each prime.

Geometric Langlands: Sheaves as Eigenfunctions

Finally, we mention the geometric counterpart of the Langlands program. Recall from section 10.2 that the Fourier basis $\{e^{ikx}\}$ simultaneously diagonalizes all translation-invariant operators. In the Langlands program, the Hecke operators play this role, and a Hecke eigenform is a simultaneous eigenvector.

The *geometric Langlands conjecture*, proved in 2024 by Gaitsgory, Raskin, and collaborators in the de Rham setting, replaces:

- Functions on Bun_G (the moduli stack of G-bundles on a curve X) with $\mathcal{D}$*-modules* (sheaves).

- Hecke eigenvalues with *Hecke eigensheaves*: a $\mathcal{D}$-module $\mathcal{F}$ on Bun_G is a Hecke eigensheaf with “eigenvalue” E (a ${}^LG$-local system on X) if the Hecke operators act on $\mathcal{F}$ by tensoring with the fibers of E.

The conjecture, now a theorem, establishes an equivalence of categories:

$$D\text{-mod}(\mathrm{Bun}_G) \;\simeq\; \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LS}_{{}^LG})$$

relating $\mathcal{D}$-modules on Bun_G to ind-coherent sheaves on the stack of Langlands parameters. This is the ultimate expression of the principle that eigenbases and spectral decompositions are two sides of a categorical duality—the same principle that, in its simplest form, gives us the Fourier transform.

Summary: The Langlands Rosetta Stone

We conclude by summarizing the dictionary between harmonic analysis and number theory that has emerged. Every concept in the Fourier theory of the preceding section has a direct number-theoretic counterpart:

Harmonic Analysis	**Number Theory / Langlands**
Domain G	Reductive group over a global field
Function space $L^2(G)$	$L^2(G(K)\backslash G(\mathbb{A}_K))$
Translation operator τ_a	Hecke operator T_p
Characters $\chi : G \to S^1$	Automorphic representations π
Peter-Weyl decomposition	Spectral decomposition (Eisenstein series)
Fourier transform	Langlands functorial transfer
Fourier coefficients = eigenvalues	$a_p(f) = \lambda_p(f)$
Pontryagin dual $\hat{G}$	Langlands dual group ${}^LG$
Poisson summation	Selberg-Arthur trace formula
Additive Fourier	Fourier transform on $\mathbb{A}_K$
Multiplicative Fourier (Mellin)	Integration over $\mathbb{A}_K^\times$
Convolution $\to$ multiplication	Euler product of L-function
Functional equation of $\hat{f}$	Functional equation of $L(s, \pi)$
Plancherel theorem (L^2-isometry)	Plancherel formula for $G(\mathbb{A})$
Gelfand transform of $L^1(G)$	Satake isomorphism

The Langlands program is, at its core, the assertion that the harmonic analysis of automorphic forms on reductive groups is *controlled by Galois representations*—that the spectral decomposition of $L^2(G(K)\backslash G(\mathbb{A}))$ is indexed not by an arbitrary set, but by homomorphisms from the absolute Galois group into the Langlands dual group. Every technique in the preceding sections—the duality map J_p, the musical isomorphism, the Fourier transform as a cotangent-space exchange, the Peter-Weyl theorem, the representation-theoretic eigenbasis—reappears in the Langlands program in a deeper and more arithmetic form. The journey from L^p duality to the Langlands correspondence is long, but the underlying principle is one and the same: *find the symmetry, decompose into irreducibles, and read off the invariants.*

11

Distribution Theory

Distributions are a generalization of functions that allow us to give more general solutions than classical solutions, similar to how we found a weak solution last week. Locally integrable functions will be distributions (in the appropriate interpretation), but we'll see that there are generalized solutions that don't even fall under the notion of "function".

More broadly, distribution theory is a further exploration of the "dual theory" philosophy that has been a recurring theme in our study of analysis. In functional analysis, we repeatedly found that passing to the dual space reveals structure invisible in the original space: the Riesz representation theorem identified $(L^p)^*$ with L^q, and the Hahn–Banach theorem showed that the dual space is always rich enough to separate points. Distribution theory pushes this idea one step further. Rather than studying functions directly, we study what they do to test functions—that is, we study the linear functionals they induce. By broadening the class of allowable functionals beyond those induced by honest functions, we arrive at a theory in which every distribution is infinitely differentiable, every linear constant-coefficient PDE has a solution, and the Fourier transform becomes an isomorphism. In short, duality buys us completeness of operations that the classical function spaces lack.

Another use we will soon see is the intimate connection between distributions and the Fourier transform. The theory of tempered distributions will allow us to extend the Fourier transform to objects far beyond L^1 or L^2, giving us a powerful and unified framework for studying partial differential equations, signal processing, and harmonic analysis.

11.1 Intuition

For now, let $f \in L^p$. Then as we know, $(L^p)^* \cong L^q$ is an isometric isomorphism where $1 \leq p < \infty$ and q is the conjugate exponent. In particular, f is mapped to the bounded linear functional

$$g \mapsto \int fg.$$

Using the Lebesgue differentiation theorem, by choosing $\varphi_r = m(B_r)^{-1}\chi_{B_r}$, where B_r is a ball of radius r centered at x, we can recover the value of $f(x)$ for a.e. x, as $\lim_{r\to 0} \int f\varphi_r$. In this way, we can think of L^p as being embedded into $(L^q)^*$: the function f is completely determined (up to a.e. equivalence) by the values of the functional it defines.

Now, let us modify the above by letting f be merely locally integrable (in L^1_{loc}), but restricting our test functions to $\varphi \in C_c^\infty$. Then it is clear that $\varphi \mapsto \int f\varphi$ is still a well-defined linear functional on C_c^∞: the integral converges since φ has compact support and f is integrable on compact sets. Moreover, the pointwise values of f can be recovered a.e. from this functional by the same approximation argument (see Theorem 8.15 in Folland, the Borel Representation Theorem). So locally integrable functions embed faithfully into the space of linear functionals on C_c^∞.

Here is the key observation: unlike in the case for L^p spaces, there are in fact *many* linear functionals on C_c^∞ that are *not* of the form $\varphi \mapsto \int f\varphi$ for any locally integrable f. The most familiar example is the Dirac delta, $\varphi \mapsto \varphi(0)$, which "evaluates at a point"—something no honest L^1_{loc} function can do. These functionals, subject to some mild continuity conditions we will soon specify, will be our "generalized functions", or *distributions.*

11.2 Definitions

We will first rapid-fire some important definitions to be up to speed with terminology. Recall that $C_c^\infty(E)$ where $E \subseteq \mathbb{R}^n$ is the set of all smooth functions whose support is compact and contained in E. If $U \subseteq \mathbb{R}^n$ is open, $C_c^\infty(U)$ is the union of spaces $C_c^\infty(K)$ where K ranges over all compact subsets of U. Each $C_c^\infty(K)$ is a Fréchet space with the topology defined by the family of seminorms

$$\varphi \mapsto \|\partial^\alpha \varphi\|_u \qquad (\alpha \in \{0, 1, 2, \ldots\}^n).$$

Thus, a sequence $\{\varphi_i\}$ converges in $C_c^\infty(K)$ if and only if $\partial^\alpha\varphi_i \to \partial^\alpha\varphi$ uniformly for all multi-indices α. This is a strong form of convergence: we demand that not only the functions themselves, but all of their derivatives of every order, converge uniformly. The completeness of $C_c^\infty(K)$ under this topology is established through Exercise 9 in Folland, Section 5.1 (the key idea is that a uniform limit of smooth functions, where all derivatives also converge uniformly, is again smooth). With these in hand, we take the following definitions:

1. A sequence $\{\varphi_i\}$ in $C_c^\infty(U)$ *converges in* $C_c^\infty(U)$ to φ if there exists a compact set $K \subseteq U$ such that $\{\varphi_i\} \subseteq C_c^\infty(K)$ and $\varphi_i \to \varphi$ in the topology of $C_c^\infty(K)$, that is, $\partial^\alpha\varphi_i \to \partial^\alpha\varphi$ uniformly for all α. Note the requirement that all the φ_i are supported in a *single* compact set K; the supports are not allowed to "escape to infinity".

2. If X is a locally convex topological vector space and $T : C_c^\infty(U) \to X$ is a linear map, T is *continuous* if $T|_{C_c^\infty(K)}$ is continuous for each compact $K \subseteq U$, that is, if $T(\varphi_i) \to T(\varphi)$

whenever $\varphi_i \to \varphi$ in $C_c^\infty(K)$ for some compact $K \subseteq U$. In other words, continuity is tested "one compact set at a time".

3. A linear map $T : C_c^\infty(U) \to C_c^\infty(U')$ is *continuous* if for each compact $K \subseteq U$, there is a compact $K' \subseteq U'$ such that $T(C_c^\infty(K)) \subseteq C_c^\infty(K')$ and T is continuous from $C_c^\infty(K)$ to $C_c^\infty(K')$. The first condition says that T does not spread out the support in an uncontrolled way; the second says that the map is continuous in the Fréchet topology.

With these, we will define a distribution:

Definition 11.2.1: Distribution

A *distribution* on U is a continuous linear functional on $C_c^\infty(U)$. The space of all distributions on U is denoted by $\mathcal{D}'(U)$, and we set $\mathcal{D}' = \mathcal{D}'(\mathbb{R}^n)$. We impose the weak* topology on $\mathcal{D}'(U)$, that is, the topology of pointwise convergence on $C_c^\infty(U)$.

A couple of remarks are in order. First, the $\mathcal{D}'$ notation comes from the fact that Schwartz used the notation $\mathcal{D}$ for C_c^∞, and so $\mathcal{D}'$ denotes the continuous dual of C_c^∞—the space of all continuous linear functionals on it. Next, there is in fact a locally convex topology on $C_c^\infty(U)$ (the so-called *inductive limit topology*) with respect to which sequential convergence is given by the convergence defined above, and the continuity of linear maps $T : C_c^\infty \to X$ and $T : C_c^\infty \to C_c^\infty$ agrees with the definitions in the list above. However, defining this topology rigorously requires the machinery of inductive limits of locally convex spaces, and the construction is somewhat involved without contributing much additional insight to the current exposition of distributions, so it will be omitted. The interested reader may consult Rudin's *Functional Analysis*, Chapter 6, for the full treatment. Furthermore, when we talk about convergence of distributions $F_i \to F$, we will always mean pointwise convergence: $\langle F_i, \varphi\rangle \to \langle F, \varphi\rangle$ for every $\varphi \in C_c^\infty(U)$.

Before giving some examples of distributions, we will remind the reader how to check for continuity of a linear functional in the setting of Fréchet spaces. The following proposition gives a very concrete, "hands-on" criterion that avoids all the topological machinery.

Proposition 11.2.2: Distribution Continuity Check

Let $u : C_c^\infty(U) \to \mathbb{R}$ (or $\mathbb{C}$) be linear. Then $u \in \mathcal{D}'(U)$ if and only if for every compact set $K \subseteq U$, there exist constants $C_K > 0$ and $N_K \in \mathbb{N}$ such that

$$|\langle u, \varphi\rangle| \le C_K \sum_{|\alpha| \le N_K} \|\partial^\alpha \varphi\|_{L^\infty}$$

for all $\varphi \in C_c^\infty(K)$.

***Proof*:**
Suppose first that the estimate holds. We wish to show that u is continuous on each $C_c^\infty(K)$. Let $\varphi_i \to \varphi$ in $C_c^\infty(K)$. Then for every multi-index α, $\partial^\alpha \varphi_i \to \partial^\alpha \varphi$ uniformly, which means $\|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \to 0$ for each α. Applying the estimate to $\varphi_i - \varphi$:

$$|\langle u, \varphi_i\rangle - \langle u, \varphi\rangle| = |\langle u, \varphi_i - \varphi\rangle| \le C_K \sum_{|\alpha| \le N_K} \|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \to 0,$$

so $\langle u, \varphi_i \rangle \to \langle u, \varphi \rangle$, and u is continuous on $C_c^\infty(K)$. Since K was arbitrary, u is a distribution.

Conversely, suppose u is continuous on $C_c^\infty(K)$ but no such estimate holds. Then for each $j \in \mathbb{N}$, there exists $\varphi_j \in C_c^\infty(K)$ with

$$|\langle u, \varphi_j \rangle| > j \sum_{|\alpha| \le j} \|\partial^\alpha \varphi_j\|_{L^\infty}.$$

Set $\psi_j = \varphi_j / |\langle u, \varphi_j \rangle|$. Then $\langle u, \psi_j \rangle = 1$ for all j (taking absolute values and adjusting the sign if needed, we can arrange $|\langle u, \psi_j \rangle| = 1$). But for $|\alpha| \le j$:

$$\|\partial^\alpha \psi_j\|_{L^\infty} < \frac{1}{j},$$

so $\psi_j \to 0$ in $C_c^\infty(K)$. By continuity of u, we should have $\langle u, \psi_j \rangle \to \langle u, 0 \rangle = 0$, contradicting $|\langle u, \psi_j \rangle| = 1$.

The way I like to think about the continuity check is as follows: we will soon see that there are many distribution spaces, for example the Schwartz space $\mathcal{S}$ will have a parallel $\mathcal{S}'$ (the *tempered distributions*). What the continuity condition tells us is that a distribution respects the convergence structure of its test function space. If $\varphi_j \to \varphi$ in the test function topology, then $\langle u, \varphi_j \rangle \to \langle u, \varphi \rangle$. Hence, we can translate convergence information from the test function space to information about the distribution. This principle—that convergence in the test function space implies convergence of the pairing—is the thread that ties all distribution spaces together.

Example 11.2.3: Distributions

1. *Locally integrable functions.* For every $f \in L^1_{\text{loc}}(U)$, we can define a distribution on U, namely the functional $\varphi \mapsto \int f\varphi$. This is indeed continuous: if $\varphi \in C_c^\infty(K)$, then

$$\left| \int f\varphi \right| \le \|f\|_{L^1(K)} \|\varphi\|_{L^\infty},$$

so the estimate in Proposition 11.2.2 holds with $N_K = 0$ and $C_K = \|f\|_{L^1(K)}$. Moreover, two functions f and g define the same distribution precisely when they are equal a.e. (this follows from the fact that C_c^∞ is dense in L^q for the appropriate q, or more directly from the Lebesgue differentiation theorem). In this way, $L^1_{\text{loc}}(U)$ embeds into $\mathcal{D}'(U)$, and we often identify a locally integrable function with the distribution it defines.

2. *Radon measures.* Every Radon measure μ on U defines a distribution $\varphi \mapsto \int \varphi \, d\mu$. The continuity check is similar: for $\varphi \in C_c^\infty(K)$, we have $|\int \varphi \, d\mu| \le |\mu|(K) \|\varphi\|_{L^\infty}$, where $|\mu|(K)$ is the total variation of μ on K. Again, this satisfies the estimate with $N_K = 0$. Note that this generalizes the previous example, since a locally integrable function f defines a Radon measure $d\mu = f\, dm$ via the Radon–Nikodym theorem.

3. *Derivatives of point masses.* If $x_0 \in U$ and α is a multi-index, the map $\varphi \mapsto \partial^\alpha \varphi(x_0)$ is a distribution. The continuity estimate is immediate: $|\partial^\alpha \varphi(x_0)| \le \|\partial^\alpha \varphi\|_{L^\infty}$. This distribution does not arise from a locally integrable function: no L^1_{loc} function can "see" the value of a derivative at a single point. It arises from a Radon measure μ precisely when $\alpha = 0$, in which case $\mu = \delta_{x_0}$ is the point mass at x_0. For $|\alpha| \ge 1$, these distributions are genuinely "beyond" measures—they are the first examples of distributions that cannot be represented by any kind of integration.

As mentioned earlier, we often associate a function f with the distribution it defines, and even label the distribution by f. For this reason, it might be confusing to write $f(\varphi)$ to mean the distribution defined by f evaluated on φ, and so for historical reasons we use the bracket notation $\langle f, \varphi\rangle$ for the distributional pairing. This notation is suggestive of an inner product (and indeed, when $f \in L^2$ and $\varphi \in C_c^\infty$, it literally is the L^2 inner product), though the reader should keep in mind that the pairing is between two different spaces.

Let's next establish some common notation:

1. First, it is often enough that we will reflect a function. Thus, if we put a tilde over a function, we mean:
$$\tilde{\varphi}(x) = \varphi(-x).$$
This notation will appear frequently in the context of convolutions, where the reflection arises naturally from the change of variables $y \mapsto -y$.

2. Second, the point-mass (or Dirac delta) is a very common distribution, which we will denote by δ:
$$\langle \delta, \varphi\rangle = \varphi(0).$$
More generally, δ_{x_0} denotes the point mass at x_0, defined by $\langle \delta_{x_0}, \varphi\rangle = \varphi(x_0)$. When $x_0 = 0$, we simply write δ.

As an example of the use of the delta function, and to illustrate how convergence of distributions works in practice:

Proposition 11.2.4: Convergence To Delta Function

Let $f \in L^1(\mathbb{R}^n)$ with $\int f = a$, and for $t > 0$, let $f_t(x) = t^{-n} f(t^{-1}x)$. Then $f_t \to a\delta$ in $\mathcal{D}'$ as $t \to 0$.

Before the proof, let us unpack what this says. The rescaling $f_t(x) = t^{-n} f(t^{-1}x)$ concentrates the mass of f near the origin as $t \to 0$: the factor t^{-n} ensures that $\int f_t = \int f = a$ for all t, while the argument $t^{-1}x$ compresses the support. So f_t becomes a taller and narrower spike near 0, with constant total mass a. The proposition says that in the distributional sense, this spike converges to a times the delta function—exactly what our intuition would predict.

***Proof*:**
Let $\varphi \in C_c^\infty$. By Theorem 8.15 (Folland), we have:
$$\langle f_t, \varphi\rangle = \int f_t(x)\varphi(x)\,dx = \int t^{-n} f(t^{-1}x)\varphi(x)\,dx.$$
Performing the substitution $y = t^{-1}x$ (so $x = ty$ and $dx = t^n\,dy$):
$$\int t^{-n} f(t^{-1}x)\varphi(x)\,dx = \int f(y)\varphi(ty)\,dy.$$
Now we recognize the right-hand side as $f_t * \tilde{\varphi}(0)$. As $t \to 0$, $\varphi(ty) \to \varphi(0)$ pointwise, and since $|\varphi(ty)| \le \|\varphi\|_{L^\infty}$ and $f \in L^1$, the dominated convergence theorem gives:
$$\langle f_t, \varphi\rangle = \int f(y)\varphi(ty)\,dy \to \varphi(0)\int f(y)\,dy = a\varphi(0) = a\langle \delta, \varphi\rangle,$$
completing the proof.

11.2.1 Equality and Support

For equality of distributions, note that it makes sense to say that $F = G$ on an open set $V \subseteq U$ if and only if $\langle F, \varphi \rangle = \langle G, \varphi \rangle$ for all $\varphi \in C_c^\infty(V)$. If F and G are continuous functions, this implies they are pointwise equal on V, and if F and G are only locally integrable, this means they are equal a.e. on V.

Since a function in $C_c^\infty(V_1 \cup V_2)$ need not be supported in either V_1 or V_2 individually, it is not immediately obvious that if $F = G$ on V_1 and on V_2, then $F = G$ on $V_1 \cup V_2$. However, this is indeed the case, thanks to partitions of unity:

Proposition 11.2.5: Equality On Support

Let $\{V_\alpha\}$ be a collection of open subsets of U and let $V = \bigcup_\alpha V_\alpha$. If $F, G \in \mathcal{D}'(U)$ and $F = G$ on each V_α, then $F = G$ on V.

***Proof*:**
It suffices to show that $F - G = 0$ on V, so we may assume $G = 0$ and show that $F = 0$ on V. Let $\varphi \in C_c^\infty(V)$. Then $\operatorname{supp}(\varphi)$ is a compact subset of $V = \bigcup_\alpha V_\alpha$. By compactness, finitely many of the V_α cover $\operatorname{supp}(\varphi)$, say $V_{\alpha_1}, \ldots, V_{\alpha_m}$. Let $\{\chi_1, \ldots, \chi_m\}$ be a smooth partition of unity subordinate to this finite cover, so that each $\chi_j \in C_c^\infty(V_{\alpha_j})$, $0 \leq \chi_j \leq 1$, and $\sum_{j=1}^m \chi_j = 1$ on a neighborhood of $\operatorname{supp}(\varphi)$. Then $\varphi = \sum_{j=1}^m \chi_j \varphi$, and each $\chi_j \varphi \in C_c^\infty(V_{\alpha_j})$. Since $F = 0$ on each V_{α_j}, we have $\langle F, \chi_j \varphi \rangle = 0$ for each j, and by linearity:

$$\langle F, \varphi \rangle = \sum_{j=1}^m \langle F, \chi_j \varphi \rangle = 0.$$

Since $\varphi \in C_c^\infty(V)$ was arbitrary, $F = 0$ on V.

As a consequence of Proposition 11.2.5, if $F \in \mathcal{D}'(U)$, there is a largest open set on which $F = 0$, namely the union of all open subsets of U on which $F = 0$. The complement of this set in U is called the *support* of F, denoted $\operatorname{supp}(F)$. When F is a locally integrable function, this agrees with the usual notion of support (up to a set of measure zero). The set of all distributions on U with compact support is denoted $\mathcal{E}'(U)$; it turns out (though we will not prove it here) that $\mathcal{E}'(U)$ is precisely the dual of $C^\infty(U)$—we no longer need to require that the test functions have compact support, because the distribution itself is compactly supported.

11.3 Operations on Distributions

One of the amazing facts about distributions is that even though they are not "functions" in the classical sense, we can perform many of the same operations on them: differentiation, multiplication by a smooth function, convolution, and translation! To achieve this, we use a single unifying principle that we now describe.

11.3.1 The Transpose Principle

Suppose U and V are open subsets of $\mathbb{R}^n$ and T is a linear map from some subspace $\mathcal{V}$ of $L^1_{\mathrm{loc}}(U)$ into $L^1_{\mathrm{loc}}(V)$. Suppose that there is another linear map $T' : C_c^\infty(V) \to C_c^\infty(U)$ (called the *transpose* of T) such that

$$\int (Tf)\varphi = \int f(T'\varphi) \qquad (f \in \mathcal{V},\ \varphi \in C_c^\infty(V)).$$

Suppose further that T' is continuous in the sense defined earlier (i.e., for each compact $K' \subseteq V$, there exists a compact $K \subseteq U$ with $T'(C_c^\infty(K')) \subseteq C_c^\infty(K)$ and T' continuous from $C_c^\infty(K')$ to $C_c^\infty(K)$). Then T can be *extended* to a map from $\mathcal{D}'(U)$ to $\mathcal{D}'(V)$, still denoted by T, by defining:

$$\langle TF, \varphi\rangle = \langle F, T'\varphi\rangle \qquad (F \in \mathcal{D}'(U),\ \varphi \in C_c^\infty(V)).$$

Why does this work? First, TF is a well-defined linear functional on $C_c^\infty(V)$ since $T'\varphi \in C_c^\infty(U)$ and F is a linear functional on $C_c^\infty(U)$. Second, TF is continuous: if $\varphi_i \to \varphi$ in $C_c^\infty(V)$, then $T'\varphi_i \to T'\varphi$ in $C_c^\infty(U)$ by continuity of T', and so $\langle TF, \varphi_i\rangle = \langle F, T'\varphi_i\rangle \to \langle F, T'\varphi\rangle = \langle TF, \varphi\rangle$ since F is a distribution. Third, this definition is *consistent* with the original map on functions: if F is a locally integrable function in $\mathcal{V}$, then $\langle TF, \varphi\rangle = \langle F, T'\varphi\rangle = \int f(T'\varphi) = \int (Tf)\varphi$, which is exactly the distribution defined by the function Tf.

The continuity of T' also guarantees that the extended map $T : \mathcal{D}'(U) \to \mathcal{D}'(V)$ is continuous with respect to the weak* topology on distributions: if $F_\alpha \to F$ in $\mathcal{D}'(U)$, then for any $\varphi \in C_c^\infty(V)$, $\langle TF_\alpha, \varphi\rangle = \langle F_\alpha, T'\varphi\rangle \to \langle F, T'\varphi\rangle = \langle TF, \varphi\rangle$, so $TF_\alpha \to TF$ in $\mathcal{D}'(V)$.

This "transpose trick" is the engine behind all of the operations we now define.

11.3.2 Operations

1. *(Differentiation)* Let $Tf = \partial^\alpha f$, defined initially on $C^{|\alpha|}(U)$. If $\varphi \in C_c^\infty(U)$, integration by parts gives

   $$\int (\partial^\alpha f)\varphi = (-1)^{|\alpha|} \int f(\partial^\alpha \varphi);$$

   there are no boundary terms since φ has compact support. Hence $T' = (-1)^{|\alpha|}\partial^\alpha|_{C_c^\infty(U)}$, and we can define the *distributional derivative* $\partial^\alpha F \in \mathcal{D}'(U)$ of any $F \in \mathcal{D}'(U)$ by:

   $$\langle \partial^\alpha F, \varphi\rangle := (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle.$$

   By this procedure, every distribution is infinitely differentiable! We can define the derivatives of arbitrary locally integrable functions, even when they are not differentiable in the classical sense. When F happens to be a $C^{|\alpha|}$ function, the distributional derivative agrees with the classical one (by integration by parts), so this truly is a generalization. We will compute the distributional derivatives of some important locally integrable functions shortly.

   A particularly important consequence: distributional derivatives commute. That is, $\partial^\alpha\partial^\beta F = \partial^{\alpha+\beta} F = \partial^\beta\partial^\alpha F$ for all distributions F and all multi-indices α, β. This is immediate from the definition and the fact that classical derivatives of smooth functions commute.

2. *(Multiplication by a smooth function)* Let $\psi \in C^\infty(U)$ and define $Tf = \psi f$. Then $T' = T|_{C_c^\infty(U)}$ (that is, $T'\varphi = \psi\varphi$), since $\int (\psi f)\varphi = \int f(\psi\varphi)$. Evidently T' maps $C_c^\infty(U)$ to $C_c^\infty(U)$ and is continuous (since multiplication by a smooth function preserves both support and uniform

convergence of derivatives, by the Leibniz rule). Thus we define $\psi F \in \mathcal{D}'(U)$ for any $F \in \mathcal{D}'(U)$ by:

$$\langle \psi F, \varphi \rangle := \langle F, \psi\varphi \rangle.$$

Note that we require ψ to be C^∞; multiplying a distribution by a merely continuous or L^∞ function is in general not well-defined. The Leibniz rule extends to distributions: $\partial^\alpha(\psi F) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} (\partial^{\alpha-\beta}\psi)(\partial^\beta F)$, which is proved by testing against $\varphi \in C_c^\infty$ and using the classical Leibniz rule on the test function side.

3. *(Translation)* Let $y \in \mathbb{R}^n$ and let $V = U + y = \{x + y : x \in U\}$. Take $T = \tau_y$ (recall that $\tau_y f(x) = f(x - y)$). Via the substitution $x \mapsto x + y$, we get

$$\int f(x-y)\varphi(x)\,dx = \int f(x)\varphi(x+y)\,dx,$$

and so $T' = \tau_{-y}|_{C_c^\infty(V)}$. This is evidently continuous, and so for a distribution $F \in \mathcal{D}'(U)$, we define:

$$\langle \tau_y F, \varphi \rangle := \langle F, \tau_{-y}\varphi \rangle.$$

As a quick sanity check: $\langle \tau_y \delta, \varphi \rangle = \langle \delta, \tau_{-y}\varphi \rangle = (\tau_{-y}\varphi)(0) = \varphi(y) = \langle \delta_y, \varphi \rangle$. So translating the delta function by y gives the point mass at y, exactly as expected.

4. *(Composition with linear maps)* Let S be an invertible linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. Let $V = S^{-1}(U)$ and $Tf = f \circ S$. By the change of variables formula (see Chapter 4), for $\varphi \in C_c^\infty(V)$:

$$\int (f \circ S)(x)\varphi(x)\,dx = \int f(y)\varphi(S^{-1}y)|\det S|^{-1}\,dy.$$

So $T'\varphi = |\det(S)|^{-1}\varphi \circ S^{-1}$. For $F \in \mathcal{D}'(U)$, we define:

$$\langle F \circ S, \varphi \rangle := |\det S|^{-1}\langle F, \varphi \circ S^{-1} \rangle.$$

As a special case, taking $S = -I$ (the map $x \mapsto -x$) gives $\langle \tilde{F}, \varphi \rangle = \langle F, \tilde{\varphi} \rangle$, which defines the *reflection* of a distribution.

5. *(Convolution I: as a function)* Let $\psi \in C_c^\infty$, and let

$$V = \{x : x - y \in U \text{ for all } y \in \operatorname{supp}(\psi)\}.$$

Note that V is open (but may be empty). If $f \in L^1_{\mathrm{loc}}(U)$, then the integral

$$f * \psi(x) = \int f(x-y)\psi(y)\,dy = \int f(y)\psi(x-y)\,dy = \langle f, \tau_x\tilde{\psi} \rangle$$

is well-defined for all $x \in V$. Since the last expression involves only the distributional pairing, we can define for $F \in \mathcal{D}'(U)$:

$$F * \psi(x) = \langle F, \tau_x\tilde{\psi} \rangle.$$

This is well-defined because $\tau_x\tilde{\psi} \in C_c^\infty(U)$ for $x \in V$. Moreover, $F * \psi$ is a *smooth* function on V: since $\tau_x\tilde{\psi} \to \tau_{x_0}\tilde{\psi}$ in C_c^∞ as $x \to x_0$ (and similarly for all derivatives), continuity of F gives $F * \psi(x) \to F * \psi(x_0)$. More precisely, one can show by differentiating under the pairing that $\partial^\alpha(F * \psi) = F * (\partial^\alpha\psi)$ for all multi-indices α, which establishes that $F * \psi \in C^\infty(V)$.

A pertinent example is the following:

$$\delta * \psi(x) = \langle \delta, \tau_x \tilde{\psi} \rangle = \tau_x \tilde{\psi}(0) = \tilde{\psi}(-x) = \psi(x),$$

so δ is the identity element for convolution! This is one of the most important properties of the delta function and lies at the heart of the theory of Green's functions: if L is a differential operator and E is a distribution satisfying $LE = \delta$ (a *fundamental solution*), then $u = E * f$ formally solves $Lu = f$.

6. *(Convolution II: as a distribution)* Let ψ, $\tilde{\psi}$, and V be as above. If $f \in L^1_{\text{loc}}(U)$ and $\varphi \in C_c^\infty(V)$, then by Fubini's theorem:

$$\int (f*\psi)(x)\varphi(x)\,dx = \iint f(y)\psi(x-y)\varphi(x)\,dy\,dx = \int f(y)\left(\int \psi(x-y)\varphi(x)\,dx\right)dy = \int f(y)(\varphi*\tilde{\psi})(y)\,dy.$$

Thus $Tf = f * \psi$ maps $L^1_{\text{loc}}(U)$ to $L^1_{\text{loc}}(V)$ with transpose $T'\varphi = \varphi * \tilde{\psi}$. For a distribution $F \in \mathcal{D}'(U)$, we therefore get:

$$\langle F * \psi, \varphi \rangle = \langle F, \varphi * \tilde{\psi} \rangle.$$

We should verify that the two definitions of convolution are consistent:

Proposition 11.3.1: Consistency of Convolution

Let $F \in \mathcal{D}'(U)$ and $\psi \in C_c^\infty$, and let V be as above. Then the smooth function $x \mapsto \langle F, \tau_x \tilde{\psi} \rangle$ (Convolution I) and the distribution defined by $\langle F, \varphi * \tilde{\psi} \rangle$ (Convolution II) agree, in the sense that Convolution II is the distribution defined by the locally integrable function from Convolution I.

***Proof*:**
We need to show that for all $\varphi \in C_c^\infty(V)$:

$$\int \langle F, \tau_x \tilde{\psi} \rangle \varphi(x)\,dx = \langle F, \varphi * \tilde{\psi} \rangle.$$

The left-hand side can be written as the limit of Riemann sums. More precisely, for φ supported in a compact set $K' \subseteq V$, the function $x \mapsto \tau_x \tilde{\psi}$ is a continuous map from K' into $C_c^\infty(K)$ for some compact $K \subseteq U$. One checks (using the Riemann sum approximation and continuity of F) that the integral of a continuous $C_c^\infty(K)$-valued function against a scalar function can be "passed inside" the distributional pairing. But the integral $\int \tau_x \tilde{\psi}\, \varphi(x)\,dx$, computed pointwise, is exactly the function $y \mapsto \int \psi(x-y)\varphi(x)\,dx = (\varphi * \tilde{\psi})(y)$. Hence:

$$\int \langle F, \tau_x \tilde{\psi} \rangle \varphi(x)\,dx = \left\langle F, \int \tau_x \tilde{\psi}\, \varphi(x)\,dx \right\rangle = \langle F, \varphi * \tilde{\psi} \rangle.$$

11.4 Density Results

Next, we will show that smooth compactly supported functions $C_c^\infty(U)$ are dense in $\mathcal{D}'(U)$. This is a remarkable fact: it says that every distribution, no matter how singular, can be approximated (in the weak* topology) by honest smooth functions. We begin with a preparatory lemma.

Lemma 11.4.1: Build-Up For Density

Suppose that $\varphi \in C_c^\infty$, $\psi \in C_c^\infty$ with $\int \psi = 1$, and let $\psi_t(x) = t^{-n}\psi(t^{-1}x)$ for $t > 0$. Then:

1. Given any open neighborhood U of $\mathrm{supp}(\varphi)$, we have $\mathrm{supp}(\varphi * \psi_t) \subseteq U$ for sufficiently small $t > 0$.
2. $\varphi * \psi_t \to \varphi$ in C_c^∞ as $t \to 0$.

***Proof*:**
(a) Note that $\mathrm{supp}(\varphi * \psi_t) \subseteq \mathrm{supp}(\varphi) + \mathrm{supp}(\psi_t) = \mathrm{supp}(\varphi) + t \cdot \mathrm{supp}(\psi)$. Since $\mathrm{supp}(\varphi)$ is compact, the set $\mathrm{supp}(\varphi) + t \cdot \mathrm{supp}(\psi)$ is the Minkowski sum of $\mathrm{supp}(\varphi)$ with a set of diameter shrinking to 0 as $t \to 0$. Since U is an open neighborhood of the compact set $\mathrm{supp}(\varphi)$, there exists $\varepsilon > 0$ such that the ε-neighborhood of $\mathrm{supp}(\varphi)$ is contained in U. For t small enough that $t \cdot \mathrm{supp}(\psi)$ lies within a ball of radius ε, we get $\mathrm{supp}(\varphi * \psi_t) \subseteq U$.

(b) We need to show that $\partial^\alpha(\varphi * \psi_t) \to \partial^\alpha \varphi$ uniformly for every multi-index α. Since differentiation commutes with convolution, $\partial^\alpha(\varphi * \psi_t) = (\partial^\alpha \varphi) * \psi_t$. So it suffices to show that $g * \psi_t \to g$ uniformly whenever $g \in C_c^\infty$. We have:

$$(g * \psi_t)(x) - g(x) = \int [g(x-y) - g(x)]\psi_t(y)\, dy = \int [g(x-tz) - g(x)]\psi(z)\, dz,$$

using the substitution $y = tz$. Since g is smooth and compactly supported, it is uniformly continuous, so for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|g(x-tz) - g(x)| < \varepsilon$ whenever $t|z| < \delta$. For $z \in \mathrm{supp}(\psi)$ and t small enough, this is satisfied, giving $|g * \psi_t(x) - g(x)| \leq \varepsilon \int |\psi|$ for all x. By part (a), all the $\varphi * \psi_t$ are eventually supported in a single compact set, so the convergence is in C_c^∞.

Proposition 11.4.2: Density Of Smooth Compactly Supported Functions In Distributions

Let $U \subseteq \mathbb{R}^n$ be open. Then $C_c^\infty(U)$ is dense in $\mathcal{D}'(U)$ in the weak* topology.

***Proof*:**
Let $F \in \mathcal{D}'(U)$; we must find a net (or sequence) $\{\varphi_i\}$ in $C_c^\infty(U)$ such that $\langle \varphi_i, \eta\rangle \to \langle F, \eta\rangle$ for all $\eta \in C_c^\infty(U)$, where $\langle \varphi_i, \eta\rangle = \int \varphi_i \eta$.

Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\int \psi = 1$ and let $\psi_t(x) = t^{-n}\psi(t^{-1}x)$. We also need a family of cutoff functions: let $\{\chi_j\}_{j=1}^\infty$ be a sequence in $C_c^\infty(U)$ with $0 \leq \chi_j \leq 1$ and $\chi_j \to 1$ pointwise on U (e.g., take $\chi_j = 1$ on $\{x \in U : |x| \leq j \text{ and } \mathrm{dist}(x, U^c) \geq 1/j\}$ and smooth it out). Consider the distributions $F_j = \chi_j F$ (defined by $\langle F_j, \eta\rangle = \langle F, \chi_j \eta\rangle$). Each F_j has compact support in U, and $F_j \to F$ in $\mathcal{D}'(U)$ since for fixed $\eta \in C_c^\infty(U)$, $\chi_j \eta \to \eta$ in $C_c^\infty(U)$ for large j (eventually $\chi_j = 1$ on $\mathrm{supp}(\eta)$).

Now consider $F_j * \psi_t$. By our earlier discussion, this is a C^∞ function. For t sufficiently small, it has compact support in U (since F_j has compact support in U and the convolution only slightly enlarges the support). By Proposition 11.2.4 and the properties of convolution, $F_j * \psi_t \to F_j$ in

$\mathcal{D}'$ as $t \to 0$: for any $\eta \in C_c^\infty(U)$,

$$\langle F_j * \psi_t, \eta \rangle = \langle F_j, \eta * \tilde{\psi}_t \rangle \to \langle F_j, \eta \rangle$$

by Lemma **??**, since $\eta * \tilde{\psi}_t \to \eta$ in C_c^∞. By a diagonal argument (choosing $t_j \to 0$ slowly enough), we obtain a sequence $g_j = F_j * \psi_{t_j} \in C_c^\infty(U)$ with $g_j \to F$ in $\mathcal{D}'(U)$.

11.5 Distributional Derivatives of Classical Functions

We promised earlier that the distributional derivative lets us differentiate functions that are not classically differentiable, and now we deliver on that promise. The computations in this section are among the most important examples in all of distribution theory, and the reader should internalize them thoroughly.

Recall that for $F \in \mathcal{D}'(\mathbb{R}^n)$ and a multi-index α, the distributional derivative is defined by

$$\langle \partial^\alpha F, \varphi \rangle = (-1)^{|\alpha|} \langle F, \partial^\alpha \varphi \rangle.$$

Example 11.5.1: Derivative Of The Heaviside Function

Let $H : \mathbb{R} \to \mathbb{R}$ be the Heaviside step function, $H(x) = \mathbf{1}_{[0,\infty)}(x)$. Then $H \in L^1_{\text{loc}}(\mathbb{R})$ and hence defines a distribution. Classically, H is differentiable everywhere except at $x = 0$, where it has a jump discontinuity of height 1. Let us compute H' in the distributional sense. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\langle H', \varphi \rangle = -\langle H, \varphi' \rangle = -\int_0^\infty \varphi'(x)\,dx = -[\varphi(x)]_{x=0}^{x=\infty} = -\big(\underbrace{\varphi(\infty)}_{=0} - \varphi(0)\big) = \varphi(0) = \langle \delta, \varphi \rangle.$$

Therefore $H' = \delta$ in $\mathcal{D}'(\mathbb{R})$. This is the prototypical example: the distributional derivative detects the jump discontinuity and places a point mass there. More generally, if f is piecewise C^1 with jump discontinuities of height c_j at points x_j, then $f' = f'_{\text{classical}} + \sum_j c_j \delta_{x_j}$ in $\mathcal{D}'$.

Example 11.5.2: Derivative Of $|x|$ And The Sign Function

Consider $f(x) = |x|$ on $\mathbb{R}$. This function is continuous but not differentiable at $x = 0$. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\begin{aligned}
\langle f', \varphi \rangle = -\langle f, \varphi' \rangle &= -\int_{-\infty}^\infty |x| \varphi'(x)\,dx \\
&= -\int_{-\infty}^0 (-x)\varphi'(x)\,dx - \int_0^\infty x\varphi'(x)\,dx \\
&= \int_{-\infty}^0 x\varphi'(x)\,dx - \int_0^\infty x\varphi'(x)\,dx.
\end{aligned}$$

Integrating by parts on each piece (the boundary terms at $\pm\infty$ vanish since φ has compact support, and the boundary terms at 0 cancel):

$$= \Big[x\varphi(x)\Big]_{-\infty}^0 - \int_{-\infty}^0 \varphi(x)\,dx - \Big[x\varphi(x)\Big]_0^\infty + \int_0^\infty \varphi(x)\,dx = -\int_{-\infty}^0 \varphi + \int_0^\infty \varphi = \int_{-\infty}^\infty \operatorname{sgn}(x)\varphi(x)\,dx,$$

where $\operatorname{sgn}(x) = \mathbf{1}_{(0,\infty)}(x) - \mathbf{1}_{(-\infty,0)}(x)$ is the sign function. Hence $|x|' = \operatorname{sgn}(x)$ in $\mathcal{D}'$. Note that $\operatorname{sgn}(x) = 2H(x) - 1$, which is consistent: differentiating again gives $|x|'' = \operatorname{sgn}'(x) = 2H'(x) = 2\delta$.

The next computation requires introducing an important distribution that is not given by a locally integrable function in the standard sense.

Definition 11.5.3: Principal Value Distribution

The *principal value of* $1/x$, denoted p.v.$(1/x)$, is the distribution on $\mathbb{R}$ defined by

$$\left\langle \text{p.v.}\frac{1}{x}, \varphi \right\rangle = \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.$$

We should verify that the limit exists and that this defines a distribution. Since φ is smooth, we can write $\varphi(x) = \varphi(0) + x\psi(x)$ where $\psi(x) = \int_0^1 \varphi'(tx)\, dt$ is also smooth. Then

$$\int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx = \underbrace{\varphi(0)\int_{|x|>\varepsilon} \frac{dx}{x}}_{=0 \text{ by symmetry}} + \int_{|x|>\varepsilon} \psi(x)\, dx \to \int_{-\infty}^{\infty} \psi(x)\, dx$$

as $\varepsilon \to 0$. Since ψ is smooth and φ has compact support, the integral converges. The continuity estimate $|\langle \text{p.v.}(1/x), \varphi\rangle| \le C(\|\varphi\|_\infty + \|\varphi'\|_\infty)$ on any fixed compact set follows from this decomposition, confirming that p.v.$(1/x) \in \mathcal{D}'(\mathbb{R})$. Notice that the order of this distribution is 1 (it requires control of one derivative), unlike the distributions arising from locally integrable functions, which have order 0.

Example 11.5.4: Derivative Of $\log|x|$

The function $\log|x|$ is locally integrable on $\mathbb{R}$ (since $\int_{-1}^1 |\log|x||\, dx < \infty$) and hence defines a distribution. We claim that its distributional derivative is p.v.$(1/x)$. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\begin{aligned}\langle (\log|x|)', \varphi\rangle = -\langle \log|x|, \varphi'\rangle &= -\int_{-\infty}^{\infty} \log|x|\, \varphi'(x)\, dx \\ &= -\lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \log|x|\, \varphi'(x)\, dx.\end{aligned}$$

Integrating by parts (with the boundary terms at $\pm\infty$ vanishing):

$$\begin{aligned} &= -\lim_{\varepsilon\to 0^+} \left(\Big[\log|x|\, \varphi(x)\Big]_{|x|=\varepsilon}^{|x|=\infty} - \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx \right) \\ &= -\lim_{\varepsilon\to 0^+} \big(-\log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)]\big) + \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx. \end{aligned}$$

Since $\varphi(\varepsilon) + \varphi(-\varepsilon) \to 2\varphi(0)$ and $\varepsilon\log\varepsilon \to 0$, the first term requires more care. We have $\log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)] = 2\varphi(0)\log\varepsilon + O(\varepsilon\log\varepsilon)$. But this term appears with a minus sign outside, so:

$$= \lim_{\varepsilon\to 0^+} \log\varepsilon\, [\varphi(\varepsilon) + \varphi(-\varepsilon)] + \lim_{\varepsilon\to 0^+} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.$$

For the boundary term, note that $[\log|x|\, \varphi(x)]_{-\infty}^{-\varepsilon} + [\log|x|\, \varphi(x)]_{\varepsilon}^{\infty} = -\log\varepsilon\, \varphi(-\varepsilon) - \log\varepsilon\, \varphi(\varepsilon)$.

Thus:

$$= \lim_{\varepsilon\to 0^+}\Big[\log\varepsilon\big(\varphi(\varepsilon)+\varphi(-\varepsilon)\big) + \int_{|x|>\varepsilon}\frac{\varphi(x)}{x}\,dx\Big].$$

Since $\varphi(\varepsilon)+\varphi(-\varepsilon) = 2\varphi(0)+O(\varepsilon^2)$ (the odd part cancels) and $\int_\varepsilon^1 dx/x = -\log\varepsilon$ (with the symmetric contribution from $[-1,-\varepsilon]$ cancelling), the $\log\varepsilon$ terms cancel exactly, and we are left with

$$\langle(\log|x|)', \varphi\rangle = \lim_{\varepsilon\to 0^+}\int_{|x|>\varepsilon}\frac{\varphi(x)}{x}\,dx = \left\langle \text{p.v.}\frac{1}{x}, \varphi\right\rangle.$$

Therefore $(\log|x|)' = \text{p.v.}(1/x)$ in $\mathcal{D}'(\mathbb{R})$.

It is also instructive to see how multiplication interacts with the principal value:

Proposition 11.5.5: Multiplication By x

As distributions on $\mathbb{R}$, we have $x\cdot\text{p.v.}(1/x) = 1$, that is, the distribution $x\cdot\text{p.v.}(1/x)$ agrees with the constant function 1.

***Proof*:**
For $\varphi\in C_c^\infty(\mathbb{R})$:

$$\langle x\cdot\text{p.v.}(1/x),\, \varphi\rangle = \left\langle \text{p.v.}\frac{1}{x},\, x\varphi\right\rangle = \lim_{\varepsilon\to 0^+}\int_{|x|>\varepsilon}\frac{x\varphi(x)}{x}\,dx = \int_{-\infty}^{\infty}\varphi(x)\,dx = \langle 1,\varphi\rangle,$$

since φ is integrable and the integrand $\varphi(x)$ has no singularity.

This is a distributional identity with no classical analogue: the "function" $1/x$ is not locally integrable, so the product $x\cdot(1/x)$ is not defined pointwise in L^1_{loc}. But the distributional version makes perfect sense and gives the expected answer. On the other hand, note that $x\cdot\delta = 0$ (since $\langle x\delta,\varphi\rangle = \langle\delta, x\varphi\rangle = 0$), so x is a zero-divisor in $\mathcal{D}'$—a phenomenon that does not occur for ordinary functions.

11.6 Tempered Distributions

In the Fourier transform chapter, we established the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ as the natural domain for the Fourier transform: the map $\varphi\mapsto\hat{\varphi}$ is an isomorphism of $\mathcal{S}$ onto itself. We now extend this theory to distributions, obtaining a powerful framework that allows us to take the Fourier transform of objects like δ, constant functions, and polynomials.

The key observation is that $C_c^\infty \subsetneq \mathcal{S}$, and so the dual of $\mathcal{S}$ is *smaller* than $\mathcal{D}'$: every continuous linear functional on $\mathcal{S}$ restricts to one on C_c^∞, but not every distribution extends continuously to $\mathcal{S}$ (since $\mathcal{S}$ has a stronger topology than C_c^∞). The distributions that do extend are precisely those of "at most polynomial growth", which is exactly the right condition for the Fourier transform to make sense.

Definition 11.6.1: Tempered Distribution

A *tempered distribution* is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. The space of all tempered distributions is denoted $\mathcal{S}'(\mathbb{R}^n)$, equipped with the weak* topology.

Concretely, $F \in \mathcal{S}'$ means: $F : \mathcal{S} \to \mathbb{C}$ is linear, and for each sequence $\varphi_j \to \varphi$ in $\mathcal{S}$ (meaning $\sup_x |x^\alpha \partial^\beta(\varphi_j - \varphi)(x)| \to 0$ for all α, β), we have $\langle F, \varphi_j \rangle \to \langle F, \varphi \rangle$. The analogue of Proposition 11.2.2 holds: $F \in \mathcal{S}'$ if and only if there exist $C > 0$ and $N \in \mathbb{N}$ such that

$$|\langle F, \varphi \rangle| \leq C \sum_{|\alpha|,|\beta| \leq N} \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta \varphi(x)|$$

for all $\varphi \in \mathcal{S}$. The proof is analogous to the one for $\mathcal{D}'$ (using the Schwartz space seminorms $\|\varphi\|_{\alpha,\beta} = \sup |x^\alpha \partial^\beta \varphi|$ in place of the $C_c^\infty(K)$ seminorms).

Since $C_c^\infty(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n)$ is dense, every tempered distribution restricts to a distribution, giving an inclusion $\mathcal{S}' \hookrightarrow \mathcal{D}'$. The inclusion is strict: for example, e^{x^2} defines a distribution (via $\varphi \mapsto \int e^{x^2} \varphi$, which converges for $\varphi \in C_c^\infty$ since φ has compact support) but not a tempered distribution (the integral $\int e^{x^2} \varphi$ may diverge for $\varphi \in \mathcal{S}$ that merely decays rapidly).

Example 11.6.2: Tempered Distributions

1. *L^p functions.* Every $f \in L^p(\mathbb{R}^n)$ for $1 \leq p \leq \infty$ defines a tempered distribution via $\varphi \mapsto \int f\varphi$. Indeed, $|\int f\varphi| \leq \|f\|_p \|\varphi\|_q$ by Hölder, and $\|\varphi\|_q$ is controlled by finitely many Schwartz seminorms. More generally, any function f with $|f(x)| \leq C(1 + |x|)^N$ for some C, N (i.e., of polynomial growth) and locally integrable defines a tempered distribution.

2. *The delta function and its derivatives.* Since $|\partial^\alpha \varphi(0)| \leq \sup |\partial^\alpha \varphi| \leq \|\varphi\|_{0,\alpha}$, the distributions $\varphi \mapsto \partial^\alpha \varphi(x_0)$ are tempered.

3. *p.v.$(1/x)$.* We showed earlier that $|\langle \text{p.v.}(1/x), \varphi \rangle| \leq C(\|\varphi\|_\infty + \|\varphi'\|_\infty)$. Since these are Schwartz seminorms (with $\alpha = 0$), $\text{p.v.}(1/x) \in \mathcal{S}'(\mathbb{R})$.

4. *Polynomials.* Any polynomial $p(x)$ defines a tempered distribution. The estimate is $|\int p\varphi| \leq C \sup |x^N \varphi(x)|$ for large enough N.

11.6.1 The Fourier Transform on $\mathcal{S}'$

The Fourier transform on $\mathcal{S}$ satisfies $\int \hat{f} g = \int f \hat{g}$ for $f, g \in \mathcal{S}$ (this is immediate from Fubini's theorem and the definition $\hat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi}\, dx$). This identity is exactly of the transpose form we exploited in Section 3: the "transpose" of the Fourier transform is the Fourier transform itself (on $\mathcal{S}$). This motivates:

Definition 11.6.3: Fourier Transform On $\mathcal{S}'$

For $F \in \mathcal{S}'(\mathbb{R}^n)$, define $\hat{F} \in \mathcal{S}'(\mathbb{R}^n)$ by

$$\langle \hat{F}, \varphi \rangle = \langle F, \hat{\varphi} \rangle \qquad (\varphi \in \mathcal{S}).$$

This is well-defined because $\hat{\varphi} \in \mathcal{S}$ whenever $\varphi \in \mathcal{S}$ (by Corollary 10.3.4), and F is a continuous functional on $\mathcal{S}$. Moreover, $\hat{F}$ is continuous on $\mathcal{S}$ since $\varphi_j \to \varphi$ in $\mathcal{S}$ implies $\hat{\varphi}_j \to \hat{\varphi}$ in $\mathcal{S}$ (the Fourier transform is a continuous map on $\mathcal{S}$).

The operational rules of the Fourier transform extend immediately to tempered distributions by the transpose principle:

Proposition 11.6.4: Properties Of The Fourier Transform On $\mathcal{S}'$

Let $F \in \mathcal{S}'(\mathbb{R}^n)$. Then:

1. $\widehat{\partial^\alpha F}(\xi) = (2\pi i\xi)^\alpha \hat{F}(\xi)$ as tempered distributions.
2. $\widehat{x^\alpha F} = \left(\frac{-1}{2\pi i}\right)^{|\alpha|} \partial^\alpha \hat{F}$.
3. (Fourier inversion) If we define $\check{F}$ by $\langle \check{F}, \varphi\rangle = \langle F, \check{\varphi}\rangle$ where $\check{\varphi}(\xi) = \varphi(-\xi)$, then $\check{\hat{F}} = \hat{\check{F}} = F$.
4. The Fourier transform is an isomorphism from $\mathcal{S}'$ to $\mathcal{S}'$.

***Proof*:**
Each identity is verified by testing against $\varphi \in \mathcal{S}$ and using the corresponding identity on $\mathcal{S}$.

For (1): $\langle \widehat{\partial^\alpha F}, \varphi\rangle = \langle \partial^\alpha F, \hat{\varphi}\rangle = (-1)^{|\alpha|}\langle F, \partial^\alpha \hat{\varphi}\rangle$. But by the properties of the Fourier transform on $\mathcal{S}$, $\partial^\alpha \hat{\varphi}(\xi) = \widehat{(-2\pi i x)^\alpha \varphi}(\xi)$, so:

$$= (-1)^{|\alpha|}\langle F, \widehat{(-2\pi i x)^\alpha \varphi}\rangle = (-1)^{|\alpha|}\langle \hat{F}, (-2\pi i x)^\alpha \varphi\rangle = \langle (2\pi i x)^\alpha \hat{F}, \varphi\rangle.$$

Replacing the dummy variable x by ξ, this reads $\widehat{\partial^\alpha F} = (2\pi i \xi)^\alpha \hat{F}$.

Part (2) is proved similarly, and (3) follows from Fourier inversion on $\mathcal{S}$ (Theorem 10.3.2): $\langle \check{\hat{F}}, \varphi\rangle = \langle \check{F}, \hat{\varphi}\rangle = \langle F, \check{\hat{\varphi}}\rangle = \langle F, \varphi\rangle$, using $\check{\hat{\varphi}} = \varphi$. Part (4) follows from (3).

The real payoff of this theory is that we can now compute Fourier transforms that were previously out of reach. Let us do so for the most important cases.

Example 11.6.5: Fourier Transforms Of Important Distributions

1. *The delta function.* For $\varphi \in \mathcal{S}$:

$$\langle \hat{\delta}, \varphi\rangle = \langle \delta, \hat{\varphi}\rangle = \hat{\varphi}(0) = \int \varphi(x)\, dx = \langle 1, \varphi\rangle.$$

   Therefore $\hat{\delta} = 1$: the Fourier transform of a point mass is the constant function. By Fourier inversion, $\hat{1} = \delta$: the Fourier transform of the constant function 1 is the delta function. This beautifully encodes the principle that "maximally concentrated in space $\Leftrightarrow$ maximally spread in frequency".

2. *The sign function.* From $\operatorname{sgn}'(x) = 2\delta$ and the identity $\widehat{f'} = 2\pi i\xi \hat{f}$, we get $2\pi i\xi \cdot \widehat{\operatorname{sgn}} = 2\hat{\delta} = 2$, and so $\widehat{\operatorname{sgn}}(\xi) = \frac{1}{\pi i \xi}$ in $\mathcal{S}'(\mathbb{R})$. More precisely, $\widehat{\operatorname{sgn}} = \frac{1}{\pi i}\,\text{p.v.}(1/\xi)$.

3. *The Heaviside function.* Since $H = \frac{1}{2}(\operatorname{sgn} + 1)$:

$$\hat{H} = \frac{1}{2}\widehat{\operatorname{sgn}} + \frac{1}{2}\hat{1} = \frac{1}{2\pi i}\,\text{p.v.}\frac{1}{\xi} + \frac{1}{2}\delta.$$

4. *Polynomials.* From $\hat{1} = \delta$ and the identity $\widehat{x^\alpha f} = (2\pi i)^{-|\alpha|}(-\partial)^\alpha \hat{f}$, we get

$$\widehat{x^\alpha} = \left(\frac{-1}{2\pi i}\right)^{|\alpha|} \partial^\alpha \delta.$$

So the Fourier transform of a monomial is a derivative of δ.

11.7 Distributions With Compact Support

We introduced the notation $\mathcal{E}'(U)$ earlier for the set of distributions on U with compact support. This space turns out to have a very clean characterization: it is the dual of $C^\infty(U)$ (with *no* support restriction on the test functions).

Proposition 11.7.1: Characterization Of $\mathcal{E}'$

Let $F \in \mathcal{D}'(U)$. The following are equivalent:

1. F has compact support.
2. F extends (uniquely) to a continuous linear functional on $C^\infty(U)$.

Moreover, every continuous linear functional on $C^\infty(U)$ arises in this way.

***Proof*:**
(1) $\Rightarrow$ (2): Suppose $\operatorname{supp}(F) \subseteq K$ for some compact $K \subseteq U$. Choose $\chi \in C_c^\infty(U)$ with $\chi = 1$ on a neighborhood of K. For any $\psi \in C^\infty(U)$, define $\langle F, \psi \rangle := \langle F, \chi\psi \rangle$. This is well-defined since $\chi\psi \in C_c^\infty(U)$, and it is independent of the choice of χ: if χ' is another such cutoff, then $\chi - \chi' = 0$ on a neighborhood of $K = \operatorname{supp}(F)$, so $\langle F, (\chi - \chi')\psi \rangle = 0$. Continuity on $C^\infty(U)$ follows from the continuity estimate on $C_c^\infty(\operatorname{supp}(\chi))$.

(2) $\Rightarrow$ (1): Suppose F extends to a continuous functional on $C^\infty(U)$ but $\operatorname{supp}(F)$ is not compact. Then for each j, there exists $\varphi_j \in C_c^\infty(U)$ with $\operatorname{supp}(\varphi_j) \cap \{|x| > j\} \neq \emptyset$, $\operatorname{supp}(\varphi_j) \cap \operatorname{supp}(\varphi_k) = \emptyset$ for $j \neq k$ (by choosing supports far apart), and $\langle F, \varphi_j \rangle = 1$. But $\psi = \sum_j \varphi_j / j$ converges in $C^\infty(U)$ (since the supports are locally finite), and $\langle F, \psi \rangle = \sum_j \langle F, \varphi_j \rangle / j = \sum 1/j = \infty$, contradicting continuity.

The "moreover" part: given a continuous linear functional L on $C^\infty(U)$, its restriction to $C_c^\infty(U)$ is a distribution (since $C_c^\infty(K) \hookrightarrow C^\infty(U)$ is continuous for each compact K). The argument above shows this distribution has compact support, and L is its unique extension.

Proposition 11.7.2: Finite Order Of Compactly Supported Distributions

Every $F \in \mathcal{E}'(U)$ has *finite order*: there exists $N \in \mathbb{N}$ such that the continuity estimate

$$|\langle F, \varphi\rangle| \leq C \sum_{|\alpha|\leq N} \|\partial^\alpha \varphi\|_{L^\infty}$$

holds for *all* $\varphi \in C^\infty(U)$ (not just those supported in a particular compact set). The smallest such N is called the *order* of F.

***Proof*:**
Since $\mathrm{supp}(F) \subseteq K$ for some compact K, we have $\langle F, \varphi\rangle = \langle F, \chi\varphi\rangle$ for any cutoff $\chi \in C_c^\infty(U)$ with $\chi = 1$ near K. By the distribution continuity check (Proposition 11.2.2) applied on $\mathrm{supp}(\chi)$:

$$|\langle F, \varphi\rangle| = |\langle F, \chi\varphi\rangle| \leq C_{\mathrm{supp}(\chi)} \sum_{|\alpha|\leq N} \|\partial^\alpha(\chi\varphi)\|_{L^\infty}.$$

By the Leibniz rule, $\|\partial^\alpha(\chi\varphi)\|_{L^\infty} \leq C' \sum_{|\beta|\leq|\alpha|} \|\partial^\beta\varphi\|_{L^\infty(\mathrm{supp}(\chi))}$. The result follows.

The inclusion chain $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$ now has a clean interpretation:

1. $\mathcal{D}' = (C_c^\infty)'$: all distributions, no restrictions.

2. $\mathcal{S}' = (\mathcal{S})'$: "at most polynomial growth" distributions—these are the ones compatible with the Fourier transform.

3. $\mathcal{E}' = (C^\infty)'$: compactly supported distributions—these are the most "localized".

A fact we will use later: if $F \in \mathcal{E}'(\mathbb{R}^n)$, then $\hat{F}$ is not merely a tempered distribution but an actual smooth function. In fact, $\hat{F}(\xi) = \langle F_x, e^{-2\pi i x\cdot\xi}\rangle$ (where the subscript indicates the variable of the pairing) is well-defined for all ξ since $x \mapsto e^{-2\pi i x\cdot\xi}$ is in C^∞ and F extends to C^∞. Moreover, $\hat{F}$ extends to an *entire* function on $\mathbb{C}^n$, and satisfies the Paley–Wiener estimate $|\hat{F}(\xi + i\eta)| \leq C(1 + |\xi| + |\eta|)^N e^{2\pi R|\eta|}$ when $\mathrm{supp}(F) \subseteq B_R$ (compare with Corollary 10.2.15, which established the analogous result for functions).

11.8 Singular Integrals And Calderón–Zygmund Theory

In Section 11.5, we encountered the principal value distribution p.v.$(1/x)$. This object is the kernel of one of the most important operators in harmonic analysis: the *Hilbert transform*. In this section, we study the Hilbert transform from the distributional perspective and then generalize to the Calderón–Zygmund theory of singular integral operators.

11.8.1 The Hilbert Transform

Definition 11.8.1: Hilbert Transform

For $f \in \mathcal{S}(\mathbb{R})$ (or more generally $f \in L^p(\mathbb{R})$ for $1 \le p < \infty$), the *Hilbert transform* of f is defined by

$$Hf(x) = \text{p.v.}\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{f(y)}{x-y}\,dy = \frac{1}{\pi}\lim_{\varepsilon\to 0^+}\int_{|x-y|>\varepsilon} \frac{f(y)}{x-y}\,dy.$$

Equivalently, $Hf = f * K$ where $K = \frac{1}{\pi}\,\text{p.v.}(1/x)$ is the distributional convolution.

The Hilbert transform arises naturally in complex analysis (it relates the real and imaginary parts of a function holomorphic in the upper half-plane), in signal processing (it performs a 90ř phase shift), and in PDE theory (it is the prototypical example of a singular integral operator). Its distributional nature is essential: the kernel $1/(\pi x)$ is not locally integrable, so the convolution must be interpreted in the principal value sense.

The Fourier-analytic characterization of H is remarkably clean:

Theorem 11.8.2: Fourier Multiplier Characterization Of H

For $f \in \mathcal{S}(\mathbb{R})$:

$$\widehat{Hf}(\xi) = -i\,\text{sgn}(\xi)\,\hat{f}(\xi).$$

That is, H is a Fourier multiplier with symbol $m(\xi) = -i\,\text{sgn}(\xi)$.

***Proof*:**
Since $Hf = \frac{1}{\pi}\text{p.v.}(1/x) * f$ and convolution becomes multiplication under the Fourier transform (for tempered distributions), we have $\widehat{Hf} = \frac{1}{\pi}\widehat{\text{p.v.}(1/x)} \cdot \hat{f}$. From Example 11.6.5, $\widehat{\text{sgn}}(\xi) = \frac{1}{\pi i}\text{p.v.}(1/\xi)$. Taking the inverse Fourier transform of both sides: $\text{sgn}(x) = \frac{1}{\pi i}\,(\text{p.v.}\check{(1/\cdot)})(x)$, which gives $\widehat{\text{p.v.}(1/x)}(\xi) = -\pi i\,\text{sgn}(\xi)$. Therefore

$$\widehat{Hf} = \frac{1}{\pi}(-\pi i)\,\text{sgn}(\xi)\hat{f}(\xi) = -i\,\text{sgn}(\xi)\hat{f}(\xi).$$

Since $|-i\,\text{sgn}(\xi)| = 1$ for $\xi \ne 0$, an immediate consequence is:

Corollary 11.8.3: L^2 Boundedness Of H

The Hilbert transform is an isometry on $L^2(\mathbb{R})$: $\|Hf\|_{L^2} = \|f\|_{L^2}$. Moreover, $H^2 = -\text{Id}$ on L^2, so $H^{-1} = -H$.

***Proof*:**
By Plancherel: $\|Hf\|_{L^2}^2 = \|\widehat{Hf}\|_{L^2}^2 = \int |\,\text{sgn}(\xi)|^2|\hat{f}(\xi)|^2\,d\xi = \|\hat{f}\|_{L^2}^2 = \|f\|_{L^2}^2$. For the second claim: $\widehat{H^2f} = (-i\,\text{sgn})^2\hat{f} = -\hat{f}$, so $H^2f = -f$.

The deeper result, which lies at the heart of Calderón–Zygmund theory, is that H extends to a bounded operator on L^p for all $1 < p < \infty$. This can be proved using the Riesz–Thorin interpolation theorem (theorem 6.4.2) to interpolate between the L^2 bound and a suitable weak-type estimate.

Theorem 11.8.4: L^p Boundedness Of The Hilbert Transform

For $1 < p < \infty$, the Hilbert transform extends to a bounded operator $H : L^p(\mathbb{R}) \to L^p(\mathbb{R})$. That is, there exists $C_p > 0$ such that $\|Hf\|_{L^p} \leq C_p\|f\|_{L^p}$ for all $f \in L^p(\mathbb{R})$. The operator H is *not* bounded on L^1 or L^∞.

***Proof*:**
(Sketch) We already know H is bounded on L^2 with constant 1. For $p = 1$, one proves the *weak-type* $(1, 1)$ *estimate*: $m(\{x : |Hf(x)| > \lambda\}) \leq C\|f\|_{L^1}/\lambda$ for all $\lambda > 0$. This is the delicate part, and it uses the Calderón–Zygmund decomposition of f into a "good" part (bounded by λ) and a "bad" part (concentrated on a set of measure $\lesssim \|f\|_{L^1}/\lambda$). Once this is established, the Marcinkiewicz interpolation theorem (or direct Riesz–Thorin interpolation between L^2 and the weak L^1 endpoint) gives boundedness on L^p for $1 < p \leq 2$. Duality (since $H^* = -H$) extends this to $2 \leq p < \infty$. The failure at $p = 1$ and $p = \infty$ can be seen from explicit examples (e.g., $H(\chi_{[0,1]})$ has a logarithmic singularity).

11.8.2 Calderón–Zygmund Kernels

The Hilbert transform is the one-dimensional prototype of a much larger class of operators. In $\mathbb{R}^n$ for $n \geq 2$, the analogous objects are the *Riesz transforms* $R_jf = c_n\,\text{p.v.}(x_j/|x|^{n+1}) * f$, which are Fourier multipliers with symbol $-i\xi_j/|\xi|$. The general framework encompassing all of these is the Calderón–Zygmund theory.

Definition 11.8.5: Calderón–Zygmund Kernel

A *Calderón–Zygmund kernel* on $\mathbb{R}^n$ is a function $K : \mathbb{R}^n \setminus \{0\} \to \mathbb{C}$ satisfying:

1. (Size condition) $|K(x)| \leq C|x|^{-n}$ for all $x \neq 0$.
2. (Smoothness condition) $|\nabla K(x)| \leq C|x|^{-n-1}$ for all $x \neq 0$.
3. (Cancellation condition) $\sup_{0<r<R} \left|\int_{r<|x|<R} K(x)\,dx\right| < \infty$.

The associated *Calderón–Zygmund operator* is $Tf = \text{p.v.}(K) * f$, interpreted in the distributional sense.

The size condition says K has a singularity of order $|x|^{-n}$ at the origin—exactly on the boundary of local integrability in $\mathbb{R}^n$, which is why the principal value is necessary. The smoothness condition controls the regularity away from the origin, and the cancellation condition ensures that the principal value limit exists.

Theorem 11.8.6: Calderón–Zygmund Theorem

Let T be a Calderón–Zygmund operator on $\mathbb{R}^n$. If T is bounded on $L^2(\mathbb{R}^n)$, then T extends to a bounded operator on $L^p(\mathbb{R}^n)$ for all $1 < p < \infty$, and satisfies the weak-type $(1,1)$ estimate

$$m(\{x : |Tf(x)| > \lambda\}) \leq \frac{C}{\lambda}\|f\|_{L^1}.$$

The proof of this theorem is one of the landmark achievements of 20th-century harmonic analysis. The key tool is the *Calderón–Zygmund decomposition*, which splits a function into "good" and "bad" parts at a given height λ, and then estimates each part separately. The good part is handled by the assumed L^2 bound, while the bad part is estimated using the smoothness and cancellation conditions on the kernel K. The full proof can be found in Stein's *Singular Integrals and Differentiability Properties of Functions*, Chapter II. The connection to distribution theory is fundamental: the operators are defined by convolution with singular distributions, and their Fourier multiplier characterization (obtained via the Fourier transform on $\mathcal{S}'$) is the primary tool for establishing L^2 boundedness.

11.8.7 From Hilbert Transforms to the Langlands Program The Hilbert transform and Calderón–Zygmund theory may seem like purely analytic tools, but they have deep connections to number theory and representation theory. In the theory of automorphic forms, the intertwining operators that relate different representations of a reductive group are singular integrals of Calderón–Zygmund type, defined on the group manifold rather than $\mathbb{R}^n$. The L^p boundedness of these operators is closely related to the analytic continuation of Eisenstein series, a central construction in the Langlands program. In this way, the distributional framework developed here—principal values, Fourier multipliers, and singular kernels—extends far beyond classical analysis into the arithmetic world.

11.9 Sobolev Spaces

We now introduce one of the most important families of function spaces in analysis and PDE theory: the Sobolev spaces. These spaces quantify "how many derivatives a function has" in the L^2 sense, and they form the natural setting for weak solutions to differential equations. The theory of tempered distributions and the Fourier transform from the preceding sections will play a central role.

Definition 11.9.1: Sobolev Spaces (H^s)

For $s \in \mathbb{R}$, the *Sobolev space* $H^s(\mathbb{R}^n)$ is defined by

$$H^s(\mathbb{R}^n) = \left\{ f \in \mathcal{S}'(\mathbb{R}^n) : (1+|\xi|^2)^{s/2}\hat{f}(\xi) \in L^2(\mathbb{R}^n) \right\},$$

equipped with the inner product

$$\langle f, g \rangle_{H^s} = \int_{\mathbb{R}^n} \hat{f}(\xi)\overline{\hat{g}(\xi)}(1+|\xi|^2)^s \, d\xi$$

and corresponding norm $\|f\|_{H^s} = \langle f, f \rangle_{H^s}^{1/2}$.

The weight $(1+|\xi|^2)^{s/2}$ is called the *Japanese bracket* and is sometimes written $\langle \xi \rangle^s$. The intuition is: requiring $(1+|\xi|^2)^{s/2}\hat{f} \in L^2$ means that $\hat{f}$ must decay like $|\xi|^{-s}$ faster than a typical L^2 function. Since high-frequency components of $\hat{f}$ correspond to rapid oscillations (i.e., derivatives) of f, imposing more decay on $\hat{f}$ at high frequencies is equivalent to requiring f to have more derivatives. When s is a non-negative integer, this coincides with the classical requirement that f and its first s partial derivatives all lie in L^2.

Proposition 11.9.2: Properties of H^s

1. $H^s(\mathbb{R}^n)$ is a Hilbert space for every $s \in \mathbb{R}$.
2. $H^0(\mathbb{R}^n) = L^2(\mathbb{R}^n)$, with equality of norms (by Plancherel).
3. If $s > t$, then $H^s(\mathbb{R}^n) \hookrightarrow H^t(\mathbb{R}^n)$ continuously: $\|f\|_{H^t} \le \|f\|_{H^s}$.
4. $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$ for every s.
5. For $s \ge 0$ an integer, $f \in H^s(\mathbb{R}^n)$ if and only if $\partial^\alpha f \in L^2(\mathbb{R}^n)$ for all $|\alpha| \le s$ (where derivatives are taken in the distributional sense). The norms are equivalent:
$$\|f\|_{H^s} \asymp \left(\sum_{|\alpha| \le s} \|\partial^\alpha f\|_{L^2}^2 \right)^{1/2}.$$
6. (Duality) $(H^s)^* \cong H^{-s}$ via the L^2 pairing $\langle f, g \rangle = \int \hat{f}\overline{\hat{g}}$.
7. Differentiation is a bounded operator: $\partial^\alpha : H^s \to H^{s-|\alpha|}$ with $\|\partial^\alpha f\|_{H^{s-|\alpha|}} \le C\|f\|_{H^s}$.

***Proof*:**
(1) The map $f \mapsto (1+|\xi|^2)^{s/2}\hat{f}$ is an isometry from H^s into L^2. Since L^2 is complete and the map is an isometry onto a closed subspace (in fact, onto all of L^2, since for any $g \in L^2$ the function f with $\hat{f} = g/(1+|\xi|^2)^{s/2}$ lies in H^s and maps to g), H^s is complete.

(2) is immediate from the definition: $(1+|\xi|^2)^0 = 1$, so $\|f\|_{H^0} = \|\hat{f}\|_{L^2} = \|f\|_{L^2}$.

(3) For $s > t$: $(1+|\xi|^2)^{t/2} \le (1+|\xi|^2)^{s/2}$, so $\|f\|_{H^t} \le \|f\|_{H^s}$.

(4) Given $f \in H^s$, let $\chi_R \in C_c^\infty$ with $\chi_R = 1$ on B_R and $0 \le \chi_R \le 1$. Define f_R by $\hat{f}_R = \chi_R \hat{f}$. Then $\hat{f}_R \in C_c^\infty \subset \mathcal{S}$, so $f_R \in \mathcal{S}$ (by the Fourier isomorphism on $\mathcal{S}$). And $\|f - f_R\|_{H^s}^2 = \int_{|\xi|>R} |\hat{f}|^2(1+|\xi|^2)^s \, d\xi \to 0$ as $R \to \infty$ by dominated convergence.

(5) For integer $s \ge 0$: $\widehat{\partial^\alpha f}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$, so $\|\partial^\alpha f\|_{L^2}^2 = (2\pi)^{2|\alpha|} \int |\xi^\alpha|^2 |\hat{f}|^2 \, d\xi$. Since $\sum_{|\alpha|\le s} |\xi^\alpha|^2 \asymp (1+|\xi|^2)^s$ (both are polynomials of degree $2s$ with the same leading behavior), the norm equivalence follows.

(6) The pairing $\langle f, g\rangle = \int \hat{f}\hat{g}$ gives $|\langle f, g\rangle| \le \|f\|_{H^s}\|g\|_{H^{-s}}$ by Cauchy–Schwarz. That every continuous linear functional on H^s is of this form follows from the Riesz representation theorem on the Hilbert space L^2 after conjugating by the isometry $f \mapsto \langle\xi\rangle^s \hat{f}$.

(7) $\|\partial^\alpha f\|_{H^{s-|\alpha|}}^2 = \int |(2\pi i\xi)^\alpha|^2 |\hat{f}|^2 (1+|\xi|^2)^{s-|\alpha|} \, d\xi \le C \int |\hat{f}|^2 (1+|\xi|^2)^s \, d\xi = C\|f\|_{H^s}^2$, since $|\xi^\alpha|^2(1+|\xi|^2)^{-|\alpha|} \le C$.

The most important result about Sobolev spaces is the embedding theorem, which says that if s is large enough, then functions in H^s are not just L^2 but actually continuous (and even classically differentiable).

Theorem 11.9.3: Sobolev Embedding Theorem

If $s > k + n/2$ for some non-negative integer k, then $H^s(\mathbb{R}^n) \hookrightarrow C^k(\mathbb{R}^n)$ (the space of k-times continuously differentiable functions that vanish at infinity). Moreover, the embedding is continuous: there exists $C = C(s, k, n) > 0$ such that

$$\sum_{|\alpha|\le k} \|\partial^\alpha f\|_{L^\infty} \le C\|f\|_{H^s}$$

for all $f \in H^s(\mathbb{R}^n)$.

***Proof*:**

It suffices to prove the case $k = 0$ (i.e., $s > n/2$ implies $H^s \hookrightarrow C^0$ with $\|f\|_\infty \le C\|f\|_{H^s}$); the general case follows by applying this to $\partial^\alpha f \in H^{s-|\alpha|}$ with $s - |\alpha| > n/2$ (which holds when $s > k + n/2$ and $|\alpha| \le k$).

By the Fourier inversion formula (applied to $f \in \mathcal{S}$, then extended by density): $f(x) = \int \hat{f}(\xi) e^{2\pi i x\cdot\xi} \, d\xi$, and so

$$|f(x)| \le \int |\hat{f}(\xi)| \, d\xi = \int |\hat{f}(\xi)|(1+|\xi|^2)^{s/2} \cdot (1+|\xi|^2)^{-s/2} \, d\xi.$$

By Cauchy–Schwarz:

$$|f(x)| \le \left(\int |\hat{f}(\xi)|^2 (1+|\xi|^2)^s \, d\xi\right)^{1/2} \cdot \left(\int (1+|\xi|^2)^{-s} \, d\xi\right)^{1/2} = \|f\|_{H^s} \cdot C_s.$$

The constant $C_s = \left(\int (1+|\xi|^2)^{-s} \, d\xi\right)^{1/2}$ is finite precisely when $2s > n$, i.e., $s > n/2$: switching to polar coordinates, the integral behaves like $\int_0^\infty r^{n-1}(1+r^2)^{-s} \, dr$, which converges if and only if $2s - n + 1 > 1$.

This gives $\|f\|_\infty \le C_s\|f\|_{H^s}$ for $f \in \mathcal{S}$. Since $\mathcal{S}$ is dense in H^s, every $f \in H^s$ is the H^s-limit of a sequence $f_j \in \mathcal{S}$, and the estimate $\|f_j - f_k\|_\infty \le C_s\|f_j - f_k\|_{H^s} \to 0$ shows that f_j converges uniformly. The uniform limit is a continuous function that agrees with f as an element of L^2. The vanishing at infinity follows from the density of C_c^∞ in H^s.

The threshold $s = n/2$ is sharp, as the following example shows.

Example 11.9.4: The Step Function (The Boundary Case)

On $\mathbb{R}$, the threshold for $H^s(\mathbb{R}) \hookrightarrow C^0(\mathbb{R})$ is $s > 1/2$. Consider the Heaviside function $H(x) = \mathbf{1}_{[0,\infty)}(x)$. We showed that $\hat{H}(\xi) = \frac{1}{2}\delta(\xi) + \frac{1}{2\pi i}\text{p.v.}(1/\xi)$, and more precisely, for a smooth truncation $H_R(x) = H(x)\chi_{[-R,R]}(x)$ (where χ is a smooth cutoff), one has $|\hat{H}_R(\xi)| \sim 1/|\xi|$ for large $|\xi|$.

Then $\|H_R\|_{H^s}^2 \sim \int |\xi|^{-2}(1+|\xi|^2)^s\,d\xi \sim \int |\xi|^{2s-2}\,d\xi$, which converges near $\xi = 0$ (no issue) but diverges at infinity when $2s - 2 \ge -1$, i.e., $s \ge 1/2$.

Therefore $H \notin H^{1/2}(\mathbb{R})$, and a fortiori $H \notin H^s(\mathbb{R})$ for $s \ge 1/2$. The jump discontinuity at $x = 0$ prevents H from lying in H^s for s at or above the embedding threshold. This illustrates the sharpness of the Sobolev embedding theorem: at the boundary $s = n/2$, the embedding into C^0 fails, and discontinuous functions can sneak in.

11.9.5 Hearing the Shape of a Fractal Mark Kac's famous question "Can one hear the shape of a drum?" asks whether the eigenvalues of the Laplacian on a bounded domain Ω determine Ω up to isometry. In 1911, Hermann Weyl showed that the eigenvalue counting function $N(\lambda) = \#\{\lambda_j \le \lambda\}$ satisfies the asymptotic law $N(\lambda) \sim c_n\,\text{vol}(\Omega)\lambda^{n/2}$ as $\lambda \to \infty$—so one can at least "hear" the volume and dimension.

Sobolev spaces enter the picture through the connection between eigenvalue asymptotics and regularity. The Laplacian $-\Delta$ on Ω with Dirichlet boundary conditions is a self-adjoint operator on $L^2(\Omega)$, and its eigenfunctions lie in $H^s(\Omega)$ for all s (by elliptic regularity). The Sobolev embedding theorem then controls the L^∞ norms of eigenfunctions in terms of their H^s norms, which in turn controls the growth of the eigenvalue counting function.

When $\partial\Omega$ is a fractal (say, the von Koch snowflake), Weyl's law acquires a correction term involving the Minkowski content and dimension of the boundary. In Lapidus's theory of "fractal drums", the modified Weyl law takes the form $N(\lambda) = c_n\,\text{vol}(\Omega)\lambda^{n/2} - c_{n-1}\mathcal{M}^d(\partial\Omega)\lambda^{d/2} + o(\lambda^{d/2})$, where d is the Minkowski dimension of $\partial\Omega$ and $\mathcal{M}^d$ is the Minkowski content. The spectral theory of the Laplacian, Sobolev regularity, and the fractal geometry of the boundary are thus intimately intertwined. In this way, distribution theory reaches from the abstract world of generalized functions all the way to the geometry of fractals.

11.10 Applications To Partial Differential Equations

Distributions were invented by Laurent Schwartz in the 1940s precisely to provide a rigorous framework for the generalized solutions that physicists had been using informally for decades. In this section, we develop the basic theory of constant-coefficient linear PDEs from the distributional perspective, culminating in the Malgrange–Ehrenpreis theorem: every such equation has a fundamental solution.

11.10.1 Fundamental Solutions

Consider a constant-coefficient linear differential operator

$$P(\partial) = \sum_{|\alpha| \le m} a_\alpha \partial^\alpha, \qquad a_\alpha \in \mathbb{C}.$$

The associated *symbol* is the polynomial $p(\xi) = \sum_{|\alpha|\le m} a_\alpha (2\pi i \xi)^\alpha$, chosen so that $\widehat{P(\partial)f} = p \cdot \hat{f}$ for $f \in \mathcal{S}'$. Given a source term g, we wish to solve the equation $P(\partial)u = g$.

Definition 11.10.1: Fundamental Solution

A *fundamental solution* (or *Green's function*) for $P(\partial)$ is a distribution $E \in \mathcal{D}'(\mathbb{R}^n)$ satisfying

$$P(\partial)E = \delta.$$

If such an E exists and g is a distribution for which the convolution $E*g$ is well-defined (for example, if $g \in \mathcal{E}'$ or $g \in L^p$ with compact support), then $u = E*g$ satisfies $P(\partial)u = P(\partial)(E*g) = (P(\partial)E)*g = \delta * g = g$, so u is a distributional solution. Thus, the problem of solving all constant-coefficient PDEs reduces to finding a single distribution E.

Example 11.10.2: Fundamental Solutions Of Classical Operators

1. *The Laplacian in $\mathbb{R}^n$ ($n \ge 3$).* For $\Delta = \sum_{j=1}^n \partial_j^2$, the fundamental solution is $E(x) = c_n |x|^{2-n}$, where $c_n = 1/(n(2-n)\omega_n)$ and ω_n is the volume of the unit ball. This is locally integrable (since $2-n > -n$ for $n \ge 3$), and the identity $\Delta E = \delta$ is verified by a direct computation: for $\varphi \in C_c^\infty$, $\langle \Delta E, \varphi\rangle = \langle E, \Delta\varphi\rangle = c_n \int |x|^{2-n}\Delta\varphi(x)\,dx$, and integrating by parts on $\{|x| > \varepsilon\}$ and letting $\varepsilon \to 0$ yields $\varphi(0)$ (the boundary terms on the sphere $|x| = \varepsilon$ contribute $\varphi(0)$ in the limit by the mean value property and the computation of the surface integral).
2. *The Laplacian in $\mathbb{R}^2$.* Here $E(x) = \frac{1}{2\pi}\log|x|$, and the verification is similar.
3. *The heat operator.* For $P(\partial) = \partial_t - \Delta_x$ on $\mathbb{R}^{n+1}$, the fundamental solution is the heat kernel $E(x,t) = (4\pi t)^{-n/2} e^{-|x|^2/4t}$ for $t > 0$, extended by 0 for $t < 0$.

11.10.2 The Malgrange–Ehrenpreis Theorem

The examples above might suggest that finding fundamental solutions requires ad hoc computations tailored to each operator. The remarkable theorem of Malgrange (1955) and Ehrenpreis (1954), proved independently, shows that this is not the case: *every* nonzero constant-coefficient operator has a fundamental solution. This is one of the great triumphs of distribution theory.

The proof uses the Fourier transform to convert the PDE $P(\partial)E = \delta$ into an algebraic problem: taking Fourier transforms gives $p(\xi)\hat{E}(\xi) = 1$, so we need to "divide by p" in $\mathcal{S}'$. The difficulty is that p may have real zeros. The key algebraic lemma shows that this division is always possible.

Lemma 11.10.3: Division By Linear Factors In $\mathcal{S}'(\mathbb{R})$

Let $a \in \mathbb{C}$. For every $f \in \mathcal{S}'(\mathbb{R})$, the equation $(\xi - a)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R})$.

***Proof*:**
If $a \notin \mathbb{R}$, then $1/(\xi - a)$ is a C^∞ function on $\mathbb{R}$ with all derivatives bounded by powers of $1/|\operatorname{Im}(a)|$, so $u = f/(\xi - a)$ (defined by $\langle u, \varphi\rangle = \langle f, \varphi/(\xi - a)\rangle$) is a well-defined tempered distribution.

If $a \in \mathbb{R}$, we may assume $a = 0$ by translation (replacing $\varphi(\xi)$ by $\varphi(\xi + a)$). We must solve $\xi \cdot u = f$, that is:

$$\langle u, \xi\varphi(\xi)\rangle = \langle f, \varphi(\xi)\rangle \qquad \text{for all } \varphi \in \mathcal{S}.$$

The map $\varphi \mapsto \xi\varphi$ sends $\mathcal{S}$ into itself, but is not surjective: its range is $\mathcal{S}_0 = \{\psi \in \mathcal{S} : \psi(0) = 0\}$, a closed subspace of codimension 1.

We define u on $\mathcal{S}_0$ as follows. For $\psi \in \mathcal{S}_0$, the function $\psi(\xi)/\xi$ is smooth on $\mathbb{R} \setminus \{0\}$ and extends smoothly to all of $\mathbb{R}$ (since $\psi(0) = 0$: write $\psi(\xi) = \xi \int_0^1 \psi'(t\xi)\,dt$, so $\psi(\xi)/\xi = \int_0^1 \psi'(t\xi)\,dt \in C^\infty$). Moreover, $\psi/\xi \in \mathcal{S}$: for large $|\xi|$, $|\xi^k \partial^j(\psi/\xi)| \lesssim |\xi|^{k-1}|\partial^j\psi| + \text{lower} \to 0$ since $\psi \in \mathcal{S}$, and near $\xi = 0$, smoothness was just established.

So for $\psi \in \mathcal{S}_0$, define $\langle u, \psi\rangle = \langle f, \psi/\xi\rangle$. This satisfies $\langle u, \xi\varphi\rangle = \langle f, \varphi\rangle$ for all $\varphi \in \mathcal{S}$ (since $\xi\varphi \in \mathcal{S}_0$ and $(\xi\varphi)/\xi = \varphi$).

To extend u to all of $\mathcal{S}$, fix any $\varphi_0 \in \mathcal{S}$ with $\varphi_0(0) = 1$. Every $\psi \in \mathcal{S}$ decomposes uniquely as $\psi = (\psi - \psi(0)\varphi_0) + \psi(0)\varphi_0$, where $\psi - \psi(0)\varphi_0 \in \mathcal{S}_0$. Define

$$\langle u, \psi\rangle = \langle f, (\psi - \psi(0)\varphi_0)/\xi\rangle + c \cdot \psi(0)$$

for an arbitrary constant $c \in \mathbb{C}$. This is continuous on $\mathcal{S}$ (both $\psi \mapsto \langle f, (\psi - \psi(0)\varphi_0)/\xi\rangle$ and $\psi \mapsto \psi(0)$ are continuous), and satisfies $\xi u = f$ by construction. The constant c reflects the non-uniqueness: if u_1 and u_2 both solve $\xi u = f$, then $\xi(u_1 - u_2) = 0$, so $u_1 - u_2$ is supported at $\{0\}$, hence is a linear combination of δ and its derivatives—but the equation $\xi v = 0$ in $\mathcal{S}'(\mathbb{R})$ forces $v = c\delta$ for some c (since $\langle v, \xi\varphi\rangle = 0$ for all φ, meaning v vanishes on $\mathcal{S}_0$, and $\mathcal{S}_0$ has codimension 1 with the complementary functional being evaluation at 0, i.e., δ).

Lemma 11.10.4: Division By Polynomials In $\mathcal{S}'(\mathbb{R})$

Let $p(\xi)$ be a nonzero polynomial on $\mathbb{R}$. For every $f \in \mathcal{S}'(\mathbb{R})$, the equation $p(\xi)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R})$.

***Proof*:**
Factor $p(\xi) = c\prod_{j=1}^m(\xi - a_j)$ where $a_1, \ldots, a_m \in \mathbb{C}$ are the roots (counted with multiplicity). Apply Lemma **??** inductively: first solve $(\xi - a_m)u_1 = f$, then $(\xi - a_{m-1})u_2 = u_1$, and so on. After m steps, we obtain $u = u_m/c$ with $p(\xi)u = f$.

To extend this to several variables, we use the following observation: division by a polynomial in $\mathbb{R}^n$ can be reduced to division in one variable by treating the remaining variables as parameters.

Lemma 11.10.5: Division By Polynomials In $\mathcal{S}'(\mathbb{R}^n)$

Let $p(\xi) = p(\xi_1, \ldots, \xi_n)$ be a nonzero polynomial on $\mathbb{R}^n$. For every $f \in \mathcal{S}'(\mathbb{R}^n)$, the equation $p(\xi)u = f$ has a solution $u \in \mathcal{S}'(\mathbb{R}^n)$.

***Proof*:**
We proceed by induction on n. The case $n = 1$ is Lemma **??**.

For $n \geq 2$, write $\xi = (\xi_1, \xi') \in \mathbb{R} \times \mathbb{R}^{n-1}$. After a linear change of coordinates (which preserves $\mathcal{S}'$), we may assume that p has degree m in ξ_1:

$$p(\xi_1, \xi') = a_m(\xi')\xi_1^m + a_{m-1}(\xi')\xi_1^{m-1} + \cdots + a_0(\xi')$$

where a_m is a nonzero polynomial in ξ' (this can always be arranged by a generic rotation).

We proceed in two stages. First, perform polynomial long division in ξ_1: using the Euclidean algorithm for polynomials in ξ_1 (with coefficients that are distributions in ξ'), we can reduce to the case where we divide by $a_m(\xi')$, which is a polynomial in $n-1$ variables, and then divide by a monic polynomial in ξ_1 (whose roots are the roots of p as a polynomial in ξ_1).

More precisely, here is the argument. Define the "partial Fourier transform" in ξ_1 and use the one-dimensional division lemma for each fixed ξ'. The subtlety is ensuring the result depends measurably (and in fact, in a tempered fashion) on ξ'.

We present the clean version. The polynomial p can be written as

$$p(\xi_1, \xi') = a_m(\xi') \prod_{j=1}^{m} (\xi_1 - r_j(\xi')),$$

where $r_j(\xi')$ are the roots in $\mathbb{C}$ (which are algebraic functions of ξ'). To solve $pu = f$, it suffices to solve $a_m(\xi')v = f$ (where v and f are tempered distributions on $\mathbb{R}^n$, and $a_m(\xi')$ acts by multiplication in the ξ' variables), and then solve $\prod_j(\xi_1 - r_j(\xi'))w = v$ by dividing by each linear factor in ξ_1 successively.

For the first step: $a_m(\xi')$ is a nonzero polynomial on $\mathbb{R}^{n-1}$. By the inductive hypothesis, the division $a_m(\xi')v = g$ can be solved in $\mathcal{S}'(\mathbb{R}^{n-1})$ for any tempered distribution g on $\mathbb{R}^{n-1}$. Extending this to $\mathbb{R}^n$ (treating ξ_1 as a parameter) solves $a_m(\xi')v = f$ in $\mathcal{S}'(\mathbb{R}^n)$.

For the second step: each factor $(\xi_1 - r_j(\xi'))$ involves division in the ξ_1 variable. When $r_j(\xi')$ is not real for a given ξ', the division is trivial (divide by a nonvanishing smooth function). When $r_j(\xi')$ is real, we apply the one-dimensional division lemma. The argument from Lemma **??** extends to the parametric setting: the decomposition $\psi(\xi_1) = (\psi(\xi_1) - \psi(r_j(\xi'))\varphi_0(\xi_1)) + \psi(r_j(\xi'))\varphi_0(\xi_1)$ depends smoothly on the parameter ξ' (using the smooth dependence of roots on coefficients away from coalescence points, and handling the coalescence points by a partition of unity argument).

The full technical details require verifying that each step produces a tempered distribution (not merely a distribution), which follows from the polynomial growth of all quantities involved. We refer to Theorem 7.3.2 in Hörmander's *The Analysis of Linear Partial Differential Operators I* for the complete argument in the parametric case.

With the division lemma in hand, the Malgrange–Ehrenpreis theorem is an immediate consequence.

Theorem 11.10.6: The Malgrange–Ehrenpreis Theorem

Let $P(\partial) = \sum_{|\alpha| \le m} a_\alpha \partial^\alpha$ be a nonzero constant-coefficient linear differential operator on $\mathbb{R}^n$. Then there exists a distribution $E \in \mathcal{D}'(\mathbb{R}^n)$ such that

$$P(\partial)E = \delta.$$

***Proof*:**
Let $p(\xi) = \sum_{|\alpha| \le m} a_\alpha (2\pi i \xi)^\alpha$ be the symbol of $P(\partial)$, so that $\widehat{P(\partial)f} = p \cdot \hat{f}$ for $f \in \mathcal{S}'$. Taking Fourier transforms, the equation $P(\partial)E = \delta$ becomes

$$p(\xi)\hat{E}(\xi) = \hat{\delta}(\xi) = 1.$$

By Lemma **??**, the equation $p(\xi)u = 1$ has a solution $u \in \mathcal{S}'(\mathbb{R}^n)$ (taking $f = 1$, which is a tempered distribution). Define $E = \check{u}$ (the inverse Fourier transform of u). Then $E \in \mathcal{S}' \subset \mathcal{D}'$ and

$$\widehat{P(\partial)E} = p \cdot \hat{E} = p \cdot u = 1 = \hat{\delta},$$

so $P(\partial)E = \delta$ by the injectivity of the Fourier transform on $\mathcal{S}'$.

Let us pause to appreciate what we have proved. The Malgrange–Ehrenpreis theorem says that the purely algebraic problem of dividing by a polynomial in $\mathcal{S}'$ translates, via the Fourier transform, into the analytical statement that every constant-coefficient PDE has a distributional fundamental solution. The entire proof rests on three pillars: the Fourier transform on $\mathcal{S}'$ (Section 11.6), the algebraic structure of polynomial division, and the one-dimensional division lemma (Lemma **??**), which itself is a simple consequence of the codimension-1 structure of the subspace $\{\psi \in \mathcal{S} : \psi(0) = 0\}$.

Corollary 11.10.7: Solvability Of Constant-Coefficient PDEs

Let $P(\partial)$ be a nonzero constant-coefficient operator and let $g \in \mathcal{E}'(\mathbb{R}^n)$ (a compactly supported distribution). Then the equation $P(\partial)u = g$ has a solution $u \in \mathcal{D}'(\mathbb{R}^n)$.

***Proof*:**
Let E be the fundamental solution from the Malgrange–Ehrenpreis theorem. Since $g \in \mathcal{E}'$, the convolution $u = E * g$ is well-defined as a distribution (the compact support of g ensures that $\langle E * g, \varphi \rangle = \langle E_x, \langle g_y, \varphi(x+y) \rangle \rangle$ converges for every $\varphi \in C_c^\infty$). Then

$$P(\partial)u = P(\partial)(E * g) = (P(\partial)E) * g = \delta * g = g.$$

Example 11.10.8: Revisiting The Laplacian
For $P(\partial) = \Delta = \sum_j \partial_j^2$ on $\mathbb{R}^n$, the symbol is $p(\xi) = -4\pi^2|\xi|^2$. The equation $-4\pi^2|\xi|^2\hat{E} = 1$ cannot be solved by simply writing $\hat{E} = -1/(4\pi^2|\xi|^2)$, since this function is not locally integrable near $\xi = 0$ for $n \le 2$. However, the division lemma constructs a distributional solution. In this case, the solution can also be obtained by regularization: for instance, $\hat{E}(\xi) = \lim_{\varepsilon \to 0^+} -1/(4\pi^2(|\xi|^2 + \varepsilon))$, where the limit is taken in $\mathcal{S}'$. The inverse Fourier transform recovers the classical fundamental solution: $E(x) = c_n|x|^{2-n}$ for $n \ge 3$ and $E(x) = \frac{1}{2\pi}\log|x|$ for $n = 2$.

11.10.9 Distribution Theory and the Broader Landscape The Malgrange–Ehrenpreis theorem is an existence theorem: it guarantees a fundamental solution exists, but says nothing about its regularity, uniqueness (fundamental solutions are not unique in general—one can always add a solution of the homogeneous equation), or how it depends on the operator. For *elliptic* operators (those whose symbol satisfies $|p(\xi)| \geq c|\xi|^m$ for large $|\xi|$), the theory of elliptic regularity shows that distributional solutions are automatically smooth wherever the source term is smooth. For *hyperbolic* operators (like the wave operator), the fundamental solution has singularities propagating along characteristic surfaces. These deeper questions—regularity, propagation of singularities, and the fine structure of solutions—form the subject of microlocal analysis, which extends distribution theory by localizing in both position and frequency. The wavefront set of a distribution, a subset of the cotangent bundle $T^*\mathbb{R}^n$, provides the precise language for these phenomena, and opens the door to the modern theory of pseudodifferential operators and Fourier integral operators.

12

More Measures and Integrals

This last chapter will cover some additional settings in which notions of measure and integration is defined. We will examine the case of measure being defined on locally compact abelian groups and on low dimensional subsets of $\mathbb{R}^n$.

12.1 Topological Groups and Haar Measure

Definition 12.1.1: Topological Group

Let G be a group. Then G is a *topological group* if there is a topology on G such that the binary operation $(x, y) \mapsto xy$ and inversion $x \mapsto x^{-1}$ is continuous. In other words, topological groups are the group objects of **Top**.

Can you come up with an example to see why it is necessary to add this condition? Since we have a binary operation, we have a few more natural operations we may do on sets:

$$xH = \{xh \mid h \in H\} \qquad Ax = \{hx \mid h \in H\}$$
$$H^{-1} = \left\{h^{-1} \mid h \in H\right\} \qquad HK = \{hk \mid h \in H,\ k \in K\}$$

If $A \subseteq G$ such that $A = A^{-1}$, we'll say A is *symetric*

Example 12.1.2: Topological Group

1. For any group G, if G is equipped with the discrete topology it is a topological group, and all homomorphisms are continuous between any other topological group. In this way, the theory of topological groups can be thought as subsuming group theory. Any group with the indiscrete topology (also known as trivial topology) is also a topological group.

2. If $H \leqslant G$ is also a topological space, then H is a topological group with the subspace topology.

3. The real numbers $\mathbb{R}$ and $\mathbb{R}^{\times}$ under multiplication is a topological group (can you prove continuity of $+$, $\cdot$, and $(-)^{-1}$?). More generally, $\mathbb{R}^n$ is a topological group, or any *toplogical vector-space* (a vector-space where the action is continuous as well) is a topological group

4. The circle S^1, and the n-torus $(S^1)^n$ are topological groups. show that $\mathbb{R}/\mathbb{Z}$ is both homeomorphic and isomorphic to S^1. If we take S^1 as a subset of $\mathbb{C}$, then we $S^1 = U(1) = \{x \in \mathbb{C} \mid |x| = 1\}$.

5. The general linear group over the reals $\mathrm{GL}_n(\mathbb{R})$ is a topological group. Similarly, the special linear group,$\mathrm{SL}_n(\mathbb{R}) = \{A \in \mathrm{GL}_n \mid \det(A) = 1\}$, is a topological group.

6. Take $O(n) \subseteq \mathrm{GL}(n)$, where $O(n) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid AA^t = I\}$. This is a topological group known as the *orthogonal group*, since it consists of all orthonormal basis of $\mathbb{R}^n$. Similarly, we can define $U(n) = \left\{A \in \mathrm{GL}_n(\mathbb{R}) \,\middle|\, A\overline{A}^t = I\right\}$ is a topological group known as the *unitary group*. Similarly to $\mathrm{SL}_n(\mathbb{R})$, we have:

$$\mathrm{SO}(n) = \mathrm{SL}_n(\mathbb{R}) \cap O(n) \qquad \mathrm{SU}(n) = \mathrm{SL}_n(\mathbb{R}) \cap U(n)$$

   Note that all these groups are *compact*.

7. (If you know some Differential Geomtery) Any Lie group is naturally a topological group. An example of a topological group that is not a Lie group is $\mathbb{Q}$ with the subspace topology induced by $\mathbb{R}$.

8. The p-addic integers, $\mathbb{Z}_p$ has the induced inverse-limit topology (in this case, the p-addic topology). This group is compact, homeomorphic to the Cantor set, but is totally disconnected (hence cannot be a Lie Group)

9. Similarly to $\mathbb{Z}_p$, giving $\mathbb{Q}$ a different p-addic metric, we can take a different completion to form $\mathbb{Q}_p$, which is also a Lie Group.

10. In commutative algebra, it is common to define a topology on a group in the following way: Let G be a commutative group and take a descending sequence of subgroups[a]:

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n \supseteq \cdots$$

   Then let the collection of (G_n) define the *local neighbhorhood* of 0, and $x + G_n$ be an open neighbhorhood of x. Then this generate's a topology, which we'll call the *subgroup topology*. This topology is particularly important when we talk about the *completion* of groups; more on that at the end of this section.

[a] If G were not commutative, we would take a descending sequence of normal subgroups

Proposition 12.1.3: Properties Of Topological Groups

Let G be a topological group.

1. The topology of G is translation invariatn: If $U \subseteq G$ is open, then Ux and xU is open
2. For every open neighborhood of e, say for example U, there is a symmetric neighborhood V of e such that $V \subseteq U$
3. For every neighborhood U of e, ther is a neighborhood V of e with $VV \subseteq U$
4. if H is a subgroup of G, so it $\overline{H}$
5. Every open subgroup of G is also closed
6. If K_1, K_2 are compact subset of G, so is K_1K_2

Proof:

1. exercise (use continuity of binary operation)

2-3. exercise (use continuity of both operations at the identity)

4. Let $x, y \in \overline{H}$ so that there is nets $\langle x_\alpha \rangle_{\alpha \in A}$ and $\langle y_\beta \rangle_{\beta \in B}$ in H that converges to x and y. Then $x_\alpha^{-1} \to x^{-1}$ and $x_\alpha y_\beta \to xy$ (with the usual product ordering on $A \times B$), so x^{-1} and xy belong to $\overline{H}$.
5. Since H is an open subgroup, so are al xH, so $G \setminus H = \bigcup_{x \notin H} xH$ is open, hence H is closed
6. K_1K_2 is the iamge of the comacpt set $K_1 \times K_2$ under the binary operation.

Definition 12.1.4: Left And Right Translation

Let f be a continuous function on G and $y \in G$. Then a *left* translation (resp. right translation) of f is a function L_y (resp. R_y) that maps f to:

$$L_y f(x) = f(y^{-1}x) \qquad R_y f(x) = f(xy)$$

We put y^{-1} instead of y so that translation is a homomorphism: $L_{yz} = L_y L_z$ and $R_{yz} = R_y R_z$.

Definition 12.1.5: Left And Right Uniform Continuity

Let $f : G \to \mathbb{R}$ (or $\mathbb{C}$). Then f is said to be *left* (resp. *right*) *uniformally continuous* if

$$\forall \epsilon > 0,\ \exists V(e \in V),\ \|L_y f - f\| < \epsilon$$

for $y \in V$ (resp. $\|R_y f - f\|_u < \epsilon$).

Proposition 12.1.6: Compact Domain And Left/Right Uniform Continuity

Let $f \in C_c(G)$. Then f is left and right uniformly continuous.

***Proof*:**
Folland p.340

We will mainly bo working with Hausdorff groups. The following shows how this is not much of a restriction

Proposition 12.1.7: Topological Groups And Hausdorfness

Let G be a topological group. TheN

1. If G is T_1, G is Hausdorff
2. If G is not T_1, let H be the closure of $\{e\}$. Then H is a normal subgroup and G/H (with respect to the quotient topology) is hausdorff.

***Proof*:**
Folland p.341

1. here
2. We first show H is normal. Evidently, H is the smallest (or minimal) closed subgroup of G (since $\{e\}$ is the smallest subgroup of G). Then H has to be normal, since if it werent then If $H' = gHg^{-1}$ was some conjugate such that $H \neq H'$[a], then $H' \cap H$ would be a closed subgroup smaller than H. Hence, G/H is well-defined, and it is a simple to check that it indeed is a topological group with respect to the quotient topology. Then $\{\overline{e}\}$ is closed since its preimage of H. Finally, by continuity of the binary operation, the pre-image of a closed set is closed so the set in the preimage:
$$(\overline{x}, \overline{x}^{-1}) \mapsto \overline{e}$$
is closed closed in the product topology and so each $\overline{x}$ is closed making G Hausdorff.

[a]Note that H' is closed since left and right multiplication is a homeomorphism

As a consequence of proposition 12.1.7(2), every Borel measurable function on G is constant on the cosets of H, and hence is effectively already a function on G/H, and so for the most parrt we may as well work with the Hausdorff group G/H. For many of the following results, we'll require the additional restraint that G be locally compact.

Definition 12.1.8: Left/Right Invariant Borel Measure

Let G be a locally compact group. Then a Borel measure μ on G is called *left-invariant* (resp. *right-invariant*) if $\mu(xE) = \mu(E)$ (resp. $\mu(Ex) = \mu(E)$) for all $x \in G$ and $E \in \mathcal{B}_G$. Similarly, a linear function I on $C_c(G)$ is called left- or right-invariatn if $I(L_x(f)) = I(f)$ or $I(R_x f) = I(f)$ respectively for all f.

Definition 12.1.9: Haar Measure

Let G be a locally compact topological group. Then a *left* (resp. *right*) *Haar measure* on G is a nonzero left-invariant (resp. right-invariant) Radon measure μ on G.

Example 12.1.10: Haar Measure

1. Most famously, the Lebesgue measure is a (left- and right-invariant) Haar measure on $\mathbb{R}^n$
2. The counting measure is a (left and right) Haar meausre on any group with the discrete topology
3. Let $\mathbb{T}$ be the circel group, and consider $f : [0, 2\pi] \to \mathbb{T}$ defined by

$$f(t) = (\cos(t), \sin(t))$$

then we may define the Haar measure by:

$$\mu(S) = \frac{1}{2\pi} m(f^{-1}(S))$$

where m is the Lebesgue meaure on $[0, 2\pi]$. We multiplied by $(2\pi)^{-1}$ so that $\mu(T) = 1$

4. Let $G = \mathbb{R}^\times$. Then a Haar measure μ can be given by

$$\mu(S) = \int_S \frac{1}{t} dt$$

For any Borel subset $S \subseteq \mathbb{R}^\times$. For example, if $S = [a, b]$, then $\mu(S) = \log(b/a)$. This measure is indeed translatoin invariant since:

$$\mu(gS) = \mu([ga, gb]) = \log(gb/ga) = \log(b/a) = \mu(S)$$

and similarly for Sg. We may define a similar measure by taking

$$\mu(S) = \int_S \frac{1}{|t|} dt$$

5. Let $G = \mathrm{GL}_n(\mathbb{R})$. Then one may define a (left and right) Harr measure to be

$$\mu(S) = \int_S \frac{1}{|\det(X)|^n} dX$$

where X is the Lebesgue meaure on $\mathbb{R}^{n^2}$ identified with the set of all $n \times n$ matrices (this follows from the change of variables formula).

With the Haar measure, we may naturally define Haar integrals of functions, namely

$$\int f(x)d\mu(x)$$

where μ is the Haar measure.

12.2 Pontryagin Duality

In the 19th century, Joseph Fourier was trying to find out how to model the movement of heat through solid bodies, in particular model how heat seems to dissipate and homogenise within its medium. His efforts lead to the development of *Fourier series* and the *Fourier Transfrom* of periodic function with sufficient regularity properties, namely that $f \in L^2(\mathbb{T}^n)$ so that $\mathcal{F}(f) = \hat{f} \in \ell^2(\mathbb{Z}^n)$. Under the right conditions, we may generalize the Fourier transform to functions on $\mathbb{R}^n$ (usually, $f \in L^1(\mathbb{R}^n)$ or $f \in \mathcal{S}(\mathbb{R}^n)$). In this paper, we will extend the Fourier transform to locally compact abelian groups and show that many classical results of Fourier analysis hold.

In order to define the Fourier transformation on groups, we require the notion of *topological groups* which will allow us to define a Borel σ-algebra on which we will define a measure known as the *Haar measure*. With a measure, we may define the notion of *integrable function on a group G*. We next require the generalization of the domain of $\hat{f}$. When working with $\mathbb{T}^n$, the answer was "obvious" in that we had $\hat{f} : \mathbb{Z}^n \to \mathbb{R}$ since $\hat{f}$ enumerated the orthonormal basis of $L^2(\mathbb{T}^n)$. When we generlized the Fourier transform to functions $f \in L^p(\mathbb{R}^n)$ (for $1 \leq p \leq 2$), we had that the domain of $\hat{f}$ was $\mathbb{R}^n$, and in general the domain of $\hat{f}$ transformed like *functionals*. We will explore this concept when we define *Pontryagin Duality*.

After having introduced the relevant concepts, we shall introduce Fourier transforms on locally compact (Hausdorff) abelian group. Due to the Pontryagin duality, many classical results of Fourier Analysis are readily recovered. We shall not go over too much detail as this is more an essay than the beginning of a course in Harmonic Analysis, however some interesting highlights will be added.

The paper will end with some interesting uses and consequence of the Fourier transform and of the Pontryagin duality, including how it demonstrates compact abelian groups are just as hard to classify as abelian groups, how the Fourier Transform shows up in Representation Theory of finite groups, how Pontryagin Duality allows us to classify locally compact abelian groups through their characters.

A quick disclaimer that I will be assuming some 1st-year graduate analysis knowledge for this paper as I will have to touch on topics covered in those courses.

Let G be a locally compact (Hausdorff) abelian group. If G was also a topological vector space (i.e. a topological field k acts continuously on G so that G becomes a topological vector space), then we would have a natural notion of duality since G would be a vector space over some underlying field k. For general group, no such "natural dualizing" object exists. Instead, we may try to find some interesting group so that the contravariant functor between locally compact abelian groups (which we'll denote **LCA**):

$$\mathrm{Hom}(-, H) : \mathbf{LCA} \to \mathbf{LCA} \tag{12.1}$$

has properties similar to what we are used to in the dual theory we have in Functional Analysis (ex. reflexivity or properties of adjoints) and that we recover some classical results from Fourier Theory. Thanks to work by Lev Pontryagin, we have that a natural choice for H in equation (12.1) is the circle group with the usual subspace topology of $\mathbb{R}^2$. We shall henceforth label this group as $\mathbb{T}$.

Definition 12.2.1: Pontryagin Dual

Let G be a locally compact abelian group. Then

$$\widehat{G} := \operatorname{Hom}(G, \mathbb{T})$$

where $\operatorname{Hom}(G, \mathbb{T})$ is the collection of all continuous homomorphisms, is the *Pontryagin dual* of G endowed with the topology of uniform convergence of compact sets[a]. An element of $\widehat{G}$ is called a *character* of G.

[a]If you are familiar with complex analysis, this topology is commonly used for the convergence of holomorphic functions

For a better understanding of the topology on $\widehat{G}$, consider The collection of sets

$$W_K^V := \left\{ \chi \in \widehat{G} \;\middle|\; \chi(K) \subseteq V \right\}$$

where $K \subseteq G$ is a compact subset and $V \subseteq \mathbb{T}$ is a neighborhood containing the identity. Then the basis for the topology of $\widehat{G}$ is the collection of all W_K^V along with their translations. For examples of such convergence, recall power series centered at 0 ([1]) on $\mathbb{C}$ with radius R converge uniformly in $\overline{B_r(0)}$ when $r < R$, but it does not necessarily converge uniformly on $r = R$. Hence, power series technically converge in the *compact-open* topology[2].

Example 12.2.2: Pontryagin Dual

1. Let $G = \mathbb{Z}/n\mathbb{Z}$ with the discrete topology. Then:

$$\widehat{\mathbb{Z}/n\mathbb{Z}} = \operatorname{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{T}) = \mathbb{Z}/n\mathbb{Z}$$

To see this, notice that any $f \in \widehat{G}$ is determined by where it maps 1. Since G has finite order, $f(1)$ must have finite order. Necessarily, $f(1)$ maps to a root of unity $e^{\frac{2\pi i}{k}}$ where $1 \leq k \leq n$. Now it is easy to finish the computations to see that $\widehat{Z/n\mathbb{Z}} = \mathbb{Z}/n\mathbb{Z}$

2. Let $G = \mathbb{R}$. Then:

$$\widehat{\mathbb{R}} = \operatorname{Hom}(\mathbb{R}, \mathbb{T}) = \mathbb{R}$$

To see this, we will slightly modify a proof presented in Foland's *Real Analysis* Theorem 8.19. Let $\chi \in \widehat{\mathbb{R}}$. Let $a \in \mathbb{R}$ such that $\int_0^a \chi(t)dt \neq 0$. Such an a exists, or else by the Lebesgue Differentiation Theorem $\chi = 0$ a.e. Setting $A = \left(\int_0^a \chi(t)dt\right)^{-1}$, we get:

$$\chi(x) = \chi(x)AA^{-1} = A\int_0^a \chi(x)\chi(t)dx = A\int_0^a \chi(x+t)dt = A\int_x^{x+a} \chi(t)dt$$

Since the function inside the integral continuous, χ is in fact C^1 (in fact C^∞, but we will not need this fact). Now, notice that:

$$\chi'(x) = A[\chi(x+a) - \chi(x)] = B\chi(x)$$

[1]they may be centered anywhere, but we will center them at zero for simplicity
[2]To be precise, they are locally uniformly convergent, which since $\mathbb{C}$ is locally compact is equivalent to uniform convergence on compact sets

where $B = A(\chi(a) - 1)$. This is now a simple ODE, which can be solved to show we get an exponential (and is unique by the uniqueness of a solution to an ODE). For an alternative approach, notice that

$$\frac{d}{dx}(e^{-Bx}\chi(x)) = 0$$

so $e^{-Bx}\chi(x)$ is constant. Since $\chi(0) = 1$, we have $\chi(x) = e^{Bx}$, and since $|\chi(x)| = 1$, B must be purely imaginary meaning $B = 2\pi i\xi$ for some $\xi \in \mathbb{R}$, completing the proof.

We may easily generalize this proof for when $G = \mathbb{R}^n$. In this case, the character's will be of the form $\chi(x) = e^{2\pi i\xi\cdot x}$ where $\xi \cdot x$ is the dot product.

3. Let $G = \mathbb{Z}$. Then:

$$\widehat{\mathbb{T}} = \mathrm{Hom}(\mathbb{T}, \mathbb{T}) = \mathbb{Z}$$

and similarly:

$$\widehat{\mathbb{Z}} = \mathrm{Hom}(\mathbb{Z}, \mathbb{T}) = \mathbb{T}$$

For the first, recall that $\mathbb{T} \cong \mathbb{R}/\mathbb{Z}$. Then looking at the above proof, we would consider χ to be periodic with period 1 and $e^{2\pi i\xi} = 1$ if and only if $\xi \in \mathbb{Z}$. A similar argument works for the second case.

Looking at this from a second perspective, from usual Fourier Analysis we know that the Fourier transform is a map $\mathcal{F} : L^2(\mathbb{T}) \to \ell^2(\mathbb{Z})$, and so it should be expected that $\mathbb{Z}$ and $\mathbb{T}$ are Pontryagin dual to each other.

You may ask if it is always the case that that $\widehat{\widehat{\mathbb{Z}}} = \mathbb{Z}$ just like for vector-spaces. We know that in general duality theory, this is not the case (the dual of the dual may be much larger than the original object, think of $C_b(X)$). If it is the case, that means that the dual $\widehat{G}$ of G contains all the required information about G. For Pontryagin duality, it is always reflexive:

Theorem 12.2.3: Pontryagin Duality Theorem

Let G be a locally compact abelian group. Then there is a natural isomorphism:

$$G \mapsto \widehat{\widehat{G}}$$

mapping $g \mapsto \mathrm{ev}_g$ where $\mathrm{ev}_g : \widehat{G} \to G$ is defined by $\mathrm{ev}_g(\chi) = \chi(g)$.

We shall delay the proof until we have the notion of Fourier Transform for groups, and due to its complexity only present some interesting parts of it. Note that in general, $G \ncong \widehat{G}$ (think of $L^p(X) \ncong L^q(X) = (L^p(X))^*$), however if G is finite and abelian then $G \cong \widehat{G}$ (though not canonically). The essential take away from Pontryagin duality is that $\widehat{G}$ contains enough information on G to recover G.

12.2.4 Remark The map in theorem 12.2.3 can in fact be augmented to be a covariant functor (even an exact functor, i.e. it preserves short exact sequences). Hence, we may look at the study of extending group or the homology of groups in the Pontryagin dual.

The statement that the Fourier transform changes the basis from "positions" $\{\delta_x\}$ to "waves" $\{e^{ikx}\}$

hints at a broader duality: it changes the very nature of the "points" that constitute our space.

Let us define a function f by its values at geometric points. In the language of distributions, we can write any function as a continuous sum of Dirac deltas:

$$f(x) = \int_{\mathbb{T}^1} f(y)\delta_y(x)\,dy$$

Here, the building blocks are points $y \in \mathbb{T}^1$. A "point" is simply a linear functional that evaluates a function at a specific location: $\mathrm{ev}_y(f) = f(y)$.

Now consider the Fourier Transform. It maps a function f on the group G to a function $\hat{f}$ on the dual group $\widehat{G}$ (the set of characters). What are the "points" of this new space? They are the irreducible representations (the frequencies).

$$\hat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\,dx$$

When we transform a geometric point δ_y, it becomes a function on the dual group:

$$\widehat{\delta_y}(\chi) = \overline{\chi(y)}$$

This reveals that the Fourier Transform is an isomorphism between two different kinds of spaces:

- The *Physical Space* G, whose points are geometric locations x.
- The *Spectral Space* $\widehat{G}$, whose points are irreducible representations χ.

This is a specific instance of a grander theme in mathematics known as *Gelfand Duality*. In this framework, any commutative C^*-algebra (like the algebra of continuous functions) corresponds to a topological space (its spectrum). The Fourier Transform is the map that identifies the algebra of functions under convolution ($L^1(G)$) with the algebra of functions under pointwise multiplication ($C_0(\widehat{G})$). Effectively, we trade the geometry of the group for the geometry of its representation theory.

12.2.1 Generalizing the Fourier Transform

We may now combine all we have defined into the following

Definition 12.2.5: L^1 Fourier Transform On Group

Let $f \in L^1(G)$. Then define the *Fourier Transform* of f to be $\widehat{f} : \widehat{G} \to \mathbb{C}$ via

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}d\mu(x)$$

This definition is the proper generalization of Fourier transform. Let's say $G = \mathbb{R}$ with the usual Lebesgue measure. Then as we saw in example 12.2.2, χ is the map $\xi \mapsto e^{-2\pi i\xi}$ so that

$$\widehat{f}(\chi) = \int_{\mathbb{R}} f(x)e^{2\pi i\xi}dx$$

12.2.6 Remark the Fourier transform $f \mapsto \widehat{f}$ is the *Galfand Transformation* of f. If we take $A(\widehat{G}) = \left\{ \widehat{f} \;\middle|\; f \in L^1(G) \right\}$, then $A(\widehat{G})$ is a separating sub-algebra of $C_0(\widehat{G})$. It is also not to hard to see that it is self-adjoint, and so by the Stone-Weiestrass theorem it is dense in $C_0(\widehat{G})$.

By our usual Fourier theory, we recover some important properties, namely:

Proposition 12.2.7: Properties Of Abstract Fourier Transform

Let $f \in L^1(G)$ where G is a locally compact (Hausdroff) abelian group with a Haar measure μ. Then:

1. $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$
2. $\widehat{L_x f(\chi)} = \hat{f}(\chi)$
3. The Riemann-Lebesgue lemma holds

***Proof*:**

1. This is from direct computation and applying Fubini's Theorem and properties of the character:

$$\begin{aligned}
\widehat{(f * g)}(\xi) &= \int \int f(x-y)g(y)\overline{\chi(x)}dydx \\
&= \int \int f(x-y)\overline{\chi(x-y)}g(y)\overline{\chi(y)}dxdy \\
&= \hat{f}(\xi) \int g(y)\overline{(\chi(y))}dy \\
&= \hat{f}(\xi)\hat{g}(\xi)
\end{aligned}$$

2. Usually, $\mathcal{F}(L_y f)(x) = e^{-2\pi i y x} f(x)$. For the Fourier transform on groups, we get that its translation invariant since

$$\widehat{L_x f(\chi)} = \int_G f(x^{-1}y)\overline{\chi}(y)dy = \int_G f(y)\overline{\chi(y)}dy = \hat{f}(\chi)$$

3. (Riemann-Lebesgue Lemma) Notice that the map $\mathcal{F}$ mapping f to $\widehat{f}$ maps $L^1(G)$ to bounded continuous functions. Since the integral is finite and the image of the characters is $\mathbb{T}$ (namely we may bound it with $\sup_{x\in G}(\chi(x))$), $\widehat{f}$ will vanish at infinity. In fact, $\mathcal{F}$ is norm decreasing and hence is continuous. Hence, we may write

$$\mathcal{F} : L^1(G) \to BC_0(G)$$

where $BG_0(G)$ is the collection of bounded continuous functions that vanish at infinity.

One of the most important results in Fourier Analysis is the ability to recover f from $\widehat{f}$ under suitable conditions, that is the *Fourier Inversion Theorem*. This result naturally generalizes. Note that since $f \in L^1(G)$, we may treat f as being define μ-a.e., and so we may only recover f μ-a.e. in the following theorem unless f has more regularity:

Theorem 12.2.8: Fourier Inversion Theorem For Groups

Let G be a locally compact abelian group with Haar measure μ. Then there is a unique Haar measure $\widehat{\mu}$ on $\widehat{G}$ such that whenever $f \in L^1(G)$ and $\widehat{f} \in L^1(\widehat{G})$, then μ-a.e. we have:

$$f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(x)d\widehat{\mu}(x)$$

If f is continuous, then is is true for all $x \in G$

Proof:
See:

D. Ramakrishnan and R. J. Valenza, Fourier analysis on number fields (Graduate Texts in Mathematics, vol. 186, Springer, 1999), chapter 3.

Another very important result is Fourier Analysis on L^2;

Theorem 12.2.9: Plancherel Inversion Theorem For Groups

Let G be a locally compact abelian group, μ a Haar measure on G, and $\widehat{\mu}$ the dual measure on $\widehat{G}$ given in theorem 12.2.8. If $f : G \to \mathbb{C}$ is continuous with compact support, then $\widehat{f} \in L^2(\widehat{G})$ and

$$\int_G |f(x)|^2 d\mu(x) = \int_{\widehat{G}} \left|\widehat{f}(\chi)\right|^2 d\widehat{\mu}(\chi)$$

In particular, The Fourier transform is an L^2_μ isometry from $C_c(G)$ to $L^2_{\widehat{\mu}}(\widehat{G})$

Proof:
See *Fourier Analysis on Groups* by Walter Rudin, Chapter 3.

The Plancherel formula allows us to generalize the space on which the Fourier transform is defined. To see this, first note that $C_c(G)$ and $L^2(G)$ are Banach spaces. By a classical result in Real analysis, $C_c(G)$ is L^2-dense in $L^2(G)$. By proposition 2.1.11 in *Analysis Now*, there is a unique extension from $C_c(G)$ to $L^2(G)$ of the Fourier transform so that we get a *unitary operator*

$$\mathcal{F} : L^2_\mu(G) \to L^2_{\widehat{\mu}}(\widehat{G})$$

If G is compact, then $L^2(G) \subseteq L^1(G)$, and hence we get this result for general L^2-functions. If G is *not* locally compact, then to define the L^2 Fourier transform, we would start on some dense subspace like $C_c(G)$ and then extend isometrically to the whole space using Plancherel's Theorem.

Finally, we can recover the inverse Fourier transform of the dual group for the L^2 Fourier transform

Theorem 12.2.10: Dual Group Fourier Transform

Let G be a locally compact abelian group with Haar measure μ and let $\mathcal{F}$ be the L^2 Fourier transform. Then the adjoint of the Fourier Transform restricted to continuous functions on compact support is the Inverse Fourier Transform

$$L^2_{\widehat{\mu}}(\widehat{G}) \to L^2_{\mu}(G)$$

***Proof*:**
See *Fourier Analysis on Groups* by Walter Rudin, Chapter 3.

Notable example would be when $G = \mathbb{T}$ so that $\widehat{G} = \mathbb{Z}$ and the Fourier transform computes the Fourier series of periodic functions. If G is finite, then we will get the *discrete Fourier Transform* from this process.

12.2.2 Proof Pontryagin Dual

We come back now to Pontryagin duality and prove a special case of it:

***Proof*:**
(of Pontryagin Duality)

This theorem is rather difficult to prove. We shall prove that α is injective, where $\alpha : G \to \widehat{\widehat{G}}$ is defined by $g \mapsto \text{ev}_g$ (the evaluation map at g). The proof can be completed by showing that α is in fact a dense embedding onto it's image, which since α is over a locally compact Hausdorff spaces will make α an open map, and hence must map to the entire codomain, completing the proof.

Let $f \in L^1(G)$ and $\chi \in \widehat{G}$. Recall that $\widehat{L_x f} = \widehat{f}$, and so by the Fourier inversion theorem, $L_x f = f$ for all $f \in C_c^+(G)$ (the set of all continuous positive operators with compact support). Let $x \in G \setminus \{e\}$. Suppose for the sake of contradiction that $\alpha(x) = 1$, that is $\chi(x) = 1$ for all $\chi \in \widehat{G}$. We shall show that this contradicts $L_x f = f$. Since G is Hausdorff, there exists an open neighborhood U of e such that

$$U \cap (x^{-1}U) = \emptyset$$

Without loss of generality we may choose U so that it lies in a compact neighborhood of the identity. Then BY Urysohn's lemma, there exists a continuous positive definition function $f \neq 0$ with support in U. Since $f \in L^1(G)$ an f is compactly supported and positive, $f \in C_c^+(G)$. Then the support of f and $L_x f$ are disjoint, but that contradict $L_x f = f$, completing the proof.

For more information, see https://www.math.uh.edu/~haynes/files/topgps8.pdf

12.2.3 Interesting Uses

Pontryagin has a couple of very interesting consequences, perhaps a good starting point is that the classification of compact abelian groups is just as difficult as the classification of general abelian groups using the following result:

Theorem 12.2.11: Compactness Criterion For LCA Group

Let G be a locally compact abelian group. Then G is compact if and only if $\widehat{G}$ is discrete. Conversely, G is discrete if and only if $\widehat{G}$ is compact.

***Proof*:**
The fact that G is compact implies $\widehat{G}$ is discrete and that G is discrete implies $\widehat{G}$ is compact comes straight from the topology of G and $\widehat{G}$ (namely that it is the compact-open topology). To get the converse of both, we simply apply Pontryagin duality.

Proposition 12.2.12: Unit In $L^1(G)$

Let G be a locally compact abelian group with Haar measure μ. Then $L^1(G)$ has a unit if and only if G is discrete.

***Proof*:**
We know that $L^1(G)$ always has an approximate identity (think of good kernels). The identity of $L^1(G)$ would be an element such that $f * \mathrm{id} = \mathrm{id} * f = f$, so $\widehat{f} \cdot \widehat{\mathrm{id}} = \widehat{f}$. We see through this formula that $\widehat{\mathrm{id}}$ would have to be the Dirac-delta distribution at 1 with norm 1. This distribution is a function if and only if G is discrete.

Representation Theory

Another place this comes in is within representation theory. Recall in [9, chapter 24.4] there was a quick word on Fourier transforms in representation theory. In classical representation theory of finite groups, we define the character to be $\chi : G \to \mathbb{C}^*$. Since G is finite, G maps to a root the roots of unity in $\mathbb{T}$, and hence the notion of character in classical representation theory aligns with our definitions. Then we may define the inner product of characters to be:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_1(g) \overline{\chi_2}(g)$$

which will give a way to decompose representations into irreducible components via irreducible characters. If G is abelian, then may translate the decomposition theorem for representations of finite abelian groups to the decomposition of$L^2(G)$ via

$$L^2(G) = \bigoplus_{i=1}^{n} E_n$$

and in the infinite case we would have take the closure

$$L^2(G) = \widehat{\bigoplus_{i \in \mathbb{N}} E_n}$$

where E_n are the subspaces given by the usual orthonormal basis for L^2 when doing Fourier analysis. Hence, representation theory and Fourier analysis are linked!!

Class Field Theory

For those how have seen some class field theory, they have seen the Artin Reciprocity map. Recall that the idele group is a locally compact group, and so it is precisely the context in which we expect Pontryagian duality to be applied! In the local case, we have the Artin map $K^* \cong \mathrm{Gal}_K(K^{\mathrm{ab}})$ between two locally compact groups; taking the Pontryagin dual will preserve this mapping and let us work with the characters of these groups! In the global case using the Adèle ring and Idele group, we shall not get an isomorphism in the Pontryagin dual but a perfect pairing.

Part IV

Appendix, Index, Bibliography

A

Decay vs. Smooth Table for Fourier

This table represents the relationship between the smoothness of a function and the decay of its Fourier transform:

Function Space	**Symbol**	**Property of $f(x)$**	**Property of $\hat{f}(\xi)$ (Dual)**
The "Nice" Spaces (High Regularity)			
Real Analytic	$C^\omega(\mathbb{R})$	**Holomorphic Extension:** Analytic in a strip	**Exponential Decay:** $\lvert\hat{f}(\xi)\rvert \leq Ce^{-a\lvert\xi\rvert}$ ($\iff$)
Compact Smooth	$C_c^\infty(\mathbb{R})$	**Extreme Locality:** Zero outside a box	**Extreme Regularity:** Entire Analytic Function
Schwartz	$\mathcal{S}(\mathbb{R})$	**Rapid Decay:** Faster than polynomial	**Rapid Decay:** Faster than polynomial ($\iff$)
Finite Smoothness	$C^k(\mathbb{R})$	**Finite Regularity:** k derivatives exist	**Polynomial Decay:** Decays like $1/\lvert\xi\rvert^k$
The "Rough" Spaces (Low Regularity)			
Wiener Algebra	$A(\mathbb{R})$	**Continuous + Integrable:** f is bounded, continuous	**Absolute Convergence:** $\hat{f} \in L^1$ ($\implies f \in C^0$)
Square Integrable	$L^2(\mathbb{R})$	**Finite Energy:** Isometry	**Finite Energy:** Isometry ($\iff$ Plancherel)
Integrable	$L^1(\mathbb{R})$	**Finite Area:** (Riemann-Lebesgue)	**Slow Decay:** Continuous, $\to 0$ ($\not\Longrightarrow L^1$)
Distributions (Compact Support)	$\mathcal{E}'(\mathbb{R})$	**Singular:** Dirac Deltas, etc.	**Polynomial Growth:** $\lvert\hat{f}(\xi)\rvert \leq C(1+\lvert\xi\rvert)^N$ ($\iff$)

Table A.1: The Duality of Regularity and Decay in Real Analysis

This next table illustrates the "Hierarchy of Continuous Regularity". It answers the question: How jagged is the function, and how much high-frequency energy is required to describe that jaggedness?

As you move down the table, the function becomes "rougher" (more singular), and consequently, its Fourier Transform $\hat{f}(\xi)$ decays slower.

Function Class	**Property**	**Example** $f(x)$	**Decay of** $\hat{f}(\xi)$
High Regularity (Fast Decay)			
Real Analytic (C^ω)	Holomorphic Strip (Super-Smooth)	$e^{-\pi x^2}$ (Gaussian)	**Exponential** $O(e^{-a\|\xi\|})$
Smooth (C_c^∞)	infinitely diff.	$e^{-1/(1-x^2)}$ (Bump Func)	**Rapid Decay** $O(\|\xi\|^{-N})$ for all N
Finite Smoothness (C^k)	k derivatives (Continuous $f^{(k)}$)	$x^k\|x\|$ (Splines)	**Polynomial** $o(\|\xi\|^{-k})$
Medium Regularity (Algebraic Decay)			
Lipschitz ($C^{0,1}$)	Bounded Derivative (Finite Slope)	$\|x\|$ (Triangle Wave)	**Linear** $O(\|\xi\|^{-1})$
Hölder ($C^{0,\alpha}$)	Infinite Derivative (Vertical Cusp)	$\|x\|^\alpha$ ($0 < \alpha < 1$)	**Fractional** $O(\|\xi\|^{-\alpha})$
Bounded Variation (BV)	Finite Total Variation (Step / Monotonic)	$\text{sign}(x)$ (Square Wave)	**Linear** $O(\|\xi\|^{-1})$
Low Regularity (Slow Decay)			
Wiener Algebra ($A(\mathbb{T})$)	Abs. Conv. Fourier (Always Continuous)	$\sum \frac{\sin(nx)}{n^2}$ (Weierstrass)	**Summable** $\hat{f} \in \ell^1$
Continuous (C^0)	**Not** BV (Wild Oscillations)	$x\sin(1/x)$ (Damped Osc.)	**Arbitrarily Slow** $o(1)$ (Riemann-Lebesgue)
Integrable (L^1)	Finite Area	–	**Arbitrarily Slow** $o(1)$ (Riemann-Lebesgue)
Distributions ($\mathcal{D}'$)	Singular (Not a function)	$\delta(x)$ (Impulse)	**Growth** $O(\|\xi\|^N)$

Table A.2: The Precise Spectrum of Continuity and Fourier Decay (Corrected)

In this hierarchy, the decay rate $1/|\xi|^s$ tells us exactly how "smooth" the function is.

1. Integer Powers ($s = k$): These correspond to existence of Weak Derivatives. A function has a weak derivative if its slope is well-behaved enough to have finite energy (L^2), even if it has kinks or jumps. For example, the absolute value function $|x|$ has no classical derivative at 0, but its "weak derivative" is the Step Function (which is perfectly valid in physics as a force field).

2. Fractional Powers ($s = k.5$): These correspond to the Geometry of the Singularity. They tell us whether a sharp point is a "needle" (Cusp), a "corner" (Lipschitz), or a "smooth turn" (Curvature). The higher the power s, the more "damped" the high frequencies are, and the harder it is for the function to make sudden changes.

Decay Rate	**Smoothness Class**	**Allowed Singularity**	**Physical Intuition**
$1/\lvert\xi\rvert$	L^2 (Finite Energy) Discontinuous	**Jump** (Step Function)	**Teleportation** Position breaks instantly.
$1/\lvert\xi\rvert^{1.5}$	C^0 (Continuous) Infinite Slope	**Cusp** (Vertical Tangent)	**Lightning Rod** Sharpest possible point.
$1/\lvert\xi\rvert^2$	$C^{0,1}$ (Lipschitz) Finite Slope	**Corner** (Kink)	**Brick Wall** Velocity breaks instantly.
$1/\lvert\xi\rvert^{2.5}$	C^1 (Smooth) Continuous Slope	**Curvature Singularity** (Infinite Curvature)	**Tight Turn** Acceleration explodes.
$1/\lvert\xi\rvert^3$	$C^{1,1}$ (Bounded Curvature) Continuous Velocity	**Acceleration Jump** (Finite Curvature Jump)	**Kick** Force turns on instantly.

Table A.3: The Fine Structure of Fourier Decay and Singularities

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Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Local Langlands*. 2026. URL: https://nathanaelsrawley.com/.

[2] Pelczynski A. Bessaga C. "On bases and unconditional convergence of series in Banach spaces". eng. In: *Studia Mathematica* 17.2 (1958), pp. 151–164. URL: http://eudml.org/doc/216910.

[3] Marián; Habala Fabian and Marián J. Fabian. *Banach Space Theory: The Basis for Linear and Nonlinear Analysis*. eng. Springer New York, 2011. ISBN: 978-1-4419-7514-0 978-1-4419-7515-7.

[4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Partial Differential Equations*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_PDEs.pdf.

[5] Nathanael Chwojko-Srawley. *Everyting You Need To Know About Geometric Representation-Theory*.

[6] Per Enflo. "On the nonexistence of uniform homeomorphisms between Lp-spaces". en. In: 8 (2 1970), pp. 103–105. ISSN: 0004-2080. DOI: 10.1007/BF02589549. URL: http://projecteuclid.org/euclid.afm/1485894407.

[7] Per Enflo. "A counterexample to the approximation problem in Banach spaces". en. In: 130 (0 1973), pp. 309–317. ISSN: 0001-5962. DOI: 10.1007/BF02392270. URL: http://projecteuclid.org/euclid.acta/1485889774.

[8] Ezra Miller. "GEOMETRY AND MEASURE ON CAT(k) SPACES". en. In: (2024).

[9] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. 2026. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.